
SAFETY MANUAL

**Please read this safety manual carefully
before starting operation.**

Brother Industries, Ltd.

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What is the Safety Manual?

Please read

The Safety Manual should be read by all personnel involved in the installation, operation, setup, machining and maintenance of this machine. The Safety Manual refers to all people involved in any way with this product as “Users”.

Purpose

This product consists of components such as high-voltage electronic circuits, high-speed rotating tools, and a table and ATC magazine which operate with tremendous force. The machine is equipped with a variety of safety functions, but there is a strong possibility that accidents involving serious injury may occur if the machine is not handled correctly.

The purpose of this Safety Manual is to protect users from hazards associated with this product. As a result, the Safety Manual provides a description of warnings and instructions.

Warning: Warns the user of the type of danger

Safety directions to avoid danger: Tells the user what to do to avoid that danger

Safety Manual Content and Use with Other Manuals

A safety manual is included in all instruction manuals (excluding the programming manuals) that are specific to each worker or operator.

Read the Operation Manual to learn more about the machine functions or operation procedures.

1. Operation Manual (For general operators)

Safety Manual

Door Interlock Function

Operation Manual (How to operate the machine necessary for basic operations)

	Japanese	English	German	French	Chinese
Part code:	698419001	698420001	698424001	698425001	698421001

2. Operation Manual

Safety Manual

Door Interlock Function

Operation Manual (Machine operation procedures required for setup and machining)

		Japanese	English	Chinese
(Operation)	Part code:	698414001	698415001	698416001
(Data)	Part code:	698443001	698444001	698445001

3. Installation Manual

Safety Manual

Door Interlock Function

Installation Manual (Installing and assembly procedure for the machine and options)

	Japanese	English	Chinese
Part code:	698409001	698410001	698411001

4. Programming Manual

Programming Manual (Procedure for creating machining programs)

		Japanese	English	Chinese
(Conversation)	Part code:	693727001	693728001	693729001
(NC)	Part code:	693715001	693716001	693717001

Safety Labels

There are particularly important notes and items in the Safety Manual that are affixed to the machine as safety labels. A description of each safety label can be found at the end of the Safety Manual. If any of the safety labels on the machine are peeling off or worn out, get a replacement label and re-affix it.

Language

The Safety Manual and safety labels for machines shipped domestically are in Japanese. Depending on the country, the manual and labels are available in 6 different languages (English, German, French, Chinese, Portuguese and Korean) for machines shipped internationally. Request a manual and labels in a language that is understood by operators and staff. Contact the dealer where the machine was purchased for support in other languages.

How to Obtain a Safety Manual and Safety Labels

Contact the dealer where the machine was purchased if you have lost your Safety Manual or safety labels. If you can't find a dealer, use the information below to contact the dealer nearest you.

1-5, Kitajizoyama, Noda-cho, Kariya-shi, Aichi-ken, 448-0803, Japan.

BROTHER INDUSTRIES, LTD.

Phone +81-566-95-0075

Fax +81-566-25-3721

Handling Your Safety Manual

Be careful not to lose your Safety Manual and always make sure that it is available and on hand. In addition, if the owner of the machine changes, please give the manual to the new owner along with the machine.

Hazard Level

The hazard level is classified into three categories depending upon the level of damage that may occur when warnings are not observed.

1. DANGER



If this is not observed, death or an unrecoverable physical disability may occur.

2. WARNING



If this is not observed, serious injury may occur.

Order of Explanation

An explanation of the dangers and warnings are provided in the following order.

1. Keyword (danger, warning or note) and pictograph which express the hazard level.
2. Type of danger (subject)
3. Potential damage or injury
4. Safety directions to avoid danger

Pictograph Meaning

The warnings and instructions in the safety labels and Safety Manual use pictographs to warn the user of the intuitive dangers and to provide the user with safety directions to avoid those dangers. The section here shows the pictographs and their meanings.

1. Danger



Slippery



Objects Shoot Out



Risk of Falling



Flammable



Keep Hands Clear



Uneven Surface



Electric Shock Risk



Beware of Moving Parts



Burn Hazard



Inhalation Hazard



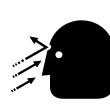
Keep Clear of Moving Parts



Rotating Tool



Moving Blade



Objects Shoot Out



Falling Objects

2. Safety Directions to Avoid Danger



Ear Protection Required



Gloves Required



Safety Boots Required



Hard Hats Required



Goggles Required



Disconnect Power Plug



Connect Ground



Task Requires 2 People



Prohibited Act



Do Not Disassemble



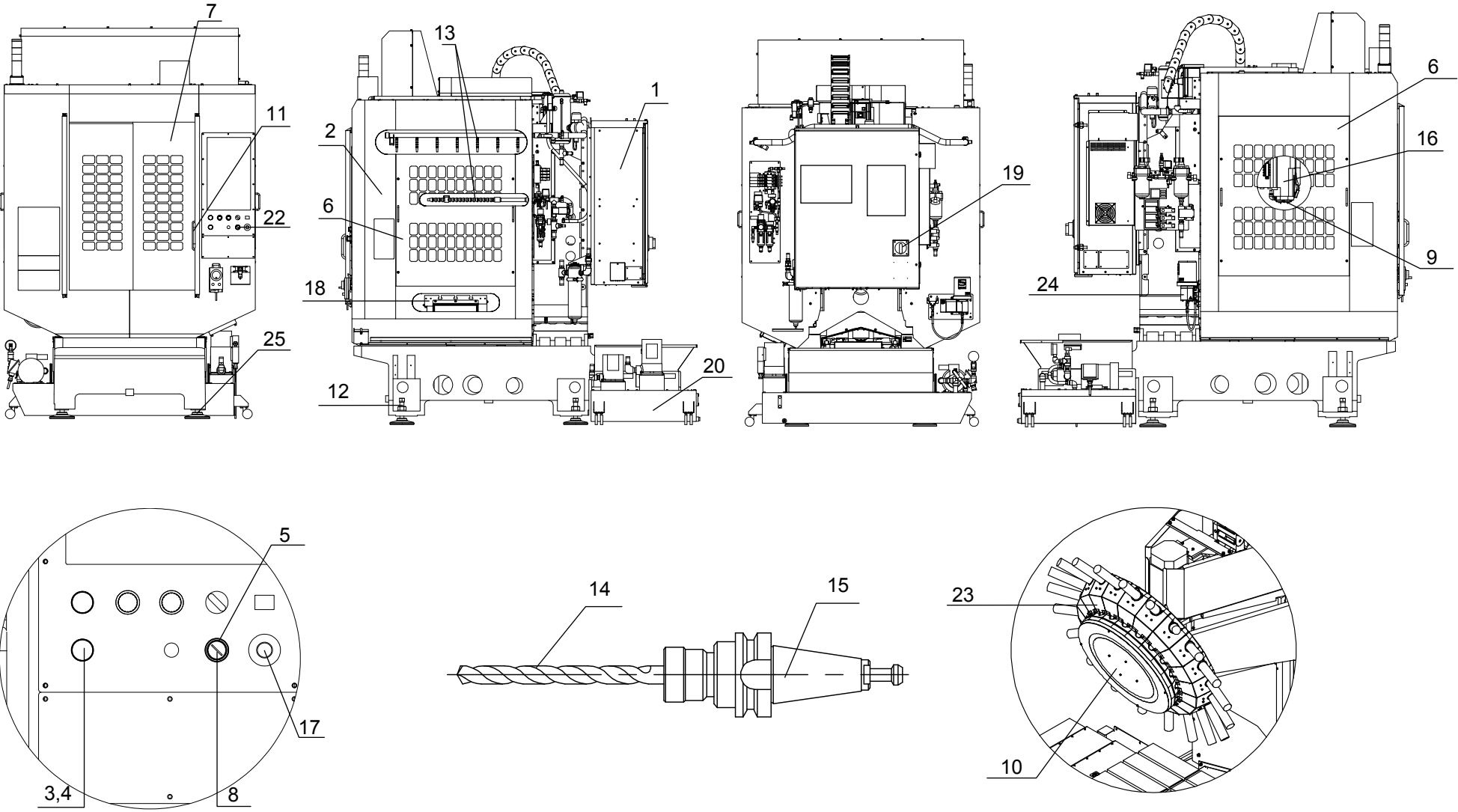
No Gloves



Do Not Touch



Keep Off



No.	各部の名称 日本語	Part names English	部件名称 中文
1	制御箱	control box	控制箱
2	機械カバー	machine cover	机械外罩
3	フィードホールドスイッチ	feed hold switch	进给保持开关
4	フィードホールドランプ	feed hold lamp	进给保持指示灯
5	ドアインタロックスイッチ	door interlock switch	操作门连锁开关
6	メンテナンスカバー	maintenance cover	维修罩盖
7	正面扉	front door	正门
8	ドアインタロックキー	door interlock key	操作门连锁钥匙
9	スピンドル	spindle	主轴
10	ATC マガジン	ATC magazine	ATC 刀库
11	取っ手	handle	把手
12	レベルリングボルト	leveling bolt	水平调节螺栓
13	クーラントノズル	coolant nozzle	冷却液喷嘴
14	工具	tool	刀具
15	工具ホルダー	tool holder	刀具夹
16	スピンドルヘッド	spindle head	主轴头
17	非常停止スイッチ	emergency stop switch	紧急停止开关
18	テーブル	table	工作台
19	主電源ブレーカ	main power breaker	主电源开关
20	クーラントタンク/チップコンベア	coolant tank /chip conveyor	冷却液罐/排屑器
21	側面扉	side door	侧门
22	操作パネル	operation panel	操作板
23	グリップカバー	grip cover	夹钳套
24	自動間欠給油ポンプ	automatic intermittent lubricating unit	自动间歇加油泵
25	アンカープレート	anchor plate	地基板

Warnings & Safety Directions to Avoid Danger

Numbers inside the parentheses or brackets, such as (1) and (2), correspond to the numbers for their respective part names.

DANGER



- D1 If any modifications are carried out, such as disabling the limit switch for the door interlock, the safety devices may stop operating. Therefore, the machine may operate even when a door (7) (21) (26) (27) is open, and you may get caught or drawn into the machine.
- D1-1 Do not modify the safety devices. Do not secure the safety devices in such a way that will prevent them from operating.
- D1-2 If modifications are necessary, contact Brother Industries to obtain written approval before proceeding.
- D1-3 If an operator finds that the machine has been modified, they should notify a supervisor immediately, without operating the machine.
- D1-4 If there is an inner door (26), lock the inner door (26) with the key before using the machine.
- D1-5 If there is an inner door (26), do not open the inner door (26) while the axis in the machining room is operating.



- D2 If you put any parts of your body inside the machine while the maintenance cover (6) has been removed for setup or maintenance work, you may get caught or drawn into the machine if the machine is operated by mistake.
- D2-1 Do not put any part of your body inside the machine while it is operating.
- D2-2 Turn OFF the main power breaker (19) before removing the maintenance cover (6).
- D2-3 Attach a padlock to the main power breaker (19) so that the power cannot be turned ON.
- D2-4 A sign or notice should be placed near the operation panel (22) of the machine to warn others that work is in progress.



- D3 If the emergency stop switch (17) is damaged, the emergency stop may not function.
- D3-1 Check the emergency stop switch (17) operation before starting machine operation.
- D3-2 If the emergency stop switch (17) is damaged or does not function, it should be replaced.



- D4 If oil-based coolants are used during cutting, the cutting area may become hot and sparks may be generated.
- D4-1 When using oil-based coolants, a fire alarm box and fire extinguishing equipment must always be on hand for use.
- D4-2 An operator must always monitor the machine while cutting is in progress.
- D4-3 Do not use a combustible coolant.



D5 Using water-soluble coolants on combustible metals (for example, magnesium, titanium or aluminum) can produce hydrogen, and when the hydrogen ignites, it can lead to an explosion and fire.

D5-1 Use coolant for combustible metals or oil-based coolant.



D6 There is risk of compromising the safety when disassembling and modifying parts which are not described in the manual.

D6-1 Do not disassemble or modify parts which are not described in the manual.

⚠ DANGER (For installer / machine setup specialist)

The following warnings and precautions are for installers and machine setup specialists.
These warnings and precautions are not for work performed by operators.



- ID1 High-voltage components are present inside the control box. There is risk of electric shock if you touch these components by mistake.
- ID1-1 Maintenance and inspection of electrical components must only be carried out by a qualified electrician who is trained in electrical safety and who has thorough knowledge of the electronic circuits in this machine.
- ID1-2 Turn OFF the main power breaker (19), and then wait at least 20 minutes before carrying out work.
- ID1-3 Attach a padlock to the main power breaker (19) so that the power cannot be turned ON.
- ID1-4 A sign or notice should be placed near the operation panel (22) of the machine to warn others that work is in progress.
- ID1-5 If leaving the machine unattended, close the control box (1) and secure it with screws.
- ID1-6 Use a step or step-ladder to provide a secure foot hold when the working position or conditions are awkward or out of reach.

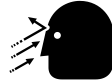


- ID2 If the power supply is not grounded, there is risk of electric shock because the leakage current breaker will not operate.
- ID2-1 Connect the ground according to the specified method.
- ID2-2 The PE line for the power line is longer than the other lines (L1, L2 and L3), and therefore, all of the slack should be taken up when the line is connected.

WARNING



- W1 If the door interlock switch (5) is disabled, the machine will operate even when a door (7) (21) (26) (27) is open, and you may get caught or drawn into the machine.
- W1-1 The installer or setup personnel must never leave the machine unattended while the door interlock switch (5) is disabled.
- W1-2 The supervisor must not let the operator use the machine when the door interlock switch (5) is disabled.
- W1-3 The operator must check that the door interlock switch (5) is enabled before operating the machine.
- W1-4 The operator must inform the supervisor if the door interlock key (8) is attached.
- W1-5 The door interlock key (8) should be kept by the supervisor.
- W1-6 If there is an inner door (26), lock the inner door (26) with the key before using the machine.
- W1-7 If there is an inner door (26), do not open the inner door (26) while the axis in the machining room is operating.



- W2 If the maintenance cover (6) is removed, coolant coming from inside the machine when it is operating may get into your eyes, or the tools (14) or workpieces may shoot out causing injury.
- W2-1 The operator should visually check to make sure that the maintenance cover (6) is attached before turning ON the power.
- W2-2 The operator should warn the supervisor if the maintenance cover (6) is not attached.
- W2-3 The supervisor must attach the maintenance cover (6).
- W2-4 Take the necessary precautions even when the maintenance cover (6) is attached, because dangerous situations, such as an explosion caused by combustible materials, cannot always be prevented.



- W3 If tools and other items are left in or around the machine, they may fall down or shoot out when the machine is operating, and injury may occur.
- W3-1 The installer should not leave tools or other items in or around the machine.
- W3-2 The operator must not operate the machine if tools or other items have been left in or around the machine. If this is not observed, the supervisor should be notified.



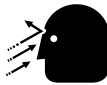
- W4 If this is not observed, someone may trip over the machine, or nearby equipment, or slip on coolant and fall over while setup or maintenance work is being carried out.
- W4-1 Protective equipment such as safety footwear and helmets must be worn while working.
- W4-2 The area around the machine must be organized, tidy and clean before performing work.
- W4-3 Any excess length of cable should be removed so that there is no slack. If cables are running along the floor, they must be covered.



- W5 There are movable parts inside the machine such as the spindle head (16) and table (18), and your body may get caught on them if you go inside the machine.
- W5-1 Do not go inside the machine while the machine is operating.
- W5-2 The operator should visually check to make sure that there is nobody inside the machine before starting the machine.
- W5-3 Do not go inside the machine. If you need to go inside the machine, the supervisor must be notified.
- W5-4 If you go inside the machine, turn OFF the main power breaker (19), and then attach a padlock to the main power breaker (19) so that the power cannot be turned ON.
- W5-5 A sign or notice should be placed near the operation panel (22) of the machine to warn others that work is in progress.



- W6 When opening and closing a door (7) (21) (26) (27), do not insert your fingers into the space between the door panels or between the door and the machine cover (2). Otherwise, your fingers may get caught causing injury.
- W6-1 Always be sure to hold the handle (11) when opening and closing a door (7) (21) (26) (27).
- W6-2 If you insert your fingers in between a door (7) (21) (26) (27) and the machine cover (2), they may get caught causing injury.



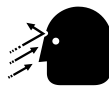
- W7 The coolant may shoot out and injure your eyes while work is being carried out.
- W7-1 Always be sure to wear protective goggles.
- W7-2 If coolant gets into eyes, rinse with clean water and then seek medical advice.



- W8 When replacing the setup equipment and the tool (14), the blade of the tool (14) or other parts may cut your hands.
- W8-1 Do not hold the tool (14) by the blade. Wear leather gloves and hold the tool holder (15) with both hands.
- W8-2 Do not insert your fingers in between the tool pots (28).
- W8-3 Do not insert your fingers in between the tool (14) or the tool holder (15) and the ATC magazine (10).
- W8-4 Wear leather gloves when working near the tool (14).
- W8-5 Use a step or step-ladder to provide a secure foot hold when the working position or conditions are awkward or out of reach.



- W9 If items such as the tool (14) and workpieces are not properly attached, or if cutting conditions are not within the tool (14) limits, items may shoot out from inside the machine and strike personnel causing injury.
- W9-1 Always read the tool labels and instruction manuals for information on tool (14) limits.
- W9-2 Check that the tips are securely attached before installing the tool (14).
- W9-3 Workpieces should be securely attached.
- W9-4 Do not rotate the lathe spindle while it is unbalanced.
- W9-5 Close the doors (7) (21) (26) (27) before operating the machine.
- W9-6 Take the necessary precautions even when the doors (7), (21), (26) and (27) are closed, because dangerous situations, such as an explosion caused by combustible materials, cannot always be prevented.



- W10 If an air blower or similar equipment is used to remove shavings, the shavings may shoot out and get into your eyes.
- W10-1 Always wear protective goggles when using an air blower or similar equipment.
- W10-2 Shavings should not be blown off the machine using an air blower or similar equipment.



- W11 Working in noisy conditions can cause hearing loss.
- W11-1 Always be sure to wear ear plugs when working in noisy conditions.



- W12 Keep away from objects which are moving or gyrating, otherwise you may get caught by them causing injury.
- W12-1 Keep away from the tool (14), spindle (9), ATC magazine (10) and chip auger (31) while they are rotating.
- W12-2 Do not hold the rotating parts.
- W12-3 Wear tight-fitting clothing. Long hair should be tucked away securely underneath caps or helmets.
- W12-4 Do not wear leather gloves except when performing setup work while the machine is stopped.
- W12-5 Do not wear accessories.
- W12-6 Do not modify the safety devices. Do not secure the safety devices in such a way that will prevent them from operating.
- W12-7 If modifications are necessary, contact Brother Industries to obtain written approval before proceeding.
- W12-8 If an operator finds that the machine has been modified, they should notify a supervisor immediately, without operating the machine.
- W12-9 If there is an inner door (26), lock the inner door (26) with the key before using the machine.
- W12-10 If there is an inner door (26), do not open the inner door (26) nor the side door (21) while the axis in the machining room is operating.



W13 You may fall down and injure yourself if you climb onto the machine, the chip conveyor (20) or the coolant tank (20).

W13-1 Do not climb onto the machine, the chip conveyor (20) or the coolant tank (20).



W14 If the machine is set up without taking into account the clearance when the table (18) rotates relative to the jig, etc., a partition, etc., may interfere with the jig. As a result, the jig may break off and shoot out causing injury.

W14-1 The operator should visually check to make sure that the maintenance cover (6) is attached before turning ON the power.

W14-2 The operator should warn the supervisor if the maintenance cover (6) is not attached.

W14-3 The supervisor must attach the maintenance cover (6).

W14-4 Take the necessary precautions even when the maintenance cover (6) is attached, because dangerous situations, such as an explosion caused by combustible materials, cannot always be prevented.



W15 Touching the rotary joint while it is hot may lead to a burn injury.

W15-1 Wait at least 1 hour after machine operation has stopped before touching the rotary joint directly with your hands.



W16 Touching the tool, chips, shavings or workpieces immediately after machining may lead to a burn injury due to the heat generated from cutting.

W16-1 Do not use your hand to directly touch parts that generate or emit heat.



W17 If an error occurs on the pneumatic control device, the worker may get caught in or wedged by the tool pot (28) and pot shutter (29), causing injury.

W17-1 Do not put your hand to movable parts on the tool pot (28) and the pot shutter (29) at the ATC standby position.

⚠ WARNING (For installer / machine setup specialist)

The following warnings and precautions are for installers and machine setup specialists. These warnings and precautions are not for work performed by operators.



- IW1 If the control box (1) and operation panel (22) are wet, an electric shock or short-circuit may cause a fire.
- IW1-1 The cover to the control box (1) must be closed while the machine is operating.
- IW1-2 The control box (1) and the operation panel (22) must be kept free of coolant, water and shavings.
- IW1-3 Do not touch the control box (1) and the operation panel (22) with wet hands.



- IW2 If this warning is not heeded, personnel may accidentally bump into the machine or nearby equipment, while setup and maintenance work is being carried out, and cause injury.
- IW2-1 Protective equipment such as safety footwear and helmets must be worn while working.
- IW2-2 Turn OFF the main power breaker (19), and then attach a padlock to the main power breaker (19) so that the power cannot be turned ON.
- IW2-3 A sign or notice should be placed near the operation panel (22) of the machine to warn others that work is in progress.



- IW3 While setup or maintenance work is being carried out, coolant may shoot out of hoses which have been removed or which are damaged, and it may get into your eyes.
- IW3-1 Always be sure to wear protective goggles.
- IW3-2 Release the coolant pressure before carrying out such work.
- IW3-3 When replacing parts or disconnecting hoses from the coolant equipment, the coolant may shoot out and injure your eyes.
- IW3-4 Handle the hoses carefully so that they are not subjected to any impacts.
- IW3-5 If coolant gets into eyes, rinse with clean water and then seek medical advice.



- IW4 High-pressure air escaping from damaged sections of air hoses or from valves while setup or maintenance work is being carried out may cause injury to your eyes or ears.
- IW4-1 Always be sure to wear protective goggles.
- IW4-2 Always be sure to wear ear plugs.
- IW4-3 Connecting and changing over air hoses containing high-pressure air must only be performed a qualified technician who has been trained in handling high-pressure air with thorough knowledge of this machine.
- IW4-4 Always be sure to disconnect the pressure source and reduce the pressure before connecting high-pressure air hoses.



- IW5 When carrying heavy objects, be careful not to drop anything on your feet, strain your back or lose your balance and fall over.
- IW5-1 Protective equipment such as safety footwear and helmets must be worn when working.
- IW5-2 Ask for assistance when carrying heavy objects.



- IW6 If using a three wire type sensor for power supply lines, an open circuit or short-circuit may cause a fire.
- IW6-1 Installers should always use two wire type sensors instead of three wire type sensors.
- IW6-2 If the installer must use a three wire type sensor, a separate fuse (1 A or less) should be added to the power supply line.



- IW7 If terminal connections on the PCBs or terminal block screws become loose, poor contacts may cause overheating or a fire.
- IW7-1 There should be no loose screws when connecting the wiring.
- IW7-2 Do not replace fuses and electrical parts that have been soldered to the inside of the unit.



- IW8 If thermal settings are changed, the protection equipment may not operate and may cause a fire.
- IW8-1 The installer should always check the set values when replacing parts after delivery.



- IW9 If the machine is moved while the cables are still connected, an open circuit may cause electric shock.
- IW9-1 The installer should disconnect all cables from the main machine unit, the coolant tank (20), the chip conveyor (20) and the peripheral equipment before moving the machine.



- IW10 When lifting or performing leveling work during machine setup, the machine may overturn or the jack may become loose, and a person may become trapped by the machine.
- IW10-1 Only a forklift with long forks that can bear the full weight of the machine should be used.
- IW10-2 The machine must be set up on a stable, level surface.
- IW10-3 The installer must make sure that the floor meets all of the following requirements.
- The floor must be able to bear a weight equal to the weight of a machine without any problem.
 - The floor must be level, with no ledges or depressions.
 - The floor surface must be sufficiently rigid, and able to withstand an applied pressure of 1.0 MPa (10 kgf/cm²) without deforming.
 - The floor must be in good condition and not deteriorate even after prolonged use.
- IW10-4 The installer should perform leveling work on the floor, and should use tools to perform adjustments and should never place his or her hands underneath the machine.
- IW10-5 The installer should use a jack at the center-front part of the machine's base.



- IW11 If the machine is used in conditions where coolant mist may be generated and there is no mist collector present or the mist collector has low performance, the coolant mist may be inhaled and have adverse effects on your health.
- IW11-1 When using CTS equipment or if coolant mist is being generated, always be sure to use a mist collector.
- IW11-2 Clean the filter for the mist collector regularly.



- IW12 If it is dark inside the machine, you may cut your hand on the jig or the tool (14).
- IW12-1 If the illumination on the tip of the tool (14) is less than 500 lux, a light should be placed inside the machine.



- IW13 If you touch cables which have been crushed or damaged by heavy objects, there is risk of electric shock.
- In addition, short-circuits may occur if cables are crushed or damaged by heavy objects.
- IW13-1 Cables should be gathered up or arranged to prevent them from being crushed. If a cable has become damaged, it should be replaced with a new one.



- IW14 Do not climb onto the machine or other nearby equipment when performing installation or maintenance, otherwise you may fall down and injure yourself.
- IW14-1 Do not climb onto the machine or other nearby equipment, otherwise you may fall down and injure yourself. Do not climb onto the machine or nearby equipment.



- IW15 Do not touch shavings directly with your hands when cleaning them away. Otherwise, you may injure yourself by cutting your hand on the shavings.
- IW15-1 Do not touch the shavings and workpieces directly with your hands.
- IW15-2 Wear leather gloves and use a brush when cleaning away the shavings.
- IW15-3 Ask for assistance when carrying heavy objects, such as the chip receiver.



- IW16 There is risk of electric shock if you touch high-voltage components by mistake while performing work inside the control box (1) with the main power breaker (19) turned ON.
- IW16-1 Maintenance and inspection of electrical components must only be carried out by a qualified electrician who is trained in electrical safety and who has thorough knowledge of the electronic circuits in this machine.
- IW16-2 A sign or notice should be placed near the operation panel (22) of the machine to warn others that work is in progress.
- IW16-3 If leaving the machine unattended, close the control box (1) and secure it with screws.
- IW16-4 Use a step or step-ladder to provide a secure foot hold when the working position or conditions are awkward or out of reach.



- IW17 Your hand may get drawn into the machine if you put it into the discharge port on the conveyor while the chips are cleaned during automatic operation.
- IW17-1 A chip bucket with a high edge on at least 3 sides must be set up at the bottom of the chip discharge port for the chip conveyor (20).
- IW17-2 The chip conveyor (20) must be stopped first before replacing the chip bucket.
- IW17-3 Mark the floor (black/yellow safety tape) so that others can identify the setup location of the chip bucket.



- IW18 If you stand on the chip conveyor (20), chip auger (31) or the coolant tank (20), you may slip or trip on the pumps and fall down.
- IW18-1 Take the necessary precautions to avoid the equipment from overturning if performing work while standing on the chip conveyor (20), chip auger (31) or the coolant tank (20).



- IW19 Your hand may get drawn into the machine if you put it into the chip feed port on the chip conveyor (20) or chip auger (31).
- IW19-1 Do not put your hand inside the chip feed port on the conveyor when the power is turned ON.



- IW20 Touching the control box or operations box during installation or maintenance work may lead to a burn injury due to the machine heating up.
- IW20-1 Wait 20 minutes after turning OFF the breaker before touching parts inside the control box or operations box.



- IW21 Touching the motor immediately after operation when performing installation or maintenance work may lead to a burn injury due to the machine heating up.
- IW21-1 Wait 30 minutes after operation before touching parts that generate or emit heat.



- IW22 If the machine cover (2) or the middle partition is modified, coolant coming from inside the machine when it is operating may shoot out and get into your eyes, or the tools (14) or workpieces may shoot out causing injury.
- IW22-1 Do not modify the machine cover (2) or the middle partition.
- IW22-2 If modifications are necessary, contact Brother Industries to obtain written approval before proceeding.



- IW23 When the machine is equipped with the lathe machining function and the ceiling cover is removed, tools or workpieces may shoot out from inside the machine during operation causing injury.
- IW23-1 The installer should visually check to make sure that the ceiling cover is securely attached before turning on the power.
- IW23-2 The supervisor must ensure that the ceiling cover is attached.



- IW24 Your hand may get drawn into the machine if you put it into the discharge port on the chip auger (31).
- IW24-1 Do not put your hand inside the discharge port on the conveyor when the power is turned ON.
- IW24-2 During automatic operation, keep the discharge port covered on the chip conveyor (and other conveyors) and ensure there are no gaps so that a worker cannot insert his/her hand.



- IW25 There are pneumatic drive devices (28)(29) inside the machine, and your body may get caught on them if you go inside the machine.
- IW25-1 The operator should visually check to make sure that there is nobody inside the machine before connecting the primary pressure source.
- IW25-2 When working close to the pneumatic drive devices (28)(29), use the pressure adjustment knob on the air source regulator (30) to set the pressure to zero, and attach the cover to the regulator (30).
- IW25-3 Turn OFF the primary pressure source.
- IW25-4 A sign or notice should be placed near the regulator (30) on the machine to warn others that work is in progress.

Precautions Related to Machine Tool Fires

Always read before use.

1. Usage Precautions

- 1-1 Handling coolant

We recommend using water-soluble coolant in order to prevent fires from starting and/or spreading. However, there is danger of an explosion and fire if ignition occurs because using water-soluble coolants when machining on combustible metals (for example, magnesium, titanium or aluminum) can produce hydrogen. Therefore, use metalworking or oil-based coolants. We recommend installing an automatic fire extinguishing system when using metalworking or oil-based coolants. Contact the manufacturer for the automatic fire extinguishing system for details on the installation.
(Note) Oil-based coolant cannot be used on the CTS.
- 1-2 Support during machining

Do not let the machine operate unattended.
Regularly monitor the machine for normal operation during machining. In addition, to be prepared in the event of a fire, we recommend installing an automatic fire extinguishing system on the machine unit and on peripheral devices when necessary. Assuming that there may be circumstances in which a fire cannot be extinguished, install a fire detection and alarm system at the location where the machine is installed.
- 1-3 Cutting conditions

When machining under cutting conditions that lead to an excessive load on the machine, the chips or shavings become hot and can cause a fire. Machine using cutting conditions that are appropriate for the workpiece and tool combination. In addition, we recommend replacing the tools regularly in order to maintain the appropriate cutting conditions.
- 1-4 NC program check

Avoid programming which uses tool path instructions that cause interference, or which use continuous operation or cutting instructions that lead to an excessive load on the machine and thus result in overheating or a fire. In addition, execute a dry run before starting the program, and make sure that there are no problems before machining in order to avoid overloads or interference caused by program errors or the wrong tool offset amount. Furthermore, use tool life management in the program for optimum tool use, because tool wear or breakage can prevent the cutting conditions from being met and thus cause an overload.
- 1-5 Machine setup check

Be sure to check the machine setup carefully. If you fail to check the machine setup, there is risk of the spindle and workpieces colliding or an overload may occur, which can lead to overheating and/or a fire. Be sure to check and manage the protruding length of the tool and the tool diameter. In addition, make sure that the attachment position of the workpiece, the selected jig and the clamp position of the workpiece match the NC program.
- 1-6 Tool breakage and abnormal wear support

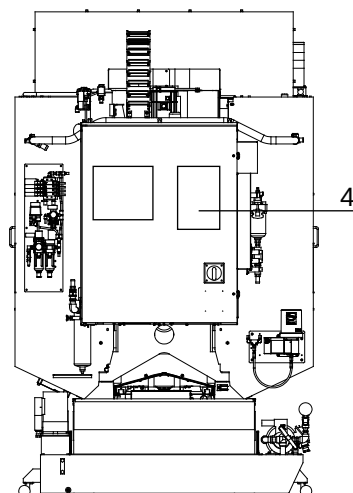
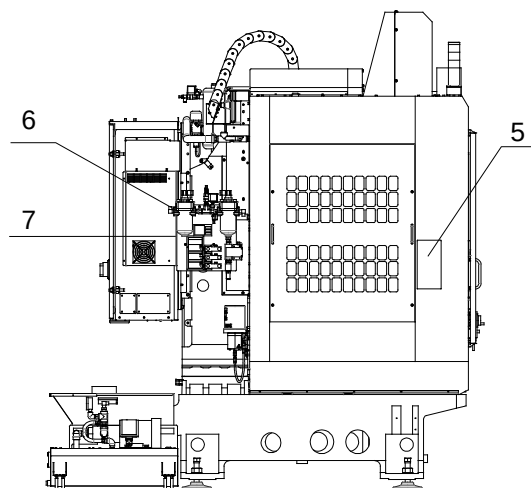
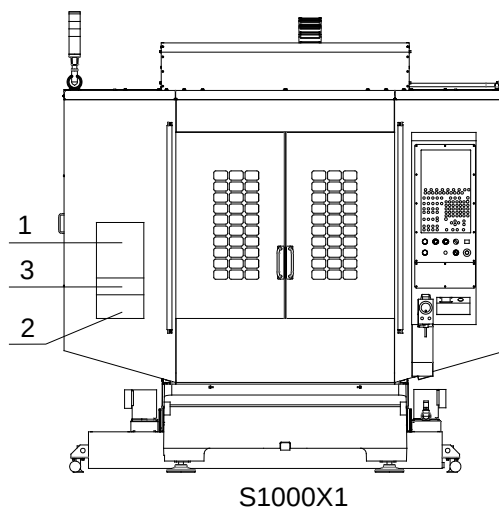
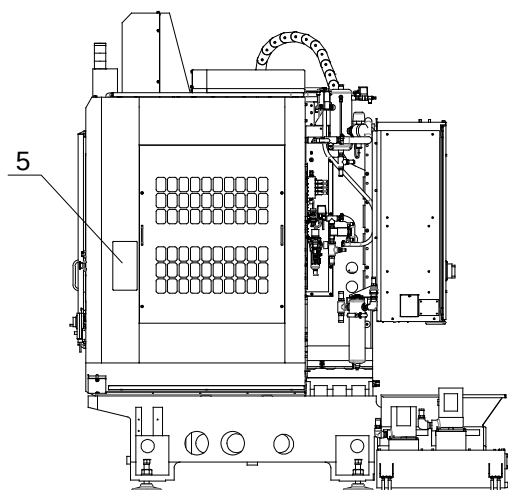
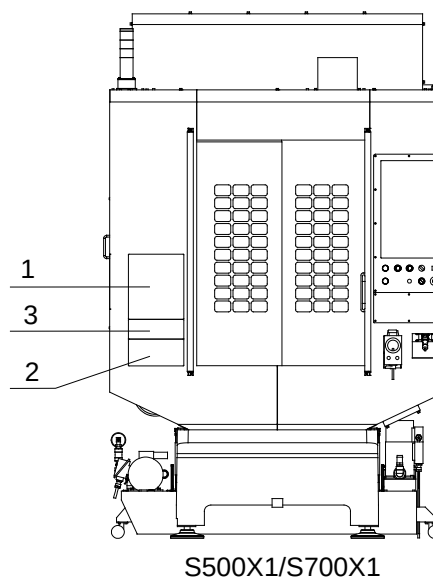
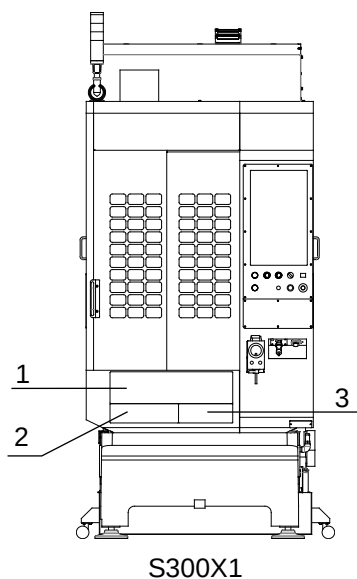
If the tool breaks or if the tool teeth suffer abnormal wear, the load on the teeth increases and the teeth overheat, and it may generate sparks on the teeth. Therefore, the tool life must be managed. In addition, we recommend installing a tool breakage detector.
- 1-7 Chip and shaving removal support

When the chips or shavings build up inside the machine, clean up the chips or shavings right away otherwise it can lead to a fire. In addition, when using coolant, adjust the direction of the nozzles so it hits the machining points. When managing and disposing of the ejected chips or shavings, be sure to process them appropriately.

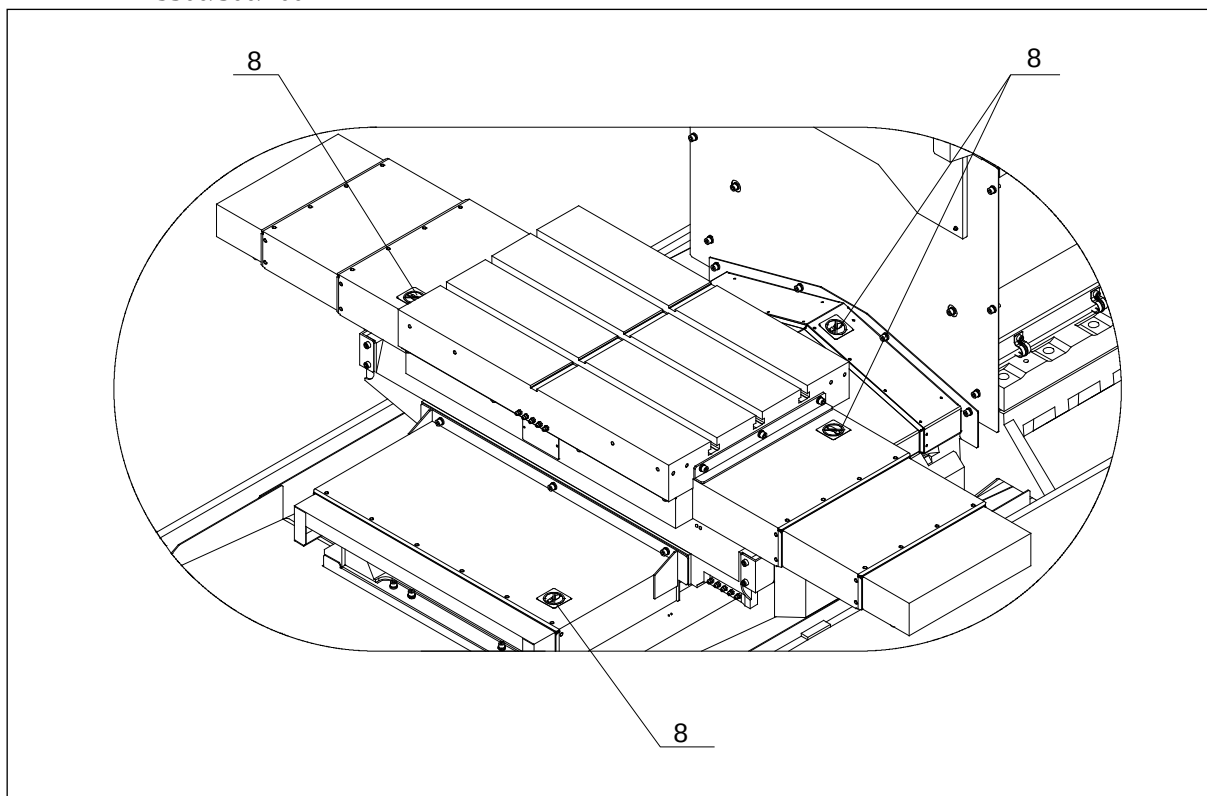
-
- 1-8 Refilling the coolant
When the coolant supply is weak, the coolant may not reach and hit the machining points sufficiently and therefore lead to a fire. Clean the inside of the tank regularly and check the coolant level inside the tank in order to ensure a stable coolant supply. In addition, inspect the discharge volume from the nozzle on a daily basis.
- 1-9 Support for changing the coolant
When changing the coolant from a water-soluble coolant to an oil-based one, perform the following countermeasures because there is a heightened risk of a fire.
- When the oil-based coolant has a high viscosity, replace the pump with a larger capacity one to ensure the appropriate discharge volume.
 - Install a mist collector in order to remove mist which can lead to a fire.
 - Install an automatic fire extinguishing system that can quickly extinguish a fire in the event that one breaks out.
 - The burning or combustion conditions and fire extinguishing method varies depending on the machining materials and coolant types. Therefore, be sure to check the safety data sheet (SDS) issued by the material manufacturer and coolant manufacturer, and use the appropriate coolant.
- (Note) Oil-based coolant cannot be used on the CTS.
Be sure to use a special mist collector when machining combustible materials. If an appropriate mist collector is not used, a fire or explosion may occur.
In addition, always be sure to install a fire damper when using a mist collector.**
- 1-10 General notes
- Do not smoke near the machine and do not perform work that generates sparks, such as welding work.
 - Do not put objects that are highly flammable close to the machine, such as wood chips, paper, cloth or gas cylinders. In addition, do not insert those objects inside the machine or mix them with the chips or shavings.
 - When using an oil-based coolant, select types with a high flash point and supply coolant at a sufficient flow rate on the cutting positions.
 - If a fire erupts while a user is operating the machine, be sure to alert the other workers about the fire and immediately report the fire to the Fire Department.
 - If a fire erupts while a user is operating the machine, take the appropriate measures to ensure your safety and at the same time take steps to extinguish the fire early on using fire extinguishing equipment or manually activating the automatic fire extinguishing system when necessary.
 - Be careful when extinguishing an electrical fire because pouring water on places where the insulation is damaged on the electrical circuits can cause electric shock due to a ground-fault or short-circuit.
 - Make sure that there is fire extinguishing equipment with fire retardant available for electrical fires and use it appropriately.
 - Hazards from modifications performed by the machine tool user
There are potential hazards as noted below if the machine tool user performs modifications. Therefore, do not modify the machine tool.
 - (1) Changing the wiring material or adding or changing a unit can cause an overcurrent and lead to an electrical fire.
In addition, using a fire retardant insulating film for the wiring material can lead to the fire spreading.
 - (2) Using the machine while the detector for the automatic fire extinguishing system, the fire alarm box or the fire extinguishing function is disabled can hamper or prevent a fire from being extinguished.
In addition, the seal to the machining chamber becomes compromised if the machine cover is removed on equipment with a gas type automatic fire extinguishing system or if there is an opening on the machining chamber cover or window. As a result, the ability to extinguish a fire becomes hampered and the user may be unable to extinguish the fire.

Checking Safety Labels

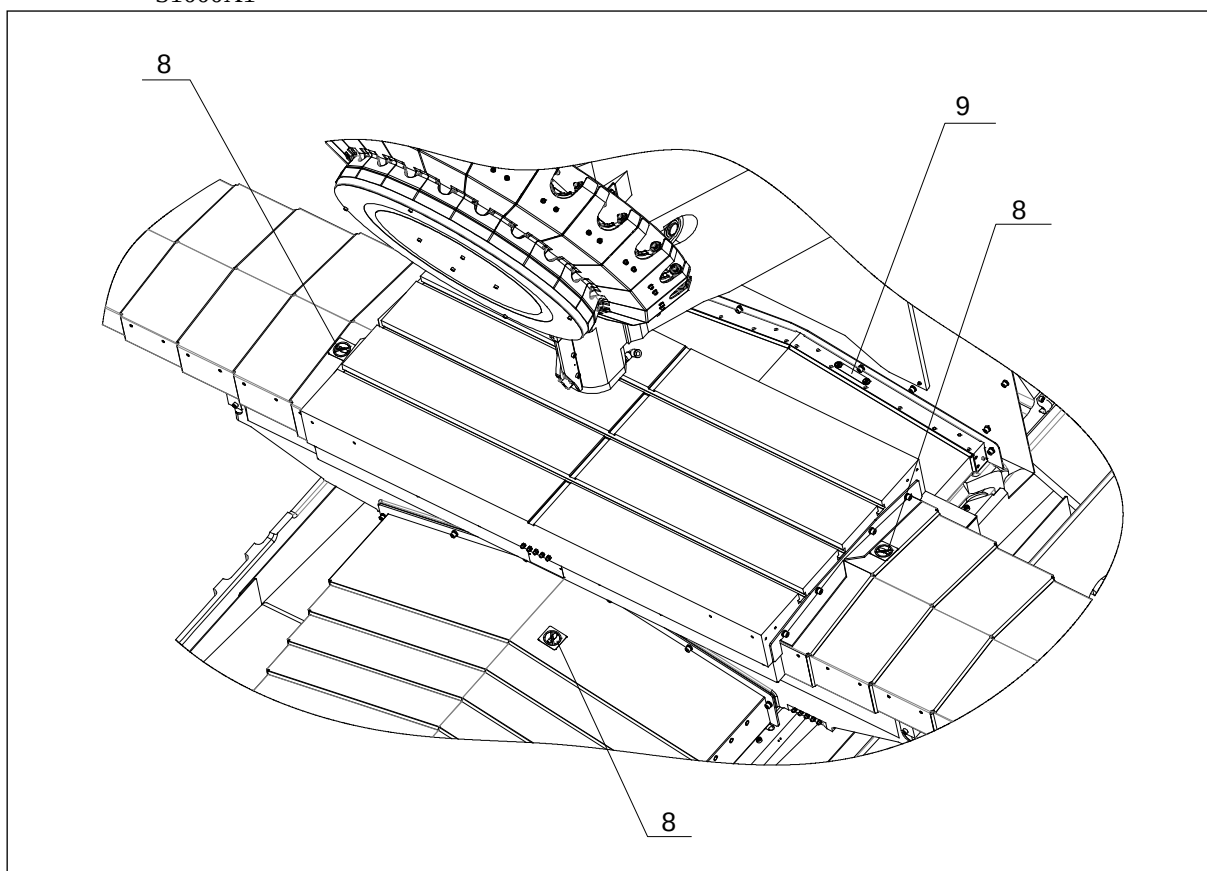
The safety labels are affixed to the machine as shown in the figures below. Check and make sure that there are not safety labels missing. If there is a safety label missing, obtain a new label and re-affix it.



S300/500/700X1



S1000X1



[1] Front safety label

S500X1/S700X1/S1000X1

Language

Japanese, Chinese, English
English, German, French

Part code

693508001
693504001

This machine has hazards.

Read instruction manuals to understand hazards and avoid them before operation.

Diese Maschine birgt Gefahren.

Um Risiken zu erkennen und vermeiden, vor dem Einschalten die Betriebsanleitung lesen.

L'utilisation de cette machine comporte des risques.

Lisez les manuels d'instruction pour connaître les risques et les éviter avant d'utiliser cette machine.

 DANGER	 WARNING	 WARNING
		
Machine in operation will suddenly move and crush you. Keep away from moving parts.	Broken tool or workpiece can hit and wound you. Shut door before starting machine.	Rotating parts can catch and injure you. Keep away from rotating tool, spindle, and magazine. Never hold moving parts. Do not wear gloves while operating machine.
 GEFAHR	 WARNUNG	 WARNUNG
Maschine in Betrieb kann plötzlich laufen und Sie stoßen. Beim Betrieb der Maschine von allen bewegenden Teilen sich fernhalten, um Verletzungen zu vermeiden.	Gebrochene Werkzeuge oder Werkstücke könnten Sie treffen und verletzen. Vor dem Start der Maschine Tür schließen.	Umlaufende Teile könnten Sie fassen und verletzen. Von umlaufendem Werkzeug, Spindel und Magazin sich fernhalten. Rotierende Teile nicht berühren. Während des Bedienens keine Handschuhe tragen.
 DANGER	 AVERTISSEMENT	 AVERTISSEMENT
En opération, cette machine bougera soudainement et provoquera des risques d'écrasement. Ne vous approchez jamais des parties mobiles.	Les outils cassés ou les pièces usinées peuvent être projetés et risquent de vous frapper et de vous blesser. Fermez la porte avant de faire démarrer la machine.	Les parties tournantes peuvent vous accrocher et vous blesser. Ne vous approchez jamais des outils tournant, de la broche et du magasin. Ne saisissez jamais les parties en mouvement. Ne portez pas de gants pendant l'utilisation.

693504001 / 0906(1)

S300X1

Language

Japanese, Chinese, English
English, German, French

Part code

693509001
693505001

This machine has hazards.
Read instruction manuals to understand hazards and avoid them before operation.

⚠ DANGER	⚠ WARNING	⚠ WARNING
		
Machine in operation will suddenly move and crush you. Keep away from moving parts.	Broken tool or workpiece can hit and wound you. Shut door before starting machine.	Rotating parts can catch and injure you. Keep away from rotating tool, spindle, and magazine. Never hold moving parts. Do not wear gloves while operating machine.

Diese Maschine birgt Gefahren. Um Risiken zu erkennen und vermeiden, vor dem Einschalten die Betriebsanleitung lesen.
L'utilisation de cette machine comporte des risques.
Lisez les manuels d'instruction pour connaître les risques et les éviter avant d'utiliser cette machine.

⚠ GEFAHR	⚠ WARNUNG	⚠ WARNUNG
		
Maschine in Betrieb kann plötzlich laufen und Sie stoßen. Beim Betrieb der Maschine von allen bewegenden Teilen sich fernhalten, um Verletzungen zu vermeiden.	Gebrochene Werkzeuge oder Werkstücke könnten Sie treffen und verletzen. Vor dem Start der Maschine Tür schließen.	Umlaufende Teile könnten Sie fassen und verletzen. Von umlaufendem Werkzeug, Spindel und Magazin sich fernhalten. Rotierende Teile nicht berühren. Während des Bediens keine Handschuhe tragen.
⚠ DANGER	⚠ AVERTISSEMENT	⚠ AVERTISSEMENT
En opération, cette machine bougera soudainement et provoquera des risques d'écrasement. Ne vous approchez jamais des parties mobiles.	Les outils cassés ou les pièces usinées peuvent être projetés et risquent de vous frapper et de vous blesser. Fermez la porte avant de faire démarrer la machine.	Les parties tournantes peuvent vous accrocher et vous blesser. Ne vous approchez jamais des outils tournant, de la broche et du magasin. Ne saisissez jamais les parties en mouvement. Ne portez pas de gants pendant l'utilisation.

693509001 / 090611

[2] Front tool label

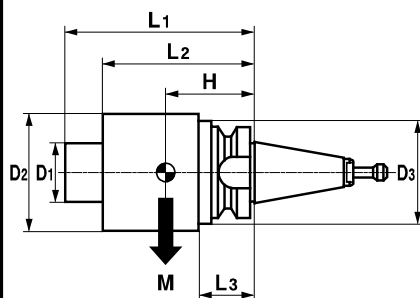
The content noted on the label varies depending on the maximum tool storage capacity in the magazines, the spindle speed and the spindle specifications.

S500X1/S700X1

BT type

If use beyond the limitation of the tool and spindle speed, machine may be broken. Check the details by the operation manual before operation.
 工具、主軸回転数の制限を超えて使用すると、機械が破損する場合があります。操作前に、取扱説明書で詳細を確認して下さい。
 如果使用的刀具超过了刀具夹持器，刀具平衡或主轴速度的限制可能损坏机器。错误选择标准刀具与特大刀具也会损坏机器。

Max Spindle Rotation Speed	最高主軸回転速度	最高主軸回転速	10000 min ⁻¹ / 16000 min ⁻¹ / 27000 min ⁻¹		
Spindle Taper	主軸テーパ	主軸錐度	7 / 24 No.30		
Tool Shank	ツールシャク	刀柄	MAS-BT		
Retention Knob	プルスタッド	拉釘	MAS-P30T-2 (30°)		
Total in Magazine	マガジン全工具合計	刀架中总重量	M total 25 kg (14Tools) / 35 kg (21Tools)		
Limitation of Tool	工具の制限	刀具限制	D1 ≤ 40 mm L1 ≤ 250 mm D2 ≤ 110 mm L2 ≤ 160 mm D3 ≤ 46 mm L3 ≤ 30 mm M ≤ 3 kg MxH ≤ 180 kg·mm	D1 ≤ 40 mm L1 ≤ 250 mm D2 ≤ 55 mm L2 ≤ 160 mm D3 ≤ 46 mm L3 ≤ 30 mm M ≤ 2 kg MxH ≤ 100 kg·mm	D1 ≤ 40 mm L1 ≤ 200 mm D2 ≤ 50 mm L2 ≤ 160 mm D3 ≤ 46 mm L3 ≤ 30 mm M ≤ 2 kg MxH ≤ 50 kg·mm
Limitation of Tool Balance	工具バランス制限	刀具平衡限制	100 g·mm	50 g·mm	10 g·mm
Limitation Spindle Rotation Speed	主軸回転数制限	主軸转速限制	10000 min ⁻¹	16000 min ⁻¹	27000 min ⁻¹



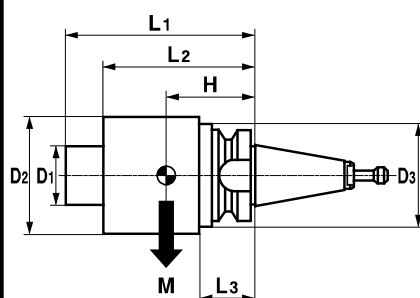
693713001 / 1308(2)

Part code: 693713001

BBT type

If use beyond the limitation of the tool and spindle speed, machine may be broken. Check the details by the operation manual before operation.
 工具、主軸回転数の制限を超えて使用すると、機械が破損する場合があります。操作前に、取扱説明書で詳細を確認して下さい。
 如果使用的刀具超过了刀具夹持器，刀具平衡或主轴速度的限制可能损坏机器。错误选择标准刀具与特大刀具也会损坏机器。

Max Spindle Rotation Speed	最高主軸回転速度	最高主軸回転速	10000 min ⁻¹ / 16000 min ⁻¹ / 27000 min ⁻¹		
Spindle Taper	主軸テーパ	主軸錐度	7 / 24 No.30 (BIG-PLUS)		
Tool Shank	ツールシャク	刀柄	MAS-BT / BBT30		
Retention Knob	プルスタッド	拉釘	MAS-P30T-2 (30°)		
Total in Magazine	マガジン全工具合計	刀架中总重量	M total 25 kg (14Tools) / 35 kg (21Tools)		
Limitation of Tool	工具の制限	刀具限制	D1 ≤ 40 mm L1 ≤ 250 mm D2 ≤ 110 mm L2 ≤ 160 mm D3 ≤ 46 mm L3 ≤ 30 mm M ≤ 3 kg MxH ≤ 180 kg·mm	D1 ≤ 40 mm L1 ≤ 250 mm D2 ≤ 55 mm L2 ≤ 160 mm D3 ≤ 46 mm L3 ≤ 30 mm M ≤ 2 kg MxH ≤ 100 kg·mm	D1 ≤ 40 mm L1 ≤ 200 mm D2 ≤ 50 mm L2 ≤ 160 mm D3 ≤ 46 mm L3 ≤ 30 mm M ≤ 2 kg MxH ≤ 50 kg·mm
Limitation of Tool Balance	工具バランス制限	刀具平衡限制	100 g·mm	50 g·mm	10 g·mm
Limitation Spindle Rotation Speed	主軸回転数制限	主軸转速限制	10000 min ⁻¹	16000 min ⁻¹	27000 min ⁻¹



693714001 / 1308(2)

Part code: 693714001

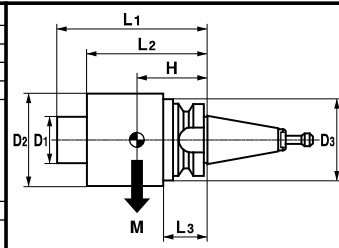
BT DUAL CONTACT

S300X1

BT type

If use beyond the limitation of the tool and spindle speed, machine may be broken. Check the details by the operation manual before operation.
 工具、主軸回転数の制限を超えて使用すると、機械が破損する場合があります。操作前に、取扱説明書で詳細を確認して下さい。
 如果使用的刀具超过了刀具夹持器，刀具平衡或主轴速度的限制可能损坏机器。错误选择标准刀具与特大刀具也会损坏机器。

Max Spindle Rotation Speed	最高主軸回転速度	最高主軸回転速度	10000 min ⁻¹ / 16000 min ⁻¹ / 27000 min ⁻¹		
Spindle Taper	主軸テーパ	主軸錐度	7 / 24 No.30		
Tool Shank	ツールシャンク	刀柄	MAS-BT		
Retension Knob	ブルスタッド	拉釘	MAS-P30T-2 (30°)		
Total in Magazine	マガジン全工具合計	刀架中总重量	M total 25 kg (14 Tools) / 35 kg (21 Tools)		
Limitation of Tool	工具の制限	刀具限制	D1 ≤ 40mm	D3 ≤ 46mm	D1 ≤ 40mm
			L1 ≤ 250mm (14 Tools)	L3 ≥ 30mm	L1 ≤ 250mm (14 Tools)
			L1 ≤ 160mm (21 Tools)	M ≤ 3kg	L1 ≤ 160mm (21 Tools)
			D2 ≤ 110mm	MxH ≤ 180kgmm	D2 ≤ 110mm
Limitation of Tool Balance	工具/バランス制限	刀具平衡限制	L2 ≤ 160mm (14 Tools)		L2 ≤ 160mm (14 Tools)
			L2 ≤ 145mm (21 Tools)		L2 ≤ 145mm (21 Tools)
				100 g·mm	
				50 g·mm	
Limitation Spindle Rotation Speed	主軸回転数制限	主軸转速限制			10 g·mm



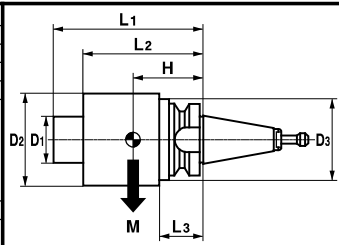
693842001 / 1308(2)

Part code: 693842001

BBT type

If use beyond the limitation of the tool and spindle speed, machine may be broken. Check the details by the operation manual before operation.
 工具、主軸回転数の制限を超えて使用すると、機械が破損する場合があります。操作前に、取扱説明書で詳細を確認して下さい。
 如果使用的刀具超过了刀具夹持器，刀具平衡或主轴速度的限制可能损坏机器。错误选择标准刀具与特大刀具也会损坏机器。

Max Spindle Rotation Speed	最高主軸回転速度	最高主軸回転速度	10000 min ⁻¹ / 16000 min ⁻¹ / 27000 min ⁻¹		
Spindle Taper	主軸テーパ	主軸錐度	7 / 24 No.30 (BIG-PLUS)		
Tool Shank	ツールシャンク	刀柄	MAS-BT / BBT30		
Retension Knob	ブルスタッド	拉釘	MAS-P30T-2 (30°)		
Total in Magazine	マガジン全工具合計	刀架中总重量	M total 25 kg (14 Tools) / 35 kg (21 Tools)		
Limitation of Tool	工具の制限	刀具限制	D1 ≤ 40mm	D3 ≤ 46mm	D1 ≤ 40mm
			L1 ≤ 250mm (14 Tools)	L3 ≥ 30mm	L1 ≤ 250mm (14 Tools)
			L1 ≤ 160mm (21 Tools)	M ≤ 3kg	L1 ≤ 160mm (21 Tools)
			D2 ≤ 110mm	MxH ≤ 180kgmm	D2 ≤ 110mm
Limitation of Tool Balance	工具/バランス制限	刀具平衡限制	L2 ≤ 160mm (14 Tools)		L2 ≤ 160mm (14 Tools)
			L2 ≤ 145mm (21 Tools)		L2 ≤ 145mm (21 Tools)
				100 g·mm	
				50 g·mm	
Limitation Spindle Rotation Speed	主軸回転数制限	主軸转速限制			10 g·mm



693843001 / 1308(1)

BT DUAL CONTACT

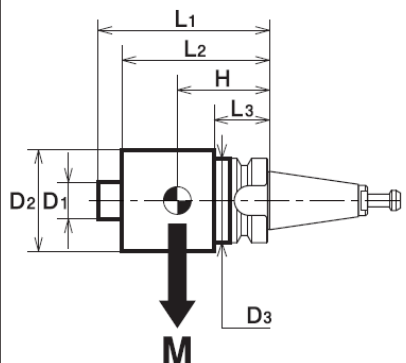
Part code: 693843001

S1000X1

BT type

If use beyond the limitation of the tool and spindle speed, machine may be broken. Check the details by the operation manual before operation.
 工具、主軸回転数の制限を超えて使用すると、機械が破損する場合があります。操作前に、取扱説明書で詳細を確認して下さい。
 如果使用的刀具超过了刀具夹持器，刀具平衡或主轴速度的限制可能损坏机器。错误选择标准刀具与特大刀具也会损坏机器。

Max Spindle Rotation Speed	最高主軸回転速度	最高主軸旋轉速	10000 min ⁻¹ / 16000 min ⁻¹	
Spindle Taper	主軸テーパ	主軸錐度	7 / 24 No.30	
Tool Shank	ツールシャンク	刀柄	MAS-BT	
Retension Knob	ブルスタッド	拉釘	MAS-P30T-2(30°)	
Total in Magazine	マガジン全工具合計	刀架中总重量	M total 25kg (14Tools) / 35kg (21Tools)	
Limitation of Tool	工具の制限	刀具限制	D ₁ ≤ 40mm L ₁ ≤ 250mm D ₂ ≤ 110mm L ₂ ≤ 160mm D ₃ ≤ 46mm L ₃ ≥ 30mm M ≤ 3kg MxH ≤ 180kg·mm	D ₁ ≤ 40mm L ₁ ≤ 250mm D ₂ ≤ 55mm L ₂ ≤ 160mm D ₃ ≤ 46mm L ₃ ≥ 30mm M ≤ 2kg MxH ≤ 100kg·mm
Limitation of Tool Balance	工具バランス制限	刀具平衡限制	100 g·mm	50g·mm
Limitation Spindle Rotation Speed	主軸回転数制限	主軸转速限制	10000 min ⁻¹	16000 min ⁻¹



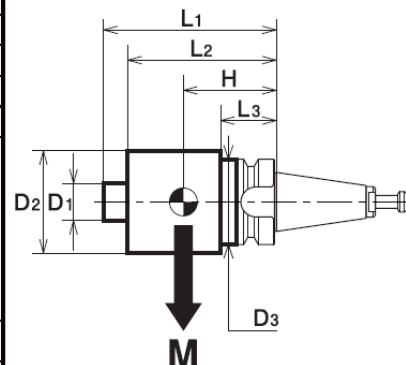
698140001/1408(1)

Part code: 698141001

BBT type

If use beyond the limitation of the tool and spindle speed, machine may be broken. Check the details by the operation manual before operation.
 工具、主軸回転数の制限を超えて使用すると、機械が破損する場合があります。操作前に、取扱説明書で詳細を確認して下さい。
 如果使用的刀具超过了刀具夹持器，刀具平衡或主轴速度的限制可能损坏机器。错误选择标准刀具与特大刀具也会损坏机器。

Max Spindle Rotation Speed	最高主軸回転速度	最高主軸旋轉速	10000 min ⁻¹ / 16000 min ⁻¹	
Spindle Taper	主軸テーパ	主軸錐度	7 / 24 No.30 (BIG-PLUS)	
Tool Shank	ツールシャンク	刀柄	MAS-BT / BBT30	
Retension Knob	ブルスタッド	拉釘	MAS-P30T-2(30°)	
Total in Magazine	マガジン全工具合計	刀架中总重量	M total 25kg (14Tools) / 35kg (21Tools)	
Limitation of Tool	工具の制限	刀具限制	D ₁ ≤ 40mm L ₁ ≤ 250mm D ₂ ≤ 110mm L ₂ ≤ 160mm D ₃ ≤ 46mm L ₃ ≥ 30mm M ≤ 3kg MxH ≤ 180kg·mm	D ₁ ≤ 40mm L ₁ ≤ 250mm D ₂ ≤ 55mm L ₂ ≤ 160mm D ₃ ≤ 46mm L ₃ ≥ 30mm M ≤ 2kg MxH ≤ 100kg·mm
Limitation of Tool Balance	工具バランス制限	刀具平衡限制	100 g·mm	50g·mm
Limitation Spindle Rotation Speed	主軸回転数制限	主軸转速限制	10000 min ⁻¹	16000 min ⁻¹



698141001/1408(1)

Part code: 698141001

BT DUAL CONTACT

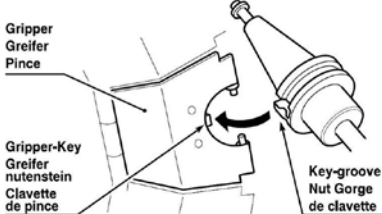
[3] Tool removal label

Language

Japanese, Chinese, English
English, German, French

Part code

693515001
693513001

	⚠ WARNING Scharfe werkzeugsschneide könnte die Finger verletzen. Beim Aus- und Einbau der Werkzeuge immer Handschuhe tragen.	After installed the tool-holder, Please certify aligning its key-groove with gripper key. Nach dem Einsetzen des Werkzeughalters prüfen, daß der Greifer-nutenstein in der Nut sitzt. Après l'installation du porteoutil, vérifier l'alignement de sa gorge de clavette avec la clavette de la pince.
⚠ WARNING Sharp tool edge can cut finger. Use gloves to detach or put tools on.	⚠ AVERTISSEMENT La pointe coupante d'outil peut vous couper au doigt. Lorsque vous attachez ou détachez les outils, portez toujours des gants de protection.	

693513001 / 0906(1)

[4] Rear safety label

Language

Japanese, Chinese, English
English, German, French

Part code

693502001
693500001

⚠ DANGER 	⚠ WARNING 
Hazardous voltage will cause severe injury or death. Never attempt to adjust or repair electrical circuits unless you are familiar with circuits and qualified to work safely on electrical circuits. Turn off main power before working. Post notice you are working.	Spilt coolant, lubricant, or chips are slippery and can cause injury. Turn off main power and clean in and around machine before servicing. Wear head protector and safety shoes while servicing.
⚠ GEFAHR Hochspannung kann Sie schwer oder tödlich verletzen. Nicht versuchen, Einstellungen oder Reparaturen am elektrischen Stromkreis ohne Fachkenntnisse oder Qualifikation vorzunehmen. Vor dem Eingreifen Hauptschalter ausschalten und für Dritte kenntlich machen, daß an der Maschine gearbeitet wird.	⚠ WARNUNG Auf Kühlmittel oder Schmierstoffen, bzw. Spänen könnten Sie ausrutschen und sich verletzen. Vor dem Warten der Maschine Hauptschalter ausschalten und den Innenraum sowie die Umgebung der Maschine säubern. Während der Wartung Kopfschutz und Sicherheitsschuhe tragen.
⚠ DANGER Haute tension causera des blessures graves et des risques de mort. Les réparations et les réglages des circuits électriques ne doivent être effectués que par du personnel qualifié. Coupez l'alimentation principale avant de travailler sur les circuits électriques. Apposez une affiche spécifiant que vous êtes en train de travailler.	⚠ AVERTISSEMENT Les huiles de coupe ou de lubrification et les copeaux peuvent vous faire glisser et vous blesser. Coupez l'alimentation principale et nettoyez dans et autour de la machine avant de mettre en service. Portez un casque et des chaussures de sécurité en mettant en service.

693500001 / 0906(1)

[5] Side cover safety label

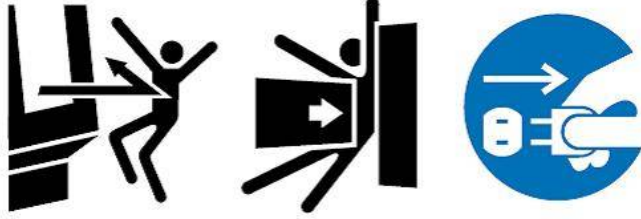
Language

Japanese, Chinese, English
English, German, French

Part code

693512001
693510001

⚠ DANGER



Machine in operation will suddenly move, broken tool or workpiece can hit and crush you.

Turn off main power before opening maintenance door.

⚠ GEFAHR

Ist die Maschine in Betrieb, kann sie sich plötzlich von der Stelle bewegen; ein abgebrochenes Werkzeug oder ein Werkstück kann Personen treffen oder zerdrücken.

Vor dem Öffnen der Wartungstür, Hauptschalter abschalten.

⚠ DANGER

Attention aux outils cassés ou aux pièces travaillées qui peuvent voler et vous écraser si la machine se met brusquement en marche.

Coupez l'alimentation électrique avant d'ouvrir la porte d'entretien.

693510001 / 0906(1)

[6] CTS safety label 1

Language

Japanese, Chinese, English
English, German, French

Part code

693001001
693002001

⚠ WARNING



**Splashed coolant
can cause eye damage.**

⊕ **Press the manual air valve button
and lower the pressure of the pipe
connected to the pump to 0 MPa
before performing maintenance.** ⊕

⚠ WARNING

**Kühlmittelspritzer
können Augenverletzungen verursachen.**

**Vor der Wartung den manuellen
Knopf des Luftventils drücken,
um den Druck der Rohrleitung
zur Pumpe auf 0 MPa zu senken.**

⚠ AVERTISSEMENT

**Les éclaboussures de réfrigérant
sont dangereuses pour les yeux.**

**Avant toute intervention d'entretien,
appuyer sur le bouton de soupape
d'air manuelle pour faire baisser la
pression du tuyau relié à la pompe
jusqu'à 0 MPa.**

693002001 / 0301(1)

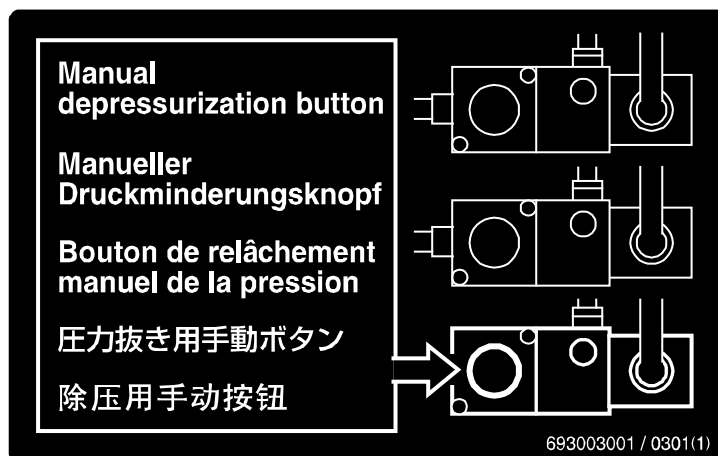
[7] CTS Label 2

Language

English, German, French, Japanese, Chinese

Part code

693003001



[8] Keep off label 40

Part code

693178001



[9] Keep off label S

Part code

693243001



Door Interlock Function

- 1 Handling Precautions**
- 2 External View**
- 3 Startup Inspection**
- 4 Specification Details**

1 Handling Precautions

DANGER

If any modifications are carried out, such as disabling the limit switch for the door interlock, the safety devices may stop operating. Therefore, the machine may operate even when a door is open, and you may get caught or drawn into the machine.

Do not modify the safety devices. Do not secure the safety devices in such a way that will prevent them from operating.

If modifications are necessary, contact Brother Industries to obtain written approval before proceeding.

If an operator finds that the machine has been modified, they should notify a supervisor immediately, without operating the machine.

WARNING

If the door interlock switch is disabled, the machine will operate even when a door is open, and you may get caught or drawn into the machine.

The installer or setup personnel must never leave the machine unattended while the door interlock switch is disabled.

The supervisor must not let the operator use the machine when the door interlock switch is disabled.

The operator must check that the door interlock switch is enabled before operating the machine.

The operator must inform the supervisor if the door interlock key is attached.

The door interlock key should be kept by the supervisor.

WARNING

There are movable parts inside the machine such as the spindle head and table, and your body may get caught on them if you go inside the machine.

Do not go inside the machine while the machine is operating.

The operator should visually check to make sure that there is nobody inside the machine before starting the machine.

Do not go inside the machine. If you need to go inside the machine, the supervisor must be notified.

If you go inside the machine, turn OFF the main power breaker, and then attach a padlock to the main power breaker so that the power cannot be turned ON.

A sign or notice should be placed near the operation panel of the machine to warn others that work is in progress.

** WARNING**

Keep away from objects which are moving or gyrating, otherwise you may get caught by them causing injury.

Keep away from the tool, spindle and ATC magazine while they are rotating.

Do not hold the rotating parts.

Wear tight-fitting clothing. Long hair should be tucked away securely underneath caps or helmets.

Do not wear leather gloves except when performing setup work while the machine is stopped.

Do not wear accessories.

Do not modify the safety devices. Do not secure the safety devices in such a way that will prevent them from operating.

If modifications are necessary, contact Brother Industries to obtain written approval before proceeding.

If an operator finds that the machine has been modified, they should notify a supervisor immediately, without operating the machine.

Door Interlock Function

Purpose

The door interlock function is used to protect the maintenance workers from getting caught on a rotating tool or moving and gyrating parts.

Installation description

The installation varies depending on the setup, but basically it restricts machine operation and locks the door, etc.

Structure

The door interlock device is composed of the following items:

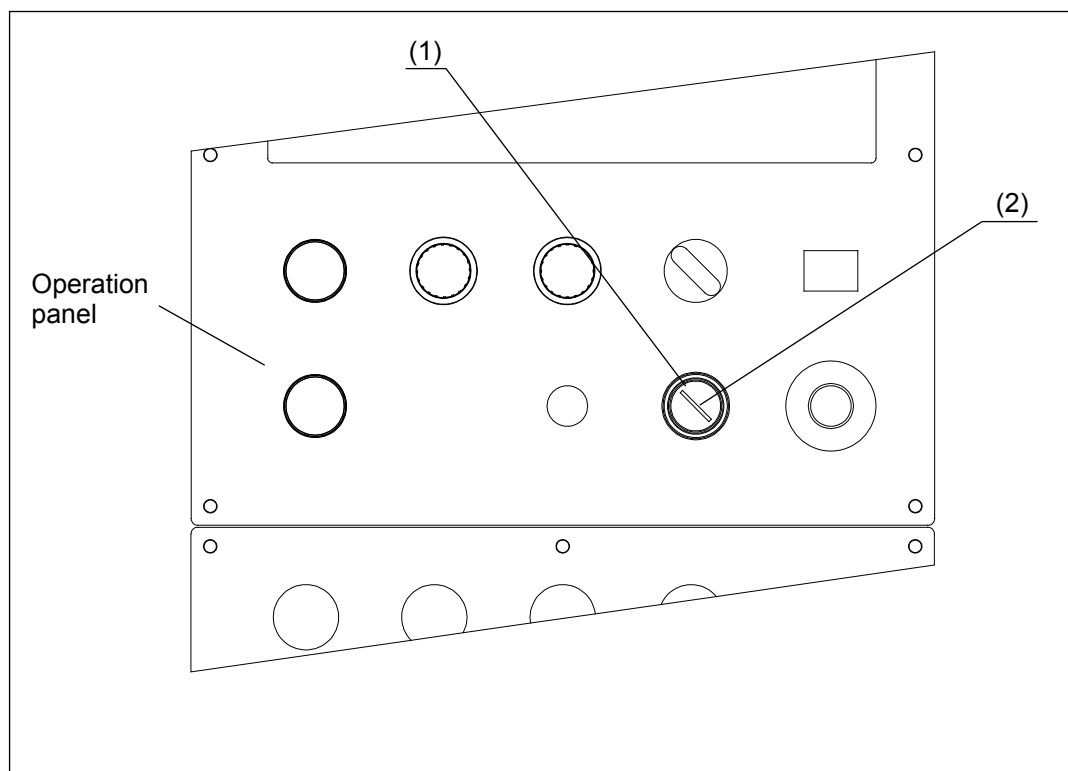
1. Door limit switch
2. Door lock

Usage

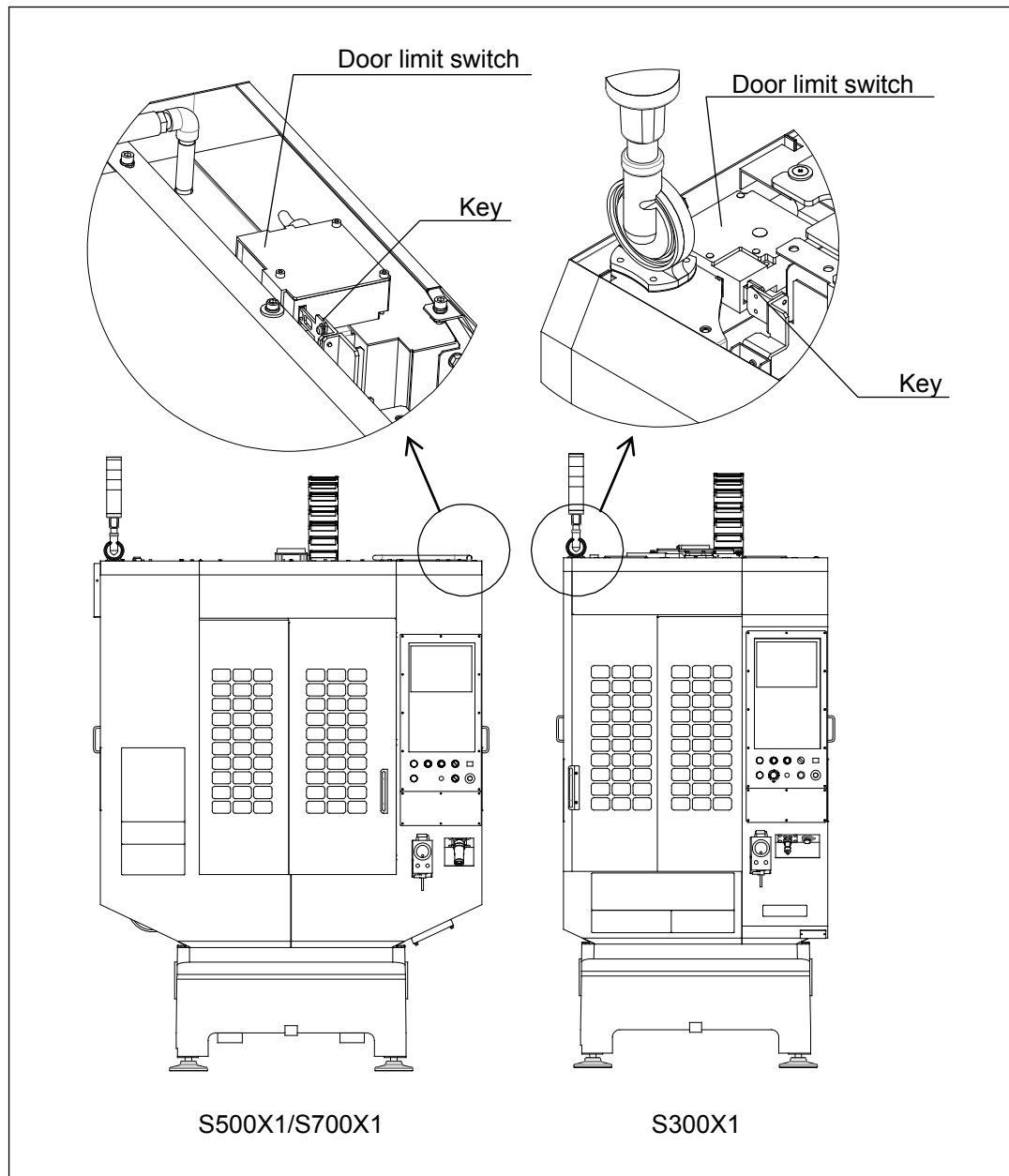
1. Normal
The door interlock function switch (1) is always enabled.
2. During maintenance work
Disable the switch only when necessary during maintenance work. First, read the “Safety manual for workers performing setup/tool changes” and the “Safety manual for trained maintenance workers” before engaging in maintenance work.

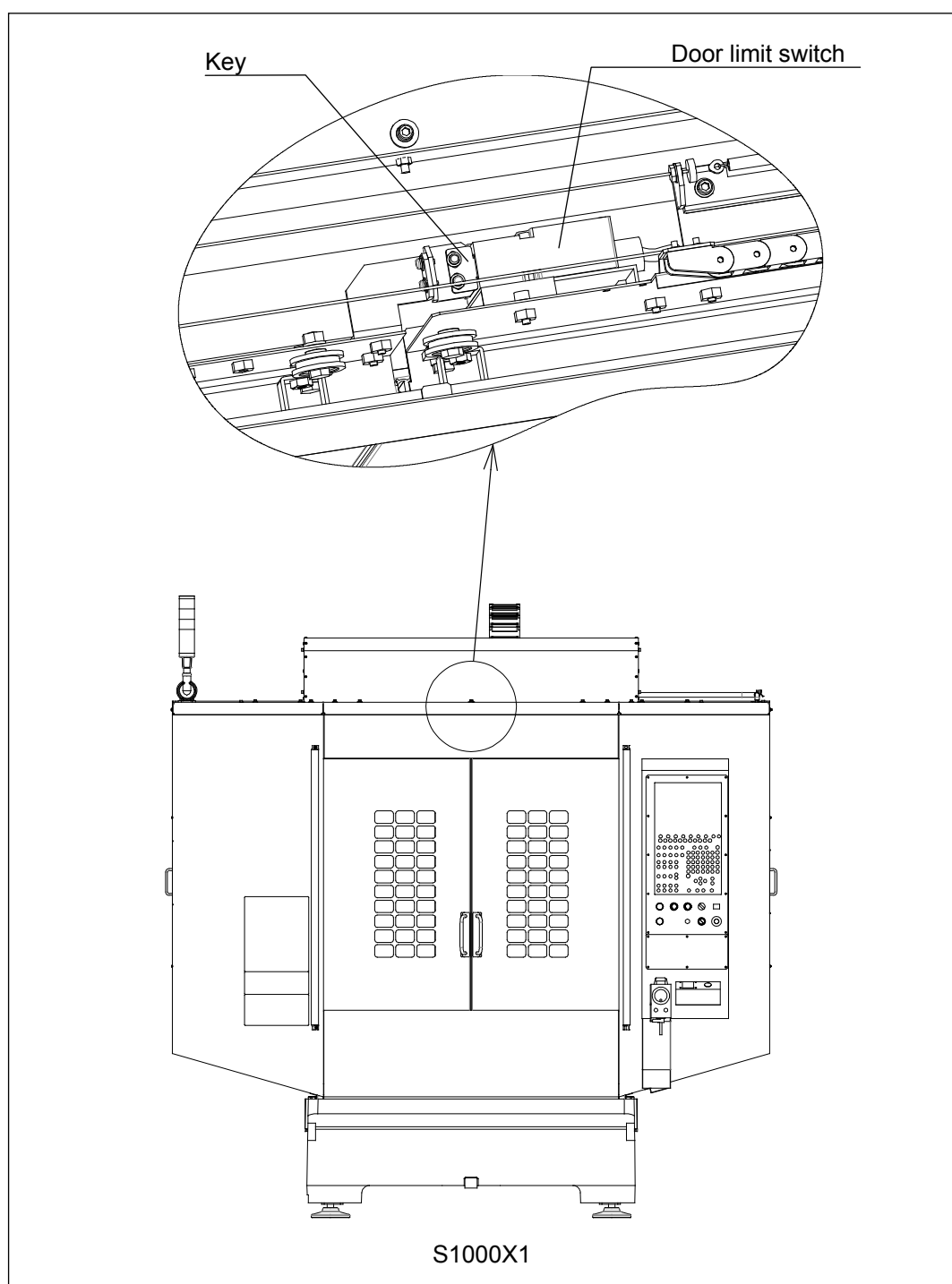
Always follow the items below when disabling the interlock:

- Take the necessary steps to avoid getting caught on the rotating tool or moving and gyrating parts.
- Always enable the switch again after performing the work task.
- Return the key (2) to the supervisor.

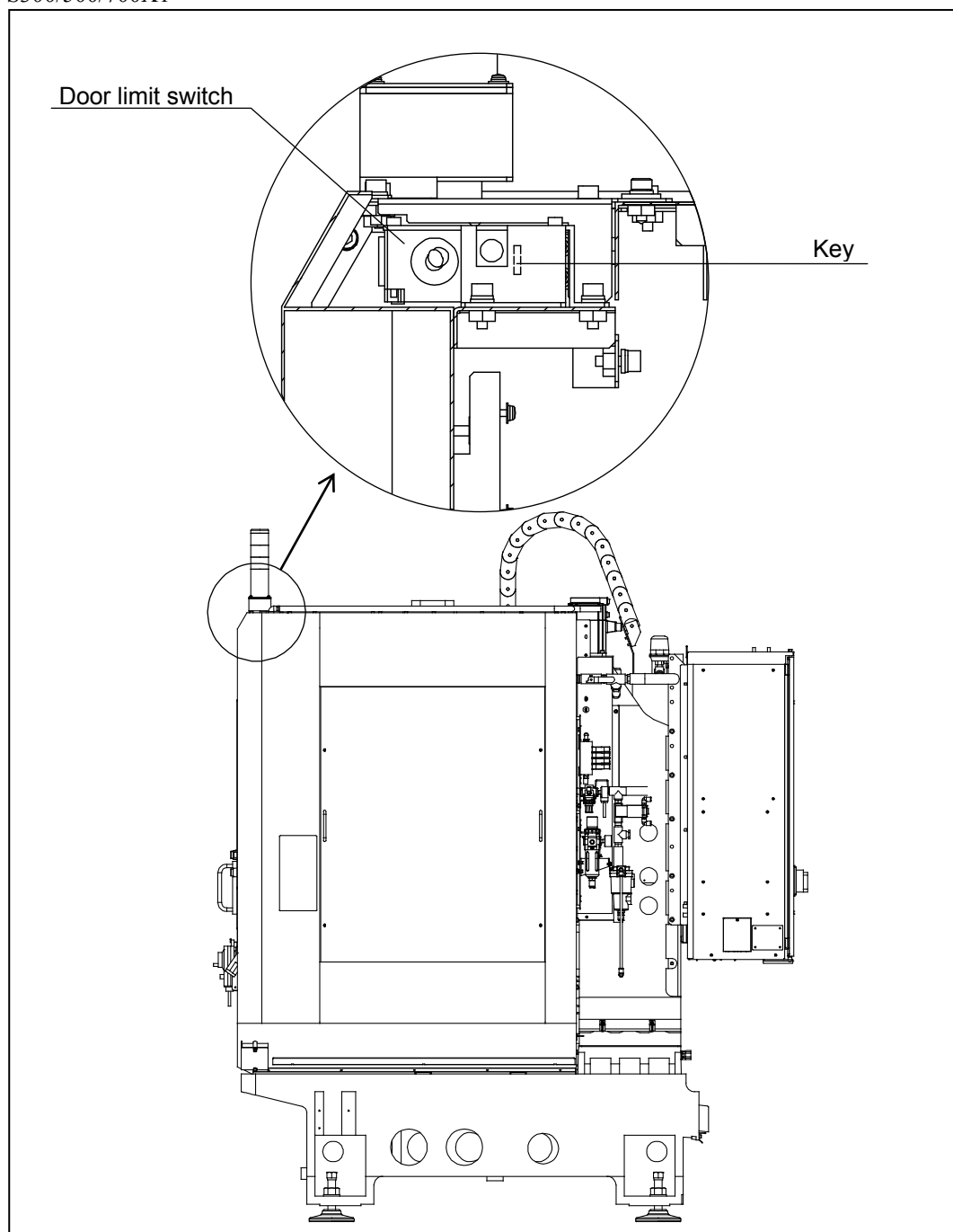


2 External View

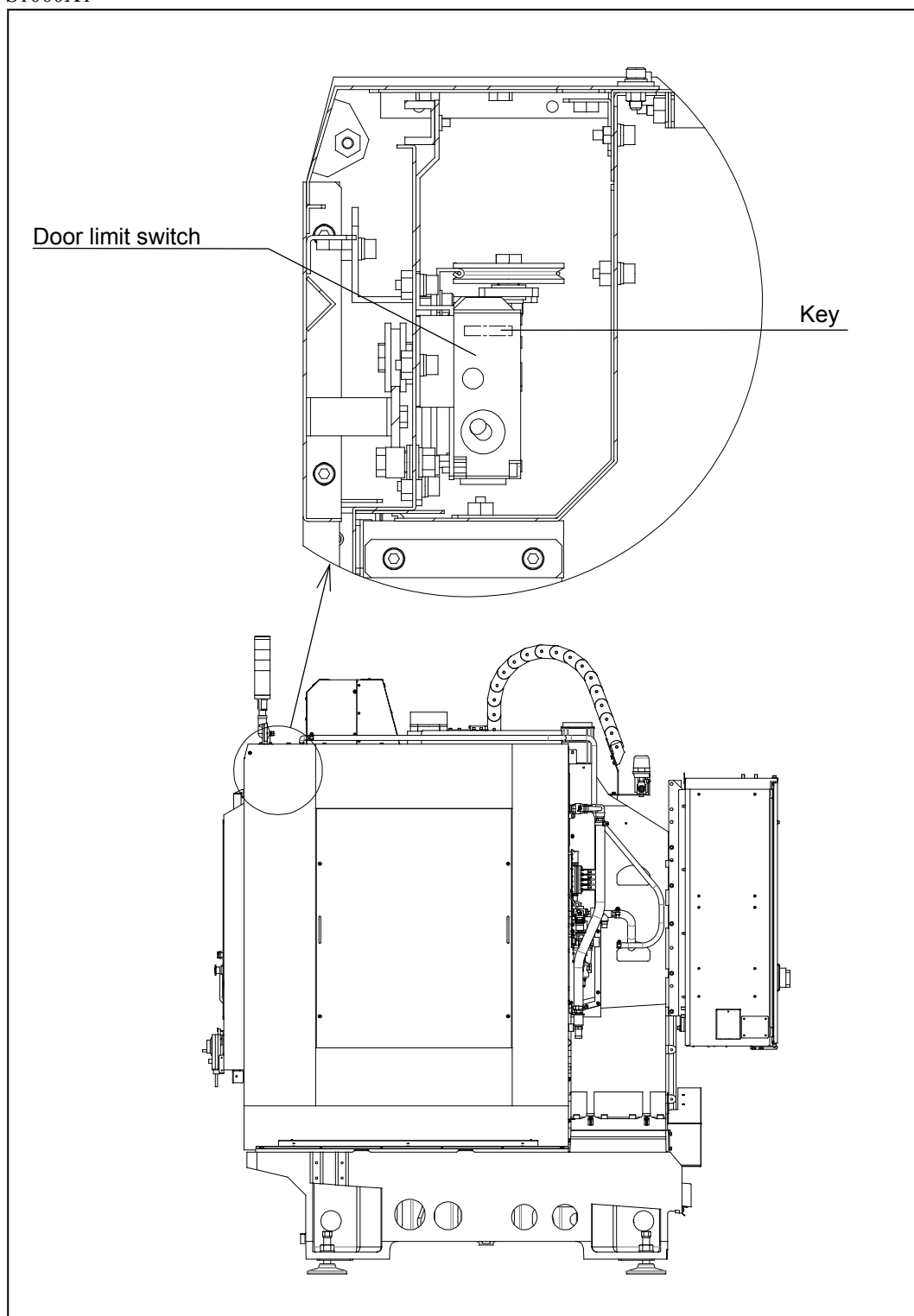




S300/500/700X1



S1000X1



3 Startup Inspection

Always be sure to inspect the door interlock device before starting operation.
If not functioning properly, report it immediately to the supervisor.

Door interlock device inspection procedure

- Make sure that the device does not operate when the door is open
- 1. Make sure that the [DOOR INTERLOCK] switch is enabled, and turn ON the power.
- 2. First, close all doors and covers before performing a machine zero return operation.
- 3. Open the front door.
- 4. While in manual mode, press the [-X] key on the operation panel to move the X-axis.
- 5. Make sure that the alarm <<*Door open>> is triggered and the axis does not move. If the alarm is displayed, the door interlock device is functioning properly.
- Door lock inspection
- 6. Close the front door.
- 7. Press the [SP.CW] key. The spindle should rotate.
- 8. Open the front door. Make sure that the front door is locked.
If the front door locks, the door interlock device is functioning properly.
- 9. Press the [SP.STOP] key. The spindle should stop.

Emergency stop inspection procedure

1. Press the [EMERGENCY] switch on the machine's operation panel.
2. Make sure that the alarm <<*Emergency button activated>> appears on the screen.
3. Turn the [EMERGENCY] switch button to the right to reset the button.
4. Press the [RST] key on the operation panel.
5. Make sure that the screen alarm disappears.

4 Specification Details

Door (*6)	Door interlock switch	Door lock (*6)	Operation
Closed	Disabled	• ATC in progress while in MDI/memory operation mode • Additional axis for lathe spindle is operating (*7)	Full operation is possible
	Enabled	Machine operating (*4)	
Open	Disabled	Lock does not engage	Spindle rotation, magazine turn, and ATC and tapping operation are not possible. (*2) Spindle stop and spindle orientation are possible. Axis travel is limited to operating under the conditions given in "Limits on operation when doors are open" (below).
	Enabled		Full operation is not possible. (*2) (*5)
Closed ↓ Open	Disabled	-	Operation (also including spindle rotation) immediately stops. (*3)
	Enabled		Operation (also including spindle rotation) immediately stops. (*5)
Open	Disabled → Enabled		

- Limits on operation when doors are open

Manual mode	Low travel speed and low rotation speed (*1). Only the following are allowed: jogging travel, step travel and travel controlled by the manual pulse generator. For the operations below, the alarm <<Door open>> is triggered. [Z.RTN], [ATC], [MAG.F], [MAG.R] and [SP.CW] keys The maximum step travel amount for the rotation axis is up to "1°". If a step travel amount that exceeds this is set, the alarm <<Step travel is excessive>> is triggered and travel will not be carried out.
MDI operation mode	Low travel speed and low rotation speed (*1). Operation of additional axis for lathe spindle is not possible. MDI operation and automatic centering only are possible (block stops). * Manual operations in MDI manual mode follow manual mode operation.
Memory operation mode	Low travel speed and low rotation speed (*1). Operation of additional axis for lathe spindle is not possible. Single operation during memory operation only is possible. Operation is not carried out even if the following external input signals are input. (including PLC and communication) External zero return, external Z-axis machine zero return, external no.* reference zero return (*: 1 – 6) and external no.* zero return (*: 2 – 3)
All modes	PLC axis operation is not carried out even if a PLC axis command start signal is input.

- *1 The rapid feed travel speed that is used is set in the <Rapid feedrate (door open)> setting under the machine parameter. The cutting travel speed is also restricted by the <Rapid feedrate (door open)> setting. The same is true for the additional axes (A-, B- and C-axes); the speed is restricted by the <4th- to 8th-axis rapid feedrate(door open)> setting in the machine parameter.
- *2 When an attempt is made to operate, the error <<Door open>> is triggered if the front door is open.
- *3 The block stops during tap, spindle orientation, ATC and magazine turning operations. If the door is opened during spindle rotation, the error <<Spindle stopped with door open>> is triggered and the spindle stops. Thereafter, if the door is closed and operation is restarted, the rotation prior to the spindle stopping will resume.
However, when the following operations are performed, even if the error <<Spindle stopped with door open>> turns off, the door is closed and operation is restarted, the rotation prior to the spindle stopping will not resume.
 - The [RST] key is pressed.
 - The spindle is operated in manual mode.
- *4 Machine is operating.
The manual pulse generator is enabled during axis travel, spindle rotation, turning spindle rotation and MDI/memory operation (including temporary stops and block stops, but excluding the end of the program and program stops).
- *5 When the door interlock is enabled, the servo turns OFF if the door is opened.
If the door is opened before starting memory operation or during a program stop (M00), the error <<*Door open>> is triggered and the servo turns OFF.
Closing the door or disabling the door interlock will clear the error and turn ON the servo.
However, the error <<Machine stopped with *door open>> is triggered if the door is opened during operation in memory operation mode.
This error cannot be cleared by closing the door alone. The [RST] key must be also pressed.
- *6 The door status is considered closed only when the front door and the side door are closed. In addition, the front door and the side door are locked as noted in the lock conditions above.
- *7 Including when the manual pulse generator is enabled.

OPEARTION MANUAL (OPERATION)

Read this manual before performing work.

brother

Introduction

Thank you for purchasing this machine.

Always be sure to read this manual carefully first, in order to use the machine functions properly and safely.

This machine manual is divided into the following sections.

- **Operation Manual**
This manual describes the operation procedure for the machine.
- **Installation Manual**
This manual describes the machine's installation procedure and inspections.
- **Programming Manual**
This manual provides a program description.

Brother is not responsible or liable for accidents that occur during special machine use or handling that does not follow the general safety usage guidelines.

Keep this manual for future reference.

Attach this manual to the machine if it is resold.

Contact the nearest Brother sales office or Brother approved service dealer if this manual or the safety labels are damaged, lost or missing. (Charges apply)

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We provide source code to customers who wish to use GPL or LGPL software. Contact Brother for further details. (A separate fee and service charge applies.)

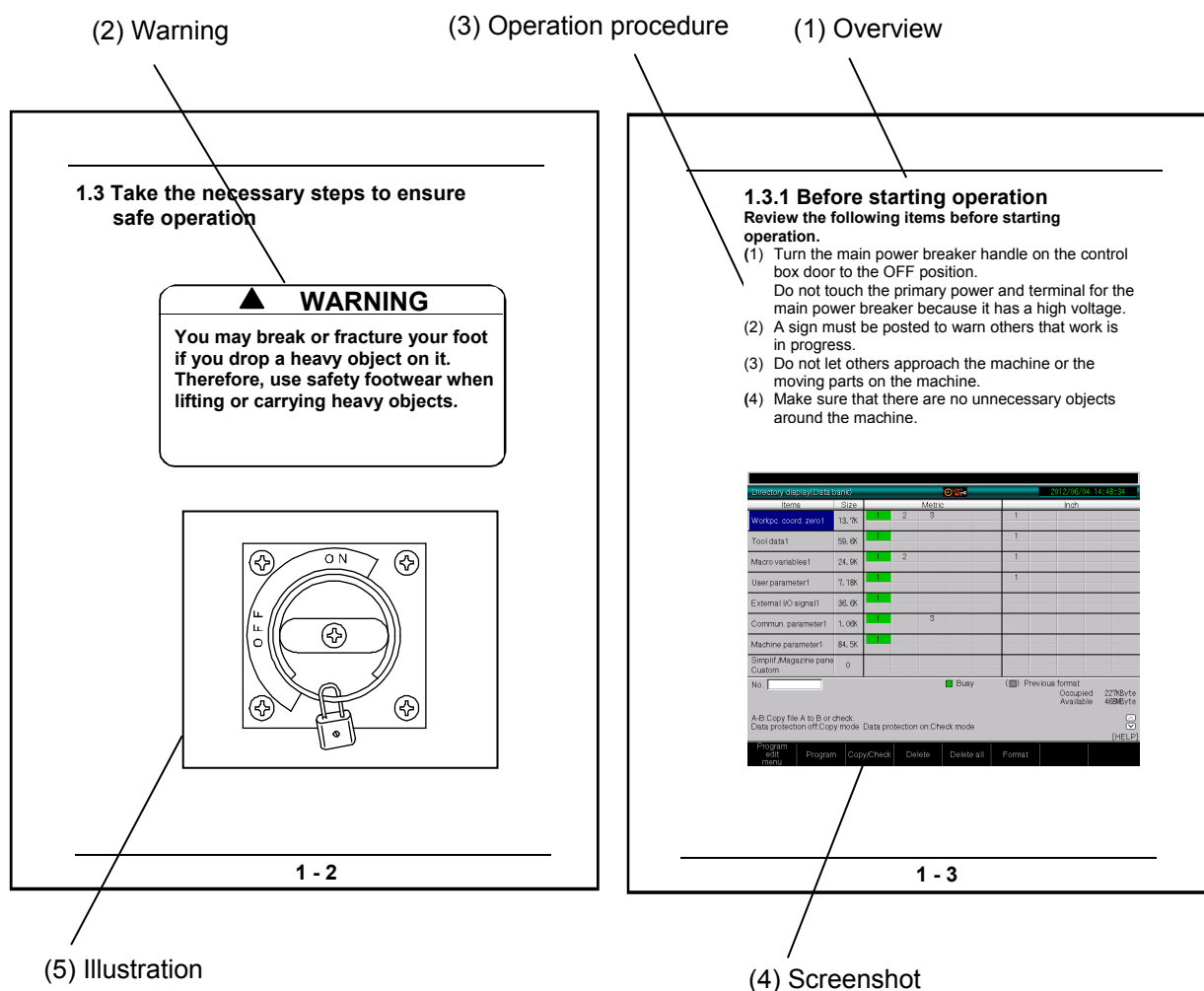
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How to Read This Manual

This manual is divided into the following sections.

- (1) **Overview** ----- A summary of the content is provided for the corresponding section.
- (2) **Warning**----- A warning is provided for any hazards that could potentially cause serious bodily injury, death or damage to the machine.
The hazards are described in the following order.
- (2-1) Hazard level
 - (2-2) Type of hazard
 - (2-3) Potential damage
 - (2-4) Safety directions to avoid danger
- (3) **Operation procedure** --- The procedure describes how to operate each function.
- (4) **Screenshot** --- A screenshot is inserted into places to highlight certain points in the operation procedure. The screenshot messages are shown at an approximate position and may differ slightly from the actual position of the line or column. The same applies to the font.
- (5) **Illustration** ---- Illustrations, such as explanatory drawings, diagrams that show the dimensions, positioning, ranges, figures or configurations, are used in certain places where a written explanation alone may be hard to understand.



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*1 The description in this chapter is omitted because this product is not equipped with this function.

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CHAPTER 4 SCREEN DISPLAY**4**

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CHAPTER 1

HANDLING PRECAUTIONS

- 1.1 Important Precautions**
- 1.2 Safety Precautions**
- 1.3 Authorized Workers and Operators**

1.1 Important Precautions

DANGER

If any modifications are carried out, such as disabling the limit switch for the door interlock, the safety devices may stop operating. Therefore, the machine may operate even when a door is open, and you may get caught or drawn into the machine.

Do not modify the safety devices. Do not secure the safety devices in such a way that will prevent them from operating.

If modifications are necessary, contact Brother Industries to obtain written approval before proceeding.

If an operator finds that the machine has been modified, they should notify a supervisor immediately, without operating the machine.

DANGER

If you put any parts of your body inside the machine while the maintenance cover has been removed for setup or maintenance work, you may get caught or drawn into the machine if the machine is operated by mistake.

Do not put any part of your body inside the machine while it is operating.

Turn OFF the main power breaker before removing the maintenance cover.

Attach a padlock to the main power breaker so that the power cannot be turned ON.

A sign or notice should be placed near the operation panel of the machine to warn others that work is in progress.

DANGER

If the emergency stop switch is damaged, the emergency stop may not function.

Check the emergency stop switch operation before starting machine operation.

If the emergency stop switch is damaged or does not function, it should be replaced.

DANGER

If oil-based coolants are used during cutting, the cutting area may become hot and sparks may be generated.

When using oil-based coolants, a fire alarm box and fire extinguishing equipment must always be on hand for use.

An operator must always be standing beside the machine while cutting is in progress.

⚠ WARNING

If the door interlock switch is disabled, the machine will operate even when a door is open, and you may get caught or drawn into the machine.

The installer or setup personnel must never leave the machine unattended while the door interlock switch is disabled.

The supervisor must not let the operator use the machine when the door interlock switch is disabled.

The operator must check that the door interlock switch is enabled before operating the machine.

The operator must inform the supervisor if the door interlock key is attached.

The door interlock key should be kept by the supervisor.

⚠ WARNING

There are movable parts inside the machine such as the spindle head and table, and your body may get caught on them if you go inside the machine.

Do not go inside the machine while the machine is operating.

The operator should visually check to make sure that there is nobody inside the machine before starting the machine.

Do not go inside the machine. If you need to go inside the machine, the supervisor must be notified.

If you go inside the machine, turn OFF the main power breaker, and then attach a padlock to the main power breaker so that the power cannot be turned ON.

A sign or notice should be placed near the operation panel of the machine to warn others that work is in progress.

1.1.1 Before Using the Machine

This machine can be used for drilling, tapping, facing and turning processes for metals such as steel and aluminum.

Be sure to read this instruction manual thoroughly before using the machine, in order to understand the details about dangers and how to avoid them.

If any modifications are carried out, such as disabling the limit switch for the door interlock, the safety devices may stop operating. Therefore, the machine may operate even when the front door is open, and you may get caught or drawn into the machine. As a result, using the machine in particular ways which are not mentioned in the instruction manual or in ways which depart from general safe behavior is extremely dangerous and may lead to personal injury or damage to the machine.

Before operating the machine, make sure that the control box door, the machine's front door and the maintenance cover are all securely closed.

1.1.2 Axis Travel

Axis travel for the machine's table or spindle head occurs at a high speed.

Before performing axis travel, make sure that there are no obstacles in the travel area and that there is no interference among tools, workpiece and fixtures.

Never go inside the machine through the machine cover during operation.

1.1.3 Indexing the Magazine

The magazine rotates at a high speed.

Securely attach a holder onto the magazine grip.

Never go inside the machine through the machine cover while the magazine is rotating.

1.1.4 Automatic Operation

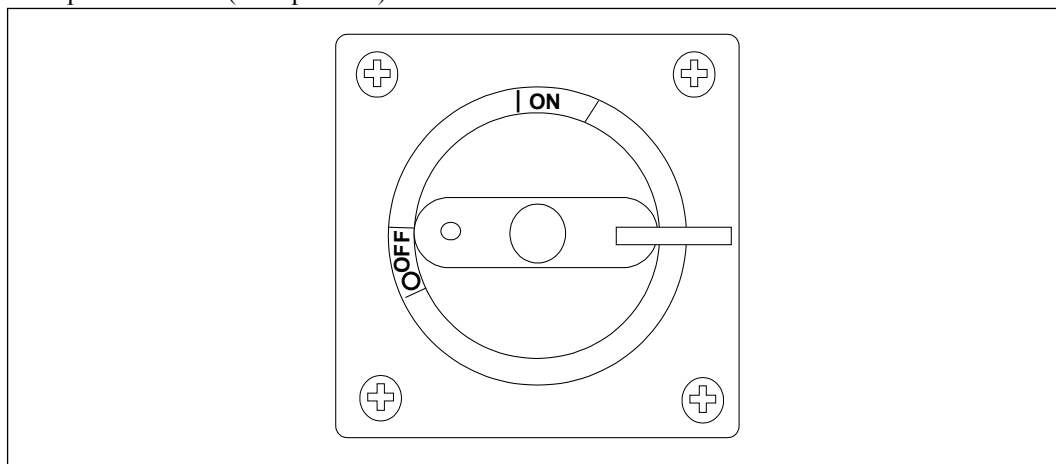
Before operating the machine, make sure that the control box door and the maintenance cover door are both securely closed.

Never directly or indirectly touch any moving parts during operation.

1.1.5 Working Inside the Machine

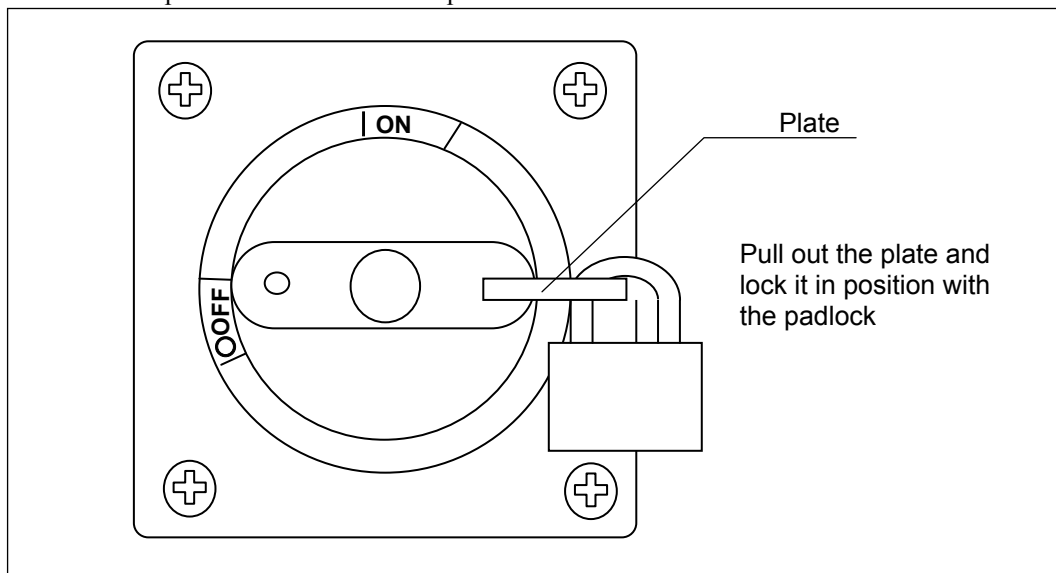
When working inside the machine near moving parts, always turn OFF the main power breaker handle on the control box door.

Main power breaker (OFF position)



Lock the handle with a padlock.

Lock the main power breaker in the OFF position



1.1.6 Signs of Danger or Machine Trouble

Press the [EMERGENCY] switch to stop all machine operations. Press the switch if you sense any machine abnormality or danger in the slightest.

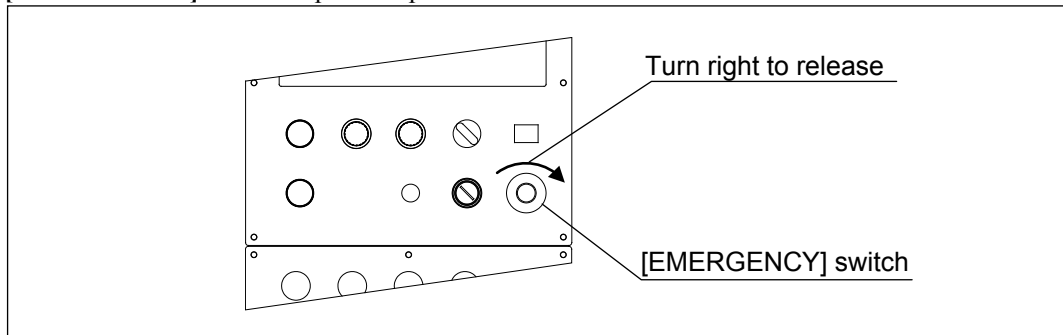
Refer to “Chapter 7, 7.12 Emergency Stop” for further details on the release procedure.

Refer to “Chapter 11, 11.9 ATC Recovery Procedure” for further details on the recovery procedure after an emergency stop during a tool change operation.

Refer to “Chapter 5, 5.11.1 Synchronized Tap Return” for further details on the recovery procedure when the [EMERGENCY] switch is pressed because the tap becomes stuck in the workpiece.

If objects are hung from the [EMERGENCY] switch, it may prevent the switch from being pressed, and injury may occur. Do not hook or hang any objects from the switch.

[EMERGENCY] switch – operation panel



1.1.7 If the Door Interlock Switch is Disabled

If the door interlock switch is disabled, the machine will operate even when the front door is open, and a worker may get caught or drawn into the machine. When performing maintenance, pay attention to the machine operation, and ensure that it can be stopped at any time before re-enabling the switch after maintenance is completed.

Do not operate the machine when the door interlock device switch is disabled, unless you are a specially trained maintenance worker.

1.1.8 Machining

If oil-based coolants are used during cutting, the cutting area may become hot and sparks may be generated. When using oil-based coolants, a fire alarm box and fire extinguishing equipment must always be on hand for use, and an operator must always be standing beside the machine while cutting is in progress.

If workpieces and jigs are not correctly installed, they may shoot out and hit you, causing injury. Install them securely to the table.

Loud sounds or noises can occur depending on the cutting conditions. Review the cutting conditions to lower the level of the noises, and use protective gear if necessary.

1.2 Safety Precautions

WARNING

If tools and other items are left in or around the machine, they may fall down or shoot out when the machine is operating, and injury may occur.

The installer should not leave tools or other items in or around the machine.

The operator must not operate the machine if tools or other items have been left in or around the machine. If this is not observed, the supervisor should be notified.

WARNING

If this is not observed, someone may trip over the machine, or nearby equipment, or slip on coolant and fall over while setup or maintenance work is being carried out.

Protective equipment such as safety footwear and helmets must be worn while working.

The area around the machine must be organized, tidy and clean before performing work.

Any excess length of cable should be removed so that there is no slack. If cables are running along the floor, they must be covered.

WARNING

The coolant may shoot out and injure your eyes while work is being carried out.

Always be sure to wear protective goggles.

If coolant gets into eyes, rinse with clean water and then seek medical advice.

WARNING

If items such as the tool and workpieces are not properly attached, or if cutting conditions are not within the tool limits, items may shoot out from inside the machine and strike personnel causing injury.

Always read the tool labels and instruction manuals for information on tool limits.

Check that the tips are securely attached before installing the tool.

Workpieces should be securely attached.

Close the doors before operating the machine.

WARNING

If an air blower or similar equipment is used to remove shavings, the shavings may shoot out and get into your eyes.

Always wear protective goggles when using an air blower or similar equipment.

Shavings should not be blown off the machine using an air blower or similar equipment.

⚠ WARNING

**Working in noisy conditions can cause hearing loss.
Always be sure to wear ear plugs when working in noisy conditions.**

⚠ WARNING

**The shavings may be flammable depending on the material being processed (for example, magnesium or wood).
Clean shavings away as soon as possible.
Keep fire extinguishing equipment on hand at all times during processing, and never let the machine operate unattended.**

1

These are the basic precautions to be taken to ensure operation safety when handling the machine.

1.2.1 Check Before Turning ON the Power

1. Make sure that the control box door is closed. Make sure that there are no unnecessary objects around the machine.
2. Make sure that there is nobody in or around the range of motion of any moving parts.

1.2.2 Check Before Starting Operation

1. Make sure that it is clean inside and around the machine.
2. Make sure that there are no hand tools left inside the machine.
3. Make sure there is no dust or damage on the taper parts of the spindle, tool in use, or pull stud. Make sure that the tool used is not worn.
4. Make sure that the tools are securely set on the magazine and that the pull studs are not loose.
5. Make sure that no objects are hanging on the [EMERGENCY] switch.
6. Make sure that the safety devices are mounted securely and correctly.
7. Make sure that the safety devices operate correctly.

1.2.3 Precautions During Operation

1. If the door interlock function switch is disabled, the machine moves even when the door is open.
2. Do not operate the machine without attaching the machine cover and maintenance cover. Always be sure to operate the machine with the front door closed. Turn OFF the power when removing the maintenance cover.
3. Before machining the first actual workpiece after setup, etc., check the machine motion using the program. In addition, make sure that there is no interference between the machine and the workpiece, jig etc.
4. Pay particular attention to safety during spindle rotations, axis travel and movements, and tool changes.
5. Always be sure to follow and adhere to the spindle speeds and tool limits when operating.
6. Do not put your hand inside the machine cover during spindle rotation. Do not put your hand on chips or workpiece. Do not use your hand or a tool to stop spindle rotation.
7. Wear safety footwear.
8. Wear ear plugs, or you may suffer from hearing loss.
9. Do not put your hand or any other part of your body under the spindle head.
10. Do not put your fingers in between the grip cover.
11. Fix the workpiece and jig securely on the table.
12. When using an oil based coolant, a fire may occur due to the heat generated from the machining section. Install a fire detector and automatic extinguisher when carrying out automated operation.
13. Wear protective goggles because coolant may splash into your eyes.
14. The shavings may be flammable depending on the material being processed (for example, magnesium or wood). Clean up shavings or chips as soon as possible. Keep fire extinguishing equipment on hand at all times during processing, and never let the machine operate unattended.

1.2.4 Precautions During Operation If Machine Trouble Occurs

1. To perform an emergency stop, press the red [EMERGENCY] switch on the operation panel.
2. Check and identify the cause of the problem or error, and then take the appropriate steps.
3. When two or more operators are working together, communicate with each other out loud to ensure safety.

1.2.5 Precautions When Turning OFF the Power

After work, always be sure to turn OFF the main power breaker before leaving the machine.

1.2.6 Precautions When Changing Operation Mode

When changing the operation mode, first make sure that there is no one inside the machine before operating.

1.2.7 Other Precautions

1. Make sure that the control box and operation panel are not exposed to oil, water or shavings.
2. Draw the table toward the operator for machine setup.
3. Stop the machine and turn OFF the main power breaker before cleaning the machine.
4. Do not use compressed air or a cleaning gun for cleaning and removing chips from the machine body (control box, operation panel and moving parts such as ball screw, linear guide, magazine, spindle, etc).
5. Do not touch the switches with a wet hand or with anything containing water.
6. If chips or shavings stick to the tool, be sure to stop the spindle first and then remove them.
7. Do not put any heavy objects such as the mist collector on the ceiling cover.

1.3 Authorized Workers and Operators

This manual is written for personnel who work on and operate the machine.

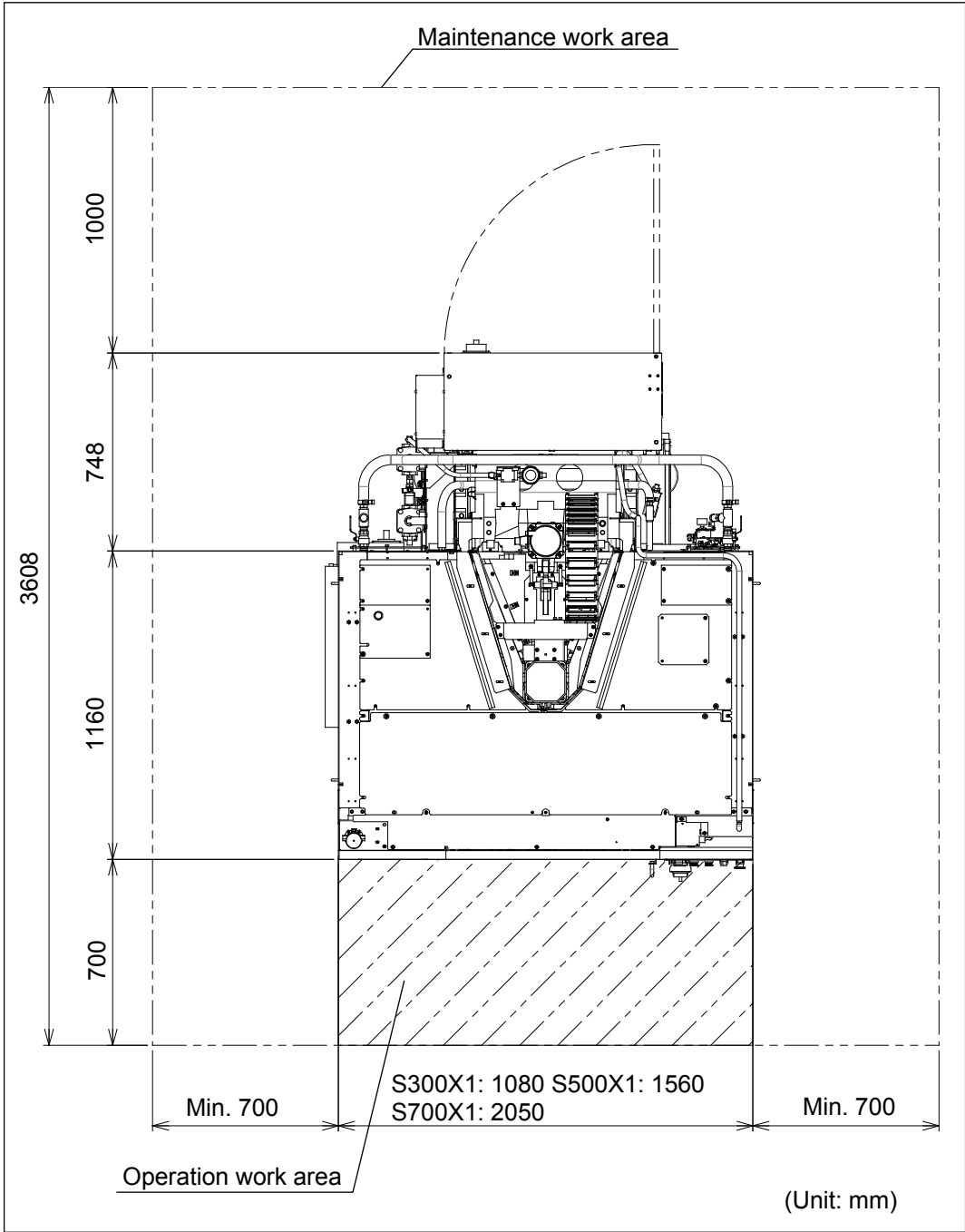
1.3.1 Qualified Personnel

It is prohibited for children to operate or work on this machine.
Only trained personnel are qualified and authorized to operate this machine.

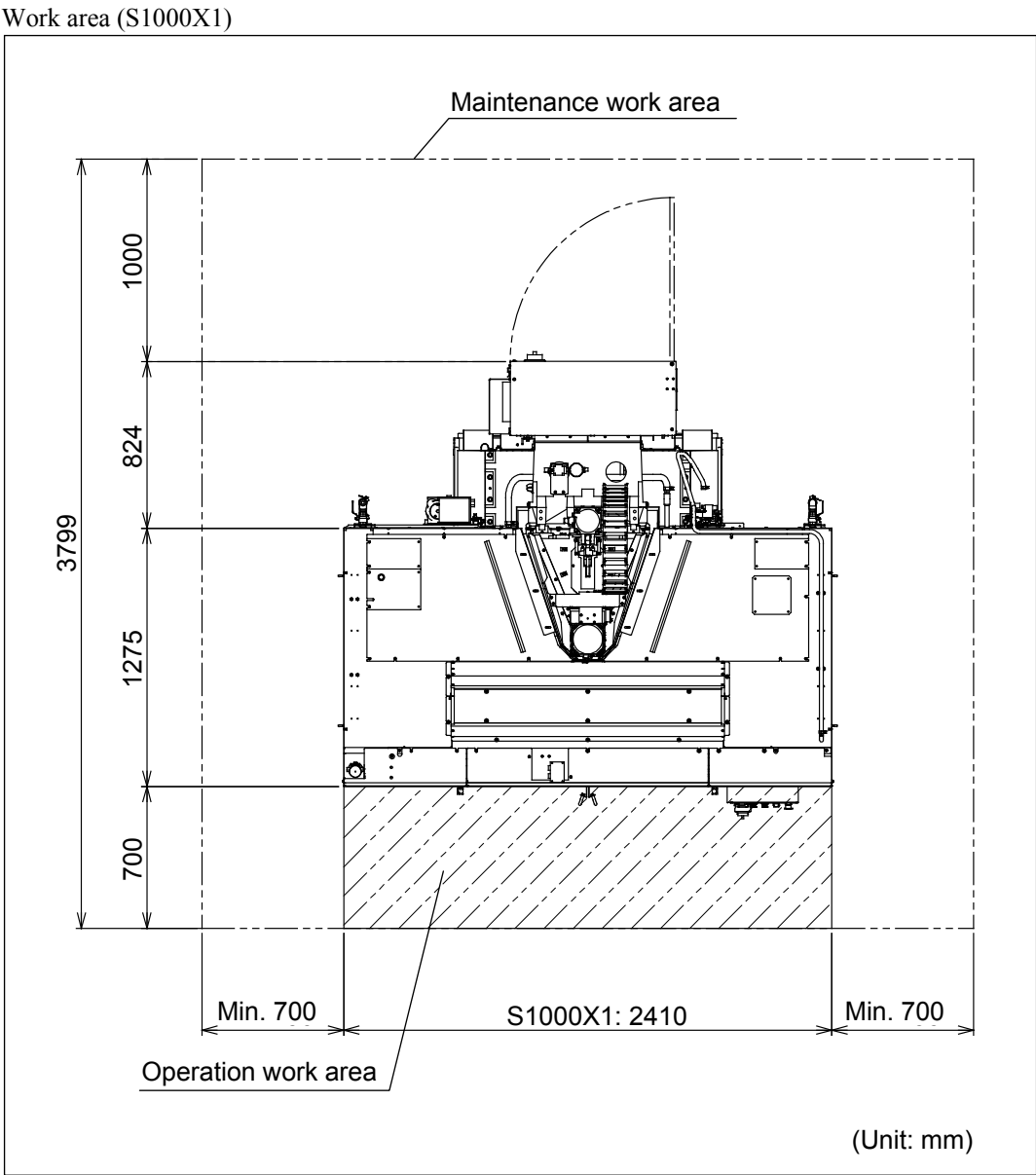
1.3.2 Work Area

Before performing work, be sure to make enough space for the “Operation work area” and “Maintenance work area” as indicated by the two-dotted dashed line in the diagram below.

Work area (S300X1/S500X1/S700X1)



1



CHAPTER 2

OVERVIEW

- 2.1 Main Features**
- 2.2 Basic Structure**
- 2.3 Table Dimensions and Machining Range**
- 2.4 CNC Unit Specifications**
- 2.5 Machining Capabilities**
- 2.6 Safety Devices**
- 2.7 Coolant**

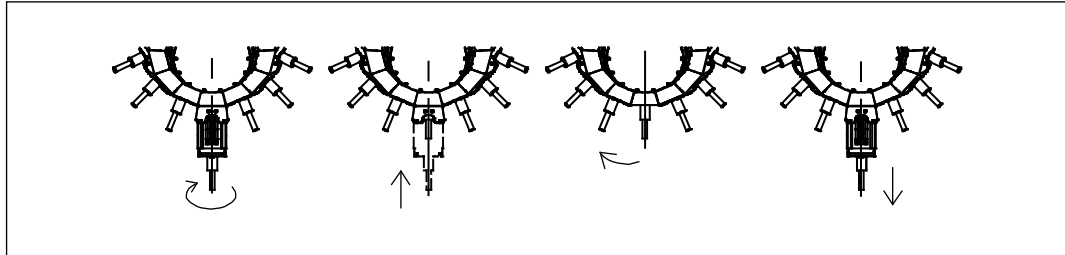
2.1 Main Features

The main features of this machine are described below.

2.1.1 Automatic Tool Changer (ATC)

1. The unique armless mechanism allows simple and quick tool change, thus reducing non-cutting time.
2. The magazine rotates fast and takes the shorter path to index the tool to be used.
3. The X- and Y-axis positioning is carried out before machining while the magazine rotates, thus reducing non-cutting time.

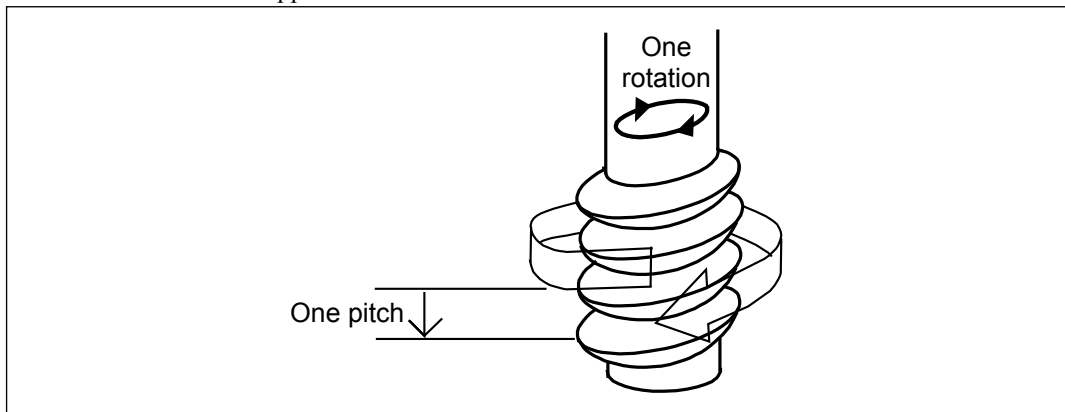
Motion of automatic tool changer (ATC)



2.1.2 Tapping Cycle

Fully synchronized control of spindle rotation with the Z-axis feed makes it unnecessary to use a tapping holder. As a result, the air cutting required in conventional tapping or drilling can be reduced.

Part names and external appearance



2.2 Basic Structure

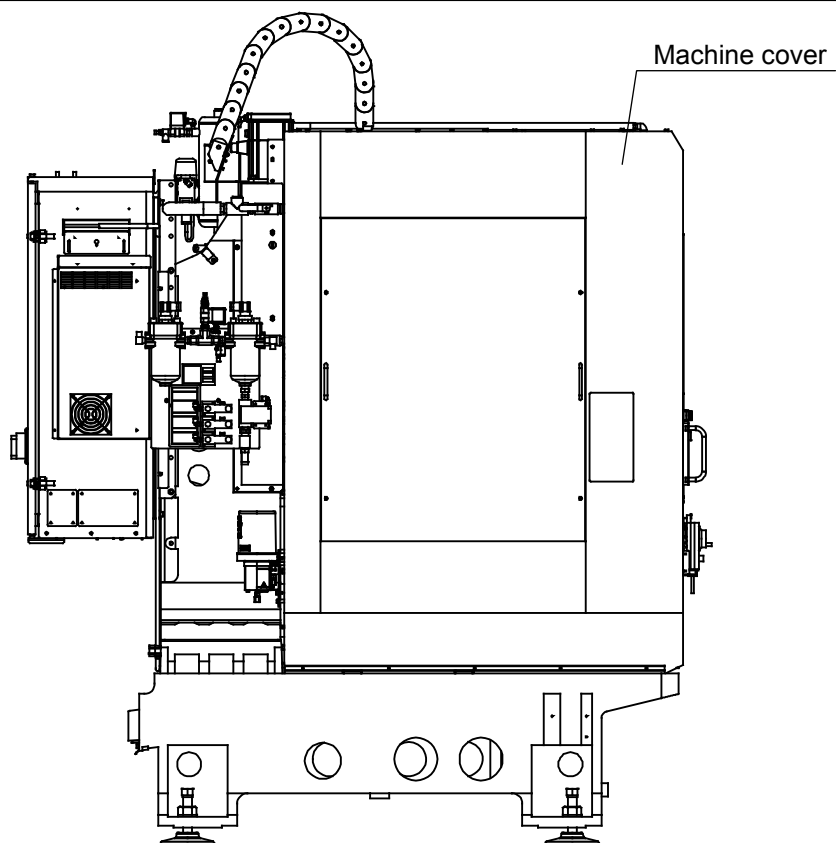
The basic structure of this machine is described below.

2.2.1 Basic Structure

1. It has a vertical spindle head.
2. The X- and Y-axes travel along the table.
3. The Z-axis is for spindle head movement.
4. The shortest random access path and unique armless ATC mechanism facilitate quick tool changing.
5. The control box is located at the rear of the machine. It requires minimal space for installation space because it integrates mechanical and electrical control.

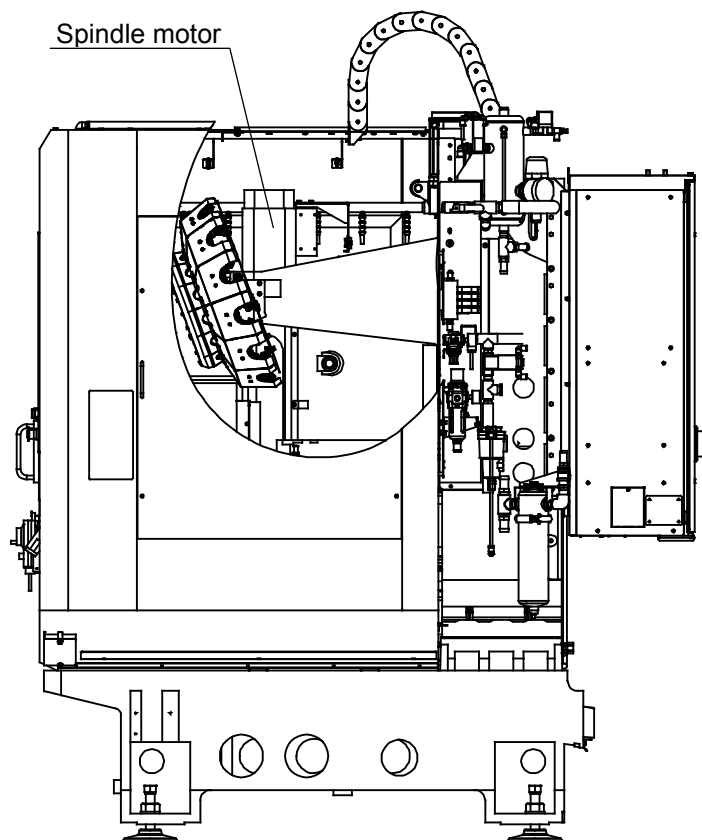
Part names and external appearance

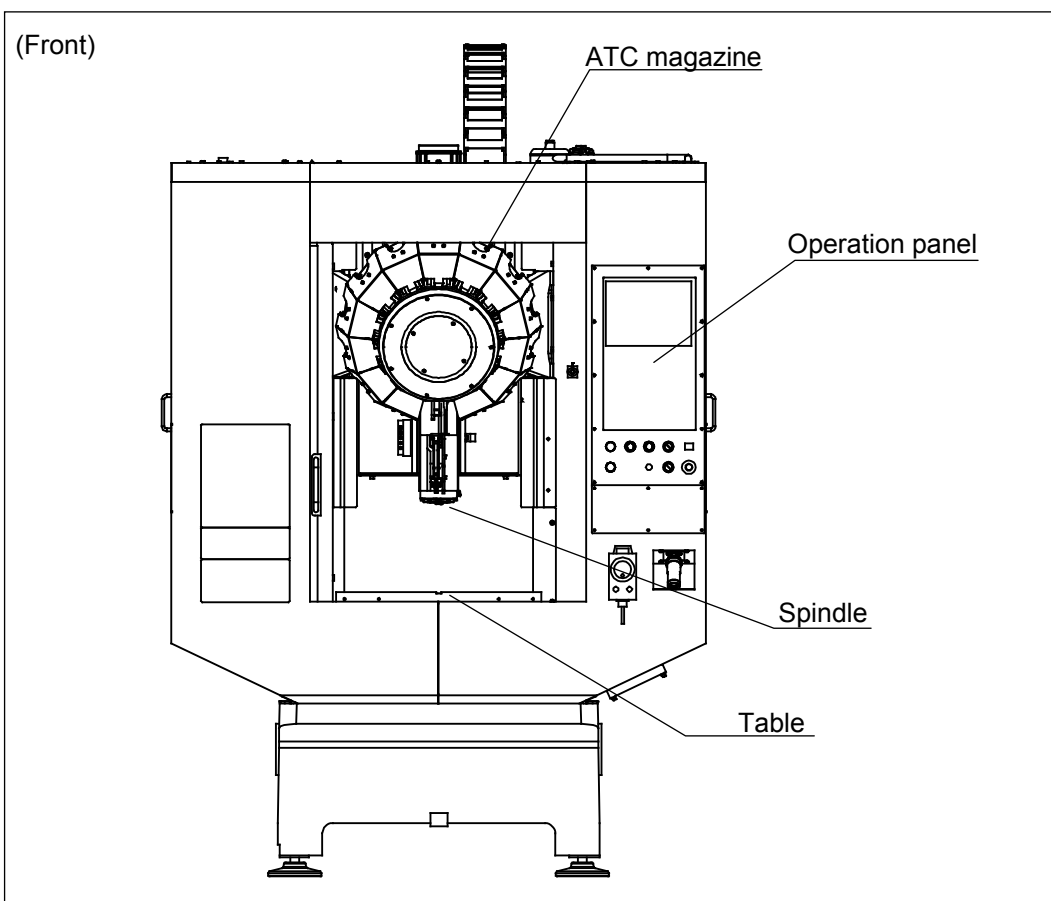
(Left side)



(Right side)

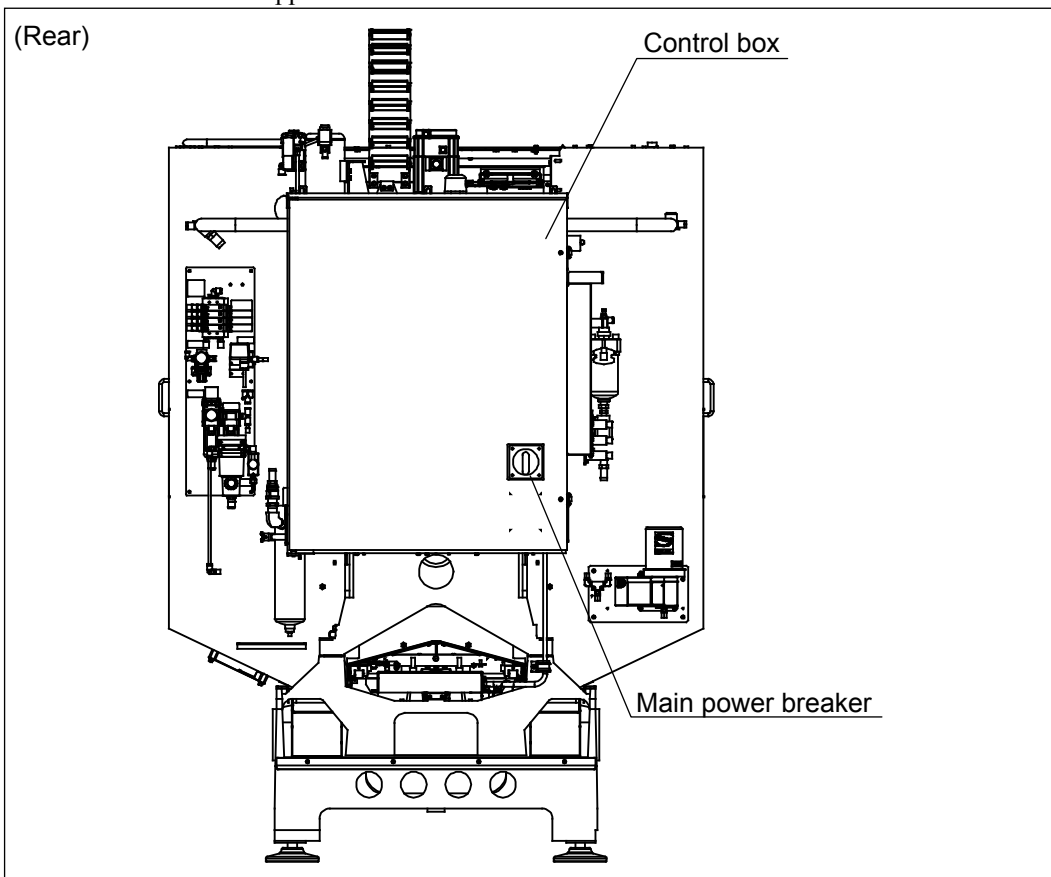
Spindle motor





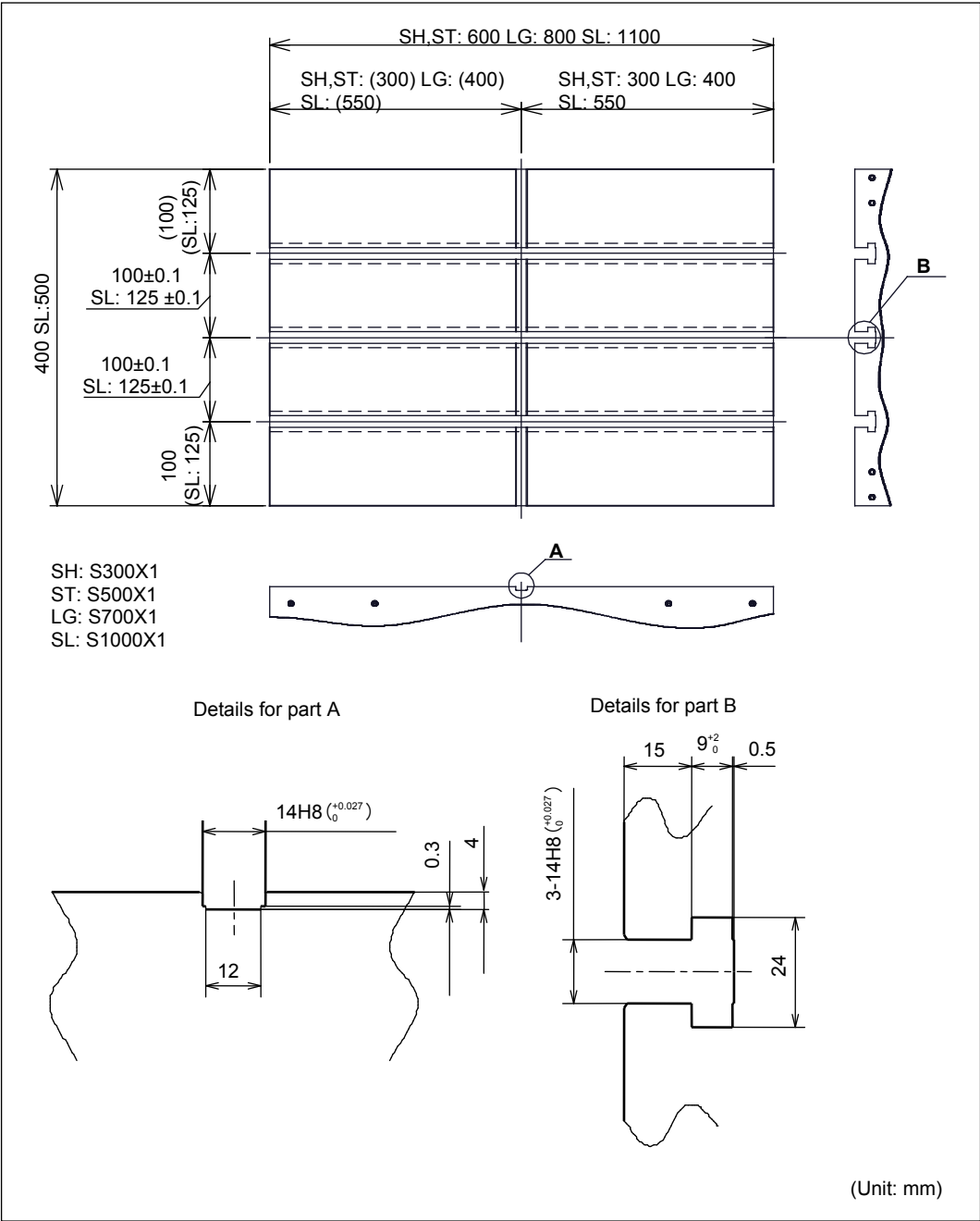
2.2.2 Rear of Machine

Part names and external appearance

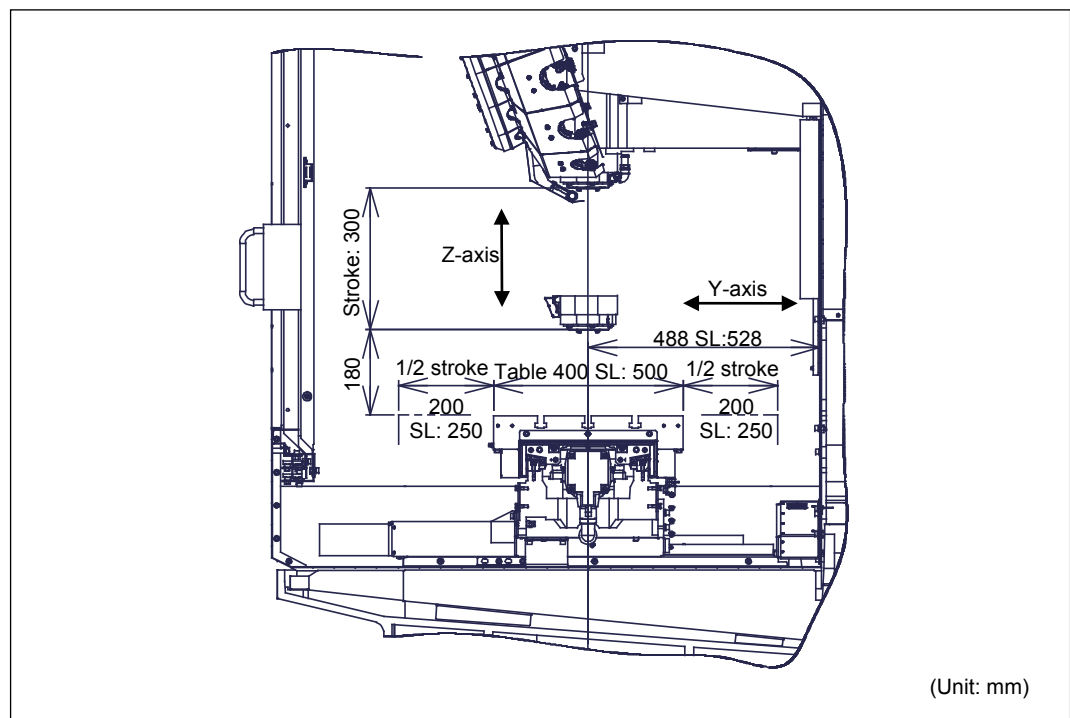
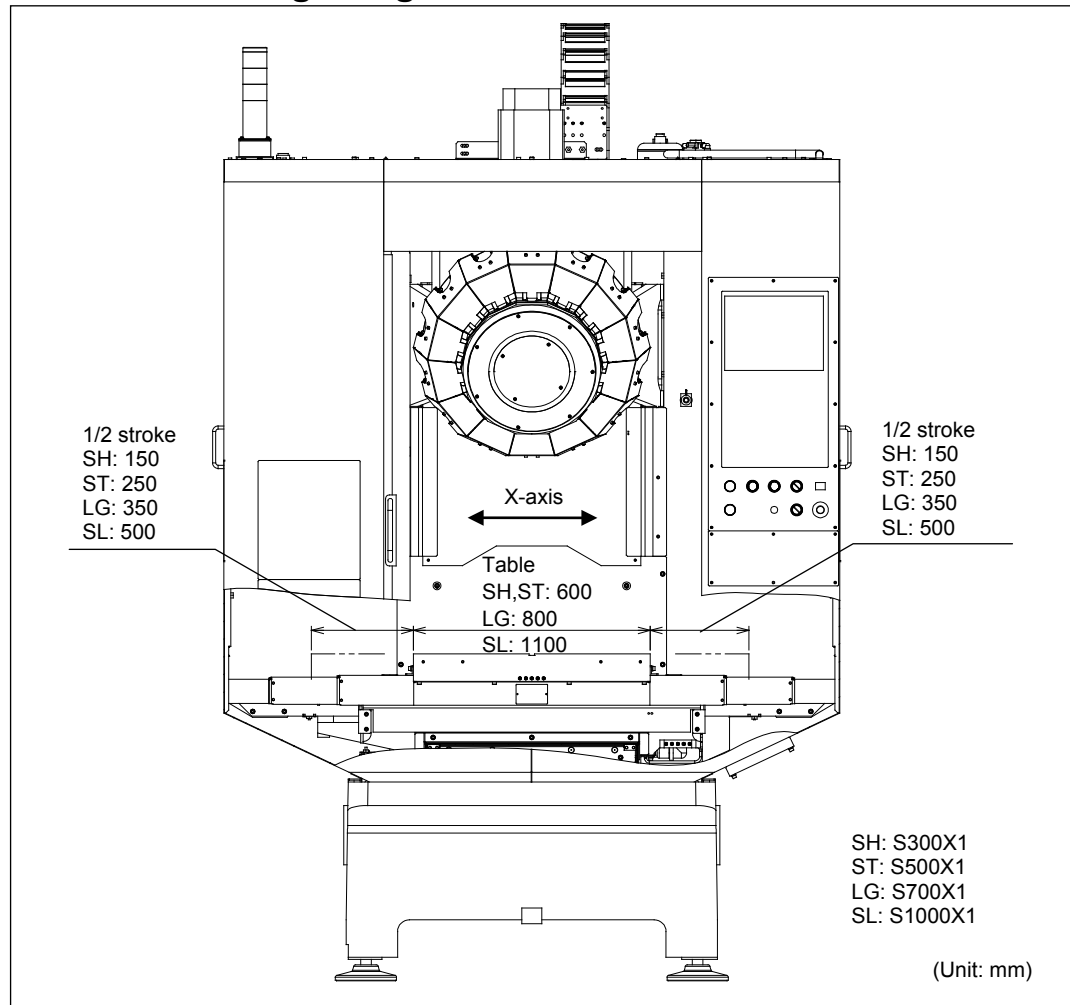


2.3 Table Dimensions and Machining Range

2.3.1 Table Top Dimensions



2.3.2 Machining Range



2.4 CNC Unit Specifications

CNC unit specifications (1)

Function	Description
Control axes	5 axes (X, Y, Z, A and B)
Simultaneously controlled axes	Positioning: 5 axes (X, Y, Z, A and B) Interpolation: 4 axes for linear interpolation (X, Y, Z, A or B) (2 linear axes plus 2 rotational axes, such as X-, Y-, A- and B-axes, cannot be selected.) 2 axes for arc interpolation 3 axes for helical / conical interpolation (X, Y and Z)
Minimum setting unit	0.001 mm (0.0001 inch) Option: 0.0001 mm (0.00001 inch)
Minimum travel unit	0.001 mm (0.0001 inch) Option: 0.0001 mm (0.00001 inch)
Programming	Absolute / Incremental programming
Maximum command value	±9999.999 mm (±999.9999 inch) Option: ±9999.9999 mm (±999.99999 inch) When issuing a radius command for circular interpolation: ±99999.999 mm (±9999.9999 inch) Option: ±99999.9999 mm (±9999.99999 inch)
Display	12.1 inch LCD
Decimal point input	Decimal points can be used for data input.
MDI	NC program data can be input via keyboard.
Machine zero return	Machine zero return is carried out automatically via keyboard input.
Position display	Coordinates for the current position on each axis can be displayed.
Coordinate system setting	A coordinate system can be set for a specified position on X-, Y- and Z-axes.
Stroke limit	The travel area can be set or restricted using parameters or programming commands for each axis.
Emergency stop	All commands and machine operations are immediately stopped.
High-speed travel (Manual operation - RPD)	Axes travel at 10%, 25%, 50%, 75%, or 100% of the maximum speed. However, the maximum speed is restricted during manual operation.
Low-speed travel (Manual operation - JOG)	Axes travel at a speed that ranges between 50 and 4000 mm/min (2.0 to 157.4 inch/min).
Step travel (Manual operation - STEP)	Axes travel in steps: 4 different steps that range between 0.001 mm to 1.0 mm (0.0001 to 0.1 inch/step).
Pause (temporary stop)	Axis travel is temporarily stopped. Press the START switch to resume operation.
Automatic acceleration/deceleration	Axis travel is automatically accelerated or decelerated.
Synchronized tapping	Screw pitch is completely synchronized with 1 spindle rotation.
Interpolation function	Linear interpolation, circular interpolation, helical thread cutting interpolation, focal interpolation and conical interpolation.
Canned cycle	Boring, tapping and other operations are automatically executed in a fixed sequence.
Program restart	Operation can be started from a specific part in the middle of a program.
Pitch error compensation	Mechanical error resulting from screw feed on additional axes (4 th to 10 th axes) is corrected.
Dry run	Cutting feed rate is switched to high speed travel in manual operation in order to check the program.
Dwell	Operation pauses or stops temporarily for a specified time.
Backlash compensation	Lost motion in the mechanical system is compensated by pulses that move each axis within range between 0 and 0.127 mm (0.0050 inch).
Tool length setting	Tool length measured on the machine is automatically registered in the memory as a data setting.

Function	Description
Background programming	A program can be edited during a memory operation.
Program number search	Any program in the memory can be called using a 4-digit program number.
Self-diagnosis alarm	CNC unit performs various checks and displays the status on the screen if a problem is found.
External program and data call	External program or data can be transferred or deleted using an external communication device.
Restart	Program operation can be repeated in memory mode.
Block skip	The block command containing a slash (/) can be skipped. (NC language only)
Optional stop	The program can be stopped at a M01 block. (NC language only)
Machine lock	A program can be checked without operating the machine.
Automatic spindle speed setting	Spindle speed is automatically determined according to the workpiece material, cutting conditions and tool diameter that are set in the data bank. (Conversation only)
Automatic tool selection	The tools required for a given program are automatically selected from the tools registered in the data bank. (Conversation only)
Scheduled program	Up to 99 programs can be combined and executed consecutively. (Conversation only)
Motion program	A motion sequence for your machining requirements can be registered. This programming enables you to machine a sequence even though it is not set in the conversational programs or in the data bank. (Conversation only)

2.5 Machining Capabilities

Data varies depending on conditions or factors such as the type of tool, tool status, cutting conditions and the working environment inside the factory. Therefore, use the following data as a guide or a reference.

Machining capabilities

Specification: Spindle rotation speed 10000 min⁻¹

Machining	Conditions	S45C	FC200	ADC
Drilling	Diameter (mm)	φ25	φ28	φ32
	Feed (mm/rev)	0.1	0.15	0.2
Tapping	Diameter (mm)	M16	M24	M27
	Pitch (mm)	2.0	3.0	3.0
Facing φ50 carbide 4-blade (Cutting width: 40)	Cutting amount (cm ³ /min)	100	137	-
	Cutting width (mm)	40	40	-
	Infeed amount (mm)	5.2	6.0	-
	Feed rate (mm/min)	484	573	-
Facing φ125 carbide 5-blade (Cutting width: 100)	Cutting amount (cm ³ /min)	-	-	960
	Cutting width (mm)	-	-	100
	Infeed amount (mm)	-	-	3.2
	Feed rate (mm/min)	-	-	3000

Specification: Spindle rotation speed 16000 min⁻¹

Machining	Conditions	S45C	FC200	ADC
Drilling	Diameter (mm)	φ18	φ22	φ24
	Feed (mm/rev)	0.1	0.15	0.2
Tapping	Diameter (mm)	M14	M18	M22
	Pitch (mm)	2.0	2.5	2.5
Facing φ50 carbide 4-blade (Cutting width: 40)	Cutting amount (cm ³ /min)	48	73	-
	Cutting width (mm)	40	40	-
	Infeed amount (mm)	2.5	3.2	-
	Feed rate (mm/min)	484	573	-
Facing φ125 carbide 5-blade (Cutting width: 100)	Cutting amount (cm ³ /min)	-	-	660
	Cutting width (mm)	-	-	100
	Infeed amount (mm)	-	-	2.2
	Feed rate (mm/min)	-	-	3000

Specification: Spindle rotation speed 10000 min⁻¹ with high torque

Machining	Conditions	S45C	FC200	ADC
Drilling	Diameter (mm)	φ30	φ34	φ40
	Feed (mm/rev)	0.15	0.15	0.2
	Diameter (mm)	φ26	φ26	φ30
	Feed (mm/rev)	0.25	0.4	0.7
Tapping	Diameter (mm)	M27	M33	M39
	Pitch (mm)	3.0	3.5	4.0
Facing φ50 carbide 4-blade (Cutting width: 40)	Cutting amount (cm ³ /min)	100	137	-
	Cutting width (mm)	40	40	-
	Infeed amount (mm)	5.2	6.0	-
	Feed rate (mm/min)	484	573	-
Facing φ125 carbide 5-blade (Cutting width: 100)	Cutting amount (cm ³ /min)	-	-	1700
	Cutting width (mm)	-	-	100
	Infeed amount (mm)	-	-	5.7
	Feed rate (mm/min)	-	-	3000

Specification: Spindle rotation speed 27000 min⁻¹

Machining	Conditions	S45C	FC200	ADC
Drilling	Diameter (mm)	φ17	φ19	φ20
	Feed (mm/rev)	0.10	0.15	0.20
Tapping	Diameter (mm)	M12	M18	M22
	Pitch (mm)	1.75	2.5	2.5
Facing φ50 carbide 4-blade (Cutting width: 40)	Cutting amount (cm ³ /min)	24	45	-
	Cutting width (mm)	40	40	-
	Infeed amount (mm)	1.2	2.0	-
	Feed rate (mm/min)	484	573	-
Facing φ125 carbide 5-blade (Cutting width: 100)	Cutting amount (cm ³ /min)	-	-	600
	Cutting width (mm)	-	-	100
	Infeed amount (mm)	-	-	2.0
	Feed rate (mm/min)	-	-	3000

2.6 Safety Devices

DANGER

If the emergency stop switch is damaged, the emergency stop may not function.

Check the emergency stop switch operation before starting machine operation.

If the emergency stop switch is damaged or does not function, it should be replaced.

WARNING

If the door interlock switch is disabled, the machine will operate even when a door is open, and you may get caught or drawn into the machine.

The installer or setup personnel must never leave the machine unattended while the door interlock switch is disabled.

The supervisor must not let the operator use the machine when the door interlock switch is disabled.

The operator must check that the door interlock switch is enabled before operating the machine.

The operator must inform the supervisor if the door interlock key is attached.

The door interlock key should be kept by the supervisor.

WARNING

If the maintenance cover is removed, coolant coming from inside the machine when it is operating may get into your eyes, or the tools or workpieces may shoot out causing injury.

The operator should visually check to make sure that the maintenance cover is attached before turning ON the power.

The operator should warn the supervisor if the maintenance cover is not attached.

The supervisor must attach the maintenance cover.

This machine is equipped with safety devices to ensure that the safety of the operator or worker as well as to protect the machine.

2.6.1 Main Safety Devices

1. Covers (Machine cover and maintenance cover)
They protect the operators and workers by preventing the chips or shavings from shooting out during operation.
Do not operate the machine without the covers attached.
2. Door limit switch
If a door is opened during automatic operation, the door sensor is activated and machine operation is stopped.
In addition, automatic operation will only start if the door is closed.
3. Emergency stop switch
Pressing this switch stops all machine operation instantly.
Press this button right away if you sense any machine abnormality or danger in the slightest.
Refer to “Chapter 7, 7.12 Emergency Stop” for further details on the release procedure.
4. Alarm message display
If a machine error occurs, an alarm message is displayed on the operation panel screen.
If multiple errors occur, the most important alarm message is displayed.
Press the **[ALARM]** key on the operation panel to display all the alarm messages. Up to 18 messages can be displayed for current alarms, which are shown in order of importance.

2.7 Coolant

- (Note 1)** To select a coolant, ask the coolant dealer about the coolant's lubrication quality, corrosion prevention, bubbling prevention and safety.
Do not use a chemical solution (synthetic coolant). They provide poor lubrication and strip coating off the machine, possibly leading to machine damage.
- (Note 2)** Do not use any coolant that hardens or swells the nitrile rubber (NBR) or fluorocarbon rubber.
- (Note 3)** Water soluble coolant that has been broken down by bacteria, etc., can cause a bad smell, environmental deterioration and other problems. It may also cause rust, due to deteriorating quality, which can lead to machine damage. If the coolant has deteriorated, stop using it and replace all the coolant with new coolant.

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CHAPTER 3

3

OPERATION

- 3.1 Turning Power ON**
- 3.2 Turning Power OFF**
- 3.3 External Appearance of Operation Panel**
- 3.4 Screen Layout**
- 3.5 Switch and Lamp Functions**
- 3.6 Key Functions**
- 3.7 Important Notes When Attaching Workpieces, Jigs and Tools**
- 3.8 Attach Tool to Magazine and Remove Tool from Magazine**

3.1 Turning Power ON

WARNING

If the door interlock switch is disabled, the machine will operate even when a door is open, and you may get caught or drawn into the machine.

The installer or setup personnel must never leave the machine unattended while the door interlock switch is disabled.

The supervisor must not let the operator use the machine when the door interlock switch is disabled.

The operator must check that the door interlock switch is enabled before operating the machine.

The operator must inform the supervisor if the door interlock key is attached.

The door interlock key should be kept by the supervisor.

WARNING

If the maintenance cover is removed, coolant coming from inside the machine when it is operating may get into your eyes, or the tools or workpieces may shoot out causing injury.

The operator should visually check to make sure that the maintenance cover is attached before turning ON the power.

The operator should warn the supervisor if the maintenance cover is not attached.

The supervisor must attach the maintenance cover.

WARNING

If tools and other items are left in or around the machine, they may fall down or shoot out when the machine is operating, and injury may occur.

The installer should not leave tools or other items in or around the machine.

The operator must not operate the machine if tools or other items have been left in or around the machine. If this is not observed, the supervisor should be notified.

WARNING

If items such as the tool and workpieces are not properly attached, or if cutting conditions are not within the tool limits, items may shoot out from inside the machine and strike personnel causing injury.

Always read the tool labels and instruction manuals for information on tool limits.

Check that the tips are securely attached before installing the tool.

Workpieces should be securely attached.

Close the doors before operating the machine.

The following procedure is for turning ON the power.

3.1.1 Check Before Turning ON the Power

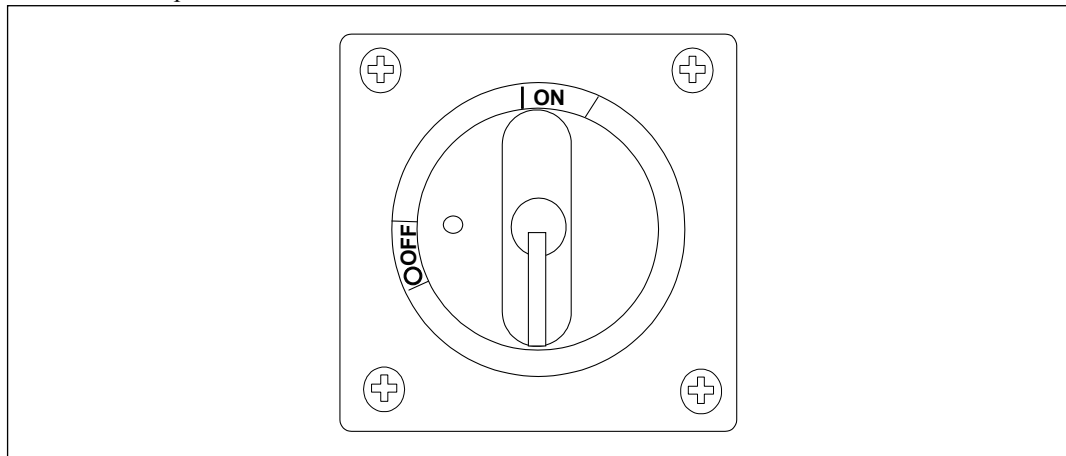
1. Make sure that the main power breaker handle is OFF.
2. Make sure that the machine and its surroundings are free from obstacles.
3. Make sure that there are no tools or objects left on the machine table.
4. Make sure that the control box door is closed.
5. Make sure that there is no dust or damage on the taper parts of the spindle.
6. Make sure that the taper parts on the tools are clean without pitching or wear and that the pull studs are not loose.
7. Make sure that the tool is securely attached to the magazine gripper.
8. Make sure that no chips or shavings are left inside the machine cover from the day before.
9. Make sure that no chips or shavings are left in the chip pan from the day before.
10. Make sure that there is enough coolant.
11. Make sure that the air is supplied with the specified pressure.
12. Make sure that the front door is closed.
13. Make sure that the power switch is not pressed down.

3

3.1.2 Turning Power ON

1. Connect the machine cord to the factory side power outlet, and turn ON the factory side power.
2. Turn ON the main power breaker in the control box located at the rear of the machine.
3. Make sure that the main power lamp on the operation panel is flashing.

When the main power breaker is turned ON:

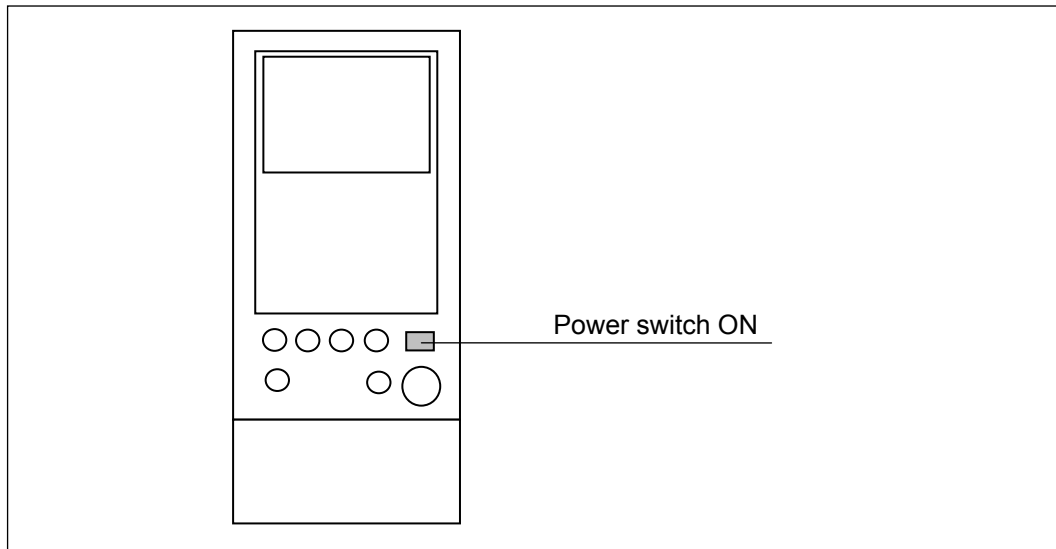


4. Turn ON the power on the operation panel.
When the power switch is pressed, the green lamp lights up and the initial screen is displayed. At the same time, the control box fan rotates, and the servo lock is engaged on the X-, Y- and Z- axes as well as on the magazine motor.
- Make sure that the fan on the side of the control box is rotating.
 - The screen display may take about 15 seconds to appear after turning ON the power, depending on the status of the machine. (The power switch flashes at this time.) In addition, it takes about 50 seconds until the machine can be operated.
 - If the power is turned ON while the door interlock is enabled and the front door is left open, the alarm <<*door open>> is triggered and the servo lock will not engage.
 - Press the power switch with the memory card inserted into the USB connector. If the system does not start up normally, turn the power OFF, remove the memory card and press the power switch again.

Checks

1. Make sure that the magazine rotates smoothly and that the tools can be installed or removed properly.
2. Make sure that the cooling fan on the side of the control box is operating properly.
3. Make sure that there are no abnormal noises while the spindle is rotating.
4. Make sure that the spindle is properly oriented during the positioning operation.

Power switch ON



3.2 Turning Power OFF

The following procedure is for turning OFF the power.

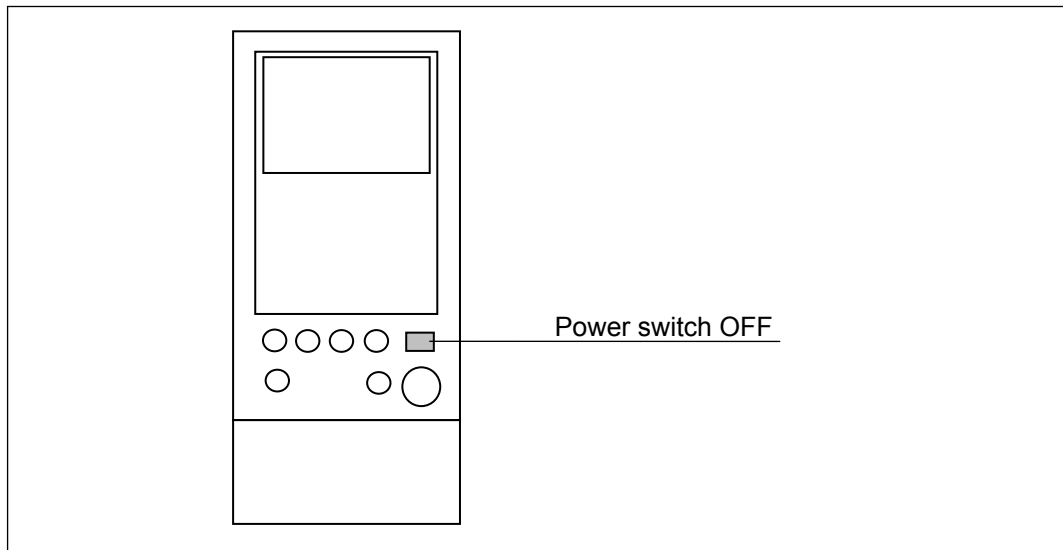
3.2.1 Check Before Turning OFF the Power

1. Make sure that all moving parts have stopped on the machine.
2. Make sure that all the operations such as program editing, data changes, and the input and output with external devices have been completed.

3.2.2 Turning Power OFF

1. After the power switch on the operation panel is pressed, the green lamp flashes and the screen turns off.

Power switch OFF

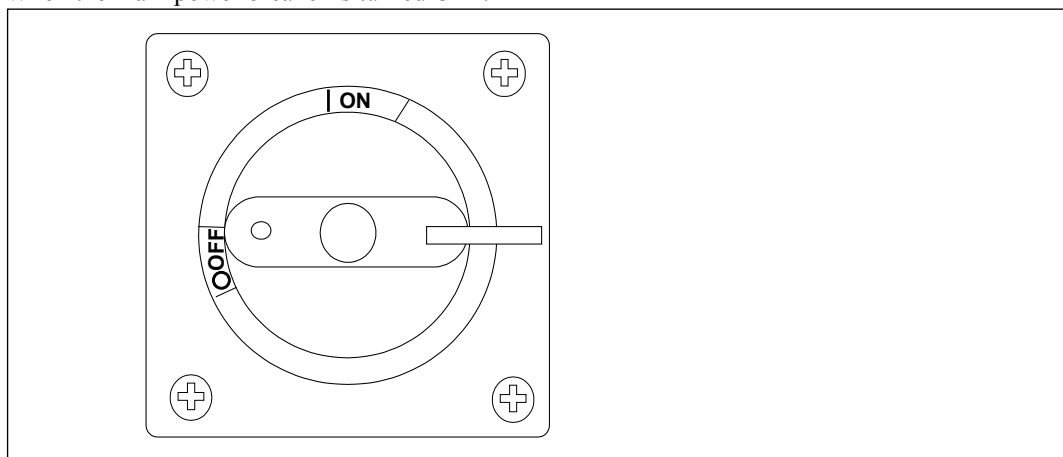


3

2. After the screen turns off completely, turn OFF the main power breaker (handle) in the control box located at the rear of the machine.

(Note) If the main power breaker (handle) is turned OFF before the screen turns off, it may corrupt the data.

When the main power breaker is turned OFF:



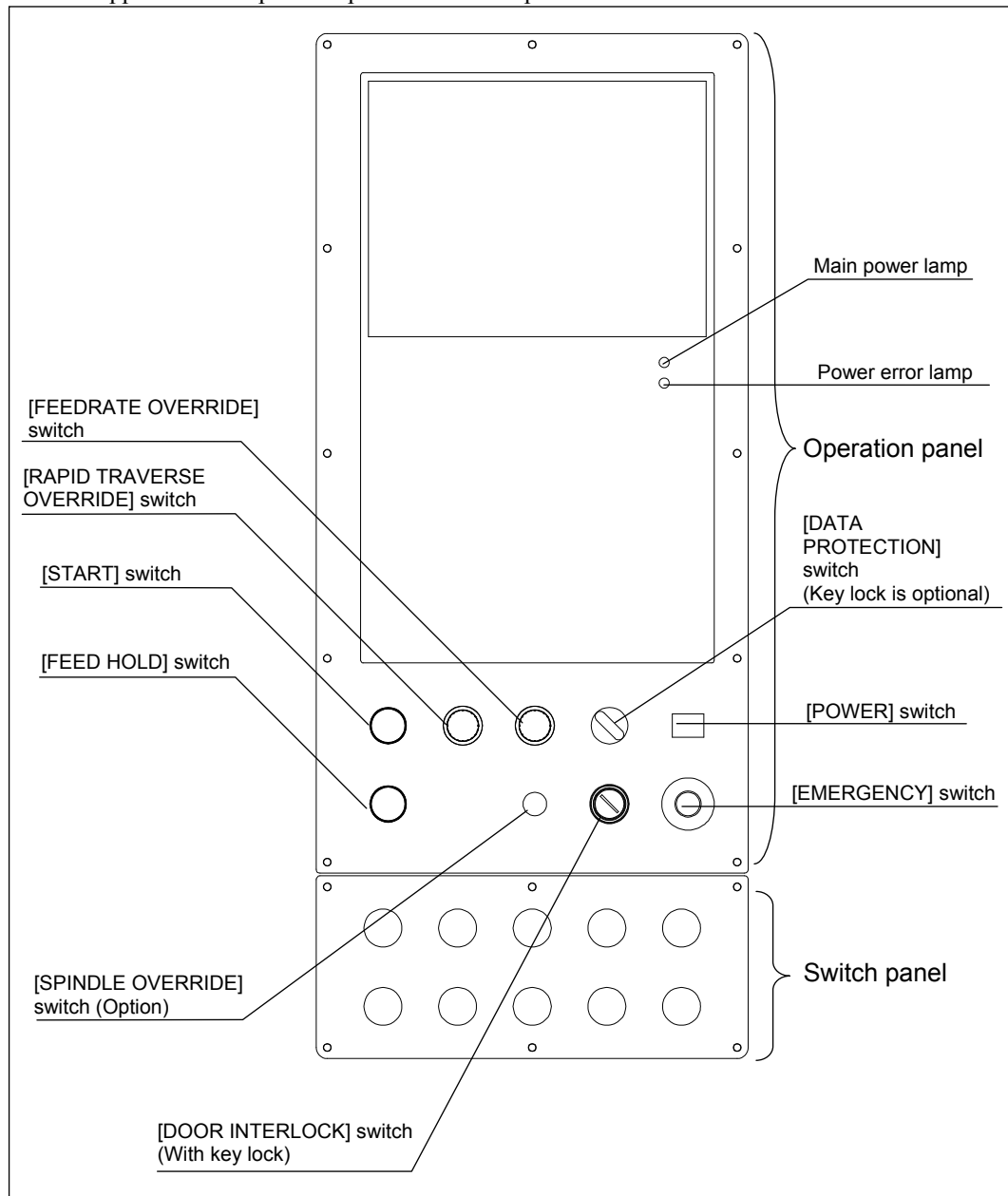
3. Disconnect the machine side power cord that is connected to the factory's power outlet. In addition, turn OFF the power on the factory's switchboard, etc.

Turn the main power breaker OFF in the control box when the machine stops due to a power failure. After power is restored, turn it ON again following the procedure "3.1 Turning Power ON" in the Operation Manual.

3.3 External Appearance of Operation Panel

The NC operation panel consists of an operation panel and switch panel.

External appearance of operation panel and switch panel



3.4 Screen Layout

The following shows a description of the layout for the operation panel screen.

The screenshot displays the operation panel screen layout with the following components and callouts:

- (1) Alarm / Operator message area:** Displays messages such as "7514 Spindle override on" and "IO0500 *Emergency button activated".
- (2) Title area:** Displays the title "ATC tool(details)" and the current date and time "2012/06/04 14:54:38".
- (3) Program display area:** Displays a table of tool data for Program(9900/0515/01). The table includes columns for Tool No., Tool name, Tool data, Main tool, Tool life, Tool type, and Color.
- (4) Input request area / Instructions area:** Displays a form for tool selection, including fields for Tool No., O:Cap setting, and Program(0001). It also includes a [HELP] button and a :Set tool button.
- (5) Function key area:** Displays a row of function keys: Set program No., Set tool, List, Initialize life, and Next tool.

Tool No.	Tool name	Tool data	Main tool	Tool life	Tool type	Color
01	40 Drill	10.000		*****	STD tool	
02	02 Drill	1.000		10000(H)	STD tool	
03	28 Center drill	3.000x90.000		*****	STD tool	
04	95 Rolled tap	M3.000x0.500		*****	STD tool	
05	45 CBR tool	12.000		*****	STD tool	
06	26 Rolled tap	M3.000x0.500		0min	STD tool	
07	01 Center drill	1.000x90.000		50000min	STD tool	
08	36 Reamer	10.000		10000pcs.	STD tool	
09	41 Chamfer tool	90.000x12.000		*****	STD tool	
10	98 Centering TL	10.000x120.000		*****	STD tool	
11					STD tool	

Set program No.	Set tool	List	Initialize life	Next tool
01 28 Center drill	05	09		
02 40 Drill	06	10		
03 45 CBR tool	07	11		
04 41 Chamfer tool	08	12		

1. Alarm / Operator message area
This area displays current alarms and operator messages.
2. Title area
This area displays the title of the screen that is currently displayed. In addition, the right side displays the current date and time as well as pictographs that show the machine status.
3. Program display area
This area provides data required for machine operation and a program description.
4. Input request area / Instructions area
If data needs to be entered, the input request is displayed in this area.
Based on input request details, the user enters number values and selects the required items from the instructions area.
5. Function key area
This area displays function keys to select items required for machine operation, etc.

Pictographs

The following pictographs or icons are displayed to indicate the machine status.

	Lathe spindle selection	Displays when the lathe spindle is currently selected.
	Auto-power OFF function enabled	Displays when the automatic power OFF function is enabled.
	Door interlock disabled	Displays when the door interlock is disabled.
	Door open	Displays when the door is open.
	Spindle override	Displays when the spindle override is not 100%.
	Cutting feedrate override	Displays when the cutting feedrate override is not 100%.
	Rapid traverse override	Displays when the rapid traverse override is not 100%.
	External I/O signal key inputting	Displays when the signal from an external I/O signal key is inputting. ^{*1}
	Slave communication	Displays when communicating with a slave via a serial port or Ethernet. ^{*1}
	Encoder amplifier battery expired	Displays when the battery has expired. The alarm <<Time to change batteries>> (Stop level 1 and cancel level 1) is triggered at the same time. In addition, if the battery has expired while turning ON the power, the alarm <<Time to change batteries>> (Stop level 2 and cancel level 2) is triggered at the same time.

^{*1} When this event occurs at the same time, the “External I/O signal key inputting” takes precedence.

3.4.1 Character String Display

For places where data for the following character strings (character string data) can be entered, the characters will be displayed in Japanese when Japanese is selected as the language setting.

- <Tool name> for tool data
- <Name> of the auxiliary function for user registration in the user parameter
- <External alarm ** message> for external I/O signal
- NC program comments section
- * Japanese cannot be entered using key operations on this machine. If data is output using an external I/O to a computer and then the character string data is edited using the Japanese language setting, it will be displayed in Japanese when that data is input into this machine again.
- * When the machine parameter <Display language> is configured to another setting besides <0: Japanese> and the character string data was set in Japanese, the text and characters will not display properly and be illegible.
If this occurs, use the alphabet to correct the corresponding text.

3.4.2 Switching Unit Systems

3.4.2.1 Switching Metric ↔ Inch

The unit system can be switched to metric or inch by changing the <Machine unit system> in the user parameter.

- The unit system changes for the X-, Y-, Z-axes as well as the PLC-axis (for the linear axis setting).
- The unit system for the additional axis and the PLC-axis (for rotation axis setting) is always set to “degrees” and cannot be changed.
- The metric system is always used for machine parameters.
- The metric system is always used when the magazine axis is displayed using coordinates.

3.4.2.2 Minimum Unit Settings

The minimum unit setting can be changed. (Type 2 is an option)

- Refer to the table below for the minimum unit settings.
- The minimum unit setting for all axes (X-, Y- and Z-axes, additional axis, magazine axis and PLC axis) can be changed.
- Conversation language cannot be used for Type 2.

The alarm <<User param. setting error (switch 1)>> (Cancel level 2) is triggered when the user parameter <Conversation/NC language change> is set to <0: Conversation>.

The screen is displayed in NC language when the alarm mentioned above is triggered.

This manual, the Data Manual, the Installation Manual and the Programming Manual are written assuming that the machine is Type 1. If the machine is Type 2, refer to the conversion table below.

Type	Minimum unit setting	Minimum travel unit	Maximum stroke	Minimum speed unit
Type 1	0.001 mm	0.001 mm	±9999.999 mm	0.01 mm/min 0.0001 mm/rev
	0.0001 inch	0.0001 inch	± 999.9999 inch	0.001 inch/min 0.00001 inch/rev
	0.001 deg	0.001 deg	±9999.999 deg	0.01 deg/min 0.0001 deg/rev
Type 2 (Option)	0.0001 mm	0.0001 mm	±9999.9999 mm	0.001 mm/min 0.00001 mm/rev
	0.00001 inch	0.00001 inch	± 999.99999 inch	0.0001 inch/min 0.000001 inch/rev
	0.0001 deg	0.0001 deg	±9999.9999 deg	0.001 deg/min 0.00001 deg/rev

3.5 Switch and Lamp Functions

A description of the functions for the operation panel and switch panel's switches is provided.

1. **[POWER]** switch
 When the main power breaker is ON, it flashes slowly. This switch turns the power ON/OFF. The main operations and corresponding lamp status are shown below.
 Main power breaker ON : Flashes (Approximately 9 second intervals)
 Power ON : Lit
 Power OFF : Flashes (Approximately 0.5 second intervals)
 When pressed, the power turns ON and the green lamp lights up. If the switch is pressed again while it is green, it flashes in short intervals. The power turns OFF and the lamp flashes slowly again.
 Press the power switch and if the green lamp flashes, the continuous control circuit for inrush currents is activated. When the power is turned OFF and ON within 15 seconds after the previous machine operation, the control circuit is activated. After 15 seconds has elapsed since the previous machine operation, the power turns ON again.
 2. **[START]** switch
 Use this switch to begin MDI operation and memory operation.
 The switch lamp stays lit until the operation that was started is complete.
 3. **[FEED HOLD]** switch
 This switch temporarily stops (pauses) MDI operation or memory operation.
 The **[FEED HOLD]** switch lamp lights up when operation has paused or stopped temporarily. Spindle rotation does not stop. During a tool change or tapping process, it does not stop until these operations are completely finished.
 4. **[DATA PROTECTION]** switch
 Turn OFF (disable) this switch to input or to change the program and the data, and turn ON the switch (enable) to prevent making changes or deletions to the program and data.
 Disabled...Program editing, data input and changes to data are possible.
 Enabled...Programs stored in the memory and all data are protected from changes or deletions.
- * The **[DATA PROTECTION]** switch does not have a lock. Instead of a lock, a password is used.

Procedure to change the data protection status using a password

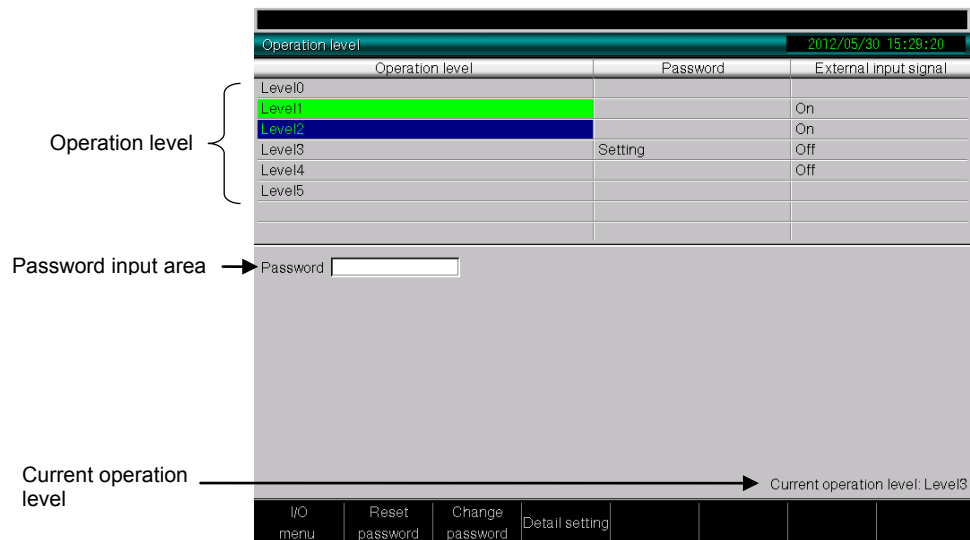
(1) Overview

The data protection status can be switched by changing the operation level using the previously set password.

This is where the procedure to change the status to “Data protection ON / mode change not possible” (Programs saved in the memory and data cannot be changed or deleted nor can the mode be changed).

(2) Password setting

- i) Select the “10. Operation level” from the I/O menu to display the operation level screen.



<Level 0>...Data protection ON / mode change not possible

<Level 4>...Data protection OFF

- * Set the [DATA PROTECTION] switch on the operation panel to “O” (disable).

- ii) Move the cursor to the <Level 4> position.

- iii) Press the [F2] (Change password) key.

- iv) Enter the password into the password input field, and press the [F1] (Change) key.

- * Use 4 to 10 alphanumeric characters to set the password.

- * This password is used as a lock. Keep it in a safe place so you don't forget it.

(3) Changing data protection status (when the power is ON)

Select the “10. Operation level” from the I/O menu to display the operation level screen.

When changing to <Level 0> from <Level 4>

... Press the [F1] (Reset password) key.

When changing to <Level 4> from <Level 0>

... Move the cursor to the <Level 4> position, enter the password into the password input field (2) and press the [ENT] key.

(4) Changing data protection status (when the power is starting up)

The user will be requested to enter a password.

When you wish to use <Level 4>

... Enter the password into the password input field (2) and press the [ENT] key.

When you wish to use <Level 0>

... Press the [F1] (Reset password) key, or press the [RST] key.

- * When the power is turned OFF in operation level 0, your password is not required.

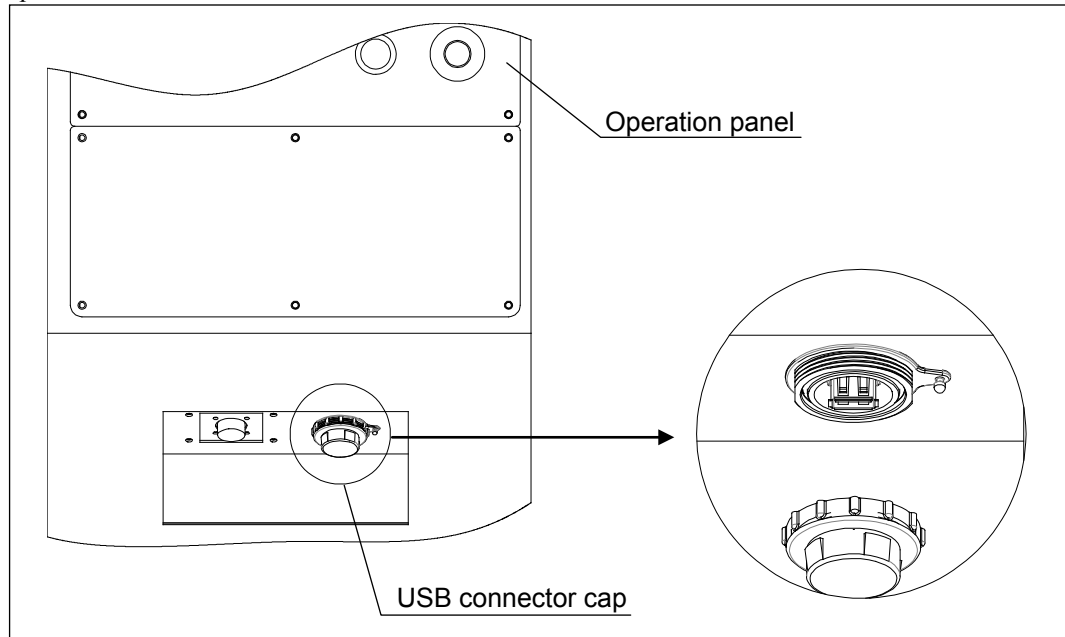
More advanced settings for the operation level can be used and configured. Refer to the “Chapter 6 Data Protection” in the Operation Manual (Data) for further details.

5. **[EMERGENCY]** switch
Use this switch to stop the machine in the event of an emergency or to stop machine malfunction due to an input error. If pressed during machining, machining operation stops immediately and the <<Emergency button activated >> display flashes on the screen.
6. **[DOOR INTERLOCK]** switch
Use this switch together with the door interlock device.
Enable: The door interlock function is enabled.
Disable: The door interlock function is disabled.
7. Main power lamp
This lamp turns to green when the main power breaker on the control box is turned ON.
This lamp flashes when an abnormal voltage continues to be input.
8. Power error lamp
The power automatically turns OFF when this lamp lights up.
This lamp turns red when the interphase input voltage is 15% greater than the rated voltage. In addition, it flashes when the phase-ground voltage for the input voltage is abnormally high.
Turn OFF the main power breaker on the control box to reset the error.
9. **[RAPID TRAVERSE OVERRIDE]** switch
Use this switch to override positioning operations.
4 different speeds can be set in the user parameter <Rapid traverse override speed 1 - 4>.
10. **[FEEDRATE OVERRIDE]** switch
Use this switch to cutting feed operations.
The ratio range is 0 to 200%.
When the user parameter (switch 1) <Stop using feedrate override during rapid feed> is set to <1: Enabled> and the **[FEEDRATE OVERRIDE]** switch is set at "0%", rapid feed stops regardless of the **[RAPID TRAVERSE OVERRIDE]** switch setting.
11. **[SPINDLE OVERRIDE]** switch (Option)
Use this switch to override spindle rotation.
The ratio range is 50 to 200%.

3.5.1 Memory Card Interface

3.5.1.1 Operation Procedure

Refer to the “Chapter 5 Communication” in the Operation Manual (Data) for further details about operation.



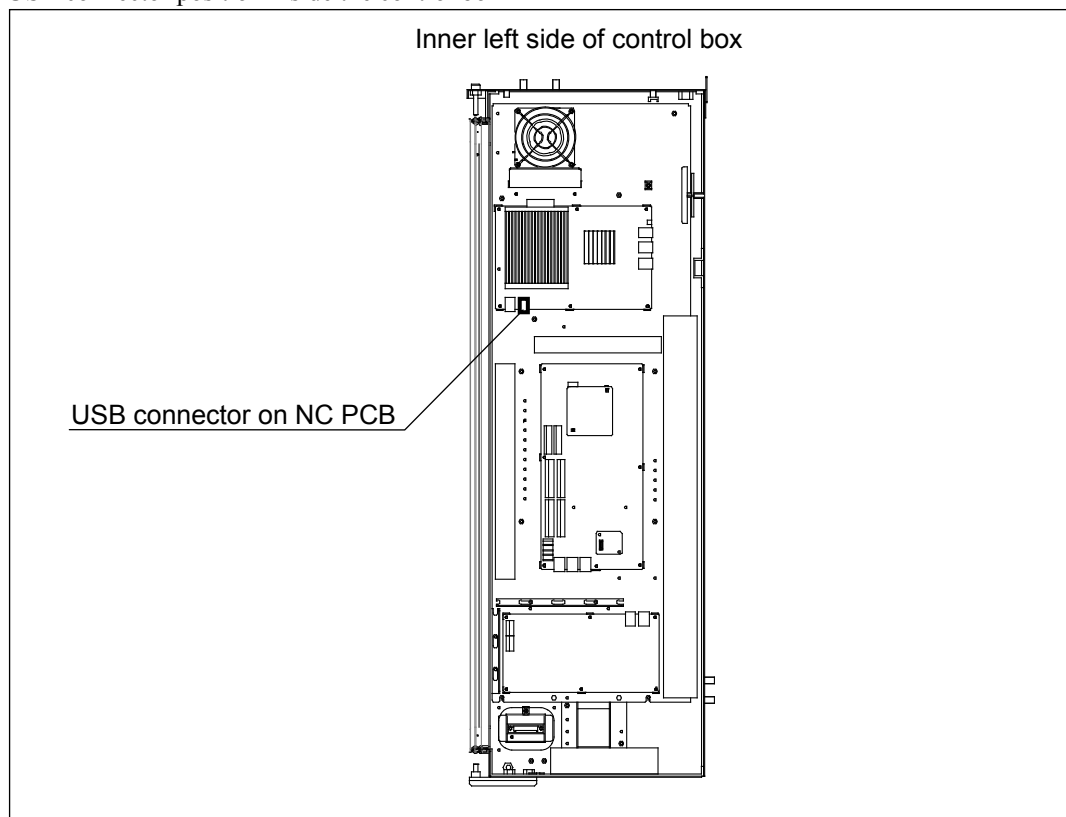
Opening and closing the USB connector cap
Turn left to open the cap, and turn right to close it.

3.5.1.2 Operation Precautions

- (Note 1) A memory card is not provided with this machine.
The customer needs to procure one.
- (Note 2) The memory card file format only supports FAT16 and FAT32.
- (Note 3) While the memory card write protection is ON, input from the memory card to the machine is enabled while output from the machine to the memory card is disabled. In addition, when the memory card is formatted for FAT32, input/output is available only when the write protection is OFF.
- (Note 4) Disengage or unlock the security feature before using the memory card. Otherwise the memory card may not be recognized.
- (Note 5) While the machine is accessing the memory card, the access lamp for the memory card lights up. Remove the memory card only after the necessary operations have been completed and the access lamp has turned off. Removing the memory card while the machine is accessing it may cause the machine to fail to recognize the memory card until the power is turned on again.
- (Note 6) Always remove the memory card from the machine when it is not used. Otherwise, the memory card could be accidentally hit, causing damage to the USB connector or the memory card.
- (Note 7) When removing the memory card, always be sure to close the USB connector cap. Otherwise, oil mist in the atmosphere can cause and lead to a poor contact.
- (Note 8) Even if the media has more than 2GB of free space, the screen cannot display a value exceeding 2GB.

- (Note 9) If 9 or more types of media during power-on are inserted and removed, the 9th and any subsequent media will not be recognized. Turn the power OFF and ON to reset this count.
- (Note 10) Even if the medium has enough free space, only a maximum of 512 files can be created.
- (Note 11) Use media with a single drive.
Using a memory card divided into multiple drives or using a memory card through an adapter supporting multiple media may not be recognized by the machine.
- (Note 12) When you wish to use this machine for a long period with the memory card inserted (tape operation, etc.), use a USB connector that is installed on the NC PCB inside the control box, because a poor contact may develop in an environment with oil mist, etc.
Refer to “USB connector position inside the control box” in the diagram.

USB connector position inside the control box



3.5.1.3 Compatible Memory Devices

Commercially available USB memory devices for Ver.1.1 and those compatible with both Ver.1.1 & 2.0 can be used on the machine.

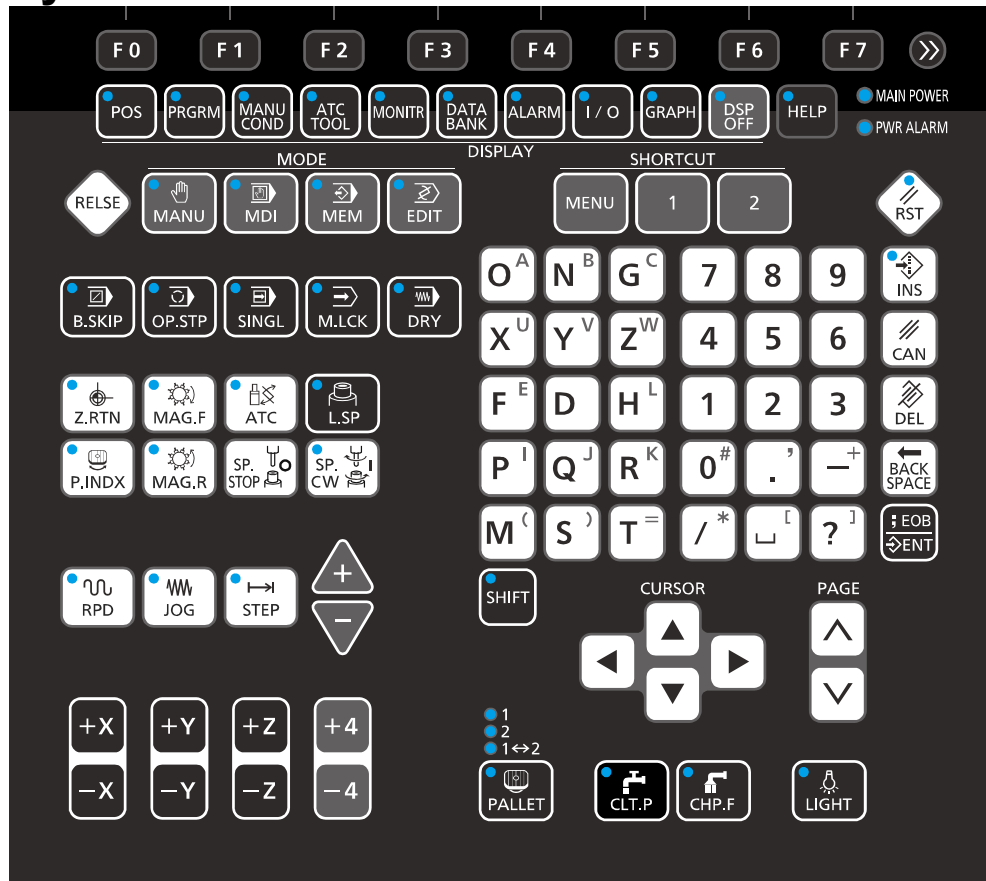
However, some memory devices may not be compatible.

Other media (compact flash, smart media and SD cards, etc.) can be used similarly by attaching an adapter permitting interface to the USB.

However, malfunction may occur depending on the adapter or media used.

Be aware that not all media commercially available is compatible with this machine.

3.6 Key Functions



A description of the panel key functions is provided.

[RST] key

Use this key to return the control unit to the default status.

- When this key is pressed, the alarm/operator message status for cancel level 1, 2 or 4 is reset. (The alarm/operator message without an asterisk (*) and the alarm/operator message with an asterisk (*) on the screen can be reset. All the alarm/operator messages other than the power OFF cancel level are reset.)
- If the user parameter <Cancel editing when resetting> is set <1: Yes> and this key is pressed during the setting of a program or data, all the input data become invalid.
- When this key is pressed during memory operation, the machine stops immediately together with X, Y, Z, 4, 5, 6, 7, 8, and the spindle. However, during a tool change or tapping process, it does not stop until the operation is completed.

[MANU] key

Use this key to change to manual operation mode.

The following operations can be performed in manual operation mode.

- Machine zero return
- Pallet index
- Single tool change (ATC)
- Magazine forward/reverse rotation
- High-speed travel
- Low-speed travel
- Step travel
- Spindle normal rotation and stop
- Operation using the handle
- Spindle selection

[MDI] key

Use this key to change to MDI operation mode.

[MEM] key

Use this key to change to memory operation mode.

[EDIT] key

Use this key to change to program edit mode.

[Z.RTN] key

This key is used in manual operation mode to return X, Y, Z, 4, 5, 6, 7 and 8 to the machine zero point.

After the machine zero return, the machine coordinate for X, Y, 4, 5, 6, 7 and 8 is 0. And, the value for the Z-axis is set in the machine parameter <Distance to Z-axis zero>.

However, the pallet (QT-axis) does not operate for models equipped with a quick table.

[MAG.F] key

Use this key to rotate the magazine in manual operation mode.

Press the key once to rotate the magazine clockwise for 1 index position.

This key is only valid when the Z-axis is at the ATC zero point.

[MAG.R] key

Use this key to rotate the magazine in reverse in manual mode.

Press the key once to rotate the magazine counterclockwise for 1 index position.

This key is only valid when the Z-axis is at the ATC zero point.

[ATC] key

Use this key to change the magazine tool in manual operation mode.

Each time the key is pressed, positioning is performed to the ATC zero point on the Z-axis, the magazine rotates and positioning is performed to the Z-axis zero point. Then, the tool change operation is completed.

[P.INDX] key (only models equipped with a quick table)

Use this key to index pallet 1 and pallet 2, alternating in sequence, while in manual operation mode.

When this key is pressed, the Z-axis moves up to its machine zero position to index a pallet on the opposite side.

[L. SP] key (NC language)

This key is only valid when the machine is equipped with a lathe spindle.

This key is used to select either the spindle (tool side rotates) or the lathe spindle (workpiece side rotates). When the LED is turned off, the spindle is selected, and when the LED lights up, the lathe spindle is selected.

Press the **[L. SP]** key to stop the spindle/lathe spindle.

[SP.CW] key

Use this key to rotate the spindle clockwise in manual operation mode.

The direction of rotation is clockwise when looking from above the spindle motor.

The spindle can rotate at a speed between 100 min-1 and the <Max. spindle speed> set in the machine parameter.

The default setting is 100 min-1 when the power is turned ON.

When the machine is equipped with a lathe spindle and it is selected, the spindle can rotate between 50 min-1 and the <Turning spindle maximum rotational frequency> set in the machine parameter.

The default setting is 50 min-1 when the power is turned ON.

[SP.STOP] key

Use this key to stop rotation in manual operation mode.

[RPD] key

Press this key to move the axis fast when using the axis travel keys to move each axis. The axis can move at a speed between 10-100% of the speed value set in the machine parameter <Manual rapid feedrate>. The default setting is 10% when the power is turned ON.

[JOG] key

Press this key to move the axis slowly when using the axis travel keys to move each axis. The X-, Y- or Z-axis can move at a speed between 50 and 4000 mm/min, and the 4th-, 5th-, 6th-, 7th-, or 8th-axis can move at a speed between 0.1 and 7.5 min⁻¹. If the machine parameter <P1 to 4-axis type> is set to <0: Linear axis>, the PLC-axis can move at a speed between 50 and 4000 mm/min. If it is set to <1: Rotation axis>, the PLC-axis can move at a speed between 0.1 and 7.5 min⁻¹. The default settings are 50 mm/min and 0.1 min⁻¹, respectively, when the power is turned ON.

[STEP] key

Press this key to move the axis one step at a time (fixed distance) when using the axis travel keys to move each axis.

The X-, Y- or Z-axis can be moved between 0.001 and 1.0 mm, and the 4th-, 5th-, 6th-, 7th-, or 8th-axis can be moved between 0.001 and 180°.

If the machine parameter <P1 to 4-axis type> is set to <0: Linear axis>, the PLC-axis can be moved between 0.001 and 1.0 mm. If it is set to <1: Rotation axis>, the PLC-axis can be moved between 0.001 and 180°.

The default settings are 0.001 mm and 0.001°, respectively, when the power is turned ON.

[+] key

Use this key to increase the setting value in increments for high speed travel (RPD), low speed travel (JOG), step travel (STEP) and for the spindle speed while in the manual operation mode.

[-] key

Use this key to decrease the setting value in increments for high speed travel (RPD), low speed travel (JOG), step travel (STEP) and for the spindle speed while in the manual operation mode.

[RELSE] key

Use this key to release or reset a magazine position error or ATC motion error.

The [+X] or [-X] key is used to move the X-axis, the [+Y] or [-Y] key is used to move the Y-axis, the [+Z] or [-Z] key is used to move the Z-axis, the [+4] or [-4] key is used to move the 4th, 5th, 6th, 7th, 8th, or PLC-axis in the + or – direction in the manual operation mode.

[B.SKIP] key (NC language)

Use this key to ignore or skip the command for a block that has a slash (/).

[OP.STP] key (NC language)

Use this key to stop the program at a M01 block.

[SINGL] key

Use this key to run a program that is called in the memory operation mode for a specific motion. When the LED is off, a cycle operation is executed.

[M.LCK] key

Use this key to run a program virtually without operating the machine. This function checks the program contents. Spindle motion, tool changes, and M code operation cannot be executed.

[DRY] key

Use this key for a dry run to check the program and machine operations.

In this mode, positioning and automatic tool changes are executed as programmed, and the cutting feed operates at a high speed in manual operation mode.

The Z-axis can operate at a higher position that is set in the user parameter <Dry run offset>.

[POS] key

Use this key to display the remaining distance between the coordinate position on the current axis (X, Y, Z, 4, 5, 6, 7, 8 and PLC) and the command coordinate value. In addition, it can be used to reference the current feedrate and the spindle speed as well.

[PRGRM] key

Use this key to display the program when in program edit mode or in memory operation mode. And, it displays the block operation screen when in MDI operation mode.

[MANU COND] key

Use this key to display the screen for setting each travel speed and the spindle speed in manual operation mode.

[ATC TOOL] key

Use this key to set and display which magazine number to attach the tool, which is registered in the tool data for the data bank.

[MONITR] key

Use this key to set the displays for the workpiece counter, time, communication monitor, measurement results, automatic thermal distortion compensation, etc.

[DATA BANK] key

Use this key to change to the screen where data and information required for machine operation can be displayed.

[ALARM] key

Use this key to change to the screen that shows the current alarm(s) and the alarm history.

[I/O] key

Use this key to display the machine control status for troubleshooting or repairing.

[GRAPH] key

Use this key to change to the graph screen. It displays tool path simulation and operation graphs.

[DSP OFF] key

Use this key to turn off the screen display.

Press any screen selection key to turn on the screen display. The screen selection keys are [POS] key, [ALARM] key, [PRGRM] key, [I/O] key, [MANU COND] key, [ATC TOOL] key, [MONITR] key, [DATA BANK] key and [GRAPH] key.

[EOB/ENT] key

Use this key to set the data that is in the input request field.

It is also used to input the end of block code (;) while inputting a program in NC language.

[HELP] key

Use this key to display help directions for screen that is currently displayed.

Pressing the key once display the help screen. When the key is pressed again, the help screen disappears and it returns to the previous screen.

[F0] to [F7] keys

The key functions change depending on the screen that is currently displayed. The respective functions are displayed when needed.

[>>] (Next function) key

Press this key to display the function which is not shown on the current screen.

[INS] key

Use this key to add data to a program while in program edit mode.

[CAN] key

Use this key to cancel the data in the input request field.

[DEL] key

Use this key to delete data where the cursor is while in program edit mode.

[BACK SPACE] key

Use this key to delete the data in front of the cursor in program edit mode or in the input request field.

Number keys and character keys

Use these keys to enter data (number values).

[SHIFT] key

Use this key to change the characters when entering text.

If a character key is pressed while the **[SHIFT]** key is being pressed, the upper character on the character key is input when the user parameter <Shift key lock> is set to <0: No>.

Every time this key is pressed, the ON and OFF status changes when user parameter <Shift key lock> is set to <1: Yes>. If you wish to input the upper character on the character key, turn the status ON and press the character key. If the LED is lit up, the status is ON.

[CURSOR] key

Use these keys to move the cursor in the direction of the arrow. These four arrows are referred to as the cursor keys.

[PAGE UP] key

Use this key to go back to the previous page of a program or to previous data when there are multiple pages.

[PAGE DOWN] key

Use this key to go to the next page of a program or the next set of data when there are multiple pages.

[PALLET] key (only models equipped with a quick table)

Use this key to select a pallet used in memory operation.

Each time the **[PALLET]** key is pressed, the status changes as follows: "OFF" → "1" → "2" → "1↔2" → "OFF." When the status is "OFF," the LED light turns off, and when the status is "1," "2" or "1↔2" the LED for the corresponding status lights up.

OFF: No pallet is used.

1: Only pallet 1 is used for machining.

2: Only pallet 2 is used for machining.

1↔2: Pallets 1 and 2 are used, alternating in sequence, for machining.

[CLT.P] key

Use this key to start the pump or stop it.

Press once: The coolant pump switch turns ON and the lamp lights up.

Press again: The coolant pump switch turns OFF and the lamp turns off.

When this switch is turned OFF, no coolant is supplied even if a coolant supply command is issued using key input or program operation.

When the power is turned ON, the status of the switch is maintained from when the power was turned OFF.

The switch may automatically turn OFF depending on the user parameter setting for <Automatic coolant off time>. Refer to section "2.5 User Parameter" in the Operation Manual (Data) for further details.

[CHP.F] key

The coolant unit is used together as a set with the chip shower (option). Refer to "Chapter 12 (1) Coolant Unit" for further details.

Press once: The chip shower motor starts. The lamp lights up.

Press again: The chip shower motor turns off. The lamp turns off.

When the power is turned ON, the status of the switch is maintained from when the power was turned OFF.

[LIGHT] key

It is used together with the machine light (option) as a set. It turns the light on and off.

Press once: The machine light turns on. The lamp lights up.

Press again: The machine light turns off. The lamp turns off.

When the power is turned ON, the status of the switch is maintained from when the power was turned OFF.

The lamp may automatically turn off depending on the user parameter setting for <Automatic machine light off time>. Refer to section “2.5 User Parameter” in the Operation Manual (Data) for further details.

Shortcut **[MENU]** key

Use this key to display the screen for registering screens that are used frequently.

Shortcut **[1]** & **[2]** keys

Use these keys to call a screen that is registered.

3.7 Important Notes When Attaching Workpieces, Jigs and Tools

WARNING

When replacing the setup equipment and the tool, the blade of the tool and the grip cover may cut your hands.

Do not hold the tool by the blade. Wear leather gloves and hold the tool holder with both hands.

Do not insert your fingers into the grip cover.

Do not insert your fingers in between the tool or the tool holder and the ATC magazine.

Wear leather gloves when working near the tool.

Use a step or step-ladder to provide a secure foot hold when the working position or conditions are awkward or out of reach.

WARNING

If items such as the tool and workpieces are not properly attached, or if cutting conditions are not within the tool limits, items may shoot out from inside the machine and strike personnel causing injury.

Always read the tool labels and instruction manuals for information on tool limits.

Check that the tips are securely attached before installing the tool.

Workpieces should be securely attached.

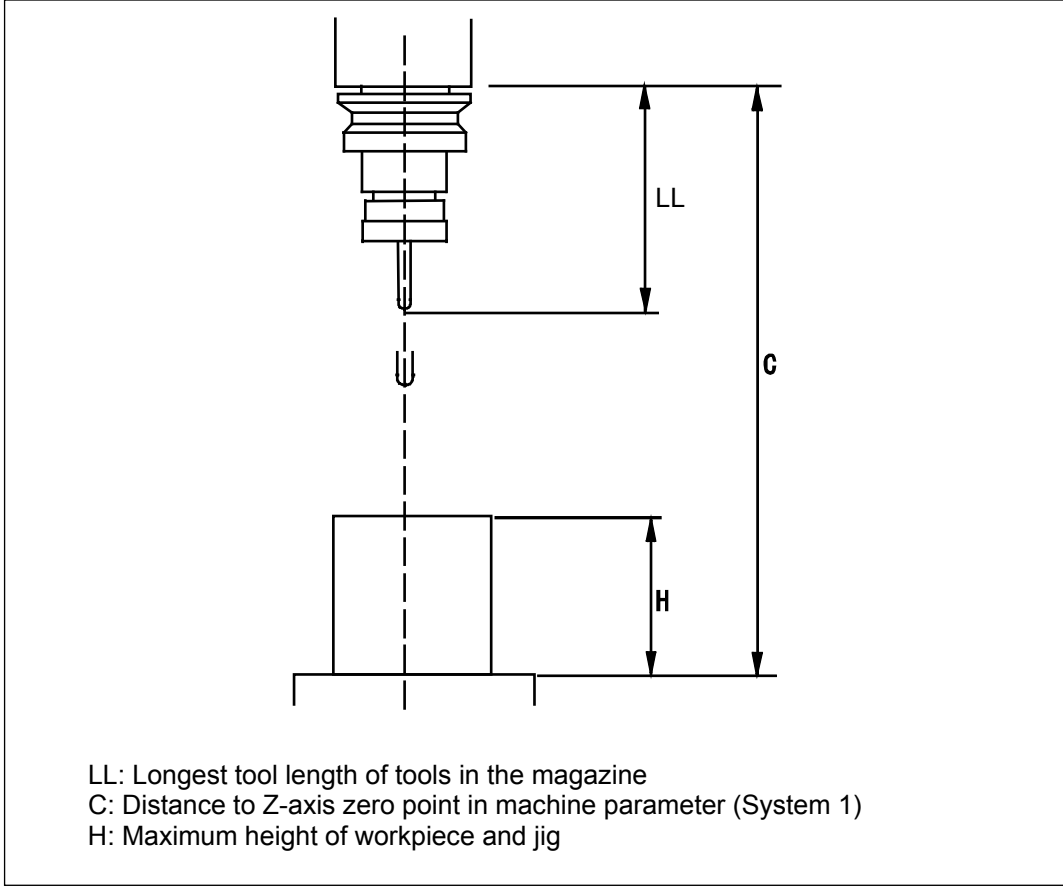
Close the doors before operating the machine.

(Note) When mounting workpieces and tools, there are limits or restrictions when setting the workpieces, jig height, tool length and tool holder, etc. Use the following procedure to check the restrictions and limits before mounting a workpiece and tool.

3.7.1 Workpiece, Jig Height and Tool Length Dimensional Limits

Before mounting a workpiece on the machine, make sure that there is no interference between the tool and workpiece jig during a tool change by referring to the dimensional formulas below. Apply the dimensions for the workpiece to be used into LL, C, and H in the figure below.

Workpiece, jig height and tool length dimensions and conditions



When the setup conditions are $H \leq C - LL - 10 \text{ mm}$	When the setup conditions are $H > C - LL - 10 \text{ mm}$
No interference.	There is interference.
Mount the workpiece using the confirmed setup conditions.	There may be interference depending on magazine rotational direction, tool mounting position and/or machining order, so only mount the workpiece after making sure there is no interference.

3.7.2 Tool Holder Limits

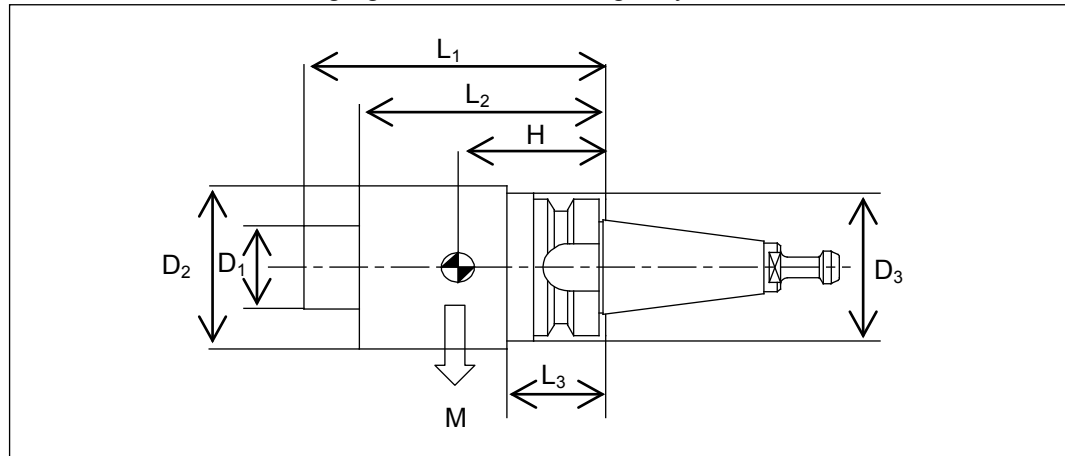
1. Tool dimensional limits and spindle speed restrictions

The spindle speed restrictions vary depending on the tool, as shown in the table below.
Make sure that the tool settings meet all the conditions.

Lower the speed when the conditions are not met.

M: Total tool weight (kg)

H: Distance from the gauge line to the center of gravity of the whole tool



Maximum spindle speed		10000 min ⁻¹	16000 min ⁻¹	27000 min ⁻¹ *3
Spindle taper		BT specification: 7/24 No.30 BBT specification: 7/24 No.30 BIG-PLUS		
Tool shank		BT specification: MAS-BT30 BBT specification: MAS-BT30 / BBT30		
Pull stud		MAS-P30T-2 (30°)*1		
Tool limits	S500X1 S700X1 S1000X1 S300X1 (Specification with 14 magazines)	D ₁ ≤ 40 mm L ₁ ≤ 250 mm D ₂ ≤ 110 mm *2 L ₂ ≤ 160 mm D ₃ ≤ 46 mm L ₃ ≥ 30 mm M ≤ 3 kg M×H ≤ 180 kg·mm	D ₁ ≤ 40 mm L ₁ ≤ 250 mm D ₂ ≤ 55 mm L ₂ ≤ 160 mm D ₃ ≤ 46 mm L ₃ ≥ 30 mm M ≤ 2 kg M×H ≤ 100 kg·mm	D ₁ ≤ 40 mm L ₁ ≤ 200 mm D ₂ ≤ 50 mm L ₂ ≤ 160 mm D ₃ ≤ 46 mm L ₃ ≥ 30 mm M ≤ 2 kg M×H ≤ 50 kg·mm
	S300X1 (Specification with 21 magazines)	D ₁ ≤ 40 mm L ₁ ≤ 160 mm D ₂ ≤ 110 mm *2 L ₂ ≤ 145 mm D ₃ ≤ 46 mm L ₃ ≥ 30 mm M ≤ 3 kg M×H ≤ 180 kg·mm	D ₁ ≤ 40 mm L ₁ ≤ 160 mm D ₂ ≤ 55 mm L ₂ ≤ 145 mm D ₃ ≤ 46 mm L ₃ ≥ 30 mm M ≤ 2 kg M×H ≤ 100 kg·mm	D ₁ ≤ 40 mm L ₁ ≤ 160 mm D ₂ ≤ 50 mm L ₂ ≤ 145 mm D ₃ ≤ 46 mm L ₃ ≥ 30 mm M ≤ 2 kg M×H ≤ 50 kg·mm
Tool balance limit		100 g·mm	50 g·mm	10 g·mm
Spindle speed limit		10000 min ⁻¹	16000 min ⁻¹	27000 min ⁻¹
Total for all magazine tools		M total ≤ 25 kg (14 Tools) M total ≤ 35 kg (21 Tools)		

*1 Use a pull stud that complies with Brother CTS specifications.

When mounting a pull stud, always refer to the recommended clamping torque value in the tool maker's Pull Stud Manual. If the clamping torque is above the standard value, the taper part may swell and negatively impact the tool runout accuracy and machining performance.

*2 A Disc-Cutter-S and Disc-Cutter-Pro (OSG) with a tool diameter D₂ that is 125 mm or smaller can be used. However, there is risk of interference if the Disc-Cutter-S and the Disc-Cutter-Pro (OSG) are used side by side. Therefore, mounting them next to one another is not possible.

*3 There is no 27000 min⁻¹ specification setting for S1000X1.

2. Tool balance.

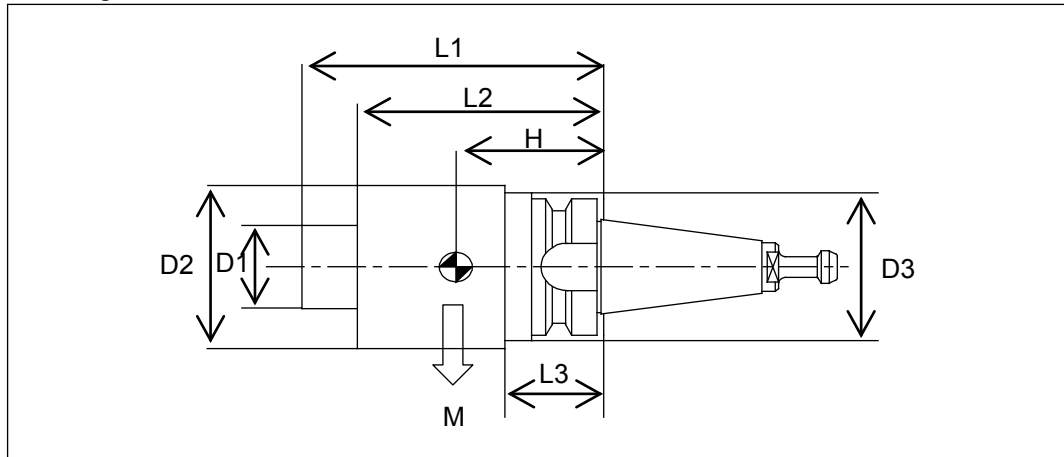
**(Note) Using unbalanced tools may damage the spindle.
Always use well-balanced tools.**

Use a tool and tool holder that are centrally symmetric with the respect to the spindle rotation center.

Pay particular attention to the tool balance because the spindle rotates at high speed.

Using unbalanced tools may damage the spindle.

If the spindle speed is $22,000 \text{ min}^{-1}$ or higher, using a tool that is balanced to G2.5 (single tool) or higher is recommended.



Consult Brother or the nearest dealer authorized for further information on tool balance.

Tool holders (BT30*-DMC**-**HS) made by Yukiwa Seiko Inc. are recommended as well balanced tool holders.

Refer to “3.7.2 Tool holder limits” and make sure that the tool setup meets all requirements.

Using an incorrect tool holder or one that is out of specification may cause the tool holder to detach from the spindle, resulting in machine damage.

3. Shape of pull stud for CTS (Coolant-through-Spindle) tool

(Note) Use a pull stud for CTS tools specified by Brother. Using pull studs not specified by Brother can cause an accident such as the tool detaching and falling off, etc.

(1) Shape of coolant-through-spindle pull stud (partial change in MAS II types) specified by Brother Industries, Ltd.

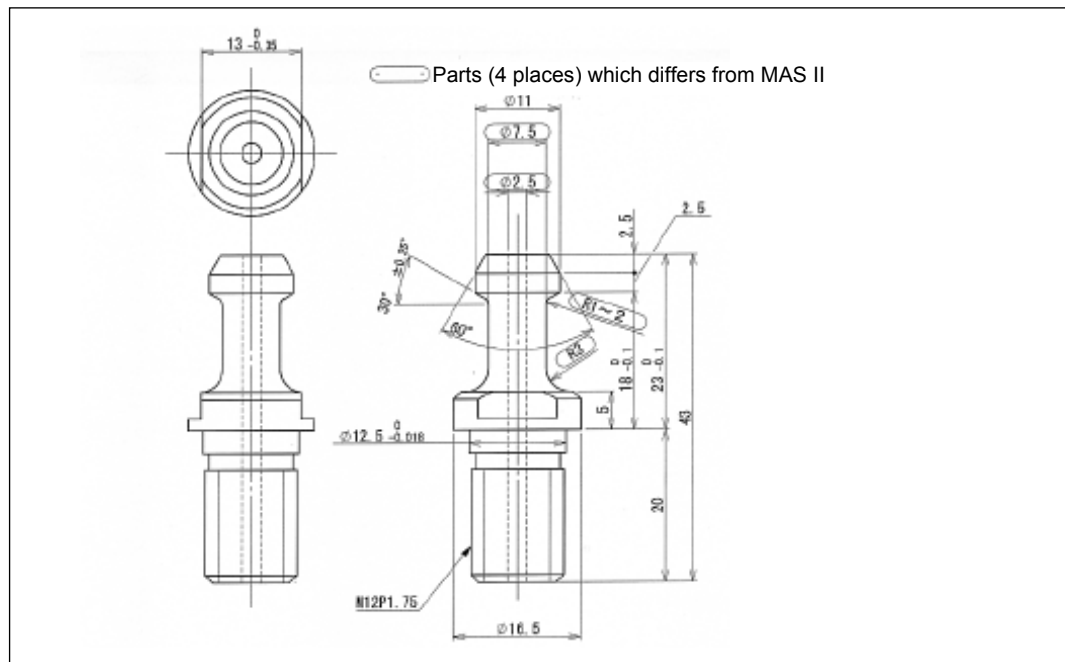
The coolant-through-spindle is limited to this shape.

(2) The dimensions and tolerances (except the changed part) must comply with the MAS II type standards.

(3) Make sure that it can withstand a tensile yield force of 56000 N or higher.

(4) The pull stud material that is generally used may not be strong enough.

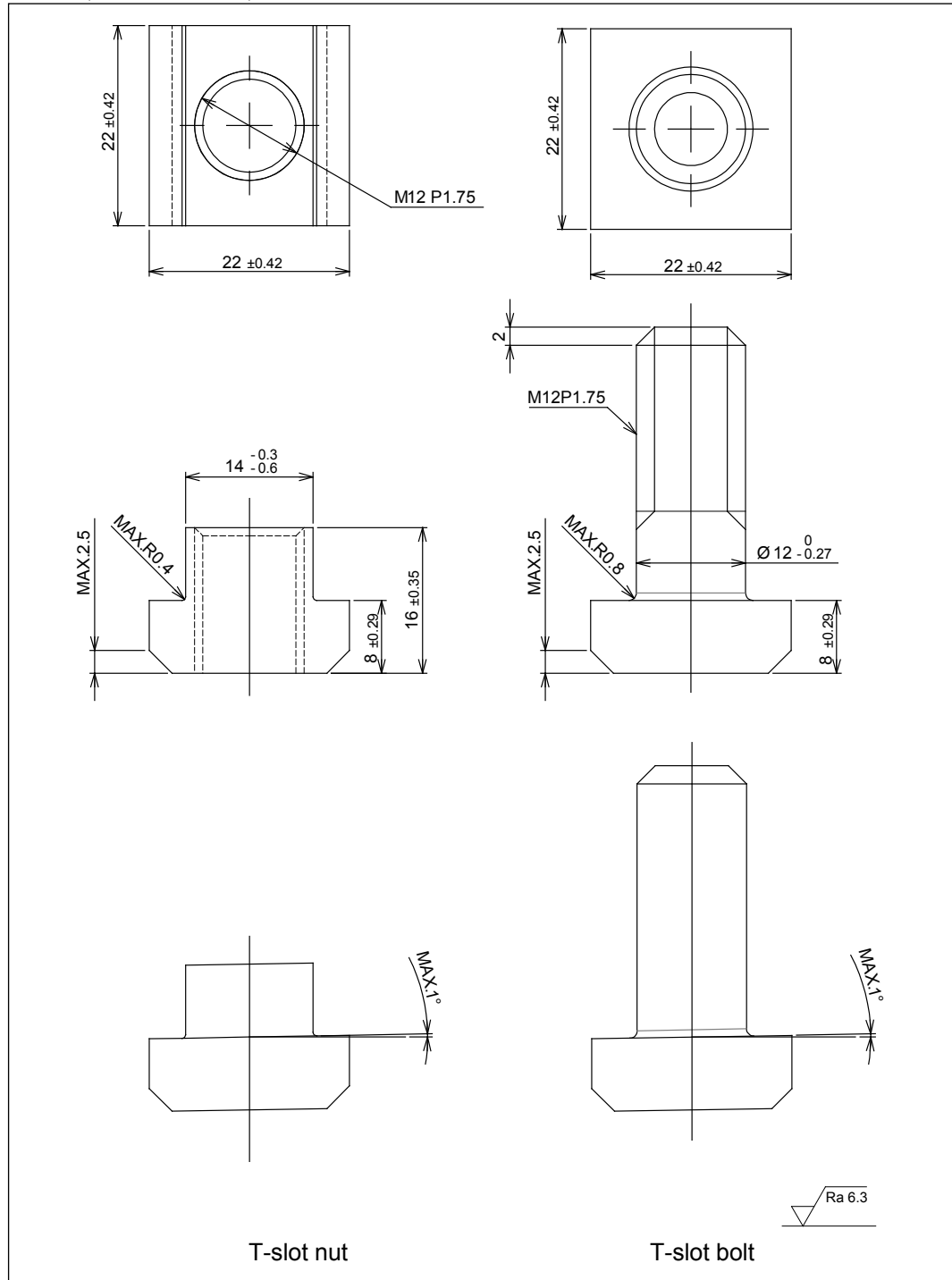
Reference material: High strength material like SNCM420 (JIS: Tensile strength 980 N/mm² or greater).



Reference: Manufacturers and products that comply with Brother specifications
 Nikken Kosakusho Works: PS-130E
 Yukiwa Seiko: P30TB-101
 Big Daishowa Seiki: P30T-2H3
 NT Tool Corporation: S/PSB-OA-CH

3.7.3 Special Notes When Attaching Workpieces and Jigs

1. Make sure that the jig is securely mounted on the table.
2. Make sure that the workpiece is securely attached to the jig. When using a power source for air pressure and oil pressure, etc., make sure that the required pressure is supplied.
(Note) If machining when the workpiece and jig are not securely attached, the workpiece or jig may become detached and shoot out causing injury.
3. To secure the jig to the table, a T-slot bolt (nominal size: 14) defined in JIS B 1166 or a T-slot nut (nominal size: 14) defined in JIS B 1167 can be used.



4. Make sure the accuracy of the mounting surface, where the jig is secured to the table, is equivalent to the accuracy of the table (Flatness: Less than 0.01). If the dimensions are not accurate, be careful because it can deform the surface of the table or negatively impact the jig attachment.

3.8 Attach Tool to Magazine and Remove Tool from Magazine

WARNING

When replacing the setup equipment and the tool, the blade of the tool and the grip cover may cut your hands.
 Do not hold the tool by the blade. Wear leather gloves and hold the tool holder with both hands.
 Do not insert your fingers into the grip cover.
 Do not insert your fingers in between the tool or the tool holder and the ATC magazine.
 Wear leather gloves when working near the tool.
 Use a step or step-ladder to provide a secure foot hold when the working position or conditions are awkward or out of reach.

WARNING

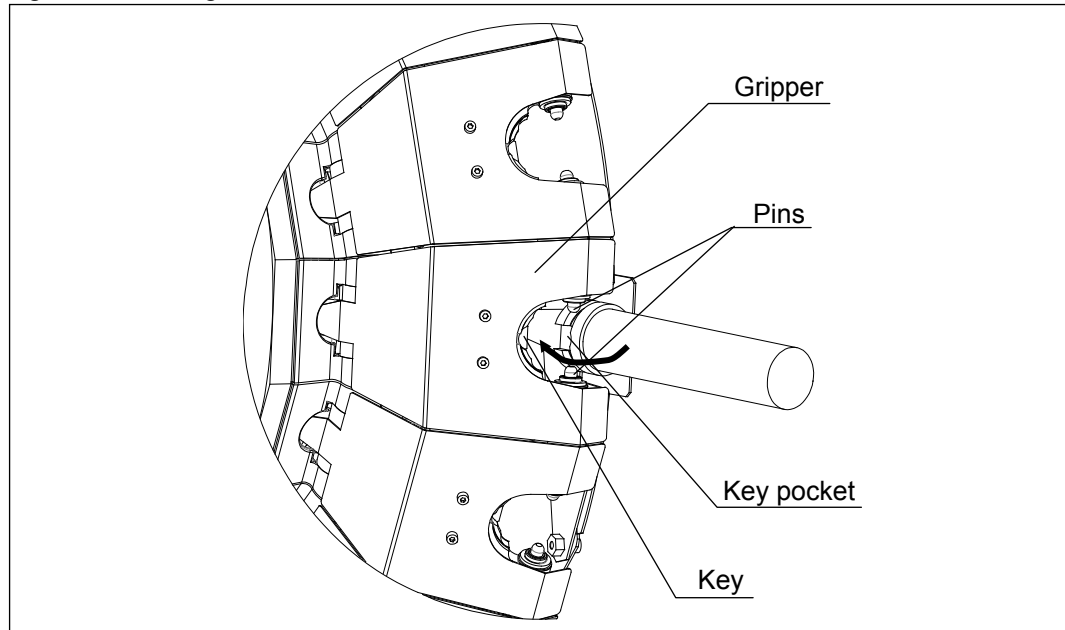
If items such as the tool and workpieces are not properly attached, or if cutting conditions are not within the tool limits, items may shoot out from inside the machine and strike personnel causing injury.
 Always read the tool labels and instruction manuals for information on tool limits.
 Check that the tips are securely attached before installing the tool.
 Workpieces should be securely attached.
 Close the doors before operating the machine.

(Note) The tool can be attached or removed at the tool change position on the magazine.
 When removing a tool, wear leather gloves and be careful not to hold the tool by the blade.
 If the tool is not securely attached, it may fall off and/or damage the machine.

3.8.1 Attach Tool to Magazine

1. Press the [MANU] key to switch to manual operation mode.
2. Press the [ATC] key. The Z axis rises to the machine zero point and the spindle is positioned at the same time.
3. Press the [ATC] key. The Z-axis moves to the ATC zero point.
4. Press the [MAG.F] key or the [MAG.R] key to rotate the magazine.
 Move the magazine number, where you wish to attach the tool, to the tool change position for the magazine.
5. Attach the tool to the magazine. When attaching the tool, insert the gripper key into the tool holder key pocket. Then, fit the gripper pin onto the V-groove for the tool holder, and push all the way into the gripper to secure it.
 In addition, avoid holding it near the key pocket or V-groove because there is a risk of pinching your hand or getting it caught when holding the key pocket or V-groove part.
6. Press the [ATC] key. The Z-axis lowers to the machine zero point.

Figure for attaching the tool

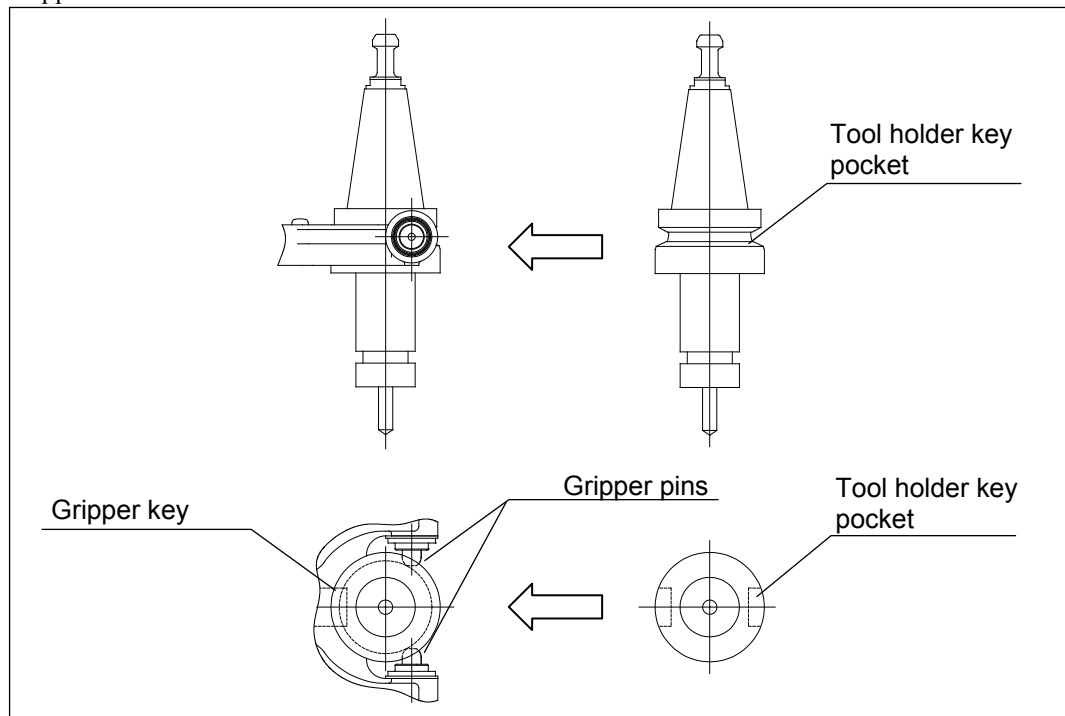


When attaching the tool, attach a tool that weighs about the same as the gripper that faces the magazine, in order to maintain good balance.

Ex: S500X1 (14 magazine specification)

Magazine pot No.	Tool weight (kg)	Magazine pot No.	Tool weight (kg)
1	3.0	8	3.0
2	3.0	9	3.0
3	2.5	10	2.5
4	1.0	11	1.0
5	1.0	12	1.0
6	0.5	13	0.5
7	1.0	14	1.0
		Total	24.0

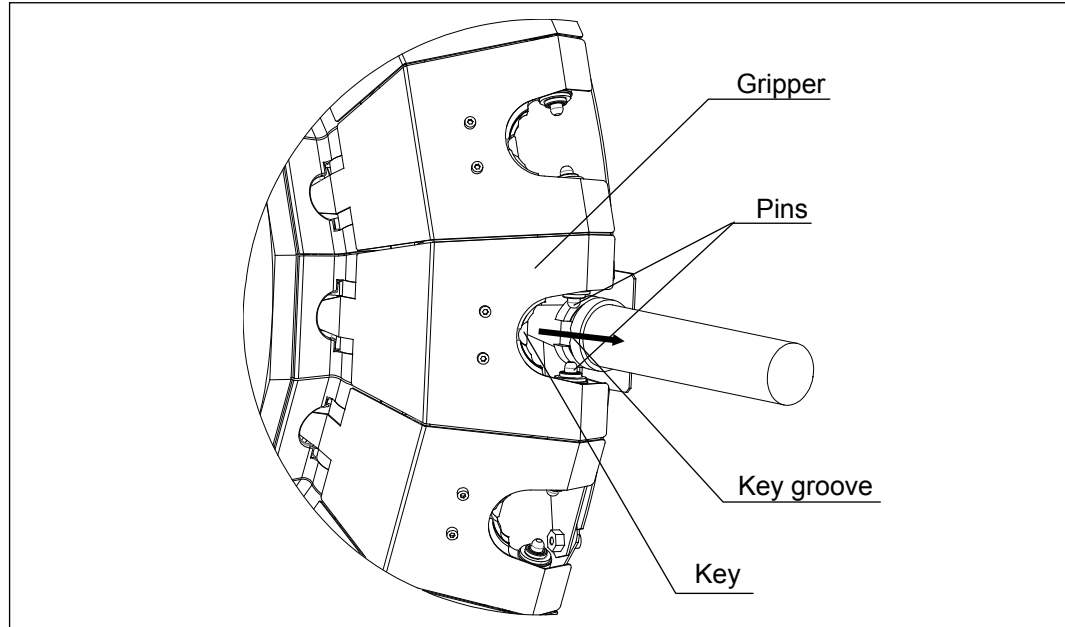
Gripper and tool holder



3.8.2 Tool Removal from Magazine

1. Move the gripper for removing the tool to the tool change position.
Refer to “Chapter 3-3.8.1 Attach Tool to Magazine” for further details on magazine rotation.
2. Remove the tool from the magazine. When removing a tool from the magazine, push the tool holder in the direction of the arrow and pull it as shown below.

Figure for removing the tool



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CHAPTER 4

SCREEN DISPLAY

4

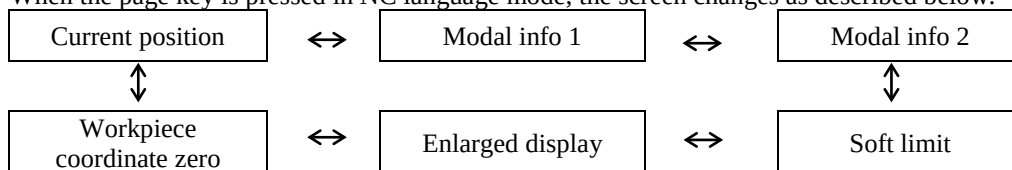
- 4.1 Position
- 4.2 Program Display
- 4.3 Manual Conditions
- 4.4 ATC Tools
- 4.5 Production Monitor
- 4.6 Data Bank
- 4.7 Alarm Display
- 4.8 Input/Output
- 4.9 Graph
- 4.10 Display OFF
- 4.11 Shortcuts
- 4.12 Screen Functions

4.1 Position

Press the **[POS]** key to display information about the coordinate position, etc.

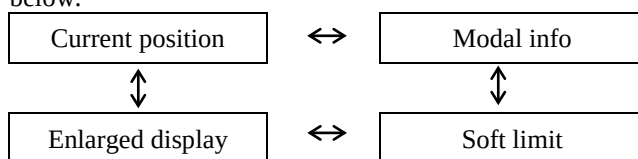
- Changing screens in NC language mode

When the page key is pressed in NC language mode, the screen changes as described below.



- Changing screens in conversation language mode

When the page key is pressed in conversation language mode, the screen changes as described below.



4.1.1 Current Position Screen

This screen is used in both NC language and conversation language modes.

The screen displays the following information during machine operation: the current position of each axis, the override status, the door status (open/close), the door interlock status, the inner pallet number, the load monitor, the spindle speed, the feedrate and the magazine conditions.

Ex: (Current position screen in NC language mode)

Current position				 2012/06/04 15:29:43	
Program(1234)					
Relative coord. pos.		Absolute coord. pos.		Load monitor	
X	-100.000	X	-100.000	Rapid feed	100 %
Y	-100.000	Y	-100.000	Cut feed	100 %
Z	480.000	Z	480.000	Spindle	100 %
C	0.300	C	0.300		
A	0.000	A	0.000		
B	0.000	B	0.000		
A		A			
B		B			
Machine coord. pos.		Remaining distance			
X	-100.000	X	0.000	Door	Spindle speed
Y	-100.000	Y	0.000	Door	Close 0 min ⁻¹
Z	480.000	Z	0.000		Feedrate
C	0.300	C	0.000		
A	0.000	A	0.000	Door interlock	Magazine
B	0.000	B	0.000	Invalid	Tool No. 01 Center drill
A		A			Magazine No. 07
B		B			
☐ Workpiece coordinate zero ☒ Modal 1					
[HELP]					
Current position	Modal 1	Modal 2	Soft limit	Enlargement	Workpiece coordinate zero
					Select add axes/change display
					Counter reset

Function display on current position screen in conversation language mode

Current position	Modal	Soft limit	Enlargement	Select add. axes/change display	Counter reset		
------------------	-------	------------	-------------	---------------------------------	---------------	--	--

- Axes that are displayed for the relative coordinate position, machine coordinate position, absolute coordinate position, remaining distance and load monitor
The additional axes and PLC axes that are displayed are set in the user parameters <4th row display axis> to <12th row display axis>.
Press the function key <Select add. axes/change display> to display more axes (9th axis or more).
- PLC axes
The digits displayed for the PLC axes vary depending on the machine parameter setting <P1-axis (to P4-axis) type>.

	Metric	Inch
<0: Linear axis>	-9999.999 to 9999.999	-999.9999 to 999.9999
<1: Rotation axis>	With stroke -9999.999 to 9999.999 Without stroke 0 to 359.999	(Same as left)

Program

The program number that is currently selected in memory operation is displayed.

Selected program	Display format
No.1000 in the NC root directory is selected	1000
No.2000 in the NC directory named "ABCDEFGF" is selected	ABCDEFGF/2000
Program No.3000 is selected in tape operation	3000
Program without a program number is selected in tape operation	????
Program No.5000 is selected and is the 4th program registered in the schedule program No.9901 located in the NC root directory (conversation language)	9901/5000/04
Program No.5000 is selected and is the 4th program registered in the schedule program named "ABCDEFGF" located in the NC root directory (conversation language)	ABCDEFGF/9901/5000/04

Relative coord. pos.

The coordinate can be set arbitrarily. The current coordinate is displayed in a coordinate system where a position set by counter reset is designated as the start point.

Machine coord. pos.

The current coordinate value is displayed in a coordinate system that is unique to the machine. The position refers to the distance from the machine zero point on the X- and Y-axes, and the distance from the table surface to the end of the spindle on the Z-axis.

Absolute coord. pos.

The current coordinate value is displayed in the coordinate system in which the workpiece zero point is arbitrarily set as the start point.

- Displays the coordinate value on the Z-axis taking the tool length of the spindle into account when using conversation language.
- When the (H tool number) is displayed on the right while using the NC language, the display indicates that the tool length offset/tool position compensation is being used. The following are taken in account when displaying values:
 - Z-axis tool length offset is taken into account when using tool length offset.
 - Tool position compensation is taken in account for the tool number displayed for each axis when using tool position compensation.

In addition, (R) is displayed on the right because the rotational transformation is in progress and the coordinate value before rotation is displayed.

The display method for the coordinates, during tool length offset, tool position compensation and rotational transformation, may vary depending on the user parameters (switch 1): <XYZ-axis absolute coord. display> and <Display coordinates during rotational transformation>.

- When using feature coordinate machining mode (G68.2 modal), the <Absolute coordinate display format> can be selected in the user parameter (feature coordinate setting) to either show the values based on the workpiece coordinate system or the values based on the feature coordinate system.

While (F) is displayed on the right, the coordinate value from the feature coordinate system is displayed.

Remaining distance

The remaining distance (amount of travel not yet executed) is displayed for memory operation or MDI operation travel commands.

(Additional axis name display)

The name display for the additional axis changes depending on whether the machine is equipped with a quick table or not.

When the machine is equipped with a quick table, the pallet number is displayed after the axis name such as "B1" or "B2".

When the machine is not equipped with a quick table, only “B” is displayed as the axis name.

Override

The status is displayed for the switches [RAPID TRAVERSE OVERRIDE], [FEEDRATE OVERRIDE] and [SPINDLE OVERRIDE].

Door

The status of the door is displayed (open or closed). The door name varies depending on the machine model.

Door interlock

The status of the door interlock is displayed.

Inner pallet

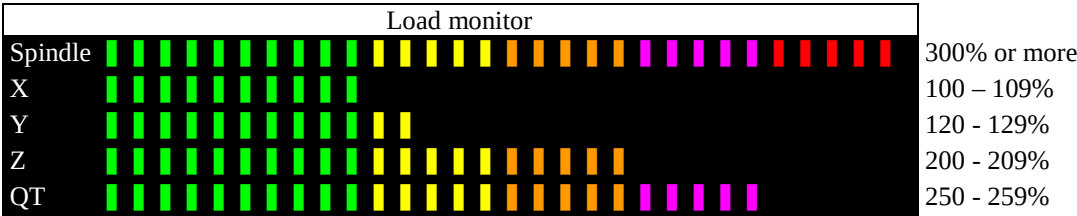
The inner pallet number is displayed. (Only machine models equipped with QT)

Load monitor

The load on each axis is displayed.

(The scale is divided into increments of 10% with a maximum of 300%.)

10% to 100% on the scale is displayed in green, 110% to 150% is displayed in yellow, 160% to 200% is displayed in orange, 210% to 250% is displayed in purple and 260% or more is displayed in red.



Spindle speed

The speed of the spindle currently rotating is displayed.

The actual spindle speed is displayed for memory operation, MDI operation or manual operation mode.

Feedrate

The actual feedrate is displayed for memory operation or MDI operation mode.

<Positioning> is displayed during axis travel for positioning.

This display is shown only during axis travel.

Magazine

The status of the ATC is displayed. The display varies depending on the type of ATC.

The display for machine models with a double arm mechanism is shown in the screenshot example.

Tool No.: Displays the information for the tool (registered as the spindle on the ATC tool screen) mounted on the current spindle.

Next tool number: Displays the information for the tool mounted on the next tool pot.

Pot No.: Displays the magazine number for the next tool pot.

Machine models with an armless mechanism are shown below.

Magazine	
Tool No.	01 Center drill
Magazine No.	02

Magazine No.: Displays the magazine number for the spindle that is currently indexed.

Select add. axes/change display (same function key on position screen)

In manual operation mode, press the <Select add. axes/change display> function key to specify the axis to be operated during jog operation or handle operation or to change the display axis.

The names of additional axes and the PLC axes are displayed in the function area. They are displayed in the order that they are set in the parameters <4th row display axis> to <12th row display axis>.

Pressing <Select add. axes/change display> function key changes the display back and forth between “4th - 8th row display axis” and “9th - 12th row display axis” in the function area. The axis names displayed as additional axes can be changed using the machine parameter settings: <Optional 4th (-8th) -axis>, <5th (-8th) -axis position> and <4th (-8th) -axis address>. The axis names displayed as the PLC axes can be changed using the machine parameter setting <Optional P1(-4) -axis>.

Display axes for 4th through 8th rows

Exit	QT-axis	A1-axis	A2-axis	B1-axis	B2-axis	Select add. axes/change display	
------	---------	---------	---------	---------	---------	---------------------------------	--

Press **[F1]** through **[F5]** to select an axis and use **[+4]** and **[-4]** in manual mode to operate it. Press **[F6]** to change the mode and select the axes displayed from the 9th to the 12th rows. Press **[F0]** to exit the mode for changing the additional axis and display.

Display axes for 9th through 12th rows

Exit	P1-axis	P2-axis	P3-axis	P4-axis		Select add. axes/change display	
------	---------	---------	---------	---------	--	---------------------------------	--

4

Note, when changing between the “4th - 8th row display axis” and the “9th - 12th row display axis”, the coordinate display, including the machine coordinates, also changes.

When all of the axis options for <9th to 12th row display axis> are set to < Not set>, the **[F6]** (Select add. axes/change display) key is grayed out and disabled.

When the same axis is set to multiple user parameters for <4th - 12th row display axis>, even if one of the function keys showing that axis is pressed, the **[+4]** and **[-4]** keys will operate that axis. Furthermore, all of the function keys showing that axis are selected.

Counter reset

Press this function key to reset the relative coordinate position. The screen below is displayed. Use the cursor keys to select the axis to be changed, and set a new position.

The screenshot shows the 'Counter reset' screen with the following data:

Relative coord. pos.		Absolute coord. pos.		Override		Load monitor	
X	-100.000	X	-100.000	Rapid feed	100 %	SP	
Y	-100.000	Y	-100.000	Cut feed	100 %	X	
Z	480.000	Z	480.000	Spindle	100 %	Y	
C	0.300	C	0.300			Z	
A	0.000	A	0.000			C	
B	0.000	B	0.000			A	
A		A				B	
B		B				A	
						B	
Machine coord. pos.		Remaining distance		Door		Spindle speed	
X	-100.000	X	0.000	Door	Close		0 min ⁻¹
Y	-100.000	Y	0.000				
Z	480.000	Z	0.000				Feedrate
C	0.300	C	0.000				
A	0.000	A	0.000	Door interlock			Magazine
B	0.000	B	0.000	Invalid			Tool No. 01 Center drill
A		A					Magazine No. 07
B		B					

Reset value: [HELP]

Back Set all axes to zero

Press the up and down cursor keys to select an axis.

The relative coordinates for the selected axis can be changed by entering the desired coordinates and pressing **[ENT]**.

[F1] (Set all axes to zero): Sets the relative coordinates for all axes to zero.

(Note) Pressing **[F1]** (Set all axes to zero) sets the relative coordinates for all axes to zero, including additional axes and PLC axes which are not displayed on the screen.

4.1.2 Software Limit

This screen is used in both NC language and conversation language modes.

Ex: Software limit screen in NC language mode

Soft limit				2013/04/24 16:37:11			
Program(0450)							
Stroke limit 1		Stroke 1		Stroke 3			
(-)	(+)	(-)	(+)	(-)	(+)		
X	-21.6535 0.0000	X	-19.6850 0.0000	X	0.0000 0.0000		
Y	-21.6535 0.0000	Y	-15.7480 0.0000	Y	0.0000 0.0000		
Z	0.0000 25.3937	Z	0.0000 23.6220	Z	0.0000 0.0000		
P1		P1	-999.9999 999.9999	Stroke 4			
P2		P2	-999.9999 999.9999				
P3		P3	-999.9999 999.9999				
P4		P4	-999.9999 999.9999				
Stroke limit 2		Stroke 2		Stroke 5			
(-)	(+)	(-)	(+)	(-)	(+)		
X	0.0000 0.0000	X	0.0000 0.0000	X	0.0000 0.0000		
Y	0.0000 0.0000	Y	0.0000 0.0000	Y	0.0000 0.0000		
Z	0.0000 0.0000	Z	0.0000 0.0000	Z	0.0000 0.0000		
[HELP] ☒ Modal 2 ☒ Enlargement							
Current position	Modal 1	Modal 2	Soft limit	Enlargement	Workpiece coordinate zero	Select add. axes/change display	

Function display on software limit screen in conversation language mode

Current position	Modal	Soft limit	Enlargement	Select add. axes/change display			Stroke 3~5
------------------	-------	------------	-------------	---------------------------------	--	--	------------

Stroke limit 1

The travel range for each axis is displayed in the machine coordinates for memory operation and MDI operation.

The effective limit is displayed, using either the stroke limit set in the user parameter <Stroke limit 1 for X (-8th) -axis> or the stroke limit set by G22. When G23 mode is used, nothing is displayed for the X-, Y- and Z-axes.

Stroke limit 2

The restricted travel range is displayed in the machine coordinate for memory operation and MDI operation.

The user parameter <Stroke limit 2 for X (-Z) -axis> is displayed.

Stroke 1

The travel range for each axis is displayed in the machine coordinates.

The machine parameters <X (-Z) -axis stroke 1> and <(4th-8th, P1 - P4) axis stroke> are displayed.

Stroke 2 - 10

The restricted travel range is displayed in the machine coordinates.

The machine parameter <X-(-Z, Addnl.) axis stroke 2(-10)> is displayed.

Press the [F7] key on the software limit screen to change the display for strokes 3 to 10.

The additional axes and PLC axes that are displayed are set in the user parameters <4th row display axis> to <12th row display axis>.

4.1.3 Enlarged Display

This screen is used in both NC language and conversation modes. The absolute coordinate value or relative coordinate position on the current position screen are magnified and displayed.

Ex: Enlarged display screen in NC language mode

Enlarged display							2012/06/04 15:30:58	
Program(1234)								
Absolute coord. pos.								
X	-100.000							
Y	-100.000							
Z	480.000							
C	0.300	P1	0.000					
A	0.000	P2	0.000					
B	0.000	P3	0.000					
A	P4		0.000					
B								
					☐ Soft limit ☒ Workpiece coordinate zero			
[HELP]								
Current position	Modal 1	Modal 2	Soft limit	Enlargement	Workpiece coordinate zero	Select add. axes/change display	Relative coordinates	

Function display on enlarged display screen in conversation language mode

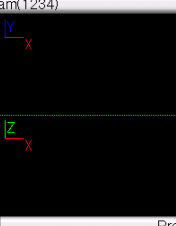
Current position	Modal	Soft limit	Enlargement	Select add. axes/change display	Relative coordinates		
------------------	-------	------------	-------------	---------------------------------	----------------------	--	--

Press the function key to change the display between the absolute coordinate position and relative coordinate position. (The absolute coordinate position and relative coordinate position are the same as those on the current position screen.)

The displayed axes include the X-, Y- and Z-axes as well as the additional axes and PLC axes that are set in the user parameters <4th row display axis> to <12th row display axis>.

4.1.4 Modal Info 1

This screen is displayed only in NC language mode.

Modal info 1				Program(1234)		2012/06/04 15:32:41	
Absolute coord. pos.		Remaining distance					
X	-100.000	X	0.000				
Y	-100.000	Y	0.000				
Z	480.000	Z	0.000				
C	0.300	C	0.000				
A	0.000	A	0.000				
B	0.000	B	0.000				
A		A					
B		B					
G code		S/F		Program			
00	17	22	40	S	T1; >> N0100 G100 X-20. Y-20. Z12. G43 H1 S6000 M3(CE T2; N0101 G81 Z8.5 R12.F1000; N0102 X-40. ; N-103 X-60. ; N0200 G100 X-20. Y-20. Z12. G43 H2 S6000 M3(DR T3;		
49	50	50.1	64	F			
54	67	68	80	Tool			
90	94	97	99	T001	TAP M6 P1.0 H Wear offset D Wear offset		
M code							
05	08	97	141				
221	222	223	224				
230	269	270	305				
401	495						
				☐ Current position ☒ Modal 2			
Current position	Modal 1	Modal 2	Soft limit	Enlargement	Workpiece coordinate zero	Select add axes/change display	Relative coordinates

The screen displays the absolute coordinates, relative coordinates, remaining distance, modal values for G and M codes, command values (S, F, T, H, and D), operation graphs and program data.

Press the function key to change the display between the absolute coordinate position and relative coordinate position.

(The absolute coordinate position, relative coordinate position and remaining distance are the same as those on the current position screen.)

- Axes displayed for absolute coordinate position, relative coordinate position and remaining distance

The 4th through 8th axes that are displayed are the additional axes and PLC axes which are set in the user parameters <4th row display axis> to <12th row display axis>.

G code

The current modal values for G code are displayed in groups.

16 types of modals can be displayed using the user parameter (switch 3) settings: <G code 1 for modal display> to <G code 16 for modal display>.

M code

The current modal values for M code are displayed in groups.

16 types of modals can be displayed using the user parameter (switch 3) settings: <M code 1 for modal display> to <M code 16 for modal display>.

S/F

The current modal values are displayed for S and F.

Tool

The current modal values for T, D and H as well as their actual corrected values are displayed.

T:

The tool number (or group number or pot number) where a command was issued with a T address is displayed. (Ex: "T001") In addition, the tool name for that tool number is displayed.

H and D:

The tool number where a command was issued with an H or D address is displayed. (Ex: "H002") In addition, the offset or compensated value for that tool number is displayed.

Program

The block being executed in MDI operation mode is displayed.

The same display applies to the memory operation screen when in memory operation mode.

(Note 1) The automatic centering status is displayed when performing automatic centering.

(Note 2) The memory operation status is displayed when in manual operation mode.

4.1.5 Modal Info 2

This screen is displayed only in NC language mode.

Modal Info 2										2019/04/01 19:21:43	
Program(YOKOI0008)											
Positioning end check distance(M270)						Rotational transformation					
X	0.300					Center X					
Y	0.300					Center Y					
Z	0.300					Center Z					
QT	0.300					Angle					
A1	0.300					Mirror image					
B1	0.300					Reference X					
A2	0.300					Reference Y					
B2	0.300					Reference Z					
Tool offset amount						High accuracy		Scaling			
X						Off		Center		Scaling	
Y								X			
Z						Load monitor		Y			
Tool wear offset amount						Off		Z			
X						Spindle speed clamp		Fixed outer ax ctrl			
Y						S					
Z						Q					
Macro variables											
☒ Modal 1 ☒ Soft limit											
[HELP]											
Current position	Modal 1	Modal 2	Soft limit	Enlargement	Workpiece coordinate zero	Select add axes/change display					

The screen displays the positioning end check distance, tool offset amount, tool wear offset amount, rotational transformation, mirror image, scaling, high accuracy, machining load monitor, spindle speed clamp, fixed outer axis control and macro variables.

Positioning end check distance

The modals for groups M270-279 and the positioning end check distance for each axis are displayed. The machine parameters <P1 (- P4) -axis positioning end check distance/angle> are displayed for the PLC axes listed for “Positioning end check distance”.

Tool offset amount

The compensation is displayed during tool length offset and tool position compensation. Next to the compensation for the X-, Y- and Z-axes, the tool number corresponding to each axis is added and displayed as in “(H12)”.

Tool wear offset amount

The wear compensation is displayed during tool length offset and tool position compensation.

Rotational transformation

The rotation center coordinate and rotation angle are displayed during rotational transformation.

Mirror image

The mirror reference axis is displayed when mirror imaging is executed.

High accuracy

Displays the high accuracy mode status (OFF, standard, rough, medium rough, medium rough S, finishing, finishing S, adjustment A, adjustment B, adjustment C, accuracy specification A, accuracy specification B, accuracy specification C, M260 - M267, M280 - M287).

Machining load monitor

Machining load monitor status (OFF and in use parameter number) is displayed.

Scaling

The scaling center coordinate and scaling magnification rate are displayed when scaling is executed.

Spindle speed clamp

The maximum spindle speed (S) and the minimum spindle speed (Q) are displayed when the spindle speed clamp (G92) is specified.

Constant peripheral speed control axis

If the constant peripheral speed is being controlled, the constant peripheral speed control axis (X-, Y- or Z-axis) is displayed.

Macro variables

The macro variables for the position screen display are shown when variable(s) are selected in the user parameter.

4.1.6 Workpiece Coordinate Zero

This screen is displayed only in NC language mode.
The display changes when in normal operation mode and when in feature coordinate machining mode.

(Ex: When in normal operation mode)

Workpiece coordinate zero					2014/07/10 11:20:29		
Program(CAPT01,0001)							
G54		G92		G52		External	
X	-100.000	X	0.000	X	0.000	X	
Y	-220.000	Y	0.000	Y	0.000	Y	
Z	0.000	Z	0.000	Z	0.000	Z	
A	0.000	A	0.000	A	0.000	A	
B	0.000	B	0.000	B	0.000	B	
C	0.000	C	0.000	C	0.000	C	
						Workpiece coordinate zero	
X		X		X		X	-100.000
Y		Y		Y		Y	-220.000
Z		Z		Z		Z	0.000
A						A	0.000
B						B	0.000
C						C	0.000
				☒ Enlargement			
				☒ Current position			
[HELP]							
Current position	Modal 1	Modal 2	Soft limit	Enlargement	Workpiece coordinate zero	Select add. axes/change display	

(Ex: When in feature coordinate machining mode)
This is displayed while in feature coordinate machining mode (G68.2 modal in progress).
When the feature coordinate machining mode is cancelled (G69 modal), the screen changes back to the display: “G54.2 P0” “G54.2(Current)”.

Refer to "Chapter 12 (19) Feature Coordinate Setting Function” for further details on each display item.

Workpiece coordinate zero				2018/04/26 09:59:25			
Program(0033)							
G58		G92		G52		External	
X	0.000	X	0.000	X	0.000	X	
Y	0.000	Y	0.000	Y	0.000	Y	
Z	0.000	Z	0.000	Z	0.000	Z	
A	0.000	A	0.000	A	0.000	A	
B	0.000	B	0.000	B	0.000	B	
C	0.000	C	0.000	C	0.000	C	
G68.2						Workpiece coordinate zero	
X	-10.000					X	-10.000
Y	-10.000					Y	-10.000
Z	400.000					Z	400.000
A	0.000					A	0.000
B	0.000					B	0.000
C	0.000					C	0.000
G53.1							
R							
				☒ Enlargement			
				☒ Current position			
[HELP]							
Current position	Modal 1	Modal 2	Soft limit	Enlargement	Workpiece coordinate zero	Select add. axes/change display	

G54 (G54 to G59, G54.1 P01 to G54.1 P48)
The coordinate zero point for the current workpiece is displayed.

G92
The amount of shift set with the G92 command is displayed.

G52
The amount of shift from the workpiece coordinate zero point set with the G52 command is displayed.

External

The zero point offset for the external workpiece is displayed.
The field is blank when the offset is disabled.

(When in normal operation mode)

G54.2 P0 (G54.2 P0 to G54.2 P8) (Option)

The reference rotary fixture offset that is selected is displayed. The field is blank when the rotary fixture offset is canceled (G54.2P0).

G54.2 (Current) (Option)

The current rotary fixture offset amount is displayed. The reference rotary fixture offset that is selected is displayed. The field is blank when the rotary fixture offset is canceled (G54.2P0).

Workpiece coordinate zero

The machine coordinate that is displayed adds in the workpiece zero point offset (G54 to G59, G54.1 P01 to G54.1 P48) and the shift amount for G92, G52, external and G54.2 (current).

(When in feature coordinate machining mode)

G68.2

The workpiece coordinate value (shift amount from workpiece coordinate zero) at zero in the feature coordinate system is displayed.

G53.1

R shows the rotation amount (angle) from XY in the workpiece coordinate system for the XY in the feature coordinate system.

Workpiece coordinate zero

The machine coordinate is displayed for the zero point position in the workpiece coordinate system that is currently being used for control.

The zero point in the workpiece coordinate system - the feature coordinate system setting is based on - and the machine coordinates that include each shift amount are displayed.

4.1.7 Modal Info

This screen is displayed only in conversation language mode.
The screen displays the machine status during memory operation.



The screen displays the absolute coordinates, relative coordinates, remaining distance, rotational transformation, positioning check distance, modal information, operation graphs and current operation (next operation).

Press the function key to change the display between the absolute coordinate position and relative coordinate position. (The absolute coordinate position, relative coordinate position and remaining distance are the same as those on the current position screen.)

- Axes displayed for absolute coordinate position, relative coordinate position and remaining distance
The 4th through 8th axes that are displayed are the additional axes and PLC axes which are set in the user parameters <4th row display axis> to <12th row display axis>.

Rotational trans.

The rotation center coordinate and rotation angle are displayed when the rotational transformation is executed.

Nothing is displayed until the first rotational transformation is performed.

The center is displayed in machine coordinates.

(Note) After the operation is executed, the results remain displayed until the next rotational transformation is executed.

Pos. check dist.

The group status for signal output 270 to 279 and the positioning end check distance for each axis are displayed.

High accuracy

Displays the high accuracy mode status (OFF, standard, rough, medium rough, medium rough S, finishing, finishing S, adjustment A, adjustment B, adjustment C, accuracy specification A, accuracy specification B, accuracy specification C, signal output 260 - 267, signal output 280 - 287).

Modal info

The information for the job being executed is displayed.

1. Job

The job name to be executed is displayed. (Cursor position on the memory operation screen)

The job number and job type are displayed.

Ex: Job 01 Tap

| | Job type
| | Job number

2. Coord.
The job number in the coordinate setting is displayed.
No job or the job number is displayed.
Ex.1: After executing the job in the coordinate setting...Coord. 01
Ex.2: No job in the coordinate setting...Coord. [No job]
(Note) When the job in the coordinate setting is executed for operation, the job number is displayed.
The job number remains displayed until the next job in the coordinate setting is executed.
3. Comd.
The axis travel status is displayed.
The status is displayed as absolute or incremental.
4. Motion
The operating motion is displayed.
One of the following operations is displayed: positioning, line, arc CW (clockwise), arc CCW (counterclockwise), drill, tap, thread cutting CW or thread cutting CCW.
Ex: Arc CW (1000)
| _____ Feedrate
(Note) The feedrate is displayed for all items except for [Positioning].
5. Mode
The operating mode is displayed.
The mode is displayed as position check or cutting.
6. Cutting mode
The mode is displayed as standard or high accuracy A.
7. Tool
The spindle tool information is displayed.
The tool information includes the tool number and tool name.
Ex: 01 Center drill
| _____ Tool name
| _____ Tool number
8. Length
The spindle tool length is displayed.
9. Spindle
The spindle status is displayed.
One of the following operations is displayed: stop, servo on, CW (clockwise), CCW (counterclockwise) or orientation.
(Note) The spindle speed is also displayed for CCW (counterclockwise) and CW (clockwise).

Ex: CCW (10000)
| _____ Spindle speed

• The angle is also shown when orientation is displayed.
Ex: Orientation (180.000)
| _____ Angle
10. Workpiece height comp.
The workpiece height compensation value is displayed, which is set while preparing for program operation.
11. Tool wear offset (dia.)
The tool wear offset (dia.) value is displayed, which is set for the tool while preparing for program operation.

12. Tool wear offset (length)
The tool wear offset (length) value is displayed, which is set for the tool while preparing for program operation.
13. Cutter compensation
The cutter compensation for the spindle tool is displayed.
Displays:
Right (***.***)
Left (***.***)
Center
. ... Compensation

Current motion and next motion

A summary of the next motion is displayed before the operation. During operation, the current motion and the next motion are displayed.

The display is the same as the motion on the memory operation screen.

- After changing the program number, resetting, turning the power ON or ending the program on the memory operation screen, the following applies for each item:
Machining coordinates, tool and spindle...Current data is maintained.
Coord. ... No job
Cmd. ... Absolute
Mode ... Position check
Rotational trans. ... All data cleared
Current motion, cutter compensation, workpiece height comp., tool wear offset (length) and tool wear offset (dia.) ... Data cleared
Job, motion and next motion ... Displays depending on the program number on the memory operation screen.
- After performing a tool change in manual or MDI operation mode:
The tool displayed changes as the tool changes. Cutter compensation, tool wear offset (length) and tool wear offset (dia.) are cleared.

4.2 Program Display

The information displayed is different depending on the mode selection:

1. Operation program is displayed when in memory operation mode.
2. Display is invalid when in manual operation mode.
3. MDI operation mode screen is displayed when in MDI operation mode.
4. Program edit mode screen is displayed when in program edit mode or operation program edit mode.

4.3 Manual Conditions

When the [MANU COND] key is pressed, the screen shown below is displayed for setting the travel speeds and the spindle speed in manual operation mode.

The screenshot shows the 'Manual conditions' screen with a title bar at the top containing 'Manual conditions', a status icon, and the date/time '2012/06/04 15:37:47'. The main area is divided into several sections:

- Manual conditions table:**

High travel speed(XYZ/P)	10	%
High rotation speed(4/P)	10	%
Low travel speed(XYZ/P)	50.00	mm/min
Low rotation speed(4/P)	0.100	min ⁻¹
Step travel(XYZ/P)	0.001	mm
Step rotation amount(4/P)	0.001	°
Spindle speed	100	min ⁻¹
- Machine coord. pos. table:**

X	-100.000
Y	-100.000
Z	480.000
C	0.300
A	0.000
B	0.000
A	
B	
- Absolute coord. pos. table:**

X	-100.000
Y	-100.000
Z	420.000
C	0.300
A	0.000
B	0.000
A	
B	
- Relative coord. pos. table:**

X	-100.000
Y	-100.000
Z	480.000
C	0.300
A	0.000
B	0.000
A	
B	

At the bottom, there is a [HELP] button and a row of function keys: C-axis, A-axis, B-axis, A-axis, B-axis, Change additional axes display, and Recover/Inspection Menu.

(Note) When manual operation mode is not enabled, the manual condition data can only be referenced and cannot be changed, even though the additional axis function keys are enabled.

4.4 ATC Tools

4.4.1 ATC Tool (Details) Screen

When the [ATC TOOL] key is pressed, the screen shown below is displayed for setting the tools attached to the magazine and the different types of tools.

(Note) The display varies depending on the type of ATC.
 A list of spindle tools is displayed as shown in the screenshot example below for machine models with a double arm mechanism. A list of spindle tools is not displayed for machine models with an armless mechanism.

Ex: When using NC language mode

Tool No.	Tool name	Tool data	Group	Tool life	Tool type	Color
01	CENTER DRILL1	100.000x1.000		100000(H)	STD tool	
02	DRILL1	80.000x0.000		100000(H)	STD tool	
03	CARBIDE DRILL1	78.000x0.000		1000(G)	STD tool	
04	Cap			*****	STD tool	
05					STD tool	
06					STD tool	
07					STD tool	
08					STD tool	
09					STD tool	
10					STD tool	

Program No.

Current folder /

[HELP]

Set program No. Set tool List Program list Initialize life Next tool

Ex: When using conversation language mode

Tool No.	Tool name	Tool data	Main tool	Tool life	Tool type	Color
01	Center drill	0.000x90.000		500(H)	STD tool	Blue
02	Drill	4.300		10000(H)	STD tool	Red
03	Point tap	M5.000x0.800		5000(H)	STD tool	Magenta
04	Cap			*****	STD tool	
05	Chamfer tool	90.000x10.000		*****	STD tool	Green
06	End mill (F)	1.000		*****	STD tool	Cyan
07	End mill (R)	1.000		*****	STD tool	Yellow
08					STD tool	
09					STD tool	
10					STD tool	

Program No.

Current folder /

[HELP]

Set program No. Set tool List Program list Initialize life Next tool

Current folder

The folder is displayed for the program which is currently selected in memory operation mode.

(Note 1) Tool data can only be referenced and cannot be changed in modes other than MDI operation mode. However, if there is one or more tools with a tool life that has expired, press the [F5] (Initialize life) key to display the <Initialize life> screen in order to initialize the life of the tool.

(Note 2) For NC language mode, when the user parameter <Select memory operation> is set to <1:Tape>, the tool setting screen is displayed. The program number cannot be specified.

(Note 3) When changing the setting for the user parameter <Conversation/NC language change>, the tools which were set before the language change are displayed in red and cannot be used with the newly set language.

Set program No.

When the program number is specified, the tool number (registration number in tool list) and the tool name are displayed. This tool number refers to a tool that is required for running the specified program and is displayed in the order specified in the program. The background turns green for those tools that are already registered in the ATC tools.

To specify a program, the program number can be directly input, or it can be selected from the list.

- Entering the program No.
Enter the program number in the input field, and press the **[ENT]** key. The program in the “Current folder,” which is displayed in the instructions area, is selected.
- Selecting from the list
Press **[F3]** (Program list) and select a program from the displayed list. Use the cursor to select the program from the displayed list, or enter the program No., and then press the **[ENT]** key.

Tool No.				[HELP]	
Program(0003)				[Set tool]	
01	01	Center drill	05		
02	02	Drill	06		
03	04	Chamfer tool	07		
04	03	Point tap	08		
				09	
				10	
				11	
				12	
Set	Set tool	List	Program list	Initialize life	Next tool
program No.					

(Note 1) A given program can also be specified as an extended memory operation.

(Note 2) In conversation language mode, the alarm message <<No tools assigned>>, <<? not calculated>> or <<Memory operation data not created>> is displayed for a program to which tools are not assigned, and it cannot be specified.

For a schedule program, the operator message <<Program number error>> is displayed and it cannot be specified.

(Note 3) In NC language mode, when the user parameter <Select memory operation> is set to <1:Tape>, **[F0]** (Set program No.) is grayed out and cannot be pressed.

(Note 4) When the folder selected in memory operation mode is changed after a program number has been specified, the program number does not set.

- * This applies when a program in another folder is selected in memory operation, and also when the folder is changed in pallet program operation, external program selection and operation using a slave command, etc.

The number of tool numbers displayed in the tool list is the same as the magazine capacity (for machine models with a double arm ATC, the magazine capacity + 1).

Pressing **[F7]** (Next tool) displays the tool(s) not currently displayed in the list starting from the next number. After the last tool is displayed, the first tool number in the list is displayed again.

If the tool to be used in a program is not registered in the tool data (tool list for conversation language mode), register it before running the program.

For NC language mode, if a tool is specified with the tool group number, the group number is displayed in brackets after the tool number.

Ex: When tool number 12 is registered in group 03 → 12 (03)

- * When tools are specified with the group number in a program, the tools that have not expired yet are searched and displayed in the order that they were registered to the group. When the tool life has expired for all of the tools, the tool registered last to the group is displayed.

When a tool is not registered in the tool group to be used in the program, register it before running the program.

Set tool

Enter a program number and press the [ENT] key (program setting mentioned above) or press the [F1] (Set tool) key to display the tool setting screen.

Ex: When using NC language mode

ATC tool (details)						
2014/07/10 11:13:34						
Program(CAPT01/0001)						
Tool No.	Tool name	Tool data	Group	Tool life	Tool type	Color
Spindle	01 CENTER DRILL1	100.000x1.000		100000(H)	STD tool	
01	02 DRILL1	80.000x0.000		100000(H)	STD tool	
02	03 CARBIDE DRILL1	78.000x0.000		1000(C)	STD tool	
03	Cap			*****	STD tool	
04	04 SPIRAL TAP1	50.000x0.000		100000min	STD tool	
05					STD tool	
06					STD tool	
07					STD tool	
08					STD tool	
09					STD tool	
10					STD tool	
Tool No. [] [HELP]						
Program(CAPT01/0001) [Set tool]						
01 01	CENTER DRILL1	05		09		
02 02	DRILL1	06		10		
03 03	CARBIDE DRILL1	07		11		
04 04	SPIRAL TAP1	08		12		
Set program No.	Set tool	List	Program list	Initialize life		Next tool

The cursor is positioned at the spindle for double arm machine models.
The cursor is positioned at 01 for armless machine models.

Function key description

- [F0] Changes to program number setting request mode.
- [F1] Changes to tool number setting mode.
- [F2] Changes to the ATC tool list display screen.
- [F3] Changes to the screen for selecting a program from the program list in the program No. setting mode.
- [F7] Displays the next tool in display area.

Tool No.

Register the number of the tool to be stored in the magazine.

- Place the cursor on the magazine number to be set.
- Set the tool number (1 to 99) registered to the <Tool list> in the data bank.

When 0 is set instead of a tool number, the cap is specified to the magazine number. The magazine number where the cap is specified is not used during a tool change.

(Note) Be sure to match the tool that is set on the ATC tool screen with the tool actually installed to the magazine. Failure to do so may cause an accident or damage to the machine.

- Place the cursor on the magazine number to be set.
- Set a tool number (1 to 99) that is registered to “Tool data (Tool list for conversation mode)” in the data bank, or specify the cap (0).

When a tool number is set, the <Tool name>, <Tool data> and <Tool life> for the tool are displayed.

<Tool life> is highlighted if the life of the set tool is expired or if a tool life warning is issued for the tool.

(Note 1) If a tool number is changed in NC language mode and the tool number is not registered to the same tool group as before the change, the <Group> field data is also deleted at the same time. If a tool number is changed in conversation mode, the <Main tool> field data is deleted at the same time.

(Note 2) The alarm <<This tool is already registered>> is triggered if an attempt is made to set the same tool number in two places.

(Note 3) The alarm <<No tool in tool list>> is triggered if an attempt is made to set a tool number that is not registered to the <Tool list> in conversation language mode.

Tool group number (for NC language mode)

This is used when using a tool registered as a tool group.

Register the group number for the tool to be stored in the magazine.

1. Move the cursor to <Group> of the number of the magazine to which the tool to be used as a tool group is installed.
2. Set the tool group number where the tool to be used is registered.

When the life of the tool being used is expired, the next tool registered in the tool group is automatically used.

- The tool group must be set to the tool list on the data bank screen.
- T command in an NC program must include tool group numbers.

ATC tool(details) 2014/07/10 11:34:26						
Program(CAPT01/0001)						
Tool No.	Tool name	Tool data	Group	Tool life	Tool type	Color
01	01 CENTER DRILL1	100.000x1.000	01	100000(H)	STD tool	
02	02 DRILL1	80.000x0.000	02	100000(H)	STD tool	
03	Cap			*****	STD tool	
04	03 CARBIDE DRILL1	78.000x0.000	03	1000(C)	STD tool	
05	04 SPIRAL TAP1	50.000x0.000		100000mIn	STD tool	
06	05 CENTER DRILL2	100.000x1.000	01	100000(H)	STD tool	
07					STD tool	
08	06 DRILL2	80.000x0.000	02	100000mIn	STD tool	
09	08 SPIRAL TAP2	50.000x0.000	03	100000mIn	STD tool	
10					STD tool	
11					STD tool	

Tool No.	[HELP]
0:Cap setting	:Set tool
Program(CAPT01/0001)	

01 01 CENTER DRILL1	05		09	
02 02 DRILL1	06		10	
03 03 CARBIDE DRILL1	07		11	
04 04 SPIRAL TAP1	08		12	

Set program No.	Set tool	List	Program list	Initialize life	Next tool
-----------------	----------	------	--------------	-----------------	-----------

- The timing and motion to change to a spare tool varies depending on the user parameter settings <Spare tool change> and <Check tool life on first use>.

•		
M6 T901;	←	(1) When the tool life has expired at this point and there is a spare tool
:		(2) When the tool life has expired at this point and there is no spare tool
M6 T902;		
:		
M6 T901;	←	(3) When the tool life has expired at this point and there is a spare tool
:		(4) When the tool life has expired at this point and there is no spare tool
M30;		

Case	Setting for <Spare tool change>			
	<0: At operation start>		<1: At tool change>	
	Setting for <Check tool life on first use>			
	<0: No>	<1: Yes>	<0: No>	<1: Yes>
(1)	Change with spare tool	Change with spare tool	Change with spare tool	Change with spare tool
(2)	No alarm	Expired tool life alarm	Expired tool life alarm	Expired tool life alarm
(3)	No alarm	No alarm	Change with spare tool	Change with spare tool
(4)	No alarm	No alarm	Expired tool life alarm	Expired tool life alarm

- Spare tools are selected only when the tool life has not expired in the order that they are registered in the tool group.
- When the tool number is not set, the alarm <<Tool number not set>> is triggered.
- When the tool number is already set but is not registered to the tool group, the alarm <<No tool registered in group>> is triggered.

Main /spare tool (for conversation language mode)

This is used to register spare tools to the magazine.

Register the number of the main tool to be stored in the magazine.

1. Move the cursor to <Main tool> for the magazine number where the tool to be used as a spare tool is installed.
2. Set the spare tool using the tool number.

Main tool refers to the tool which is used before a spare tool.

Spare tool refers to a tool that is automatically used when the tool life of the main tool has expired.

Register the main tool number to the magazine where the spare tool is attached.

The screenshot shows the 'ATC tool(details)' screen with a date/time stamp of 2014/07/10 11:32:46. It displays a table of tools with columns: Tool No., Tool name, Tool data, Main tool, Tool life, Tool type, and Color. The tools listed are: 01 Center drill, 02 Drill, 03 Point tap, 04, 05 09 Drill, 06, 07, 08, 09, 10, and 11. Below the table, there is a 'Main tool' input field and a 'Program(0003)' section. The 'Program(0003)' section shows a grid of tool numbers and names: 01 01 Center drill, 02 02 Drill, 03 04 Chamfer tool, 04 03 Point tap, 05, 06, 07, 08, 09, 10, 11, and 12. At the bottom, there are buttons for 'Set program No.', 'Set tool', 'List', 'Program list', 'Initialize life', and 'Next tool'.

- The timing and motion to change to a spare tool varies depending on the user parameter setting <Spare tool change>.

Case	Setting for <Spare tool change>	
	<0: At operation start>	<1: At tool change>
When the tool life has expired and there is a spare tool at the start of operation	Change with spare tool	Change with spare tool
When the tool life has expired and there is no spare tool at the start of operation	Expired tool life alarm	Expired tool life alarm
When the tool life has expired and there is a spare tool for the tool change in the middle of operation	No alarm	Change with spare tool
When the tool life has expired and there is no spare tool for the tool change in the middle of operation	No alarm	Expired tool life alarm

- Before cutting, after cutting, miscellaneous functions, tool wear offset (dia.), length and dwell specified for the tool assignment also apply to the spare tool.
- When the tool number is not set, the alarm <<No tool in magazine>> is triggered.
- When specifying a tool number that is already set as the main tool for another spare tool, the alarm <<This tool is already registered.>> is triggered.
- Alarm at spare tool change
If a tool is specified with a tool group number in a program or the spare tool is set, when the tool life has expired and the tool is changed to the spare tool automatically, the alarm <<Spare tool used>> is triggered.

Tool type

There are four types of tools: <STD tool>, <Large tool>, <Medium> and <Small tool>. Refer to the tool label on the machine to set the tools correctly.

For machine models that can use large tools, <Large tool> can be specified for the <Tool type>. When a large tool is set to a magazine, the adjacent pots are marked with an “X” (restricted use) and no tools other than a cap can be registered.

- When a tool has been set to an adjacent pot, setting a large tool triggers the alarm <<Tool is registered to next pot>>. Therefore, the tool cannot be changed to a large tool.
- When a tool number other than a cap is specified for the pots marked with an “X” (restricted use), the alarm <<Invalid pot>> is triggered and the setting is not possible.
- When a tool change operation is performed on a pot marked with an “X” (restricted use), the alarm <<Invalid pot>> is triggered and the tool change operation does not execute. (Performs the same operation as the cap specified pot.)

For machine models that can use medium tools, <Medium> can be specified for the <Tool type>. For machine models that can use small tools, <Small tool> can be specified for the <Tool type>.

ATC tool (details)						
2014/07/10 11:11:34						
Program(CAPT01/0001)						
Tool No.	Tool name	Tool data	Group	Tool life	Tool type	Color
Spindle						
01	02 DRILL1	100.000x1.000	01	100000(H)	STD tool	
02	Cap	80.000x0.000	02	100000(H)	STD tool	
03	03 CARBIDE DRILL1	78.000x0.000		1000(C)	STD tool	
04	04 SPIRAL TAP1	50.000x0.000		100000min	STD tool	
05	05 CENTER DRILL2	100.000x1.000	01	100000(H)	STD tool	
06	06 DRILL2	80.000x0.000	02	100000min	STD tool	
07					STD tool	
08	07 CARBIDE DRILL2	78.000x0.000	03	10000(C)	Large tool	
09					STD tool	
10					STD tool	
Tool No.						
Program(CAPT01/0001) [HELP] .Set tool						
01 01	CENTER DRILL1	05		09		
02 02	DRILL1	06		10		
03 03	CARBIDE DRILL1	07		11		
04 04	SPIRAL TAP1	08		12		
Set program No.	Set tool	List	Program list	Initialize life	Next tool	

The tool type varies depending on the size of the tool's outer diameter.

STD tool: Tool with an outer diameter that is 110 mm or smaller

(When using S300X1/S500X1/S700X1/S1000X1, S300X2/S500X2/S700X2 or F600X1(14MG))

Tool with an outer diameter that is 80 mm or smaller

(When using R450X1/R650X1, R450X2/R650X2(14/22MG), M140X2 or F600X1(22MG))

Tool with an outer diameter that is 55 mm or smaller

(When using R650X2(40MG))

Large tool: Tool with an outer diameter that is 125 mm or smaller

(Note 1) The standard tool can be replaced without worrying about interference with the adjacent pot.

(Note 2) When setting a large tool, make sure that no tool is registered to the adjacent pots.

When a large tool is set, the tool change is carried out so that there is no interference with other tools.

(Note 3) Configure the settings when using an OSG made disc cutter (outer diameter: 125 mm). When using the adjacent pot, a tool with an outer diameter larger than 110 mm will interfere with the spindle. Interference can be prevented by configuring the settings.

Color

Colors used to draw a program path on the graphics screen can be set.

ATC tool (details)							2014/07/10 11:08:36	
Program(CAPT01/0001)								
	Tool No.	Tool name	Tool data	Group	Tool life	Tool type	Color	
Spindle	01	CENTER DRILL1	100.000x1.000	01	100000(H)	STD tool		
01	02	DRILL1	80.000x0.000	02	100000(H)	STD tool		
02	06	SPIRAL TAP2	50.000x0.000		100000mIn	STD tool		
03	03	CARBIDE DRILL1	78.000x0.000		1000(C)	Medium		
04	Cap				*****	STD tool		
05	05	CENTER DRILL2	100.000x1.000	01	100000(H)	STD tool		
06	06	DRILL2	80.000x0.000	02	100000mIn	STD tool		
07						STD tool		
08	07	CARBIDE DRILL2	78.000x0.000	03	10000(C)	Large tool		
09						STD tool		
10						STD tool		

Color

[0:No color 1:BLU 2:R 3:PP 4:GRN 5:LBLU 6:Y 7:W] [HELP]

Program(CAPT01/0004) ■ Set tool

01 01 CENTER DRILL1	05 05 CENTER DRILL2	09
02 02 DRILL1	06 06 DRILL2	10
03 03 CARBIDE DRILL1	07 07 CARBIDE DRILL2	11
04 04 SPIRAL TAP1	08 08 SPIRAL TAP2	12

Set program No.	Set tool	List	Program list	Initialize life	Next tool
-----------------	----------	------	--------------	-----------------	-----------

4.4.2 ATC Tool List Screen

Press the [F2] key to display the tool list screen.

ATC tool (list)							2014/07/10 11:07:37	
Program(CAPT01/0001)								
	Tool No.	Tool name		Tool No.	Tool name		Tool No.	Tool name
Spindle	01	CENTER DRILL1						
01	02	DRILL1						
02	06	SPIRAL TAP2						
03	03	CARBIDE DRILL1						
04	Cap							
05	05	CENTER DRILL2						
06	06	DRILL2						
07								
08	07	CARBIDE DRILL2						
09								
10								
11								
12								
13								
14								

[HELP]

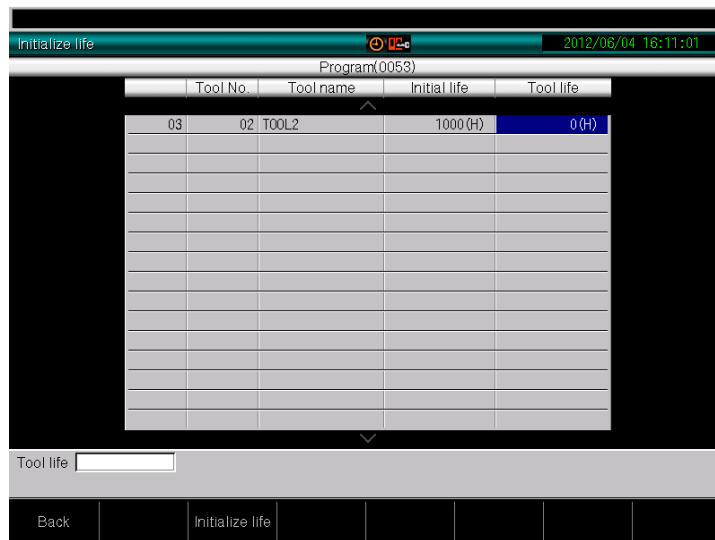
Set program No.	Set tool	Details	Program list	Initialize life	Next tool
-----------------	----------	---------	--------------	-----------------	-----------

(Note) The display varies depending on the type of ATC. The display for machine models with a double arm mechanism is shown in the screen above. Spindle tools are not displayed for machine models with an armless mechanism.

4.4.3 Initialize Tool Life

Press the **[F5]** (Initialize life) key to display the initialize life screen.

On this screen, the tool life can be initialized for tools registered on the ATC tool screen that have an expired tool life.



Only tools with an expired tool life are displayed on this screen.

If there is no tool with an expired tool life, this screen is not displayed.

Press the cursor key to move the cursor to the tool that you wish to initialize.

When the **[F2]** (Initialize life) key is pressed, the tool at the cursor position is initialized.

(For the countdown method, the initial life is displayed. For the count up method, 0 is displayed.)

The tool life can be entered directly.

In this situation, enter the tool life value that you wish to set, and then press the **[ENT]** key.

Press the **[F0]** (Back) key to return to the ATC tool screen.

- (Note 1) The initialize life screen does not display while editing the tool data and during communication.
- (Note 2) The initialize life screen does not display during memory operation.
- (Note 3) Tool life cannot be initialized if the [DATA PROTECTION] switch is enabled.

4.5 Production Monitor

4.5.1 Workpiece Counter

Perform one of the following key operations on the production monitor menu screen to display the screen below:

- Press [1] and then [ENT].
- Move the cursor to menu number 1 and press [ENT].
- Press [F1] (Workpiece counter).

Workpiece counter

2012/06/04 16:11:44

	Counter 1	Counter 2	Counter 3	Counter 4
Count	1	1	0	0
Current value	49	50	1	1
Target value	50	50	0	0
End signal value	48	48	0	0
End / Signal	Ending	Completed		

Count

0 - 999
0: Does not count

Production monitor menu

Clear current value

Input field and description of items

Count	0 to 999 0: Does not count
Current value	0 to 99999
Target value	0 to 99999 0: Continues operation
End signal value	0 to 99999 0: Does not issue a warning when ending

(Note) Input is not possible during memory operation. The information can only be referenced.

Press the [F4] (Clear current value) key to initialize the current value to 0 for the counter where the cursor is.

(Note) Initialization is not possible during memory operation.

Workpiece counter

2012/06/04 16:12:12

	Counter 1	Counter 2	Counter 3	Counter 4
Count	1	1	0	0
Current value	49	50	1	1
Target value	50	50	0	0
End signal value	48	48	0	0
End / Signal	Ending	Completed		

The current value of counter 1 is cleared.
OK to clear?

Yes

No

Press the [F0] (Yes) key to initialize to 0, and the [F1] (No) key to cancel the initialization.

Count

Set the value to be added to the workpiece counter at the end of the program (NC language mode: M30 or M02, conversation language mode: program end operation). When 0 is set, no value is added.

Current value

When a value other than 0 is set for the count and the counter is used in a program, the value set for the count is added each time the program ends (NC language mode: M30 or M02, conversation language mode: program end operation).

The value is not added if the user parameter <Workpiece counter when program is repeated> is off when the program is repeated.

For conversation language mode, when using a schedule program, the value is added at the end of the last program in the schedule.

Target value

Set the value at which the counter stops counting.

At the end of the program (NC language mode: M30 or M02, conversation language mode: program end operation), the operator message <Counter 1 (- 4) end> is displayed when the current value exceeds the setting here.

In addition, the external signal (count end signal) turns ON.

The message <Completed> is also displayed on the screen.

When the machine is equipped with an optional indicator, the yellow or red indicator lights up depending on the user parameter setting <Count end lamp status>.

End signal value

Set the value on the counter to indicate when the end of the count is approaching.

At the end of the program (NC language mode: M30 or M02, conversation language mode: program end operation), the operator message <Counter 1 (- 4) approaching setting> is displayed when the current value exceeds the setting here.

In addition, the external signal (count end approaching signal) turns ON.

The message <Ending> is also displayed on the screen.

Press the **[CAN]** key or perform a key operation to clear the alarm message. However, the count end approaching signal remains ON.

Restarting memory operation is possible even when the operator message is triggered.

Perform one of the following to turn the count end approaching signal OFF:

- Set current value so that it is smaller than the end signal value.
- Set the end signal value so that it is larger than current value.
- Set "0" for the end signal value.
- Set "0" for count.

4.5.1.1 Workpiece Counter ON Command in Program (NC Language)

When the M code commands below are issued in memory operation or MDI operation, the count value is added to the counter per the commands M02 or M30.

M211	Workpiece counter 1 ON
M212	Workpiece counter 2 ON
M213	Workpiece counter 3 ON
M214	Workpiece counter 4 ON

When M211 through M214 are executed, the counter* on the workpiece counter screen is highlighted. The screen returns to normal when the workpiece counter is disabled.

When the M code commands below are issued in memory operation or MDI operation, the corresponding counter settings are cancelled.

M221	Workpiece counter 1 OFF
M222	Workpiece counter 2 OFF
M223	Workpiece counter 3 OFF
M224	Workpiece counter 4 OFF

The counter is cancelled when the power is turned ON, when the **[RST]** key is pressed, when M02 or M30 is executed, when operation is reset or when the M221 through M224 are executed (workpiece counter is cancelled).

4.5.1.2 Workpiece Counter ON Command in Program (Conversation Language)

Set one of the following signal output codes in a signal output job for machining data or using a miscellaneous function in a motion call job.

Signal output code 211	Workpiece counter 1 ON
Signal output code 212	Workpiece counter 2 ON
Signal output code 213	Workpiece counter 3 ON
Signal output code 214	Workpiece counter 4 ON

Example of signal output job:

Job 01 signal output	Code 211
----------------------	----------

Example of motion call job - miscellaneous function:

Job 01 motion call	Pattern	OFF
	Tool No.	01
	Motion 01 miscellaneous function	Code 211
	Motion end	

The signal output code can also be set for the miscellaneous function on the <Tool assignment> screen in the operation preparation menu.

- After the memory operation starts up, the workpiece counter is enabled as long as at least one of the signal output codes 211 to 214 is set before the program ends.
In addition, multiple counters can be enabled at the same time by setting multiple signal output codes: 211 to 214.
- The workpiece counter is disabled when the power is turned ON, when the **[RST]** key is pressed or when the program end operation is executed.
To highlight the counter when the program starts, set the signal output code in job 01.

Setting the code during a program varies depending on the machining sequence.
When using the tool assignment screen, set the code to the tool to be used first in the machining.

In the example below, the counter is not highlighted until 02 drill is used.

Tool assignment		2012/06/04 17:34:14					
Program(0002)		**** ????					
Job01 Tap	1. Center hole	Tool No.	01 Center drill				
		Tool length offset	Cutter compensation	**			
		Misc.func.before cutting	Misc.func.after cutting				
		Dwell time					
	2. Hole	Tool No.	14 Drill	2.500			
		Tool length offset	Cutter compensation	**			
		Misc.func.before cutting	Misc.func.after cutting	211			
		Dwell time					
	3. Tapping	Tool No.	43 Point tap	M3.000			
		Tool length offset	Cutter compensation	**			
		Misc.func.before cutting	Misc.func.after cutting				
		Dwell time					
	Job02 Tap	1. Center hole	Tool No.	02 Drill	1.000		
			Tool length offset	Cutter compensation	**		
			Misc.func.before cutting	Misc.func.after cutting			
Dwell time							
Miscellaneous function code after cutting							
Signal output:1 - 99,900 - 999,800 - 899(PLC)							
Excluding 3,4,30,Tool change code(501 - 599,701 - 721)							
Tool assignment is completed. ? calculation was performed.							
[HELP]							
Completion mode	Previous tool	Next tool	Change function	Calculation result	Search	Tool check	Assign tools

3. When signal output codes 211 to 214 are executed, the counter* on the workpiece counter screen is highlighted. The screen returns to normal when the workpiece counter is disabled. To disable the workpiece counter, set the following signal output codes:

Signal output code 221	Workpiece counter 1 OFF
Signal output code 222	Workpiece counter 2 OFF
Signal output code 223	Workpiece counter 3 OFF
Signal output code 224	Workpiece counter 4 OFF

4.5.2 Time Display

4.5.2.1 Time Display

Perform one of the following key operations on the production monitor menu screen:

- Press [2] and then [ENT].
- Move the cursor to menu number 2 and press the [ENT] key.
- Press [F2] (Time display).

Then, <Machine data> related to machine operation is displayed, for example, information related to the cycle time under <Current> or information related to the <Previous value>.

When operation starts, the <Current> cycle time starts to record again.

The cycle time for the <Previous value> updates the value according to the user parameter (switch 1) setting for <Reset cycle time value>.

The screen below is displayed on machine models with a QT (quick table).

Time display		2018/10/09 19:03:41	
Machine data			
Pallet 1 program		Curr.????	Previous value????
Cycle time		0000:00:00.0	0000:00:00.0
Cutting time		0000:00:00.0	0000:00:00.0
Non cutting time		0000:00:00.0	0000:00:00.0
Cutting time / cycle time		0 %	0 %
Operation end date and time		-	
Pallet 2 program		Curr.????	Previous value????
Cycle time		0000:00:00.0	0000:00:00.0
Cutting time		0000:00:00.0	0000:00:00.0
Non cutting time		0000:00:00.0	0000:00:00.0
Cutting time / cycle time		0 %	0 %
Operation end date and time		-	
Operation end counter			0205
Operation time		0154:20:24	
Power on time		00123:57:54	
Total operation time		00001:08:48	
Date		2018/10/09 19:03:41	
Production monitor menu	Time display	Operation time log	Initialize operation end counter

Description

Pallet 1 program: Displays the program folder name and program number.

Pallet 1 or 2 information is displayed when equipped with a quick table.

When the operation is complete, the <Previous value> for the pallet side that was used is updated in the <Current> data.

The screen below is displayed on machine models not equipped with a QT (quick table).



Time display		2018/10/10 11:47:21
Machine data		
	Curr.	Previous value
Program	????	????
Cycle time	0000:00:00.0	0000:00:00.0
Cutting time	0000:00:00.0	0000:00:00.0
Non cutting time	0000:00:00.0	0000:00:00.0
Cutting time / cycle time	0 %	0 %
Operation end date and time		-
Operation end counter		0002
Operation time	1906:04:56	
Power on time	00017:39:45	
Total operation time	00000:00:00	
Date	2018/10/10 11:47:21	
Production monitor menu	Time display	Operation time log
		Initialize operation end counter

Description

Program: Displays the program folder name and program number.

Cycle time	Displays the program cycle time.
Cutting time	Displays the cutting time in the program.
Non cutting time	Displays the non-cutting time in the program.
Cutting time/cycle time	Displays the ratio of the cutting time to the cycle time.
Operation end date and time	Only displays the previous value. Displays the date and time when the program operation ended. The <Current> value does not display during operation.
Operation end counter	Counts when the <Previous value> is updated.
Operation time	Displays the total operation time to date.
Power on time	Displays the total power ON time to date.
Total operation time	Displays the total operation time to date.
Date	Displays the current date and time.

(Note) When the machine lock or dry run is enabled, the actual machining time may vary.

Press the [F7] key (Initialize operation end counter) to reset the <Operation end counter> to 0 (zero).

4.5.2.2 Operation Time Log

The operation time, operation cycles and operation rate are displayed. Press the **[F2]** (Operation time log) key on the time display screen.

Operation time log				2012/06/04 16:21:34
Date	Operation time	Operation cycles	Oper. rate (total oper. time/power on time)	
2012/05/08	00:00:00	0	0 %	
2012/05/09	00:00:00	0	0 %	
2012/05/10	00:00:00	0	0 %	
2012/05/11	00:00:00	0	0 %	
2012/05/14	00:03:34	6	0 %	
2012/05/15	00:00:00	0	0 %	
2012/05/16	00:00:00	0	0 %	
2012/05/17	00:00:00	0	0 %	
2012/05/18	00:00:00	0	0 %	
2012/05/21	00:00:24	98	0 %	
2012/05/22	00:00:00	0	0 %	
2012/05/23	00:00:00	0	0 %	
2012/05/24	00:08:01	9	0 %	
2012/05/25	00:00:00	0	0 %	
2012/05/28	00:00:00	0	0 %	
2012/05/29	01:10:44	6	4 %	
2012/05/30	01:34:38	6	8 %	
2012/05/31	00:00:00	0	0 %	
2012/06/01	00:02:26	13	0 %	
2012/06/04	00:00:05	19	0 %	
Production monitor menu	Time display	Operation time log		

4.5.3 Communication Monitor

Perform one of the following key operations on the production monitor menu screen to display the screen below:

- Press **[3]** and then **[ENT]**.
- Move the cursor to menu number 3 and press **[ENT]**.
- Press **[F3]** (Communi. monitor).

Communication monitor		2012/06/04 16:22:06	
Port	Message	Date and time	

A summary of serial communication and Ethernet communication (except HTTP connection) logs is displayed. A maximum of 20 communication logs are shown in chronological order.

The <Signal line monitor> displays the signal line status for the serial port. When the signal is ON, it is highlighted in blue.

The <Ethernet connection status> shows the client's IP address that is connected to the machine. The HTTP connection is also displayed.

The data is valid only while the power is ON. When the power is turned OFF, the data is cleared.

- Port
The connection ports (serial port, Ethernet, and Ethernet (FTP)) used for sending and receiving data are displayed.
- Message
The messages that are received and sent are displayed.
A summary of the data that is received and sent per file is displayed.
The log does not record the FTP protocols per command.

- ## Ethernet information

[illegible]

The following information is displayed in the Ethernet details (NC).
Host name, IP address, subnet mask, default gateway, DHCP, DHCP server address, auto acquisition of DNS server address, DNS server address 1, DNS server address 2 and MAC address

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4.5.4 Measurement Results

Perform one of the following key operations on the production monitor menu screen to display the screen below:

- Press **[4]** and then **[ENT]**.
- Move the cursor to the menu and press **[ENT]**.
- Press **[F4]** (Measurement results).

To change the value in the measurement results, move the cursor to the position that you wish to change and enter a value.

Note that the measurement type cannot be changed. In addition, only the most recent measurement result can be changed.

Also, no changes can be made during operation.

Measurement result screen in NC language mode

Measurement result 1				2012/06/04 16:26:24				
Latest		G121	Previous 1		G121	Previous 2		G121
X		-99.818	X		-103.431	X		-103.270
Y		-151.689	Y		-146.630	Y		-146.701
Z			Z			Z		
Rotation		3.622	Rotation		343.259	Rotation		343.348
Time	2012/05/29	19:34:02	Time	2012/05/29	19:33:21	Time	2012/05/29	19:32:20
Previous 3		G121	Previous 4		G121	Previous 5		G121
X		-101.521	X		-104.102	X		-102.789
Y		-147.434	Y		-150.723	Y		-148.365
Z			Z			Z		
Rotation		343.127	Rotation		359.811	Rotation		346.461
Time	2012/05/29	19:31:36	Time	2012/05/29	19:30:18	Time	2012/05/29	19:29:14
Previous 6		G121	Previous 7			Previous 8		
X		-105.832	X			X		
Y		-147.930	Y			Y		
Z			Z			Z		
Rotation		350.302	Rotation			Rotation		
Time	2012/05/29	19:28:28	Time			Time		
X								
Production monitor menu	Meas. result 1	Meas. result 2	Meas. result 3	Meas. result 4				Delete all

Measurement result screen in conversation language mode

Measurement result 1				2012/06/04 17:27:03			
Latest		Parallel	Previous 1		Parallel	Previous 2	
X	1.721		X	0.188		X	
Y			Y			Y	
Z			Z			Z	
Rotation			Rotation			Rotation	
Center X			Center X			Center X	
Center Y			Center Y			Center Y	
Time	2012/06/04 17:25:47		Time	2012/06/04 17:25:10		Time	
Previous 3			Previous 4		Corner	Previous 5	
X			X	3.644		X	
Y			Y	-1.138		Y	
Z			Z			Z	
Rotation			Rotation			Rotation	
Center X			Center X			Center X	
Center Y			Center Y			Center Y	
Time			Time	2012/06/04 17:22:49		Time	
X							
Production monitor menu	Meas. result 1	Meas. result 2	Meas. result 3	Meas. result 4			Delete all

Press one of the function keys between **[F1]** (Meas. result 1) and **[F4]** (Meas. result 4) to display the measurement results 1 through 4, respectively.

- Delete all measurement results

The measurement results that are no longer needed can be deleted.

1. Press **[F7]** (Delete all) on the measurement result screen to delete all the data. A message is displayed on the screen to confirm the deletion.



2. Press **[F0]** (Yes) to delete the data, and press **[F1]** (No) to cancel the deletion.
3. Press the function key to return to the measurement result screen after deleting the data (or after canceling the deletion). Data cannot be deleted during operation.

(Note) All measurement results are deleted when the user parameter setting <Machine unit system> is changed.

4.5.5 G and M Code Lists

4.5.5.1 G and M Code Lists

The lists are displayed only in NC language mode.

Perform one of the following key operations on the production monitor menu screen to display the screen below:

- Press **[6]** and then **[ENT]**.
- Move the cursor to menu number 6 and press **[ENT]**.
- Press **[F6]** (GM code list).

Lists are displayed for the G codes, M codes and user defined M codes.

G code list

2016/10/07 11:13:49

G code list

00	Positioning	10	Reference rotary fixture offset setting
01	Linear interpolation	12	Circular cutting CW
02	Circular/helical interpolation CW	13	Circular cutting CCW
02	Spiral/conical interpolation CW	17	XY plane selection
03	Circular/helical interpolation CCW	18	ZX plane selection
03	Spiral/conical interpolation CCW	19	YZ plane selection
04	Dwell	22	Programmable stroke limit valid
09	Exact stop check	23	Programmable stroke limit invalid
10	Workpiece coord. zero setting	28	Reference position return
10	Workpiece coord. zero setting(meas.)	29	Return from reference position
10	Extended workpc. coord. zero setting	30	2nd to 6th reference position return
10	Extended workpc. coord. zero setting(meas.)	31	Skip function
10	Tool length offset setting	31	Continuous skip function
10	Cutter compensation setting	33	Equal lead thread cutting
10	Tool length wear offset setting	36	Coordinate calculation function(bolt hole)
10	Cutter wear offset setting	37	Coordinate calculation function(line(angle))
10	Tool information setting	38	Coordinate calculation function(line(X, Y))
10	Tool offset data setting	39	Coordinate calculation function(line(grid))
10	Tool offset wear data setting	40	Cutter/nose R compensation cancel

Some G codes cannot be used with some models or options

Production
monitor menu

G code list

M code list

User
defined
M code list

4.5.5.2 M Code List

This list is displayed only in NC language mode.

Perform one of the following key operations on the production monitor menu screen to display the screen below:

- Press **[6]** and then **[ENT]**.
- Move the cursor to menu number 6 and press **[ENT]**.
- Press **[F6]** (M code list).

Lists are displayed for the M codes and the user defined M codes.

4.5.5.3 User Defined M Code List

The signals and names are displayed for the <Miscellaneous function (1-20)> registered in the user parameter <User registered miscellaneous functions>.

[illegible]

4.5.6 Status Log

Perform one of the following key operations on the production monitor menu screen to display the screen below:

- Press **[7]** and then **[ENT]**.
- Move the cursor to menu number 7 and press **[ENT]**.
- Press **[F7]** (Status log).

Status log										2012/06/04 16:28:35	
Date		Total time									
06/04 (Today)		Off	10:10	Standby	04:21	Error	00:49	Operating	00:00	Stopped	01:05
06/03 (Yesterday)		Off	24:00	Standby	00:00	Error	00:00	Operating	00:00	Stopped	00:00
06/02 (2 days ago)		Off	24:00	Standby	00:00	Error	00:00	Operating	00:00	Stopped	00:00
06/01 (3 days ago)		Off	02:42	Standby	15:19	Error	03:47	Operating	00:02	Stopped	02:08
05/31 (4 days ago)		Off	00:00	Standby	22:31	Error	01:28	Operating	00:00	Stopped	00:00
05/30 (5 days ago)		Off	00:00	Standby	09:27	Error	12:09	Operating	01:34	Stopped	00:48
05/29 (6 days ago)		Off	00:00	Standby	20:40	Error	01:02	Operating	01:10	Stopped	01:05

To display details: Use the cursor to select the day and press [ENT].

Production monitor menu	Today	Yesterday	2 days ago	3 days ago	4 days ago	5 days ago	6 days ago
-------------------------	-------	-----------	------------	------------	------------	------------	------------

The status log is saved in 1 day units for a maximum of 7 days (may be less depending on the amount of data) of machine use.

Move the cursor to a specific day and press **[ENT]**, or press any key between **[F1]** (Today) and **[F7]** (6 days ago). A detailed log of the selected day is displayed.

Press the **[F1]** (Sort by status /Sort by time) key on the detailed log screen to switch the displayed order (Sort by status /Sort by time) of the data.

For the <Error No./Program No.>, the error number is displayed when an error is triggered, and the program number is displayed during operation. However, only the program number is displayed when the operating program is in the root folder. In all other situations, it is displayed in the “Folder name/Program No”.

Status log											
Date		06/04 (Today)						2012/06/04 16:29:12			
Status		Time		Error No./Program No.							
Power is off		00:00:00 - 10:10:25									
Standby mode		10:10:25 - 10:10:25									
Error occurred		10:10:25 - 10:42:44		NC0025 **							
Standby mode		10:42:44 - 10:57:43									
Error occurred		10:57:43 - 10:57:58		SM2020 *							
Standby mode		10:57:58 - 10:58:52									
Error occurred		10:58:52 - 10:59:10		SM2020 *							
Standby mode		10:59:10 - 11:07:03									
Error occurred		11:07:03 - 11:08:10		SM2055 *							
Standby mode		11:08:10 - 11:59:51									
Running		11:59:51 - 11:59:51		0604							
Stopped		11:59:51 - 11:59:53									
Running		11:59:53 - 11:59:53		0604							
Stopped		11:59:53 - 11:59:54									
Running		11:59:54 - 11:59:54		0604							
Stopped		11:59:54 - 11:59:56									
Running		11:59:56 - 11:59:56		0604							
Stopped		11:59:56 - 11:59:59									
Back	Sort by status										

In addition, it is possible to delete all the data for the status logs. Press the [>>] (Next function) key when the status log list is displayed. Then, press the [F1] (Delete data) key. Press the [F0] (Yes) key to delete the data for the status log, or the [F1] (No) key to cancel the deletion.



4.5.7 Other

This list is displayed only in conversation language mode. Perform one of the following key operations on the production monitor menu screen to display the screen below:

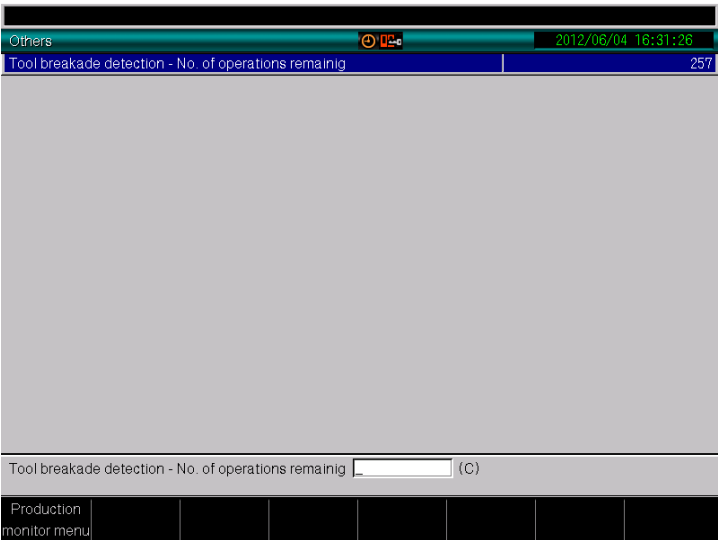
- Press [8] and then [ENT].
- Move the cursor to menu number 8 and press [ENT].
- Press [F8] (Other).

The number of memory operations remaining until the tool breakage detection is performed is set and displayed.

When “1” is displayed, the tool breakage detection is performed for the next operation. When the memory operation is completed, the setting for the “User parameter” <Tool breakage detection execution interval (No. of operations)> is applied.

The setting range is: 1 to 999.

The setting cannot be changed during memory operation.



4.5.8 Maintenance Notice

Perform one of the following key operations on the production monitor menu screen to display the screen below:

- Press **[8]** (**[9]** for conversation language mode) and then **[ENT]**.
- Move the cursor to menu number 8 (9 for conversation language mode) and press **[ENT]**.
- Use the **[>>]** (Function) key to change the displayed functions and then press the **[F1]** key (**[F2]** for conversation language mode) (Maintenance notice).

The maintenance notice function triggers an operator message to notify the operator if a consumable part etc. needs to be replaced when the tool change count, etc., exceeds the set value. The operator is notified via an operator message, a screen display alert and a notice from an external signal output.

No	Type (unit)	Message	Function	Current value	Target value	Notice	Status
1	Z-axis travel distance (m)	GREASING X YZ AXIS	Valid	0	100000	Notified	Normal
2	None						
3	None						
4	None						
5	None						
6	None						
7	None						
8	None						

Type []

0:None 5:4th-axis spd.10:No. of tool changes 15:Side door closed 20:P1 mov. amnt
 1:Spindle spd. 6:5th-axis spd.11:Magazine turn pitches 16:Inner door closed 21:P2 mov. amnt
 2:X-axis travel dist.7:6th-axis spd.12:Coolant ON period 17:Servo control ON frequency22:P3 mov. amnt
 3:Y-axis travel dist.8:7th-axis spd.13:Coolant ON frequency 18:Power ON frequency 23:P4 mov. Amnt
 4:Z-axis travel dist.9:8th-axis spd.14:Outer/front door closed19:Power ON period

Production monitor menu

[F0] Production monitor menu

Press this key to return to the production monitor menu screen.

“Type, message, function, current value, target value and notice” can be set on the above screen. A maximum of 40 items can be set.

The setting procedure is described below.

Type

A type can be selected from the items below. (The same type can be specified.)

Set value	Display (Unit)	Description
0	-	Nothing is displayed from <Message> to <Status>.
1	Spindle speed (×1000 rpm)	Displays the total number of spindle revolutions.
2	X-axis travel distance (m)	Displays the total travel distance for X-, Y- and Z-axes. (Inch: × 100 inches)
3	Y-axis travel distance (m)	
4	Z-axis travel distance (m)	
5	4th-axis speed (rotation)	Displays the total number of revolutions for the index (index device). (Note) The display unit is ×1000 rpm when the machine parameter <Optional *-axis> is set to < Equipped (lathe spindle)>.
6	5th-axis speed (rotation)	
7	6th-axis speed (rotation)	
8	7th-axis speed (rotation)	
9	8th-axis speed (rotation)	

Set value	Display (Unit)	Description
10	No. of tool changes (times)	Displays the total count for G100 and M6. (Note) • Tool changes in manual mode are not counted. • When there is a tool change from a standard to a large tool on an ATC machine model equipped with an arm, the arm rotates twice but the operation is only counted as once. • Changing to the same tool is not counted. • Tool changes during automatic centering in conversation language are counted. • Tool attachment and removal during MDI operation is counted in conversation language on ATC machine models equipped with a turret. • Tool changes are counted when setting the start point in conversation language or resuming an NC program.
11	Magazine turn pitches (pitch)	Displays the total number of pitches that the magazine has turned. (Note) • Magazine turns during JOG operation in maintenance mode are not counted. • Operations that turn the magazine are not counted when pressing the “MAG.F” and “MAG.R” keys while holding down the “RELSE” key in manual mode.
12	Coolant ON period (hhmmss)	Displays the total time period when the center-through-coolant is ON.
13	Coolant ON frequency (times)	Displays the total number of times that the center-through-coolant coolant is turned ON.
14	Outer/front door closed (times)	Displays the total number of times that the outer or front door is closed.
15	Side door closed (times)	Displays the total number of times that the side door is closed.
16	Inner door closed (times)	Displays the total number of times that the inner door is closed.
17	Servo relay control ON frequency (times)	Displays the total number of times that the servo relay control is changed from OFF to ON.
18	Power ON frequency (times)	Displays the total number of times that the power is turned ON.
19	Power ON period (hhmmss)	Displays the total time period while the power for the machine is ON.
20	P1-axis moving amount	Displays the total distance traveled for the PLC axis. (Note) • The travel amount refers to distance or rotations depending on the machine parameter setting <P1 (-P4)-axis type>. - When set to “Linear axis”, the travel amount refers to the total distance traveled (m). (Inch: × 100 inches) - When set to “Rotation axis”, the travel amount refers to the total number of rotations (rotation). • Even if the machine parameter settings <P1 (-P4)-axis type> are changed during counting, the counting continues but it uses the counting method that corresponds to the new axis type (i.e., after the type is changed).
21	P2-axis moving amount	
22	P3-axis moving amount	
23	P4-axis moving amount	

Message

The messages set here are used as operator messages.

A maximum of 20 characters can be entered using the characters available on the touch panel.

Function

The maintenance notice function is enabled or disabled.

When set to <0: Invalid>, the current value is not updated, and <Normal> is displayed in the <Status> column.

Current value

The current value is set.

Set the current value to "0" when enabling the maintenance notice function.

Even after the target value is reached (even if an operator message is triggered), the current value is updated.

Values exceeding the upper limit of the setting range are not counted.

Setting range

When the unit is in "hhmmss" format: 0 to 99999:59:59

Other unit format: 0 to 999999999 (9 digits)

Target value

The target value is set.

When the current value is equal or greater than the target value, an operator message is triggered but operation continues.

Setting range

When the unit is in "hhmmss" format: 0 to 99999:59:59

Other unit format: 0 to 999999999 (9 digits)

Notice (operator message)

This notice setting enables or disables the aforementioned operator message that is triggered when the current value is equal or greater than the target value.

Ex: When <GREASING XYZ AXIS> is set in the <Message> field for No.1, if the <Current value> exceeds the <Target value>, then the message <<Maintenance (GREASING XYZ AXIS)>> is displayed.

When the <Message> field is not set, the message <<Maintenance ()>> is displayed.

Status

<Normal>: Displays when the current value is less than the target value or when the maintenance notice function is disabled.

<Tgt val over>: Displays when the maintenance notice function is enabled and the current value is equal or greater than the target value.

Blank: Blank when the type is not set.

The external output signal MNTALM turns ON when the <Tgt val over> is displayed for any item.

4.6 Data Bank

NC language mode

The data bank screens are divided into the following categories.

- Workpiece coordinate zero
- Tool data
- Macro variables
- User parameter
- External I/O signal
- Communication parameter
- Machine parameter
- G/M code macro
- Special settings

Refer to the Operation Manual (Data) for further details.

Conversation language mode

The data bank screens are divided into the following categories.

- Tool list
- Tool pattern
- Tapping drill diameter
- Cutting conditions
- User parameter
- External I/O signal
- Communication parameter
- Machine parameter
- Special settings

Refer to the Operation Manual (Data) for further details.

4.7 Alarm Display

Refer to “Chapter 11 Alarms” for further details.

4.8 Input/Output

Refer to the Installation Manual for further details.

4.9 Graph

Refer to the Operation Manual (Data) for further details.

4.10 Display OFF

Press the **[DSP OFF]** key to turn off the screen display.

Press any key to turn on the screen display again.

When the “User parameter” <Automatic screen dimming time> is set, the screen automatically dims when no operation has been performed for a specified time.

4.11 Shortcuts

The shortcut function is used to register frequently-used screens beforehand in order to quickly call and access simple operations thereafter.

4.11.1 Basic Operations

This section provides a description of general operations used to register shortcuts and to call and access screens.

4.11.1.1 Data Protection

When the **[DATA PROTECTION]** switch is enabled, the data registered for shortcuts is protected in order to avoid accidentally changing the data.

Therefore, disable the **[DATA PROTECTION]** switch before registering or arranging the shortcuts.

4.11.1.2 Screen Configuration

The shortcut function is used on a popup screen.

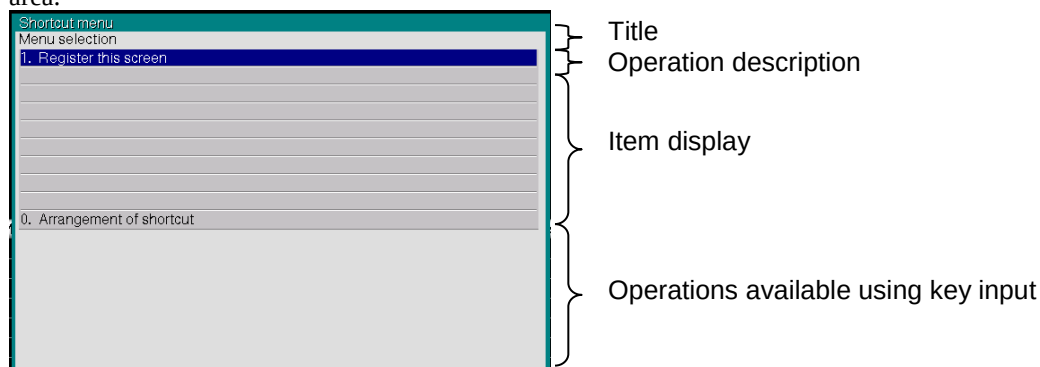
The screen is configured as follows:

1st line: Title

2nd line: Operation description

3rd line and thereafter: Display area (item selection, data input field, etc.)

In addition, operations that are available using key input are displayed on the bottom of the display area.



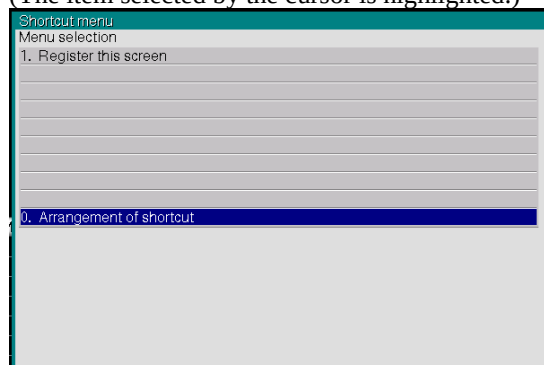
The information displayed on the screen under the popup menu is updated even while the popup is displayed.

4.11.1.3 Item Selection

Press the number and letter keys, or move the cursor using the **[CURSOR]** keys and press the **[ENT]** key to select items displayed from the popup menu.

For example, press the **[0]** key, or use the **[CURSOR]** keys to move the cursor to <0. Arrangement of shortcut> and press the **[ENT]** key in order to select <0. Arrangement of shortcut> from the popup menu below.

(The item selected by the cursor is highlighted.)



4.11.1.4 Cancel Operation

Press the shortcut **[MENU]** key, shortcut **[1]** key or shortcut **[2]** key to stop operation in the middle of an operation.

Operation can also be cancelled by pressing the **[RST]** key, but the operations being executed under the popup menu, such as memory operation and communication, are also reset.

4.11.1.5 Special Notes

- (Note 1)** Operations (such as entering values in the data input field, moving the cursor or selecting functions, etc.) cannot be performed on screens displayed under the popup menu. For example, when the [F1] key is pressed while the popup menu is displayed over the production monitor menu screen, the workpiece counter screen does not display.
- (Note 2)** The shortcut operation is cancelled if an operation is performed such as changing to another screen under the popup menu (screen selection using the [MONITOR], [ALARM] keys, etc. or mode selection using the [MANU] or [MEM] keys).
- (Note 3)** When turning on the screen again, if the screen under the popup menu is different from the one before the screen was turned off, the shortcut operation is cancelled.

4.11.2 Register Shortcuts

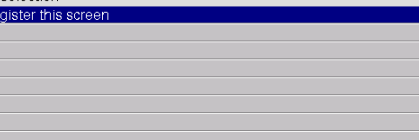
A maximum of 10 shortcuts can be registered and assigned to the shortcut **[1]** and **[2]** keys.

4.11.2.1 Register Screens

1. Display the screen to be registered and assigned.

Tool list						2012/06/04 16:38:14	
Tool No	01 (P:02)	02 (P:03)	03 (P:04)	04 (P:05)	05 (P:06)		
Tool name	TAP M6 P1.0	TOOL2	TOOL3	TOOL4	TOOL5		
Tool length offset	50.000	10.000	0.000	0.000	0.000		
T length wear offset	0.000	0.000	0.000	0.000	0.000		
Cutter comp.	10.000	5.000	0.000	0.000	0.000		
Cutter wear offset	0.000	0.000	0.000	0.000	0.000		
Initial life	*****	1000 (H)	*****				
Life warning	*****	100 (H)	*****				
Tool life	*****	1000 (H)	*****				
Tool No	06	07	08	09	10		
Tool name							
Tool length offset	0.000	0.000	0.000	0.000	0.000		
T length wear offset	0.000	0.000	0.000	0.000	0.000		
Cutter comp.	0.000	0.000	0.000	0.000	0.000		
Cutter wear offset	0.000	0.000	0.000	0.000	0.000		
Initial life							
Life warning							
Tool life							
Tool name							

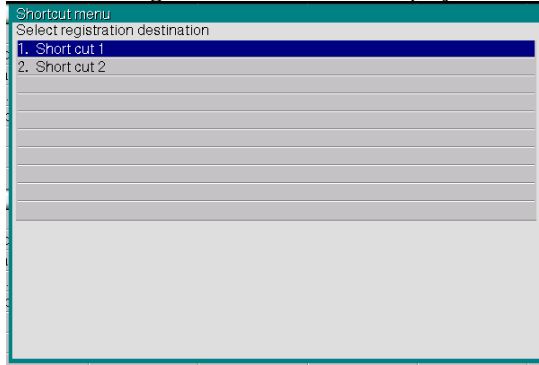
2. Press the shortcut **[MENU]** key to display the shortcut menu.



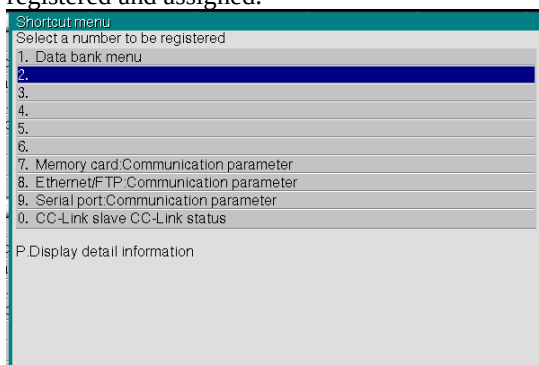
The screenshot shows a presentation slide with a blue header bar containing the text "Shortcut menu". Below the header, the text "Menu selection" is displayed. A list of menu items is shown, with "1. Register this screen" highlighted in blue. Below this, there are several empty menu items. At the bottom, the text "0. Arrangement of shortcut" is visible.

Menu selection
1. Register this screen
0. Arrangement of shortcut

3. Select <1. Register this screen> to display the following screen.

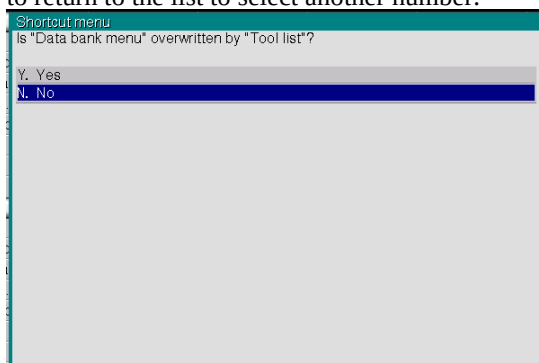


4. Select <1. Short cut 1> to register and assign to the shortcut **[1]** key. Select <2. Short cut 2> to register and assign to the shortcut **[2]** key.
5. When a list of the shortcuts currently registered is displayed, select the number to be registered and assigned.



If a name exceeds 64 half-width characters (32 full-width characters), an ellipsis “...” is displayed at the end of the character string. Move the cursor to the **[P]** (Display detail information) key and press it to see more information. Refer to "4.11.3 Call shortcuts" for further details.

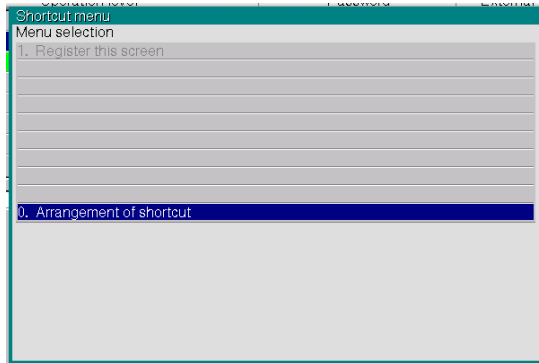
6. When a number is selected that is already registered and assigned to another screen, a confirmation message is displayed. Select <Y. Yes> to overwrite the data, or select <N. No> to return to the list to select another number.



7. The popup menu disappears.

4.11.2.2 Screens that Cannot be Registered

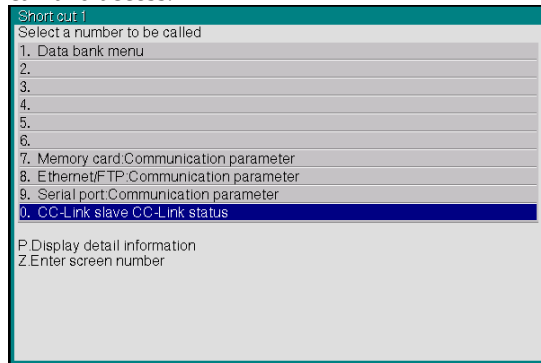
When the shortcut [MENU] key is pressed on a screen that cannot be registered, <1. Register this screen> cannot be selected. Refer to “4.11.5 Screen list” for screens that can be registered.



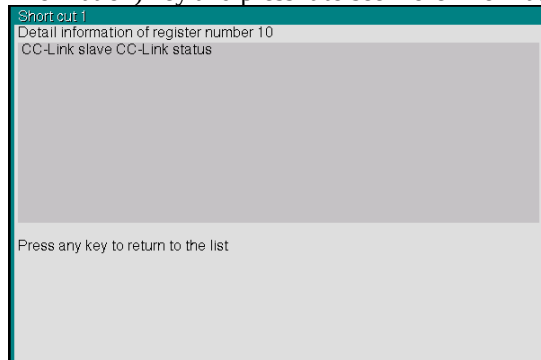
4.11.3 Call Shortcuts

4.11.3.1 When Calling Screens that are Registered

1. Press shortcut key [1] or [2] for the registered shortcut that you wish to call and access.
2. When a list of the shortcuts currently registered is displayed, select the screen that you wish to call and access.



If a name exceeds 64 half-width characters (32 full-width characters), an ellipsis “...” is displayed at the end of the character string. Move the cursor to the [P] (Display detail information) key and press it to see more information.

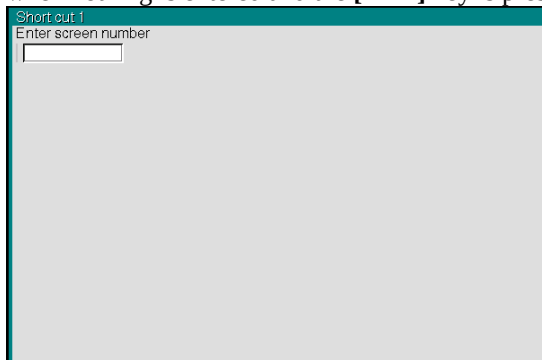


Press a number or letter key on the detail information screen to return to the list.

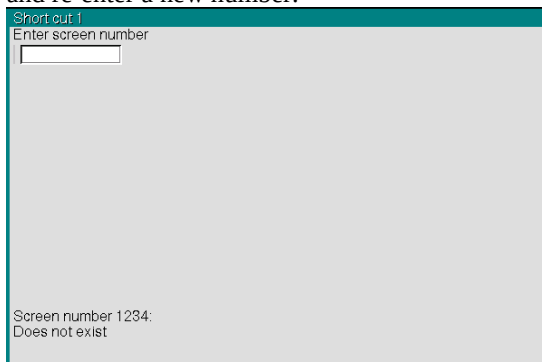
4.11.3.2 When Calling a Screen by Specifying the Screen Number

Each screen is assigned a unique screen number. Enter that number to call and access that screen. Refer to “4.11.5 Screen list” for further information about screen numbers.

1. Press shortcut key **[1]** or **[2]**.
2. When a list of the shortcuts currently registered is displayed, press the **[Z]** key (Enter screen number).
3. Use the keyboard to enter a screen number, and then press the **[ENT]** key. (Proceeds to step 4 when nothing is entered and the **[ENT]** key is pressed.)



4. If the corresponding screen exists, that screen is displayed. If it does not exist, the message “Screen number xxxx does not exist” (“xxxx” refers to the screen number entered) is displayed at the bottom of the popup menu. Use the **[CAN]** key to delete the invalid number and re-enter a new number.

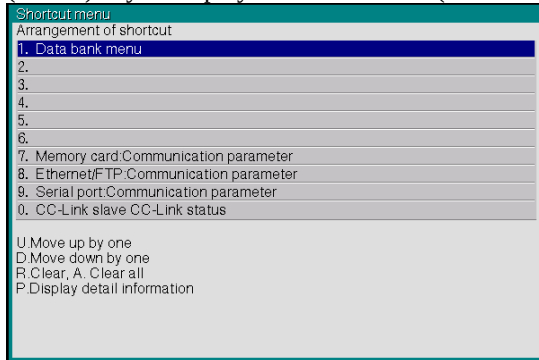


4.11.3.3 Call Special Screens

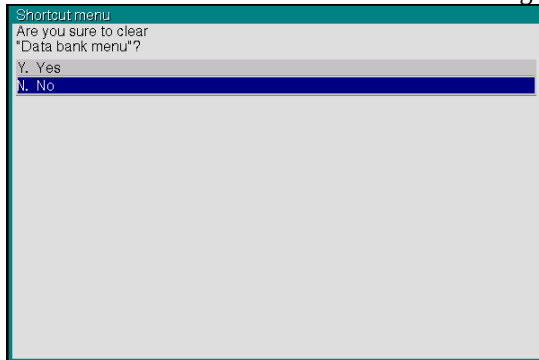
After calling the position import screen for tool data (automatic setting screen), the position import screen for the spindle tool is displayed, not the position import screen that was registered.

4.11.4 Arrange Shortcuts

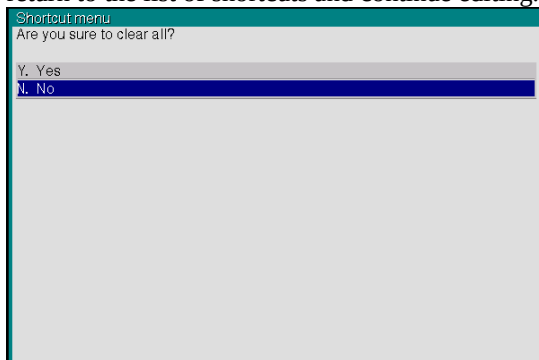
1. Press the shortcut **[MENU]** key to display the shortcut menu.
2. Select <0. Arrangement of shortcut>.
3. Select <1. Short cut 1> to arrange and make changes to the shortcut **[1]** key. Select <2. Short cut 2> to arrange and make changes to the shortcut **[2]** key.
4. When the list of registered shortcuts is displayed, move the cursor to the item you wish to change. When the 64 half-width character limit is exceeded, move the cursor and press the **[P]** (details) key to display the continuation (or next screen).



5. Use the **[U]**, **[D]**, **[R]** and **[A]** keys to edit the list. (**[U]**: Move up by one, **[D]**: Move down by one, **[R]**: Clear and **[A]**: Clear all)
6. When clear is selected in step “5”, a confirmation message is displayed. If you wish to proceed and clear the data, select <Y. Yes>. If not, select <N. No>. In either case, you can return to the list of shortcuts and continue editing.



7. When clear all is selected in step “5”, a confirmation message is displayed. If you wish to proceed and clear all data, select <Y. Yes>. If not, select <N. No>. In either case, you can return to the list of shortcuts and continue editing.



4.11.5 Screen List

The screen numbers and screen names used for shortcuts and the shortcut restrictions are shown in the following table. Note that when the following notes are written in the notes column, the corresponding restriction applies.

Notes	Restriction
Conversation language mode only	The screen only changes in conversation language mode.
NC language mode only	The screen only changes in NC language mode.
Conversation/NC language mode	The screen change is different depending on the language mode.
Mode specific	This screen is specific to manual, MDI operation, memory operation and edit modes. If the mode is different, the shortcut is disabled.
Screw type	The screen does not change when the screw type is different.

4.11.5.1 Position

There are no restrictions based on the mode or based the status of the position screen before changing screens.

No.	Shortcut name	Notes
001	Current position	
002	Modal info 1	NC language mode only
003	Modal info 2	NC language mode only
004	Soft limit	
005	Enlarged display	
006	Workpiece coordinate zero: Position	NC language mode only
007	Counter reset	
008	Modal info	Conversation language mode only

4.11.5.2 ATC Tools

There are no restrictions based on the mode or based the status of the ATC tool screen before changing screens.

No.	Shortcut name	Notes
050	ATC tool (Details)	
051	ATC tool (List)	
052	Initialize life: ATC tool	The screen changes only when the tool life has expired. Otherwise, the ATC tool (details) screen or ATC tool (list) screen will be displayed.

4.11.5.3 Input/Output

There are no restrictions based on the mode.

On the <PLC> screen for input and output, the user cannot change to another input and output screen from any of the following screens (only from the <PLC I/O> screen).

In addition, the user also cannot change to another input and output screen from any sub screens below.

- <PLC> screen
- <Parameter setting> screen
- <Power failure retention/write range setting> screen
- <Conversion mode setting> screen

No.	Shortcut name	Notes
100	I/O menu	
101	I/O (Main)	
102	I/O (Local)	
103	I/O (Slave status display)	
104	I/O (Extension I/O)	
105	I/O (Fieldbus)	
106	I/O (Version)	
107	I/O servo (Spindle)	
108	I/O servo (X-axis)	
109	I/O servo (Y-axis)	
110	I/O servo (Z-axis)	

No.	Shortcut name	Notes
111	I/O servo (4th-axis)	
112	I/O servo (5th-axis)	
113	I/O servo (6th-axis)	
114	I/O servo (7th-axis)	
115	I/O servo (8th-axis)	
116	I/O servo (Magazine)	
117	I/O servo (P1-axis)	
118	I/O servo (P2-axis)	
119	I/O servo (P3-axis)	
120	I/O servo (P4-axis)	
121	External I/O signal (Input 1)	
122	External I/O signal (Input 2)	
123	External I/O signal (Input 3)	
124	External I/O signal (Output 1)	
125	External I/O signal (Output 2)	
126	External I/O signal (Output 3)	
127	Soft switch	
128	Machine diagnostics	
129	PLC	
130	PLC I/O	
131	Global OM (External input/output X/Y)	
132	Global OM (Internal relay M)	
133	Global OM (Data register D)	
134	Global OM (NC input/output W BX/BY)	
135	Global OM (NC input/output BDX/BDY)	
136	Global OM (Status)	
137	High speed ladder 1 (Local relay LM)	
138	High speed ladder 1 (Local data register LD)	
139	High speed ladder 1 (Local timer LT)	
140	High speed ladder 1 (Local counter LC)	
141	High speed ladder 1 (Special relay SM)	
142	High speed ladder 1 (Special register SD)	
143	Standard ladder 1 (Local relay LM)	
144	Standard ladder 1 (Local data register LD)	
145	Standard ladder 1 (Local timer LT)	
146	Standard ladder 1 (Local counter LC)	
147	Standard ladder 1 (Special relay SM)	
148	Standard ladder 1 (Special register SD)	
149	High speed ladder 2 (Local relay LM)	
150	High speed ladder 2 (Local data register LD)	
151	High speed ladder 2 (Local timer LT)	
152	High speed ladder 2 (Local counter LC)	
153	High speed ladder 2 (Special relay SM)	
154	High speed ladder 2 (Special register SD)	
155	Standard ladder 2 (Local relay LM)	
156	Standard ladder 2 (Local data register LD)	
157	Standard ladder 2 (Local timer LT)	
158	Standard ladder 2 (Local counter LC)	
159	Standard ladder 2 (Special relay SM)	
160	Standard ladder 2 (Special register SD)	
161	Registration monitor: PLC	
162	Running counters	
163	Operation log, Today	
164	Operation log, Previous day	
165	Operation log, 2 days ago	
166	Operation log, 3 days ago	
167	Operation log, 4 days ago	
168	Operation log, 5 days ago	
169	Operation log, 6 days ago	

No.	Shortcut name	Notes
170	Current waveform	
171	Current waveform (Set)	
172	Current waveform (Save)	Changing screens is possible only when the function is valid (when the function is invalid, the screen changes to <Current waveform>).
173	Current waveform (Zoom)	Changing screens is possible only when the function is valid (when the function is invalid, the screen changes to <Current waveform>).
175	Saved data waveform	Changing screens is possible only when the function is valid (when the function is invalid, the screen changes to <Current waveform>).
176	Saved data waveform (List)	Changing screens is possible only when the function is valid (when the function is invalid, the screen changes to <Current waveform>).
177	Saved data waveform (Save)	Changing screens is possible only when the function is valid (when the function is invalid, the screen changes to <Current waveform>).
178	Saved data waveform (Zoom)	Changing screens is possible only when the function is valid (when the function is invalid, the screen changes to <Current waveform>).
180	Operation level	
181	I/O (Communication error counter)	
182	Request code	
183	I/O (Saved value for comm. error counter)	
184	Current waveform (Save/Log)	Changing screens is possible only when the function is valid (when the function is invalid, the screen changes to <Current waveform>).
185	Current waveform (Save/Log) file name change	Changing screens is possible only when the screen can be changed to <Current waveform (Save/Log)> (when not possible, the screen changes to <Current waveform>).
186	Input/output (Loading system)	
187	I/O servo (AT-axis)	

4.11.5.4 Data Bank

There are no restrictions based on the mode.

If a data bank file is being edited on the data bank screen, changing to an editable screen is not possible.

If the communication parameter is displayed in edit mode, changing screens to the communication parameter is not possible.

Changing screens is not possible if a data bank file for that screen is being transferred.

No.	Shortcut name	Notes
200	Data bank menu	
201	Workpiece coordinate zero	NC language mode only
202	Workpiece coordinate zero (Extended)	NC language mode only
203	Tool data menu	
204	Tool list	
206	Tool list (Extension information)	Conversation language mode only
207	Tool list (Extension 1)	NC language mode only
208	Tool list (Extension 2)	NC language mode only Only machine models (specifications) with this function

No.	Shortcut name	Notes
209	Automatic setting (Tool length offset)	NC language mode only Changes to the automatic setting screen for the spindle tool or the indexing magazine tool.
210	Automatic setting (S/F command value)	NC language mode only Changes to the automatic setting screen for the spindle tool or the indexing magazine tool.
211	Tool group	NC language mode only
212	Tool data range (Tool length)	NC language mode only
213	Tool data range (Tool diameter)	NC language mode only
214	Tool data range (Tool position X)	NC language mode only Only machine models (specifications) with this function
215	Tool data range (Tool position Y)	NC language mode only Only machine models (specifications) with this function
216	Tool length setting	Conversation language mode only
217	Tool pattern: Data bank	Conversation language mode only
218	Metric coarse thread: Tapping drill diameter	Conversation language mode only
219	Metric fine pitch thread: Tapping drill diameter	Conversation language mode only
220	Unified coarse thread: Tapping drill diameter	Conversation language mode only
221	Unified fine pitch thread: Tapping drill diameter	Conversation language mode only
222	Sewing machine thread: Tapping drill diameter	Thread type, conversation language mode only
223	Miniature screw thread: Tapping drill diameter	Thread type, conversation language mode only
224	Pipe thread PS: Tapping drill diameter	Thread type, conversation language mode only
225	Pipe thread PT: Tapping drill diameter	Thread type, conversation language mode only
226	Pipe thread PF: Tapping drill diameter	Thread type, conversation language mode only
227	NPS: Tapping drill diameter	Thread type, conversation language mode only
228	BSP: Tapping drill diameter	Thread type, conversation language mode only
229	BSW: Tapping drill diameter	Thread type, conversation language mode only
230	BSF: Tapping drill diameter	Thread type, conversation language mode only
231	BA: Tapping drill diameter	Thread type, conversation language mode only
232	NPT: Tapping drill diameter	Thread type, conversation language mode only
233	Material menu: Cutting conditions	Conversation language mode only
234	S45C: Cutting condition 1 (Drill)	Conversation language mode only
235	S45C: Cutting condition 2 (Cutter)	Conversation language mode only
236	S45C: Cutting condition 3 (w/o tools)	Conversation language mode only
237	SCM435: Cutting condition 1 (Drill)	Conversation language mode only
238	SCM435: Cutting condition 2 (Cutter)	Conversation language mode only
239	SCM435: Cutting condition 3 (w/o tools)	Conversation language mode only
240	FC20: Cutting condition 1 (Drill)	Conversation language mode only
241	FC20: Cutting condition 2 (Cutter)	Conversation language mode only
242	FC20: Cutting condition 3 (w/o tools)	Conversation language mode only
243	ADC12: Cutting condition 1 (Drill)	Conversation language mode only
244	ADC12: Cutting condition 2 (Cutter)	Conversation language mode only
245	ADC12: Cutting condition 3 (w/o tools)	Conversation language mode only
246	SUS304: Cutting condition 1 (Drill)	Conversation language mode only
247	SUS304: Cutting condition 2 (Cutter)	Conversation language mode only

No.	Shortcut name	Notes
248	SUS304: Cutting condition 3 (w/o tools)	Conversation language mode only
249	Special material 1: Cutting condition 1 (Drill)	Conversation language mode only
250	Special material 1: Cutting condition 2 (Cutter)	Conversation language mode only
251	Special material 1: Cutting condition 3 (w/o tools)	Conversation language mode only
252	Special material 2: Cutting condition 1 (Drill)	Conversation language mode only
253	Special material 2: Cutting condition 2 (Cutter)	Conversation language mode only
254	Special material 2: Cutting condition 3 (w/o tools)	Conversation language mode only
255	Special material 3: Cutting condition 1 (Drill)	Conversation language mode only
256	Special material 3: Cutting condition 2 (Cutter)	Conversation language mode only
257	Special material 3: Cutting condition 3 (w/o tools)	Conversation language mode only
258	Special material 4: Cutting condition 1 (Drill)	Conversation language mode only
259	Special material 4: Cutting condition 2 (Cutter)	Conversation language mode only
260	Special material 4: Cutting condition 3 (w/o tools)	Conversation language mode only
261	Special material 5: Cutting condition 1 (Drill)	Conversation language mode only
262	Special material 5: Cutting condition 2 (Cutter)	Conversation language mode only
263	Special material 5: Cutting condition 3 (w/o tools)	Conversation language mode only
264	Macro variable number 100s	NC language mode only
265	Macro variable number 500s	NC language mode only
266	Macro variable number 600s	NC language mode only
267	Macro variable number 700s	NC language mode only
268	Macro variable number 800s	NC language mode only
269	Macro variable number 900s	NC language mode only
270	Local variable: Macro variables	NC language mode only
271	Switch 1: User parameter	
272	Switch 2: User parameter	
273	Switch 3: User parameter	
274	Reference position/Conversation zero point: User parameter	
275	User registered miscellaneous functions: User parameter	
276	Tool breakage detection: User parameter	
277	Automatic workpiece measurement: User parameter	
278	Automatic thermal distortion compensation: User parameter	
279	Quick table: User parameter	
280	High accuracy A: User parameter	
281	High accuracy B: User parameter	
282	High accuracy common: User parameter	
283	PLC: User parameter	
284	PROFIBUS DP: User parameter	
285	DeviceNet: User parameter	
286	CC-Link slave: User parameter	
287	External input signal	
288	External output signal	
289	External alarm	
290	PROFIBUS DP slave PROFIBUS DP -> NC	
291	PROFIBUS DP slave NC -> PROFIBUS DP	
292	PROFIBUS DP slave PROFIBUS DP status	
293	DeviceNet slave DeviceNet -> NC	
294	DeviceNet slave NC -> DeviceNet	
295	DeviceNet slave DeviceNet status	
296	CC-Link slave RY/RWw (CC-Link -> NC)	
297	CC-Link slave RX/RWr (NC -> CC-Link)	
298	CC-Link slave CC-Link status	
299	External panel 1	
300	External panel 2	
301	Serial port: Communication parameter	
302	Ethernet/FTP: Communication parameter	

No.	Shortcut name	Notes
303	Memory card: Communication parameter	
304	Rotary fixture offset: User parameter	NC language mode only
305	Reference rotary fixture offset	NC language mode only
306	Common parameter: G/M code macro	NC language mode only
307	G code macro: G/M code macro	NC language mode only
308	M code macro: G/M code macro	NC language mode only
309	Special setting selection: Special settings	
310	Rapid feed adjustment: Special settings	
311	Rapid feed adjustment - Linear interpolation type: Special settings	
312	3D smoothing (M380-383): User parameter	NC language mode only
313	3D smoothing (M384-387): User parameter	NC language mode only
314	3D smoothing (common): User parameter	NC language mode only
315	High accuracy (Adjustment): User parameter	Only machine models (specifications) with this function
316	High accuracy (Accuracy specification): User parameter	Only machine models (specifications) with this function
317	Auto notification function: Communication parameter	
318	Coordinate setting for tilt plane: User parameter	NC language mode only
324	Parameter (List): Machining load monitor	
319	CC-Link master: User parameter	Used for master station and intelligent device stations
320	CC-Link master CC-Link configuration	Used for master station and intelligent device stations
321	CC-Link master CC-Link → NC (master/ID)	Used for master station and intelligent device stations
322	CC-Link master NC (master/ID) → CC-Link	Used for master station and intelligent device stations
323	CC-Link master CC-Link status	Used for master station and intelligent device stations
325	Common parameter:Machining load monitor	

4.11.5.5 Alarms

There are no restrictions based on the mode. There are no restrictions based on the status of the alarm screen before changing screens.

No.	Shortcut name	Notes
340	Current alarm/Operator message	
341	Alarm log (List)	
342	List of alarms/operator messages	
343	Collision log (List)	
344	Loading system alarm	

4.11.5.6 Production Monitor

There are no restrictions based on the mode. There are no restrictions based on the status of the production monitor screen before changing screens.

No.	Shortcut name	Notes
350	Production monitor menu	
351	Workpiece counter	
352	Time display	
353	Operation time log	
354	Communication monitor	
355	Measurement result 1	
356	Measurement result 2	
357	Measurement result 3	
358	Measurement result 4	
359	Automatic thermal distortion compensation (X-axis)	

No.	Shortcut name	Notes
360	Automatic thermal distortion compensation (Y-axis)	
361	Automatic thermal distortion compensation (Z-axis)	
362	G code list	NC language mode only
363	M code list	
364	User defined M code list	
365	Status log (List)	
366	Status log, Today	
367	Status log, Previous day	
368	Status log, 2 days ago	
369	Status log, 3 days ago	
370	Status log, 4 days ago	
371	Status log, 5 days ago	
372	Status log, 6 days ago	
373	Other	Conversation language mode only
374	Maintenance notice	
375	Ethernet details (NC)	

4.11.5.7 Graph

There are no restrictions based on the mode. There are no restrictions based on the status of the graph screen before changing screens.

No.	Shortcut name	Notes
400	Operation graph	
401	Operation graph (Zoom)	
402	Operation graph (Position setting (center))	
403	Operation graph (Position setting (opposite corner))	
405	Operation graph (Graph setting)	
406	Operation graph (Workpiece coordinate zero)	NC language mode only
407	Operation graph (Extended workpiece coordinate zero)	NC language mode only
408	Operation graph (Tool data)	NC language mode only
409	Operation graph (Macro variables)	NC language mode only
410	Tool path simulation	
411	Tool path simulation (Zoom)	
412	Tool path simulation (Position setting (center))	
413	Tool path simulation (Position setting (opposite corner))	
415	Tool path simulation (Graph setting)	
416	Tool path simulation (Workpiece coordinate zero)	NC language mode only
417	Tool path simulation (Extended workpiece coordinate zero)	NC language mode only
418	Tool path simulation (Tool data)	NC language mode only
419	Tool path simulation (Macro variables)	NC language mode only
420	Program list: Tool path simulation	
421	Detailed program list: Tool path simulation	
422	Operation graph (Tool data: Extension 1)	NC language mode only
423	Operation graph (Tool data: Extension 2)	NC language mode only
424	Tool path simulation (Tool data: Extension 1)	NC language mode only
425	Tool path simulation (Tool data: Extension 2)	NC language mode only
426	Operation graph (Reference rotary fixture offset)	NC language mode only
427	Tool path simulation (Rotary fixture)	NC language mode only

4.11.5.8 Manual Conditions

The screen can only be changed to the manual conditions screen in MDI operation, memory operation and edit mode.

In addition, the screen is given priority and displayed, such as when recovery operation is enabled or when a recovery operation is required.

No.	Shortcut name	Notes
450	Manual conditions	
452	Recovery/Inspection menu	
453	Synchronized tap return	
454	ATC arm position recovery	Only machine models with this function
455	Tool clamp failure recovery	Only machine models with this function
456	Spindle test run	Only machine models (specifications) with this function
457	To measure motor insulation resistance	
458	Motor insulation resistance measurement history	
459	Expired battery recovery	
460	Reserved	

4.11.5.9 Programs

1. MDI operation

Changing screens is only possible in MDI operation mode.

During operation, the screen changes to one that is operating.

No.	Shortcut name	Notes
500	MDI operation	Changing screens is not possible when in MDI operation mode.
501	Automatic centering (Corner)	
502	Automatic centering (Parallel)	
503	Automatic centering (Circle (hole))	
504	Automatic centering (Circle (boss))	
505	Automatic centering (Z-axis height)	
506	Automatic centering (Probe compensation (corner))	
507	Automatic centering (Probe compensation (3-point circle))	

2. Memory operation

Changing screens is only possible in memory operation mode.

During operation, the screen changes to memory operation screen.

No.	Shortcut name	Notes
600	Memory operation	
601	Program list: Memory operation	Screen changes to the memory operation screen when using tape operation (general communications device).
602	Detailed program list: Memory operation	Screen changes to the memory operation screen when using tape operation (general communications device).
603	Program restart	NC language mode only Screen changes to the memory operation screen when the function is invalid.
604	Sequence search	NC language mode only Screen changes to the program restart (NC) screen when the function is invalid.
605	Start point setting	Conversation language mode only

3. Edit

Changing screens is only possible in program edit mode (including editing during operation). Changing screens to another program screen is not possible while the program is being edited.

Changing screens is not possible when a file is being transferred for that screen.

Changing screens is not possible when a maintenance menu such as a screen below is displayed.

Changing to a screen that can be edited in another program screen is not possible while the program is being edited.

Changing to the screen is not possible when a program in that screen is communicating .

No.	Shortcut name	Notes
700	Program edit menu	
701	Program list: Program edit	NC language mode only
702	Create program list folder: Program edit	NC language mode only Changes to the program list screen when in the sub directory.
703	Edit: Program	NC language mode only
704	Program list: Machining data	Conversation language mode only
705	Create program list folder creation: Machining data	Conversation language mode only Changes to the program list screen when in the sub directory.
706	Machining data	Conversation language mode only
707	Program list: Machining sequence	Conversation language mode only
708	Machining sequence: Machining data	Conversation language mode only
709	Program list: Tool pattern	Conversation language mode only
710	Tool pattern: Machining data	Conversation language mode only
711	Program list: Cutting condition	Conversation language mode only
712	Cutting condition 1 (Drill): Machining data	Conversation language mode only
713	Cutting condition 1(Drill): Machining data	Conversation language mode only
714	Cutting condition 2 (w/o tools): Machining data	Conversation language mode only
715	Program list: Operation preparation	Conversation language mode only
716	Operation preparation menu: Machining data	Conversation language mode only
717	Tool assignment: Machining data	Conversation language mode only
718	Cutting position compensation: Machining data	Conversation language mode only
719	Workpiece height compensation: Machining data	Conversation language mode only
720	Tool change position: Machining data	Conversation language mode only
721	Directory display, Program	
722	Directory display, Operate between folders	
723	Directory display, G code conversion	Conversation language mode only
724	Directory display, Data bank	
725	External I/O menu	
726	Program (External I/O)	
727	Change connection device	
728	Parameter: External I/O-Program	Changes to a screen specified in the current directory.
729	Folder operation (External I/O program)	
730	Data bank (External I/O)	
731	Parameter: External I/O-Data bank	Changes to a screen specified in the current directory.
732	Folder operation (External I/O) data bank	
733	Maintenance menu	
734	Preparations menu	NC language mode only
735	Program list (3D smoothing)	NC language mode only
736	3D smoothing preparations	NC language mode only When there is a program that has not been checked, it changes to the <3D smoothing program check> screen.

4.12 Screen Functions

4.12.1 Screenshot Function

4.12.1.1 Enable conditions

When the user parameter [switch 1] <Screenshot> is set to <1: Enable> and the[SHIFT] and [RELSE] keys are pressed, a screenshot of the current screen display is saved in a PNG image format to the memory card that is connected.

After the screenshot is saved, the operator message <<Screenshot was taken.>> appears.

When the parameter is set to <0: Disable>, a screenshot cannot be taken even if the[SHIFT] and [RELSE] keys are pressed.

4.12.1.2 Location for saving image

A folder is created in the memory card that is connected to save the screenshot image.

A folder is automatically created under the folder name “yyyymmdd (date created)” inside the “CAPTURE” folder. (Even if there is no “CAPTURE” folder on the memory card, it is automatically created.)

The file name to be saved follows the format: “yyyymmdd_****” (date created + a sequential number between 0000 and 9999).

The sequential number in the file name increases sequentially and uses the next largest number when saving. If a “yyyymmdd_9999” file already exists, the alarm <<External output file cannot be created.>> is triggered and the screenshot cannot be saved.

4.12.1.3 Restrictions

The screenshot function cannot be used when the following applies. An alarm such as <<Memory card error>> is triggered or the operator message appears depending on the situation.

- When the memory card is not connected.
- When performing an input/output all data transfer, when POPEN is processing in memory operation, or when communicating with memory card.
- When tape operation is executing in the memory card.
- When the keys are locked.
- When the NC is starting up or when a normal startup in the NC is not complete.

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CHAPTER 5

MANUAL OPERATION

5

- 5.1 Manual Operation**
- 5.2 Machine Zero Return**
- 5.3 Jog & Step Feed**
- 5.4 Spindle Selection**
- 5.5 Spindle Rotation**
- 5.6 Single Tool Change (ATC)**
- 5.7 Magazine Turn**
- 5.8 Column Travel During Tool Change**
- 5.9 Recovery & Inspection Operations**
- 5.10 Operational Error Prevention Function**
- 5.11 Details on Recovery & Inspection Operations**

5.1 Manual Operation

Press the **[MANU]** key to switch to manual operation mode and to perform the following operations.

- Machine zero return
- Jog and step feed
- Spindle rotation (clockwise/stop)
- Single tool change (ATC)
- Magazine forward/reverse rotation
- Pallet turn
- Operation using the handle
- Recovery and inspection operations
- Spindle selection

5.1.1 Manual Conditions Screen

After the power is turned ON, the following screen is first displayed when the user parameter <Select mode> is set to manual mode or when changing to manual operation mode.



When one of the following keys is pressed: **[RPD]**, **[JOG]**, **[STEP]**, **[SP.CW]** or **[SP.STOP]**, the cursor moves to the corresponding item (cursor moves to "Spindle speed" when one of the following keys is pressed: **[SP.CW]**, **[SP.STOP]**, and **[L.SP]**). In addition, press the up or down **[CURSOR]** key once to move the cursor to the item above or below, respectively.

Press the **[F7]** key (Recover/Inspection Menu) to display the <Recover / Inspection menu> screen.

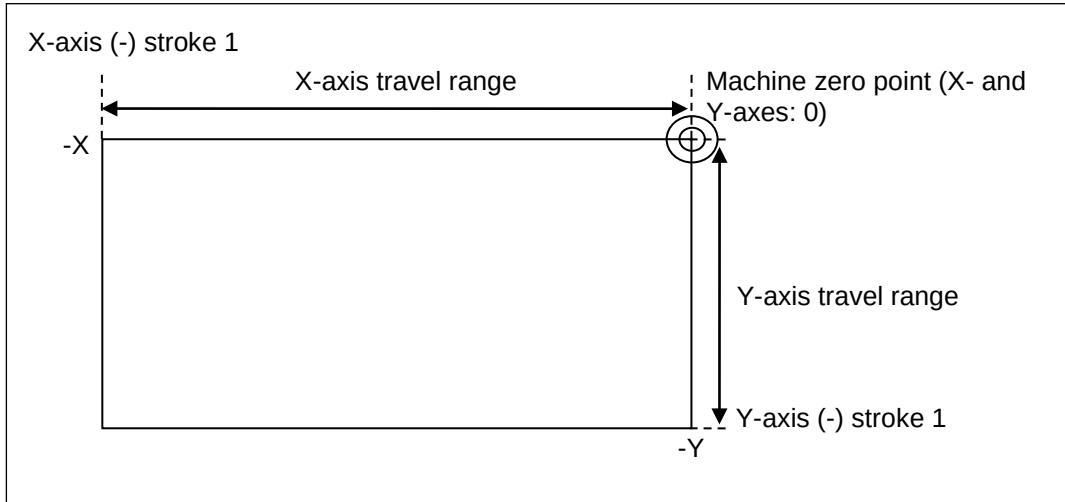
Press the **[F5]** (Change additional axes display) key to change the function display between the user parameters <4th-8th row display axis> and the <9th-12th row display axis>. In addition, this display change also changes the coordinate display, such as the machine coordinates, etc. However, refer to the PLC System Manual for further details on the PLC-axes.

- (Note 1) When all the machine parameters <Optional 4th - 8th-axis> and <Optional P1 - P4-axis> are set to <Not equipped>, the cursor does not stop at the item for the 4th-axis.
- (Note 2) When all the axis options in the user parameters <9th row display axis> through <12th row display axis> are set to <Not set>, the **[F5]** (Change additional axes display) key is grayed out and disabled.
- (Note 3) Use the machine parameter settings <4th through 8th-axis address> and <5th through 8th-axis position> to assign the 4th through the 8th-axes from the following options: A1-axis, A2-axis, B1-axis, B2-axis, C1-axis and C2-axis. The display order for the additional axes on the <Manual conditions> screen changes according to these settings. (In the screenshot shown above, the settings are configured as follows→
4th-axis: QT-axis, 5th-axis: A1-axis, 6th-axis: B1-axis, 7th-axis: A2-axis, 8th-axis: B2-axis.)

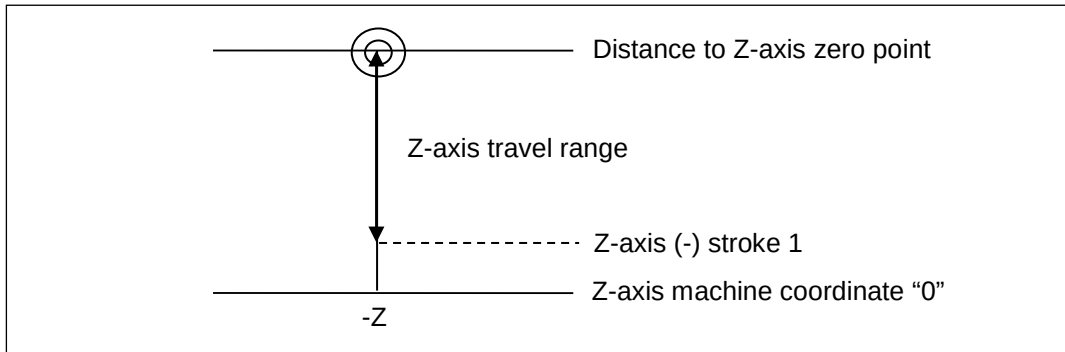
5.1.2 Axis Travel and Stroke

The axis travel range in manual operation is shown below.

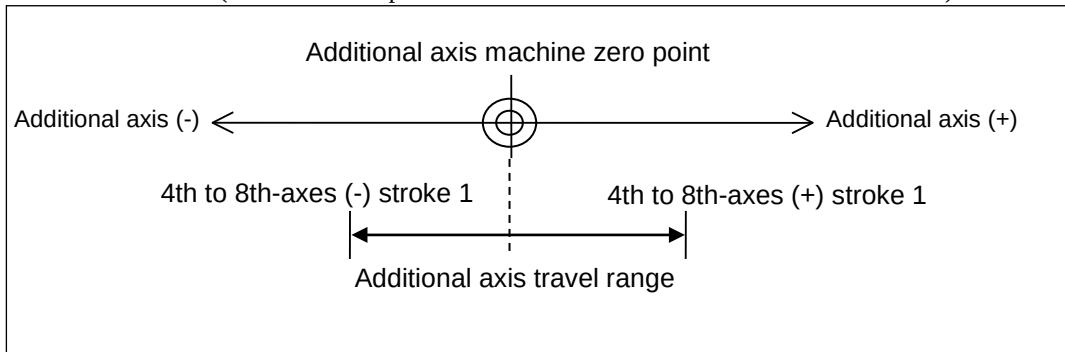
1. X- and Y-axes



2. Z-axis



3. Additional axis (When the user parameter <*-axis stroke control> is set to <1: Yes>)



(Note 1) The axes cannot travel outside of the ranges noted above even when an axis attempts to travel outside of the range from a key operation. In this situation, no specific alarm is triggered.

(Note 2) During the ATC (single tool change) operation, the axes travel outside of the ranges noted above. Refer to section “5.6 Single Tool Change” for further details.

(Note 3) Refer to the PLC System Manual for further details on the PLC-axes.

5.2 Machine Zero Return

The function of the machine zero return is to return each axis (X-, Y-, Z- and additional axes) to the machine's zero point for each axis. After the machine zero return, the machine coordinates for X-, Y- and additional axes is 0. The machine coordinate for the Z-axis is set in the machine parameter <Distance to Z-axis zero>.

5.2.1 Machine Zero Return Operation

1. Press the [MANU] key to switch to manual operation mode.
2. After pressing the [Z.RTN] key, the lamp lights up and the return operation starts.

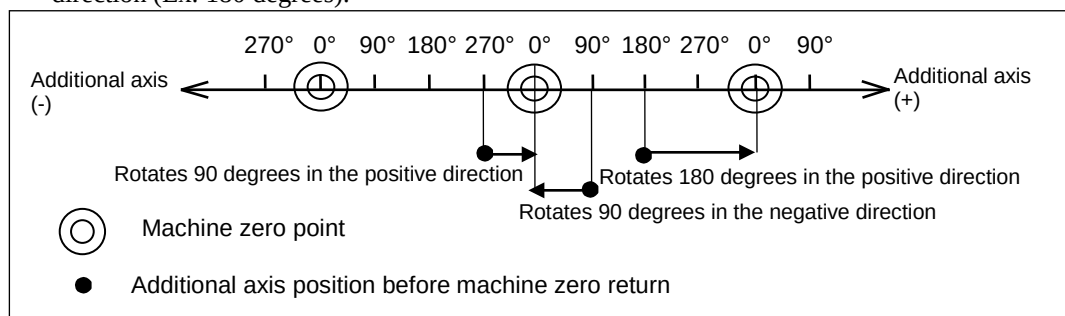
The return operation first raises the Z-axis and then performs the machine zero return. After the Z-axis machine zero return, the X- and Y-axes perform the machine zero return at the same time. The additional axes perform the machine zero return according to the order set in the machine parameters <4th-to 8th-axis zero return order>. When the machine zero return is complete, the lamp for the [Z.RTN] key turns off.

- (Note 1) For a machine equipped with a quick table, the QT-axis does not travel when the [Z.RTN] key is pressed.
- (Note 2) When the [Z.RTN] key is pressed, the machine zero return is not performed for PLC-axes.
- (Note 3) When the machine is equipped with a lathe spindle and it is selected, the additional axis that corresponds to the lathe spindle does not perform a machine zero return operation when the [Z.RTN] key is pressed.
- (Note 4) A non-linear interpolation type motion is carried out regardless of the type that is selected in the user parameter <Positioning method>.

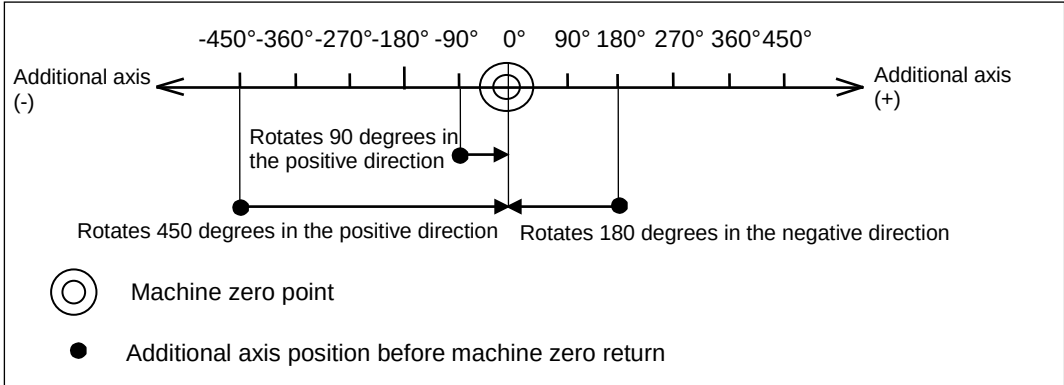
5.2.1.1 Machine Zero Return Operation on Additional Axes

The motion for machine zero return on the additional axes varies depending on the user parameter setting <*-axis stroke control (0:No 1:Yes)>.

1. When <*-axis stroke control> is set to <0: No>
The machine coordinates change to a value within the range between 0 to 359.999 degrees. As a result, there are multiple machine zero points for the additional axes for every 360 degrees. When the machine zero return operation is executed, the additional axis rotates in the direction that is closest to the 0 degree position for the machine coordinate. However, the axis rotates in the positive direction when the rotating distance is the same for the negative or positive direction (Ex: 180 degrees).



2. When <*-axis stroke control> is set to <1: Yes>
The machine coordinates change to a value within the range between -9999.999 to 9999.999 degrees. As a result, there is only one machine zero point for the additional axes.
When the machine zero return operation is executed, the additional axis rotates to the 0 degree position for the machine coordinate.



5.3 Jog & Step Feed

Use the [+X], [-X], [+Y], [-Y], [+Z] and [-Z] keys to move the X-, Y- and Z-axes. Use the [+4] and the [-4] keys to change the speeds to high speed, low speed or step travel on the QT-axis, additional axis (C1-axis, B1-axis, A1-axis, C2-axis, B2-axis, A2-axis) or PLC-axis.

However, when using the [+4] and [-4] keys to operate the additional axis or PLC-axis, after the <Manual conditions> screen is displayed, press one of the functions keys [F0] through [F4] to operate the desired axis. Or, press the [F5] function key (Change additional axes display) and then press one of the function keys [F0] through [F3] to operate the desired axis.

- *1 When selecting an additional axis and PLC-axis, only 1 axis from the additional and PLC-axes can be selected. (An axis from both the additional and PLC-axes cannot be selected at the same time.)
- *2 When the same axis is set to multiple the user parameters <4th-12th row display axis>, even if one of the function keys for that axis is pressed, the axis is controlled by using the [+4] and [-4] keys. Furthermore, all of the function keys for that axis are selected and highlighted.
- *3 When the machine is equipped with a lathe spindle and the lathe spindle is selected, if an additional axis has been assigned as the lathe spindle (which is controlled using the [+4] or [-4] keys) and it is selected, the [+4] or [-4] key operation is ignored and the operator message <<The lathe spindle is being selected>> appears.

Term definitions

- Axis travel keys
General keys: [+X], [-X], [+Y], [-Y], [+Z], [-Z], [+4] and [-4]

(Note) Refer to the PLC Manual for further details on the PLC-axes.

5.3.1 High-speed Travel

The machine parameter <Manual rapid feedrate> can be used to set the travel speed on the X-, Y- and Z-axes to 5 different levels: 10%, 25%, 50%, 75% and 100%. In addition, the machine parameters <5th through 8th-axis manual rapid feedrate> can be used to select and set the rotation speed on the additional axes to 5 different levels: 10%, 25%, 50%, 75% and 100%.

The axis travel keys can be used to move the X-, Y-, Z-, and additional axes in the desired direction. When changing the travel speed, move the cursor on the <Manual conditions> screen to the setting <High travel speed (XYZ/P)> or <High speed rotation (4/P)>, and use the [+] or [-] key to change the high travel speed.

(Note) When changing the high travel speed during axis travel, the change is immediately reflected in the travel speed.

5.3.2 Low-speed Travel (JOG)

The travel speed for the X-, Y- and Z-axes can be set to 22 different levels from 50 to 4000 mm/min (2.0 to 157.4 inch/min). In addition, the rotation speed for the additional axis can be set to 19 different levels from 0.1 to 7.5 min⁻¹.

The axis travel keys can be used to move the X-, Y-, Z- and additional axes in the desired direction. When changing the travel speed, move the cursor on the <Manual conditions> screen to the setting <Low travel speed (XYZ/P)> or <Low speed rotation (4/P)>, and use the [+] or [-] key to change the low travel speed.

(Note) When changing the low travel speed during axis travel, the change is immediately reflected in the travel speed.

5.3.3 Step Travel

The X-, Y-, Z- and additional axes can be moved in steps. Step travel can be set to 4 different levels: 0.001, 0.01, 0.1 and 1.0 mm/step (0.0001, 0.001, 0.01 and 0.1 inch/step).

The axis travel keys can be used to move the X-, Y-, Z-, and additional axes in the desired direction. When changing the level or amount for the travel step, move the cursor on the <Manual conditions> screen to the setting <Step travel (XYZ/P)> or <Step rotation amount (4/P)>, and use the [+] or [-] key to change the travel amount for the step.

5.4 Spindle Selection

When the machine is equipped with a lathe spindle, select the spindle (tool side rotates) or the lathe spindle (workpiece side rotates).

When the [L.SP] key is turned OFF, the spindle is selected. When it is turned ON, the lathe spindle is selected.

Press the [L. SP] key to stop the spindle/lathe spindle.

5.5 Spindle Rotation

The spindle can be rotated in a clockwise direction and rotation can be stopped. The spindle speed can be increased or decreased between 100 min⁻¹ and the maximum spindle speed (machine parameter). The range between 100 and 4000 min⁻¹ can be increased and decreased in smaller increments with a total of 20 different levels, and after 4000 min⁻¹, the speed can be increased and decreased in 2000 min⁻¹ steps.

When the machine is equipped with a lathe spindle and it is selected, the spindle speed can be increased or decreased between 50 min⁻¹ and the lathe maximum spindle speed (machine parameter). The range between 50 and 1000 min⁻¹ can be increased and decreased in smaller increments with a total of 20 different levels, and after 1000 min⁻¹, the speed can be increased and decreased in 500 min⁻¹ steps.

Use the [SP.CW] key to rotate the spindle. To change the speed, move the cursor on the <Manual conditions> screen to <Spindle speed>, and use the [+] or [-] key to change the spindle speed. Press the [SP.STOP] key to stop spindle rotation.

When the machine is equipped with a lathe spindle and the power is turned ON, the spindle speed is automatically set to the minimum value (Spindle: 100 min⁻¹, Lathe spindle: 50 min⁻¹).

Note, the spindle speed is saved when the spindle is changed. Therefore, after changing back to the spindle again, it is set to the saved spindle speed.

(Note1) If the [SP.CW] key is pressed while constant peripheral speed is controlled (G96 modal), the operator message <<Constant peripheral speed is being controlled>> appears and the operation is invalid. Cancel the constant peripheral speed control (G97 modal) and proceed with the operation.

(Note 2) The [+] and [-] keys are enabled only while the lamp for the [SP.CW] key is lit up.

5.6 Single Tool Change (ATC)

The tool in the next standby pot and the spindle tool can be changed or switched. The motion varies depending on the machine model.

Term definitions

- ATC zero point
Height of the set values for machine parameters <Distance to Z-axis zero> + <Distance to ATC zero>
- Spindle orientation
Function that stops the spindle at a pre-set angle/position
- Column travel for tool change
This can be used on the R450X1, R650X1, F600X1, R450X2, and R650X2 (14/22 MG) models.
The column can be moved automatically to a specified position on the X- and Y-axes during a single tool change (ATC operation) to make it easier to mount or remove the tool.
Refer to “Chapter 5 - 5.8 Column Travel for Tool Change” for further details.

(Note) Even if the [ATC] key is pressed and held down, only one motion is carried out. After the [ATC] key lamp turns off, press the [ATC] key again to perform the next motion. When another motion is not required, depending on the machine operation, that motion is skipped and the next operation is carried out.

In addition, when carrying out an operation for multiple axes simultaneously using an ATC (single tool change) operation, the axes travel in a non-linear interpolation type motion regardless of the type that is selected in the user parameter <Positioning method>.

When using R450X1, R650X1, F600X1, R450X2 and R650X2 (14/22 MG)

1. Press the [MANU] key to switch to manual operation mode.
2. When the [ATC] key is pressed, the Z-axis is raised the distance to the Z-axis zero point, and at the same time, the spindle orientation operation is carried out at a position where the tool can be changed.

(If column travel is enabled during a tool change, the [ATC] key needs to be pressed twice up to this point.)

2-1. When the [ATC] key is pressed, the Z-axis is raised the distance to the Z-axis zero point, and at the same time, the spindle orientation operation is carried out at a position where the tool can be changed.

2-2. When the [ATC] key is pressed, the column travels for a tool change.
Refer to “5.8 Column Travel During Tool Change” for further details.

3. When the [ATC] key is pressed, the Z-axis is raised again to the ATC zero point. At this time, the tool that is mounted on the spindle is removed.
4. When the [ATC] key is pressed, the magazine rotates and indexes the next tool. At this time, the magazine rotates in a clockwise direction.
5. When the [ATC] key is pressed, the Z-axis lowers the distance to the Z-axis zero point. At this time, a new tool can be mounted onto the spindle.

For R650X2 (40 MG)

1. Press the [MANU] key to switch to manual operation mode.
2. When the [ATC] key is pressed, the Z-axis is raised to the ATC zero point, and at the same time, the spindle orientation operation is carried out at a position where the tool can be changed. In addition, the pot shutter opens, and the standby pot for the next tool lowers.
3. When the [ATC] key is pressed, the X-axis and the Y-axis travel to the ATC position (set values for the machine parameters (system 1) <X-axis ATC position> and <Y-axis ATC position>). When already at the above position, this operation is skipped and it proceeds to the next operation.
4. When the [ATC] key is pressed, the arm rotates, moves vertically and the tool is changed.
5. When the [ATC] key is pressed, the pot raises. The X-axis and Y-axis return to the machine zero point. The Z-axis returns to the “Distance to Z-axis zero”. Then, the pot shutter closes.

When not using R450X1, R650X1, F600X1, R450X2, R650X2 (14/22 MG) and R650X2 40MG

1. Press the **[MANU]** key to switch to manual operation mode.
2. When the **[ATC]** key is pressed, the Z-axis is raised the distance to the Z-axis zero point, and at the same time, the spindle orientation operation is carried out at a position where the tool can be changed.
3. When the **[ATC]** key is pressed, the Z-axis is raised again to the ATC zero point. At this time, the tool that is mounted on the spindle is removed.
4. When the **[ATC]** key is pressed, the magazine rotates and indexes the next tool. At this time, the magazine rotates in a clockwise direction.
5. When the **[ATC]** key is pressed, the Z-axis lowers the distance to the Z-axis zero point. At this time, a new tool can be mounted onto the spindle.

5.7 Magazine Turn

The magazine rotates or turns when indexing a tool. Press the **[MAG.F]** key to rotate the magazine clockwise. Press the **[MAG.R]** key to rotate the magazine counterclockwise. The magazine rotates to the next pot every time this key is pressed.

If equipped with the **[MAGAZINE TURN]** switch, press the **[MAGAZINE TURN]** switch while it is set to **[MAG.F]** to rotate the magazine in a clockwise direction. Press the **[MAGAZINE TURN]** switch while it is set to **[MAG.R]** to rotate the magazine in a counterclockwise direction.

For other models besides R650X2 (40 MG)

1. Press the **[MANU]** key to switch to manual operation mode.
2. Press the **[ATC]** key. The Z axis raises to the machine zero point and at the same time the spindle orientation is carried out to a position where the tool can be changed.
3. Press the **[ATC]** key to move to the ATC zero point.
4. Press the **[MAG.F]** key or the **[MAG.R]** key. The magazine rotates and indexes the next tool.
5. After the operation is complete, press the **[ATC]** key to return the Z-axis the distance to the Z-axis zero point.

(Note) The **[MAG.F]** and **[MAG.R]** keys are valid only when the Z-axis is at the ATC zero point position. If the **[MAG.F]** or the **[MAG.R]** key is pressed while the Z-axis is not at the ATC zero point position, the alarm <<*Magazine position error>> is displayed.

For R650X2 (40 MG)

1. Press the **[MANU]** key to switch to manual operation mode.
2. Press the **[MAG.F]** key or the **[MAG.R]** key. The magazine rotates and indexes the next tool. When the pot on standby for the next tool does not rise, the alarm <<Pot position error>> is triggered, preventing operation.

5.8 Column Travel During Tool Change

Column travel is enabled on the R450X1, R650X1, F600X1, R450X2 and R650X2 (14/22MG) models.

The column can be moved automatically to a specified position on the X- and Y-axes during a single tool change (ATC operation) to make it easier to mount or remove the tool.

(Note) When using an F600X1, only the X-axis column can be moved automatically.

Procedure to enable column travel during a tool change

1. Set the user parameter (switch 1) <XY axes movement when changing tool in manual mode> to <1: Valid>.
2. Press the [MANU] key to switch to “Manual operation mode”.
3. Press the [F6] (Enable XY axes movement) key on the manual conditions screen. Then, the following message is displayed on the screen: “XY axes movement during ATC is enabled”.

After a single operation, the X- and Y-axes travel is disabled. Perform step 3 to enable travel again.

In addition, if the following operations are performed while travel is enabled, the disabled message is displayed.

- Reset
- Mode change
- Recovery operation is enabled/disabled

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Column travel during a tool change

The timing of the column travel during a tool change varies depending on the machine model. Refer to “Chapter 5.6 Single Tool Change (ATC)” for further details on the operation timing.

In order to use and press the [ATC] key consecutively, set the user parameter (switch 1): <X-axis position when changing tool in manual mode> and <Y-axis position when changing tool in manual mode>. Then, when the [ATC] key is pressed at the prescribed time, and the column travels to the given position.

5.9 Recovery & Inspection Operations

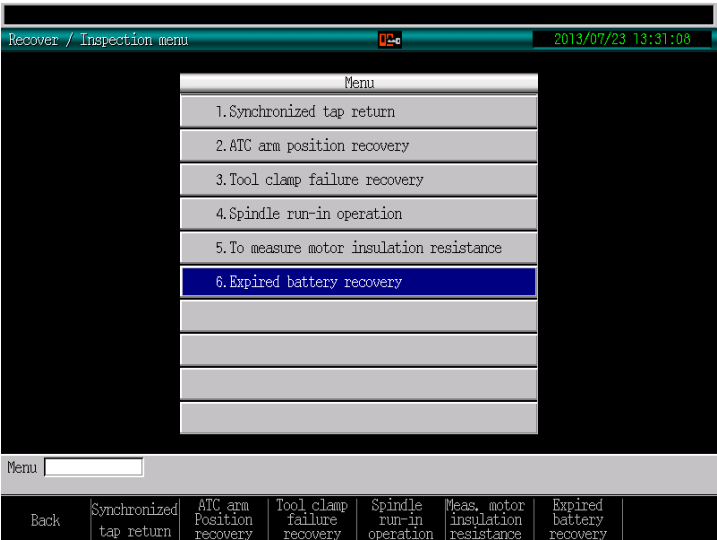
Recovery operation is carried out when a tap operation is stopped while cutting into a workpiece, or when an ATC operation is stopped in the middle of the operation, due to a power failure or an alarm.

In addition, an inspection operation is carried out when measuring the motor insulation resistance.

Press the **[F7]** key (Recover / Inspection Menu) on the <Manual conditions> screen to display the recover/inspection menu screen. The following recovery operations can be selected from the recover/inspection menu screen.

(The recover menu cannot be selected on some machine models and specifications. In this situation, the operator message <<There is no corresponding function>> (Stop level 1, cancel level 1) is triggered and the screen will not change to the recover / inspection menu screen.)

- Synchronized tap return
- ATC arm position recovery
- Tool clamp failure recovery
- Spindle run-in operation
- To measure motor insulation resistance
- Expired battery recovery



Press the **[F0]** (Back) key on the recover / inspection menu screen to return to the <Manual conditions> screen. When manual operation mode is changed to another mode on the recover / inspection menu screen and then changed back to manual operation mode, the <Manual conditions> screen is displayed.

Refer to section “5.11 Details on Recovery & Inspection Operations” for further details on recovery and inspection operations.

5.10 Operational Error Prevention Function

5.10.1 Z-axis Height Where Additional Axis Operation is Restricted

Conditions can be set to prohibit or restrict additional axis (including QT-axis, excluding PLC-axes) rotation and trigger the alarm <<Additional axis operation prohibited range error>> when using the [+4] or [-4] key in manual mode or when using handle operation to rotate the additional axis.

Current Z-axis machine coordinate value is smaller than the set value* for the user parameter <Z-axis height to prohibit travel of additional axes> or <Z-axis height to prohibit travel of additional axes (pallet 2)>.

- *1 Z-axis height to prohibit travel of additional axes: Valid when the inner pallet is set to 1 and the machine is not a QT machine model.
- *2 Z-axis height to prohibit travel of additional axes (pallet 2): Valid when the inner pallet is set to 2. (Only used on QT machine models.)
- *3 On a QT machine model, when the quick table is not indexed, the Z-axis machine coordinate is compared with the larger value between the <Z-axis height to prohibit travel of additional axes> and the <Z-axis height to prohibit travel of additional axes (pallet 2)>.

If the Z-axis is at a height where it may interfere with the rotation of the index jig, the user can enter the height of the index jig in the parameters <Z-axis height to prohibit travel of additional axes> and <Z-axis height to prohibit travel of additional axes (pallet 2)>, which prohibit or restrict additional axis rotation when using the [+4] or [-4] key or handle operation, and thereby prevents an operational error and accidental interference.

However, when these parameters are set to 0, this function is disabled and the additional axis rotates normally regardless of the Z-axis height.

5.10.2 Automatic Screen Change When Using Handle to Move Additional/PLC Axis

If the user parameter <Auto screen change at additional/PLC axis handle operation> is set to <1: Yes>, the screen display automatically changes to the manual conditions screen when the [Axis selection] switch for the handle is set to <4> in manual mode. Since the screen changes automatically, you can easily check if the additional axis or PLC axis rotates when turning the handle on the pulse generator. When the parameter is set to <0: No>, the screen does not change.

5.10.3 Restrict Operation of External Additional Axis (Only QT Machine Model)

If the user parameter <Prevent external operation for additional axes> is set to <1: Yes>, when the [-4] or [+4] key is pressed in manual mode or the handle is used to operate the external additional axis (excluding PLC-axes), the operator message <<Prohibition of operating an add. outside axis ON>> is displayed and the rotation operation is restricted for the external additional axis. Note, if the parameter is set to <0: No>, this function is disabled.

(Note 1) This function is valid only on QT machine models and is not used on the other models.

(Note 2) The [OUTSIDE INDEX A] and [OUTSIDE INDEX B] switches are valid regardless of the above parameter setting.

5.11 Details on Recovery & Inspection Operations

5.11.1 Synchronized Tap Return

A tapping tool that has been stopped while cutting into the workpiece, due to a power failure or emergency stop, can be pulled out.

5.11.1.1 Operation Procedure

When a stop level 5 alarm is triggered during a tapping process

1. If a stop level 5 alarm is triggered during a tapping process, the operator message <<Switch to manual operation mode for restoration>> is displayed at the same time.
2. Press the [MANU] key to change to manual operation mode, and then the synchronized tap return screen is displayed.

Synchronized tap return		2018/05/10 18:50:57	
Machine coord. pos.	Z 480.000	Recovery procedure	
Max. speed	100	(Note) If the tapping tool does not get stuck in the workpiece, press [F3] (Cancel synchronized tap return).	
Remaining distance	Z 0.000	1. Check pitch, sp. speed, Z return pos. and thread dir.	
Tool No.	02	2. Press [F7] key to make recovery operation valid.	
Tool name	B	3. Tap return is started.	
Specify synchronized tap return		[RELSE]+[H] Return up to "Z axis return coordinate"	
Pitch/number of threads	Pitch 1.000 Threads	[RELSE]+[F] Return every "rotation angle"	
Spindle speed	100		
Z axis return coord. (mach. coord.)	402.000		
Tap return rotation angle	360		
Thread direction	Right thread		
Tapping date and time	2018/04/06 17:34:30	Recovery operation is disabled.	
Recover/Inspection Menu		Cancel synchronized tap return	Edit valid
			Enable recovery operation

Function key description

- [F0]** (Recover/Inspection Menu) key
Returns to the recover/inspection menu screen.
- [F3]** (Cancel synchronized tap return) key
Cancels the automatic change to synchronized tap return screen (after pressing the [RST] key, changing the mode or turning the power OFF and ON).
In addition, after the [F3] (Cancel synchronized tap return) key is pressed, the operator message <<Check the tapping tool in the spindle>> appears.
*Refer to the special notes for further details.
- [F5]** (Edit valid/invalid) key
Enables or disables the edit operation.
When edit operation is <Valid>, the values that specify the synchronized tap return can be edited.
The [F0] and [F3] keys are disabled (grayed out).
When edit operation is <Invalid>, the values that specify the synchronized tap return cannot be edited.
- [F7]** (Enable/Disable recovery operation) key
Enables or disables recovery operation.
Recovery operation is possible when <Enabled>. The other function keys are disabled (grayed out). When changing to a mode that is not manual mode and a number key or letter key is input, the following operator message appears:
<<Invalidate the restoration operation>> and the key input is ignored.
The recovery operation is ignored when <Disabled>.

3. Press the **[F7]** (Enable/Disable recovery operation) key to enable recovery operation and reset the alarm.

Synchronized tap return		2018/05/10 18:51:22		
Machine coord. pos.	Z 480.000	Recovery procedure		
Max. speed	Remaining distance	(Note) If the tapping tool does not get stuck in the workpiece, press [F3] (Cancel synchronized tap return). 1. Check pitch, sp. speed, Z return pos. and thread dir. 2. Press [F7] key to make recovery operation valid. 3. Tap return is started. [RELSE] + [H] Return up to "Z axis return coordinate" [RELSE] + [F] Return every "rotation angle"		
100	Z 0.000			
Tool No.	Tool name			
02	B			
Specify synchronized tap return				
Pitch/number of threads	Pitch 1.000			
	Threads			
Spindle speed	100			
Z axis return coord. (mach. coord.)	402.000			
Tap return rotation angle	360			
Thread direction	Right thread			
Tapping date and time	2018/04/06 17:34:30	Recovery operation is enabled.		
Recover/ Inspection Menu		Cancel synchronized tap return	Edit valid	Disable recovery operation

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The current Z-axis machine coordinate position, the maximum rotational frequency (value set in the user parameter <Max. tapping speed (when returning tap)>), the remaining distance to the Z-axis return coordinate (*1), the spindle tool information, and the specified conditions for the synchronized tap return are displayed on the left side of the screen. A guide for the recovery procedure is displayed on the right side.

- * To change the values that specify the synchronized tap return, press the **[F7]** key to disable recovery operation. Then, press the **[F5]** (Edit valid) key. The data input field is displayed and the values can be edited. At this time, the **[F5]** key changes to <Edit invalid>. (*2) Use the cursor keys to move the cursor to each value and edit accordingly. To restore operation after editing, press the **[F7]** key to enable recovery operation. At this time, the values remaining in the data input field are cleared.

Synchronized tap return		2018/05/10 18:51:47		
Machine coord. pos.	Z 480.000	Recovery procedure		
Max. speed	Remaining distance	(Note) If the tapping tool does not get stuck in the workpiece, press [F3] (Cancel synchronized tap return). 1. Check pitch, sp. speed, Z return pos. and thread dir. 2. Press [F7] key to make recovery operation valid. 3. Tap return is started. [RELSE] + [H] Return up to "Z axis return coordinate" [RELSE] + [F] Return every "rotation angle"		
100	Z 0.000			
Tool No.	Tool name			
02	B			
Specify synchronized tap return				
Pitch/number of threads	Pitch 1.000			
	Threads			
Spindle speed	100			
Z axis return coord. (mach. coord.)	402.000			
Tap return rotation angle	360			
Thread direction	Right thread			
Tapping date and time	2018/04/06 17:34:30	Recovery operation is disabled.		
Pitch Pitch				
Recover/ Inspection Menu		Cancel synchronized tap return	Edit invalid	Enable recovery operation

4. When the **[H]** key is pressed with the **[RELSE]** key held down, the remaining distance to the Z-axis return coordinate is displayed and the tapping tool is pulled out to the Z-axis return coordinate. (*3)
 In addition, when the **[F]** key is pressed with the **[RELSE]** key held down, the tapping tool is pulled out to the Z-axis return coordinate following the specified tap return rotation angle (1 to 360). (*4)
5. When the tapping tool is completely pulled out (reaching the Z-axis return coordinate), the operator message <<Restoration completed, so invalidate restoration>> appears.

6. When recovery operation is disabled using the [F7] key, the operator message <<Check the tapping tool in the spindle>> appears.
7. If the power is turned OFF during the following period, the operator message <<Check the tapping tool in the spindle>> will appear when the power is turned ON the next time.
 - From the time when a stop level 5 alarm is triggered during tapping until the step 6 operation is completed.

When power is turned OFF during tapping due to a power failure, etc.

After the power is turned ON, the synchronized tap return screen automatically displays. In addition, the operator message <<Check the tapping tool in the spindle>> also appears at the same time. The subsequent operation procedure is the same as the one performed when a stop level 5 alarm is triggered.

- *1 0 is displayed as the remaining distance until step 5 (press the [RELSE] + [H] keys) is performed.
- *2 If the [F5] (Edit valid) key is pressed when the key display is <Edit invalid>, the display changes to <Edit valid> again and the values for synchronized tap return cannot be edited.
- *3 If the [RELSE] + [H] keys are pressed only once, operation is restored to the Z-axis return coordinate. (The recovery operation is not carried out only while the keys are pressed.)
- *4 When the [RELSE] + [F] keys are pressed once, operation is restored in steps following the specified tap return rotation angle. When the keys are pressed several times, operation is restored to the Z-axis return coordinate. (The recovery operation is not carried out only while the keys are pressed.)

Special notes

- (Note 1) When tapping is stopped due to a stop level 5 alarm or the power turning OFF:
- Press the [RELSE] + [H] keys to complete the synchronized tap return.
 - Press the [RELSE] + [F] keys several times to execute the synchronized tap return until operation is restored to the Z-axis return coordinate.
 - Press the [F3] key (Cancel synchronized tap return).
 - Start MDI or memory operation.
- The synchronized tap return screen is the default screen for manual operation mode until one of the above operations is performed. Therefore, if the [RST] key is pressed in manual operation mode, or if the mode is changed to another mode and then changed back to manual operation mode again, the synchronized tap return screen is displayed.
- (Note 2) The [F3] key is valid when a stop level 5 alarm is triggered during tapping or when tapping is stopped due to the power turning OFF. The key is invalid (grayed out) when the synchronized tap return is completed or when the [F3] key is pressed.
- (Note 3) During the synchronized tap return operation, the Z-axis only moves or raises in the upward direction. If the [RELSE] + [H] keys or the [RELSE] + [F] keys are pressed when the current Z-axis coordinate is greater than the Z-axis return coordinate, the Z-axis does not lower. At this time, the operator message <<Synchr. tapping return restoration position error>> appears and the key operation is ignored.
- (Note 4) If the [F7] (Enable recovery operation) key is pressed when the ATC arm is not positioned at the zero point, the following operator message appears: <<Restore the ATC arm position>> and key operation is ignored.
- (Note 5) If the [F7] (Enable recovery operation) key is pressed when the tool is not properly clamped, the following operator message appears: <<Recover tool clamp failure>> and key operation is ignored.
- (Note 6) If a manual operation is attempted before recovery using the synchronized tap return, the operator message <<Operation other than restoration op is prohibited>> appears.
- (Note 7) Even if the synchronized tap return operation is executed, the modal does not change.

- (Note 8) During the synchronized tap return operation, the [F0], [F3], [F5] and [F7] keys are invalid (grayed out) and key input is not possible.
- (Note 9) When the synchronized tap return operation is attempted while the machine is locked, the operator message <<Press [M.LCK] to disable machine lock>> appears and the synchronized tap return operation cannot be executed.
- (Note 10) If [F7] (Enable recovery operation) is pressed while constant peripheral speed is controlled (G96 modal), the message <<Constant peripheral speed is being controlled>> appears and the operation is invalid. Cancel the constant peripheral speed control (G97 modal) and proceed with the operation.
- (Note 11) If [F7] (Enable recovery operation) is pressed when the alarm <<Changed to the ATC maintenance mode>> is triggered, the operator message <<Changing to ATC maintenance mode>> appears and it becomes invalid. Exit ATC maintenance mode and then perform the operation.
- (Note 12) If [F7] (Enable recovery operation) is pressed during manual mode, the alarm <<Manual operation mode>> appears and the operation becomes invalid. Exit manual mode and then perform the operation.
- (Note 13) If [F7] (Enable recovery operation) is pressed during spindle rotation, the alarm <<Spindle rotating>> appears and the operation becomes invalid. Stop spindle rotation and then perform the operation.
- (Note 14) If [F7] (Enable recovery operation) is pressed during lathe spindle rotation, the alarm <<Lathe spindle is rotating>> appears and the operation becomes invalid. Stop spindle rotation and then perform the operation.
- (Note 15) If the [F7] (Enable recovery operation) key is pressed and the encoder position information does not match the machine coordinates because the battery has expired, the following operator message appears: <<Perform the expired battery recovery operation>> and key operation is ignored.
- (Note 16) Even if the ([RELSE] key + [F] key) are pressed to return to each “Rotation angle”, a stop level 5 alarm may be triggered. In that situation, press the ([RELSE] key + [H] key) to return to the “Z-axis return coordinate”. Even if the ([RELSE] key + [H] key) are pressed to return to the “Z-axis return coordinate”, a stop level 5 alarm may trigger while returning to the Z-axis return coordinate. In that situation, press the ([RELSE] key + [H] key) again to return to the “Z-axis return coordinate”.

5.11.1.2 Specifying Synchronized Tap Return

When the synchronized tap return screen is displayed, the values that specify the synchronized tap return are displayed. There are five values for the “Specify synchronized tap return”:

- Pitch/number of threads
 - Spindle speed
 - Z-axis return coordinate
 - Tap return rotation angle
 - Thread direction
1. Pitch/number of threads
The pitch/number of threads for the most recent tapping command is displayed. *1
 2. Spindle speed
The smaller value is displayed between the value for the most recent tapping command and the value which is set in the user parameter (switch 1) <Max. tapping speed (when returning tap)>.
 - When the spindle speed is changed at the synchronized tap return screen, the value entered by the user is displayed until the next tapping operation in memory operation or MDI operation is performed or until the power is turned ON. However, if the value entered is larger than user parameter (switch 1) <Max. tapping speed (when returning tap)>, the speed is clamped at the <Max. tapping speed (when returning tap)> during operation.
 3. Z-axis return coordinate
The Z-axis coordinate (machine coordinate) shown at the start of most recent tapping command is displayed.

4. Tap return rotation angle
Setting unit: 1 to 360 (degrees)
The value that was previously entered for the tap return rotation angle at the synchronized tap return screen is displayed.
5. Thread direction
The thread direction (Right thread or left thread) for the most recent tapping command is displayed.

All the values displayed are updated at the start of tapping operation. The date and time of the update is displayed in the "Tapping date and time". The command to update the synchronized tap return is as follows:

NC language

- G74*2
- G84*2
- G77
- G78
- G133
- G134
- G177
- G178

Conversation language

- Tapping job
- Counter boring tap job
- Rolled tap job
- Counter boring rolled tap job
- Motion call job (Motion: Cutting, Motion type: Tap)
- Motion call job (Motion: Cutting, Motion type: Single direction tapping)

In addition, items 1 to 4 can be edited. Move the cursor to the item to be edited. Then, use the number keys to edit it and press the **[ENT]** key. To edit the thread direction, enter "1" for the right thread and "2" for the left thread, and press the **[ENT]** key.

If the **[RELSE]** + **[H]** keys are pressed when the four items above are not set, the operator message <<Setting data error>> appears.

If the feedrate calculated from the items 1 and 2 exceeds the limit value, the alarm <<Feedrate error>> is triggered when the **[RELSE]** + **[H]** keys are pressed.

- *1 If the synchronized tap return operation is carried out when the values are set for both the pitch and threads, the operation is performed with the pitch value.
- *2 When the G74 and G84 commands are issued, the pitch is calculated from the F and S commands. The result is set as the values that are specified for the synchronized tap return.

- (Note 1)** If the tapping command above is executed during machine lock, the values specified for the synchronized tap return are not updated.
- (Note 2)** If the tapping command above is executed during a dry run, the values specified for the synchronized tap return are updated. However, the dry run offset amount is added to the Z-axis return coordinate.
- (Note 3)** When the screw pitch is less than the <Minimum tapping pitch> in the machine parameter (system 1), the alarm <<Pitch data error>> is triggered.

5.11.1.3 Example of Synchronized Tap Return Operation

1. For NC language

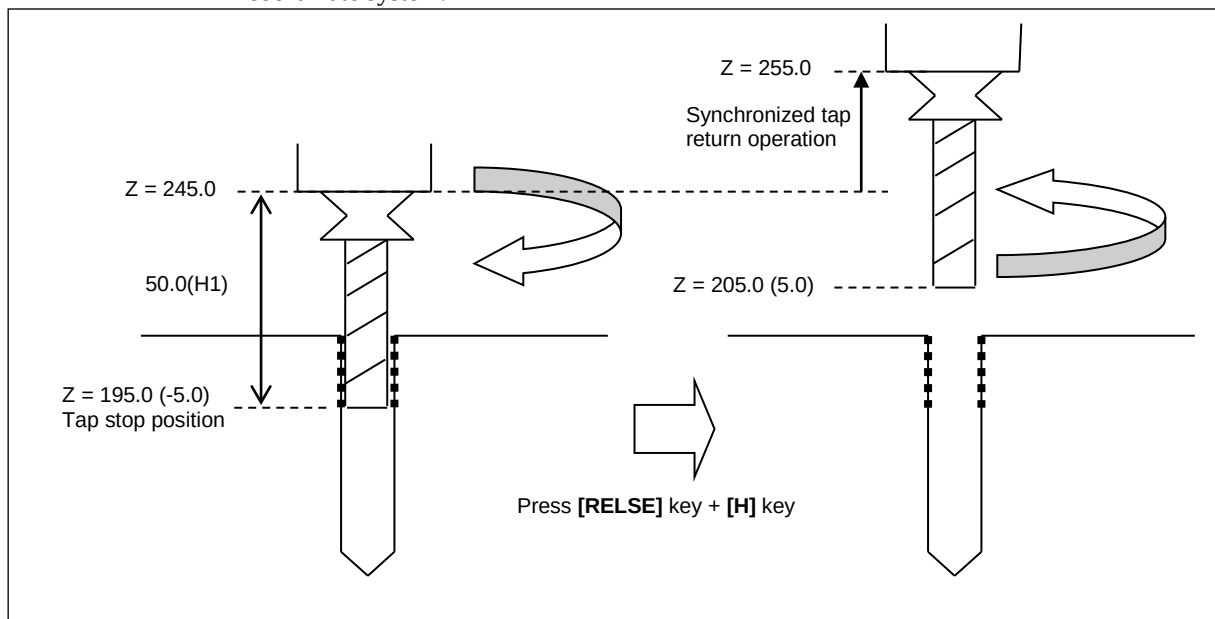
```
N1 G54 G90 G99 (G54 Z:200.000)
N2 G100 T1 G43 H1 Z10. (H1:50.000)
N3 G77 R5. Z-20. I1. S6000
N4 M30
```

If N3 tapping is stopped midway at Z = -5.0 (absolute coordinate) due to a power failure, etc., the synchronized tap return screen is displayed as follows:

Synchronized tap return		2018/05/01 16:54:01	
Machine coord. pos.	Z	245.000	
Max. speed	Remaining distance		
100	Z	0.000	
Tool No.	Tool name		
03	C		
Specify synchronized tap return			
Pitch/number of threads	Pitch	1.000	
	Threads		
Spindle speed		100	
Z axis return coord. (mach. coord.)		255.000	
Tap return rotation angle		360	
Thread direction		Right thread	
Tapping date and time		2018/05/01 16:50:05	
Recovery procedure			
(Note) If the tapping tool does not get stuck in the workpiece, press [F3] (Cancel synchronized tap return).			
1. Check pitch, sp. speed, Z return pos. and thread dir.			
2. Press [F7] key to make recovery operation valid.			
3. Tap return is started.			
[RELSE]+[H] Return up to "Z axis return coordinate"			
[RELSE]+[F] Return every "rotation angle"			
Recovery operation is disabled.			
Recover/ Inspection Menu		Cancel synchronized tap return	Edit valid
			Enable recovery operation

Follow the guide and press the **[H]** key with the **[RELSE]** key held down to display the remaining distance (Initial value: 15.000). The Z-axis raises from the current coordinate 245.000 up to 255.000 with a pitch of 1.0 mm, a spindle speed of 100 min⁻¹ and using reverse rotation.

* The Z-axis coordinate in brackets () indicates the coordinate value in the G54 absolute coordinate system.



2. For conversation language

```

Workpiece zero      X -200.000 Y -200.000 Z 300.000
                   A      ? B      ?
Return height (job change) Height 50.000
Workpiece           1
Workpiece material   ADC12
Job 001 Tapping     Torsion      Right thread
                   Thread        M6.000×1.000
                                   Prepared hole depth 5.100 Chamfering 0.000
                   Pattern        Point
                   Z-axis position Bottom of hole Blind Depth 10.000
                                   Height 0.000 Return height 2.000
                   XY position    X 0.000 Y 0.000
                   Peripheral speed Peripheral speed 100% Feed 100%

Main program end

```

After the primary hole drilling is finished, if tapping (tapping tool length: 100.000) is stopped midway at Z = -5.0 (absolute coordinate) due to a power failure, etc., the synchronized tap return

screen is displayed as follows:

Synchronized tap return		2018/05/07 10:38:12	
Machine coord. pos.		Z	395.000
Max. speed	Remaining distance		
100	Z	0.000	
Tool No.	Tool name		
01	Spiral tap		
Specify synchronized tap return			
Pitch/number of threads	Pitch	1.000	
	Threads		
Spindle speed		100	
Z axis return coord. (mach. coord.)		402.000	
Tap return rotation angle		360	
Thread direction		Right thread	
Tapping date and time		2018/04/06 17:34:30	
(Note) If the tapping tool does not get stuck in the workpiece, press [F3] (Cancel synchronized tap return). 1. Check pitch, sp. speed, Z return pos. and thread dir. 2. Press [F7] key to make recovery operation valid. 3. Tap return is started. [RELSE]+[H] Return up to "Z axis return coordinate" [RELSE]+[F] Return every "rotation angle"			
Recovery operation is disabled.			
Recover/ Inspection Menu		Cancel synchronized tap return	Edit valid Enable recovery operation

Follow the guide and press the **[H]** key with the **[RELSE]** key held down to display the remaining distance (Initial value: 7.000). The Z-axis raises from the current coordinate 395.000 up to 402.000 with a pitch of 1.0 mm, a spindle speed of 100 min⁻¹ and using reverse rotation.

5.11.2 ATC Arm Position Recovery

This function is enabled for machine models equipped with the arm type ATC mechanism. Refer to the recovery procedure in "11.9.2 Arm type ATC" for further details.

5.11.3 Tool Clamp Failure Recovery

This function is not used on this machine.

5.11.4 Spindle Run-in Operation (Spindle 27,000 specification only)

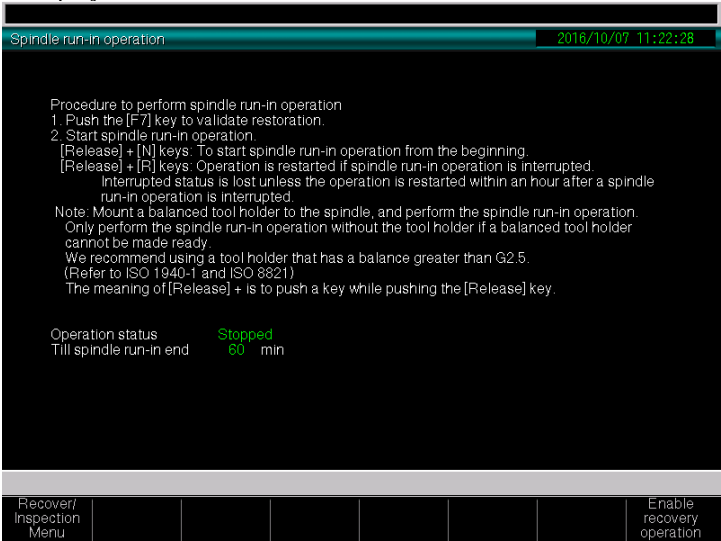
On a high-speed spindle, the grease distribution in the spindle bearing may change during transportation and storage. If the spindle is rotated at a high speed under said conditions, the spindle may seize or burn out.

Thus, a run-in operation is required before actual machining if the power has been turned OFF for a long period of time.

If the spindle run-in operation is required, an alarm is triggered which prompts the user to perform a run-in operation when the power is turned ON. When this alarm is triggered, perform the spindle run-in operation.

5.11.4.1 Spindle Run-in Procedure

- 1. Run-in operation when prompted by an alarm
When a spindle run-in operation is required, an alarm is triggered that prompts the user to <<Carry out spindle run-in in manual operation mode>>. Follow the procedure below to execute a spindle run-in operation.
 - (1) When the power is turned ON, an alarm is triggered that prompts the user to <<Carry out spindle run-in in manual operation mode>>.
The alarm above is triggered in the following situations:
 - The spindle run-in operation has never been performed before.
 - The power has remained OFF for 720 hours or longer.
 - When the power is turned ON with time/date setting values that are older than when the power was turned OFF.
 - (2) Press the **[RST]** key.
 - (3) Switch to manual operation mode.
The message <Perform Spindle run-in operation (see help for operation method)> is displayed in the instructions area on the <Manual conditions> screen.
 - (4) Press **[F0]** (Recover / Inspection Menu).
 - (5) Select <Spindle run-in operation> from the recover / inspection menu. The screen below is displayed.



- Function key description
 - [F0]** (Recover/Inspection Menu) key
Returns to the recover/inspection menu screen.
 - [F7]** (Enable/Disable recovery operation) key
Enables or disables recovery operation.
Recovery operation is possible when <Enabled>. The other function keys are disabled (grayed out). When changing to a mode that is not manual mode and a number key or letter key is input, the following operator message appears: <Invalidate the restoration operation> and the key input is ignored. The recovery operation is ignored when <Disabled>.
- Description for balanced tool holder
We recommend a grade level of G2.5 or higher for the tool holder. (Refer to ISO1940-1 and ISO8821)

- When performing a spindle run-in operation without a tool holder
The rotation noise for the spindle may get louder but this is characteristic of the specification and not a problem with performance or function.
 - <Stopped> is displayed for the <Operation status>.
- (6) Press the **[F7]** (Enable recovery operation) key.
 - The **[F0]** (Recover / Inspection Menu) key is grayed out.
 - (7) Press the **[RELSE]** + **[N]** keys to start the spindle run-in operation.
 - <Stopped> displays for the <Operation status>.
 - The **[F7]** (Disable recovery operation) key is grayed out.
 - The time displayed for <Till spindle run-in end> gradually decreases.
 - (8) When the <Operation status> changes to <Operation end>, the spindle run-in operation is complete.
 - Recovery operation is disabled. The **[F0]** and **[F7]** keys are valid.
2. Spindle run-in operation in manual operation mode
The spindle run-in operation can be performed at any time regardless of the alarm status. Perform the procedure from step 3 in the previous section.
 3. Stopping spindle run-in operation
If one of the following operations is performed or if a status below applies during the spindle run-in operation, the operation is stopped and the <Operation status> changes to <Stopped>.
 - **[SP.STOP]** key is pressed
 - Reset
 - Machine lock is enabled or disabled
 - Stop level 4 alarm or higher is triggered

To restart the spindle run-in operation, press the **[RELSE]** + **[R]** keys.

 - If run-in operation is not restarted within one hour (including when the power is turned OFF) after operation has stopped, the run-in operation starts from the beginning again even if the **[RELSE]** + **[R]** keys are pressed.
 - If memory operation is performed while the spindle run-in operation has stopped, the run-in operation starts from the beginning again even if the **[RELSE]** + **[R]** keys are pressed.
 4. Special notes on spindle run-in operation
 - (1) When recovery operation is <Enabled> on the spindle run-in screen, if the mode is changed or a number key or letter key is input, the following operator message appears: <<Invalidate the restoration operation>> and the key input is ignored.
 - (2) When recovery operation is "Enabled" on the spindle run-in screen, manual mode operations, besides **[SP.STOP]**, are ignored and the following operator message appears: <<Operation other than restoration op is prohibited>>.
 - (3) Spindle run-in operation cannot be started under the following conditions:
 - The machine lock is ON (<<Press **[M.LCK]** to disable machine lock>> appears.)
 - The SPLOCK signal is ON (<<Spindle lock signal on>> appears.)
 - When the door is open (<<Door open>> appears.)
 - A stop level 2 alarm or higher is triggered.
 - (4) The recovery operation cannot be enabled under the following conditions even when the **[F7]** (Enable recovery operation) key is pressed.
 - The lathe spindle is selected (Alarm: <<Selection of a spindle is abnormal.>>).
 - Constant peripheral speed is being controlled (Alarm: <<Constant peripheral speed is being controlled>>).
 - Manual operation mode is being used (Alarm: <<Manual operation mode>>).
 - The spindle is rotating (Alarm: <<Spindle rotating>>).
 - The lathe spindle is rotating (Alarm: <<Lathe spindle is rotating>>).
 - ATC maintenance mode is being used (Alarm: <<Changing to ATC maintenance mode>>).
 - Synchronized tap return in progress (Alarm: <<Execute synchronized tap return>>).
 - Expired battery recovery in progress (Alarm: <<Perform the expired battery recovery operation>>).
 - Changing to mode for enabling and disabling servo (Alarm <<Changing to the mode for enabling and disabling servo>> is triggered.)
 - (5) The spindle override is disabled (set at 100%) during the spindle run-in operation.

- (6) The memory operation cannot be started while the alarm <<Carry out spindle run-in in manual operation mode>> is triggered.
- (7) If a spindle rotation command is issued while the alarm <<Carry out spindle run-in in manual operation mode>> is triggered, the message <<The spindle running-in has not been completed yet>> appears.

5. Screen display description

Item name	Description	
Operation status	Stopped	Displayed before the spindle run-in operation has started or when the operation has stopped midway.
	Running	Displayed during the spindle run-in operation.
	Operation end	Displayed when the spindle run-in operation is complete.
Till spindle run-in end	Displays the time it takes until the spindle run-in operation is complete.	

5.11.4.2 List of Alarms and Operator Messages

Alarm description	Stop	Recovery	Cause	Recovery procedure
Carry out spindle run-in in manual operation mode. (*1)	4	2	Immediately after machine installation or after not operating the machine for a long period, the spindle run-in operation must be executed.	Press the [RST] key. Change to manual mode, and execute the spindle run-in operation.
The spindle running-in has not been completed yet.	4	2	Spindle rotation command was specified even though the spindle run-in operation was not completed.	Press the [RST] key. Change to manual mode, and execute the spindle run-in operation.
Carry out spindle run-in in manual operation mode. (*2)	1	1	Immediately after machine installation or after not operating the machine for a long period, the spindle run-in operation must be executed.	Change to manual mode, and execute the spindle run-in operation.
Manual operation mode	1	1	An attempt was made to enable the recovery operation during manual operation.	Enable recovery operation after manual operation is finished.
Spindle rotating	1	1	An attempt was made to enable the recovery operation while the spindle was rotating.	Enable recovery operation after manual operation is finished.
Lathe spindle is rotating	1	1	1. M410/M411 command was specified during lathe spindle rotation. 2. Recovery/inspection operation was attempted during lathe spindle rotation.	1. Issue the command after the lathe spindle has stopped. 2. Perform recovery/inspection operation after the lathe spindle has stopped.

*1 This alarm is only triggered when the power is turned ON.

*2 This alarm can be triggered at any time until the spindle run-in is complete. (It does not display during the spindle run-in operation.)

The message <Perform Spindle run-in operation> is displayed in the instructions area on the <Manual conditions> screen.

Operator message description	Stop	Recovery	Cause	Recovery procedure
Perform the expired battery recovery operation	1	1	An attempt was made to enable recovery operation, which was not an expired battery recovery operation, even though the encoder position data does not match the machine coordinates (because the battery is expired).	First, perform the expired battery recovery operation.
Execute synchronized tap return	1	1	Recovery operation other than synchronized tap return was executed when the synchronized tap return was not restored yet.	1. First, perform a synchronized tap return. 2. If a synchronized tap return cannot be performed, first, press the [F3] (Cancel synchronized tap return) key on the synchronized tap return recovery screen to reset the synchronized tap recovery. Then, perform the necessary recovery operations.
Constant peripheral speed is being controlled	1	1	An operation was performed that cannot be executed while the constant peripheral speed is being controlled.	Cancel the constant peripheral speed control.
Selection of a spindle is abnormal.	1	1	An operation was attempted that cannot be executed on the spindle that is currently selected.	Check the spindle that is currently selected.
Changing to ATC maintenance mode	1	1	An operation was performed that cannot be executed while changing to ATC maintenance mode.	Exit ATC maintenance mode.
Operation other than restoration op is prohibited.	1	1	An operation other than recovery was commanded when recovery must be performed.	Complete the recovery before issuing a command for another operation.
Invalidate the restoration operation.	1	1	The mode was switched to another mode besides manual operation mode, or a numerical or a character key was pressed, while the recovery/inspection function was enabled.	Disable the recovery/inspection function first before operating.
Changing to the mode for enabling and disabling servo	1	1	An operation was performed that cannot be executed while changing to the mode for enabling and disabling the servo.	Exit the mode for enabling and disabling the servo.

5.11.4.3 Signals

The following signal is turned ON when the alarm <<Carry out spindle run-in in manual operation mode>> is triggered.

External output signal

Signal name	Symbol	Mode that controls output				Function
		Manual	MDI	Memory	Edit	
Spindle run-in execute request	SPIALM	○	○	○	○	Turns ON when the spindle run-in operation needs to be executed.

NC → PLC output signal

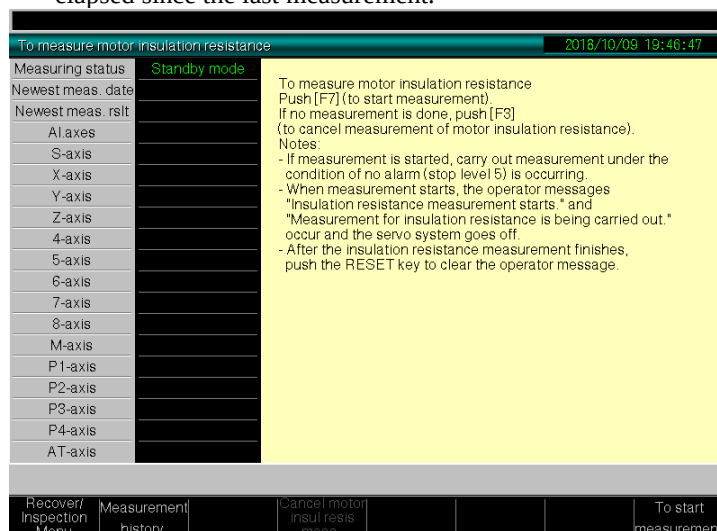
User program	Signal name	Symbol	Conversation / NC	Signal description
	Spindle run-in execute request	SPIALM	Common	Same as external output signal

5.11.5 Measure Motor Insulation Resistance

Motor insulation resistance measurement function is used to measure the motor insulation resistance. Per the parameter setting, an operator message prompting the user to measure the insulation resistance appears when the power is turned ON. In this situation, the user must measure the insulation resistance.

5.11.5.1 Procedure for Measuring the Motor Insulation Resistance

- Measurement when prompted by an operator message
If motor insulation resistance needs to be measured, the operator message <<Measure the motor insulation resistance>> appears. Use the following procedure to measure the insulation resistance.
 - When the power is turned ON, the operator message <<Measure the motor insulation resistance>> appears and the screen below is displayed at the same time.
 - The operator message appears when the following occurs:
 - * The user parameter <Motor insulation resistance measurement interval> has elapsed since the last measurement.



- <Standby mode> is displayed for <Measuring status>.
- The date for the last insulation resistance measurement is displayed for the <Newest meas. date>.
- The result for the last insulation resistance measurement is displayed for the <Newest meas. rslt>.
- The values that were monitored during the last insulation resistance measurement are displayed for <All axes> and <S-axis> to <AT-axis>.
- To cancel the motor insulation resistance measurement, press the [F3] (Cancel motor insul resis measurement) key.

- (2) If an alarm (Stop level 5) is triggered, reset the alarm.
 - Start the insulation resistance measurement when the servo is turned ON. (When the door interlock is enabled, close the door.)
 - (3) Press **[F7]** (To start measurement).
 - When a confirmation message is displayed, select <Yes>.
 - The <Measuring status> changes to <During measurement> and then the measurement starts.
 - The operator messages <<Motor insulation resistance measurement start>> and <<During measurement of motor insulation resistance>> appear and the servo turns OFF.
 - During the measurement, all of the functions are grayed out and cannot be used.
 - (4) When the <Operation status> changes to <Measurement end>, the measurement is complete.
 - The <Newest meas. date> and <Newest meas. rslt> are updated.
 - The values that are monitored during the measurement are displayed for <All axes> and <S-axis> to <AT-axis>.
 - (<S-axis> to <AT-axis> are displayed only if there is an error in the measurement of all axes.)
 - (5) Press the **[RST]** key to reset the operator message.
 - The <Manual conditions> screen is displayed.
2. Measure motor insulation resistance in manual operation mode
The motor insulation resistance can be measured at any time regardless if an operation message appears.
 - (1) Switch to manual operation mode and press **[F0]** (Recover / Inspection Menu).
 - (2) Select <To measure motor insulation resistance> from the recover/inspection menu.
Use the procedure in the previous section to perform the measurement.
 3. Stopping the motor insulation resistance measurement
Perform the operation below to force the measurement to stop.
 - Turn OFF the power
 4. Special notes on measuring the motor insulation resistance
 - (1) The mode cannot be changed on the “To measure motor insulation resistance” screen. The operator message <<During measurement of motor insulation resistance>> appears.
 - (2) Operation cannot be performed in manual operation mode on the “To measure motor insulation resistance” screen. The operator message <<During measurement of motor insulation resistance>> appears.
 - (3) While the operator message <<Measure the motor insulation resistance>> is shown, the <To measure motor insulation resistance> screen is displayed if the manual operation mode is changed.
 - (4) When the power ON time is less than 50 hours, the operator message <<Measure the motor insulation resistance>> appears even if the interval for measuring the motor insulation resistance is exceeded. (Except when the motor insulation resistance has already been measured once.)
 - (5) If the alarm <<Battery has expired. (Local)>> is triggered when turning the power ON, the operator message <<Measure the motor insulation resistance>> will not appear.
 - (6) The motor insulation resistance measurement operation will not start during operation in manual mode operation (Machine zero return, spindle rotation or lathe spindle rotation, etc.).

5. Screen display description

Item name	Description	
Measuring status	Standby mode	Status after the power is turned ON and after being reset.
	During measurement	Status when the motor insulation resistance is being measured.
	Measurement end	Status when the motor insulation resistance measurement has ended.
Newest meas. date	Displays the date when the most recent measurement was completed.	
Newest meas. rslt	Normal	Status indicating that the values that are monitored during the motor insulation resistance measurement are normal.
	Warning	Status warning the user about the values that are monitored during the motor insulation resistance measurement. This is displayed when the monitored value on any of the axes is smaller than user parameter <Motor insulation resistance monitoring acceptable value (alarm)>.
	Abnormal	Status indicating that the values that are monitored during the motor insulation resistance measurement are abnormal. This is displayed when the monitored value on any of the axes is smaller than user parameter <Motor insulation resistance monitoring acceptable value (abnormal)>. This is also displayed when the measurement is not completed within the machine parameter setting <Motor insulation resistance measurement timeout time>.
All axes	Value	The value that is monitored on all axes during the motor insulation resistance measurement. Range: 0.0 to 99.9 (When the monitored value is 100 or greater, it is displayed as "99.9".) Unit: megohm
	Timeout	Displayed when the measurement cannot be completed within the machine parameter setting <Motor insulation resistance measurement timeout time>.
	Blank	When the measurement in progress has been forced to end.
S-axis to AT-axis	Value	The value that is monitored on each axis during the motor insulation resistance measurement. Range: 0.0 to 99.9 (When the monitored value is 100 or greater, it is displayed as "99.9".) Unit: megohm Note that the measurement on each axis is executed when the measurement for all axes ends abnormally.
	Timeout	Displayed when the measurement cannot be completed within the machine parameter setting <Motor insulation resistance measurement timeout time>.
	-	Note that the measurement on each axis is executed when the measurement for all axes ends abnormally.
	Blank	When the axis does not exist or when the measurement has been forced to end. (Even in this situation, the display is blank because the PLC-axis cannot be measured due to the amplifier type.)

5.11.5.3 Parameters

User parameter (Switch 1)

Item name	Setting range	Description
Motor insulation resistance measurement interval	0 to 999 days	Set the interval to display the operator message in order to prompt the user to measure the motor insulation resistance. The operator message <<Measure the motor insulation resistance>> is displayed when the power is turned ON if the preset number of days has passed since the previous measurement. When "0" is set, the operator message does not display.
Motor insulation resistance monitoring acceptable value (all axes)	0.0 to 99.9 megohm	Set the tolerance when monitoring the insulation resistance for measuring all axes in the motor insulation resistance measurement. If the value that is monitored for the insulation resistance is smaller than the setting value, each axis is measured.
Motor insulation resistance monitoring acceptable value (abnormal)	0.0 to 99.9 megohm	Set the tolerance when monitoring the insulation resistance for measuring a single axis in the motor insulation resistance measurement. <<Motor insulation resist. monit. value is abnormal>> is displayed when the value monitored for the insulation resistance is lower than preset value.
Motor insulation resistance monitoring acceptable value (alarm)	0.0 to 99.9 megohm	Set the tolerance when monitoring the insulation resistance for measuring a single axis in the motor insulation resistance measurement. <<Motor insulation resist. monitor value is warned>> is displayed when the value monitored for the insulation resistance is lower than preset value.

Machine parameter (System 1)

Item name	Setting range	Description
Motor insulation resistance measurement timeout time	0.0 to 9.9 seconds	Set the timeout period when measuring the motor insulation resistance. The measurement is cancelled when the set time period has elapsed even if the measurement is not complete.

5.11.5.4 Alarms and Operator Messages

Alarms

Name	Stop and cancel levels	Description
Measurement impossible because alarm occurring	4, 2	An alarm (stop level 5) is triggered when the motor insulation resistance measurement has started.
Motor insulation resist. monit. value is abnormal.	5, 2	The value monitored during the motor insulation resistance measurement is abnormal.
Motor insulation resist. monitor value is warned.	5, 2	A warning is shown for the value monitored during the motor insulation resistance measurement.

Operator messages

Name	Stop and cancel levels	Description
Motor insulation resistance measurement start	5, 2	This alarm is triggered at the start of the motor insulation resistance measurement. Turn OFF the servo motors on all axes.
During measurement of motor insulation resistance	5, 2	This alarm is triggered during the motor insulation resistance measurement. Turn OFF the servo relay.
Measure the motor insulation resistance.	1, 1	This alarm is triggered after the user parameter <Motor insulation resistance measurement interval> has elapsed since the last measurement was taken. The alarm continues to display until the motor insulation resistance is measured. If the status is reset on the motor insulation resistance measurement screen, the operator message disappears. However, it appears again when the power is turned ON the next time.
During measurement of motor insulation resistance	1, 1	1. This alarm is triggered when an attempt is made to switch modes while on “To measure motor insulation resistance” screen. 2. This alarm is triggered when an attempt is made to perform a manual operation during the motor insulation resistance measurement.

5.11.6 Expired Battery Recovery

If the encoder battery expires or if the connector that is connected to the battery is left disconnected, the absolute (position) data for the encoder is cleared and the positioning will no longer match the machine coordinates. In this situation, restoring the correct positioning is possible by following the operation guide.

(Note) The positioning information may be unrecoverable if machine data 1 that is output, using the output all data function (before the battery expires), is input after the expired battery recovery. After input, move the axis, etc., to make sure that the machine coordinates and the actual position are correct.

5.11.6.1 Recovery Procedure

1. Replace the battery and reset the current position information
 - (1) When the power is turned ON without any absolute encoder data, the following happens.
 - When the user parameter <Select mode> is set to <0: Manual mode>:
The <Expired battery recovery> screen is displayed.
 - When the user parameter <Select mode> is set to <1: Memory operation mode>:
The alarm <<Switch to manual operation mode for restoration>> is displayed.
After switching to manual operation mode and then changing to the <Manual conditions> screen, the <Expired battery recovery> screen is displayed.

(Note) The alarm <<No operation is possible under the current level>> is triggered if recovery operation is not possible at the current operation level. Change to an operation level where recovery is possible.

When not using M200X3/ M300X3

Expired battery recovery 2018/10/09 19:10:58

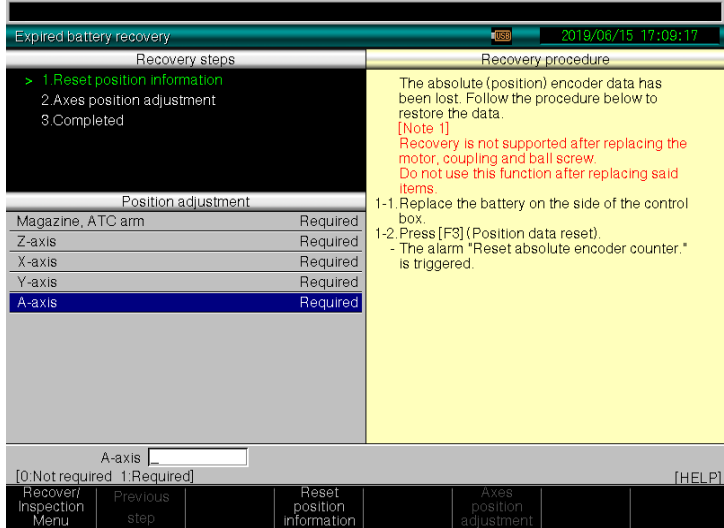
Recovery steps	Recovery procedure
> 1.Reset position information 2.Axes position adjustment 3.Completed	The absolute (position) encoder data has been lost. Follow the procedure below to restore the data. [Note 1] Recovery is not supported after replacing the motor, coupling and ball screw. Do not use this function after replacing said items. [Note 2] This does not support additional axis (excluding QT-axis) position adjustment. Check with the index manufacturer to adjust the position of the additional axis. [Note 3] After selecting the magazine and ATC arm, the magazine and ATC arm operation is restored. The ATC arm recovery is valid only when equipped with an arm type ATC mechanism.
Position adjustment	
Magazine, ATC arm Not required	
Z-axis Not required	
X-axis Not required	
Y-axis Not required	
QT-axis Not required	

Magazine, ATC arm []
[0: Not required, 1: Required]

[HELP]

Recover/Inspection Menu Previous step Reset position information Axes position adjustment

When using M200X3/M300X3



Screen description

- Recovery steps
The current recovery operation that is required is highlighted. Once that operation is carried out, the next operation is highlighted.
- Position adjustment
Each axis is displayed indicating whether position adjustment is required or not, and the user can switch the axis that is highlighted.
Required: Position adjustment needs to be performed.
Not required: Position adjustment does not need to be performed.
- Recovery procedure
A recovery procedure guide is displayed.

Function key description

- [F0]** (Recover / Inspection Menu)
Returns to the recover/inspection menu.
- [F1]** (Previous step)
Returns to the previous recovery step.
- [F3]** (Reset position information)
Resets the position information on the axis that requires position adjustment.
When **[F3]** is pressed:
<The position information will reset for the axis that requires position adjustment. Proceed?> is displayed. Select one of the following.
- [F0]** (Yes): The position information is reset.
- [F1]** (No): The position information is not reset.
The key is grayed out if none of the axes require position adjustment (<Not required>).
- [F5]** (Axes position adjustment)
Returns to the axes position adjustment screen.

- (2) Replace the battery on the side of the control box.
- (3) After pressing the **[F3]** key (Reset position information), select **[F0]** (Yes) when the confirmation message is displayed.

The position information will reset for the axis that requires position adjustment. Proceed?							
Yes	No						

(Note) When recovery operation is not possible at the current operation level, the alarm <<No operation is possible under the current level>> is triggered if **[F3]** (Reset position information) is pressed. Change to an operation level where recovery is possible.

The alarm <<Reset absolute encoder counter>> is triggered.
 The recovery procedure moves to recovery step 2.
 The recovery procedure is displayed for recovery step 2.
 The **[F0]** (Recover/Inspection menu) key is grayed out.
 The **[F1]** (Previous step) key is valid.
 The **[F3]** (Reset position information) key is grayed out.
 The **[F5]** (Axes position adjustment) key is valid.
 The cursor to select the axis for position adjustment disappears. (The user can no longer change the “Required/Not required” setting.)

When not using M200X3/ M300X3

0504 *Reset absolute encoder counter	
Expired battery recovery 2019/07/08 14:03:03	
Recovery steps	Recovery procedure
1.Reset position information > 2.Axes position adjustment 3.Completed	2-1. Press the [RST] key. - The servo error is reset. 2-2. Press [F5] (Axes position adjustment). - The screen switches to the axes adjustment screen.
Position adjustment	
Magazine, ATC arm	Required
Z-axis	Not required
X-axis	Not required
Y-axis	Not required
QT-axis	Not required
[HELP]	
Recover/ Inspection Menu	Previous step
Reset position information	Axes position adjustment

When using M200X3/M300X3

Expired battery recovery 2019/06/15 17:55:32	
Recovery steps	Recovery procedure
1.Reset position information > 2.Axes position adjustment 3.Completed	2-1. Press the [RST] key. - The servo error is reset. 2-2. Press [F5] (Axes position adjustment). - The screen switches to the axes adjustment screen.
Position adjustment	
Magazine, ATC arm	Required
Z-axis	Required
X-axis	Required
Y-axis	Required
A-axis	Required
[HELP]	
Recover/ Inspection Menu	Previous step
Reset position information	Axes position adjustment

- (4) Press the **[RST]** key.
 The alarm <<*axis servo error (Battery low)>> is cleared.
 The alarm <<Reset absolute encoder counter>> is cleared.
- (5) Press the **[F5]** (Axes position adjustment) key.
 Changes to the axes position adjustment screen.
 Refer to “5.11.6.1 2. Adjust position on each axis” for further details on the axes position adjustment procedure.

After the axes position adjustment is complete, the screen changes to the following screen.

When not using M200X3/ M300X3

Expired battery recovery		2019/07/08 14:01:51	
Recovery steps		Recovery procedure	
1.Reset position information 2.Axes position adjustment > 3.Completed		3.Recovery is complete. Press [F0] (Recovery/Inspection menu).	
Position adjustment			
Magazine, ATC arm	Not required		
Z-axis	Not required		
X-axis	Not required		
Y-axis	Not required		
QT-axis	Not required		
[HELP]			
Recover/ Inspection Menu	Previous step	Reset position information	Axes position adjustment

When using M200X3/M300X3

Expired battery recovery		2019/06/15 17:11:53	
Recovery steps		Recovery procedure	
1.Reset position information 2.Axes position adjustment > 3.Completed		3.Recovery is complete. Press [F0] (Recovery/Inspection menu).	
Position adjustment			
Magazine, ATC arm	Not required		
Z-axis	Not required		
X-axis	Not required		
Y-axis	Not required		
A-axis	Not required		
[HELP]			
Recover/ Inspection Menu	Previous step	Reset position information	Axes position adjustment

(6) Recovery is complete. Press the **[F0]** (Recover / Inspection Menu) key.

2. Adjust position on each axis
- Press the **[F5]** (Axes position adjustment) key at the <Expired battery recovery> screen to display the <Position adjustment> screen for each axis.
- Only the axes that require position adjustment are displayed, and the screen changes in the following order depending on the machine model and mechanism.
- For turret ATC mechanism:
Magazine → Z-axis → X-axis → Y-axis (→ QT-axis (when equipped with a QT))
(→ A-axis (when using M200X3/M300X3))
- For arm type ATC mechanism:
Magazine → AT-axis → Z-axis → X-axis → Y-axis (→ QT-axis (when equipped with a QT))

Magazine and ATC arm position adjustment screen

Position adjustment (Magazine, ATC arm)		2019/07/08 13:58:25
Recovery steps	Recovery procedure	
> 1. Set magazine number 2. Release alarm	1. Enter the magazine number that is currently indexed, and press [ENT]. - When equipped with an arm type ATC mechanism, the ATC arm position is adjusted at the same time as the magazine. The current arm position becomes the arm's zero point when operation is restored. - The alarm "Adjusted abso. data has been set." is triggered.	
Magazine, ATC arm		
Magazine No. 07		
Magazine No.		
[HELP]		
Previous step	Position adjustment for next axis	

5

Screen description

- Magazine No.
The magazine number that is identified by the control unit is displayed. (Not indexed if there is no display)
After the magazine number is entered and the position adjustment is carried out, the number that was entered appears.

Any screen item not mentioned in the description above is the same as the <Expired battery recovery> screen.

Function key description

- [F5]** (Position adjustment for next axis)
If there is another axis that requires position adjustment, the screen changes to the position adjustment screen for the next axis.
If there is no other axis that requires position adjustment, the recovery is complete. Then, the screen returns to the <Expired battery recovery> screen.

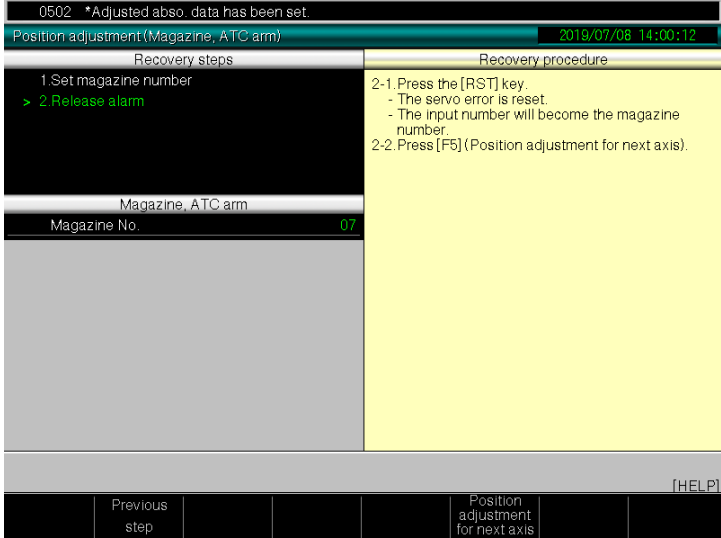
Any screen item not mentioned in the description above is the same as the <Expired battery recovery> screen.

Description of recovery procedure

- (1) Enter the magazine number that is currently indexed and press the **[ENT]** key. Then, when the confirmation message appears, select **[F0]** (Yes).

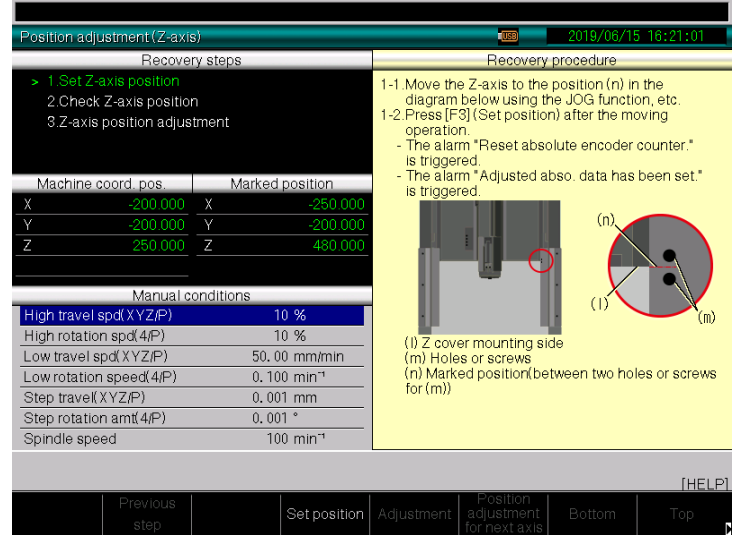
The machine position will be adjusted. Proceed?							
Yes	No						

The alarm <<Adjusted abso. data has been set>> is triggered.
 The recovery procedure moves to recovery step 2.
 The recovery procedure is displayed for recovery step 2.
 The **[F1]** (Previous step) key is valid.
 The **[F5]** (Position adjustment for next axis) key is valid.



- (2) Press the **[RST]** key.
 The alarm <<Adjusted abso. data has been set>> is cleared.
 The number that is entered appears as the magazine number.
- (3) Press the **[F5]** (Position adjustment for next axis) key.
 If there is another axis that requires position adjustment, the screen changes to the position adjustment screen for the next axis.
 If there is no other axis that requires position adjustment, the screen returns to the <Expired battery recovery> screen.

Z-axis position adjustment screen (S300X1/S500X1/S700X1/S1000X1/S300X2/S500X2/S700X2)



Z-axis position adjustment screen (M140X1/M140X2/M200X3/M300X3/R450X1/R450X2)

Position adjustment (Z-axis)		2019/06/15 17:09:50	
Recovery steps > 1. Set Z-axis position 2. Check Z-axis position 3. Z-axis position adjustment		Recovery procedure 1-1. Move the Z-axis to the position (n) in the diagram below using the JOG function, etc. 1-2. Press [F3] (Set position) after the moving operation. - The alarm "Reset absolute encoder counter." is triggered. - The alarm "Adjusted abso. data has been set." is triggered.	
Machine coord. pos.	Marked position		
X	-54.561	X	-100.000
Y	-51.693	Y	-245.000
Z	480.000	Z	455.000
Manual conditions			
High travel spd(XYZ/P)	10 %		
High rotation spd(4/P)	10 %		
Low travel spd(XYZ/P)	50.00 mm/min		
Low rotation speed(4/P)	0.100 min ⁻¹		
Step travel(XYZ/P)	0.001 mm		
Step rotation amt(4/P)	0.001 °		
Spindle speed	100 min ⁻¹		
[HELP]			
Previous step	Set position	Adjustment	Position adjustment for next axis
			Bottom Top

Z-axis position adjustment screen (R650X1/R650X2)

Position adjustment (Z-axis)		2019/07/24 07:52:40	
Recovery steps > 1. Set Z-axis position 2. Check Z-axis position 3. Z-axis position adjustment		Recovery procedure 1-1. Move the Z-axis to the position (n) in the diagram below using the JOG function, etc. 1-2. Press [F3] (Set position) after the moving operation. - The alarm "Reset absolute encoder counter." is triggered. - The alarm "Adjusted abso. data has been set." is triggered.	
Machine coord. pos.	Marked position		
X	-324.488	X	-325.000
Y	-199.993	Y	-200.000
Z	455.990	Z	455.000
Manual conditions			
High travel spd(XYZ/P)	100 %		
High rotation spd(4/P)	10 %		
Low travel spd(XYZ/P)	50.00 mm/min		
Low rotation speed(4/P)	0.100 min ⁻¹		
Step travel(XYZ/P)	0.001 mm		
Step rotation amt(4/P)	0.001 °		
Spindle speed	100 min ⁻¹		
[HELP]			
Previous step	Set position	Adjustment	Position adjustment for next axis
			Bottom Top

Z-axis position adjustment screen (F600X1)

Position adjustment (Z-axis)		2019/06/15 16:56:27	
Recovery steps > 1. Set Z-axis position 2. Check Z-axis position 3. Z-axis position adjustment		Recovery procedure 1-1. Move the Z-axis to the position (n) in the diagram below using the JOG function, etc. 1-2. Press [F3] (Set position) after the moving operation. - The alarm "Reset absolute encoder counter." is triggered. - The alarm "Adjusted abso. data has been set." is triggered.	
Machine coord. pos.	Marked position		
X	-302.527	X	-300.000
Y	-197.204	Y	-200.000
Z	449.843	Z	450.000
Manual conditions			
High travel spd(XYZ/P)	75 %		
High rotation spd(4/P)	10 %		
Low travel spd(XYZ/P)	50.00 mm/min		
Low rotation speed(4/P)	0.100 min ⁻¹		
Step travel(XYZ/P)	0.001 mm		
Step rotation amt(4/P)	0.001 °		
Spindle speed	100 min ⁻¹		
[HELP]			
Previous step	Set position	Adjustment	Position adjustment for next axis
			Bottom Top

Screen description

- Machine coordinate position
The current machine coordinate position is displayed.
In addition, the screen displays the machine coordinate position for the additional axis that is currently selected.
Prior to adjustment, the positioning is out of alignment with the actual machine position.
- Marked position
The machine coordinate position is displayed as a guide when moving the adjustment axis to the marked position.
- Manual conditions
Refer to the description on the manual conditions screen.

Any screen item not mentioned in the description above is the same as the <Position adjustment (Magazine)> screen.

Function key description

- [F3]** (Set position)
Sets the axis position that is being adjusted.
When pressed, the message <The machine position will adjust to the current position. Proceed?> is displayed. Select one of the following.
[F0] (Yes): Sets the position.
[F1] (No): Returns to the previous screen.
- [F4]** (Adjustment)
The recovery procedure moves to recovery step 3.
Adjustment is performed in recovery step 3 if recovery is not possible in recovery step 2.
- [F6]** (Bottom)
Press if the Z- axis is misaligned at the bottom relative to the marked position.
Use in recovery step 3.
- [F7]** (Top)
Press if the Z- axis is misaligned at the top relative to the marked position.
Use in recovery step 3.
- [>>]** (Next function) key
The function display changes to the selection display for the additional axis.
Press the key again to change back to the original function display.

Completion	QT-axis	A-axis	B-axis	C-axis	B-axis	Change additional axes display
------------	---------	--------	--------	--------	--------	--------------------------------

[F0] to [F4] (*-axis) for **[>>]** (Next function)

Press these keys to select the additional axis or the PLC-axis, which are operated with the **[+4]** and **[-4]** keys.

付加軸を選択すると以下のように画面に機械座標が表示されます。

[F5] (Change additional axis display) for [**>>**] (Next function)

Use the **[F0]** to **[F4]** keys for [**>>**] (Next function) to change the display for the selection axis between the additional axis and the PLC-axis.

Any screen item not mentioned in the description above is the same as the <Position adjustment (Magazine)> screen.

Description of recovery procedure

- (1) Use the JOG operation, etc., to move the Z-axis so that (l) in the diagram showed in the recovery procedure comes between the 2 holes (or screws) in the Z-axis direction (n).

If an error is triggered such as a stroke over error, press the **[RELSE]** + **[+Z]** or **[-Z]** keys and move it carefully.

- (2) After travel is complete, press the **[F3]** (Set position) key. Select **[F0]** (Yes) when the confirmation message appears.

The machine position will adjust to the current position. Proceed?							
Yes	No						

(Note) Make sure that travel is complete before setting the position.

The alarm <<Reset absolute encoder counter>> is triggered.

The alarm <<Adjusted abso. data has been set>> is triggered.

The recovery procedure moves to recovery step 2.

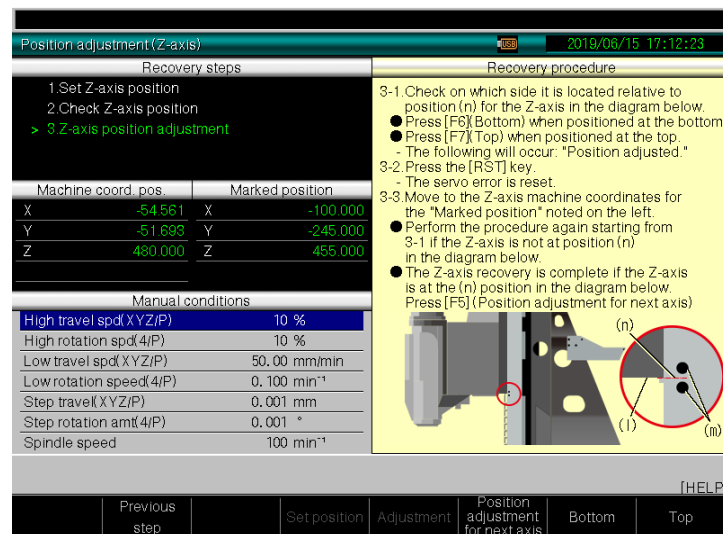
The recovery procedure is displayed for recovery step 2.

The **[F1]** (Previous step) key is valid.

The **[F3]** (Set position) key is grayed out.

The **[F4]** (Adjustment) key is valid.

The **[F5]** (Position adjustment for next axis) key is valid.



- (3) Press the **[RST]** key.

The alarm <<Reset absolute encoder counter>> is cleared.

The alarm <<Adjusted abso. data has been set>> is cleared.

- (4) Move so that the Z-axis machine coordinate matches the coordinate for the <Marked position> on the screen.

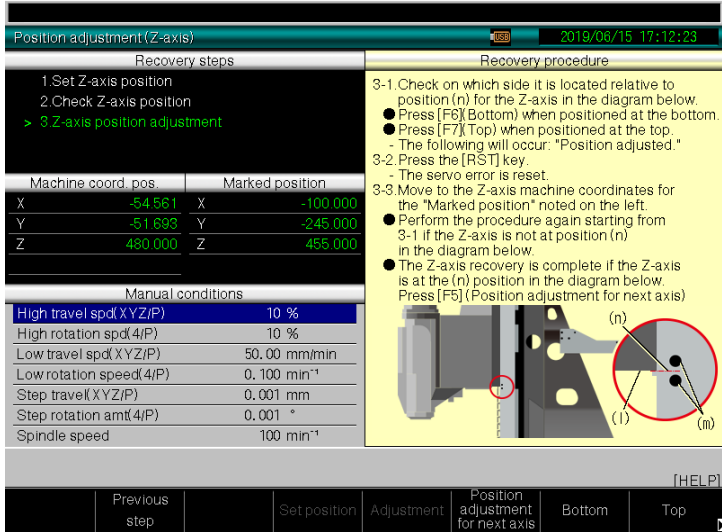
The adjustment is finished when (l) in the diagram showed in the recovery procedure is between the 2 holes (or screws) in the Z-axis direction (n).

Press the **[F5]** (Position adjustment for next axis) key.

If there is another axis that requires position adjustment, the screen changes to the position adjustment screen for the next axis.

If there is no other axis that requires position adjustment, the screen returns to the <Expired battery recovery> screen.

Press the **[F4]** (Adjustment) key when not in the above position.
 The recovery procedure moves to recovery step 3.
 The recovery procedure is displayed for recovery step 3.
 The **[F4]** (Adjustment) key is grayed out.
 The **[F6]** (Bottom) key is valid.
 The **[F7]** (Top) key is valid.



- (5) Check whether (l) in the diagram showed in the recovery procedure is above or below (n), which is between the 2 holes (or screws) in the Z-axis direction.
 Press the **[F6]** (Bottom) key when positioned on the bottom.
 Press the **[F7]** (Top) key when positioned on the top.

- (6) Select **[F0]** (Yes) when the confirmation message appears.

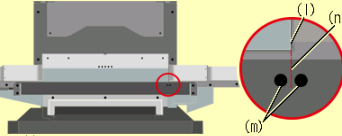
The machine position will be adjusted. Proceed?							
Yes	No						

(Note) If the top or bottom position is not adjusted properly, the misalignment from the marked position becomes more pronounced. Always check the position before setting it.
 The alarm <<Position adjusted>> is triggered.

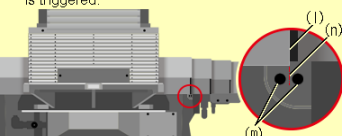
- (7) Press the **[RST]** key.
 The alarm <<Position adjusted>> is cleared.
- (8) Move so that the Z-axis machine coordinate matches the coordinate for the <Marked position> on the screen.
 If an error is triggered such as a stroke over error, press the **[RELSE]** + **[+Z]** or **[-Z]** keys and move it carefully.
 The adjustment is finished when (l) in the diagram showed in the recovery procedure is between the 2 holes (or screws) in the Z-axis direction (n).
 Press the **[F5]** (Position adjustment for next axis) key.
 If there is another axis that requires position adjustment, the screen changes to the position adjustment screen for the next axis.
 If there is no other axis that requires position adjustment, the screen returns to the <Expired battery recovery> screen.

When it is not in the above position, adjust the position again starting from procedure 3-1.

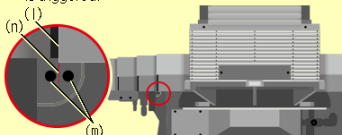
X-axis position adjustment screen (S300X1/S500X1/S700X1/S1000X1/S300X2/S500X2/S700X2)

Position adjustment (X-axis)		2019/06/15 16:21:11	
Recovery steps		Recovery procedure	
> 1. Set X-axis position 2. Check X-axis position 3. X-axis position adjustment		1-1. Move the X-axis to the position (n) in the diagram below using the JOG function, etc. 1-2. Press [F3] (Set position) after the moving operation. - The alarm "Reset absolute encoder counter." is triggered. - The alarm "Adjusted abso. data has been set." is triggered.	
Machine coord. pos.	Marked position	 (l) Mounting side for X-axis telescopic cover or end face for first section of X-axis telescopic cover (m) Holes or screws (n) Marked position (between two holes or screws for (m))	
X	-200.000		
Y	-200.000		
Z	250.000		
Manual conditions			
High travel spd(XYZ/P)	10 %		
High rotation spd(4/P)	10 %		
Low travel spd(XYZ/P)	50.00 mm/min		
Low rotation speed(4/P)	0.100 min ⁻¹		
Step travel(XYZ/P)	0.001 mm		
Step rotation amt(4/P)	0.001 °		
Spindle speed	100 min ⁻¹		
		[HELP]	
Previous step	Set position	Adjustment	Position adjustment for next axis
		Left	Right

X-axis position adjustment screen (M140X1/R450X1/R450X2)

Position adjustment (X-axis)		2019/07/17 07:57:30	
Recovery steps		Recovery procedure	
> 1. Set X-axis position 2. Check X-axis position 3. X-axis position adjustment		1-1. Move the X-axis to the position (n) in the diagram below using the JOG function, etc. 1-2. Press [F3] (Set position) after the moving operation. - The alarm "Reset absolute encoder counter." is triggered. - The alarm "Adjusted abso. data has been set." is triggered.	
Machine coord. pos.	Marked position	 (l) Mounting side for X-axis telescopic cover or end face for first section of X-axis telescopic cover (m) Holes or screws (n) Marked position (between two holes or screws for (m))	
X	-224.963		
Y	-159.982		
Z	474.356		
Manual conditions			
High travel spd(XYZ/P)	100 %		
High rotation spd(4/P)	10 %		
Low travel spd(XYZ/P)	50.00 mm/min		
Low rotation speed(4/P)	0.100 min ⁻¹		
Step travel(XYZ/P)	0.001 mm		
Step rotation amt(4/P)	0.001 °		
Spindle speed	100 min ⁻¹		
		[HELP]	
Previous step	Set position	Adjustment	Position adjustment for next axis
		Left	Right

X-axis position adjustment screen (R650X1/R650X2)

Position adjustment (X-axis)		2019/07/24 07:53:02	
Recovery steps		Recovery procedure	
> 1. Set X-axis position 2. Check X-axis position 3. X-axis position adjustment		1-1. Move the X-axis to the position (n) in the diagram below using the JOG function, etc. 1-2. Press [F3] (Set position) after the moving operation. - The alarm "Reset absolute encoder counter." is triggered. - The alarm "Adjusted abso. data has been set." is triggered.	
Machine coord. pos.	Marked position	 (l) Mounting side for X-axis telescopic cover or end face for first section of X-axis telescopic cover (m) Holes or screws (n) Marked position (between two holes or screws for (m))	
X	-324.486		
Y	-199.996		
Z	455.989		
Manual conditions			
High travel spd(XYZ/P)	100 %		
High rotation spd(4/P)	10 %		
Low travel spd(XYZ/P)	50.00 mm/min		
Low rotation speed(4/P)	0.100 min ⁻¹		
Step travel(XYZ/P)	0.001 mm		
Step rotation amt(4/P)	0.001 °		
Spindle speed	100 min ⁻¹		
		[HELP]	
Previous step	Set position	Adjustment	Position adjustment for next axis
		Left	Right

X-axis position adjustment screen (M140X2/M200X3/M300X3)

Position adjustment (X-axis)

2019/06/15 17:10:18

Recovery steps

1. Set X-axis position
 2. Check X-axis position
 3. X-axis position adjustment

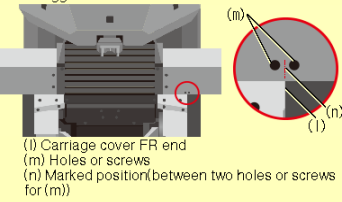
Machine coord. pos.	Marked position
X -54.561	X -100.000
Y -51.693	Y -245.000
Z 480.000	Z 455.000

Manual conditions

High travel spd(XYZ/P)	10 %
High rotation spd(4/P)	10 %
Low travel spd(XYZ/P)	50.00 mm/min
Low rotation speed(4/P)	0.100 min ⁻¹
Step travel(XYZ/P)	0.001 mm
Step rotation amt(4/P)	0.001 °
Spindle speed	100 min ⁻¹

Recovery procedure

1-1. Move the X-axis to the position (n) in the diagram below using the JOG function, etc.
 1-2. Press [F3] (Set position) after the moving operation.
 - The alarm "Reset absolute encoder counter." is triggered.
 - The alarm "Adjusted abso. data has been set." is triggered.



(l) Carriage cover FR end
(m) Holes or screws
(n) Marked position (between two holes or screws for (m))

[HELP]

Previous step

Set position

Adjustment

Position adjustment for next axis

Left

Right

X-axis position adjustment screen (F600X1)

Position adjustment (X-axis)

2019/06/15 16:56:38

Recovery steps

1. Set X-axis position
 2. Check X-axis position
 3. X-axis position adjustment

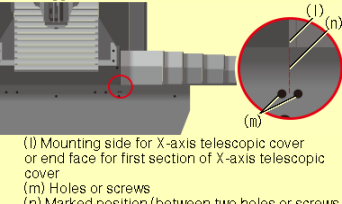
Machine coord. pos.	Marked position
X -302.527	X -300.000
Y -197.204	Y -200.000
Z 449.842	Z 450.000

Manual conditions

High travel spd(XYZ/P)	75 %
High rotation spd(4/P)	10 %
Low travel spd(XYZ/P)	50.00 mm/min
Low rotation speed(4/P)	0.100 min ⁻¹
Step travel(XYZ/P)	0.001 mm
Step rotation amt(4/P)	0.001 °
Spindle speed	100 min ⁻¹

Recovery procedure

1-1. Move the X-axis to the position (n) in the diagram below using the JOG function, etc.
 1-2. Press [F3] (Set position) after the moving operation.
 - The alarm "Reset absolute encoder counter." is triggered.
 - The alarm "Adjusted abso. data has been set." is triggered.



(l) Mounting side for X-axis telescopic cover or end face for first section of X-axis telescopic cover
(m) Holes or screws
(n) Marked position (between two holes or screws for (m))

[HELP]

Previous step

Set position

Adjustment

Position adjustment for next axis

Left

Right

Screen description

The description is the same as the <Position adjustment (Z-axis)> screen.

Function key description

The description is the same as the <Position adjustment (Z-axis)> screen.

Description of recovery procedure

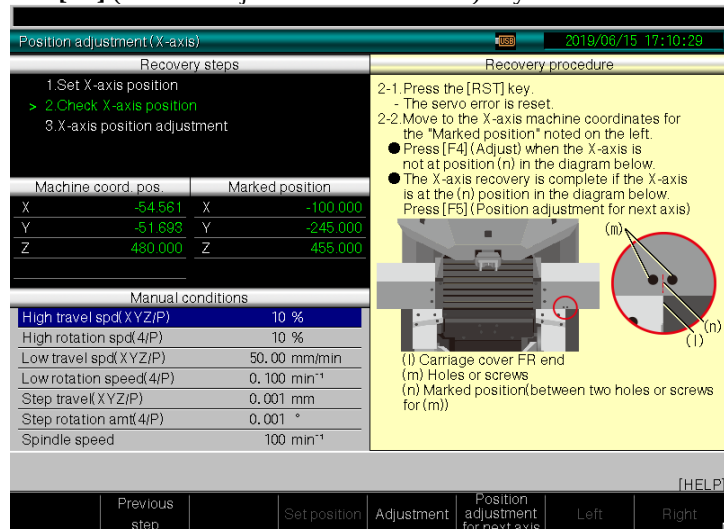
(1) Use the JOG operation, etc., to move the X-axis so that (l) in the diagram showed in the recovery procedure comes between the 2 holes (or screws) in the X-axis direction (n).
If an error is triggered such as a stroke over error, press the [RELSE] + [+X] or [-X] keys and move it carefully.

(2) After travel is complete, press the [F3] (Set position) key. Select [F0] (Yes) when the confirmation message appears.

The machine position will adjust to the current position. Proceed?							
Yes	No						

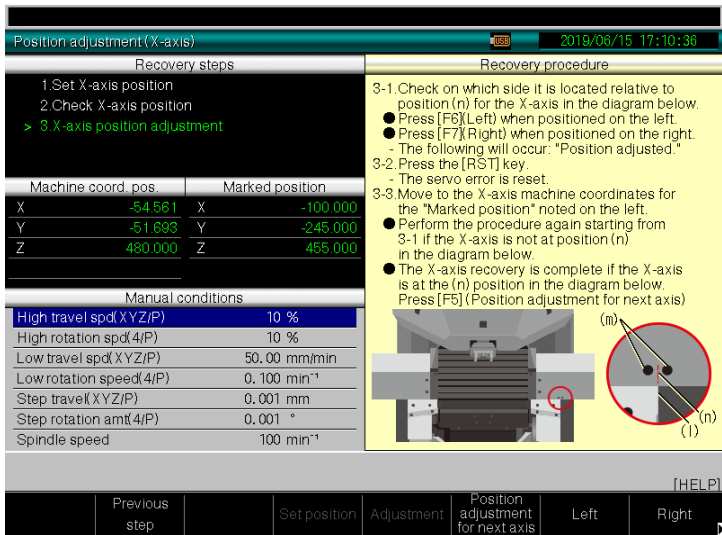
(Note) Make sure that travel is complete before setting the position.

The alarm <<Reset absolute encoder counter>> is triggered.
 The alarm <<Adjusted abso. data has been set>> is triggered.
 The recovery procedure moves to recovery step 2.
 The recovery procedure is displayed for recovery step 2.
 The **[F1]** (Previous step) key is valid.
 The **[F3]** (Set position) key is grayed out.
 The **[F4]** (Adjustment) key is valid.
 The **[F5]** (Position adjustment for next axis) key is valid.



- (3) Press the **[RST]** key.
 The alarm <<Reset absolute encoder counter>> is cleared.
 The alarm <<Adjusted abso. data has been set>> is cleared.
- (4) Move so that the X-axis machine coordinate matches the coordinate for the <Marked position> on the screen.
 The adjustment is finished when (l) in the diagram showed in the recovery procedure is between the 2 holes (or screws) in the X-axis direction (n).
 Press the **[F5]** (Position adjustment for next axis) key.
 If there is another axis that requires position adjustment, the screen changes to the position adjustment screen for the next axis.
 If there is no other axis that requires position adjustment, the screen returns to the <Expired battery recovery> screen.

Press the **[F4]** (Adjustment) key when not in the above position.
 The recovery procedure moves to recovery step 3.
 The recovery procedure is displayed for recovery step 3.
 The **[F4]** (Adjustment) key is grayed out.
 The **[F6]** (Left) key is valid.
 The **[F7]** (Right) key is valid.



- (5) Check whether (l) in the diagram showed in the recovery procedure is to the left or right of (n), which is between the 2 holes (or screws) in the X-axis direction.
 Press the **[F6]** (Left) key when positioned on the left.
 Press the **[F7]** (Right) key when positioned on the right.
- (6) Select **[F0]** (Yes) when the confirmation message appears.
- | | | | | | | | |
|---|----|--|--|--|--|--|--|
| The machine position will be adjusted. Proceed? | | | | | | | |
| Yes | No | | | | | | |
- (Note)** If the left or right position is not adjusted properly, the misalignment from the marked position becomes more pronounced. Always check the position before setting it.
- The alarm <<Position adjusted>> is triggered.
- (7) Press the **[RST]** key.
 The alarm <<Position adjusted>> is cleared.
- (8) Move so that the X-axis machine coordinate matches the coordinate for the <Marked position> on the screen.
 If an error is triggered such as a stroke over error, press the **[RELSE]** + **[+X]** or **[-X]** keys and move it carefully.
 The adjustment is finished when (l) in the diagram showed in the recovery procedure is between the 2 holes (or screws) in the X-axis direction (n).
 Press the **[F5]** (Position adjustment for next axis) key.
 If there is another axis that requires position adjustment, the screen changes to the position adjustment screen for the next axis.
 If there is no other axis that requires position adjustment, the screen returns to the <Expired battery recovery> screen.
- When it is not in the above position, adjust the position again starting from procedure 3-1.

Y-axis position adjustment screen (S300X1/S500X1/S700X1/S1000X1/S300X2/S500X2/S700X2)

Position adjustment (Y-axis)		2019/06/15 16:21:18	
Recovery steps		Recovery procedure	
> 1. Set Y-axis position 2. Check Y-axis position 3. Y-axis position adjustment		1-1. Move the Y-axis to the position (n) in the diagram below using the JOG function, etc. 1-2. Press [F3] (Set position) after the moving operation. - The alarm "Reset absolute encoder counter." is triggered. - The alarm "Adjusted abso. data has been set." is triggered.	
Machine coord. pos.	Marked position	(l) Mounting side for Y-axis telescopic cover or end face of Y guide cover (m) Holes or screws (n) Marked position (between two holes or screws for (m))	
X	-200.000		
Y	-200.000		
Z	250.000		
Manual conditions			
High travel spd(XYZ/P)		10 %	
High rotation spd(4/P)		10 %	
Low travel spd(XYZ/P)		50.00 mm/min	
Low rotation speed(4/P)		0.100 min ⁻¹	
Step travel(XYZ/P)		0.001 mm	
Step rotation amt(4/P)		0.001 °	
Spindle speed		100 min ⁻¹	
[HELP]			
Previous step	Set position	Adjustment	Position adjustment for next axis
			Front Back

Y-axis position adjustment screen (M140X1/R450X1/R450X2)

Position adjustment (Y-axis)		2019/07/17 07:58:00	
Recovery steps		Recovery procedure	
> 1. Set Y-axis position 2. Check Y-axis position 3. Y-axis position adjustment		1-1. Move the Y-axis to the position (n) in the diagram below using the JOG function, etc. 1-2. Press [F3] (Set position) after the moving operation. - The alarm "Reset absolute encoder counter." is triggered. - The alarm "Adjusted abso. data has been set." is triggered.	
Machine coord. pos.	Marked position	(l) Mounting side for Y-axis telescopic cover or end face of Y guide cover (m) Holes or screws (n) Marked position (between two holes or screws for (m))	
X	-224.963		
Y	-159.982		
Z	474.356		
Manual conditions			
High travel spd(XYZ/P)		100 %	
High rotation spd(4/P)		10 %	
Low travel spd(XYZ/P)		50.00 mm/min	
Low rotation speed(4/P)		0.100 min ⁻¹	
Step travel(XYZ/P)		0.001 mm	
Step rotation amt(4/P)		0.001 °	
Spindle speed		100 min ⁻¹	
[HELP]			
Previous step	Set position	Adjustment	Position adjustment for next axis
			Front Back

Y-axis position adjustment screen (R650X1/R650X2)

Position adjustment (Y-axis)		2019/07/24 07:53:15	
Recovery steps		Recovery procedure	
> 1. Set Y-axis position 2. Check Y-axis position 3. Y-axis position adjustment		1-1. Move the Y-axis to the position (n) in the diagram below using the JOG function, etc. 1-2. Press [F3] (Set position) after the moving operation. - The alarm "Reset absolute encoder counter." is triggered. - The alarm "Adjusted abso. data has been set." is triggered.	
Machine coord. pos.	Marked position	(l) Mounting side for Y-axis telescopic cover or end face of Y guide cover (m) Holes or screws (n) Marked position (between two holes or screws for (m))	
X	-324.486		
Y	-199.993		
Z	455.989		
Manual conditions			
High travel spd(XYZ/P)		100 %	
High rotation spd(4/P)		10 %	
Low travel spd(XYZ/P)		50.00 mm/min	
Low rotation speed(4/P)		0.100 min ⁻¹	
Step travel(XYZ/P)		0.001 mm	
Step rotation amt(4/P)		0.001 °	
Spindle speed		100 min ⁻¹	
[HELP]			
Previous step	Set position	Adjustment	Position adjustment for next axis
			Front Back

Y-axis position adjustment screen (M140X2/M200X3/M300X3)

Position adjustment (Y-axis)

2019/06/15 17:10:57

Recovery steps

> 1. Set Y-axis position

2. Check Y-axis position

3. Y-axis position adjustment

Machine coord. pos.		Marked position	
X	-54.561	X	-100.000
Y	-51.693	Y	-245.000
Z	480.000	Z	455.000

Manual conditions

High travel spd(XYZ/P)	10 %
High rotation spd(4/P)	10 %
Low travel spd(XYZ/P)	50.00 mm/min
Low rotation speed(4/P)	0.100 min ⁻¹
Step travel(XYZ/P)	0.001 mm
Step rotation amt(4/P)	0.001 °
Spindle speed	100 min ⁻¹

Recovery procedure

1-1. Move the Y-axis to the position (n) in the diagram below using the JOG function, etc.
1-2. Press [F3] (Set position) after the moving operation.
- The alarm "Reset absolute encoder counter." is triggered.
- The alarm "Adjusted abso. data has been set." is triggered.

(l) Carriage front cover end
(m) Holes or screws
(n) Marked position (between two holes or screws for (m))

[HELP]

Previous step

Set position

Adjustment

Position adjustment for next axis

Front

Back

Y-axis position adjustment screen (F600X1)

Position adjustment (Y-axis)

2019/06/15 16:56:59

Recovery steps

> 1. Set Y-axis position

2. Check Y-axis position

3. Y-axis position adjustment

Machine coord. pos.		Marked position	
X	-302.527	X	-300.000
Y	-197.204	Y	-200.000
Z	449.842	Z	450.000

Manual conditions

High travel spd(XYZ/P)	75 %
High rotation spd(4/P)	10 %
Low travel spd(XYZ/P)	50.00 mm/min
Low rotation speed(4/P)	0.100 min ⁻¹
Step travel(XYZ/P)	0.001 mm
Step rotation amt(4/P)	0.001 °
Spindle speed	100 min ⁻¹

Recovery procedure

1-1. Move the Y-axis to the position (n) in the diagram below using the JOG function, etc.
1-2. Press [F3] (Set position) after the moving operation.
- The alarm "Reset absolute encoder counter." is triggered.
- The alarm "Adjusted abso. data has been set." is triggered.

(l) Mounting side for Y-axis telescopic cover or end face for first section of Y-axis telescopic cover
(m) Holes or screws
(n) Marked position (between two holes or screws for (m))

[HELP]

Previous step

Set position

Adjustment

Position adjustment for next axis

Front

Back

Screen description

The description is the same as the <Position adjustment (Z-axis)> screen.

Function key description

The description is the same as the <Position adjustment (Z-axis)> screen.

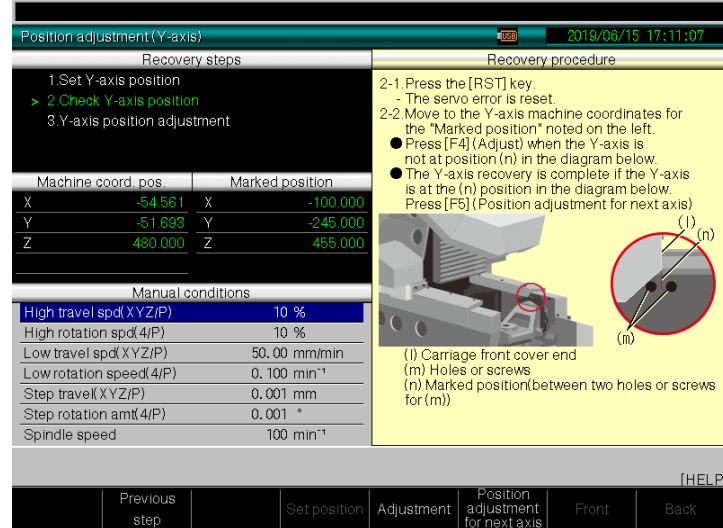
Description of recovery procedure

- (1) Use the JOG operation, etc., to move the Y-axis so that (l) in the diagram showed in the recovery procedure comes between the 2 holes (or screws) in the Y-axis direction (n).
If an error is triggered such as a stroke over error, press the **[RELSE]** + **[+Y]** or **[-Y]** keys and move it carefully.
- (2) After travel is complete, press the **[F3]** (Set position) key. Select **[F0]** (Yes) when the confirmation message appears.

The machine position will adjust to the current position. Proceed?							
Yes	No						

(Note) Make sure that travel is complete before setting the position.

The alarm <<Reset absolute encoder counter>> is triggered.
 The alarm <<Adjusted abso. data has been set>> is triggered.
 The recovery procedure moves to recovery step 2.
 The recovery procedure is displayed for recovery step 2.
 The **[F1]** (Previous step) key is valid.
 The **[F3]** (Set position) key is grayed out.
 The **[F4]** (Adjustment) key is valid.
 The **[F5]** (Position adjustment for next axis) key is valid.



(3) Press the **[RST]** key.

The alarm <<Reset absolute encoder counter>> is cleared.
 The alarm <<Adjusted abso. data has been set>> is cleared.

(4) Move so that the Y-axis machine coordinate matches the coordinate for the <Marked position> on the screen.

The adjustment is finished when (l) in the diagram showed in the recovery procedure is between the 2 holes (or screws) in the Y-axis direction (n).

Press the **[F5]** (Position adjustment for next axis) key.

If there is another axis that requires position adjustment, the screen changes to the position adjustment screen for the next axis.

If there is no other axis that requires position adjustment, the screen returns to the <Expired battery recovery> screen.

Press the **[F4]** (Adjustment) key when not in the above position.

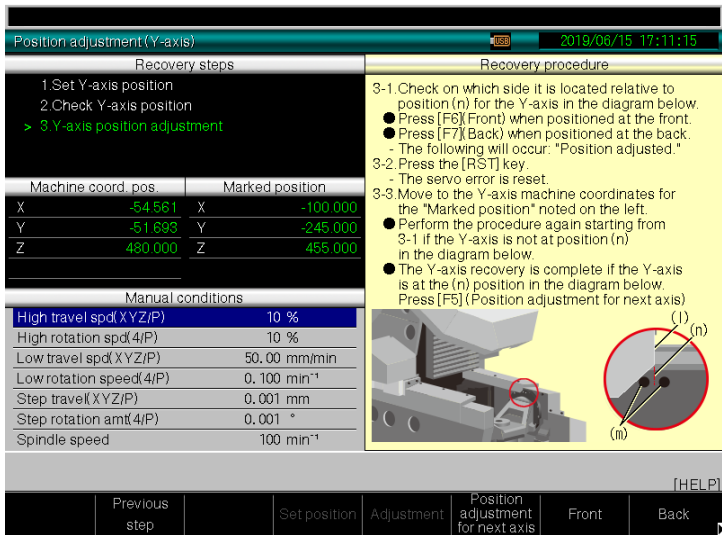
The recovery procedure moves to recovery step 3.

The recovery procedure is displayed for recovery step 3.

The **[F4]** (Adjustment) key is grayed out.

The **[F6]** (Front) key is valid.

The **[F7]** (Back) key is valid.



- (5) Check whether (l) in the diagram showed in the recovery procedure is at the front or the rear of the machine relative to (n), which is between the 2 holes (or screws) in the Y-axis direction.

Press the **[F6]** (Front) key when positioned at the front.
Press the **[F7]** (Back) key when positioned at the back.

- (6) Select **[F0]** (Yes) when the confirmation message appears.

The machine position will be adjusted. Proceed?							
Yes	No						

(Note) If the front or back position is not adjusted properly, the misalignment from the marked position becomes more pronounced. Always check the position before setting it.

The alarm <<Position adjusted>> is triggered.

- (7) Press the **[RST]** key.
The alarm <<Position adjusted>> is cleared.
- (8) Move so that the Y-axis machine coordinate matches the coordinate for the <Marked position> on the screen.
If a stroke over or other error is triggered, press the **[RELSE]** + **[+Y]** or **[-Y]** keys and move it carefully.
The adjustment is finished when (l) in the diagram showed in the recovery procedure is between the 2 holes (or screws) in the Y-axis direction (n).
Press the **[F5]** (Position adjustment for next axis) key.
If there is another axis that requires position adjustment, the screen changes to the position adjustment screen for the next axis.
If there is no other axis that requires position adjustment, the screen returns to the <Expired battery recovery> screen.

When it is not in the above position, adjust the position again starting from procedure 3-1.

QT-axis position adjustment screen (R450X1/R650X1 /R450X2/R650X2)

Position adjustment (QT-axis)		2013/10/28 12:10:53
Recovery steps		Recovery procedure
> 1. Input outer pallet number 2. Outer pallet index 3. QT-axis position adjustment 4. Check QT-axis position		1. Enter the pallet number that is close to the outer side, and press the [ENT] key.
QT-axis recovery status		
Not complete(Standby mode)		
Machine coord. pos.		
X	-251.986 QT	PALLET 2
Y	-209.965	
Z	318.001	
Manual conditions		
High travel spd(XYZ/P)	10 %	
High rotation spd(4/P)	10 %	
Low travel spd(XYZ/P)	50.00 mm/min	
Low rotation speed(4/P)	0.100 min ⁻¹	
Step travel(XYZ/P)	0.001 mm	
Step rotation amt(4/P)	0.001 °	
Spindle speed	100 min ⁻¹	
Pallet number close to the outer side		Recovery operation is disabled.
[1 Pallet 1 2 Pallet 2]		[HELP]
Previous step	Set position	Position adjustment for next axis
		Enable recovery operation

Screen description

- QT-axis recovery status
The recovery status of the QT-axis is displayed.
 <Not complete (Standby mode)>: The position adjustment for the QT-axis is not complete.
 The recovery operation is in standby mode.
 <Recovery operation in progress>: Operation is in progress for QT-axis position adjustment.
 <Completed>: The position adjustment for the QT-axis is complete.
 <Re-check>: The position adjustment for the QT-axis needs to be rechecked.

Any screen item not mentioned in the description above is the same as the <Position adjustment (Z-axis)> screen.

- Monitor effective load
The current effective load monitor is displayed for the QT-axis.
In recovery steps 2 to 4, if this value is larger than 300.0 when attempting to move the QT-axis, the alarm <<QT-axis servo error (Overload 1)>> may trigger and the QT-axis may stop during operation.
- Machine coordinate position
The current machine coordinate position is displayed.
In addition, the screen displays the machine coordinate position for the additional axis that is currently selected.
Prior to adjustment, the positioning is out of alignment with the actual machine position.

Function key description

[F7] (Enable/Disable recovery operation) key

The recovery operation can be enabled and disabled.

Recovery operation is possible when “Enabled”.

When changing to a mode that is not manual mode and a number key or letter key is input, the following operator message appears: <<Invalidate the restoration operation>> and the key input is ignored.

Recovery operation is ignored when “Disabled”.

Any screen item not mentioned in the description above is the same as the <Position adjustment (Z-axis)> screen.

Description of recovery procedure

- (1) Enter the pallet number that is close to the outside, and press the [ENT] key.
 The recovery procedure moves to recovery step 2.
 The recovery procedure is displayed for recovery step 2.
 The [F1] (Previous step) key is valid.
 The [F3] (Set position) key is valid.

The [F7] (Enable recovery operation) key is valid.

Position adjustment (QT-axis)

2019/07/17 07:58:28

Recovery steps

1. Input outer pallet number
- > 2. Outer pallet index
3. QT-axis position adjustment
4. Check QT-axis position

QT-axis recovery status

Effective load

Not complete(Standby mode)

0.0

Machine coord. pos.

X	-224.963	QT	PALLET 1
Y	-159.982		
Z	474.355		

Manual conditions

High travel spd(XYZ/P)	100 %
High rotation spd(4/P)	10 %
Low travel spd(XYZ/P)	50.00 mm/min
Low rotation speed(4/P)	0.100 min ⁻¹
Step travel(XYZ/P)	0.001 mm
Step rotation amt(4/P)	0.001 °
Spindle speed	100 min ⁻¹

Recovery procedure

[Note]

Make sure that the jig does not interfere while the X-, Y-, Z- and QT-axes automatically move during the recovery operation.

2-1. Press [F7] (Enable the recovery operation).

2-2. Wait until effective load monitor is less than 300.0, then press the [RELSE] + [P] keys.

- The "QT-axis recovery status" changes to "Recovery operation in progress."

- The X-, Y- and Z-axes will move to a position where the pallet can turn.

- The pallet corresponding to the input number will move to the outer side.

2-3. Check if the "QT-axis recovery status" changed to "Not complete (Standby mode)."

2-4. Make sure that the pallet corresponding to the input number is at the outer side, and then press [F3] (Set position).

- The alarm "Reset absolute encoder counter." is triggered.
- The alarm "Adjusted abso. data has been set." is triggered.

Recovery operation is disabled.

[HELP]

Previous
step

Set position

Position
adjustment
for next axis

Enable
recovery
operation

- (2) Press the [F7] (Enable recovery operation) key.

Recovery operation is enabled.

The [F1] (Previous step) key is grayed out.

- (3) Wait until the effective load monitor is less than 300.0, then press the [RELSE] + [P] keys.

(Note 1) Perform this operation when the [RAPID TRAVERSE OVERRIDE] switch is set to 100%.

(Note 2) If <Effective load monitor> is larger than 300.0 when attempting to move this operation, the alarm <<QT-axis servo error (Overload 1)>> may trigger and the QT-axis may stop during operation.

The <QT-axis recovery status> changes to <Recovery operation in progress>. After the recovery operation is complete, the status changes to <Not complete (Standby mode)>.

The X-, Y- and Z-axes move to a position where the pallet can be turned.

The pallet that corresponds to the number entered at recovery step 1 is moved to the outside.

- (4) Make sure that the <QT-axis recovery status> changed to <Not complete (Standby mode)>.
- (5) Make sure that the pallet that corresponds to the input number is on the outside. Then, press the [F3] (Set position) key.
Select [F0] (Yes) when the confirmation message appears.

The machine position will adjust to the current position. Proceed?							
Yes	No						

(Note) Make sure that (4) is carried out before setting the position.

The alarm <<Reset absolute encoder counter>> is triggered.

The alarm <<Adjusted abso. data has been set>> is triggered.

The recovery procedure moves to recovery step 3.

The recovery procedure is displayed for recovery step 3.

The [F3] (Set position) key is grayed out.

Position adjustment (QT-axis)



2019/07/17 08:00:57

Recovery steps

1. Input outer pallet number
2. Outer pallet index
- > 3. QT-axis position adjustment
4. Check QT-axis position

QT-axis recovery status

Effective load

Recovery operation in progress

0.0

Machine coord. pos.

X	-224.963	QT	-28.277
Y	-159.982		
Z	685.000		

Manual conditions

High travel spd (XYZ/P)	100 %
High rotation spd (4/P)	10 %
Low travel spd (XYZ/P)	50.00 mm/min
Low rotation speed (4/P)	0.100 min ⁻¹
Step travel (XYZ/P)	0.001 mm
Step rotation amt (4/P)	0.001 °
Spindle speed	100 min ⁻¹

Recovery procedure

[Note]

Make sure that the jig does not interfere while the QT-axes automatically move during the recovery operation.

3-1. Press the [RST] key.

- The servo error is reset.

3-2. Wait until effective load monitor is less than 30.0, then press the [RELSE] + [Q] keys.

- The X-, Y- and Z-axes will move to a position where the pallet can turn.
- Pallet 1 and pallet 2 will be indexed.
- The "QT-axis recovery status" changes to "Recovery operation in progress."
- The alarm "QT-axis has been set." is triggered and then proceeds to the next step.

Recovery operation is enabled

[HELP]

Previous
step

Set position

Position
adjustment
for next axis

Disable
recovery
operation

- (6) Press the [RST] key.

The alarm <<Reset absolute encoder counter>> is cleared.

The alarm <<Adjusted abso. data has been set>> is cleared.

- (7) Wait until the effective load monitor is less than 300.0, then press the [RELSE] + [Q] keys.

(Note 1) Perform this operation when the [RAPID TRAVERSE OVERRIDE] switch is set to 100%.

(Note 2) If <Effective load monitor> is larger than 300.0 when attempting to move this operation, the alarm <<QT-axis servo error (Overload 1)>> may trigger and the QT-axis may stop during operation.

The QT-axis is automatically adjusted. (After the pallet on the opposite side is indexed, the original pallet is indexed.)

The <QT-axis recovery status> changes to <Recovery operation in progress>. After the recovery operation is complete, the status changes to <Not complete (Standby mode)>.

Once the adjustment is complete, the alarm <<QT-axis automatically adjusted>> is triggered.

The recovery procedure moves to recovery step 4.

The recovery procedure is displayed for recovery step 4.

Position adjustment (QT-axis)		2019/07/17 08:02:41			
Recovery steps		Recovery procedure			
1. Input outer pallet number		[Note] Make sure that the jig does not interfere while the QT-axes automatically move during the recovery operation. 4-1. Press the [RST] key. - The servo error is reset. 4-2. Wait until effective load monitor is less than 30.0, then press the [RELSE] + [R] keys. - The X-, Y- and Z-axes will move to a position where the pallet can turn. - Pallet 1 and pallet 2 will be indexed. - The "QT-axis recovery status" changes to "Recovery operation in progress." 4-3. Check the QT-axis recovery status ● The adjusted position must be re-checked if "Re-check" appears. Perform the steps again from 4-1. ● Press [F7] (Disable recovery operation) if "Complete" appears. 4-4. Press [F5] (Position adjustment for next axis).			
2. Outer pallet index					
3. QT-axis position adjustment					
> 4. Check QT-axis position <table border="1"> <tr> <td>QT-axis recovery status</td> <td>Effective load</td> </tr> <tr> <td>Not complete(Standby mode)</td> <td>0.0</td> </tr> </table>			QT-axis recovery status	Effective load	Not complete(Standby mode)
QT-axis recovery status	Effective load				
Not complete(Standby mode)	0.0				
Machine coord. pos.					
X	-224.963 QT	PALLET 1			
Y	-159.982				
Z	474.356				
Manual conditions					
High travel spd(XYZ/P)	100 %				
High rotation spd(4/P)	10 %				
Low travel spd(XYZ/P)	50.00 mm/min				
Low rotation speed(4/P)	0.100 min ⁻¹				
Step travel(XYZ/P)	0.001 mm				
Step rotation amt(4/P)	0.001 °				
Spindle speed	100 min ⁻¹				
		Recovery operation is enabled			
		[HELP]			
Previous step	Set position	Position adjustment for next axis			
		Disable recovery operation			

- (8) Press the [RST] key.

The alarm <<QT-axis automatically adjusted>> is cleared.

- (9) Wait until the effective load monitor is less than 300.0, then press the [RELSE] + [R] keys.

(Note 1) Perform this operation when the [RAPID TRAVERSE OVERRIDE] switch is set to 100%.

(Note 2) If <Effective load monitor> is larger than 300.0 when attempting to move this operation, the alarm <<QT-axis servo error (Overload 1)>> may trigger and the QT-axis may stop during operation.

The QT-axis adjustment is checked. (After the pallet on the opposite side is indexed, the original pallet is indexed.)

The <QT-axis recovery status> changes to <Recovery operation in progress>. After the recovery operation is complete, the status changes to <Re-check> or <Complete>.

The adjustment position needs to be rechecked if the <QT-axis recovery status> changes to <Re-check>.

The alarm <<QT-axis adjustment check again>> is triggered.

Adjust the position again starting from procedure 4-1.

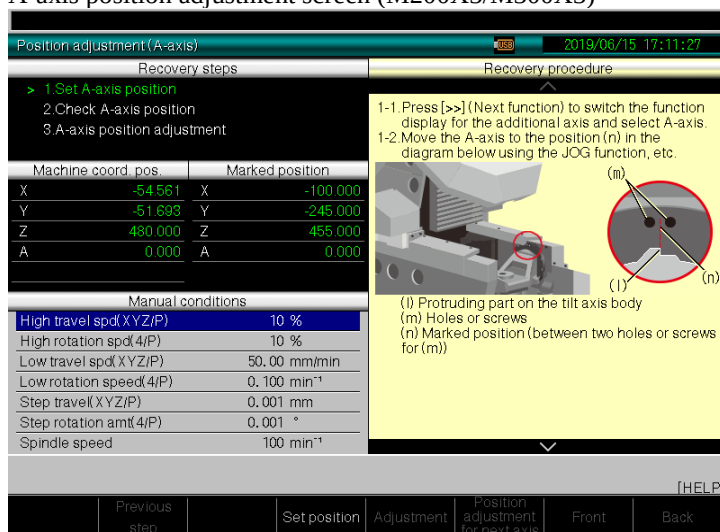
Press the [F7] (Disable recovery operation) key if the <QT-axis recovery status> changes to <Complete>.

- After the QT-axis recovery status changes to <Complete>, the alarm <<QT-axis adjustment check completed>> is triggered.
- After the [F7] (Disable recovery operation) key is pressed, the recovery operation is disabled.

- (10) Press the [F5] (Position adjustment for next axis) key.

- If there is another axis that requires position adjustment, the screen changes to the position adjustment screen for the next axis.
- If there is no other axis that requires position adjustment, the screen returns to the <Expired battery recovery> screen.

A-axis position adjustment screen (M200X3/M300X3)



Screen description

The description is the same as the <Position adjustment (Z-axis)> screen.

Function key description

The description is the same as the <Position adjustment (Z-axis)> screen.

Description of recovery procedure

- (1) Use the jog operation, etc., to move the A-axis so that (l) in the diagram showed in the recovery procedure comes between the 2 holes (or screws) in the A-axis direction (n). If an error is triggered such as a stroke over error, press the [RELSE] + [+4] or [-4] keys and move it carefully.

- (2) After travel is complete, press the **[F3]** (Set position) key. Select **[F0]** (Yes) when the confirmation message appears.

The machine position will adjust to the current position. Proceed?							
Yes	No						

(Note) Make sure that travel is complete before setting the position.

The alarm <<Reset absolute encoder counter>> is triggered.

The alarm <<Adjusted abso. data has been set>> is triggered.

The recovery procedure moves to recovery step 2.

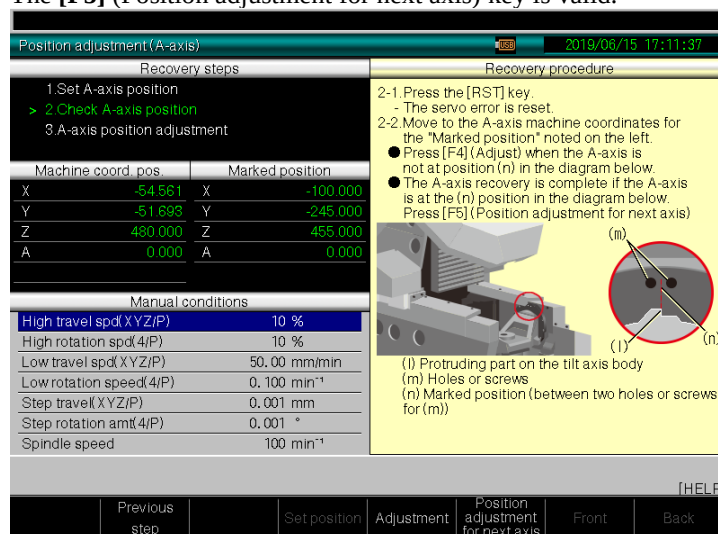
The recovery procedure is displayed for recovery step 2.

The **[F1]** (Previous step) key is valid.

The **[F3]** (Set position) key is grayed out.

The **[F4]** (Adjustment) key is valid.

The **[F5]** (Position adjustment for next axis) key is valid.



- (3) Press the **[RST]** key.

The alarm <<Reset absolute encoder counter>> is cleared.

The alarm <<Adjusted abso. data has been set>> is cleared.

- (4) Move so that the A-axis machine coordinate matches the coordinate for the "Marked position" on the screen.

The adjustment is complete when (l) in the diagram showed in the recovery procedure is between the 2 holes (or screws) in the A-axis direction (n).

Press the **[F5]** (Position adjustment for next axis) key.

If there is another axis that requires position adjustment, the screen changes to the position adjustment screen for the next axis.

If there is no other axis that requires position adjustment, the screen returns to the "Expired battery recovery" screen.

Press the **[F4]** (Adjustment) key when not in the above position.

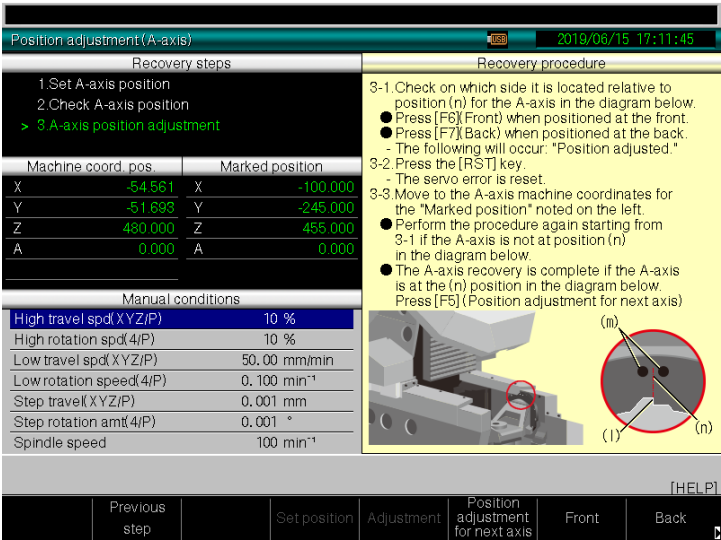
The recovery procedure moves to recovery step 3.

The recovery procedure is displayed for recovery step 3.

The **[F4]** (Adjustment) key is grayed out.

The **[F6]** (Front) key is valid.

The **[F7]** (Back) key is valid.



- (5) Check whether (I) in the diagram showed in the recovery procedure is at the front or the back relative to (n), which is between the 2 holes (or screws) in the A-axis direction.
Press the **[F6]** (Front) key when positioned at the front.
Press the **[F7]** (Back) key when positioned at the back.

- (6) Check whether (I) in the diagram showed in the recovery procedure is at the front or the back relative to (n), which is between the 2 holes (or screws) in the A-axis direction.

The machine position will be adjusted. Proceed?							
Yes	No						

(Note) If the front or back position is not adjusted properly, the misalignment from the marked position becomes more pronounced. Always check the position before setting it.

The alarm <<Position adjusted>> is triggered.

- (7) Press the **[RST]** key.
The alarm <<Position adjusted>> is cleared.
- (8) Move so that the A-axis machine coordinate matches the coordinate for the “Marked position” on the screen.
If a stroke over or other error is triggered, press the **[RELSE]** + **[+Y]** or **[-Y]** keys and move it carefully. The adjustment is complete when (I) in the diagram showed in the recovery procedure is between the 2 holes (or screws) in the A-axis direction (n).
The adjustment is complete when (I) in the diagram showed in the recovery procedure is between the 2 holes (or screws) in the Y-axis direction (n).

Press the **[F5]** (Position adjustment for next axis) key.
If there is another axis that requires position adjustment, the screen changes to the position adjustment screen for the next axis.
If there is no other axis that requires position adjustment, the screen returns to the <Expired battery recovery> screen.

When it is not in the above position, adjust the position again starting from procedure 3-1.

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CHAPTER 6

MDI OPERATION (NC LANGUAGE)

- 6.1 Program Execution
- 6.2 Call Operation Log
- 6.3 Unavailable Commands

6.1 Program Execution

After entering the data per block into the edit field, press the **[START]** switch. This executes the program that was entered. The program that is executing is displayed in the operation field, and when it ends, the program is cleared from the display.

MDI operation										2014/09/09 11:45:49	
Relative coord. pos.		G code				F/S/Tool					
X	-150.013	00	17	22	40	F					
Y	-150.010	49	50	50.1	64	S					
Z	-450.001	54P48	67	69	80	T					
A		90	94	97	98	H					
A	-0.017	M code				Wear offset					
B	0.000	05	09	97	141	D					
C	0.000	221	222	223	224	Wear offset					
B		230	269	270	305						
		401	495								

Operation

Edit

[HELP]

Automatic centering Previous operation data Next operation data Absolute coordinates Change additional axes display

The maximum number of characters that can be entered into the edit field is 127. In addition, the data in the edit area does not clear unless the power is turned OFF or the **[CAN]** key is pressed.

(The additional axes (4th to 8th axes and PLC-axis) are set in the user parameters <4th row display axis> through <8th row display axis> and in the <9th row display axis> through the <12th row display axis>. Press the **[F4]** (Change additional axes display) key to change which axes are displayed.)

6.2 Call Operation Log

The operation log saves a maximum of 10 programs on a single block executed in MDI operation. Press the **[F1]** (Previous operation data) key to call the last program that was executed. Press the **[F2]** (Next operation data) key to call the program that was executed next in the current log.

6.3 Unavailable Commands

The following commands are unavailable and cannot be executed in MDI operation.

G40, G41 and G42

G50 and G51

G50.1 and G51.1

G65, G66 and G67

G68, G69, G168, G68.2 and G53.1

G210

M00 and M01

M02 and M30

M96, M97, M98, M99 and M203

M260 through 267 and M269

M280 through 287 and M289

M298、 M299

M340 through M343

M350 through M355

M360 through M369

M380 through 387 and M389

All macro commands

Macro calls using G or M code (*)

- * When the <System G/M code calling during MDI operation> for the data bank <G/M code macro (shared parameter)> is set to <0: No call>, the alarm <<Macro program call not possible>> is triggered.

When set to <1: Call>, the operation is executed using the system G/M code.

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CHAPTER 7

MEMORY OPERATION (NC LANGUAGE)

- 7.1 Setting Tools in Magazine
- 7.2 Selecting and Operating Main Program
- 7.3 Start
- 7.4 Pause
- 7.5 Single Operation
- 7.6 Optional Block Skip
- 7.7 Optional Stop
- 7.8 Machine Lock
- 7.9 Dry Run
- 7.10 Enabling/Disabling Lock Functions and Dry Runs
- 7.11 Reset
- 7.12 Emergency Stop
- 7.13 Shifting External Workpiece Coordinates
- 7.14 Screen Display in Memory Operation
- 7.15 Program Restart Function
- 7.16 Manual Interruption During Operation and Return
- 7.17 Tool Life and Changing to Spare Tools

7.1 Setting Tools in Magazine

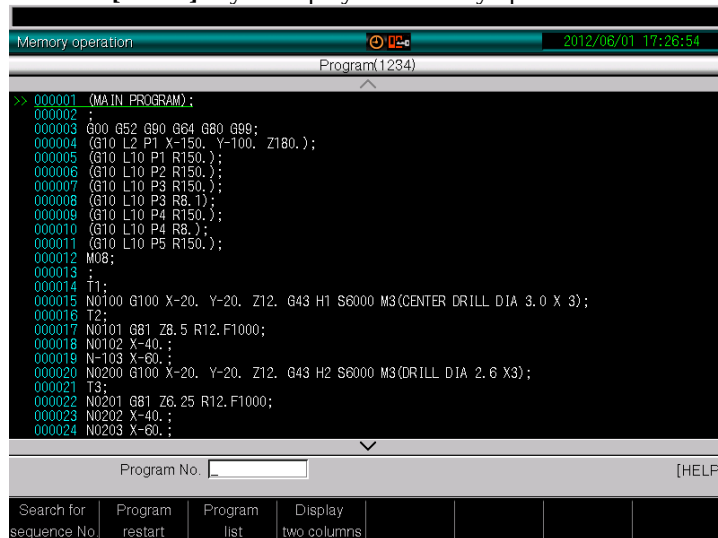
Set the tools to be used in memory operation on the ATC tool screen.
Refer to “4.4 ATC Tools” for further details.

7.2 Selecting and Operating Main Program

Select the program (main program) that you wish to operate from the memory operation screen or the program list screen.

7.2.1 Memory Operation Screen

1. Press the **[MEM]** key to display the memory operation mode.



2. Enter the program number that you wish to operate, and press the **[ENT]** key.
After the specified program is read, all the subprograms called from the main program are loaded in the NC. Then, the main program is displayed on the screen.

7.2.2 Program List Screen

1. Press the **[F2]** (Program list) key on the memory operation screen to change to the program list screen.

The screen shows the registered program number, memory size and updated date and time displayed together with the currently set program at the head of the list. In addition, the folder name is displayed when a subfolder exists.

Program list											
Program(1234)											
Current folder /											
No.	Size	Date	No.	Size	Date	No.	Size	Date	No.	Size	Date
ABC/		12/05/31	1000	99	12/05/17						
SUB256/		12/05/31	1115	47	12/05/16						
0001	111	12/05/17	1234	458	12/06/01						
0002	57	12/05/29	1235	454	12/05/29						
0050	1.77K	12/05/29	3000	2.15M	12/05/29						
0051	319	12/05/29	3500	383	12/05/21						
0052	1.20K	12/05/29									
0053	89	12/06/01									
0055	2.15M	12/05/29									
0100	15	12/06/01									
0123	53	12/05/29									
0200	248	12/05/29									
0201	249	12/05/29									
0202	43	12/05/29									
0210	244	12/05/29									
0250	254	12/05/29									
0524	2.11M	12/05/29									
Program No. 111 Byte 12/05/17 17:10:07 (00001,NC_ROOT)											
No. 0001 111 Byte 12/05/17 17:10:07 (00001,NC_ROOT)											
Running Protected											
Available 468MByte											
Details Sort Cancel											

Press the **[F1]** (Detail) key on this screen to change to the detailed program list screen. The head line of the program is displayed as a comment. (Press the **[F1]** (List) key again to return to the list screen.)

Detailed program list					2012/06/01 17:27:32	
Program(1234)						
Current folder/						
No.	Size	Comment			Date	
ABC/					12/05/31 15:03:59	
SUB256/					12/05/31 15:03:59	
0001	111	(00001,NC_ROOT)			12/05/17 17:10:07	
0002	57	(PRO0002)			12/05/29 15:14:16	
0050	1.77K	G92X0Y0Z480.			12/05/29 19:47:19	
0051	319	(CUBE)			12/05/29 20:52:19	
0052	1.20K	G92X0Y0Z480.			12/05/29 19:47:54	
0053	89	G54 G90 G99 (G54 Z200.000)			12/06/01 11:55:44	
0055	2.15M	(0055)			12/05/29 15:14:42	
0100	15	G01 X-100.			12/06/01 14:58:28	
0123	53	G121 X-100. Y-150. I10. J12. K20. D1 Z400. R450. Q1			12/05/29 13:47:13	
0200	248	(0200)			12/05/29 15:14:55	
0201	249	(0201)			12/05/29 15:15:07	
0202	43	(0202)			12/05/29 15:15:18	
0210	244	(0210)			12/05/29 15:15:32	
0250	254	(0250)			12/05/29 15:15:44	
0524	2.11M	(0524)			12/05/29 15:16:00	
Program No.		111			Available 468MByte	
ABC/		12/05/31 15:03:59  Running  Protected			 	
List	Sort					Cancel

The program that is currently set is highlighted with the cursor. When there is no program that is currently set, the program with the smallest number is highlighted with the cursor. To see other pages, press the **[PAGE UP]** and **[PAGE DOWN]** keys.

2. Enter a program number or move the cursor to select a program, and press the **[ENT]** key. After the specified program is read, all the subprograms called from the main program are loaded in the NC. Then, the main program is displayed on the memory operation screen. In addition, to select a program from another folder, move the cursor to the folder name and press the **[ENT]** key. Then, the cursor moves to that folder. (The cursor does not move even when the folder name is entered and the **[ENT]** key is pressed.) Press the **[F7]** (Cancel) key (even when not selecting a program) or the **[START]** switch to return to the memory operation screen.

7.2.3 Special Notes

- (Note 1) When the specified main program does not exist, the alarm <<No program>> is triggered. In addition, when the subprogram called from the main program does not exist, the alarm <<No subprogram>> is triggered. In this situation, the program that was previously selected is invalid.
- (Note 2) Once a program is selected, it is valid until the next program is selected.
- (Note 3) The edit program and memory operation program numbers are independent of each other, but they can be matched to each other in the user parameter settings.
For example, when using memory operation mode and the edit program number is 1000, the memory operation program number also becomes 1000. In addition, when using edit mode and the memory operation program number is 2000, the edit program number also becomes 2000. Configure this setting in the user parameter <Set same number to edit/memory operation programs>.
- (Note 4) The maximum number of programs (including the main program) that can be loaded to the NC is 256.
- (Note 5) A program number cannot be set during a pause, during a block stop or during program restart.
- (Note 6) When the program size that is loaded exceeds the size that is set in the user parameter (switch 1) <Program load size>, the machine operates in extended memory operation mode.
Extended memory operation refers to operation that is carried out using the NC memory. However, it operates using the tape operation image because the program size is larger than the memory size for operation. The NC automatically determines whether to use normal memory operation or extended memory operation based on the memory size. <Memory operation> or <Extended memory operation> is displayed on the top-left part of the program screen.
A program with 3D smoothed values is run in extended memory operation mode regardless of the program size.
When using extended memory operation, the following commands are unavailable in both tape operation as well as extended memory operation.
- Macros (WHILE DO to END)
- In addition, there are restrictions for the following commands.
- Macros (GOTO to N)
- (Note 7) If a folder is moved but no program is selected from the program list screen, the folder movement is cancelled.

7.3 Start

Press the [START] switch to start memory operation.

7.3.1 Start After Resetting

This function is only valid in memory or MDI operation mode. A program is read and executed according to the operation mode.

7.3.2 Starting After a Stop (Single Block or Program Stop, etc.)

The operation executes from the next block when in memory operation mode.

7.3.3 Starting After a Pause

Machine operation restarts at the position where operation was canceled.

7.3.4 Special Notes

- (Note 1) When the [START] switch is pressed, operation starts when the switch is released.
- (Note 2) The operation does not start even when the [START] switch is pressed if a program is loading or if a program is restarting.
- (Note 3) While the [FEED HOLD] switch is pressed, operation will not start even if the [START] switch is pressed.
- (Note 4) The LED for the [START] switch lights up during startup.

7.4 Pause

Press the [FEED HOLD] switch during startup to temporarily stop (pause) axis travel and to stop program execution.

7.4.1 Pause During Axis Travel

Axis travel is stopped immediately.

However, during a tool change or during a tapping cycle, axis travel stops once that motion is completed.

7.4.2 Pause During Dwell

Operation stops once the dwell time has elapsed.

7.4.3 Pause When Using Miscellaneous Functions (S.T.M.)

Operation stops once the miscellaneous function has been executed.

7.4.4 Special Notes

- (Note 1) The LED for the [FEED HOLD] switch lights up and the LED for the [START] switch turns off.
- (Note 2) The spindle rotation is not stopped with the [FEED HOLD] switch.

7.5 Single Operation

This function stops operation after each block is executed.

Press the [SINGL] key to turn this function ON. (The LED lights up.)

(Note 1) The operation stops several times in a block for a reference point return, canned cycle or nonstop ATC block.

(Note 2) The operation does not stop after a single block for M98 or M99 blocks.

7.6 Optional Block Skip

This function ignores the block command that includes a slash (/) at the beginning or right after the sequence number in the program.

When an operation is performed while the program is running, the changed setting does not take effect within the range up to the block for the interpretation pointer (look-ahead for program).

Press the [B.SKIP] key to turn this function ON. (The LED lights up.)

(Note 1) A command cannot be issued on a block where a 3D smoothing start code or a 3D smoothing end code is used. The message <3D smoothing start/end code is invalid.> is displayed when the [F7] (Program check) key is pressed at the <3D smoothing program check> screen.

(Note 2) A command cannot be issued inside the 3D smoothing section. <Invalid command code is being issued inside the 3D smoothing section.> is displayed when the [F7] (Program check) key is pressed at the <3D smoothing program check> screen.

7.7 Optional Stop

This function stops a program in a block where M01 is used in the program, in the same way when M00 is used.

Press the [OP.STP] key to turn this function ON. (The LED lights up.)

M01 is ignored when the [OP.STP] key is turned OFF.

(Note 1) When a program is stopped, the spindle, the lathe spindle and coolant, etc., also stop. When the user parameter (switch 1) is set to the following, machine operation recovers automatically.

- When <Spindle return method for program stop> is set to <0: Method 1>
- When <Lathe spindle return method for program stop> is set to <0: Method 1>
- When <Coolant return method for program stop> is set to <0: Method 1>

(Note 2) The optional block skip and optional stop do not affect a block that has been read ahead at the time when the switch has been pressed.

7.8 Machine Lock

When all the function operations are stopped, the display appears as if the machine is operating.

Press the [M.LCK] key to turn this function ON. (The LED lights up.)

(Note 1) The machine coordinates on the position screen do not change.

(Note 2) Press the [M.LCK] key to reset operation. However, editing or communication, etc., is not affected.

(Note 3) The miscellaneous functions (S,T and M functions) do not execute.

7.9 Dry Run

This function changes machine operation to idle (dry run) in order to check a program, etc. It is possible to ignore the feedrate command in a program and select a feedrate from the manual conditions screen by setting the high-speed travel speed (XYZ/P). In addition, this function can offset the Z-axis travel command value in the + direction by the setting the user parameter <Dry run offset>.

Press the **[DRY]** key to turn this function ON. (The LED lights up.)

When the user parameter setting <Dry run offset> is the same or less than the <Dry run offset lower limit>, the operator message <<Dry run offset: xx>> is displayed while the machine is dry-running. (xx refers to the dry run offset amount that is set).

Axis travel speed during dry run

Command type	Axis travel speed
Positioning	Parameter <*-axis rapid feedrate>
Cutting	The greater value of the values below: • F command value • <*-axis max. cutting feedrate (rotation)> × <High travel spd (XYZ/P)> on manual conditions screen
Tapping	The greater value of the values below: • F command value • <Z-axis max. cutting feedrate> × <High travel spd (XYZ/P)> on manual conditions screen However, if the above value exceeds the feedrate set in <Max. tapping speed>, the axis travel speed becomes the same feedrate as set in <Max. tapping speed>.
Thread cutting	The greater value of the values below: • Spindle speed × F command value • <*-axis max. cutting feedrate> × <High travel spd (XYZ/P)> on manual conditions screen However, the feedrate, when the peripheral speed is being controlled, is equal to: Spindle speed × F command value.

*: X-, Y-, Z- and 4th- through 8th-axes

(Note) Even if the user parameter <Dry run offset> or <Dry run offset lower limit> is changed during operation, it does not affect the operation. (If the parameter is edited and saved, the alarm <<Press the [RST] key>> is triggered.)

7.10 Enabling/Disabling Lock Functions and Dry Runs

	Memory operation	MDI operation	Manual operation
Machine lock	○	○	○
Dry run	○	○	×
Single operation	○	○	×

7.11 Reset

When the memory operation is reset, it resets to the following:

- The execution pointer returns to the beginning of the main program.
- The ON/OFF status does not change for the locks, dry run, single operation, optional block skip and optional stop.
- The cutter compensation, tool length offset, rotational transformation, mirror image, scaling, canned cycle, subprogram modal call and feature coordinate machining mode are cancelled. All other modal G codes are maintained as is.
- The spindle and the coolant stop.
- The data for the commands F, S and T do not change.

7.12 Emergency Stop

If the [EMERGENCY] switch is pressed, all operation stop immediately.

Follow the procedure below to restore operation.

1. Release the [EMERGENCY] switch. (Turn the switch in the RESET direction.)
2. Press the [RST] key.

7.13 Shifting External Workpiece Coordinates

This function changes the workpiece zero point offset using an external input signal.

Use the workpiece coordinate zero in the data bank to set how much the external workpiece coordinates shift.

When the external workpiece coordinate shift valid signal (external I/O signal: EXWORK) is ON for the current workpiece zero point offset (G54 - G59 including the extended workpiece coordinates), the external workpiece coordinate shift amount is added to create a new zero point position.

When the signal is OFF, the shift amount is not added to the zero point position.

(Note 1) This function is valid in real time and is reflected when reading the signal status (ON/OFF) while a program is being interpreted.

(Note 2) This function is also valid in MDI operation.

7.14 Screen Display in Memory Operation

7.14.1 Execution Pointer

The screen displays ">>" on the left end of the screen and the whole line is underlined when executing a block or at a block that will execute when the [START] switch is pressed.

The colors of the displayed symbols can be changed in the user parameter <Pointer display color>.

Colors: White, blue, green, purple and light blue

7.14.2 Interpretation Pointer

An exclamation point "!" is displayed at the left end of the screen for a block that is being interpreted.

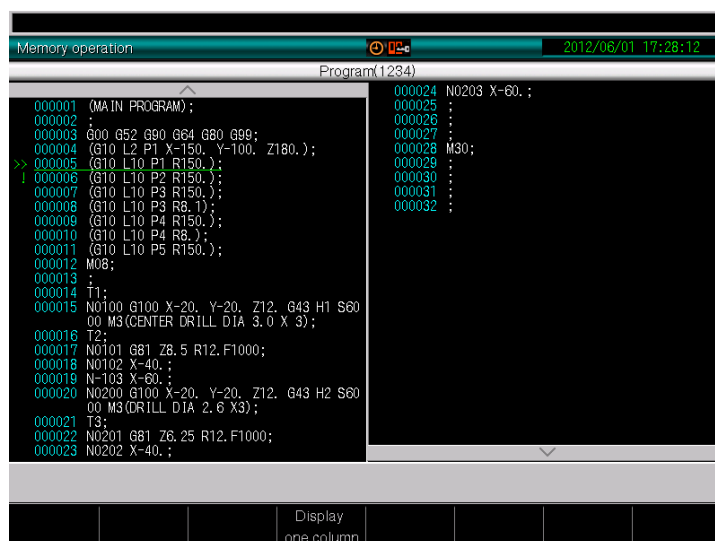
The symbol colors displayed are the same as the execution pointer.

7.14.3 Block Number

The block numbers are displayed between the pointer and the program.

The colors of the these numbers can be changed in the user parameter <Block No. display color>.

Colors: White, blue, green, purple and light blue



7.14.4 Display 1 Column / 2 Columns

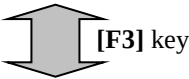
Press the **[F3]** (Display two columns/Display one column) key on the memory operation screen to change the display so the screen shows 24 lines (1 column display) or 50 lines (2 column display).

(Display one column)

```

000001 (MAIN PROGRAM);
000002 ;
000003 G00 G52 G90 G64 G80 G99;
000004 (G10 L2 P1 X-150. Y-100. Z180.);
>> 000005 (G10 L10 P1 R150.);
! 000006 (G10 L10 P2 R150.);
000007 (G10 L10 P3 R150.);
000008 (G10 L10 P3 R8.);
000009 (G10 L10 P4 R150.);
000010 (G10 L10 P4 R8.);
000011 (G10 L10 P5 R150.);
000012 M08;
000013 ;
000014 T1;
000015 N0100 G100 X-20. Y-20. Z12. G43 H1 S6000 M3(CENTER DRILL DIA 3.0 X 3);
000016 T2;
000017 N0101 G81 Z8.5 R12.F1000;
000018 N0102 X-40.;
000019 N-103 X-60.;
000020 N0200 G100 X-20. Y-20. Z12. G43 H2 S6000 M3(DRILL DIA 2.6 X3);
000021 T3;
000022 N0201 G81 Z6.25 R12.F1000;
000023 N0202 X-40.;
000024 N0203 X-60.;
    
```

Display two columns



(Display two columns)

```

000001 (MAIN PROGRAM);
000002 ;
000003 G00 G52 G90 G64 G80 G99;
000004 (G10 L2 P1 X-150. Y-100. Z180.);
>> 000005 (G10 L10 P1 R150.);
! 000006 (G10 L10 P2 R150.);
000007 (G10 L10 P3 R150.);
000008 (G10 L10 P3 R8.);
000009 (G10 L10 P4 R150.);
000010 (G10 L10 P4 R8.);
000011 (G10 L10 P5 R150.);
000012 M08;
000013 ;
000014 T1;
000015 N0100 G100 X-20. Y-20. Z12. G43 H1 S60
00 M3(CENTER DRILL DIA 3.0 X 3);
000016 T2;
000017 N0101 G81 Z8.5 R12.F1000;
000018 N0102 X-40.;
000019 N-103 X-60.;
000020 N0200 G100 X-20. Y-20. Z12. G43 H2 S60
00 M3(DRILL DIA 2.6 X3);
000021 T3;
000022 N0201 G81 Z6.25 R12.F1000;
000023 N0202 X-40.;
000024 N0203 X-60.;
000025 ;
000026 ;
000027 ;
000028 M30;
000029 ;
000030 ;
000031 ;
000032 ;
    
```

Display one column

7.15 Program Restart Function

This function starts operation from a specific part in the middle of a program.

Select type 1 or 2 from the user parameter <Program restart>.

Type 1 is the CNC-B00 compatible function and operation method. Type 2 is the CNC-A00 compatible function and operation method.

7.15.1 Program Restart Function (Type 1)

This function automatically restores operation to the block immediately before the start position specified by the user in order to restart or resume the operation seamlessly as if it had been operating. (It is possible to select whether or not to restore operation for the S, T and M codes.)

7.15.1.1 Specify Start Position

- When specifying the sequence number

- Press the **[MEM]** key.
- Press the **[F1]** (Program restart) key.
- Use the keyboard to enter the sequence number and the repeat start number.

Sequence No. + ([-] Repeat start number) + **[ENT]**

The underlined part can be omitted.

When the same sequence number command is issued multiple times, the repeat start number specifies which sequence number to restart from.

When the repeat start number is omitted, the default setting 1 is used.

The maximum repeat start number is 9999.

(Note 1) The sequence number is recognized as a numerical value. Therefore, the search stops when a number such as "0042" is specified on the keyboard or when any of the following is found in the search: N42, N042, N0042 or N00042.

(Note 2) When the **[B. SKIP]** key is turned ON, any block with a slash "/" is skipped.

- When specifying with the execution pointer

- Press the **[MEM]** key.
- Use the **[CURSOR]** key or the **[PAGE UP]** and **[PAGE DOWN]** keys to move the execution pointer.

- When specifying a saved position

Save the current position in the operation status information while operation is stopped in memory operation mode, and then resume operation from the saved position at the program restart.

- During a block stop, single block stop or pause while in memory operation mode (using the operation status information), press the **[F7]** (Save operation status) key to save the current operation status. The operator message <<The operation status has been saved.>> appears.

Specify the following operation status information:

- Current program No. and block No.
- Sub program call information

In the following situations, the **[F7]** (Save operation status) key cannot be selected. The key is grayed out.

- When using tape operation
- When using a program with 3D smoothed values
- When using an interrupt program to stop memory operation at a given position
- When the position to stop memory operation does not have a restart position that is set (after automatic workpiece measurement or in the middle of smooth path offset)

The operation status information is deleted when one of the following applies.

- When a program is selected that is different from the main program that is saved in the operation status information
- When the power is turned OFF

- (2) Program restart from the saved position
- (i) Press the **[MEM]** key.
 - (ii) Press the **[F1]** (Restart program) key.
 - (iii) Press the **[F7]** (Restart from saved position) key.
The keys above cannot be selected when the operation status information is not saved.
The keys are grayed out.
- (Note)** If a program with a saved position is a program with 3D smoothed values, the alarm <<Program restart error from saved position>> is triggered.
- (3) Specify start position
- (i) The saved operation status information is loaded, the corresponding program is displayed and the pointer used to move the start position for the corresponding block number is displayed. (Start position screen display) The pointer used to move the start position is displayed with the symbol "\$".
 - (ii) Use the **[CURSOR]** key or the **[PAGE UP ▲]** and **[PAGE DOWN ▼]** keys to move the pointer for changing the start position. The start position can only be moved with the pointer in programs that are displayed (programs with saved positions).
- (Note)** When using extended memory operation, the range only includes the three blocks above and below the position saved in the program that is displayed. When attempting to move outside of this range, the alarm <<Outside of specification range.>> is triggered.
- (iii) Press the **[F0]** (Set start position) key.
- (Note)** If operation does not restore to the saved position according to the operation status information (such as when a program is edited so that the level of the sub program call changes), the alarm <<Program restart error from saved position>> is triggered.
- (4) Specify start position again
After setting the start position in section (3), press the **[F1]** (Move start position) key to return to the start position screen and specify the start position again.

7.15.1.2 Restore Operation

After the start position is specified, press the **[START]** switch to restore operation at the block immediately before the block that is specified in the order below.

When using M140X1 or M140X2

1. Restoring to the last S and T code
If there is an S or T command, it executes that code without conditions.
2. After the Z-axis travels using rapid feed to the machine parameter <Distance to Z-axis zero>, the Y-axis travels using rapid feed to the machine zero point.
3. Tool change
It changes to the tool that was selected the last time when M06 or G100 was executed.
4. Restoring to the last M code
It restores operation following the order up to the last modal status in the M code group related to the spindle and coolant.
5. The X-axis travels to the last position on the block immediately before the block specified in the recovery search, travelling at a speed equal to "Rapid feedrate (machine parameter) × "High travel speed (%)" on manual conditions screen".
6. The additional axis travels to the last position on the block immediately before the block specified in the recovery search, travelling at a speed equal to "Rapid feedrate (machine parameter) × "High travel speed (%)" on manual conditions screen".
7. The Y-axis travels to the last position on the block immediately before the block specified in the recovery search, travelling at a speed equal to "Rapid feedrate (machine parameter) × "High travel speed (%)" on manual conditions screen".
8. The Z-axis travels to the last position on the block immediately before the block specified in the recovery search, travelling at a speed equal to "Rapid feedrate (machine parameter) × "High travel speed (%)" on manual conditions screen".
9. Press the **[START]** switch again to start operation from the specified block.

When not using M140X1 or M140X2

1. Restoring to the last S and T code
If there is an S or T command, it executes that code without conditions.
2. The Z-axis travels using rapid feed to the machine parameter <Distance to Z-axis zero>.
3. Tool change
It changes to the tool that was selected the last time when M06 or G100 was executed.
4. Restoring to the last M code
It restores operation following the order up to the last modal status in the M code group related to the spindle and coolant.
5. The X- and Y-axes travel to the last position on the block immediately before the block specified in the recovery search, travelling at a speed equal to "Rapid feedrate (machine parameter) × "High travel speed (%)" on manual conditions screen".
6. The additional axis travels to the last position on the block immediately before the block specified in the recovery search, travelling at a speed equal to "Rapid feedrate (machine parameter) × "High travel speed (%)" on manual conditions screen".
7. Next, the Z-axis travels to the last position on the block immediately before the block specified in the recovery search, travelling at a speed equal to "Rapid feedrate (machine parameter) × "High travel speed (%)" on manual conditions screen".
8. Press the **[START]** switch again to start operation from the specified block.

(Note 1) When the user parameter "S.T.M. return" is set to "0: No", the operations 3 through 4 are not carried out.

(Note 2) The interrupt signal for the interrupt type macro is received during operation (8), and for the M140X1 and M140X2, it is received after operation (9). All signals are ignored prior to this regardless if a signal is input.

(Note 3) The mode cannot be changed. "Restarting program" is displayed when the **[MANU]**, **[MDI]** and **[EDIT]** keys are pressed.

(Note 4) The override is applied to restore operation using the **[RAPID TRAVERSE OVERRIDE]** switch in operation (2) and the **[FEEDRATE OVERRIDE]** switch in operation (5) (or thereafter).

(Note 5) When the machine status has been changed or when the data bank values have been changed, the machine may not restore.

(Example of STM return)

```
N01 G90G54;
N02 T01G100X-100.Y-100.A0.;
N03 M08;
N04 G00Z300.S6000M03;
N05 G01Z250.F1000;
N06 G00Z300.;
N07 T02G100X-100.Y-50.Z300.A-90.M03;
N08 G01Z250.;
N09 G02X-100.Y-50.I0.J-50.F800;
N10 G00Z400.;
N11 X-150.Y0.M05;
N12 M09;
N13 M30;
```

When the user parameter <S.T.M. return> is set to <1: Yes> in the program above, if the program restarts from N08, the operation is restored in the following order after pressing the [START] switch.

When using M140X1 or M140X2

1. The Z-axis raises the distance to Z-axis zero point.
2. The Y-axis travels to the machine zero point.
3. The tool changes to tool number 2.
4. The spindle rotates clockwise at 6000 min⁻¹.
5. The X-axis travels to -100.
6. The A-axis travels to -90.
7. The Y-axis travels to -50.
8. The Z-axis travels to 300.

When not using M140X1 or M140X2

1. The Z-axis raises the distance to Z-axis zero point.
2. The tool changes to tool number 2.
3. The spindle rotates clockwise at 6000 min⁻¹.
4. The X- and Y-axes travel to (-100. and -50.).
5. The A-axis travels to -90.
6. The Z-axis travels to 300.

Press the [START] switch again to start operation from N08.

(Note) When the saved position is specified for the start position
If the machine is started up without pressing the function key to specify the start position, the alarm <<Enter start position.>> is triggered.

7.15.2 Program Restart Function (Type 2)

There are two methods (sequence search and restart) to restart a program.

The sequence search ignores reading the program up to the block immediately before the operation start position specified by the user. (Reading is ignored.)

The same way as in type 1, this function automatically restores operation to the block immediately before the start position specified by the user in order to restart or resume the operation seamlessly as if it had been operating. (It is possible to select whether or not to restore operation for the S, T and M codes.)

7.15.2.1 Sequence Search

Specify start position

1. When specifying the sequence number
 - (1) Press the [MEM] key.
 - (2) Press the [F0] (Sequence search) key.
 - (3) Use the keyboard to enter the sequence number and the repeat start number.
Sequence No. + ([-] Repeat start number) + [ENT]

The underlined part can be omitted.

When the same sequence number command is issued multiple times, the repeat start number specifies which sequence number to restart from (i.e. from the second or third sequence number).

When the repeat start number is omitted, the default setting 1 is used.

The maximum repeat start number is 9999.

In this situation, the operator message <<Restart by a sequence search>> is displayed to alert the user.

(Note 1) The sequence number is recognized as a numerical value. Therefore, the search stops when a number such as “0042” is specified on the keyboard or when any of the following is found in the search: N42, N042, N0042 or N00042.

(Note 2) Even if the [B. SKIP] key is turned ON, any block with a slash “/” is skipped.

(Note 3) The search is only conducted in the main program.

2. When specifying with the execution pointer
 - (1) Press the [MEM] key.
 - (2) Use the [CURSOR] key or the [PAGE UP] and [PAGE DOWN] keys to move the execution pointer. In this situation, the operator message <<Restart by a sequence search>> is displayed to alert the user.

7.15.2.2 Restart

1. Specify start position
 - (1) When specifying the sequence number
It is the same as type 1.
 - (2) When specifying a saved position
It is the same as type 1.
2. Restore operation
The same as type 1.

7.15.3 Special Notes about the Functions

(Note 1) When the user parameter <Prevent restart from middle of program> is set at <1: Yes>, the program restart function cannot be used. (If the cursor key is used to change the operation start position from the beginning of the program and the program is started, the alarm <<Restart from the middle of the program prohibited>> is triggered.)

(Note 2) Any other method other than specifying with the sequence number cannot be used in tape operation.

Any other method other than specifying with the execution pointer cannot be used in extended tape operation.

(Note 3) Special notes when 3D smoothing function is enabled

- **When there is a program restart with a specified sequence number and there is a sequence search**
In a program with 3D smoothed values, the start position cannot be specified in the 3D smoothing section. When the repeat count is specified, the sequence number inside the 3D smoothing section will not be counted.
- **When there is a program restart with a saved position**
In a program with 3D smoothed values, the operation status information cannot be saved and the start position cannot be specified.

7.16 Manual Interruption During Operation and Return

This function uses manual interruption at a given position during program operation.

7.16.1 Procedure for Manual Operation and Recovery

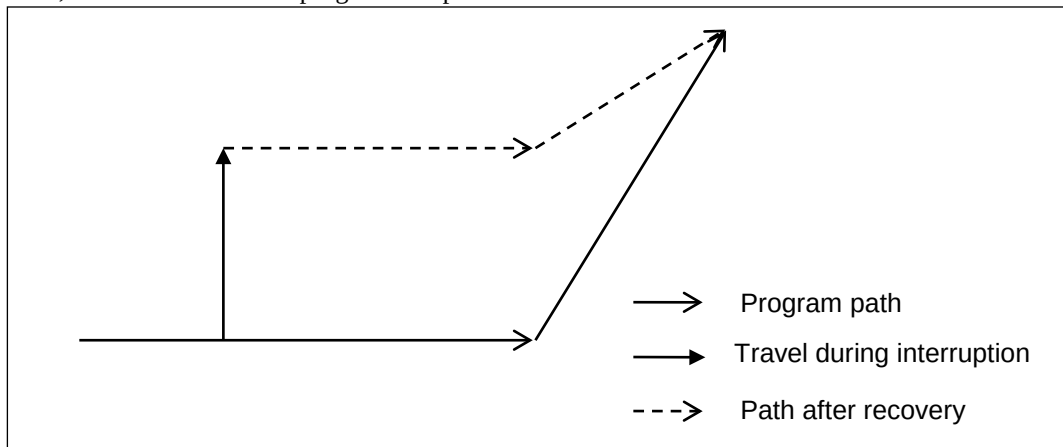
1. Press the **[FEED HOLD]** switch or turn ON the **[SINGL]** key in memory operation to stop the block.
2. Press the **[MANU]** key to switch to manual operation mode.
(If the mode is changed to manual operation mode without performing the operation in 1., the memory operation pauses.)
3. Operation in manual operation mode is possible.
4. To restore memory operation, push the **[MEM]** key to return to memory operation mode and press the **[START]** switch.

7.16.2 Axis Travel after Restoring from Manual Operation Interruption

7.16.2.1 Interruption during Axis Travel in G90 (Absolute Command) Mode

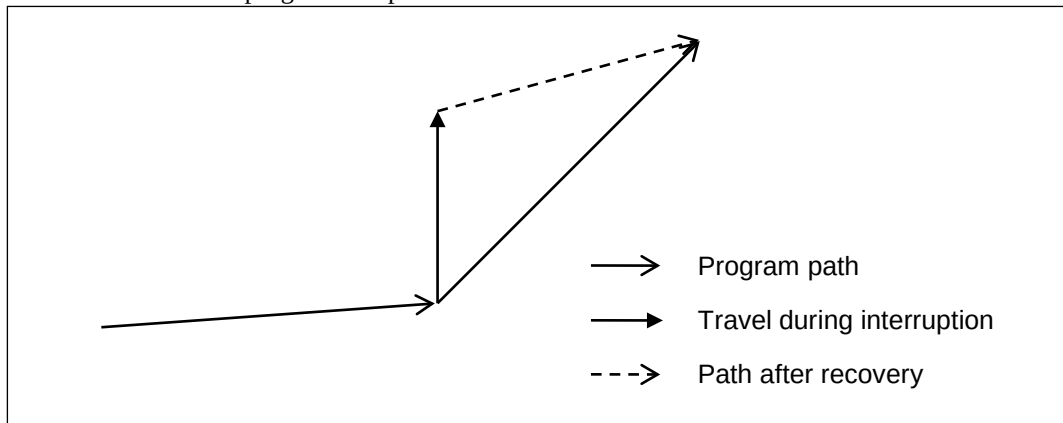
The remaining axis travel, which was not carried due to the interruption, is maintained until the block before interruption finishes travelling.

Then, the axes travel to the programmed position from the next block.



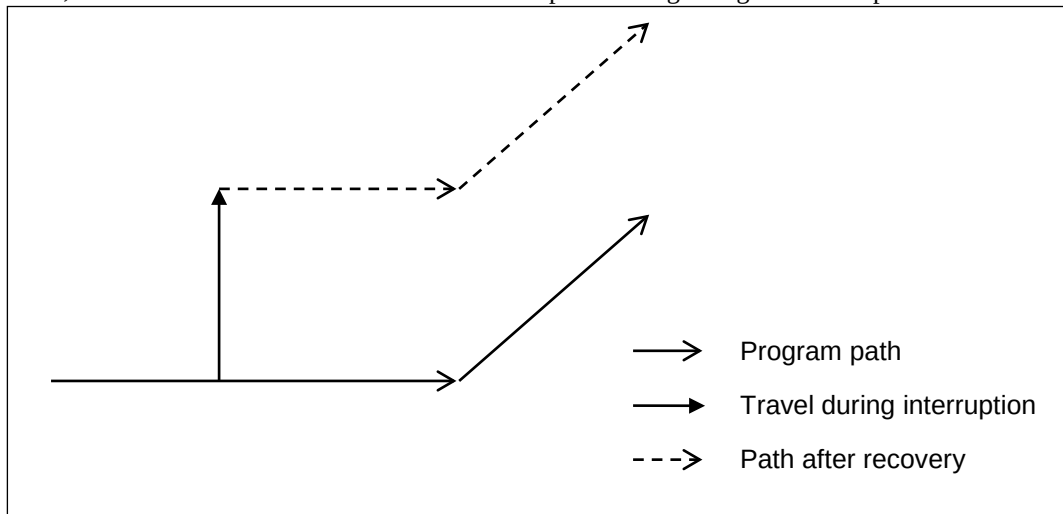
7.16.2.2 Manual Interruption during Single Block Stop in G90 (Absolute Command) Mode

The tool travels to the programmed position.



7.16.2.3 Manual Interruption in G91 (Incremental Command) Mode

The remaining axis travel, which was not carried due to the interruption, is maintained and carried out while in G91 mode. The diagram below is an example of an interruption in the middle of a block, and it is the same as when there is an interruption during a single block stop.



7.16.3 Special Notes

- (Note 1) After machine zero return, tool change, magazine normal rotation, magazine reverse rotation or pallet indexing is executed in manual operation mode, the memory operation is reset when the mode changes to the memory operation mode.
- (Note 2) Manual interruption is not possible while the workpiece is being measured. If the [MANU] key is pressed, the alarm <<During measurement>> is triggered.
- (Note 3) Manual interruption is not possible while the workpiece is being centered. If the [MANU] key is pressed, the alarm <<Automatic alignment mode>> is triggered.
- (Note 4) Manual interruption is not possible during program restart or sequence searching. If the [MANU] key is pressed, the alarm <<Restarting program>> is triggered.
- (Note 5) When returning to memory operation mode and pressing the [START] switch (after performing a spindle stop/spindle normal rotation operation in manual mode), you can configure the settings to return to the spindle (turning spindle) speed/peripheral speed that was used prior to manual operation by selecting <Spindle rotation return during restart> in the user parameter (switch 1). (When set to <1: Return>, if you return to memory operation mode after performing a spindle stop/normal rotation operation in manual mode, the following alarm appears: <<Restore the spindle speed prior to manual operation after startup>>.)
- (Note 6) The additional axis cannot travel while in feature coordinate machining mode (G68.2 modal in progress). The alarm <<Feature coordinate manufacturing mode engaged>> is displayed.

7.17 Tool Life and Changing to Spare Tools

When the tool life has expired, the tool is automatically changed to a spare tool in order to continue machining.

The tools in the same registered group are called spare tools. To use this function, a group number must be assigned to the tools to be used as spare tools in advance on the ATC tool screen.

The current tool is changed to a spare tool when a tool change code (M6, G100) is used. The T-code specifies the tool group number.

The operation varies depending on the user parameter 1setting <Check tool life on first use (0: No, 1: Yes)> and <Spare tool change (0:At operation start 1:At tool change)>.

A description of the operation is provided showing 4 possible case scenarios.

(Program)
 .
 M6 T901;
 :
 M6 T902;
 :
 M6 T901;
 :
 M30;

(1) When the tool life has expired at this point and there is a spare tool
 (2) When the tool life has expired at this point and there is no spare tool

 (3) When the tool life has expired at this point and there is a spare tool
 (4) When the tool life has expired at this point and there is no spare tool

Case	Setting for <Spare tool change>			
	<0: At operation start>		<1: At tool change>	
	Setting for <Check tool life on first use>			
	<0: No>	<1: Yes>	<0: No>	<1: Yes>
(1)	Change with spare tool	Change with spare tool	Change with spare tool	Change with spare tool
(2)	No alarm	Expired tool life alarm	Expired tool life alarm	Expired tool life alarm
(3)	No alarm	No alarm	Change with spare tool	Change with spare tool
(4)	No alarm	No alarm	Expired tool life alarm	Expired tool life alarm

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CHAPTER 8

MDI OPERATION (CONVERSATION)

- 8.1 MDI Operation
- 8.2 Practical Example of MDI Operation

8.1 MDI Operation

This is a function to move an axis as one cycle.

MDI operation

2012/06/01 17:23:14

Coord. set	1:On	Relative coord. pos.
Ext. signal		
Tool	1 - 21:Magazine No.	X -0.420
Tool change	1 - 99:Tool No.	Y -0.420
Coolant	0:Off 1:On	Z 188.976
Spindle	0:Stop 1:Turn right 2:Turn left	QT 0.000
Speed	min*	B1 0.000
Axis travel	0:Pos. setting 1:Workpiece zero return 2:Zero return 3:Zero return(No.2) 4:Zero return(No.3)	B1 0.000
X	Y	Z
A	B	Rad.
Feed	0:Positioning 1:Cutting 2:Tapping	
Feedrate	mm/min	B1 0.000

Coord. set

[HELP]

Automatic centering

Previous operation data

Next operation data

Absolute coordinates

Change additional axes display

With respect to additional axes (4th to 8th-axes and PLC-axis), the axis set in the user parameter <<4th row display axis>> to <<8th row display axis>> and the axis set in the user parameter <<9th row display axis>> to <<12th row display axis>> are alternately indicated every time when the [F4] (Change additional axes display) key is pressed.

8.1.1 Procedure from Start to Finish of One Cycle

1. Push the [MDI] key to call the MDI operation mode. Then the LED of the [PRGRM] key goes lit and the MDI operation screen appears.
2. The blinking cursor prompts to enter data for the item displayed in the data input area. Press the [CURSOR] key and the cursor moves down sequentially. Accordingly, the items indicated in the input request area also change sequentially.
3. Press the [START] switch and the Start lamp goes lit. Then the operation is executed corresponding to the data input in 2. If multiple items are set, the items are sequentially executed from the head.
4. When the operation finishes, the Start lamp goes off and the cursor returns to “Coordinate system setting”.

Push down the [F1] (pre-operation) key or the [F2] (post-operation) key to call the pre-screen or pro-screen of operation history.

8.1.2 Cycle Motion of MDI Operation

1. Coordinate system setting
Setting of coordinate system:
X- and Y -axes: Current machine coordinates are the workpiece zero point.
Z-axis: The workpiece zero point is set on the tabletop.

2. External signal

Actuation signal delivered to an external device (e.g. a preset counter. The signal is basically specified by a user but the following codes are reserved as interior use codes and can be commanded:

- 05: Spindle start
- 08: Coolant valve ON
- 09: Coolant valve OFF
- 19: Spindle orientation (0°)
- 111: Spindle orientation (180°)

The following codes are also interior use codes but cause the alarm (Input data error) when commanded in the MDI mode:

- 00: Feed hold
- 03: Spindle CW rotation
- 04: Spindle CCW rotation
- 30: Program end

3. Tool

The tool of the specified magazine number is changed. Displays as Pot number: 1 to Tool storage capacity (the tool storage capacity depends on the machine parameter <<Tool storage capacity>>.)

4. Tool change

The tool specified by number moves to the tool change position by magazine rotation.

5. Spindle

Spindle rotation/stop

6. Speed

You can freely set the rotational frequencies set in the machine parameter <<Max. spindle speed>>. For tapping, set the spindle rotational frequency in tapping.

7. Axis travel (X, Y, Z, A, B, radius)

Move the X-, Y-, Z-, A-, or B-axis to a random position, the machine zero or the machining zero. Set a radius in "Radius" when the A- or B-axis moves for cutting.

8. Feed

Set a feed method for the X-, Y-, Z-, A-, or B-axis from the following three:

0: DESIGNATE POS

The axes move quickly at the rapid traverse rate set by the machine parameter <<Rapid traverse rate *axis>>.

1: CUTTING

The axes move at a specified cutting feed rate.

2: Tap

Feed rate is determined by spindle speed and tap pitch.

9. Feedrate

Feed rate for cutting

(Note) Spindle rotation and coolant discharge do not stop even if the operation finishes. If you want to stop them, move the cursor to "Spindle" or "Coolant" and set for stopping.

8.2 Practical Example of MDI Operation

This section describes the procedures for data input and MDI operation.

8.2.1 To Set a Coordinate System

1. Press the **[MDI]** key to validate the MDI operation mode and call the MDI operation screen.
2. Using the **[CURSOR]** key, point the cursor at “Coordinate system setting”.
3. Using the numerical keys, input “1” (1: Yes) and press the **[ENT]** key. The “Coordinate system setting” means to set the machining zero.
4. Press the **[START]** switch.
For the X- or Y-axis, the current machine coordinates become the workpiece zero point. For the Z-axis, the workpiece zero point is set on the tabletop.

8.2.2 To Output an External Signal

Example of output: External signal 25

1. Press the **[MDI]** key to validate the MDI operation mode and call the MDI operation screen.
2. Using the **[CURSOR]** key, point the cursor at “External signal”.
3. Using the numerical keys, input “25” and press the **[ENT]** key.
4. Press the **[START]** switch.

8.2.3 To Change Tools

There are two operations (mounting a tool and removing a tool).

(Note) If both “Tool” and “Tool change” are set, the “Tool” motion is first carried out and then the “Tool change” motion is next carried out.

8.2.3.1 To Set a Tool in the Spindle

Screen displays vary depending on machine models as follows:

- Turret magazine machine model
Tool (1 to the number of tools: Magazine number)
- Machine models other than above
Tool (1 to the number of tools: Pot number)

Suppose that the spindle is to mount the tool of the magazine number (or pot number) 3.

1. Press the **[MDI]** key to validate the MDI operation mode and call the MDI operation screen.
2. Using the **[CURSOR]** key, point the cursor at “Tool”.
3. Enter “3” and press the **[ENT]** key.
4. Press the **[START]** switch. After the Z-axis returns to the machine zero, it returns to the ATC zero. The magazine swivels and the tool of the magazine number (pot number) 3 is mounted on the spindle. The magazine swivels in the random shortest path method.

8.2.3.2 To Mount/Remove a Tool

Example: To remove tool No. 5 from the magazine

1. Press the **[MDI]** key to validate the MDI operation mode and call the MDI operation screen.
2. Using the **[CURSOR]** key, point the cursor at "Tool change".
3. Using the numerical keys, input "5" and press the **[ENT]** key.
4. Press the **[START]** switch.

The magazine rotates to set the tool No. 5 to the tool change position. *

- *1 Set the mounting/removing position with the machine parameter "Manual tool change position offset".
- *2 For a turret magazine machine model, after the **[START]** switch is pushed, the Z-axis returns to the ATC zero. Then, the magazine swivels so that the commanded tool comes to the mounting/removing position. Then, the Z-axis returns to the machine zero. So after these motions finish, the spindle equips a tool, which is offset by the amount of the value set by the machine parameter <<Manual tool change position offset>>, from the magazine where the commanded tool exists.

8.2.4 To Discharge or Stop Coolant

Before performing this operation, be sure to turn on the **[CLT .P]** key. In addition, if you do not use coolant, turn off the **[CLT.P]** key without fail.

1. Turn on the **[CLT .P]** key.
2. Press the **[MDI]** key to validate the MDI operation mode and call the MDI operation screen.
3. Using the **[CURSOR]** key, point the cursor at "Coolant".
4. Using the numerical keys, input "1" (1: Yes) and press the **[ENT]** key.
5. Press the **[START]** switch to discharge coolant.
6. To stop discharging coolant, point the cursor at "Coolant" again, input "0" (0: Off) using the numerical keys, and press the **[ENT]** key.
7. If the **[START]** switch is pressed, coolant discharge stops.

8.2.5 Running and Stopping the Spindle

If you turn the spindle rightward at the rotational frequency 4000 min⁻¹:

1. Press the **[MDI]** key to validate the MDI operation mode and call the MDI operation screen.
2. Using the **[CURSOR]** key, point the cursor at "Spindle".
3. Using the numerical keys, input "1" (1: Turn CW) and press the **[ENT]** key. The cursor moves to "Speed".
4. Using the numerical keys, input "4000" and press the **[ENT]** key.
5. Press the **[START]** switch. The spindle turns CW at the rotational frequency 4000 min⁻¹.
6. To stop the spindle, point the cursor at "Spindle" again. Using the numerical keys, input "0" (0: Stop) and press the **[ENT]** key.
7. Press the **[START]** switch. Then the spindle stops rotations.

8.2.6 Feeding an Axis

If you move the X-axis to the -50.000 position at rapid traverse rate:

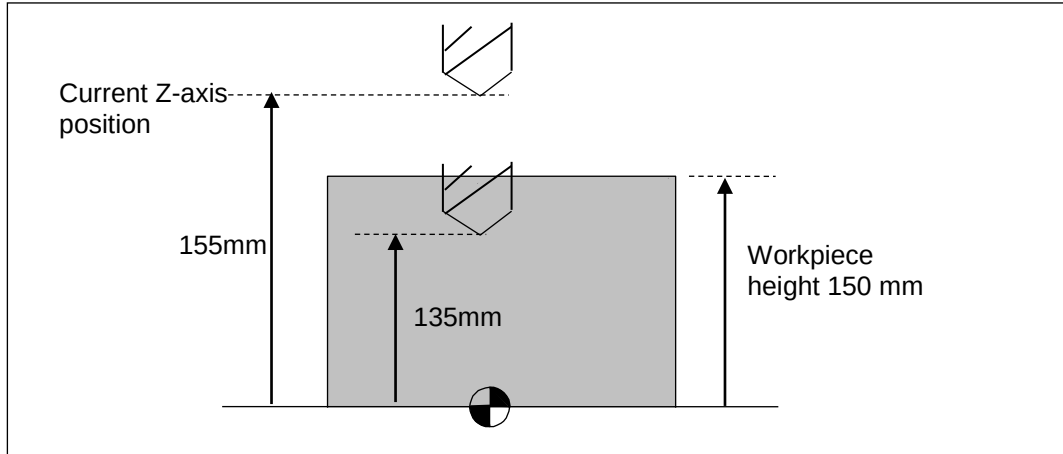
1. Press the **[MDI]** key to validate the MDI mode.
 2. Press the **[POS]** key to call the <Current position> screen, and check the current position.
 3. Press the **[PRGRM]** key to call the MDI operation screen. Using the **[CURSOR]** key, point the cursor at "Axis travel".
 4. Using the numerical keys, input "0" (0: Position command) and press the **[ENT]** key.
 5. Check that the cursor is at the "X" position. Enter -50 and press **[ENT]** key. *
 6. Using the **[CURSOR]** key, point the cursor at "Feed." Using the numerical keys, input "0" (0: Positioning) and press the **[ENT]** key.
 7. When the **[START]** switch is pressed, the X-axis moves to the position (-50.000).
 8. Push the **[POS]** key and confirm the position.
- * Also, be sure to enter a decimal point when entering a value.

8.2.7 Drilling

Here, for example, if you drill a $\phi 3.9$ hole at the 135.0 mm position on the Z-axis under the following conditions:

Spindle speed:	2000 min^{-1} , CW
Feed rate:	200 mm/min
Spindle tool:	Drill $\phi 3.9$ mm
Current Z-axis position:	155 mm (from tabletop)

Example of drilling



1. Press the **[MDI]** key to validate the MDI operation mode and call the MDI operation screen.
2. Point the cursor at "Coordinate system setting," input "1" (1: Yes) using the numerical keys and press the **[ENT]** key. Then if the **[START]** switch is pressed, the tabletop becomes the Z-axis machining zero as shown above.
3. Turn on the **[CLT.P]** key, point the cursor at "Coolant," input "1" (1: Yes) using the numerical keys, and press the **[ENT]** key.
4. Point the cursor at "Spindle," input "1" (1: Turn CW) using the numerical keys, and press the **[ENT]** key. Then, point the cursor at "Speed," input "2000" using the numerical keys, and press the **[ENT]** key.
5. Point the cursor at "Axis travel," input "0" (0: Specify position) using the numerical keys, and press the **[ENT]** key. Then, point the cursor at "Z," input "135." using the numerical keys, and press the **[ENT]** key.
6. Point the cursor at "Feed," input "1" (1: Cutting) using the numerical keys, and press the **[ENT]** key. Then, point the cursor at "Feedrate," input "200" using the numerical keys, and press the **[ENT]** key.
7. Press the **[START]** switch. After the coolant has been discharged and the spindle has started rotation, the Z-axis starts cutting from tabletop into the position 135 mm at 2000 min^{-1} , 200 mm/min.

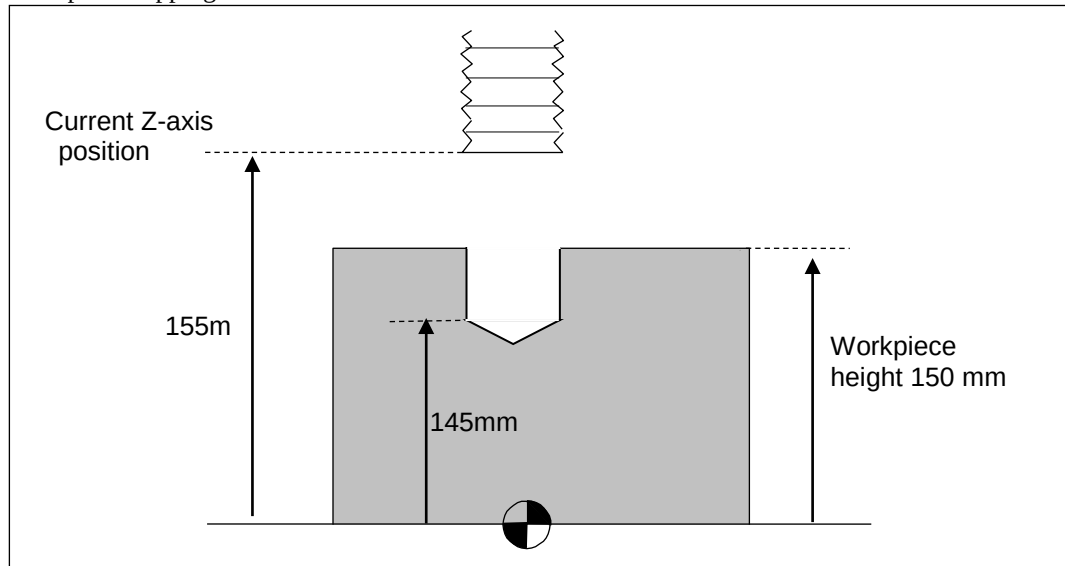
(Note) No step machining is carried out even for deep hole drilling in MDI operation.

8.2.8 Tapping

Here, for example, if you tap a $M4.5 \times P0.75$ hole at 145.0 mm in the Z-axis under the following cutting conditions:

Spindle speed: 2000 min^{-1}
 Pilot hole: $\phi 3.9 \text{ mm}$, 5 mm deep
 Spindle tool: S tapping (spiral tapping), M4.5, pitch 0.75 mm
 Current Z-axis position: 155 mm (from tabletop)

Example of tapping



1. Press the **[MDI]** key to validate the MDI operation mode and call the MDI operation screen.
2. Point the cursor at "Coordinate system setting," input "1" (1: Yes) using the numerical keys, and press the **[ENT]** key. Then, if the **[START]** switch is pressed, the tabletop becomes the Z-axis machining zero as shown above.
3. After the **[CLT.P]** key is turned on, point the cursor at "Coolant," input "1" (1: Yes) using the numerical keys, and press the **[ENT]** key.
4. Point the cursor at "Speed," input "2000" using the numerical keys, and press the **[ENT]** key.
5. Point the cursor at "Axis travel," input "0" (0: Specify position) using the numerical keys, and press the **[ENT]** key. Then, point the cursor at "Z," input "145." using the numerical keys, and press the **[ENT]** key.
6. Point the cursor at "Feed," input "2" (2: Tapping) using the numerical keys, and press the **[ENT]** key.
7. Press the **[START]** switch. Coolant is discharged. The Z-axis starts tapping from the tabletop to the position 145 mm at 2000 min^{-1} , pitch 0.75 mm. When machining finishes, the Z-axis automatic returns to the tapping start position (155 mm).

(Note) The step ratio in the tap step method always becomes 100% in the variable step method.

Variable step method

The variable step method is to decrease the tap step cut-in amount as the cut-in depth becomes deeper. The Nth tap-in amount L (N) is calculated in the following formula:

$$L(N) = \frac{\text{Cut-in coefficient} * \text{Tap outside diameter [mm]} * \text{Step rate [\%]}}{100}$$

N	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15 up
Cut-in coefficient	2.0	1.7	1.3	1.1	0.8	0.7	0.6	0.4	0.4	0.3	0.2	0.2	0.1	0.1	0.1

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CHAPTER 9

MEMORY OPERATION (CONVERSATION)

- 9.1 To Set Tools in ATC
- 9.2 Memory Operation Screen
- 9.3 Cycle Operation
- 9.4 Schedule Program Operation
- 9.5 Test Operation
- 9.6 Feed Hold
- 9.7 Resetting During Memory Operation
- 9.8 Emergency Stop
- 9.9 Start Point Setting
- 9.10 Notes on Functions

9.1 To Set Tools in ATC

On the ATC tool screen, set tools used in memory operation. For details, refer to “4.4 ATC tools”.

9.2 Memory Operation Screen

9.2.1 Memory Operation Screen

The <Memory operation> screen has two teaching displays switchable by the **[F6]** (motion display /machining position display) key. In addition, screen displays vary depending on the memory operation statuses (single operation/cycle operation, absolute mode/incremental mode, and during coordinate rotation). Further, the **[F3]** (relative coordinates/absolute coordinates) key can switch the current spindle coordinate value indication to/from the absolute coordinate system or the relative coordinate system.

9.2.1.1 For Single Operation

Machining position mode

Machining position in current operation (in Nth process) is indicated.

Memory operation		2012/06/04 13:08:18	
Program(0001)			
Workpiece zero	X	-25.000	Y -25.000 Z 100.000
	A	250.000	B 100.000
Return height(job change)	Height	55.000	
Workpieces	2		
	No. 02	X -120.000	Y -150.000
Workpiece material	ADC12		
Job01 Center hole	Center diameter	5.000	Angle 90.000
	Pattern	Circle	
	Divisions	60.000	Holes 4
	Start angle	0.000	Radius 20.000
Z-axis position	Bottom of hole	Depth *	
	Height	50.000	Return height 56.000
XY-axes position	X	-55.000	Y -55.000
Peripheral speed	Peripheral speed	100%	Feed 500%
Next machining		Absolute coordinate pos.	
Job	1	X	-0.420 A
Tool	1	Y	-0.420 B
Machining pos.	1	Z	380.000 P1 0.000
Workpiece No.	1	C	0.300 P2 0.000
Line No.	1	A	0.000 P3 0.000
Infeed	1	B	0.000 P4 0.000
Program No.		[HELP]	
Set start position	Program list	Search	Relative coordinates
		Display Operation	

Press the **[F6]** (Display Operation) key to switch the motion display mode as follows:

Operation display mode

Memory operation		2012/06/04 13:09:35	
Program(0001)			
Workpiece zero	X	-25.000	Y -25.000 Z 100.000
	A	250.000	B 100.000
Return height(job change)	Height	55.000	
Workpieces	2		
	No. 02	X -120.000	Y -150.000
Workpiece material	ADC12		
Job01 Center hole	Center diameter	5.000	Angle 90.000
	Pattern	Circle	
	Divisions	60.000	Holes 4
	Start angle	0.000	Radius 20.000
Z-axis position	Bottom of hole	Depth *	
	Height	50.000	Return height 56.000
XY-axes position	X	-55.000	Y -55.000
Peripheral speed	Peripheral speed	100%	Feed 500%
Current motion		Next motion	
Absolute command		Absolute coordinate pos.	
Coord setting	X	-25.000	X -0.420 A
	Y	-25.000	Y -0.420 B
	Z	100.000	Z 380.000 P1 0.000
	A	250.000	C 0.300 P2 0.000
	B	100.000	A 0.000 P3 0.000
			B 0.000 P4 0.000
Program No.		[HELP]	
Set start position	Program list	Search	Relative coordinates
		Display machining position	
		Next	

Press the **[F6]** (Display machining position) key to switch to the above machining position display mode.

9.2.1.2 For Cycle Operation

Machining position mode

Memory operation		2012/06/04 13:10:21	
Program(0001)			
Workpiece zero	X	-25.000	Y -25.000 Z 100.000
	A	250.000	B 100.000
Return height(job change)	Height	55.000	
Workpieces	2		
Workpiece material	No. 02	X -120.000	Y -150.000
Job01	Center hole	Center diameter	5.000 Angle 90.000
		Pattern	Circle
		Divisions	60.000 Holes 4
		Start angle	0.000 Radius 20.000
		Bottom of hole	* Depth *
		Height	50.000 Return height 56.000
		XY-axes position	X -55.000 Y -55.000
		Peripheral speed	Peripheral speed 100% Feed 500%
Current machining		Absolute coordinate pos.	
Job	1	X	-0.420 A
Tool	1	Y	-0.420 B
Machining pos.	1	Z	380.000 P1 0.000
Workpiece No.	1	C	0.300 P2 0.000
Line No.	1	A	0.000 P3 0.000
Infeed	1	B	0.000 P4 0.000
Program No.		[HELP]	
Set start position	Program list	Search	Relative coordinates
			Display Operation

Press the [F6] (Operation display) key to call the operation status mode.
The same data as for single operation are displayed.

9.2.1.3 For Absolute Mode

Memory operation		2012/06/04 13:12:20	
Program(0603)			
Workpiece zero	X	0.000	Y 0.000 Z 0.000
	A	?	B ?
Return height(job change)	Height	150.000	
Workpieces	1		
Workpiece material	S45C		
Job01	Motion call	Pattern	Off
		Tool No.	01
		Motion 01 Positioning	Specified axis
		X	-100.000 Y -100.000
		Z	400.000
		A	? B ?
		Motion 02 Abso./inc.	Absolute
		Motion 03 Positioning	XY
		X	-150.000 Y -150.000
Current motion		Next motion	
	Positioning	X	-150.000
		Y	-150.000
		Absolute coordinate pos.	
		X	-100.000 A
		Y	-100.000 B
		Z	400.000 P1 0.000
		C	0.300 P2 0.000
		A	0.000 P3 0.000
		B	0.000 P4 0.000
		Relative coordinates	Display machining position
			Next

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eCOM3OP9

9

Memory operation		① 02s		2012/06/04 13:17:41	
Program 0604)					
Workpiece zero	X	0.000	Y	0.000	Z
	A	?	B	?	
Return height(job change)	Height	120.000			
Workpieces	1				
Workpiece material	S45C				
Job01	Coord.setting				
	Measurement No.	?			
	Code	?			
	Indexing angle	A	?	B	?
	Workpiece zero	X	?	Y	?
		Z	?		
	Return H(coord.change)	Height	110.000		
	Return H(job change)	Height	120.000		
	Rotation transformation	Angle	30.000		
	Center X		-10.000	Y	-10.000
Current motion	Next motion	Absolute coordinate pos.			
	Spindle stop	X	-100.000	A	
		Y	-100.000	B	
		Z	420.000	P1	0.000
		C	0.300	P2	0.000
		A	0.000	P3	0.000
		B	0.000	P4	0.000

9.2.1.6 Absolute Coordinate Display and Relative Coordinate Display

Indication of the absolute coordinates

Coordinate values of the current spindle are indicated in the absolute coordinate system.

Memory operation		2012/06/04 13:19:23	
Program(0604)			
Workpiece zero	X	0.000	Y 0.000 Z 0.000
	A	?	B ?
Return height(job change)	Height	120.000	
Workpieces	1		
Workpiece material	S45C		
Job01 Coord. setting	Measurement No.	?	
	Code	?	
	Indexing angle	A	? B ?
	Workpiece zero	X	? Y ?
		Z	?
	Return H(coord change)	Height	110.000
	Return H(job change)	Height	120.000
	Rotation transformation	Angle	30.000
	Center	X	-10.000 Y -10.000
Next machining		Absolute coordinate pos.	
Job	1	X	-100.000 A
Tool	1	Y	-100.000 B
Machining pos.	1	Z	420.000 P1 0.000
Workpiece No.	1	C	0.300 P2 0.000
Line No.	1	A	0.000 P3 0.000
Infeed	1	B	0.000 P4 0.000

		Relative coordinates		Display Operation
--	--	----------------------	--	-------------------

If the [F3] (relative coordinates) key is pressed, the relative coordinates are called.

Relative coordinate display

Coordinate values of the current spindle are indicated in the relative coordinate system.

Memory operation		2012/06/04 13:19:44	
Program(0604)			
Workpiece zero	X	0.000	Y 0.000 Z 0.000
	A	?	B ?
Return height(job change)	Height	120.000	
Workpieces		1	
Workpiece material		S45C	
Job01 Coord. setting	Measurement No.	?	
	Code	?	
	Indexing angle	A	? B ?
	Workpiece zero	X	? Y ?
		Z	?
	Return H(coord change)	Height	110.000
	Return H(job change)	Height	120.000
	Rotation transformation	Angle	30.000
	Center	X -10.000	Y -10.000
Current motion	Next motion	Relative coordinate pos.	
	Absolute command	X	-100.000 A
	Coord setting	Y	-100.000 B
		Z	480.000 P1 -0.498
		C	0.300 P2 0.000
		A	0.000 P3 0.000
		B	0.000 P4 0.000

If the [F3] (absolute coordinates) key is pressed, the relative coordinates are called.

9.2.1.7 Axis to Indicate Coordinate Position

The axis indicated after the X/Y/Z-axis of the coordinate position can be changed by setting the user parameter <<4th row display axis>> to <<12th row display axis>>.

9.2.3 Notes

1. Edit program and memory operation program numbers are independent each other, but they can be matched by setting the user parameter.
For instance, if the memory operation mode is set when the edit program number is 1000, the memory operation program number becomes 1000. In addition, if the edit mode is set when the memory program number is 2000, the edit program number becomes 2000.
Set this with the user parameter <<Set same number to edit/memory operation programs>>.
2. Setting of program No. is not available during a temporary stop, block search, or program restart.
3. If a program which differs from a machine or thread type is used, the <<Error of the kind of screw thread>> alarm appears to disable operation. Refer to “Conversational Programming Manual, Chapter 2, 2.2.2 Hole machining”.
4. If only a folder is moved on the <Program list> screen but no program is selected, the folder movement is cancelled.

9.3 Cycle Operation

All the motions in a program are continuously operated. Use this in normal workpiece machining.

9.3.1 Procedure for Cycle Operation

1. Press the **[MEM]** key to call the memory operation mode. The Program screen appears.
 2. Input a program number or move the cursor to the desired program on the <Program list> screen and press the **[ENT]** key.
 3. When the **[START]** switch is pressed, all the jobs in the program automatically run. *
- * Tools are automatically allocated when the program starts. For tool allocation, refer to “Conversational Programming Manual, Chapter 7 Preparation for Operation”.

9.4.1 Schedule Program Operation Procedure

Memory operation						02c		2012/06/04 14:44:33	
Program(99000515/01)									
Workpiece zero		X	0.000	Y	0.000	Z	0.000		
		A	?	B	?				
Return height(job change)		Height	120.000						
Workpieces		1							
Workpiece material		S45C							
Job01	Center hole	Center diameter	3.000			Angle	118.000		
		Pattern	Point						
		Z-axis position	Bottom of hole	*	Depth	*			
		Height	90.000	Return height	120.000				
		XY-axes position	X	-200.000	Y	-200.000			
		Peripheral speed	Peripheral speed	100%	Feed	100%			
Main program end									
Sub program end									
Current motion		Next motion				Relative coordinate pos.			
		Absolute command				X -100.000 A			
		Coord.setting X				Y -100.000 B			
		Y				Z 480.000 P1 -0.498			
		Z				C 0.300 P2 0.000			
						A 0.000 P3 0.000			
						B 0.000 P4 0.000			
Program No.									
Set start position	Program list	Search	Absolute coordinates			Display machining position	Next	[HELP]	

9.4.2 To Set Schedule Program Number

9.5 Test Operation

9.5.1 Single Operation

1. This function stops the machine each time when [SINGL] key is turned ON. (The LED of the key goes lit.)

9.5.2 Machine Lock

In the machine lock status, the machine does not move but the position display updates as if the machine were operating. This function can be used for checking a program.

Machine lock is effective when the [M. LCK] key is turned ON. (The LED of the key goes lit.)

(Note 1) The machine coordinates on the Position screen do not change at this time.

(Note 2) Operation is reset by pressing the [M. LCK] key.
Editing or communication function is not affected.

(Note 3) Auxiliary functions are not executed.

9.5.3 Dry Run

This is a function to idle (dry run) a machine for checking a program or the like.

It is possible to neglect the feed rate commanded by a program to select a feed rate from the Manual conditions screen by setting the high-speed travel speed (XYZ/P). In addition, it is possible to offset the Z-axis travel command value in the + direction by the setting value of the user parameter “Dry run offset”.

Push the [DRY] key. (The LED goes lit)

If the setting value of the user parameter <<Dry run offset>> is also below the setting value of the user parameter <<Dry run offset lower limit>>, the operator message <<Dry run offset: xx>> appears while the machine is dry-running. (xx is a set dry run offset amount.)

Axis travel speed in dry run

Command	Axis travel speed
Positioning	Parameter <<Rapid traverse rate *axis>>
Cutting	Larger value than the following: • F command value • <<Cutting travel (rotation) top speed * axis>> × “High-speed travel speed (XYZ/P)” on the Manual conditions screen
Tapping	Larger value than the following: • F command value • <<Cutting travel top speed Z-axis>> × “High-speed travel speed (XYZ/P)” on the Manual conditions screen However, if the above value exceeds the feed rate of <<Tapping maximum rotational frequency>>, the axis travel speed becomes the feed rate of <<Tapping maximum rotational frequency>>.

*: Axis X,Y,Z,4~8

(Note) Even if the user parameter <<Dry run offset>> or <<Dry run offset lower limit>> is changed during operation, it is not affected to the operation. (If it is edited and saved, the alarm <<Press push the [RST] key>> appears.)

9.6 Feed Hold

By pressing the **[FEED HOLD]** switch during operation, the axis movement can be held or the program execution can be stopped.

9.6.1 Temporary Stop During Axis Movement

The axis movement stops immediately.

If motions such as tool change and tapping cycle are executed at this time, the feed hold becomes effective upon completion of such motions.

9.6.2 Feed Hold During Dwell

The feed hold becomes effective after a specified dwell time elapses.

9.6.3 Temporary Stop During Dwell

The feed hold becomes effective upon completion of such motions.

9.6.4 Notes

1. The LED of the **[FEED HOLD]** switch is lit on and the LED of the **[START]** switch is lit off.
2. The spindle rotation is not stopped by the **[FEED HOLD]** switch.

9.7 Resetting During Memory Operation

When the **[RST]** key is pressed during memory operation, the operation stops immediately. The operation start point moves to the head of the current program.

When the **[RST]** key is pressed during schedule program operation, the operation also stops. Here, the start position can be selected by the parameter <<Schedule program reset start position>>. If it is set at "0" (0: Program head), the start position returns to the head of the current machining program. If it is set at "1" (1: Schedule program head), the start position returns to the head of the schedule program.

9.7.1 Resetting During ATC

Operation stops after a tool change motion finishes. However, if the tool change is simultaneously done when an X-/Y-positioning is done, the X-/Y-positioning motion immediately stops.

9.7.2 Resetting During Tapping

Operation stops after completing the tap cycle.

9.7.3 Resetting During Other Operations

Operations stop immediately. Spindle rotation also stops.

9.8 Emergency Stop

If you press the **[EMERGENCY]** switch, all motions stop immediately.
To recover from an emergency stop, follow the steps below:

1. Release the **[EMERGENCY]** switch. (Turn it in the reset direction.)
2. Press the **[RST]** key.

9.9 Start Point Setting

The program automatically returns to the process immediately before the start position specified by users as if it were actually operated and then the operation restarts.

After the **[MEM]** key is pushed, push the **[F6]** (machining position indication) key to call the machining position display mode.

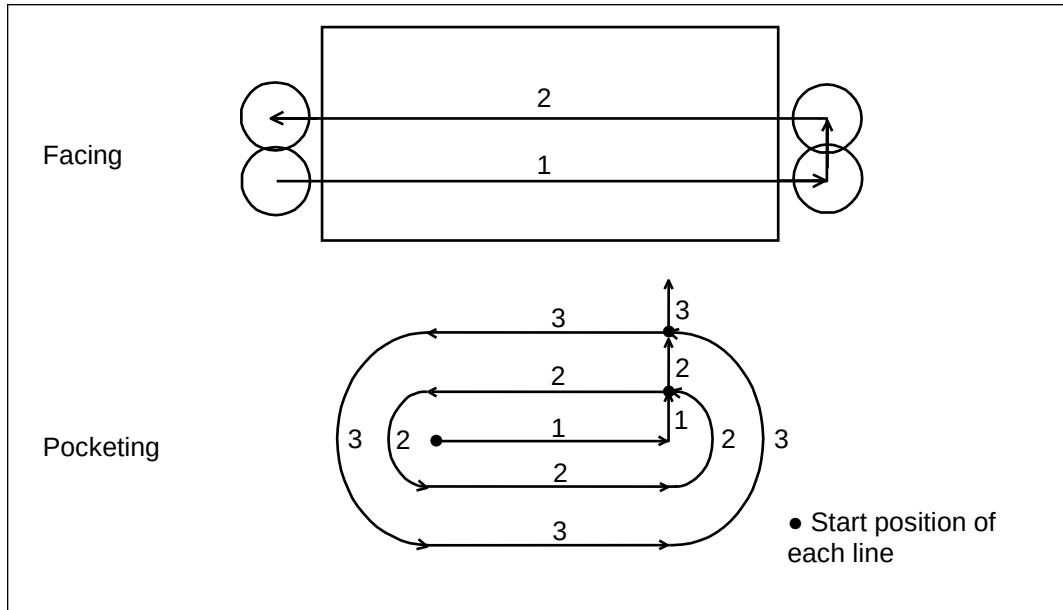
By inputting necessary data into each item indicated at the bottom left of the screen and by setting the start position, you can restart operation from middle of the program.

Memory operation		2012/06/04 14:45:04	
Program(9900/0515/01)			
Workpiece zero	X	0.000	Y 0.000 Z 0.000
	A	?	B ?
Return height(job change)	Height	120.000	
Workpieces		1	
Workpiece material		S45C	
Job01	Center hole	Center diameter	3.000 Angle 118.000
		Pattern	Point
	Z-axis position	Bottom of hole	* Depth *
		Height	90.000 Return height 120.000
	XY-axis position	X	-200.000 Y -200.000
	Peripheral speed	Peripheral speed	100% Feed 100%
Main program end			
Sub program end			
Current machining		Relative coordinate pos.	
Job	1	X	-100.000 A
Tool	1	Y	-100.000 B
Machining pos.	1	Z	480.000 P1 -0.498
Workpiece No.	1	C	0.300 P2 0.000
Line No.	1	A	0.000 P3 0.000
Infeed	1	B	0.000 P4 0.000
Program No.		[HELP]	
Set start position	Program list	Search	Absolute coordinates
			Display Operation

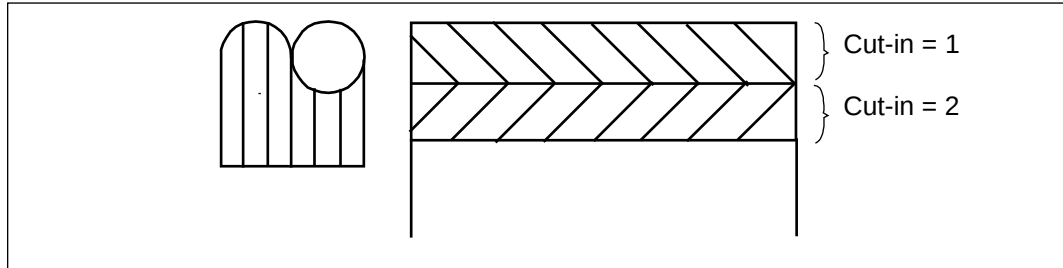
9.9.1 Start Point Display

1. Job
Indicates the current job.
2. Tool
Indicates the order of the tools used in the job. This is different from the tool pattern's tool order registered in the data bank.
3. Machining position
Indicates the order of the holes drilled using a drill pattern (circumference, square, etc.).
4. Workpiece number
Indicates the order of the workpiece currently being machined.
5. Line number
Indicates the order of the motions set in the motion call job. In the facing or pocketing process, the line numbers in machining are displayed as shown below. In the contouring or chamfering process, the line numbers in machining are indicated.
6. Cut-in
In the facing, contouring, or pocketing process, the cut-in face order is indicated.

Line number



Cut-in



9.9.2 Start Point Setting Procedure

1. Press the **[MEM]** key to call the memory operation mode.
2. Input the desired program number and press the **[ENT]** key.
3. Press the **[F0]** (Start position setting) key.
4. Using the cursor keys, move the cursor to the **[Job]** where machining starts, or call the job number by using the numeral keys, and then press **[ENT]** key.
You can use the cursor keys to move the cursor to "Job" even while setting other items. When the cursor is moved to **[Job]**, all other items are reset to "1".
5. Enter the position number where machining starts for "Tool", "Machining pos.", "Workpiece No.", "Line No.", and "Infeed". Press the **[ENT]** key for each item. (If the indication area has "Motion status", the message "Starting point is set" appears in the indication area of "Next motion".)
6. Press the **[START]** switch.

9.9.2.1 Search Function

You can search a start position by inputting the job, tool, machining position, and workpiece number in the Input request area in the order as indicated below and pressing the **[F2]** (Search) key:

“Job” [/] “Tool” [/] “Machining pos.” [/] “Workpiece No.” **[F2]** (Search)

(Note 1) Both the program number setting mode and the start point setting mode are available.

(Note 2) The order of the tools used in the job, instead of the tool number, are set for [Tool].

(Note 3) After the search completes, enter the start point setting mode, and the item next to the latest specified item is displayed in the input request area.

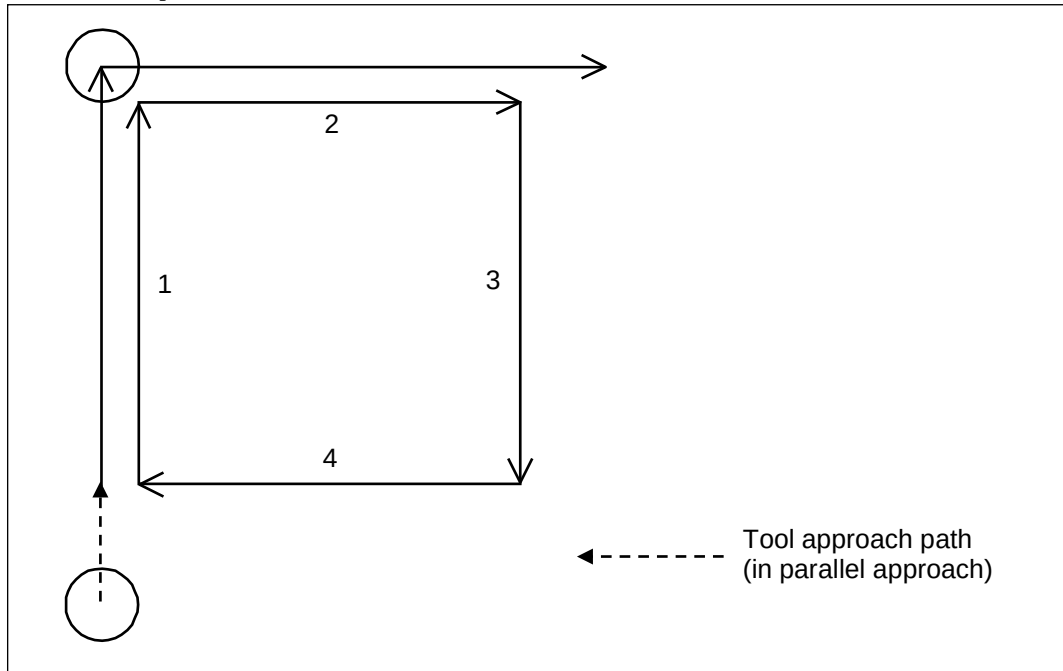
Ex.)

When specified up to a process:	Tool
When specified up to a tool:	Machining position
When specified up to a machining position:	Workpiece position
When specified up to a workpiece number:	Line number

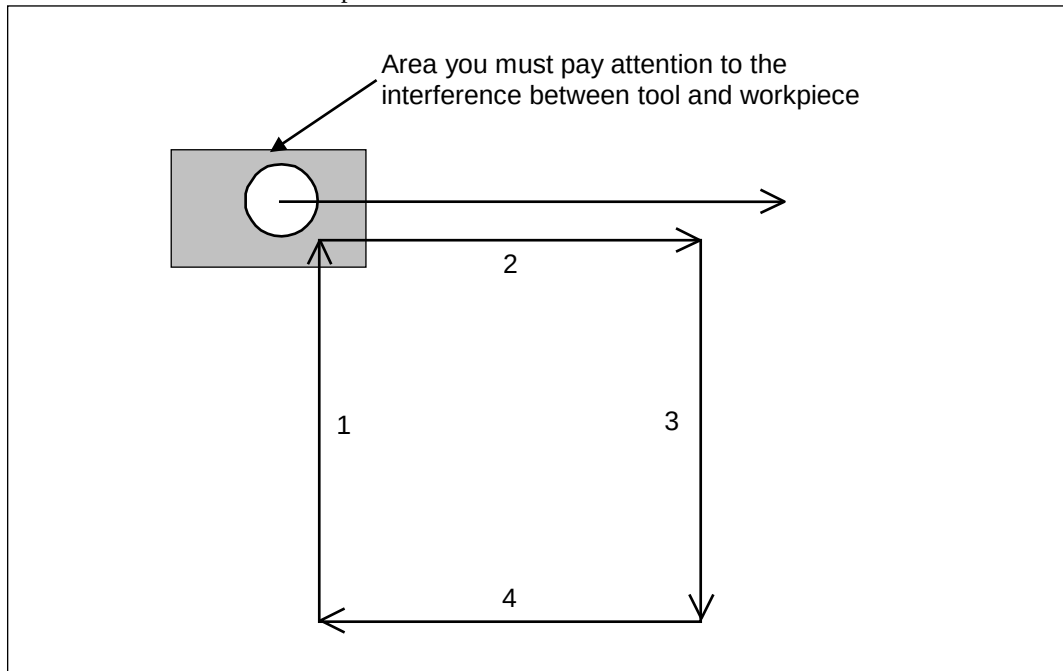
9.9.3 Precautions for Start Point Setting

1. Coordinate system setting job
Moving the cursor to Coordinate system setting is not possible when setting a start point.
2. Contouring and chamfering jobs
Multiple lines can be set in one job for contouring and chamfering. If the start point is set to a middle block, the approach must be performed first, so you must be very careful of interference between the tool and the workpiece. Accordingly, set the start point for each job, instead of setting it to the middle block.

Example
Normal travel path



When the line 2 is set as a start position



9.10 Notes on Functions

If the user parameter <<Prevent restart from middle of program>> is set with <<Prohibited>>, the above operation is unavailable. (If the start position is changed from the head by the cursor key and the like and the operation is started, the alarm <<Restart from the middle of the program prohibited>> appears.)

CHAPTER 10

QT AXIS

This function is not included in the product, and the description of this section is omitted.

CHAPTER 11

ALARMS

- 11.1 Types of Messages**
- 11.2 Message Display**
- 11.3 Message Format**
- 11.4 Stop Level and Cancel Level for Alarms and Operator Messages**
- 11.5 Display When Message Includes Additional Axis**
- 11.6 Alarm Display**
- 11.7 Alarm Code List**
- 11.8 List of servo errors**
- 11.9 ATC Recovery Procedure**
- 11.10 Collision detection function**
- 11.11 ATC Monitoring Function**
- 11.12 Tool Missing Detection Function**
- 11.13 Machining Load Monitor Function**

11.1 Types of Messages

There are two major types of CNC messages that alert the user.

One type is an “alarm” that alerts the user of an error that has been detected by the CNC. The other type is an “operator message” that warns the user or prompts the user to check.

In addition, the alarms are further divided into the categories shown in the table below according to the cause.

Category	Abbr.	Example
NC PCB hardware	NC	Watch dog, ROM/RAM error
IO/Servo communication	EC	IO/servo periodic communication time-out
IO PCB hardware	IO	Emergency switch, 200 V power supply error
Servo	SV	Servo error, *-axis position shifted In-position check timeout
System software (local)	SL	Kernel error, memory is not formatted, edit error
PLC	SP	Command code is abnormal, ladder program stopped
Local communication	CM	Communication timeout, communication buffer overflow
System software (main)	SM	Kernel error, command error
External signal	EX	External alarm
Panel communication	ES	Teaching pendant data error
Field bus	FC	Field bus watch dog, FC PCB communication error

11.2 Message Display

Alarms and operator messages that are triggered are displayed in the alarm/operator message area on the top of each screen.

If an alarm is triggered, the alarm flashes on the right side and is displayed over the operator message so that only part of the operator message can be seen. The operator message does not flash.

If multiple alarms and operator messages are triggered simultaneously, the ones with the highest priority are displayed.

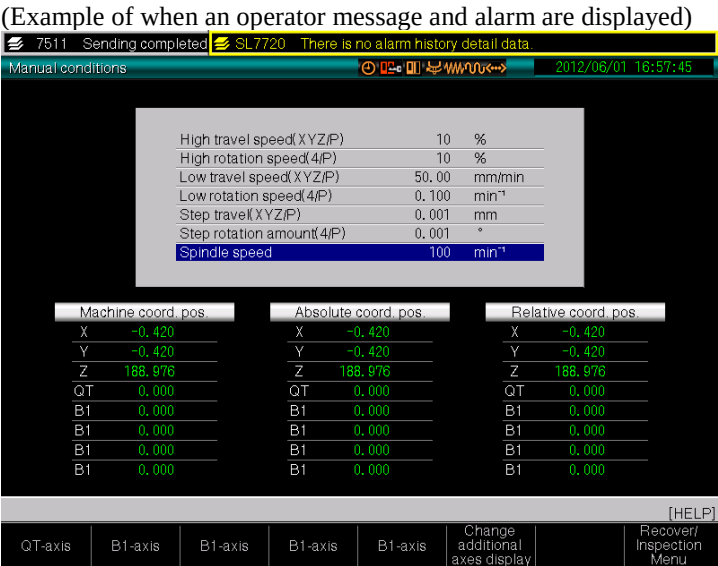
However, when a new alarm or operator message is triggered, it displays up to three seconds regardless of the priority.

When this occurs, an icon is displayed on the left of each message to notify the user that there are also other messages.

Note that the alarms and operator messages which are not displayed can be checked on the current alarm screen. (Refer to “11.6.1 Current Alarms and Operator Messages”.)

In addition, multiple icons describing the machine status are displayed to the left of the clock display on the title line.

* Refer to “Chapter 3 - 3.4 Screen Layout” for further details.



The display for the alarm/operator message area varies according to the status of alarm(s) and operator message(s) that have been triggered.

- When only an operator message is triggered

- When only an alarm is triggered

↑ Display flashes

- When an alarm and an operator message are triggered

↑ Display flashes


Lowering the priority status of an alarm or operator message:
An alarm or an operator message can be set so that it does not display when multiple alarms and operator messages are triggered simultaneously. In this way, only alarms or operator messages with higher priority are displayed in the alarm and operator message area when multiple alarms and operator messages are triggered simultaneously.
Although the priority is determined according to the importance of the alarm and the time it is triggered, the lowest priority is assigned to alarms and operator messages with a priority setting that has been lowered by the user.
To lower the priority setting of an alarm or operator message, register and set the alarm or message number in the user parameter <Low priority indication alarm No.*> (*:1 to 10).

(Note) The display priority cannot be lowered for alarms and operator messages with a cancel level 3.

11.3 Message Format

11.3.1 Alarms

`XXnnnn **alarm description` or `XXnnnn.ddd **alarm description`

XX : These two letters note the cause of the alarm. Refer to “11.1 Types of Messages”.

nnnn : These four numbers are the alarm number.

ddd : These three numbers are the auxiliary number.

** : The total number of asterisks (*) identifies the cancel level.

Alarm description : This is the name or description of the alarm.

11.3.2 Operator Messages

`nnnn **message description` or `nnnn.ddd **message description`

nnnn : These four numbers are the message number.

ddd : These three numbers are the auxiliary number.

** : The total number of asterisks (*) identifies the cancel level.

Message description : This is the name or description of the operator message.

11.4 Stop Level and Cancel Level for Alarms and Operator Messages

11.4.1 Stop Level

NC operation stops as indicated below according to each stop level.

- Stop level 5
Servo turns OFF.
- Stop level 4
The operation currently being executed stops immediately, and all operations thereafter are disabled.
- Stop level 3 (Block stop)
Operation is executed until the block currently being executed is finished, however operation does not execute thereafter. In addition, the start switch is disabled and the NC program does not execute.
- Stop level 2 (Stops at M02 and M30 commands)
Operation is executed until the program currently being executed is finished. Thereafter, the start switch is disabled and the NC program does not execute.
When the single block is ON, stop level 3 and stop level 2 are the same.
- Stop level 1
Only a warning is issued and no operations or motions are affected.

11.4.2 Cancel Level

The total number of asterisks (*) and the color of the message are used to identify the cancel level of the alarm and operator message.

- Cancel level 3: Two asterisks (**) with a red message color
- Cancel level 1: One asterisk (*) with a yellow message color
- Cancel level 1: No asterisks with a white message color

(Note) In the alarm/operator message area, the message color display for cancel levels 1 through 3 are yellow for alarms and white for operator messages.

In addition, follow the procedure below for each level to cancel the alarm and operator message.

- Cancel level 3
The power must be turned OFF to cancel the alarm or message.
(Turn the power OFF and then ON to cancel the alarm and operator message.)
- Cancel level 2
Press the **[RST]** key to cancel the alarm or message.
- Cancel level 1
Press any key to cancel the alarm or message.

11.4.3 Cancel Level Using ALMRLS Signal

If the following alarm is triggered, it can be cancelled by inputting the ALMRLS signal as an external input signal.

- When the alarm is an <External error**> (**:10 to 99) and the “Cancel level” for the external I/O signal setting is “4: ALMRLS signal”
The message display and color for the alarm are:
Message is pink with an asterisk (*)
Pressing the **[RST]** key also cancels this alarm.

11.4.4 Notes

The stop level and the cancel level are independent from each other. Therefore, they can be combined in different ways as shown below.

- Stop level 5 and cancel level 2
- Stop level 1 and cancel level 2

11.5 Display When Message Includes Additional Axis

The display for the additional axis in the message changes depending on the setting of the following machine parameters.

5th-axis position (0: No pallet 1: Table 1 2: Table 2)

6th-axis position (0: No pallet 1: Table 1 2: Table 2)

7th-axis position (0: No pallet 1: Table 1 2: Table 2)

8th-axis position (0: No pallet 1: Table 1 2: Table 2)

4th-axis address (0: A 1: B 2: C 3: QT)

5th-axis address (0: A 1: B 2: C)

6th-axis address (0: A 1: B 2: C)

7th-axis address (0: A 1: B 2: C)

8th-axis address (0: A 1: B 2: C)

		Installation position		
		0: No pallet	1: Table 1	2: Table 2
Address	0: A	A-axis	A1-axis	A2-axis
	1: B	B-axis	B1-axis	B2-axis
	2: C	C-axis	C1-axis	C2-axis
	3: QT	QT-axis		

[illegible]

- Displays the alarm number and message, or the operator message number and message, and the program number and job/block number where the alarm or operator message occurred.
- Displays alarms and operator messages in descending order of importance when multiple alarms and operator messages are triggered simultaneously.
- Displays up to 18 alarms. When alarms and operator messages are triggered simultaneously, up to two operator messages are displayed.
- Move the cursor to the alarm or operator message to reference the recovery procedure. Press the **[ENT]** key, or enter the alarm number or operator message number and press the **[ENT]** key. Then, the alarm recovery screen is displayed for the selected alarm or operator message. However, if the alarm is a servo alarm (servo error, servo warning, servo communication error, servo status error) with an auxiliary number, after selecting it with the cursor and pressing the **[ENT]** key, a servo alarm list screen is displayed.

11.6.2 Alarm Log (List)

When a critical error occurs, the alarm log records and stores that alarm in a list with a maximum storage capacity of 1000 alarms.

The alarm log screen can be used to reference the details of the alarm and operator message. Press the **[F1]** (Alarm log list) key on the current alarm and operator message screen. A screen like the one below displays.

Alarm log(List)			2016/10/07 11:24:13		
Alarm	Message	Lingo	Time	Date	
SM2124	*Carry out spindle run-in in manual operation mode.	NC	11:21:58	2016/10/07	
0506	*Enter your password.	NC	11:21:55	2016/10/07	
1500	**Turn power off.	NC	11:17:09	2016/10/07	
0503	*Press the [RST] key.	Conv.	11:16:33	2016/10/07	
0503	*Press the [RST] key.	NC	11:15:04	2016/10/07	
0506	*Enter your password.	NC	11:10:59	2016/10/07	
1500	**Turn power off.	NC	10:57:08	2016/10/07	
4000	*Press the [RST] key.	NC	10:32:31	2016/10/07	
0506	*Enter your password.	NC	10:26:33	2016/10/07	
1500	**Turn power off.	NC	10:24:17	2016/10/07	
SM0000	**Machine setting error	NC	10:21:14	2016/10/07	
SL2006.002	*Machine param. setting error(SYS 2)	NC	10:18:41	2016/10/07	
SL2006.001	*Machine param. setting error(SYS 2)	NC	10:18:41	2016/10/07	
SM0003.001	**Machine param. setting error(SYS 2)	NC	10:18:41	2016/10/07	
SL2006.002	*Machine param. setting error(SYS 2)	NC	10:16:47	2016/10/07	
4000	*Press the [RST] key.	NC	10:16:37	2016/10/07	
1500	**Turn power off.	NC	10:16:37	2016/10/07	
To display details: Move the cursor to the item and press [ENT].					
Current alarm	Alarm log list	Sort by time	Sort by number	Sort by alarms	Collision log list
					Leading

The function keys can be used to sort the alarms chronologically, by alarm number or by the frequency of the alarm.

In addition, use the **[F5]** (Collision log list) key to display the <Collision log (list)>.

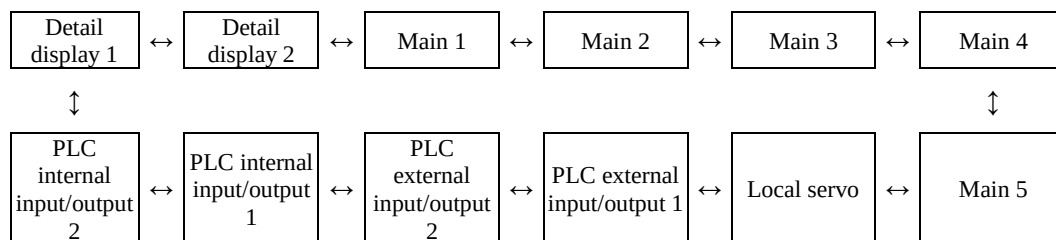
Refer to "Chapter 11 - 11.10 Collision Detection Function" for further details.

- * The alarm numbers and the stop and cancel levels are listed on the alarm log (list) screen and they show the status of the alarm that was triggered. However, if the data bank has been modified after the alarm was triggered, the following information displayed in the list is different from the original information that was recorded when the alarm was triggered.

- External alarm message
- Tool name
- Additional axis address

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eCOM3OP11-1



(Note) When the alarm log shows the operator message <<Basic axis servo data recorded>> and <<PLC axis servo data recorded>>, no signal information is displayed for the screens below.

- <Alarm log (Main 1 to 5)>
- <Alarm history (Local, Servo)>
- <Alarm log (PLC external input/output 1 to 2)>
- <Alarm log (PLC internal input/output 1 to 2)>

Detail display 1 (alarm that was triggered in NC language)

Alarm history(Detail display 1)							2019/04/02 14:48:17																																													
Alarm		Message				Lingo	Time	Date																																												
4001		*Check ATC tool.				NC	16:31:25	2019/03/26																																												
Mode		Screen	Edit program		Operation program		Block No.																																													
EDIT		PRGRM	Y0K01/4013		Y0K01/4013																																															
Machine coordinate pos.		Distance left		Execution block																																																
X -0.789		X 0.000		<table><tr><th colspan="2">Tool offset</th><th colspan="2">Cutter compensation</th></tr><tr><td colspan="2"></td><td colspan="2"></td></tr><tr><td colspan="2">X</td><td colspan="2"></td></tr><tr><td colspan="2">Y</td><td colspan="2"></td></tr><tr><td colspan="2">Z</td><td colspan="2"></td></tr><tr><th colspan="2">Spindle tool</th><th colspan="2">Spindle speed</th></tr><tr><td colspan="2"></td><td colspan="2">0 min⁻¹</td></tr><tr><th colspan="2">Override</th><th colspan="2">Feedrate</th></tr><tr><td colspan="2">Rapid feed 100 %</td><td colspan="2"></td></tr><tr><td colspan="2">Cutting feed 100 %</td><td colspan="2"></td></tr><tr><td colspan="2">Spindle 100 %</td><td colspan="2"></td></tr></table>					Tool offset		Cutter compensation						X				Y				Z				Spindle tool		Spindle speed				0 min ⁻¹		Override		Feedrate		Rapid feed 100 %				Cutting feed 100 %				Spindle 100 %			
Tool offset		Cutter compensation																																																		
X																																																				
Y																																																				
Z																																																				
Spindle tool		Spindle speed																																																		
		0 min ⁻¹																																																		
Override		Feedrate																																																		
Rapid feed 100 %																																																				
Cutting feed 100 %																																																				
Spindle 100 %																																																				
Y -0.890		Y 0.000																																																		
Z 479.434		Z 0.000																																																		
QT 0.000		QT 0.000																																																		
A1 0.000		A1 0.000																																																		
B1		B1																																																		
A2 0.000		A2 0.000																																																		
B2		B2																																																		
Magazine 314.262																																																				
P1		P1																																																		
P2		P2																																																		
P3		P3																																																		
P4		P4																																																		
AT 0.000																																																				

☒ Alarm history(PLC internal input/output 2)
☒ Alarm history(Detail display 2)

Alarm history	Current alarm					

Detail display 1 (alarm that was triggered in conversation language)

Alarm history(Detail display 1)					2019/04/02 16:36:13	
Alarm	Message			Lingo	Time	Date
4000	*Press the [RST] key			Conv.	14:54:59	2019/04/02
Mode	Screen	Edit program	Operation program	Job number		
MANU	DATA BANK	MASAK1/0002	MASAK1/9902/0001/??	01		
Machine coordinate pos		Distance left		Machining		
X	-100.010	X	0.000	Job	01	Workpiece No.
Y	-200.016	Y	0.000	Tool	01	Line No.
Z	477.031	Z	0.000	Machining position	001	Infeed
QT	-180.150	QT	0.000	Current motion		
A1	-0.376	A1	0.000			
B1		B1				
A2	38.887	A2	0.000			
B2		B2				
Magazine	254.000					
P1		P1				
P2		P2				
P3		P3				
P4		P4				
AT	0.000					
				Tool length offset		
				Cutter compensation		
				Spindle tool		
				Spindle speed		
				0 min ⁻¹		

☒ Alarm history(PLC internal input/output 2)
☒ Alarm history(Detail display 2)

Alarm history	Current alarm						Next

Detail display 2 (alarm that was triggered in NC language)

Alarm history (Detail display 2)					2019/04/02 14:49:21		
Alarm	Message			Lingo	Time	Date	
4000	*Press the [RST] key			NC	16:31:24	2019/03/26	
Mode	Screen	Edit program	Operation program	Block No.			
EDIT	PRGRM	YOK01/4013	YOK01/4013				
Machine coordinate pos		Distance left	Door	Door interlock			
X	-0.789	X 0.000	Outer door Close	Valid			
Y	-0.890	Y 0.000					
Z	479.434	Z 0.000	Side door Close				
QT	0.000	QT 0.000					
A1	0.000	A1 0.000					
B1		B1					
A2	0.000	A2 0.000					
B2		B2					
Magazine	314.262						
P1		P1					
P2		P2					
P3		P3					
P4		P4					
AT	0.000						
☐ Alarm history (Detail display 1) ☒ Alarm history (Main1)							
Alarm history list	Current alarm						

Detail display 2 (alarm that was triggered in conversation language)

Alarm history (Detail display 2)					2019/04/02 16:36:32		
Alarm	Message			Lingo	Time	Date	
4000	*Press the [RST] key			Conv	14:54:58	2019/04/02	
Mode	Screen	Edit program	Operation program	Job number			
MANU	DATA BANK	MASAK I/0002	MASAK I/9902/0001/??	01			
Machine coordinate pos		Distance left	Feedrate	Door interlock			
X	-100.010	X 0.000		Valid			
Y	-200.016	Y 0.000					
Z	477.031	Z 0.000	Override				
QT	-180.150	QT 0.000	Rapid feed 100 %	Outer door Close			
A1	-0.376	A1 0.000	Cutting feed 100 %	Side door Close			
B1		B1	Spindle 100 %				
A2	38.887	A2 0.000					
B2		B2					
Magazine	254.000						
P1		P1					
P2		P2					
P3		P3					
P4		P4					
AT	0.000						
☐ Alarm history (Detail display 1) ☒ Alarm history (Main1)							
Alarm history list	Current alarm						Next

Main 1

Alarm log (Main1)					2012/06/01 17:16:50		
Alarm	Message			Lingo	Time	Date	
4000	*Press the [Reset] key			Conv	14:39:51	2012/06/01	
Input				Output			
FEDC	BA98	7654	3210	FEDC	BA98	7654	3210
1	1101	0000	0000	0000	0000	0000	0000
2	1111	0001	0000	0100	18	0000	0000
3	0000	0011	0000	1111	19	0000	0000
4	0000	0000	0000	0000	20	0000	0000
5	0000	0011	1111	1101	21	0000	0000
6	0000	0000	0000	0101	22	0000	0000
7	0000	0000	1111	1111	23	0000	0000
8	1100	0000	0010	1111	24	0000	0000
9	0000	0000	0000	0000	25	0000	0000
10	0000	0000	0101	0101	26	0000	0000
11	0000	0000	1100	0000	27	0000	0000
12	0000	0000	0000	0000	28	0000	0000
13	0000	0000	0000	0000	29	0000	0000
14	0001	1110	0000	0000	30	0000	0000
15	0000	0000	0000	0000	31	0000	0000
16	0000	0000	0000	0000	32	0000	0000
☐ Alarm history (Detail display 2) ☒ Alarm log (Main2)							
Alarm log list	Current alarm						

Main 2

Alarm log(Main2)										2012/06/01 17:17:08				
Alarm		Message						Lingo	Time	Date				
4000		*Press the [Reset] key.						Conv.	14:39:51	2012/06/01				
Input								Output						
	FEDC	BA98	7654	3210		FEDC	BA98	7654	3210		FEDC	BA98	7654	3210
33	0000	0000	0000	0000	49	0000	0000	0000	0001	33	0000	0000	0000	0000
34	0000	0000	0000	0000	50	0000	0000	0000	0000	34	0000	0000	0000	0000
35	0000	0000	0000	0000	51	0000	0000	0000	0000	35	0000	0000	0000	0000
36	0000	0000	0000	0000	52	0000	0000	0000	0000	36	0000	0000	0000	0000
37	0000	0000	0000	0000	53	1000	0000	0000	0000	37	0000	0000	0000	0000
38	0000	0000	0000	0000	54	0000	0000	0000	0000	38	0000	0000	0000	0000
39	0000	0000	0000	0000	55	0000	0000	0000	0000	39	0000	0000	0000	0000
40	0000	0000	0000	0000	56	0000	0000	0000	0000	40	0000	0000	0000	0000
41	0000	0000	0000	0000	57	0000	0000	0100	0000	41	0000	0000	0000	0000
42	0000	1111	0000	0010	58	0000	0000	0000	0000					
43	0010	0000	0100	0010	59	0000	0000	0000	0000					
44	0000	0000	0000	0000	60	0000	0000	0000	0000					
45	0000	0000	0000	0010	61	0000	0000	0000	0000					
46	0000	1000	1101	0000	62	0000	0000	0000	0000					
47	0000	0000	0010	0001	63	0000	0000	0000	0000					
48	0000	0000	0000	0011	64	0000	0000	0000	0000					
☐ Alarm log(Main1) ☒ Alarm history (Main 3)														
Alarm log list	Current alarm													

Main 3 to 5

Alarm history (Main 3)										2012/06/01 17:17:21		
Alarm		Message						Lingo	Time	Date		
4000		*Press the [Reset] key.						Conv	14:39:51	2012/06/01		
Input												
	FEDC	BA98	7654	3210		FEDC	BA98	7654	3210			
65	0000	0000	0000	0000	81	0000	0000	0000	0000			
66	0000	0000	0000	0000	82	0000	0000	0000	0000			
67	0000	0000	0000	0000	83	0000	0000	0000	0000			
68	0000	0000	0000	0000	84	0000	0000	0000	0000			
69	0000	0000	0000	0000	85	0000	0000	0000	0000			
70	0000	0111	1111	1111	86	0000	0000	0000	0000			
71	1111	1111	1111	1111	87	0000	0000	0000	0000			
72	1111	1111	1111	1111	88	0000	0000	0000	0000			
73	0000	0000	0000	0000	89	0000	0000	0000	1001			
74	0000	0000	0000	0000	90	0000	0000	0000	0000			
75	0000	0000	0000	0000	91	0000	0000	0000	0000			
76	0000	0000	0000	0000	92	0000	0000	0000	0000			
77	0000	0000	0000	0000	93	0000	0000	0000	0000			
78	0000	0000	0000	0000	94	0000	0000	0000	0000			
79	0000	0000	0000	0000	95	0000	0000	0000	0000			
80	0000	0000	0000	0000	96	0000	0000	0000	0000			
										☐ Alarm log(Main2)		
										☒ Alarm history (Main 4)		
Alarm log list		Current alarm										

Local servo

Alarm history(Local, Servo)										2019/04/02 14:46:09		
Alarm		Message						Lingo	Time	Date		
4000		*Press the [RST] key						NC	10:47:59	2019/04/02		
Local input								Deviation				
FEDC BA98 7654 3210												
1	0000 0001 0000 0000							Spindle	0	P1		
2	0000 0000 0000 0000							X	-1	P2		
3	0000 0000 0000 0000							Y	1	P3		
4	0000 0000 0000 0000							Z	3	P4		
5	0000 0000 0000 0000							QT	-18	AT		0
								A1	-1			
								B1				
								A2	1			
								B2				
Local output								Magazine 1				
FEDC BA98 7654 3210												
1	0000 0000 1101 1101							Current limit / Current command				
2	0000 0000 0000 0011							0 / 0				
3	0000 0101 0000 0000											
4	0000 0000 0000 1111											
☐ Alarm history(Main 5) ☒ Alarm history(PLC external input/output 1)												
Alarm history list		Current alarm										

Current limit/Current command display is shown only when the machine parameter (system 2) <Quick table> is set to <2: Type 2>.

PLC external input/output 1

Alarm log (PLC external input/output 1)										2012/06/01 17:17:41									
Alarm		Message								Lingo	Time	Date							
4000		*Press the [Reset] key.								Conv.	14:39:51	2012/06/01							
Input (X000-X1FF)										Output (Y000-Y1FF)									
	FEDC	BA98	7654	3210		FEDC	BA98	7654	3210		FEDC	BA98	7654	3210		FEDC	BA98	7654	3210
000	0000	0000	0000	0000	000	0000	0000	0000	0000	000	0000	0000	0000	0000	100	0000	0000	0000	0000
010	0000	0000	0000	0000	000	0000	0000	0000	0000	010	0000	0000	0000	0000	110	0000	0000	0000	0000
020	0000	0000	0000	0000	000	0000	0000	0000	0000	020	0000	0000	0000	0000	120	0000	0000	0000	0000
030	0000	0000	0000	0000	000	0000	0000	0000	0000	030	0000	0000	0000	0000	130	0000	0000	0000	0000
040	0000	0000	0000	0000	000	0000	0000	0000	0000	040	0000	0000	0000	0000	140	0000	0000	0000	0000
050	0000	0000	0000	0000	000	0000	0000	0000	0000	050	0000	0000	0000	0000	150	0000	0000	0000	0000
060	0000	0000	0000	0000	000	0000	0000	0000	0000	060	0000	0000	0000	0000	160	0000	0000	0000	0000
070	0000	0000	0000	0000	000	0000	0000	0000	0000	070	0000	0000	0000	0000	170	0000	0000	0000	0000
080	0000	0000	0000	0000	000	0000	0000	0000	0000	080	0000	0000	0000	0000	180	0000	0000	0000	0000
090	0000	0000	0000	0000	000	0000	0000	0000	0000	090	0000	0000	0000	0000	190	0000	0000	0000	0000
0A0	0000	0000	0000	0000	000	0000	0000	0000	0000	0A0	0000	0000	0000	0000	1A0	0000	0000	0000	0000
0B0	0000	0000	0000	0000	000	0000	0000	0000	0000	0B0	0000	0000	0000	0000	1B0	0000	0000	0000	0000
0C0	0000	0000	0000	0000	000	0000	0000	0000	0000	0C0	0000	0000	0000	0000	1C0	0000	0000	0000	0000
0D0	0000	0000	0000	0000	000	0000	0000	0000	0000	0D0	0000	0000	0000	0000	1D0	0000	0000	0000	0000
0E0	0000	0000	0000	0000	000	0000	0000	0000	0000	0E0	0000	0000	0000	0000	1E0	0000	0000	0000	0000
0F0	0000	0000	0000	0000	000	0000	0000	0000	0000	0F0	0000	0000	0000	0000	1F0	0000	0000	0000	0000

☐ Alarm history (Local, Servo)
☒ Alarm log (PLC external input/output 2)

Alarm log list	Current alarm					
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PLC external input/output 2

Alarm log (PLC external input/output 2)										2012/06/01 17:17:53				
Alarm		Message								Lingo	Time	Date		
4000		*Press the [Reset] key.								Conv.	14:39:51	2012/06/01		
Input (X200-X3FF)					Output (X200-X3FF)									
	FEDC	BA98	7654	3210		FEDC	BA98	7654	3210		FEDC	BA98	7654	3210
200	0000	0000	0000	0000	300	0000	0000	0000	0000	200	0000	0000	0000	0000
210	0000	0000	0000	0000	310	0000	0000	0000	0000	210	0000	0000	0000	0000
220	0000	0000	0000	0000	320	0000	0000	0000	0000	220	0000	0000	0000	0000
230	0000	0000	0000	0000	330	0000	0000	0000	0000	230	0000	0000	0000	0000
240	0000	0000	0000	0000	340	0000	0000	0000	0000	240	0000	0000	0000	0000
250	0000	0000	0000	0000	350	0000	0000	0000	0000	250	0000	0000	0000	0000
260	0000	0000	0000	0000	360	0000	0000	0000	0000	260	0000	0000	0000	0000
270	0000	0000	0000	0000	370	0000	0000	0000	0000	270	0000	0000	0000	0000
280	0000	0000	0000	0000	380	0000	0000	0000	0000	280	0000	0000	0000	0000
290	0000	0000	0000	0000	390	0000	0000	0000	0000	290	0000	0000	0000	0000
2A0	0000	0000	0000	0000	3A0	0000	0000	0000	0000	2A0	0000	0000	0000	0000
2B0	0000	0000	0000	0000	3B0	0000	0000	0000	0000	2B0	0000	0000	0000	0000
2C0	0000	0000	0000	0000	3C0	0000	0000	0000	0000	2C0	0000	0000	0000	0000
2D0	0000	0000	0000	0000	3D0	0000	0000	0000	0000	2D0	0000	0000	0000	0000
2E0	0000	0000	0000	0000	3E0	0000	0000	0000	0000	2E0	0000	0000	0000	0000
2F0	0000	0000	0000	0000	3F0	0000	0000	0000	0000	2F0	0000	0000	0000	0000

☐ Alarm log (PLC external input/output 1)
☒ Alarm log (PLC internal input/output 1)

Alarm log list	Current alarm					
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PLC internal input/output 1

Alarm log (PLC internal input/output 1)										2012/06/01 17:18:28									
Alarm		Message								Lingo	Time	Date							
4000		*Press the [Reset] key.								Conv.	14:39:51	2012/06/01							
Input(BX000-BX1FF)										Output(BX000-BX1FF)									
	FEDC	BA98	7654	3210		FEDC	BA98	7654	3210		FEDC	BA98	7654	3210		FEDC	BA98	7654	3210
000	0000	0000	0111	0000	100	0100	0000	0000	0001	000	0000	0000	0000	0000	100	0000	0000	0000	0000
010	0000	0000	0011	1000	110	0000	0000	0000	0000	010	0000	0000	0000	0000	110	0000	0000	0000	0000
020	0000	0000	0000	0000	120	0000	0100	0000	1000	020	0000	0000	0000	0000	120	0000	0000	0000	0000
030	1000	0000	0011	1111	130	0111	0100	1000	0000	030	0000	0000	0000	0000	130	0000	0000	0000	0000
040	0000	0000	0000	0001	140	1000	0000	0000	0100	040	0000	0000	0000	0000	140	0000	0000	0000	0000
050	0000	0000	0000	0000	150	0010	0000	0000	0100	050	0000	0000	0000	0000	150	0000	0000	0000	0000
060	0000	0000	0000	0000	160	0000	0000	0000	0000	060	0000	0000	0000	0000	160	0000	0000	0000	0000
070	0000	0000	0000	0000	170	0000	0000	0000	0000	070	0000	0000	0000	0000	170	0000	0000	0000	0000
080	0000	0000	1011	0100	180	0000	0000	0000	0000	080	0000	0001	0000	0000	180	0000	0000	0000	0000
090	0000	0000	1011	0100	190	0000	0000	0000	0000	090	0000	0000	0000	0000	190	0000	0000	0000	0000
0A0	0000	0000	1011	0100	1A0	0000	0000	0000	0000	0A0	0000	0000	0000	0000	1A0	0000	0000	0000	0000
0B0	0000	0000	1011	0100	1B0	0000	0111	0011	0001	0B0	0000	0000	0000	0000	1B0	0000	0000	0000	0000
0C0	0000	0000	0000	0000	1C0	0000	0000	0000	0000	0C0	0000	0000	0000	0000	1C0	0000	0000	0000	0000
0D0	0000	0000	0000	0000	1D0	0000	0000	0000	1000	0D0	0000	0000	0000	0000	1D0	0000	0000	0000	0000
0E0	0000	0000	0001	0000	1E0	0000	0000	1000	0010	0E0	0000	0000	0100	0000	1E0	0000	0000	0000	0000
0F0	0000	0000	0000	0011	1F0	0000	0000	0000	0000	0F0	0000	0000	0000	0000	1F0	0000	0000	0000	0000

☐ Alarm log (PLC external input/output 2)
☒ Alarm log (PLC internal input/output 2)

Alarm log list	Current alarm					
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PLC internal input/output 2

Alarm log (PLC internal input/output 2)											
Message										Time	Date
Alarm											
4000	*Press the [Reset] key.										
Input (BX200-BX3FF)										Conv.	
Output (BX200-BX3FF)											
FEDC	BA98	7654	3210	FEDC	BA98	7654	3210	FEDC	BA98	7654	3210
200	0000	0000	0000	0000	300	0000	0000	0000	0000	300	0000
210	0000	0000	0000	0000	310	0000	0000	0000	0000	310	0000
220	0000	0000	0000	0000	320	0000	0000	0000	0000	320	0000
230	0000	0000	0000	0000	330	0000	0000	0000	0000	330	0000
240	0000	0000	0000	0000	340	0000	0000	0000	0000	340	0000
250	0000	0000	0000	0000	350	0000	0000	0000	0000	350	0000
260	0000	0000	0000	0000	360	0000	0000	0000	0000	360	0000
270	0000	0000	0000	0000	370	0000	0000	0000	0000	370	0000
280	0000	1000	1101	0000	380	0000	0000	0000	0000	380	0000
290	0000	0000	0010	0001	390	0000	0000	0000	0000	390	0000
2A0	0000	0000	0000	0011	3A0	0000	0000	0000	0000	3A0	0000
2B0	0000	0000	0000	0001	3B0	0000	0000	0000	0000	3B0	0000
2C0	0000	0000	0000	0000	3C0	0000	0000	0000	0000	3C0	0000
2D0	0000	0111	1111	1111	3D0	0000	0000	0000	0000	3D0	0000
2E0	1111	1111	1111	1111	3E0	0000	0000	0000	0000	3E0	0000
2F0	0000	0000	0000	0000	3F0	0000	0000	0000	0000	3F0	0000
☐ Alarm log (PLC internal input/output 1) ☒ Alarm history (Detail display 1)											
Alarm log list	Current alarm										

11.6.4 List of Alarms and Operator Messages

Press the [F2] (Alarm list) key on the current alarm and operator message screen. A screen like the one below displays.

List of alarms and operator messages				2012/06/01 17:19:04	
Alarm		Message			
^					
EX0000	**ALARM				
EX0001	**BBB				
EX0002	**CCC				
EX0003	**DDD				
EX0004	**EEE				
EX0005	**FFF				
EX0006	**GGG				
EX0007	**HHH				
EX0008	**III				
EX0009	**JJJ				
EX0010	**KKK				
EX0011	**LLL				
EX0012	**External error 22				
EX0013	**External error 23				
EX0014	**External error 24				
EX0015	**External error 25				
EX0016	**External error 26				
v					
Alarm No.					
Recovery info : Enter the alarm number or move the cursor to the number and press [ENT].					
Current alarm	Alarm log list	Alarm list	Previous category	Next category	

A list of all alarms and operator messages is displayed.
Press the [F3] (Previous category) key to display the previous category. Press the [F4] (Next category) key to display the next category.

Move the cursor to the alarm or operator message to reference the recovery procedure. Press the [ENT] key, or enter the alarm number or operator message number and press the [ENT] key. Then, the alarm and operator message recovery screen is displayed for the selected alarm or operator message.

11.6.5 Alarm and Operator Message Recovery

This screen displays the possible causes of the alarms and operator messages that were selected.

Alarm / Operator message recovery					2012/06/01 17:19:58
No.	Stop	RecoveryName	Cause		Recovery method
EX0012	5	3	External error 22	Alarm messages can be changed by user parameters. 1. External input signal EXER22 has turned ON. 2. Internal output EXER22 of built-in PLC has turned ON.	Turn the signal OFF and turn power ON again.
No.	Stop	RecoveryName	Cause		Recovery method
EX0013	5	3	External error 23	Alarm messages can be changed by user parameters. 1. External input signal EXER23 has turned ON. 2. Internal output EXER23 of built-in PLC has turned ON.	Turn the signal OFF and turn power ON again.
No.	Stop	RecoveryName	Cause		Recovery method
EX0014	5	3	External error 24	Alarm messages can be changed by user parameters. 1. External input signal EXER24 has turned ON. 2. Internal output EXER24 of built-in PLC has turned ON.	Turn the signal OFF and turn power ON again.
Current alarm					
Alarm log list		Alarm list			

When this screen is displayed, the details for the alarm that was specified are listed at the top.

Screen operations

Up and down cursor: The screen scrolls up and down.

Next and previous page: The screen scrolls to the previous screen or next screen.

- * The user can scroll down to reference other alarms in the same category.
- * If the power is turned OFF while the alarm and operator message recovery screen is displayed and there is no particular screen that takes priority over the alarm and operator message recovery screen when the power is turned ON again, the alarm and operator message recovery screen will be displayed.

11.6.6 Servo Alarm List

The cause and the recovery method are displayed for each auxiliary number for servo related alarms (servo error, servo warning, servo communication error, servo status error). Press the **[F3]** (Servo alarm list) key at the current alarm and operator message screen. A screen like the one below displays.

Servo alarm list (servo error) 2014/09/02 17:29:58				
No.	Messages	Code	Cause	Recovery method
15	NC power failure error	0Fh	NC power failure error NC is detected as being stopped.	Check the power supply, and fix any problems such as power interruptions or voltage drops.
16	Invalid frame 0	10h	Invalid frame error on port 0 Rx Invalid frame is received continuously from port 0. 1. Faulty communications cable. 2. Faulty servo amplifier or NC PCB.	1. Make sure that there are no poor or faulty contacts on the communications cable. 2. Replace the servo amplifier or NC PCB.
17	Invalid frame 1	11h	Invalid frame error on port 1 Rx Invalid frame is received continuously from port 1. 1. Faulty communications cable. 2. Faulty servo amplifier or NC PCB.	1. Make sure that there are no poor or faulty contacts on the communications cable. 2. Replace the servo amplifier or NC PCB.
18	CRC error 0	12h	CRC error on port 0 Rx Continuous error triggered on port 0 Rx. 1. Faulty communications cable. 2. Faulty servo amplifier or NC PCB.	1. Make sure that there are no poor or faulty contacts on the communications cable. 2. Replace the servo amplifier or NC PCB.
19	CRC error 1	13h	CRC error on port 1 Rx Continuous error triggered on port 1 Rx. 1. Faulty communications cable. 2. Faulty servo amplifier or NC PCB.	1. Make sure that there are no poor or faulty contacts on the communications cable. 2. Replace the servo amplifier or NC PCB.
Current alarm Servo error Servo warning Servo comm. error Servo STS error				

Screen operations

Up and down cursor: The screen scrolls up and down.

Next and previous page: The screen scrolls to the previous screen or next screen.

The types displayed in the servo alarm list can be changed using the function keys.

[F1] (Servo error) key: Displays the list of servo errors.

[F2] (Servo warning) key: Displays the list of servo warnings.

[F3] (Servo comm. error) key: Displays the list of servo communication errors.

[F4] (Servo STS error) key: Displays the list of servo status errors.

- * If the power is turned OFF while the servo alarm list screen is displayed and there is no particular screen that takes priority over the servo alarm list screen when the power is turned ON again, the servo alarm list screen will be displayed.

11.6.7 Loading System Alarm

Press the [F4] (Loading system) key at the current alarm and operator message screens. A screen like the one below displays.



- An alarm that has currently triggered on the loading system is displayed.
- The operator message <<Loading system alarm triggered>> is triggered when a loading system alarm occurs.

11.7 Alarm Code List

The list includes alarms that may occur only in certain models or models with particular specifications.

11.7.1 EX

No.	Stop level	Reset level	Alarm message	Cause	Solution
EX0000 : EX0089	5	3	External error **	Alarm messages can be changed by user parameters. 1. External input signal EXER** has turned ON. 2. Internal output EXER** of built-in PLC has turned ON.	Turn the signal OFF and turn power ON again.
EX0500 : EX0589	5	2	External error **	Alarm messages can be changed by user parameters. 1. External input signal EXER** has turned ON. 2. Internal output EXER** of built-in PLC has turned ON.	Turn the signal OFF and press the [RST] key.
EX1500 : EX1589	4	3	External error **	Alarm messages can be changed by user parameters. 1. External input signal EXER** has turned ON. 2. Internal output EXER** of built-in PLC has turned ON.	Turn the signal OFF and turn power ON again.
EX2000 : EX2089	4	2	External error **	Alarm messages can be changed by user parameters. 1. External input signal EXER** has turned ON. 2. Internal output EXER** of built-in PLC has turned ON.	Turn the signal OFF and press the [RST] key.
EX3000 : EX3089	4	1	External error **	Alarm messages can be changed by user parameters. 1. External input signal EXER** has turned ON. 2. Internal output EXER** of built-in PLC has turned ON.	Turn the signal OFF.
EX3500 : EX3589	3	3	External error **	Alarm messages can be changed by user parameters. 1. External input signal EXER** has turned ON. 2. Internal output EXER** of built-in PLC has turned ON.	Turn the signal OFF and turn power ON again.
EX4000 : EX4089	3	2	External error **	Alarm messages can be changed by user parameters. 1. External input signal EXER** has turned ON. 2. Internal output EXER** of built-in PLC has turned ON.	Turn the signal OFF and press the [RST] key.
EX4500 : EX4589	3	1	External error **	Alarm messages can be changed by user parameters. 1. External input signal EXER** has turned ON. 2. Internal output EXER** of built-in PLC has turned ON.	Turn the signal OFF.
EX5000 : EX5089	2	3	External error **	Alarm messages can be changed by user parameters. 1. External input signal EXER** has turned ON. 2. Internal output EXER** of built-in PLC has turned ON.	Turn the signal OFF and turn power ON again.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
EX5500 : EX5589	2	2	External error **	Alarm messages can be changed by user parameters. 1. External input signal EXER** has turned ON. 2. Internal output EXER** of built-in PLC has turned ON.	Turn the signal OFF and press the [RST] key.
EX6000 : EX6089	2	1	External error **	Alarm messages can be changed by user parameters. 1. External input signal EXER** has turned ON. 2. Internal output EXER** of built-in PLC has turned ON.	Turn the signal OFF.
EX6500 : EX6589	1	3	External error **	Alarm messages can be changed by user parameters. 1. External input signal EXER** has turned ON. 2. Internal output EXER** of built-in PLC has turned ON.	Turn the signal OFF and turn power ON again.
EX7000 : EX7089	1	2	External error **	Alarm messages can be changed by user parameters. 1. External input signal EXER** has turned ON. 2. Internal output EXER** of built-in PLC has turned ON.	Turn the signal OFF and press the [RST] key.
EX7500 : EX7589	1	1	External error **	Alarm messages can be changed by user parameters. 1. External input signal EXER** has turned ON. 2. Internal output EXER** of built-in PLC has turned ON.	Turn the signal OFF.

11.7.2 EC

No.	Stop level	Reset level	Alarm message	Cause	Solution
EC0000	5	3	IO/servo communication error (*)	IO/servo transmission failed.	1. The cable connection is faulty. Re-connect the cable that is connected to XNC01 and XNC02 on the NC PCB. 2. The cable may be broken or cut. Replace the cable. 3. The NC PCB is faulty. Replace the NC PCB.
EC0001	5	3	The IO equipment communication output is abnormal	1. An abnormal output is detected in IO equipment communications. 2. There is a ground fault on A1+ for KFC1. 3. KFC1 is faulty.	1. IO equipment is faulty. Contact the nearest BROTHER approved service dealer. 2. Check the A1+ wiring connection for KFC1. 3. Replace KFC1.
EC0002	5	3	The IO equipment communication input is abnormal	1. An abnormal input is detected in IO equipment communications. 2. There is a ground fault on A1+ for KFC1. 3. KFC1 is faulty.	1. IO equipment is faulty. Contact the nearest BROTHER approved service dealer. 2. Check the A1+ wiring connection for KFC1. 3. Replace KFC1.
EC0003	5	3	The servo communication output is abnormal	An abnormal output of servo communication is detected.	1. The cable connection is faulty. Re-connect the cable that is connected to X0 and X1 on the servo amplifier. 2. The cable may be broken or cut. Replace the cable. 3. The servo amplifier is faulty. Replace the servo amplifier.
EC0004	5	3	The servo communication input is abnormal	An abnormal input of servo communication is detected.	1. The cable connection is faulty. Re-connect the cable that is connected to X0 and X1 on the servo amplifier. 2. The cable may be broken or cut. Replace the cable. 3. The servo amplifier is faulty. Replace the servo amplifier.
EC0005	5	3	The IO equipment communication is abnormal (*)	An anomaly is detected in IO equipment initial communications.	IO equipment is faulty. Contact the nearest BROTHER approved service dealer.
EC0006	5	3	Servo communication is abnormal (*)	An anomaly is detected in the servo initial communications.	1. The cable connection is faulty. Re-connect the cable that is connected to X0 and X1 on the servo amplifier. 2. The cable may be broken or cut. Replace the cable. 3. The servo amplifier is faulty. Replace the servo amplifier.
EC0007	5	3	IO/servo periodic communication error (*)	1. Data frame is lost in IO/servo communications. 2. There is a ground fault on A1+ for KFC1. 3. KFC1 is faulty.	1. The LAN cable connection is faulty. 2. The NC PCB is faulty. 3. The I/O equipment is faulty. 4. Servo amplifier is faulty. 5. Check the A1+ wiring connection for KFC1. 6. Replace KFC1. Contact the nearest BROTHER approved service dealer.

No.	Stop level	Reset level	Alarm message	Cause	Solution
EC0008	5	3	No response of communication from IO equipment (*)	1. Non-responding IO equipment is detected. 2. There is a ground fault on A1+ for KFC1. 3. KFC1 is faulty.	1. Check connection of the IO equipment. Contact the maker to describe the failure. 2. Check the A1+ wiring connection for KFC1. 3. Replace KFC1.
EC0009	5	3	Unusable equipment is connected (*)	Unusable equipment is detected.	Check connection of the IO equipment. Contact the maker to describe the failure.
EC0010	5	3	Servo amplifier connection point is abnormal (*)	Servo amplifier position is not correct.	Check connection of the servo amplifier. Contact the maker to describe the failure.
EC0011	5	3	IO/servo periodic communication time-out	Failed to receive the data frame of IO/servo communication within the specified time period.	The NC PCB is defective. Contact the nearest BROTHER approved service dealer.
EC0012	5	3	Duplicated errors of expansion terminal block numbers (*)	An expansion terminal block or an ID PCB with a duplicated number was detected.	Check the number of the expansion terminal block or for the ID PCB.
EC0013	5	3	The expansion terminal block number is abnormal (*)	An expansion terminal block or ID PCB number set outside the range was detected.	Check the number of the expansion terminal block.
EC0014	5	3	Duplicated errors of servo amplifier axis number (*)	A servo amplifier of a duplicated axis number is detected.	Check the axis number of the servo amplifier.
EC0015	5	3	The servo amplifier axis number is abnormal (*)	The axis number of a servo amplifier is not within the range.	Check the axis number of the servo amplifier.
EC0016	5	3	IO equipment watch dog (*)	1. Abnormal IO equipment is detected. 2. There is a ground fault on A1+ for KFC1. 3. KFC1 is faulty.	1. IO equipment is faulty. Contact the nearest BROTHER approved service dealer. 2. Check the A1+ wiring connection for KFC1. 3. Replace KFC1.
EC0017	5	3	Servo amplifier watch dog (*)	An abnormal servo amplifier is detected.	Servo amplifier is faulty. Contact the nearest BROTHER approved service dealer.
EC0018	5	3	The field bus equipment communication output is abnormal.	An abnormal output is detected in FC PCB communications.	FC PCB is faulty. Contact the nearest BROTHER approved service dealer.
EC0019	5	3	The field bus equipment communication input is abnormal	An abnormal input is detected in FC PCB communications.	FC PCB is faulty. Contact the nearest BROTHER approved service dealer.
EC0020	5	3	IO/servo initial communication error (*)	An anomaly is detected in the IO/servo initial communications.	1. The NC PCB is defective. 2. Too many IO equipment/servo amplifiers are connected. Contact the nearest BROTHER approved service dealer.
EC0021	5	3	The detachable panel connecting position is abnormal	Reserved	Reserved
EC0022	5	3	The IO/servo connection is abnormal (*)	Connection of IO equipment/servo amplifiers is wrong.	Check connection of IO equipment/servo amplifiers.

11.7.3 SV

No.	Stop level	Reset level	Alarm message	Cause	Solution
SV0000	5	3	Spindle servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0001	5	3	X-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0002	5	3	Y-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0003	5	3	Z-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0004	5	3	*-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0005	5	3	*-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0006	5	3	*-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0007	5	3	*-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0008	5	3	*-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0009	5	3	*-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0010	5	3	*-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0011	5	3	P1-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0012	5	3	P2-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0013	5	3	P3-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0014	5	3	P4-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0015	5	3	Spindle servo communication error (*)	The cause differs depending on the auxiliary number (*). See the servo communication error list.	
SV0016	5	3	X-axis servo communication error (*)	The cause differs depending on the auxiliary number (*). See the servo communication error list.	
SV0017	5	3	Y-axis servo communication error (*)	The cause differs depending on the auxiliary number (*). See the servo communication error list.	

No.	Stop level	Reset level	Alarm message	Cause	Solution
SV0018	5	3	Z-axis servo communication error (*)	The cause differs depending on the auxiliary number (*). See the servo communication error list.	
SV0019	5	3	*-axis servo communication error (*)	The cause differs depending on the auxiliary number (*). See the servo communication error list.	
SV0020	5	3	*-axis servo communication error (*)	The cause differs depending on the auxiliary number (*). See the servo communication error list.	
SV0021	5	3	*-axis servo communication error (*)	The cause differs depending on the auxiliary number (*). See the servo communication error list.	
SV0022	5	3	*-axis servo communication error (*)	The cause differs depending on the auxiliary number (*). See the servo communication error list.	
SV0023	5	3	*-axis servo communication error (*)	The cause differs depending on the auxiliary number (*). See the servo communication error list.	
SV0024	5	3	*-axis servo communication error (*)	The cause differs depending on the auxiliary number (*). See the servo communication error list.	
SV0025	5	3	*-axis servo communication error (*)	The cause differs depending on the auxiliary number (*). See the servo communication error list.	
SV0026	5	3	P1-axis servo communication error (*)	The cause differs depending on the auxiliary number (*). See the servo communication error list.	
SV0027	5	3	P2-axis servo communication error (*)	The cause differs depending on the auxiliary number (*). See the servo communication error list.	
SV0028	5	3	P3 -axis servo communication error (*)	The cause differs depending on the auxiliary number (*). See the servo communication error list.	
SV0029	5	3	P4-axis servo communication error (*)	The cause differs depending on the auxiliary number (*). See the servo communication error list.	
SV0030	5	3	Spindle servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0031	5	3	X-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0032	5	3	Y-axis servo status error(*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0033	5	3	Z-axis servo status error(*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	

No.	Stop level	Reset level	Alarm message	Cause	Solution
SV0034	5	3	*-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0035	5	3	*-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0036	5	3	*-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0037	5	3	*-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0038	5	3	*-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0039	5	3	*-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0040	5	3	*-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0041	5	3	P1-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0042	5	3	P2-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0043	5	3	P3-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0044	5	3	P4-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0045	5	3	Spindle servo error (*)		
SV0046	5	3	X-axis servo error (*)		
SV0047	5	3	Y-axis servo error (*)		
SV0048	5	3	Z-axis servo error (*)		
SV0049	5	3	*-axis servo error (*)		
SV0050	5	3	*-axis servo error (*)		
SV0051	5	3	*-axis servo error (*)		
SV0052	5	3	*-axis servo error (*)		
SV0053	5	3	*-axis servo error (*)		
SV0054	5	3	*-axis servo error (*)		
SV0055	5	3	*-axis servo error (*)		
SV0500	5	2	Spindle servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SV0501	5	2	X-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0502	5	2	Y-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0503	5	2	Z-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0504	5	2	*-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0505	5	2	*-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0506	5	2	*-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0507	5	2	*-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0508	5	2	*-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0509	5	2	*-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0510	5	2	*-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0511	5	2	P1-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0512	5	2	P2-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0513	5	2	P3-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0514	5	2	P4-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV0515	5	2	Spindle servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0516	5	2	X-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0517	5	2	Y-axis servo status error(*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0518	5	2	Z-axis servo status error(*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0519	5	2	*-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0520	5	2	*-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	

No.	Stop level	Reset level	Alarm message	Cause	Solution
SV0521	5	2	*-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0522	5	2	*-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0523	5	2	*-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0524	5	2	*-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0525	5	2	*-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0526	5	2	P1-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0527	5	2	P2-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0528	5	2	P3-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0529	5	2	P4-axis servo status error (*)	The cause differs depending on the auxiliary number (*). See the servo status error list.	
SV0530	5	2	Thermal error (spindle motor)	The thermal relay in the spindle motor was tripped. 1. Motor is overloaded or locked. 2. Machining condition has exceeded motor's capacity.	1. Use your hands to turn the spindle to make sure that it is not heavy or sluggish. 2. Revise the machining conditions and operating pattern. Remove the above cause, and then wait 30 minutes without turning OFF the power before resuming operation. Please contact us about the status of the error.
SV0531	5	2	Spindle servo error (*)		
SV0532	5	2	X-axis servo error (*)		
SV0533	5	2	Y-axis servo error (*)		
SV0534	5	2	Z-axis servo error (*)		
SV0535	5	2	*-axis servo error (*)		
SV0536	5	2	*-axis servo error (*)		
SV0537	5	2	*-axis servo error (*)		
SV0538	5	2	*-axis servo error (*)		
SV0539	5	2	*-axis servo error (*)		
SV0540	5	2	*-axis servo error (*)		
SV0541	5	2	*-axis servo error (*)		

No.	Stop level	Reset level	Alarm message	Cause	Solution
SV0542	5	2	Thermal error (X-axis motor)	The thermal relay in the X- axis motor was tripped. 1. The motor is overloaded or locked. 2. The machining condition has exceeded the motor's capacity.	Revise the machining conditions or operating pattern. Remove the above cause, and then wait 30 minutes without turning OFF the power before resuming operation. Please contact us about the status of the error.
SV0543	5	2	Thermal error (Y-axis motor)	The thermal relay in the Y- axis motor was tripped. 1. The motor is overloaded or locked. 2. The machining condition has exceeded the motor's capacity.	Revise the machining conditions or operating pattern. Remove the above cause, and then wait 30 minutes without turning OFF the power before resuming operation. Please contact us about the status of the error.
SV0544	5	2	Thermal error (Z-axis motor)	The thermal relay in the Z- axis motor was tripped. 1. The motor is overloaded or locked. 2. The machining condition has exceeded the motor's capacity.	Revise the machining conditions or operating pattern. Remove the above cause, and then wait 30 minutes without turning OFF the power before resuming operation. Please contact us about the status of the error.
SV0545	5	2	Thermal error (*-axis motor)	The thermal relay in the 4th-axis motor was tripped. 1. The motor is overloaded or locked. 2. The machining condition has exceeded the motor's capacity.	Revise the machining conditions or operating pattern. Remove the above cause, and then wait 30 minutes without turning OFF the power before resuming operation. Please contact us about the status of the error.
SV0546	5	2	Thermal error (*-axis motor)	The thermal relay in the 5th-axis motor was tripped. 1. The motor is overloaded or locked. 2. The machining condition has exceeded the motor's capacity.	Revise the machining conditions or operating pattern. Remove the above cause, and then wait 30 minutes without turning OFF the power before resuming operation. Please contact us about the status of the error.
SV0547	5	2	Thermal error (*-axis motor)	The thermal relay in the 6th-axis motor was tripped. 1. The motor is overloaded or locked. 2. The machining condition has exceeded the motor's capacity.	Revise the machining conditions or operating pattern. Remove the above cause, and then wait 30 minutes without turning OFF the power before resuming operation. Please contact us about the status of the error.
SV0548	5	2	Thermal error (*-axis motor)	The thermal relay in the 7th-axis motor was tripped. 1. The motor is overloaded or locked. 2. The machining condition has exceeded the motor's capacity.	Revise the machining conditions or operating pattern. Remove the above cause, and then wait 30 minutes without turning OFF the power before resuming operation. Please contact us about the status of the error.
SV0549	5	2	Thermal error (*-axis motor)	The thermal relay in the 8th-axis motor was tripped. 1. The motor is overloaded or locked. 2. The machining condition has exceeded the motor's capacity.	Revise the machining conditions or operating pattern. Remove the above cause, and then wait 30 minutes without turning OFF the power before resuming operation. Please contact us about the status of the error.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SV0550	5	2	Thermal error (*-axis motor)	The thermal relay in the M-axis motor was tripped. 1. The motor is overloaded or locked. 2. The machining condition has exceeded the motor's capacity.	Revise the operating pattern. Remove the above cause, and then wait 30 minutes without turning OFF the power before resuming operation. Please contact us about the status of the error.
SV0551	5	2	Thermal error (P1-axis motor)	The thermal relay in the P1-axis motor was tripped. 1. The motor is overloaded or locked. 2. The operating conditions have exceeded the motor's capacity.	Revise the operating pattern. Remove the above cause, and then wait 30 minutes without turning OFF the power before resuming operation. Please contact us about the status of the error.
SV0552	5	2	Thermal error (P2-axis motor)	The thermal relay in the P2-axis motor was tripped. 1. The motor is overloaded or locked. 2. The operating conditions have exceeded the motor's capacity.	Revise the operating pattern. Remove the above cause, and then wait 30 minutes without turning OFF the power before resuming operation. Please contact us about the status of the error.
SV0553	5	2	Thermal error (P3-axis motor)	The thermal relay in the P3-axis motor was tripped. 1. The motor is overloaded or locked. 2. The operating conditions have exceeded the motor's capacity.	Revise the operating pattern. Remove the above cause, and then wait 30 minutes without turning OFF the power before resuming operation. Please contact us about the status of the error.
SV0554	5	2	Thermal error (P4-axis motor)	The thermal relay in the P4-axis motor was tripped. 1. The motor is overloaded or locked. 2. The operating conditions have exceeded the motor's capacity.	Revise the operating pattern. Remove the above cause, and then wait 30 minutes without turning OFF the power before resuming operation. Please contact us about the status of the error.
SV0555	5	2	Thermal error (*-axis motor)	The thermal relay in the AT-axis motor was tripped. 1. The motor is overloaded or locked. 2. The operating conditions have exceeded the motor's capacity.	Revise the operating pattern. Remove the above cause, and then wait 30 minutes without turning OFF the power before resuming operation. Please contact us about the status of the error.
SV1500	4	3	P1-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV1501	4	3	P2-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV1502	4	3	P3-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV1503	4	3	P4-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV2000	4	2	*-axis position mismatch (for positioning only)	The encoder position cannot be identified correctly.	Configure the automatic settings for the adjusted absolute values on the absolute encoder reset screen. Then, readjust the zero position.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SV2001	4	2	*-axis position mismatch (for positioning only)	The encoder position cannot be identified correctly.	Configure the automatic settings for the adjusted absolute values on the absolute encoder reset screen. Then, readjust the zero position.
SV2002	4	2	*-axis position mismatch (for positioning only)	The encoder position cannot be identified correctly.	Configure the automatic settings for the adjusted absolute values on the absolute encoder reset screen. Then, readjust the zero position.
SV2003	4	2	*-axis position mismatch (for positioning only)	The encoder position cannot be identified correctly.	Configure the automatic settings for the adjusted absolute values on the absolute encoder reset screen. Then, readjust the zero position.
SV2004	4	2	P1-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV2005	4	2	P2-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV2006	4	2	P3-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV2007	4	2	P4-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV3500	3	3	P1-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV3501	3	3	P2-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV3502	3	3	P3-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV3503	3	3	P4-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV4000	3	2	P1-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV4001	3	2	P2-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV4002	3	2	P3-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV4003	3	2	P4-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV5000	2	3	P1-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV5001	2	3	P2-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV5002	2	3	P3-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV5003	2	3	P4-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	

No.	Stop level	Reset level	Alarm message	Cause	Solution
SV5004	2	3	P1-axis servo communication error (*)		
SV5005	2	3	P2-axis servo communication error (*)		
SV5006	2	3	P3-axis servo communication error (*)		
SV5007	2	3	P4-axis servo communication error (*)		
SV5008	2	3	P1-axis servo status error (*)		
SV5009	2	3	P2-axis servo status error (*)		
SV5010	2	3	P3-axis servo status error (*)		
SV5011	2	3	P4-axis servo status error (*)		
SV5500	2	2	P1-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV5501	2	2	P2-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV5502	2	2	P3-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV5503	2	2	P4-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV6500	1	3	P1-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV6501	1	3	P2-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV6502	1	3	P3-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV6503	1	3	P4-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV7000	1	2	Spindle servo warning (*)	The cause differs depending on the auxiliary number (*). See the servo warning list.	
SV7001	1	2	X-axis servo warning (*)	The cause differs depending on the auxiliary number (*). See the servo warning list.	
SV7002	1	2	Y-axis servo warning (*)	The cause differs depending on the auxiliary number (*). See the servo warning list.	
SV7003	1	2	Z-axis servo warning (*)	The cause differs depending on the auxiliary number (*). See the servo warning list.	
SV7004	1	2	*-axis servo warning (*)	The cause differs depending on the auxiliary number (*). See the servo warning list.	

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SV7005	1	2	*-axis servo warning (*)	The cause differs depending on the auxiliary number (*). See the servo warning list.	
SV7006	1	2	*-axis servo warning (*)	The cause differs depending on the auxiliary number (*). See the servo warning list.	
SV7007	1	2	*-axis servo warning (*)	The cause differs depending on the auxiliary number (*). See the servo warning list.	
SV7008	1	2	*-axis servo warning (*)	The cause differs depending on the auxiliary number (*). See the servo warning list.	
SV7009	1	2	*-axis servo warning (*)	The cause differs depending on the auxiliary number (*). See the servo warning list.	
SV7010	1	2	*-axis servo warning (*)	The cause differs depending on the auxiliary number (*). See the servo warning list.	
SV7011	1	2	P1-axis servo warning (*)	The cause differs depending on the auxiliary number (*). See the servo warning list.	
SV7012	1	2	P2-axis servo warning (*)	The cause differs depending on the auxiliary number (*). See the servo warning list.	
SV7013	1	2	P3-axis servo warning (*)	The cause differs depending on the auxiliary number (*). See the servo warning list.	
SV7014	1	2	P4-axis servo warning (*)	The cause differs depending on the auxiliary number (*). See the servo warning list.	
SV7015	1	2	P1-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV7016	1	2	P2-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV7017	1	2	P3-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV7018	1	2	P4-axis servo error (*)	The cause differs depending on the auxiliary number (*). See the servo error list.	
SV7019	1	2	P1-axis servo status error (*)		
SV7020	1	2	P2-axis servo status error (*)		
SV7021	1	2	P3-axis servo status error (*)		
SV7022	1	2	P4-axis servo status error (*)		
SV7023	1	2	Spindle servo warning (*)		
SV7024	1	2	X-axis servo warning (*)		
SV7025	1	2	Y-axis servo warning (*)		
SV7026	1	2	Z-axis servo warning (*)		
SV7027	1	2	*-axis servo warning (*)		

No.	Stop level	Reset level	Alarm message	Cause	Solution
SV7028	1	2	*-axis servo warning (*)		
SV7029	1	2	*-axis servo warning (*)		
SV7030	1	2	*-axis servo warning (*)		
SV7031	1	2	*-axis servo warning (*)		
SV7032	1	2	*-axis servo warning (*)		
SV7033	1	2	*-axis servo warning (*)		

11.7.4 NC

No.	Stop level	Reset level	Alarm message	Cause	Solution
NC0000	5	3	Watch dog (main)	There was a circuit abnormality main CPU.	NC PCB is defective. Replace NC PCB.
NC0001	5	3	Watch dog (local)	An error has occurred in the local CPU.	1. LOCAL PCB is defective. Replace the LOCAL PCB. 2. NC PCB is defective. Replace NC PCB.
NC0002	5	3	RAM error (main)	Reading and writing using main CPU RAM is not possible.	NC PCB is defective. Replace NC PCB.
NC0003	5	3	ROM error (main)	The check sum of main CPU ROM does not match.	NC PCB is defective. Replace NC PCB.
NC0004	5	3	RAM error (main-local 1)	Reading and writing using common- RAM for main- local 1 may be not possible.	NC PCB is defective. Replace NC PCB.
NC0005	5	3	RAM error (main-local 2)	Reading and writing using common- RAM for main- local 2 may be not possible.	NC PCB is defective. Replace NC PCB.
NC0006	5	3	RAM error (main-local 3)	Reading and writing using common- RAM for main- local 3 may be not possible.	NC PCB is defective. Replace NC PCB.
NC0007	5	3	Calculation error (local)	Improper calculation is executed (local CPU).	1. LOCAL PCB is defective. Replace LOCAL PCB. 2. NC PCB is defective. Replace NC PCB.
NC0008	5	3	Main cannot be started	Main CPU has not started.	NC PCB is defective. Replace NC PCB.
NC0009	5	3	Main exception interruption (*)	Main CPU has detected incorrect interruption. The figure in brackets () indicates interruption no.	NC PCB is defective. Replace NC PCB.
NC0010	5	3	Local exception interruption (*)	Local CPU has detected incorrect interruption. The figure in brackets () indicates interruption No.	LOCAL PCB is defective. Replace LOCAL PCB.
NC0011	5	3	Kernel error (main)	An error has occurred in the main CPU operating system.	NC PCB is defective. Replace NC PCB.
NC0012	5	3	Kernel error (local)	An error has occurred in the local CPU operating system.	LOCAL PCB is defective. Replace LOCAL PCB.
NC0013	5	3	Command error	Not synchronized with the command.	NC PCB is defective. Replace NC PCB.
NC0014	5	3	Internal conflict detected (*)	An error occurred in the main CPU's system software.	NC PCB is defective. Contact the nearest BROTHER approved service dealer, telling the auxiliary number (*).
NC0015	5	3	Disk size error	The size data in the current disk does not match the size data in the NC.	1. Check the CF connections. 2. CF may be defective. Replace the CF. 3. LOCAL PCB is defective. Replace LOCAL PCB.
NC0016	5	3	Disk access error	Disk connection has failed.	1. Check the CF connections. 2. Format the memory. 3. CF may be defective. Replace the CF. 4. LOCAL PCB is defective. Replace LOCAL PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
NC0017	5	3	There is no servo communication specified axis	Data sent to nonexistent axis (servo communications).	Check servo connections.
NC0018	5	3	There is no servo signal receiving data	Data not received from servo (servo communications).	Check servo connections.
NC0019	5	3	Main system initialization error (*)	An error occurred when the system software started.	NC PCB is defective. Replace NC PCB.
NC0020	5	3	An invalid panel is connected (ES)	Reserved	Reserved
NC0021	5	3	The front simple panel cannot start	Reserved	Reserved
NC0022	5	3	The keyboard cannot be recognized	The standard panel was not recognized.	1. Check the standard panel connection. 2. The KEY PCB is faulty.
NC0023	5	3	The simple panel cannot be recognized	Reserved	Reserved
NC0024	5	3	Version consistency error (main/local)	1. Version upgrading is suspended. 2. Software version mismatch following replacement of NC PCB or compact flash.	Execute version upgrade.
NC0025	5	3	Version consistency error (main/panel)	Reserved	Reserved
NC0026	5	3	Internal conflict detected 2 (*)	An error occurred in the main CPU's system software.	NC PCB is defective. Contact the nearest BROTHER approved service dealer, telling the auxiliary number (*).
NC0027	5	3	Internal conflict detected 3 (*)	An error occurred in the main CPU's system software.	NC PCB is defective. Contact the nearest BROTHER approved service dealer, telling the auxiliary number (*).
NC0028	5	3	Internal conflict detected 4 (*)	An error occurred in the main CPU's system software.	NC PCB is defective. Contact the nearest BROTHER approved service dealer, telling the auxiliary number (*).
NC0029	5	3	PLC cannot start	The ladder program did not start within the predetermined time.	Contact the nearest BROTHER approved service dealer.
NC0030	5	3	Local CPU abnormal stop	The temperature of LOCAL PCB may have increased abnormally. LOCAL PCB may have caused abnormal termination.	1. LOCAL PCB is defective. Replace LOCAL PCB. 2. NC PCB is defective. Replace NC PCB.
NC0031	5	3	System error (Main) (*)	The NC PCB is faulty.	NC PCB is defective. Replace NC PCB.
NC0032	5	3	Version consistency error (main/magazine)	Reserved	Reserved
NC0033	5	3	Hardware compatibility error (Local-CF)	The local PCB and CF combination is wrong.	Check the type of the local PCB and CF, and select the correct combination.

No.	Stop level	Reset level	Alarm message	Cause	Solution
NC0034	5	3	Machine information does not match	The machine identification signal does not match the NC's internal information.	1. If the NC PCB has been replaced and there is an internal data bank change, check for a data setting error. 2. If the IO PCB has been replaced and this alarm still triggers, make sure that all connectors on the IO PCB have been connected or that the latest IO PCB is being used. If operation does not restore, the PCB may be faulty. Therefore, replace it with the latest IO PCB.
NC0035	5	3	Machine configuration error	1. Required OPSEL PCB is not detected. 2. There is an OPSEL PCB error. 3. There is an IO PCB error.	1. Check the connection with the OPSEL PCB. 2. Replace the OPSEL PCB. 3. Replace the IO PCB.
NC0036	5	3	Hardware compatibility error (IO)	The information in the NC and the IO PCB do not match.	If this alarm is triggered after replacing the IO PCB, check whether the IO PCB is the latest version or not. If the IO PCB is the latest version, then the PCB may be faulty. Replace the IO PCB.
NC0037	5	3	X-axis servo amplifier does not match	There is a problem with the type of servo amplifier connected to the X-axis.	Check the type of servo amplifier for the X-axis.
NC0038	5	3	Y-axis servo amplifier does not match	There is a problem with the type of servo amplifier connected to the Y-axis.	Check the type of servo amplifier for the Y-axis.
NC0039	5	3	Z-axis servo amplifier does not match	There is a problem with the type of servo amplifier connected to the Z-axis.	Check the type of servo amplifier for the Z-axis.
NC0040	5	3	*-axis servo amplifier does not match	There is a problem with the type of servo amplifier connected to the 4-axis.	Check the type of servo amplifier for the 4-axis.
NC0041	5	3	*-axis servo amplifier does not match	There is a problem with the type of servo amplifier connected to the 5-axis.	Check the type of servo amplifier for the 5-axis.
NC0042	5	3	*-axis servo amplifier does not match	There is a problem with the type of servo amplifier connected to the 6-axis.	Check the type of servo amplifier for the 6-axis.
NC0043	5	3	*-axis servo amplifier does not match	There is a problem with the type of servo amplifier connected to the 7-axis.	Check the type of servo amplifier for the 7-axis.
NC0044	5	3	*-axis servo amplifier does not match	There is a problem with the type of servo amplifier connected to the 8-axis.	Check the type of servo amplifier for the 8-axis.
NC0045	5	3	P1-axis servo amplifier does not match	There is a problem with the type of servo amplifier connected to the P1-axis.	Check the type of servo amplifier for the P1-axis.
NC0046	5	3	P2-axis servo amplifier does not match	There is a problem with the type of servo amplifier connected to the P2-axis.	Check the type of servo amplifier for the P2-axis.
NC0047	5	3	P3-axis servo amplifier does not match	There is a problem with the type of servo amplifier connected to the P3-axis.	Check the type of servo amplifier for the P3-axis.
NC0048	5	3	P4-axis servo amplifier does not match	There is a problem with the type of servo amplifier connected to the P4-axis.	Check the type of servo amplifier for the P4-axis.

No.	Stop level	Reset level	Alarm message	Cause	Solution
NC0049	5	3	X-axis motor does not match	There is a problem with the type of motor connected to the X-axis.	Check the type of motor for the X-axis.
NC0050	5	3	Y-axis motor does not match	There is a problem with the type of motor connected to the Y-axis.	Check the type of motor for the Y-axis.
NC0051	5	3	Z-axis motor does not match	There is a problem with the type of motor connected to the Z-axis.	Check the type of motor for the Z-axis.
NC0052	5	3	*-axis motor does not match	There is a problem with the type of motor connected to the 4-axis.	Check the type of motor for the 4-axis.
NC0053	5	3	*-axis motor does not match	There is a problem with the type of motor connected to the 5-axis.	Check the type of motor for the 5-axis.
NC0054	5	3	*-axis motor does not match	There is a problem with the type of motor connected to the 6-axis.	Check the type of motor for the 6-axis.
NC0055	5	3	*-axis motor does not match	There is a problem with the type of motor connected to the 7-axis.	Check the type of motor for the 7-axis.
NC0056	5	3	*-axis motor does not match	There is a problem with the type of motor connected to the 8-axis.	Check the type of motor for the 8-axis.
NC0057	5	3	P1-axis motor does not match	There is a problem with the type of motor connected to the P1-axis.	Check the type of motor for the P1-axis.
NC0058	5	3	P2-axis motor does not match	There is a problem with the type of motor connected to the P2-axis.	Check the type of motor for the P2-axis.
NC0059	5	3	P3-axis motor does not match	There is a problem with the type of motor connected to the P3-axis.	Check the type of motor for the P3-axis.
NC0060	5	3	P4-axis motor does not match	There is a problem with the type of motor connected to the P4-axis.	Check the type of motor for the P4-axis.
NC0061	5	3	Communication error with relocation detection device	Failed to communicate with relocation detection device.	1. The cable connection is faulty. 2. The cable may be broken or cut. Replace the cable. 3. The relocation detection device is faulty. Replace the relocation detection device.
NC0062	5	3	Spindle servo amplifier does not match	There is a problem with the type of servo amplifier connected to the spindle.	Check the type of servo amplifier for the spindle.
NC0063	5	3	Magazine servo amplifier does not match	There is a problem with the type of servo amplifier connected to the magazine.	Check the type of servo amplifier for the magazine.
NC0064	5	3	Spindle motor does not match	There is a problem with the type of motor connected to the spindle.	Check the type of motor for the spindle.
NC0065	5	3	Magazine motor does not match	There is a problem with the type of motor connected to the magazine.	Check the type of motor for the magazine.
NC0066	5	3	Watch dog (loading system)	Loading system error was detected.	Check the loading system status.
NC0067	5	3	Hardware compatibility error (BIO)	BIO PCB is connected to a machine where the BIO PCB is not attached.	1. If this alarm is triggered after replacing the IO PCB, make sure that the IO PCB connectors are connected. 2. Remove the BIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
NC0068	5	3	AT-axis servo amplifier does not match	There is a problem with the type of servo amplifier connected to the AT-axis.	Check the type of servo amplifier for the AT-axis.
NC0069	5	3	AT-axis motor does not match	There is a problem with the type of motor connected to the AT-axis.	Check the type of motor for the AT-axis.
NC0500	5	2	No error number	An alarm not registered in the error number table occurred. The alarm may occur when the version is inconsistent.	1. Execute version upgrade. 2. NC PCB is defective. Replace NC PCB.
NC0501	5	2	Data error (machine parameter)	Data structure of machine parameter is destroyed.	1. Press the [RST] key, and delete the destroyed data by memory display in edit mode and create new data again. 2. The NC PCB is defective. 3. The CF is defective.
NC0502	5	2	Safety check error 2	The outer pallet's additional axis moved when the door interlock was enabled, the side door was closed, the front door was open, and the QT-axis was indexed.	NC PCB is defective. Replace NC PCB.
NC0503	5	2	Safety check error 3	An axis moved when the door interlock was enabled and the side door was open, or when the door interlock was enabled, the side door was open, the front door was open, and the QT-axis was not indexed.	1. Do not change the door interlock setting from disable to enable while the axis is operating. 2. NC PCB is defective. Replace NC PCB.
NC0504	5	2	System monitor M1(*)		
NC0505	5	2	System monitor L1(*)		
NC0506	5	2	Safety check error 5	The outer pallet's additional axis failed to stop when the door interlock was enabled, the side door was closed, the front door was open, and the QT-axis was indexed (including inside door closed).	NC PCB is defective. Replace NC PCB.
NC0507	5	2	Safety check error 6	Spindle and XYZ and additional axes failed to stop when the door interlock was enabled and the side door was open (or the door was open and the QT-axis was not indexed).	NC PCB is defective. Replace NC PCB.
NC0508	5	2	Safety check error 1	The spindle was rotating when the door interlock was disabled and the side door was open, or when the door interlock was disabled, the side door was open, the front door was open, and the QT-axis was not indexed.	NC PCB is defective. Replace NC PCB.
NC0509	5	2	Safety check error 4	Spindle failed to stop when the door interlock was disabled and the side door was open (or the door was open and the C-axis was not indexed).	NC PCB is defective. Replace NC PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
NC0510	5	2	Version compatability error (Main/Teaching pendant)	The software version encountered a compatibility error because the NC PCB or the teach pendant was replaced.	1. Revert to the original part. 2. Upgrade the software
NC2000	4	2	ROM error (main)	The system area in the main CPU is damaged.	1. Execute version upgrade. 2. NC PCB is defective. Replace NC PCB.
NC2001	4	2	Edit error 1	The data was damaged during editing.	1. LOCAL PCB is defective. Replace LOCAL PCB. 2. NC PCB is defective. Replace NC PCB. 3.CF may be defective. Replace the CF.
NC2002	4	2	Edit error 2	The data was damaged during editing.	1. LOCAL PCB is defective. Replace LOCAL PCB. 2. NC PCB is defective. Replace NC PCB. 3.CF may be defective. Replace the CF.
NC2003	4	2	Edit error 3	The data cannot be edited.	1. LOCAL PCB is defective. Replace LOCAL PCB. 2. NC PCB is defective. Replace NC PCB. 3.CF may be defective. Replace the CF. 4.Decrease the program size when editing the program, or delete other programs to increase memory space, and then retry operation. Press the [RST] key.
NC2004	4	2	Parity error (user parameter)	The user parameter data is corrupt.	1.Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the corrupt data, and then create a new program. 2.NC PCB is defective. Replace NC PCB. 3.The CF is defective. Replace the CF.
NC2005	4	2	Parity error (machine parameter)	The machine parameter data is corrupt.	1.Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the corrupt data, and then create a new program. 2.NC PCB is defective. Replace NC PCB. 3.The CF is defective. Replace the CF.
NC2006	4	2	Parity error (program)	The program is corrupt.	1.Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the corrupt data, and then create a new program. 2.NC PCB is defective. Replace NC PCB. 3.The CF is defective. Replace the CF.

No.	Stop level	Reset level	Alarm message	Cause	Solution
NC2007	4	2	Parity error (soft switch)	The soft switch data is damaged.	1. Can occur when replacing NC PCB. Follow the specified procedure carefully. 2. NC PCB is defective. Replace NC PCB.
NC2008	4	2	Parity error (tool list)	The tool list data structure is damaged.	1.Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the damaged data, and then create new data. 2.NC PCB is defective. Replace NC PCB. 3.The CF is defective. Replace the CF.
NC2009	4	2	Parity error (tool pattern)	The tool pattern data structure is damaged.	1.Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the damaged data, and then create new data. 2.NC PCB is defective. Replace NC PCB. 3.The CF is defective.
NC2010	4	2	Parity error (tap hole)	The tap hole data structure is damaged.	1.Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the damaged data, and then create new data. 2.NC PCB is defective. Replace NC PCB. 3.The CF is defective. Replace the CF.
NC2011	4	2	Parity error (cutting condition)	The cutting condition data structure is damaged.	1.Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the damaged data, and then create new data. 2.NC PCB is defective. Replace NC PCB. 3.The CF is defective. Replace the CF.
NC2012	4	2	Parity error (workpc. coord. zero)	The workpiece coordinate zero point data is damaged.	1.Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the damaged data, and then create new data. 2.NC PCB is defective. Replace NC PCB. 3.The CF is defective.
NC2013	4	2	Parity error (tool data)	The tool data is damaged.	1.Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the damaged data, and then create new data. 2.NC PCB is defective. Replace NC PCB. 3.The CF is defective. Replace the CF.

No.	Stop level	Reset level	Alarm message	Cause	Solution
NC2014	4	2	Parity error (macro variables)	The macro variable data is damaged.	1. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the damaged data, and then create new data. 2. NC PCB is defective. Replace NC PCB. 3. The CF is defective. Replace the CF.
NC2015	4	2	Data error (tool list)	1. Data bank file for the current plane does not exist. 2. Data bank version of the file of current plane is obsolete. 3. Data structure is damaged.	1. Use another plane or create a new file. 2. Edit the relevant data bank and save. 3. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the corrupt data, and create a new program
NC2016	4	2	Data error (tool pattern)	1. Data bank file for the current plane does not exist. 2. Data bank version of the file of current plane is obsolete. 3. Data structure is damaged.	1. Use another plane or create a new file. 2. Edit the relevant data bank and save. 3. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the corrupt data, and create a new program
NC2017	4	2	Data error (tap hole)	1. Data bank file for the current plane does not exist. 2. Data bank version of the file of current plane is obsolete. 3. Data structure is damaged.	1. Use another plane or create a new file. 2. Edit the relevant data bank and save. 3. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the corrupt data, and create a new program
NC2018	4	2	Data error (cutting condition)	1. Data bank file for the current plane does not exist. 2. Data bank version of the file of current plane is obsolete. 3. Data structure is damaged.	1. Use another plane or create a new file. 2. Edit the relevant data bank and save. 3. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the corrupt data, and create a new program
NC2019	4	2	Data error (workpc. coord. zero)	1. Data bank file for the current plane does not exist. 2. Data bank version of the file of current plane is obsolete. 3. Data structure is damaged.	1. Use another plane or create a new file. 2. Edit the relevant data bank and save. 3. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the corrupt data, and create a new program

No.	Stop level	Reset level	Alarm message	Cause	Solution
NC2020	4	2	Data error (tool data)	1. Data bank file for the current plane does not exist. 2. Data bank version of the file of current plane is obsolete. 3. Data structure is damaged.	1. Use another plane or create a new file. 2. Edit the relevant data bank and save. 3. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the corrupt data, and create a new program
NC2021	4	2	Data error (macro variables)	1. Data bank file for the current plane does not exist. 2. Data bank version of the file of current plane is obsolete. 3. Data structure is damaged.	1. Use another plane or create a new file. 2. Edit the relevant data bank and save. 3. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the corrupt data, and create a new program
NC2022	4	2	Data error (user parameter)	1. Data bank file for the current plane does not exist. 2. Data bank version of the file of current plane is obsolete. 3. Data structure is damaged.	1. Use another plane or create a new file. 2. Edit the relevant data bank and save. 3. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the corrupt data, and create a new program
NC2023	4	2	No required data	The data has been lost during processing.	1. NC PCB is defective. Replace NC PCB. 2. CF may be defective. Replace the CF.
NC2024	4	2	Check sum error (workpc. coord. zero)	The workpiece coordinate zero point data structure is damaged.	1. Can occur when replacing NC PCB. Follow the specified procedure carefully. 2. Press the [EDIT] key to display the <Directory display> screen. Delete the damaged data, and create new data. 3. NC PCB is defective. Replace NC PCB.
NC2025	4	2	Check sum error (tool data)	The tool data structure is damaged.	1. Can occur when replacing NC PCB. Follow the specified procedure carefully. 2. Press the [EDIT] key to display the <Directory display> screen. Delete the damaged data, and create new data. 3. NC PCB is defective. Replace NC PCB.
NC2026	4	2	Check sum error (macro variables)	The macro variable data structure is damaged.	1. Can occur when replacing NC PCB. Follow the specified procedure carefully. 2. Press the [EDIT] key to display the <Directory display> screen. Delete the damaged data, and create new data. 3. NC PCB is defective. Replace NC PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
NC2027	4	2	Check sum error (soft switch)	The soft switch data is damaged.	1. Can occur when replacing NC PCB. Follow the specified procedure carefully. 2.NC PCB is defective. Replace NC PCB.
NC2028	4	2	Disk error (*)	Disk connection is faulty or disk capacity is not sufficient.	Check the CF.
NC2029	4	2	System error (MC)	An error was received but the corresponding error cannot be identified.	Press the [RST] key to clear the alarm. NC PCB is defective. Replace NC PCB.
NC2030	4	2	System monitor M2(*)		
NC2031	4	2	System monitor L2(*)		
NC2032	4	2	System error (slave)	An error was received but the corresponding error cannot be identified.	Press the [RST] key to clear the alarm. NC PCB is defective. Replace NC PCB.
NC2033	4	2	Parity error (external input/output signal)	External program I/O signal data is damaged.	1.Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the damaged data, and then create new data. 2.NC PCB is defective. Replace NC PCB. 3.CF may be defective. Replace the CF.
NC2034	4	2	Parity error (communication parameter)	Communication parameter data is damaged.	1.Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the damaged data, and then create new data. 2.NC PCB is defective. Replace NC PCB. 3.CF may be defective. Replace the CF.
NC2035	4	2	Data error (external input/output signal)	1. Data bank file does not exist. 2. Data bank version of the file is obsolete. 3. Data structure is damaged.	1. Create a new file. 2. Edit the relevant data bank and save. 3. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the corrupt data, and create a new program
NC2036	4	2	Data error (communication parameter)	1. Data bank file does not exist. 2. Data bank version of the file is obsolete. 3. Data structure is damaged.	1. Create a new file. 2. Edit the relevant data bank and save. 3. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the corrupt data, and create a new program
NC2037	4	2	Check sum error (machine data 5)	Machine data 5 is damaged.	Enter machine data 5.
NC2038	4	2	Check sum error (SYSC89)	Machine data 11 (SYSC89) is destroyed.	Enter machine data 11 (SYSC89).

No.	Stop level	Reset level	Alarm message	Cause	Solution
NC2039	4	2	Parity error (G/M code macro parameter)	G/M code macro parameter is corrupt.	1. Press the [RST] key, and delete the corrupt data in the directory display while in program edit mode. Then, create a new program. 2. The NC PCB is defective. Replace the NC PCB. 3. The CF is defective. Replace the CF.
NC2040	4	2	Data error (G/M code macro parameter)	1. Data bank version of the file is obsolete. 2. Data structure is damaged.	1. Edit the relevant data bank and save. 2. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the corrupt data, and create a new program
NC2041	4	2	Parity error (Special setting)	The special setting data is corrupt.	1. Press the [RST] key, and delete the corrupt data in the directory display while in program edit mode. Then, create a new program. 2. The NC PCB is defective. Replace the NC PCB. 3. The CF is defective. Replace the CF.
NC2042	4	2	Data error (Special setting)	1. Data bank version of the file is obsolete. 2. Data structure is damaged.	1. Edit the relevant data bank and save. 2. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the corrupt data, and create a new program
NC2043	4	2	Parity error (Machining load monitor)	Machining load monitor data is corrupt.	1. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the damaged data, and then create new data. 2. NC PCB is defective. Replace NC PCB. 3. The CF is defective. Replace the CF.
NC2044	4	2	Data error (Machining load monitor)	1. Data bank version of the file is obsolete. 2. Data structure is damaged.	1. Edit the relevant data bank and save. 2. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the corrupt data, and create a new program.
NC5500	2	2	Scheduled replacement of relocat. detect. device batteries	It is time to change the batteries on the relocation detection device.	Follow the procedure below to replace the batteries for the relocation detection device. 1. Leave the power switch turned ON, and remove all the batteries from the relocation detection device. 2. Check if the alarm <<Change batteries on relocat. detect. device with power ON>> triggers. 3. Install new batteries for the relocation detection device. 4. Press the [RST] key and check if the alarm <<Change batteries on relocat. detect. device with power ON>> resets.

No.	Stop level	Reset level	Alarm message	Cause	Solution
NC7000	1	2	Check sum error (SYSC89)	Machine data 11 (SYSC89) is destroyed.	Contact the nearest BROTHER approved service dealer.
NC7002	1	2	Version compatability error (Main/Teaching pendant)	The software version encountered a compatibility error because the NC PCB or the teach pendant was replaced.	1. Revert to the original part. 2. Upgrade the software
NC7003	1	2	Relocation was detected	The relocation detection device detected that the machine has been relocated.	Check the request code and contact the sales dealer.
NC7004	1	2	Change batteries on relocat. detect. device with power ON	The battery voltage dropped on the relocation detection device.	While the power switch is turned ON, replace all the batteries in the battery compartment (GB3) on the side of the control box. Note that turning OFF the power switch may activate the relocation detection.
NC7005	1	2	Reset the relocation detection status	The extension granted for the relocation detection status has ended.	Check the request code and contact the sales dealer.
NC7006	1	2	The input operation for the release code is locked	An invalid or incorrect release code has been input repeatedly exceeding the number of allowable attempts.	Check the request code and contact the sales dealer.
NC7007	1	2	System monitor M4(*)		
NC7500	1	1	Check sum error	Data structure is destroyed.	Contact the nearest BROTHER approved service dealer.
NC7501	1	1	Parity error	Data structure is destroyed.	Contact the nearest BROTHER approved service dealer.
NC7502	1	1	Parity error (tool list)	The tool list data is damaged.	1. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the damaged data, and then create new data. 2. NC PCB is defective. 3. The CF is defective.
NC7503	1	1	Parity error (cutting condition)	The cutting condition data is damaged.	1. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the damaged data, and then create new data. 2. NC PCB is defective. 3. The CF is defective.
NC7504	1	1	Parity error (program)	The program is damaged.	1. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the damaged data, and then create new data. 2. NC PCB is defective. 3. The CF is defective.
NC7505	1	1	Data error (data bank)	Data structure in a data bank that is not selected is damaged.	Press the [EDIT] key to display the <Directory display> screen. Delete the damaged data, and create new data.

No.	Stop level	Reset level	Alarm message	Cause	Solution
NC7506	1	1	Data error (tool list)	The tool list data structure is damaged.	1. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the damaged data, and then create new data. 2. NC PCB is defective. 3. The CF is defective.
NC7507	1	1	Data error (cutting condition)	The cutting condition data structure is damaged.	1. Press the [RST] key, and then press the [EDIT] key to display the <Directory display> screen. Delete the damaged data, and then create new data. 2. NC PCB is defective. 3. The CF is defective.
NC7508	1	1	Edit error 1	The data was damaged during editing when converting a conversation program to an NC program.	1. LOCAL PCB is defective. Replace LOCAL PCB. 2. NC PCB is defective. Replace NC PCB. 3. CF may be defective. Replace the CF.
NC7509	1	1	Open error (program)	Operation program cannot be analyzed when converting a conversation program to an NC program.	1. LOCAL PCB is defective. Replace LOCAL PCB. 2. CF may be defective. Replace the CF.
NC7510	1	1	Open error (cutting condition)	Cutting conditions used for operation cannot be referred to when converting a conversation program to an NC program.	1. LOCAL PCB is defective. Replace LOCAL PCB. 2. CF may be defective. Replace the CF.
NC7511	1	1	System monitor M3(*)		
NC7512	1	1	System monitor L3(*)		
NC7513	1	1	Cannot communicate with built-in PLC	Connect external PLC editor by communication. Cause of error, data transfer is not available with PLC.	1. Stop the external PLC editor connection. 2. Check the ladder program. 3. NC PCB is defective.
NC7514	1	1	Relocation status being detected	An attempt was made to execute a spindle rotation operation in memory operation, MDI operation or manual operation mode even though the relocation status is being detected.	Check the request code and contact the sales dealer.
NC7515	1	1	Release code is invalid	The release code is not correct or invalid.	1. Make sure that the release code is correct. 2. The request code may have been updated. Make sure that the request code is the same as the code provided to the sales dealer.
NC7516	1	1	Release code expired	The release code has expired.	Check the request code and contact the sales dealer.

11.7.5 IO

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0000	5	3	Power is turned off.	The power switch was turned off. Or the automatic power off time has elapsed.	Although this alarm is displayed at the moment when the power is turned off, it does not indicate abnormality. In case where it occurs at other situations, contact the nearest BROTHER approved service dealer for the situation it occurred.
IO0001	5	3	AC power low	1. AC input power is less than the acceptable voltage. 2. AC input power dropped instantaneously.	1. Check the power-supply voltage. - For 400 V region Change the tap connection of transformer 1. Check the voltage source capacity. Check the length and diameter of the power cable. - For 200 V region Check the voltage source capacity. Check the length and diameter of the power cable. 2. Check for momentary power failure on the plant-side power supply.
IO0002	5	3	24VDC power supply error(IO)	1. There is an overload on the 24 V power supply for TA1. 2. There is a short- circuit/ ground fault on the 24 V power supply for TA1. 3. There is a short- circuit/ ground fault on the 24 V power supply for the IO PCB. 4. There is a short- circuit/ ground fault on the 24 V power supply for the IL PCB. 5. There is a short- circuit/ ground fault on the 24 V power supply for the EXIO PCB. 6. There is a short- circuit/ ground fault on the 24 V power supply for the ID PCB.	1. Reduce the load for the added connection on the 24 V power supply. 2-1. Check for a short- circuit between the 24 V and 0 V, or between the 24 V and the ground. 2-2. Identify the short by disconnecting the connections sequentially and remove the cause of the problem. 3. Check the following wiring connections. XTIO_A pin1, XTIO_A pin3, XEX24 and XILIO 4. Check the following wiring connections. XILIO, X24 and XILKEY 5. Check the following wiring connections. XTEXPW, XPWEX1 and XPWEX2 6. Check the following wiring connections. XIDPW1 and XIDPW2
IO0003	5	3	3-phase power supply error	The three- phase power supply's phases are reversed.	Reverse the positions of the L1-phase cord and L2-phase cord on the primary side of the main breaker. Note: Phases for wiring inside the machine are fixed. Do not change the positions of cords.
IO0004	5	3	The DC power 24 V is abnormal.	1. The 24 V power supply for TA1 is faulty. 2. There is a short- circuit/ ground fault on the 24 V power supply for TA1.	1. Replace TA1. 2. Check for a short-circuit between 24 V and 0 V, or between 24 V and the ground.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0005	5	3	The door is abnormal. (Electric cylinder)	DRALM1_ERR signal (BY: Door abnormal 1 (electric cylinder)) was detected. For the factory-installed ladder program, electric cylinder abnormality was detected.	1. Check the ladder if the content has changed from the factory default settings. Consult with the person who changed the ladder program. 2. Check the light status (LED) of the electric cylinder controller and then contact us.
IO0006	5	3	Fuse meltdown error (*)	A fuse was blown by short circuit in 24 V power supply. After B52J145-4 when there is a character string 0003 or 0004 inside the parentheses (*), the following are potential causes. 1. Fuse has fallen off or is out of place. When the character string is 0003, FH1 on the IO PCB has fallen off or is out of place. When the character string is 0004, FH2 on the IO PCB has fallen off or is out of place. 2. Fuse has melted down. When an external power supply is used, if the character string is 0003, then a total current greater than 6 A ran through N100 to 107 on the XTO_B connector. If the character string is 0004, then it ran through N108 to 115 on the XTO_B connector.	1. Replace the IO PCB. The number in brackets () indicates the point where the fuse was blown. Check the wiring points corresponding to the following numbers which will be indicated in brackets (). 0001 : XTI_B N0-7 0002 : XTI_B N8-15 0003 : XTO_B N100-107 0004 : XTO_B N108-115 0005 : XTIO_B pin14,16 0006 : XTIO_B pin8,12 2. After B52J145-4, when there is a character string 0003 inside the parentheses (*), check and replace the fuse FH1 on the IO PCB. When there is a 0004, check and replace the fuse FH2 on the IO PCB. Refer to “8.2 Fuse Replacement Procedure After Fuse Meltdown Error” in the Installation Manual. 3. Replace the IO PCB.
IO0007	5	3	The 12-V DC power is overcurrent. (1) (*)	1. Overcurrent occurred in the KEY PCB. There is a 12 V power short-circuit on the KEY PCB. 2. There is a KEY PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring points corresponding to the following numbers which will be indicated in brackets (). KEY:XDOOR,XOP,XMOV (KEY PCB ASSY C00#4),XFOV (KEY PCB ASSY C00#4),XKSW,XSPOV (KEY PCB ASSY C00#4),XAB 2. Replace the KEY PCB.
IO0008	5	3	The 12-V DC power is overcurrent. (2) (*)	1. Overcurrent occurred in the KEY PCB. There is a 12 V power short-circuit on the KEY PCB. 2. There is a KEY PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring points corresponding to the following numbers which will be indicated in brackets (). KEY:All of the following points KY_STA_LD :XKSW KY_STP_LD :XKSW KY_OSTA_LD :XQT KY_OUTA_LD :XAB KY_OUTB_LD :XAB 2. Replace the KEY PCB.
IO0009	5	3	The 12-V DC power is overcurrent. (3) (*)	-	-
IO0010	5	3	The 12-V is DC power overcurrent. (1) (*)	Reserved	Reserved

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0011	5	3	The 12-V DC power is overcurrent. (2) (*)	Reserved	Reserved
IO0012	5	3	The 12-V DC power is overcurrent. (3) (*)	Reserved	Reserved
IO0014	5	3	The 12-V DC power is overcurrent. (1) (MP)	Reserved	Reserved
IO0015	5	3	The 12-V DC power is overcurrent. (2) (MP)	Reserved	Reserved
IO0016	5	3	The 12-V is DC power overcurrent. (3) (MP)	Reserved	Reserved
IO0017	5	3	The 12-V DC power is overcurrent. (4) (MP)	Reserved	Reserved
IO0018	5	3	The 24-V is DC power overcurrent. (1) (*)	1. Overcurrent occurred in the IO PCB. There is a 24 V power short-circuit on the IO PCB. 2. There is an IO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring points corresponding to the following numbers which will be indicated in brackets (). IO : XTIO_A pin13, XTIO_A pin15, XTIO_A pin19 and XTIO_A pin21 All of the following points. IO_LIGHT : XTPW pin3 IO_EXSTOUT : XTIO_B pin10 IO_STOUT : XTIO_B pin6 2. Replace the IO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0019	5	3	The 24-V DC power is overcurrent. (2) (*)	1. Overcurrent occurred in the IO PCB. There is a 24 V power short-circuit on the IO PCB. 2. There is an IO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring points corresponding to the following numbers which will be indicated in brackets (). IO: XTI_A P0-15, XTPW IO24, XTIO_A pin23 All of the following points IO_PO_100 : XTO_A P100 IO_PO_101 : XTO_A P101 IO_PO_102 : XTO_A P102 IO_PO_103 : XTO_A P103 IO_PO_104 : XTO_A P104 IO_PO_105 : XTO_A P105 IO_PO_106 : XTO_A P106 IO_PO_107 : XTO_A P107 IO_PO_108 : XTO_A P108 IO_PO_109 : XTO_A P109 IO_PO_110 : XTO_A P110 IO_PO_111 : XTO_A P111 IO_PO_112 : XTO_A P112 IO_PO_113 : XTO_A P113 IO_PO_114 : XTO_A P114 IO_PO_115 : XTO_A P115 IO : XTO_A P100 - 115 XTI_A P0 - 15 XTPW IO24 XTIO_A pin23 2. Replace the IO PCB.
IO0020	5	3	The 24-V DC power is overcurrent. (3) (*)	1. Overcurrent occurred in the IO PCB. There is a 24 V power short-circuit on the IO PCB. 2. There is an IO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring points corresponding to the following numbers which will be indicated in brackets (). IO: All of the following points IO_ARMBKRLS : XYBARM IO_SPOUT1 : XSUB3 IO_SPOUT7 : XSUB3 2. Replace the IO PCB.
IO0021	5	3	The 24-V DC power is overcurrent. (4) (*)	1. Overcurrent occurred in the IO PCB. There is a 24 V power short-circuit on the IO PCB. 2. There is an IO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring points corresponding to the following numbers which will be indicated in brackets (). IO: All of the following points IO_QTUNCV_N : XUNC1 IO_SPOUT8 : XSUB1 IO_M200 : XTOOL 2. Replace the IO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0022	5	3	The 24-V DC power is overcurrent. (5) (*)	1. Overcurrent occurred in the IO PCB. There is a 24 V power short-circuit on the IO PCB. 2. There is an IO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring points corresponding to the following numbers which will be indicated in brackets (). IO: All of the following points IO_SPOUT5 : XSUB2 IO_SPOUT6 : XSUB2 IO_SPFON : XSPKF IO_KS1ON : XSVP IO_MNUUC : XATC2 2. Replace the IO PCB.
IO0023	5	3	The 24-V DC power is overcurrent. (6) (*)	1. Overcurrent occurred in the IO PCB. There is a 24 V power short-circuit on the IO PCB. 2. There is an IO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring points corresponding to the following numbers which will be indicated in brackets (). IO: All of the following points IO_OILPMP : XLUB IO_FAN1ON : XBIO IO_FAN2ON : XBIO IO_FAN3ON : XBIO 2. Replace the IO PCB.
IO0024	5	3	The 24-V DC power is overcurrent. (7) (*)	1. Overcurrent occurred in the IO PCB. There is a 24 V power short-circuit on the IO PCB. 2. There is an IO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring points corresponding to the following numbers which will be indicated in brackets (). IO: All of the following points IO_DRCV : XADR IO_DROV : XADR IO_CTSPMPON : XCTS IO_CTSBWVON : XCTS IO_VSUPON : XCTS IO_CTSDVON : XCTS IO_SPOUT3 : XSUB1 IO_COLPMP : XBIO 2. Replace the IO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0025	5	3	The 24-V DC power is overcurrent. (8) (*)	1. Overcurrent occurred in the IO PCB. There is a 24 V power short-circuit on the IO PCB. 2. There is an IO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring points corresponding to the following numbers which will be indicated in brackets (). IO: All of the following points IO_TBvon : XYVB IO_JIGSHW : XYVJS IO_M08 : XYVC IO_AIRBLW : XYVA IO_TCVON : XYVTC IO_POTDWNV : XYVP IO_POTUPV : XYVP IO_GRN : XPAT2 IO_RED : XPAT2 IO_YEL : XPAT2 IO_DRAL_RST : XADR IO_DR_OUT5 : XADR IO_ATCASCv : XATC2 IO_ATCASOV : XATC2 IO_ATCSCv : XATC2 IO_ATCSOV : XATC2 IO_POTBKRLS : XATC2 IO_CTSVON : XCTS IO_SPOUT2 : XSUB1 IO_SPOUT4 : XSUB1 IO_CHIPPMP : XBIO 2. Replace the IO PCB.
IO0026	5	3	The 24-V DC power is abnormal. EXIO 01 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0027	5	3	The 24-V DC power is abnormal. EXIO 02 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0028	5	3	The 24-V DC power is abnormal. EXIO 03 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0029	5	3	The 24-V DC power is abnormal. EXIO 04 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0030	5	3	The 24-V DC power is abnormal. EXIO 05 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0031	5	3	The 24-V DC power is abnormal. EXIO 06 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0032	5	3	The 24-V DC power is abnormal. EXIO 07 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0033	5	3	The 24-V DC power is abnormal. EXIO 08 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0034	5	3	The 24-V DC power is abnormal. EXIO 09 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0035	5	3	The 24-V DC power is abnormal. EXIO 10 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0036	5	3	The 24-V DC power is abnormal. EXIO 11 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0037	5	3	The 24-V DC power is abnormal. EXIO 12 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0038	5	3	The 24-V DC power is abnormal. EXIO 13 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0039	5	3	The 24-V DC power is abnormal. EXIO 14 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0040	5	3	The 24-V DC power is abnormal. EXIO 15 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0041	5	3	The 24-V DC power is abnormal. EXIO 16 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0042	5	3	The 24-V DC power is abnormal. EXIO 17 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0043	5	3	The 24-V DC power is abnormal. EXIO 18 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0044	5	3	The 24-V DC power is abnormal. EXIO 19 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0045	5	3	The 24-V DC power is abnormal. EXIO 20 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0046	5	3	The 24-V DC power is abnormal. EXIO 21 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0047	5	3	The 24-V DC power is abnormal. EXIO 22 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0048	5	3	The 24-V DC power is abnormal. EXIO 23 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0049	5	3	The 24-V DC power is abnormal. EXIO 24 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0050	5	3	The 24-V DC power is abnormal. EXIO 25 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0051	5	3	The 24-V DC power is abnormal. EXIO 26 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0052	5	3	The 24-V DC power is abnormal. EXIO 27 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0053	5	3	The 24-V DC power is abnormal. EXIO 28 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0054	5	3	The 24-V DC power is abnormal. EXIO 29 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0055	5	3	The 24-V DC power is abnormal. EXIO 30 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0056	5	3	The 24-V DC power is abnormal. EXIO 31 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0057	5	3	The 24-V DC power is abnormal. EXIO 32 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0058	5	3	The 24-V DC power is abnormal. EXIO 33 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0059	5	3	The 24-V DC power is abnormal. EXIO 34 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0060	5	3	The 24-V DC power is abnormal. EXIO 35 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0061	5	3	The 24-V DC power is abnormal. EXIO 36 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0062	5	3	The 24-V DC power is abnormal. EXIO 37 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0063	5	3	The 24-V DC power is abnormal. EXIO 38 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0064	5	3	The 24-V DC power is abnormal. EXIO 39 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0065	5	3	The 24-V DC power is abnormal. EXIO 40 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0066	5	3	The 24-V DC power is abnormal. EXIO 41 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0067	5	3	The 24-V DC power is abnormal. EXIO 42 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0068	5	3	The 24-V DC power is abnormal. EXIO 43 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0069	5	3	The 24-V DC power is abnormal. EXIO 44 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0070	5	3	The 24-V DC power is abnormal. EXIO 45 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0071	5	3	The 24-V DC power is abnormal. EXIO 46 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0072	5	3	The 24-V DC power is abnormal. EXIO 47 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0073	5	3	The 24-V DC power is abnormal. EXIO 48 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0074	5	3	The 24-V DC power is abnormal. EXIO 49 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0075	5	3	The 24-V DC power is abnormal. EXIO 50 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0076	5	3	The 24-V DC power is abnormal. EXIO 51 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0077	5	3	The 24-V DC power is abnormal. EXIO 52 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0078	5	3	The 24-V DC power is abnormal. EXIO 53 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0079	5	3	The 24-V DC power is abnormal. EXIO 54 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0080	5	3	The 24-V DC power is abnormal. EXIO 55 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0081	5	3	The 24-V DC power is abnormal. EXIO 56 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0082	5	3	The 24-V DC power is abnormal. EXIO 57 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0083	5	3	The 24-V DC power is abnormal. EXIO 58 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.
IO0084	5	3	The 24-V DC power is abnormal. EXIO 59 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0085	5	3	The 24-V DC power is abnormal. EXIO 60 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error. 3. Overcurrent occurred on the ID PCB (DIP-SW for ID PCB on <60> PCB). 4. There is an ID PCB (DIP-SW for ID PCB on <60> PCB) error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB. 3. The signal causing the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the ID PCB when the text in the parentheses is 0-1. -0: Short-circuit 1. (Short-circuit on the PCB) -1: Short-circuit 2. (Short-circuit on the PCB) -2: Short-circuit 3. XIDAP -3: Short-circuit 4. XIDAP -4: Voltage drop. XPWID1 and XPWID2 4. Replace the ID PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0086	5	3	The 24-V DC power is abnormal. EXIO 61 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error. 3. Overcurrent occurred on the ID PCB (DIP-SW for ID PCB on <61> PCB). 4. There is an ID PCB (DIP-SW for ID PCB on <61> PCB) error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB. 3. The signal causing the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the ID PCB when the text in the parentheses is 0-1. -0: Short-circuit 1. (Short-circuit on the PCB) -1: Short-circuit 2. (Short-circuit on the PCB) -2: Short-circuit 3. XIDAP -3: Short-circuit 4. XIDAP -4: Voltage drop. XPWID1 and XPWID2 4. Replace the ID PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0087	5	3	The 24-V DC power is abnormal. EXIO 62 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error. 3. Overcurrent occurred on the ID PCB (DIP-SW for ID PCB on <62> PCB). 4. There is an ID PCB (DIP-SW for ID PCB on <62> PCB) error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB. 3. The signal causing the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the ID PCB when the text in the parentheses is 0-1. -0: Short-circuit 1. (Short-circuit on the PCB) -1: Short-circuit 2. (Short-circuit on the PCB) -2: Short-circuit 3. XIDAP -3: Short-circuit 4. XIDAP -4: Voltage drop. XPWID1 and XPWID2 4. Replace the ID PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0088	5	3	The 24-V DC power is abnormal. EXIO 63 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error. 3. Overcurrent occurred on the ID PCB (DIP-SW for ID PCB on <63> PCB). 4. There is an ID PCB (DIP-SW for ID PCB on <63> PCB) error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB. 3. The signal causing the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the ID PCB when the text in the parentheses is 0-1. -0: Short-circuit 1. (Short-circuit on the PCB) -1: Short-circuit 2. (Short-circuit on the PCB) -2: Short-circuit 3. XIDAP -3: Short-circuit 4. XIDAP -4: Voltage drop. XPWID1 and XPWID2 4. Replace the ID PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0089	5	3	The 24-V DC power is abnormal. EXIO 64 (*)	1. Overcurrent occurred in the EXIO PCB. There is a 24 V power short-circuit on the EXIO PCB. 2. There is an EXIO PCB error. 3. Overcurrent occurred on the ID PCB (DIP-SW for ID PCB on <64> PCB). 4. There is an ID PCB (DIP-SW for ID PCB on <64> PCB) error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the EXIO PCB for items 5 through 7 in the brackets (). 0: Short-circuit 1. XTIN_1A pin0-pin15 1: Short-circuit 2. XTIN_1A pin16-pin31 2: Short-circuit 3. XTOUT_1A pin0-pin15 3: Short-circuit 4. XTOUT_1A pin16-pin31 4: Voltage drop. XPWEX1, XPWEX2 and XTEXPW pin1-3 5: Fuse meltdown 1. XTIN_1B pin0-pin15 6: Fuse meltdown 2. XTIN_2B pin16-pin31 7: Fuse meltdown 3. XTOUT_1B pin0-pin15. XTOUT_2B pin16-pin31 2. Replace the EXIO PCB. 3. The signal causing the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). Replace the ID PCB when the text in the parentheses is 0-1. -0: Short-circuit 1. (Short-circuit on the PCB) -1: Short-circuit 2. (Short-circuit on the PCB) -2: Short-circuit 3. XIDAP -3: Short-circuit 4. XIDAP -4: Voltage drop. XPWID1 and XPWID2 4. Replace the ID PCB.
IO0090	5	3	The 24-V DC power is overcurrent. (1) (*)	1. Overcurrent occurred in the IL PCB. There is a 24 V power short-circuit on the IL PCB. 2. There is an IL PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). IL:XBKOUT2, XBKOUT3 and XBKOUT4 2. Replace the IL PCB.
IO0091	5	3	The 24-V DC power is overcurrent. (2) (*)	1. Overcurrent occurred in the IL PCB. There is a 24 V power short-circuit on the IL PCB. 2. There is an IL PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring that corresponds to the following description in the brackets (). IL:XMAING1, XMAING2, XPLT1G, XPLT2G, XPLC1G, XPLC2G, XBKOUT1 2. Replace the IL PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0092	5	3	The 24-V DC power is overcurrent. (3) (*)	1. Overcurrent occurred in the IL PCB. There is a 24 V power short-circuit on the IL PCB. 2. There is an IL PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring points corresponding to the following numbers which will be indicated in brackets (). IL: XTEMSOUT, XTEMSIN, XCHIP, XILKEY IL_CHPCON: XCHIP 2. Replace the IL PCB.
IO0093	5	3	The 24-V DC power is overcurrent. (4) (*)	1. Overcurrent occurred in the IL PCB. There is a 24 V power short-circuit on the IL PCB. 2. There is an IL PCB error.	1. The signal which caused the alarm is indicated in brackets (). Check the wiring points corresponding to the following numbers which will be indicated in brackets (). IL: XDR1, XDR2, XDR3, XDR4, XDR5, XDR6, XPLT1, XPLT2, XCLP IL_DRLK1: XDR1 IL_DRLK2: XDR2 IL_DRLK3: XDR3 IL_DRLK4: XDR4 IL_DRLK5: XDR5 IL_DRLK6: XDR6 2. Replace the IL PCB.
IO0094	5	3	AC power not detected	AC input power cannot be detected.	1. Make sure that the cable is connected to the X200 connector on the IO PCB. 2. Make sure that the motor breaker FC2 relay is ON.
IO0095	5	3	AC power freq. abnormal (out of acceptable range)	AC input power frequency is out of acceptable range.	Check the AC power frequency. Acceptable ranges: For 50 Hz region, 49 Hz to 51 Hz. For 60 Hz region, 59 Hz to 61 Hz.
IO0096	5	3	AC power freq. abnormal (detector abnormal)	The software version may be old.	Update the software to the latest version.
IO0097	5	3	AC power overvoltage	AC input power exceeded the acceptable voltage.	1. Check the power-supply voltage. 1-1. For 400 V region, check that the voltage tap of transformer 1 is connected correctly.
IO0098	5	3	DC power 12V overcurrent 4 (KEY)	1. Overcurrent occurred in the KEY PCB. There is a 12 V power short-circuit on the KEY PCB. There is a 5 V power supply short-circuit on the manual pulse generator. 2. There is a KEY PCB error.	1. Check the wiring for the manual pulse generator. Replace the manual pulse generator and cable. The input signal is short-circuited. Check the following wiring. XMOV XFOV XSPOV 2. Replace the KEY PCB.
IO0099	5	3	DC power 12V overcurrent 1 (TP)	1. Overcurrent occurred in the ES PCB on the teaching pendant. There is a 12 V power short-circuit on the ES PCB. 2. There is a ES PCB error.	1. The input signal is short-circuited. 2. Replace the ES PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0100	5	3	DC power 12V overcurrent 2 (TP)	1. Overcurrent occurred in the ES PCB on the teaching pendant. There is a 12 V power short-circuit on the ES PCB. 2. There is a ES PCB error.	1. The input signal is short-circuited. 2. Replace the ES PCB.
IO0101	5	3	DC power 12V overcurrent 3 (TP)	1. Overcurrent occurred in the ES PCB on the teaching pendant. There is a 12 V power short-circuit on the ES PCB. 2. There is a ES PCB error.	1. The input signal is short-circuited. 2. Replace the ES PCB.
IO0102	5	3	DC power 12V overcurrent 4 (TP)	1. Overcurrent occurred in the ES PCB on the teaching pendant. There is a 12 V power short-circuit on the ES PCB. 2. There is a ES PCB error.	1. The input signal is short-circuited. 2. Replace the ES PCB.
IO0103	5	3	The door is abnormal. (Electric cylinder type 2)	The DRALM1_ERR signal (BY: Door error 1 (Electric cylinder)) was detected. An electric cylinder error was detected in the factory set ladder program.	1. Check the ladder if the content has changed from the factory default settings. Consult with the person who changed the ladder program. 2. Check the light status (LED) of the electric cylinder controller and then contact us.
IO0104	5	3	24VDC power supply error (IO-EM24)	1. There is an IO PCB error. 2. There is an IL PCB error. 3. The connection is faulty on the XILIO cable between the IO PCB and the IL PCB.	1. Replace the IO PCB. 2. Replace the IL PCB. 3. Check the connection for the XILIO cable.
IO0105	5	3	24VDC power supply error (IO-ILPW)	1. There is an IO PCB error. 2. There is an IL PCB error. 3. The connection is faulty on the XILIO cable between the IO PCB and the IL PCB.	1. Replace the IO PCB. 2. Replace the IL PCB. 3. Check the connection for the XILIO cable.
IO0106	5	3	IL PCB not connected	The IL PCB is not connected.	1. Turn OFF the breaker and make sure that the LED on the PCB is turned off. Next, check the wiring between XIOO1-XILIN and between XILIO-XILIO which runs between the IO PCB and IL PCB. Make sure that everything is normal and that there is no disconnected connector, faulty connection or incorrect wiring. 2. The IO PCB or the IL PCB may be damaged if the alarm will not reset even when the wiring is normal.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0107	5	3	IO PCB not connected	The IO PCB is not connected.	1. Turn OFF the breaker and make sure that the LED on the IO PCB is turned off. Next, check the wiring between AVR and IO PCBs and the wiring for the "XSVP" connector. Also, check the LAN connection between the NC PCB and IO PCB and the wiring for the "XIOIN" connector. Make sure that everything is normal and that there is no disconnected connector, faulty connection or incorrect wiring. 2. The IO PCB or the IL PCB may be damaged if the alarm will not reset even when the wiring is normal.
IO0108	5	3	EDM signal not checked 1	There is an EDM1 signal error. The alarm <<Emergency button activated>> or <<Door open>> may also be triggered at the same time.	1. Check the wiring that connects from the XMAING1 and XMAING2 connectors on the IL PCB to the motor amplifier and motor relay. Make sure that everything is normal and that there is no faulty connection or incorrect wiring. 2. Check the jump wiring for the X3, X4 and X10 motor amplifier connectors, and make sure that is everything is normal and that the wires are not reversed. 3. If there is a loop wire for XMAING2, check the connection and fix any problems to ensure everything is normal. 4. If the directions in 1 through 3 do not return operation to normal, then the IL PCB and amplifier may be damaged.
IO0109	5	3	EDM signal not checked 3	There is an EDM3 signal error. The alarm <<Emergency button activated>> or <<Door open>> may also be triggered at the same time.	1. Check the wiring that connects from the XPLT1G connector to the motor amplifier and motor relay. Make sure that everything is normal and that there is no faulty connection or incorrect wiring. 2. Check the jump wiring for the X3 and X4 motor amplifier connectors, and make sure that is everything is normal and that the wires are not reversed. 3. If there is a loop wire for XPLT1G, XPLT2G, XPLC1G and XPLC2G, make sure that the wiring is normal and fix any problems to ensure everything is normal. 4. If the directions in 1 through 3 do not return operation to normal, then the IL PCB and amplifier may be damaged.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0110	5	3	EDM signal not checked 4	There is an EDM4 signal error. The alarm <<Emergency button activated>> or <<Door open>> may also be triggered at the same time.	1. Check the wiring that connects from the XPLT2G connector to the motor amplifier and motor relay. Make sure that everything is normal and that there is no faulty connection or incorrect wiring. 2. Check the jump wiring for the X3 and X4 motor amplifier connectors, and make sure that everything is normal and that the wires are not reversed. 3. If there is a loop wire for XPLT1G, XPLT2G, XPLC1G and XPLC2G, make sure that the wiring is normal and fix any problems to ensure everything is normal. 4. If the directions in 1 through 3 do not return operation to normal, then the IL PCB and amplifier may be damaged.
IO0111	5	3	EDM signal not checked 5	There is an EDM5 signal error. The alarm <<Emergency button activated>> or <<Door open>> may also be triggered at the same time.	1. Check the wiring that connects from the XPLC1G connector to the motor amplifier and motor relay. Make sure that everything is normal and that there is no faulty connection or incorrect wiring. 2. Check the jump wiring for the X3 and X4 motor amplifier connectors, and make sure that everything is normal and that the wires are not reversed. 3. If there is a loop wire for XPLT1G, XPLT2G, XPLC1G and XPLC2G, make sure that the wiring is normal and fix any problems to ensure everything is normal. 4. If the directions in 1 through 3 do not return operation to normal, then the IL PCB and amplifier may be damaged.
IO0112	5	3	EDM signal not checked 6	There is an EDM6 signal error. The alarm <<Emergency button activated>> or <<Door open>> may also be triggered at the same time.	1. Check the wiring that connects from the XPLC2G connector to the motor amplifier and motor relay. Make sure that everything is normal and that there is no faulty connection or incorrect wiring. 2. Check the jump wiring for the X3 and X4 motor amplifier connectors, and make sure that everything is normal and that the wires are not reversed. 3. If there is a loop wire for XPLT1G, XPLT2G, XPLC1G and XPLC2G, make sure that the wiring is normal and fix any problems to ensure everything is normal. 4. If the directions in 1 through 3 do not return operation to normal, then the IL PCB and amplifier may be damaged.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0500	5	2	Emergency button activated	1. [EMERGENCY] switch was pressed. 2. External emergency stop signal became open. -IL PCB XTEMSIN1-2,3-4,5-6 3. Alarm is triggered when the handle or teaching pendant is attached or removed. 4. There is a ground fault on A1+ for KFC1. 5. KFC1 is faulty.	1. Remove the cause of emergency stop and press the [RST] key after releasing the [EMERGENCY] switch. 2. Close the contact of the input terminal and input an external reset signal. 3. Check the short cap attachment after attaching or removing the handle or teaching pendant, and then press the [RST] key. 4. Check the A1+ wiring connection for KFC1. 5. Replace KFC1.
IO0501	5	2	Inverter error (ATC)	1. The ATC turning inverter is faulty. Check the alarm number on the front of the inverter. 2. Uv: Main circuit low voltage Flashing/Stopped 1/Operating 3. ov: Main circuit overvoltage Flashing/Stopped Lighting/Operating 4. oH: Cooling fan overheated Flashing/Stopped Lighting/Operating 5. oC: Overcurrent Overcurrent flew to inverter output. 6. oL: Overload 1 Motor 2 Inverter 3 Excessive torque detected 7. GF: Ground fault A ground fault was detected on the inverter output side. 8. FAN flashing: Cooling fan abnormal 9. F**: Control circuit abnormal	2. Check the wiring of the power supply to the inverter. Inverter side "+1 -" 3. AC power-supply voltage is too high. For 400 V specifications, check the voltage tap of transformer 1. 4-1. Check the operation of the internal fan of the inverter. 4-2. Check the cooling fan of the control panel. 5. Check for short circuit in inverter output. 6. The mechanical parts may be heavy. 7. A ground fault was caused at inverter output. Check the wiring and ATC motor. 8. The inverter fan is faulty. 9. The inverter control circuit is faulty. - 00 Memory initial error - 01 ROM error - 04 EEPROM error - 05 A/D converter error
IO0502	5	2	Door abnormal (sensor error)	DR_SN_ERR signal (BY: Door abnormal (sensor error)) was detected. For the factory-installed ladder program, both of the limit switches on the door open end and close end were turned on (automatic door specifications).	Check the ladder. For the factory-installed ladder program: 1. Make sure that there are no chips or shavings caught between the door and machine cover. 2. Check the door limit switch. 3. The door sensor wiring is short-circuited. 4. The door sensor is faulty. Replace the door sensor. 5. The IL PCB is faulty. Replace the IL PCB. 6. The IO PCB is faulty. Replace the IO PCB. After removing the causes listed above, press the [RST] key to clear the alarm.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0503	5	2	Overtravel (+*)	1. OVTR5P signal was input. 2. The 5th axis was selected and JOG operation by [RELSE]+[+4] was executed with OVTR5P signal ON.	Select the manual operation mode and press the [RST] key while holding down the [RELSE] key. Then, while holding down the [RELSE] key, move each axis to the direction opposite to that indicated by the alarm using the [Axis travel] key so that it goes out of the over travel region.
IO0504	5	2	Overtravel (-*)	1. OVTR5M signal was input. 2. The 5th axis was selected and JOG operation by [RELSE]+[-4] was executed with OVTR5M signal ON.	Select the manual operation mode and press the [RST] key while holding down the [RELSE] key. Then, while holding down the [RELSE] key, move each axis to the direction opposite to that indicated by the alarm using the [Axis travel] key so that it goes out of the over travel region.
IO0505	5	2	Overtravel (+*)	1. OVTR6P signal was input. 2. The 6th axis was selected and JOG operation by [RELSE]+[+4] was executed with OVTR6P signal ON.	Select the manual operation mode and press the [RST] key while holding down the [RELSE] key. Then, while holding down the [RELSE] key, move each axis to the direction opposite to that indicated by the alarm using the [Axis travel] key so that it goes out of the over travel region.
IO0506	5	2	Overtravel (-*)	1. OVTR6M signal was input. 2. The 6th axis was selected and JOG operation by [RELSE]+[-4] was executed with OVTR6M signal ON.	Select the manual operation mode and press the [RST] key while holding down the [RELSE] key. Then, while holding down the [RELSE] key, move each axis to the direction opposite to that indicated by the alarm using the [Axis travel] key so that it goes out of the over travel region.
IO0507	5	2	Overtravel (+*)	1. OVTR7P signal was input. 2. The 7th axis was selected and JOG operation by [RELSE]+[+4] was executed with OVTR7P signal ON.	Select the manual operation mode and press the [RST] key while holding down the [RELSE] key. Then, while holding down the [RELSE] key, move each axis to the direction opposite to that indicated by the alarm using the [Axis travel] key so that it goes out of the over travel region.
IO0508	5	2	Overtravel (-*)	1. OVTR7M signal was input. 2. The 7th axis was selected and JOG operation by [RELSE]+[-4] was executed with OVTR7M signal ON.	Select the manual operation mode and press the [RST] key while holding down the [RELSE] key. Then, while holding down the [RELSE] key, move each axis to the direction opposite to that indicated by the alarm using the [Axis travel] key so that it goes out of the over travel region.
IO0509	5	2	Overtravel (+*)	1. OVTR8P signal was input. 2. The 8th axis was selected and JOG operation by [RELSE]+[+4] was executed with OVTR8P signal ON.	Select the manual operation mode and press the [RST] key while holding down the [RELSE] key. Then, while holding down the [RELSE] key, move each axis to the direction opposite to that indicated by the alarm using the [Axis travel] key so that it goes out of the over travel region.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0510	5	2	Overtravel (-*)	1. OVTR8M signal was input. 2. The 8th axis was selected and JOG operation by [RELSE]+[-4] was executed with OVTR8M signal ON.	Select the manual operation mode and press the [RST] key while holding down the [RELSE] key. Then, while holding down the [RELSE] key, move each axis to the direction opposite to that indicated by the alarm using the [Axis travel] key so that it goes out of the over travel region.
IO0511	5	2	Thermal error (transformer)	The thermal sensor of transformer 1 operated. (400 V power supply region)	1. Check that excessive load is not connected. 2. Review the machining condition and operation pattern. Remove the above causes and resume operation after turning off the power for 30 minutes.
IO0512	5	2	Abnormal temperature rise in electrical cabinet	The temperature sensor on the NC PCB detected abnormality.	1. Check that the ambient temperature is 40 degrees or less. 2. Check that the cooling fan does not stop. 3. Check the machine parameter. 4. The NC PCB is faulty. Replace the NC PCB.
IO0513	5	2	Thermal error (chip shower)	1. Chip shower pump is overloaded. 2. The set value of the thermal relay is not correct. 3. The power supply of the pump is open phased.	1-1. Remove the causes of pump overload. - The viscosity of cutting oil is high. - Clogged piping 1-2. Press the reset button for thermal relay FC11 in the C protection unit to reset trip. 1-3. Operate after turning off the power for 20 minutes. 2. The set value of thermal relay FC11 in the C protection unit is not correct. 3-1. Check the wiring to the pump. 3-2. Press the reset button for thermal relay FC11 in the C protection unit to reset trip. *For S1000X1, be sure to check both FC11 and FC16. When there are two chip shower pumps, be sure to check both FC11 and FC16.
IO0514	5	2	Thermal error (coolant)	1. The coolant pump is overloaded. 2. The set value of the thermal relay is not correct. 3. The power supply of the pump is open phased.	1-1. Remove the causes of pump overload. - The viscosity of cutting oil is high. - Clogged piping 1-2. Press the reset button for thermal relay FC10 in the C protection unit to reset trip. 1-3. Operate after turning off the power for 20 minutes. 2. The set value of thermal relay FC10 in the C protection unit is not correct. 3-1. Check the wiring to the pump. 3-2. Press the reset button for thermal relay FC10 in the C protection unit to reset trip.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0515	5	2	Thermal error (ATC motor)	The thermal sensor in the ATC motor operated.	1. Remove the causes of the ATC motor overload. - ATC operation condition capacity is exceeded. - Motor overload, lock 2. Operate after turning off the power for 30 minutes.
IO0516	5	2	Servo relay 1 error	1. When this alarm occurs before turning ON the servo relay, - The servo relay KFS1 contact is already ON.	1. Turn on the power supply again. If the power is not turned on, the servo relay KFS1 contact is welded.
IO0518	5	2	Air pressure low	1. Air pressure decreased. 2. Pressure switch setting is high. 3. The wrong value is set for the "Spindle air blow/Tool wash control method".	1. Check the air piping and pressure. 2. Check the pressure switch setting. 3. Check the parameter setting. Machine parameter (system 1) Spindle air blow/Tool wash control method.
IO0519	5	2	Door operation timeout	DR_TOV_ERR signal (BY: Door time over) was detected. For the factory-installed ladder program, Automatic door took longer time for open/close operation than that was set for the user parameter (switch 1) "automatic door operation time".	Press the [RST] key to clear the alarm. Check the ladder. For the factory-installed ladder program: 1. Check the parameter setting. - User parameter (switch 1) automatic door operation time 2. Check the door limit switch. 3-1. Make sure that the door close valve operated. 3-2. Check the setting of the speed controller valve for the door close cylinder. 4-1. Make sure that the door open valve operated. 4-2. Check the setting of the speed controller valve for the door open cylinder. 5. Make sure that there are no chips or shavings between doors R and L. 6. Make sure that there are no chips or shavings between the door and machine cover. 7. The door sensor is faulty. Replace the door sensor. 8. The IL PCB is faulty. Replace the IL PCB. 9. The IO PCB is faulty. Replace the IO PCB. 10. The stroke learning operation may not have been executed for the automatic door. Perform the stroke learning operation again.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0520	5	2	Servo I/O error	Although a servo error was detected at I/O level, the cause was unknown.	1. Check the connections for the following connectors on the IL PCB: XMAING1, XMAING2, XPLT1G, XPLT2G, XPLC1G and XPLC2G. 2. For axes not using an additional axis, disconnect the communication cables between the NC PCB and amplifier, the IL PCB and the amplifier or between the amplifiers. 3. The IO PCB or the IL PCB may be damaged if the alarm will not reset even when the wiring is normal.
IO0521	5	2	Spindle tool unclamped (sensor 1)	1. The spindle was rotated by hand during spindle tool unclamping. 2. The spindle tool unclamp signal was turned on during spindle rotation.	1. Press the [RST] key. 2. The air circuit or sensor is faulty.
IO0522	5	2	Spindle tool unclamped (sensor 2)	1. The spindle was rotated by hand during spindle tool unclamping. 2. The spindle tool clamp signal was turned off during spindle rotation.	1. Press the [RST] key. 2. The air circuit or sensor is faulty.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0523	5	2	Tool clamp error	1. The tool holder was not clamped to the spindle normally. 2. The tool holder is incorrectly inserted in the pot. 3. The holder is faulty. 4. Chips adhere to the taper of the tool holder. 5. The spindle clamp collet malfunction 6. The ATC mechanism is faulty. 7. The tool clamp sensor was not turned on.	1. Recover in accordance with "Recovery of tool clamp error" in the instruction manual. <When clearance between spindle end face and holder end face is less than 1 mm> 1) Switch to the manual operation mode. The screen automatically switches to the release operation screen for this error. 2) Press the [F7] key. 3) Press the [ATC] key to execute ATC and return the holder to the pot. 4) Use after checking for chips and damages on the holder end face, spindle end face, and spindle taper. <When clearance between spindle end face and holder end face is 1 mm or more> Remove the ATC arm to remove the holder. Recover by referring to the installation manual "Recovery of tool clamp error". The tool holder may be installed to the pot in the reverse direction. Use a holder complying with the standard. Check the filter for end face washing. Apply great metal spray to the collet. Contact the nearest BROTHER approved service dealer for repair. 2. Check that the IO PCB XSQRNG connector is not disconnected. 3. The tool clamp sensor is faulty. 4. The IO PCB is faulty. Contact the nearest BROTHER approved service dealer for the situation it occurred.
IO0524	5	2	Chip conveyor error	BY: Chip conveyor error SCRWSMER signal was detected. For the factory-installed ladder program, 1. The conveyor slip sensor FC55 operated. 2. The conveyor motor is overloaded. 3. The set value of the thermal relay is not correct.	Check the ladder. For the factory-installed ladder program, 1-1. Remove the causes of the hinge conveyor slip. 1-2. Press Reset on the chip conveyor control box. 2-1. Remove the causes of the overload. 2-2. Reset trip on the thermal relay in the chip conveyor control box. Floor conveyor FC61 Filter conveyor FC62 Oil skimmer FC63 (option) Back wash pump FC53 3. Check the set value for the thermal relay. Reset the thermal relay trip after repair.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0525	5	2	The door is abnormal. (Electric cylinder)	DRALM2_ERR signal (BY: Door abnormal 2 (electric cylinder)) was detected. For the factory-installed ladder program, electrical cylinder error was detected.	1. Check the ladder if the content has changed from the factory default settings. Consult with the person who changed the ladder program. 2. Check the light status (LED) of the electric cylinder controller and then contact us.
IO0526	5	2	ATC arm position error 2	The ATC arm is not at the zero point.	Recover it by referring to "3.2. Operation procedure" of "ATC arm position recovery".
IO0527	5	2	Input signal error (Emergency stop switch)	Inconsistency between systems occurred in the emergency stop wiring. 1. Wiring connection is faulty for emergency stop. 2. [EMERGENCY] switch is defective. 3. IL PCB is defective. 4. There is a poor contact or disconnection on the wire for the emergency stop.	1. Check for a faulty connection and re-connect properly on the XTEMSIN, XILKEY, XILIO and XCHIP connectors on the IL PCB and on the teaching pendant connectors. 2. Replace the [EMERGENCY] switch. 3. Replace the IL PCB. 4. Replace the cable.
IO0528	5	2	Input sig. error (during interlock interruption)	Inconsistency between systems occurred in the wiring while the interlock was disengaged. 1. Teaching pendant (option) is defective. 2. Wiring connection is faulty for the interlock disable switch. 3. Interlock disable switch is defective. 4. IL PCB is defective. 5. There is a poor contact or disconnection on the wire for the interlock disable switch.	1. When equipped with a teaching pendant, remove the teaching pendant, and connect the short cap and then set the interlock disable switch to "O" (disable). If the alarm does not trigger again, then the teaching pendant is defective. Replace the teaching pendant. 2. Check for a faulty connection on the XILKEY connector for the IL PCB, and re-connect properly. 3. Replace the interlock disable switch. 4. Replace the IL PCB. 5. Replace the cable.
IO0529	5	2	Input signal error (QT clamp sensor)	Inconsistency between systems occurred in the QT clamp sensor wiring. 1. The sensor wiring for the QT clamp has a faulty connection. 2. The QT clamp sensor is defective. 3. The IL PCB is defective. 4. The QT clamp sensor wiring is disconnected and/or the contact is faulty.	Perform the following while the QT is clamped. 1. Check for a faulty connection on the XCLP connector for the IL PCB, and re-connect properly. 2. Replace the QT clamp sensor. 3. Replace the IL PCB. 4. Replace the cable.
IO0530	5	2	Input signal error (Pallet 1 index sensor)	Inconsistency between systems occurred in the pallet 1 index sensor wiring. 1. The index sensor wiring for pallet 1 has a faulty connection. 2. The pallet 1 index sensor is defective. 3. The IL PCB is defective. 4. The pallet 1 index sensor is disconnected and/or the contact is faulty.	When pallet 1 is indexed, perform the following. 1. Check for a faulty connection on the XPLT1 connector for the IL PCB, and re-connect properly. 2. Replace the pallet 1 index sensor. 3. Replace the IL PCB. 4. Replace the cable.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0531	5	2	Input signal error (Pallet 2 index sensor)	Inconsistency between systems occurred in the pallet 2 index sensor wiring. 1. The index sensor wiring for pallet 2 has a faulty connection. 2. The pallet 2 index sensor is defective. 3. The IL PCB is defective. 4. The pallet 2 index sensor is disconnected and/or the contact is faulty.	1. Check for a faulty connection on the XPLT2 connector for the IL PCB, and re-connect properly. 2. Replace the pallet 2 index sensor. 3. Replace the IL PCB. 4. Replace the cable.
IO0532	5	2	Input sig. error (Front/outer door sensor)	Inconsistency between systems occurred in the front door sensor wiring. 1. The front door wiring has a faulty connection. 2. The front door is defective. 3. The IL PCB is defective. 4. The front door wiring is disconnected and/or the contact is faulty.	Close all of the doors and perform the following. 1. Check for a faulty connection on the XDR1 connector for the IL PCB, and re-connect properly. 2. Replace the front door/outer door sensor. 3. Replace the IL PCB. 4. Replace the cable.
IO0533	5	2	Input signal error (Internal door sensor)	Inconsistency between systems occurred in the inner door sensor wiring. 1. The short connector XDR2,XDR3 on the IL PCB has a faulty connection. 2. The short connector XDR2,XDR3 on the IL PCB is disconnected and/or the contact is faulty. 3. The IL PCB is defective.	Close all of the doors and perform the following. 1. Check for a faulty connection on the XDR2,XDR3 connector for the IL PCB, and re-connect properly. 2. Replace the XDR2,XDR3 short connector on the IL PCB. 3. Replace the IL PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0534	5	2	Input signal error (Left side door sensor)	Models that are not M200X3 or M300X3 Inconsistency between systems occurred in the left side door sensor wiring. 1. The wiring for the left side door has a faulty connection. 2. The XDR3 short connector on the IL PCB is disconnected and/or the contact is faulty. 3. The left side door is defective. 4. The IL PCB is defective. 5. The wiring for the left side door is disconnected and/or the contact is faulty. When using M200X3 or M300X3 There was a mismatch between the systems in the sensor wiring for the manual side door (for the side door on the opposite side of the BV7 when equipped with the BV7 option). 1. Wiring connection is faulty for manual side door. 2. XDR3 short connector on the IL PCB is disconnected and/or the contact is faulty. 3. Manual side door is faulty. 4. The IL PCB is faulty. 5. There is a poor contact or disconnection on the wire for the manual side door.	Models that are not M200X3 or M300X3 Close all of the doors and perform the following. 1. Check for a faulty connection on the XDR3 connector for the IL PCB, and re-connect properly. 2. Replace the XDR3 short connector on the IL PCB. 3. Replace the left side door sensor. 4. Replace the IL PCB. 5. Replace the cable. When using M200X3 or M300X3 Close all doors, and carry out the following. 1. Check for a faulty connection on the XDR3 connector for the IL PCB, and re-connect properly. 2. Replace the XDR3 short connector on the IL PCB. 3. Replace the manual side door sensor. 4. Replace the IL PCB. 5. Replace the cable.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0535	5	2	Input signal error (Right side door sensor)	When not equipped with the BV7 option Inconsistency between systems occurred in the right side door sensor wiring. 1. The wiring for the right side door has a faulty connection. 2. The XDR4 short connector on the IL PCB is disconnected and/or the contact is faulty. 3. The right side door is defective. 4. The IL PCB is defective. 5. The wiring for the right side door is disconnected and/or the contact is faulty. When equipped with the BV7 option There was a mismatch between the systems in the sensor wiring for the side door on the BV7 side. 1. Wiring connection is faulty for side door on the BV7 side. 2. XDR4 short connector on the IL PCB is disconnected and/or the contact is faulty. 3. Side door on the BV7 side is faulty. 4. The IL PCB is faulty. 5. There is a poor contact or disconnection on the wire for the side door on the BV7 side. 6. DRRLY PCB is defective.	When not equipped with the BV7 option Close all of the doors and perform the following. 1. Check for a faulty connection on the XDR4 connector for the IL PCB, and re-connect properly. 2. Replace the XDR4 short connector on the IL PCB. 3. Replace the right side door sensor. 4. Replace the IL PCB. 5. Replace the cable. When equipped with the BV7 option Close all doors, and carry out the following. 1-1. Check for a faulty connection on the XDR4 connector for the IL PCB, and re-connect properly. 1-2. Check for a faulty connection on the XDR4IN and XD4OUT connectors for the DRRLY PCB, and re-connect properly. 2. Replace the XDR4 short connector on the IL PCB. 3. Replace the side door sensor on the BV7 side. 4. Replace the IL PCB. 5. Replace the cable. 6. Replace the DRRLY PCB.
IO0536	5	2	Input sig. error (Machine-external door 1 sensor)	Inconsistency between systems occurred in the external door 1 sensor wiring. 1. There is an error in the sensor system for the front door / outer door, left side door and right side door. 2. The sensor wiring for outside door 1 has a faulty connection. 3. The XDR5 short cord on the IL PCB is disconnected and/or the contact is faulty. 4. The sensor for outside door 1 is defective. 5. The IL PCB is defective. 6. The sensor wiring for outside door 1 is disconnected and/or the contact is faulty.	Close all of the doors and perform the following. 1. If one of the following alarms was triggered: IO0532, IO0534 or IO0535, reset those alarm(s) first. 2. Check for a faulty connection on the XDR5 connector for the IL PCB, and re-connect properly. 3. If bits 8,9/A and B for the 71th input line on the I/O screen are not all "11" or "00", check for a faulty short cord connection or disconnection on the XDR5 connector for the IL PCB. 4. Replace the sensor for outside door 1. 5. Replace the IL PCB. 6. Replace the cable.
IO0537	5	2	Input sig. error (PLC left side door sensor)	Inconsistency between systems occurred in the PLC left side door sensor wiring. The IL PCB is defective.	If bits A and B for the 71st input line on the I/O screen are not "11", there is an error in the sensor system for the PLC left side door. Replace the IL PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0538	5	2	Input sig. error (Machine-external door 2 sensor)	Inconsistency between systems occurred in the external door 2 sensor wiring. 1. The sensor wiring for outside door 2 has a faulty connection. 2. The outside door 2 sensor is defective. 3. The IL PCB is defective. 4. The sensor wiring for outside door 2 is disconnected and/or the contact is faulty.	Close all of the doors and perform the following. 1. If one of the following alarms was triggered: IO0532, IO0534 or IO0535, reset those alarm(s) first. 2. Check for a faulty connection on the XDR6 connector for the IL PCB, and re-connect properly. 3. If bits C,D/E and F for the 71th input line on the I/O screen are not all "11" or "00", check for a faulty short cord connection or disconnection on the XDR6 connector for the IL PCB. 4. Replace the sensor for outside door 2. 5. Replace the IL PCB. 6. Replace the cable.
IO0539	5	2	Input sig. error (PLC right side door sensor)	Inconsistency between systems occurred in the PLC right side door sensor wiring. The IL PCB is defective.	If bits E and F for the 71st input line on the I/O screen are not "11", there is an error in the sensor system for the PLC right side door. Replace the IL PCB.
IO0540	5	2	Input sig. error (Main spindle system signal)	There is a mismatch between systems in the gate ON signal from the spindle system signals that connect to the servo amplifier on XMAING1 and XMAING2 for the IL PCB. 1. There is a timing delay in the redundant contacts when a close operation is carried out on the door limit switch, when a release operation is carried out on the emergency switch, when an operation is carried out on the interlock switch, and when a connector is inserted into the teaching pendant. 2. There is a problem or error with the door limit switch, the emergency switch, the interlock switch, the QT clamp sensor (only on machine models with QT) or the pallet index sensor (only on machine models with QT). 3. The cable for the spindle system signal has a faulty connection. 4. The IL PCB is defective. 5. The wiring for spindle system signal is disconnected and/or the contact is faulty. (When also using the BV7) 6. DRRLY PCB is defective.	1. Press the [EMERGENCY] switch, release after 5 seconds and then press the [RST] key. If an alarm is triggered, proceed to number 2 below. When the alarm resets, there is no problem because there was just a timing delay in the redundant contacts when the an operation is carried out on the door limit switch, the emergency switch, the interlock switch, the QT clamp sensor (only on machine models with QT) or the pallet index sensor (only on machine models with QT), or when connecting to the teaching pendant. 2. Before this alarm triggers, follow the directions in number 1. If the alarm is triggered again, replace the door limit switch, the emergency switch, the interlock switch, or the QT clamp sensor (only on machine models with QT) that was confirmed to be the problem. If a sensor input signal error (IO0527-IO0539) triggers at the same time, first check the recovery method for a sensor input signal error. 3. Make sure that the contact is not faulty or that the connector is not loose on the spindle system signal wiring. 4. If no error is found in the switches, sensors and wiring but the error does not disappear after restarting, replace the IL PCB. 5. Replace the cable for the spindle system signal. (When also using the BV7) 6. Replace the DRRLY PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0541	5	2	Input signal error (Chip conveyor)	There is a mismatch between systems in the gate ON signal for the chip conveyor system that outputs from XCHIP on the IL PCB to the motor relay on the chip conveyor. 1. There is a timing delay in the redundant contacts when a switch operation is carried out or when a connector is inserted. When an operation is carried out (especially when an operation is slowly carried out) on the door limit switch, on the [EMERGENCY] switch or on the [DOOR INTERLOCK] switch; or when a connector is inserted for the handle or for the teaching pendant. 2. There is a problem or error with a switch or sensor. There is problem or error with the door limit switch, the [EMERGENCY] switch, the [DOOR INTERLOCK] switch, the QT clamp sensor (only on machine models with QT) or the pallet index sensor (only on machine models with QT). 3. There is a poor contact on a connector for the chip conveyor system signals. 4. There is a cable disconnection for the chip conveyor system signals. 5. IL PCB is defective.	1. Press the [EMERGENCY] switch, wait more than 5 seconds then quickly release it and press the [RST] key. If the alarm is cancelled, there is no problem because there is a timing delay in the redundant contacts. 2. Replace a switch or sensor. If a sensor input signal error (IO0527-IO0539) has also been triggered, first, follow and check the recovery procedure for the sensor input signal error. 3. Re-connect the connector for the chip conveyor spindle system signals. 4. Replace the cable used for the chip conveyor system signals. 5. Replace the IL PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0542	5	2	Input sig. error (Pallet 1 side additional axis)	There is a mismatch between systems in the gate ON signal for the additional axis signal (Pallet 1 side) that outputs from XPLT1G, XPLT2G, XPLC1G and XPLC2G on IL PCB to the additional axis amplifier for pallet 1. 1. There is a timing delay in the redundant contacts when a switch operation is carried out or when a connector is inserted. When an operation is carried out (especially when an operation is slowly carried out) on the door limit switch, on the [EMERGENCY] switch or on the [DOOR INTERLOCK] switch; or when a connector is inserted for the handle or for the teaching pendant. 2. There is a problem or error with a switch or sensor. There is problem or error with the door limit switch, the [EMERGENCY] switch, the [DOOR INTERLOCK] switch, the QT clamp sensor (only on machine models with QT) or the pallet index sensor (only on machine models with QT). 3. There is a poor contact on a connector for the additional axis signal system for pallet 1. 4. There is a cable disconnection for the additional axis signal system for pallet 1. 5. IL PCB is defective. (When also using the BV7) 6. DRRLY PCB is defective.	1. Press the [EMERGENCY] switch, wait more than 5 seconds then quickly release it and press the [RST] key. If the alarm is cancelled, there is no problem because there is a timing delay in the redundant contacts. 2. Replace a switch or sensor. If a sensor input signal error (IO0527-IO0539) has also been triggered, first, follow and check the recovery procedure for the sensor input signal error. 3. Re-connect the connector for the additional axis signal system on pallet 1. 4. Replace the cable used for the additional axis signal system on pallet 1. 5. Replace the IL PCB. (When also using the BV7) 6. Replace the DRRLY PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0543	5	2	Input sig. error (Pallet 2 side additional axis)	There is a mismatch between systems in the gate ON signal for the additional axis signal (Pallet 2 side) that outputs from XPLT1G, XPLT2G, XPLC1G and XPLC2G on IL PCB to the additional axis amplifier for pallet 2. 1. There is a timing delay in the redundant contacts when a switch operation is carried out or when a connector is inserted. When an operation is carried out (especially when an operation is slowly carried out) on the door limit switch, on the [EMERGENCY] switch, on the [DOOR INTERLOCK] switch, on the [MODE OF SAFE OPERATION] switch or on the [SERVICE MODE] switch; or when a connector is inserted for the handle or for the teaching pendant. 2. There is a problem or error with a switch or sensor. There is a problem or error with the door limit switch, the [EMERGENCY] switch, the [DOOR INTERLOCK] switch, the [MODE OF SAFE OPERATION] switch, the [SERVICE MODE] switch, the QT clamp sensor (only on machine models with QT) or the pallet index sensor (only on machine models with QT). 3. There is a poor contact on a connector for the additional axis signal system for pallet 2. 4. There is a cable disconnection for the additional axis signal system for pallet 2. 5. IL PCB is defective. (When also using the BV7) 6. DRRLY PCB is defective.	1. Press the [EMERGENCY] switch, wait more than 5 seconds then quickly release it and press the [RST] key. If the alarm is cancelled, there is no problem because there is a timing delay in the redundant contacts. 2. Replace a switch or sensor. If a sensor input signal error (IO0527-IO0539) has also been triggered, first, follow and check the recovery procedure for the sensor input signal error. 3. Re-connect the connector for the additional axis signal system on pallet 2. 4. Replace the cable used for the additional axis signal system on pallet 2. 5. Replace the IL PCB. (When also using the BV7) 6. Replace the DRRLY PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0544	5	2	Input sig. error (Mach.-ext. door 1, PLC axis)	There is a mismatch between systems in the gate ON signal for the PLC axis signal system (outside door 1 side) that outputs from XPLT1G, XPLT2G, XPLC1G and XPLC2G on IL PCB to the PLC axis amplifier for outside door 1. 1. There is a timing delay in the redundant contacts when a switch operation is carried out or when a connector is inserted. When an operation is carried out (especially when an operation is slowly carried out) on the door limit switch, on the [EMERGENCY] switch, on the [DOOR INTERLOCK] switch, on the [MODE OF SAFE OPERATION] switch or on the [SERVICE MODE] switch; or when a connector is inserted for the handle or for the teaching pendant. 2. There is a problem or error with a switch or sensor. There is a problem or error with the door limit switch, the [EMERGENCY] switch, the [DOOR INTERLOCK] switch, the [MODE OF SAFE OPERATION] switch, the [SERVICE MODE] switch, the QT clamp sensor (only on machine models with QT) or the pallet index sensor (only on machine models with QT). 3. There is a poor contact on a connector in the PLC-axis signal system on the outside door 1 side. 4. There is a cable disconnection in the PLC-axis signal system on the outside door 1 side. 5. IL PCB is defective.	1. Press the [EMERGENCY] switch, wait more than 5 seconds then quickly release it and press the [RST] key. If the alarm is cancelled, there is no problem because there is a timing delay in the redundant contacts. 2. Replace a switch or sensor. If a sensor input signal error (IO0527-IO0539) has also been triggered, first, follow and check the recovery procedure for the sensor input signal error. 3. Re-connect the connector used for the PLC-axis signal system on the outside door 1 side. 4. Replace the cable used for the PLC-axis signal system on the outside door 1 side. 5. Replace the IL PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0545	5	2	Input sig. error (Mach.-ext. door 2, PLC axis)	There is a mismatch between systems in the gate ON signal for the PLC axis signal system (outside door 2 side) that outputs from XPLT1G, XPLT2G, XPLC1G and XPLC2G on IL PCB to the PLC axis amplifier for outside door 2. 1. There is a timing delay in the redundant contacts when a switch operation is carried out or when a connector is inserted. When an operation is carried out (especially when an operation is slowly carried out) on the door limit switch, on the [EMERGENCY] switch, on the [DOOR INTERLOCK] switch, on the [MODE OF SAFE OPERATION] switch or on the [SERVICE MODE] switch; or when a connector is inserted for the handle or for the teaching pendant. 2. There is a problem or error with a switch or sensor. There is a problem or error with the door limit switch, the [EMERGENCY] switch, the [DOOR INTERLOCK] switch, the [MODE OF SAFE OPERATION] switch, the [SERVICE MODE] switch, the QT clamp sensor (only on machine models with QT) or the pallet index sensor (only on machine models with QT). 3. There is a poor contact on a connector in the PLC-axis signal system on the outside door 2 side. 4. There is a cable disconnection in the PLC-axis signal system on the outside door 2 side. 5. IL PCB is defective.	1. Press the [EMERGENCY] switch, wait more than 5 seconds then quickly release it and press the [RST] key. If the alarm is cancelled, there is no problem because there is a timing delay in the redundant contacts. 2. Replace a switch or sensor. If a sensor input signal error (IO0527-IO0539) has also been triggered, first, follow and check the recovery procedure for the sensor input signal error. 3. Re-connect the connector used for the PLC-axis signal system on the outside door 2 side. 4. Replace the cable used for the PLC-axis signal system on the outside door 2 side. 5. Replace the IL PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0546	5	2	The door is abnormal. (Door controller)	DR_CTL_ERR signal (BY: Door abnormal (door controller)) was detected. For the factory-installed ladder program, the limit switch on the door opening (closing) end is abnormal at the completion of door operation (electric cylinder type automatic door).	Press the [RST] key to reset the alarm. If the alarm occurs every time the door operates, check the ladder program. For the factory-installed ladder program: 1. Make sure that there are no chips or shavings caught between the door and the machine cover. 2. Check the limit switch on the door. 3. The door sensor is faulty. Replace the door sensor. 4. The IL PCB is faulty. Replace the IL PCB. 5. The IO PCB is faulty. Replace the IO PCB.
IO0547	5	2	The door is abnormal. (Home return)	DR_CTL_ERR signal (BY: Door abnormal (door controller)) was detected. For the factory-installed ladder program, the limit switch on the door opening (closing) end is abnormal at the completion of door operation (electric cylinder type automatic door).	Press the [RST] key to reset the alarm. If the alarm occurs every time the door operates, check the ladder program. Contact us and report the conditions of the error when it occurs in a factory-installed ladder program. 1. Check the door limit switch. 2. Make sure that there are no chips or shavings caught between the door and the machine cover. 3. The IL PCB is faulty. Replace the IL PCB. 4. The IO PCB is faulty. Replace the IO PCB.
IO0548	5	2	The 12-V DC power is overcurrent. (4) (ES)	Reserved	Reserved
IO0549	5	2	The door is abnormal (Electric cylinder type 2)	The DRALM2_ERR signal (BY: Door error 2 (Electric cylinder)) was detected. An electric cylinder error was detected in the factory set ladder program.	1. Check the ladder if the content has changed from the factory default settings. Consult with the person who changed the ladder program. 2. Check the light status (LED) of the electric cylinder controller and then contact us.
IO0554	5	2	Thermal error (Chip auger motor)	The COILCONTMLER signal (BY: Thermal error (Chip auger motor)) was detected. For the factory-installed ladder program, 1. The chip auger motor is overloaded. 2. The installation value for the thermal relay is faulty. 3. There is an open-phase on the power source for the chip auger motor.	1-1. Remove the cause of the overload on the chip auger motor. - Clogged with chips or shavings 1-2. Reset the trip for the thermal relay FC6 and FC7 located inside the C protection unit. 1-3. Operate the machine after turning the power supply OFF for 20 minutes. 2. The installation-default value is faulty for the thermal relay FC6 and FC7, located inside the C protection unit. 3-1. Check the wiring to the chip auger motor. 3-2. Reset the trip for the thermal relay FC6 and FC7 located inside the C protection unit.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO0555	5	2	Input signal error (LB side door sensor)	There was a mismatch between the systems in the side door sensor wiring. 1. The door stopped for a set amount of time at a boundary position where the side door opens/closes. 2. Wiring connection is faulty for side door. 3. The side door is defective. 4. BIO PCB is defective. 5. There is a poor contact or disconnection on the wire for the side door. (When also using the BV7) 6. There is a disconnection on the cable that connects to the XD4MON connector on the DRRLY PCB. 7. The cable that connects to the XD4MON connector on the DRRLY PCB is not connected. 8. DRRLY PCB is defective.	1. Do not stop the door close to the boundary position where the side door opens/closes. In addition, make sure that the motion speed for the TR-axis is not too slow when linking with the side door. 2. Check for a faulty connection on the XDR4IN and XD4OUT connectors for the DRRLY PCB, or on the XDR4 connector on the IL PCB. Then, re-connect properly. 3. Replace the side door sensor. 4. Replace the BIO PCB. 5. Replace the cable. (When also using the BV7) 6. Replace the cable that connects to the XD4MON connector on the DRRLY PCB. 7. Attach the cable that connects to the XD4MON connector on the DRRLY PCB. 8. Replace the DRRLY PCB.
IO1500	4	3	The door is abnormal. (Electric cylinder)	DRALM1_ERR signal (BY: Door abnormal 1 (electric cylinder)) was detected. For the factory-installed ladder program, electrical cylinder error was detected.	1. Check the ladder if the content has changed from the factory default settings. Consult with the person who changed the ladder program. 2. Check the light status (LED) of the electric cylinder controller and then contact us.
IO1501	4	3	The door is abnormal. (Electric cylinder type 2)	The DRALM1_ERR signal (BY: Door error 1 (Electric cylinder)) was detected. An electric cylinder error was detected in the factory set ladder program.	1. Check the ladder if the content has changed from the factory default settings. Consult with the person who changed the ladder program. 2. Check the light status (LED) of the electric cylinder controller and then contact us.
IO2000	4	2	ATC arm operation timeout	1. ATC arm turn was not completed within the specified time limit of ATC arm turn timeout period 1-4. 2. The ATC arm cannot turn. 3. The ATC arm turned more than 180 degrees.	1. Check the value of "ATC arm turn timeout period" of the machine parameter (system 3). Operations performed when the alarm occurred are classified into 1 to 4. Turn by manual operation → 1 Turn with medium diameter tool → 2 Turn with large diameter tool → 3 Turn with standard tool → 4 2-1. The inverter is faulty. 2-2. The ATC motor is faulty. 2-3. The mechanism is faulty. 3-1. The ATC zero point sensor is faulty. 3-2. ATC zero point sensor wiring 3-3. The IO PCB is faulty.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO2001	4	2	Pot operation timeout	1. Moving up or moving down of the pot was not completed within the specified time limit. 2 The alarm detection time is too short. 3. The pot moves slowly and takes too much time. 4. The pot sensor is not input.	1-1. Select the manual operation mode and press the [RST] key while holding down the [RELSE] key. 1-2. Press the [ATC] key while holding down the [RELSE] key to move the pot to the correct position. 2. Check the system 3 "Pot operation monitor time" of the machine parameter. 3. The speed controller of the pot cylinder is turned down excessively. 4-1. Check the connection for the XSQP or XYVP connector on the IO PCB. 4-2. Check the connections for XGUP, XGDN and XGPUD inside the relay box. 4-3. Positioning of pot up sensor and pot down sensor is faulty. 4-4. The pot up sensor or pot down sensor may be faulty. 4-5. The IO PCB is faulty.
IO2002	4	2	Pot sensor error	The pot upper end/lower end/both ends sensor is ON when the power is turned ON, when the [RST] key is pressed (except when the [RELSE] key is pressed), and when pot operation is monitored.	1. Recover using the [RST] key. 2. Check the connection for the XSQP connector on the IO PCB. 3. Check the connections for XGUP, XGDN and XGPUD inside the relay box. 4. The position adjustment is faulty for the pot amplifier sensor or the pot down sensor. 5. The pot amplifier sensor or the pot down sensor is faulty. 6. The IO PCB is faulty.
IO2003	4	2	ATC arm position error 1	1. The ATC arm is out of position and not at zero point. 2 The ATC zero point sensor is not ON.	1-1. Select the manual operation mode and press the [RST] key while holding down the [RELSE] key. 1-2. Press the [ATC] key while holding down the [RELSE] key in manual operation to recover the ATC arm to the zero point. 2-1. Check the connection of the IO PCB connector XSQA. 2-2. The ATC zero point sensor SQA is faulty. 2-3. The IO PCB is faulty.
IO2004	4	2	Magazine operation timeout	The magazine was not indexed within 5 seconds per 1 pitch.	1. When it does not move at all The magazine motor is not operating. Check the wiring to the motor. 2. When it stops in the track Check if the tool weight is too heavy.
IO2005	4	2	Detection signal off	The detection signal is not ON in measurement operation.	Check the measurement position after pressing the [RST] key.
IO2006	4	2	Detection signal on	The detection signal is not OFF in measurement operation.	Check the measurement position after pressing the [RST] key.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO2007	4	2	Signal output timeout	1. After the external signal is output, the completion signal is not turned ON even when the external signal reading time limit has been elapsed. 2. After M46X is output, the M46X signal is not turned ON or OFF even when the specified time limit has been elapsed.	1. Check the completion signal. 2. Change the value of the user parameter (switch 1). - External signal reading time limit
IO2008	4	2	Door abnormal (sensor error)	DR_SN_ERR signal (BY: Door abnormal (sensor error)) was detected. For the factory-installed ladder program, both of the limit switches on the door open end and close end were turned on (automatic door specifications).	Check the ladder. For the factory-installed ladder program: 1.1 Check the door limit switch. 1.2 Make sure that there are no chips or shavings caught between the door and machine cover. 1.3 The door sensor wiring is short-circuited. 1.4 The door sensor is faulty and defective. Replace the door sensor. 1.5 The IL PCB is faulty. Replace the IL PCB. 1.6 The IO PCB is faulty. Replace the IO PCB. After removing the causes listed above, press the [RST] key to clear the alarm.
IO2009	4	2	Pot operation time too short	The time required to move up or move down the pot was shorter than "Minimum pot operation time" of the machine parameter 3 .	1. The position adjustment may be faulty for the pot amplifier sensor or the pot down sensor. 2. The pot amplifier sensor or the pot down sensor may be faulty. 3. The operation adjustment may be faulty for the pot amplifier down cylinder. 4. Check "Pot operation monitor time" of the machine parameter 3.
IO2010	4	2	Tool breakage sensor error	1. Without tool breakage detection option The parameter setting relating to tool breakage detection is not correct. 2. With tool breakage detection option The sensor input of the tool breakage detection device is not normal.	1. Set "Tool breakage detection option" of the user parameter to (0:Not equipped). 2-1. Set "Tool breakage detection option" of the user parameter to (2:Type2). 2-2. Check for chips entangled with the detection arm. If the alarm still occurs, contact the nearest BROTHER approved service dealer. 3. The tool breakage detection device is faulty. 4. The IO PCB is faulty.
IO2011	4	2	Tool breakage detection timeout	1. Without tool breakage detection option The parameter setting relating to tool breakage detection is not correct. 2. With tool breakage detection option The sensor input of the tool breakage detection device is not normal.	1. Check for chips entangled with the detection arm. If the alarm still occurs, contact the nearest BROTHER approved service dealer. 2. The tool breakage detection device is faulty. 3. The IO PCB is faulty.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO2012	4	2	SP override connector not connected	The connector of the spindle override is not connected.	1. Make sure that the spindle override switch or the special short connector is connected to the XSPOV connector. 2. When not using the spindle override, if the "Prevent spindle override" in the user parameter (switch 1) is set to "1:Yes," then the spindle override can be disabled and this alarm can also be reset with the [RST] key, etc. 3. The KEY PCB is faulty.
IO2013	4	2	Center-through-coolant error 3	CT_ERR3 signal (BY: CT error 3) was detected. For the factory-installed ladder program, the line filter is clogged.	Check the ladder. For the factory-installed ladder program, check for clogging of the line filter. 1. Clean or replace the filter element. 2. Check the piping of the line filter. If clogged, clean or replace it.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO2014	4	2	Center-through-coolant error 4	CT_ERR4 signal (BY: CT error 4) was detected. For the factory-installed ladder program, 1. The motor breaker of the high pressure device control panel was tripped. The set value of the motor breaker is abnormal. 2. The motor breaker of the high pressure device control panel was tripped. The high pressure pump and supply pump are overloaded. 3. The motor breaker of the high pressure device control panel was tripped. An overvoltage was applied to the high pressure pump and supply pump.	Check the ladder. For the factory-installed ladder program, 1. Check the motor breakers QH10 and QH11 in the high pressure device control panel and confirm which system was tripped. 1-1. Check the set value of the motor breaker on the tripped system. - The set value of the motor breaker differs depending on the power-supply voltage and frequency. For the set value, refer to the installation manual. 1-2. Turn ON the motor breaker after removing the cause. 2. Check the motor breakers QH10 and QH11 in the high pressure device control panel and confirm which system was tripped. 2-1. Check and remove the cause of overload of the tripped system. — The cutting fluid is thick. — The piping is clogged. — The pump is abnormal. 2-2. Turn ON the motor breaker after removing the cause. 2-3. Resume operation after turning off the power for 20 minutes. 3. Measure the voltage of the input terminals on the motor breaker QH10 and QH11. If the input terminal voltage is high, check and remove the cause of the overvoltage. 3-1. Check that the wiring of the high pressure device is normal. If it is abnormal, correct it. 3-2. Turn ON the motor breaker after removing the cause. 3-3. Resume operation after turning off the power for 20 minutes.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO2015	4	2	Center-through-coolant error 5	CT_ERR5 signal (BY: CT error 5) was detected. For the factory-installed ladder program, 1. The wiring of the supply pump relay KP11 is abnormal. 2. The supply pump relay KP11 is faulty.	Check the ladder. For the factory-installed ladder program, 1. The wiring from the machine control panel to the high pressure device control panel may be broken. - Wiring connected to the TBHP PCB in the machine control panel - Wiring from the machine control panel to the high pressure device control panel - Wiring in the high pressure device control panel 1-1. Check for wiring error and correct it according to the installation manual. 1-2. Check the wiring for breakage. If broken, replace it. 2. The supply pump relay KP11 in the high pressure device control panel may be faulty. 2-1. Check the operation of KP11. Replace it when it is malfunctioning. 2-2. Check KP11 for weld. If welded, replace it.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO2016	4	2	Center-through-coolant error 9	CT_ERR9 signal (BY: CT error 9) was detected. 1. The parameter setting is wrong. 2. The wiring is abnormal. 3. The filter sensor is faulty.	Check the ladder. For the factory-installed ladder program, 1. HPERR1 was turned off with the internal control valid. 1-1. Check that the model uses internal control to control the high pressure device. 1-2. Check the machine parameter (system 1) "Center-through-coolant control" and match the parameter to the model used. - For a model not using internal control, set to "0:Type1". - For a model using internal control, set to "1:Type2". 2. HPERR1 was turned off with the internal control valid. Check the wiring according to the installation manual and remove the cause of the stop. — Wiring in the machine control panel — Wiring between the machine control panel and the line filter sensor — Wiring between the high pressure device control panel and the back wash filter sensor. — Wiring between the machine control panel and the high pressure device control panel 3. The HPERR1 signal was turned off with the internal control valid. The filter clog monitoring sensor may be damaged. Check and replace it if damaged. 3-1. Line filter sensor — Sensor contact abnormality 3-2. Back wash filter sensor — Sensor contact abnormality
IO2017	4	2	ATC shutter time is over.	1. Open/close is not completed within 5 seconds after ATC shutter open/close operation. 2. If the ATC shutter opens/closes, the open or close sensor is not input. 3. If the ATC shutter does not open/close, the cylinder does not operate.	1. Recover using the [RST] key. 2. Check the connection for the XATC2 connector on the IO PCB. 3. Check the wiring connections inside the relay box. 4. Check the adjustment position on the open end or close end sensor for the ATC shutter. 5. The open end or close end sensor for the ATC shutter is faulty. 6. The IO PCB is faulty.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO2018	4	2	The ATC shutter is not open.	The ATC shutter did not open when arm turn started.	1. Recover using the [RST] key. 2. Check the connection for the XATC2 connector on the IO PCB. 3. Check the wiring connections inside the relay box. 4. Check the adjustment position on the open end sensor for the ATC shutter. 5. The open end sensor for the ATC shutter is faulty. 6. The IO PCB is faulty.
IO2019	4	2	ATC shutter sensor error	Both of the ATC shutter open end sensor and close end sensor are ON when the power is ON and the [RST] key is pressed.	1. Check the connection for the XATC2 connector on the IO PCB. 2. Check the wiring connections inside the relay box. 3. The open end or close end sensor for the ATC shutter is faulty. 4. The cable short-circuited. 5. The IO PCB is faulty.
IO2020	4	2	Thermal error (Center-through-coolant)	CTOHERR signal (BY: Thermal error (CT coolant)) was detected. For the factory-installed ladder program, 1. The coolant pump is overloaded. 2. The set value of the thermal relay is not correct. 3. The power supply of the pump is open phased.	1-1. Remove the causes of pump overload. - The viscosity of cutting oil is high. - The piping is clogged. 1-2. Press the reset button for thermal relay FC12 in the C protection unit to reset trip. 1-3. Operate after turning off the power for 20 minutes. 2. The set value of thermal relay FC12 in the C protection unit is not correct. 3-1. Check the wiring to the pump. 3-2. Press the reset button for thermal relay FC12 in the C protection unit to reset trip.
IO2021	4	2	TB detection pos. timeout(ACKS)	Without the tool breakage detection option, the option is set to "Type2".	If the tool breakage detection option is not given, set "Tool breakage detection option" of the user parameter to "0:Not equipped". If the alarm occurs in other situations, contact the nearest BROTHER approved service dealer. 1. The tool breakage detection device is faulty. 2. The IO PCB is faulty.
IO2022	4	2	TB detection pos. timeout(MVFIN)	The IO board or the tool breakage detection device is faulty.	Recover using the [RST] key. If it occurs repeatedly, contact the nearest BROTHER approved service dealer. 1. The tool breakage detection device is faulty. 2. The IO PCB is faulty.
IO2023	4	2	Tool breakage detection pos. error	The tool breakage detection motor did not move to the specified position.	Recover using the [RST] key. If it occurs repeatedly, contact the nearest BROTHER approved service dealer. 1. The tool breakage detection device is faulty. 2. The IO PCB is faulty.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO2024	4	2	TB detection device driver error	The IO board or the tool breakage detection device is faulty.	Recover using the [RST] key. If it occurs repeatedly, contact the nearest BROTHER approved service dealer. 1. The tool breakage detection device is faulty. 2. The IO PCB is faulty.
IO2025	4	2	CTS pressure has not increased	CTSUPPER signal (BY: CTS pressurizing error) was detected. For the factory-installed ladder program, 1. Pressure in the piping does not rise when coolant is discharged. 1-1. Coolant is not in the piping. 1-2. Coolant is leaking from the piping. 1-3. Oil level of the tank is lowered. 1-4. The valve is faulty. 1-5. The pump is faulty or damaged. 1-6. The pressure switch is abnormal. 2. The wrong value has been set in the machine parameter.	1. If there is no coolant in the piping, one of the following cases may apply. 1-1-1. Immediately after piping, or at the second or third start up after an extended shutdown, coolant may not be in the piping. Repeat the procedure by pressing the [RST] key to reset the alarm and turn ON the coolant again. 1-1-2. There may be no coolant inside the piping while the back wash cycle is being executed. Check if the center-through-coolant ON command issued immediately after the program ends, or immediately after the back wash cycle command. Repeat the procedure by pressing the [RST] key to reset the alarm and turn ON the coolant again. 1-2. Check the piping for abnormality. 1-3. Check the coolant level of the tank and supply coolant. If no abnormality is found, also check the user parameter (switch 1) "CTS pressurize check delay time." 1-4. If the pump operates, check if the valve is faulty. 1-5. The pump is abnormal. 1-6. The pressure switch is abnormal. 2. Make sure that the machine parameter (system 1) <Center-through-coolant> or <Cyclone filter> is set correctly.
IO2026	4	2	CTS pressure has not lowered	CTSUNPER signal (BY: CTS depressurizing error) was detected. For the factory-installed ladder program, Pressure in the piping is not reduced. 1. The tool and tool holder are abnormal. 2. The piping is clogged. 3. The pressure switch is abnormal. 4. The valve is faulty.	1-1. Press the manual buttons on the discharge valve and drain valve of the CTS unit to lead coolant in the piping to the drain. 1-2. Check that the tool and tool holder is for CTS. 1-3. Check the hole for CTS is not clogged. 2. Check the piping for abnormality. If no abnormality is found, also check the user parameter (switch 1) "CTS depressurize check delay time". 3. The pressure switch is abnormal. 4. If the alarm occurs during back washing, check that the valve is not faulty.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO2027	4	2	Spindle tool unclamped (sensor 1)	1. The spindle was rotated by hand during spindle tool unclamping. 2. The spindle tool unclamp signal was turned on during spindle rotation. 3. Spindle command was specified with the spindle tool unclamped.	1. Press the [RST] key. 2. The air circuit or sensor is faulty. Contact the nearest BROTHER approved service dealer for the situation it occurred. 3. Execute the spindle command after clamping the spindle tool.
IO2028	4	2	Spindle tool unclamped (sensor 2)	1. The spindle was rotated by hand during spindle tool unclamping. 2. The spindle tool unclamp signal was turned on during spindle rotation. 3. Spindle command was specified with the spindle tool unclamped.	1. Press the [RST] key. 2. The air circuit or sensor is faulty. Contact the nearest BROTHER approved service dealer for the situation it occurred. 3. Execute the spindle command after clamping the spindle tool.
IO2029	4	2	Pot shutter timeout	The pot shutter did not finish opening or closing within the machine parameter (system 3) <Pot shutter timeout period>.	1. Recover using the [RST] key. 2. Check the connection for the XATC2 connector on the IO PCB. 3. Check the connections for XGOP, XGCL and XGSOC inside the relay box. 4. Check the adjustment position on the open end or close end sensor for the pot shutter. 5. The open end or close end sensor for the pot shutter is faulty. 6. The IO PCB is faulty. 7. Check the machine parameter (system 3) setting for <Pot shutter timeout period>.
IO2030	4	2	Pot shutter is not open.	The pot shutter did not open when pot operation started.	1. Recover using the [RST] key. 2. Check the connection for the XATC2 connector on the IO PCB. 3. Check the connections for XGOP and XGSOC inside the relay box. 4. Check the adjustment position on the open end sensor for the pot shutter. 5. The open end sensor for the pot shutter is faulty. 6. The IO PCB is faulty.
IO2031	4	2	Pot shutter sensor error	Both of the pot shutter open end sensor and close end sensor were ON when the power was ON and the [RST] key was pressed.	1. Check the connection for the XATC2 connector on the IO PCB. 2. Check the connections for XGOP and XGCL inside the relay box. 3. The open end or close end sensor for the pot shutter is faulty. 4. The cable is short-circuited. 5. The IO PCB is faulty.
IO2032	4	2	ATC arm shutter timeout	ATC arm shutter open/close was not completed within 5 seconds.	1. Recover using the [RST] key. 2. Check the connection for the XATC2 connector on the IO PCB. 3. Check the adjustment position on the open end or close end sensor for the ATC arm shutter. 4. The open end or close end sensor for the ATC arm shutter is faulty. 5. The IO PCB is faulty.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO2033	4	2	ATC arm shutter is not open.	The ATC arm shutter did not open when arm turn operation started.	1. Recover using the [RST] key. 2. Check the connection for the XATC2 connector on the IO PCB. 3. Check the adjustment position on the open end sensor for the ATC arm shutter. 4. The open end sensor for the ATC arm shutter is faulty. 5. The IO PCB is faulty.
IO2034	4	2	ATC arm shutter sensor error	Both of the ATC arm shutter open end sensor and close end sensor were ON when the power was ON and the [RST] key was pressed.	1. Check the connection for the XATC2 connector on the IO PCB. 2. The open end or close end sensor for the ATC arm shutter is faulty. 3. The cable is short-circuited. 4. The IO PCB is faulty.
IO2035	4	2	*-axis not clamped	The QT (4th axis) clamp confirmation input signal is not turned on even after 3 seconds have elapsed.	1. Check the air pressure. 2. Check the connection of the IO PCB connector. - CNCLP - CN4 3. Check that the solenoid YV4 is operating. 4. The IO PCB is faulty. 5. The clamp sensor SQCL is faulty.
IO2036	4	2	*-axis not unclamped	The QT (4th axis) unclamp confirmation input signal is not turned ON even after 3 seconds have elapsed.	1. Check the connection of the relay connector CNSQUCL. 2. Unclamp sensor positioning and the sensor body are faulty. 3. The IO PCB is faulty.
IO2037	4	2	*-axis clamped	1. The unclamp confirmation input signal was turned off or the clamp confirmation input signal was turned on while QT (4th axis) was turning. 2. The unclamp confirmation input signal was turned off or the clamp confirmation input signal was turned on without QT (4th axis) indexed.	1. The unclamp confirmation signal or clamp confirmation signal may be faulty. If it occurs repeatedly, contact the nearest BROTHER approved service dealer. 2. The IO PCB may be faulty. If it occurs repeatedly, contact the nearest BROTHER approved service dealer.
IO2038	4	2	*-axis unclamped	1. The clamp confirmation input signal was turned off or the unclamp confirmation input signal was turned on while QT (4th axis) stopped. 2. The clamp confirmation input signal was turned off or the unclamp confirmation input signal was turned on with QT (4th axis) indexed.	1. The clamp confirmation signal or unclamp confirmation signal may be faulty. If it occurs repeatedly, contact the nearest BROTHER approved service dealer. 2. The IO PCB may be faulty. If it occurs repeatedly, contact the nearest BROTHER approved service dealer.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO2039	4	2	*-axis not clamped	The 5th axis cannot be clamped even after 3 seconds has elapsed. 1. The unclamp output is not connected correctly. 2. The unclamp confirmation signal or clamp confirmation signal is not input correctly.	1-1. The output target or destination of the unclamp output signal is wrong. - Check the output setting target using the data bank (external I/O signal). - Check that it is connected to the terminal block which is set by the parameter. 1-2. The logic of the unclamp output signal is inverse. - Machine parameter (system 2) Unclamp output signal 2-1. The input target and destination of the unclamp confirmation signal or clamp confirmation signal are wrong. - Check the input target using the data bank (external I/O signal). - Check that it is connected to the terminal block which is set by the parameter. 2-2. The logic of the unclamp confirmation signal or the clamp confirmation signal is inverse. - Machine parameter (system 2) Unclamp confirmation input signal, clamp confirmation input signal 2-3. Use dry contacts or two-wire proximity sensors for the unclamp confirmation signal and the clamp confirmation signal. *Check the following parameters to identify which of the unclamp confirmation signal and clamp confirmation signal is faulty. - Machine parameter (system 2) Clamp confirmation input signal
IO2040	4	2	*-axis not unclamped	The 5th axis cannot be unclamped even after 3 seconds has elapsed. 1. The unclamp valve does not operate. 2. The unclamp confirmation signal is not input. 3. When the machine parameter (system 2) <Optional 5th-axis> is set to <Yes(positioning only)>, the return distance to unclamp is too big.	1. The unclamping portion is faulty. 2. The unclamp confirmation signal is faulty. 3. Check the machine parameter (system 2) settings <5th-axis rapid feedrate(door open)> and <5th-axis position check pulse>.
IO2041	4	2	*-axis clamped	The unclamp confirmation signal was turned off while the 5th axis was turning.	The unclamp confirmation signal is faulty.
IO2042	4	2	*-axis unclamped	The clamp confirmation signal was turned off or the unclamp confirmation signal was turned on while the 5th axis stopped.	1. The unclamping portion is faulty. 2. The unclamp confirmation signal or clamp confirmation signal is faulty. *Check the following parameters to identify which of the unclamp confirmation signal and clamp confirmation signal is faulty. - Machine parameter (system 2) Clamp confirmation input signal

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO2043	4	2	*-axis not clamped	The 6th axis cannot be clamped even after 3 seconds has elapsed. 1. The unclamp output is not connected correctly. 2. The unclamp confirmation signal or clamp confirmation signal is not input correctly.	1-1. The output target or destination of the unclamp output signal is wrong. - Check the output setting target using the data bank (external I/O signal). - Check that it is connected to the terminal block which is set by the parameter. 1-2. The logic of the unclamp output signal is inverse. - Machine parameter (system 2) Unclamp output signal 2-1. The input target and destination of the unclamp confirmation signal or clamp confirmation signal are wrong. - Check the input target using the data bank (external I/O signal). - Check that it is connected to the terminal block which is set by the parameter. 2-2. The logic of the unclamp confirmation signal or the clamp confirmation signal is inverse. - Machine parameter (system 2) Unclamp confirmation input signal, clamp confirmation input signal 2-3. Use dry contacts or two-wire proximity sensors for the unclamp confirmation signal and the clamp confirmation signal. *Check the following parameters to identify which of the unclamp confirmation signal and clamp confirmation signal is faulty. - Machine parameter (system 2) Clamp confirmation input signal
IO2044	4	2	*-axis not unclamped	The 6th axis cannot be unclamped even after 3 seconds has elapsed. 1. The unclamp valve does not operate. 2. The unclamp confirmation signal is not input. 3. When the machine parameter (system 2) <Optional 6th-axis> is set to <Yes(positioning only)>, the return distance to unclamp is too big.	1. The unclamping portion is faulty. 2. The unclamp confirmation signal is faulty. 3. Check the machine parameter (system 2) settings <6th-axis rapid feedrate(door open)> and <6th-axis position check pulse>.
IO2045	4	2	*-axis clamped	The unclamp confirmation signal was turned off while the 6th axis was turning.	The unclamp confirmation signal is faulty.
IO2046	4	2	*-axis unclamped	The clamp confirmation signal was turned off or the unclamp confirmation signal was turned on while the 6th axis stopped.	1. The unclamping portion is faulty. 2. The unclamp confirmation signal or clamp confirmation signal is faulty. *Check the following parameters to identify which of the unclamp confirmation signal and clamp confirmation signal is faulty. - Machine parameter (system 2) Clamp confirmation input signal

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO2047	4	2	*-axis not clamped	The 7th axis cannot be clamped even after 3 seconds has elapsed. 1. The unclamp output is not connected correctly. 2. The unclamp confirmation signal or clamp confirmation signal is not input correctly.	1-1. The output target or destination of the unclamp output signal is wrong. - Check the output setting target using the data bank (external I/O signal). - Check that it is connected to the terminal block which is set by the parameter. 1-2. The logic of the unclamp output signal is inverse. - Machine parameter (system 2) Unclamp output signal 2-1. The input target and destination of the unclamp confirmation signal or clamp confirmation signal are wrong. - Check the input target using the data bank (external I/O signal). - Check that it is connected to the terminal block which is set by the parameter. 2-2. The logic of the unclamp confirmation signal or the clamp confirmation signal is inverse. - Machine parameter (system 2) Unclamp confirmation input signal, clamp confirmation input signal 2-3. Use dry contacts or two-wire proximity sensors for the unclamp confirmation signal and the clamp confirmation signal. *Check the following parameters to identify which of the unclamp confirmation signal and clamp confirmation signal is faulty. - Machine parameter (system 2) Clamp confirmation input signal
IO2048	4	2	*-axis not unclamped	The 7th axis cannot be unclamped even after 3 seconds has elapsed. 1. The unclamp valve does not operate. 2. The unclamp confirmation signal is not input. 3. When the machine parameter (system 2) <Optional 7th-axis> is set to <Yes(positioning only)>, the return distance to unclamp is too big.	1. The unclamping portion is faulty. 2. The unclamp confirmation signal is faulty. 3. Check the machine parameter (system 2) settings <7th-axis rapid feedrate(door open)> and <7th-axis position check pulse>.
IO2049	4	2	*-axis clamped	The unclamp confirmation signal was turned off while the 7th axis was turning.	The unclamp confirmation signal is faulty.
IO2050	4	2	*-axis unclamped	The clamp confirmation signal was turned off or the unclamp confirmation signal was turned on while the 7th axis stopped.	1. The unclamping portion is faulty. 2. The unclamp confirmation signal or clamp confirmation signal is faulty. *Check the following parameters to identify which of the unclamp confirmation signal and clamp confirmation signal is faulty. - Machine parameter (system 2) Clamp confirmation input signal

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO2051	4	2	*-axis not clamped	The 8th axis cannot be clamped even after 3 seconds has elapsed. 1. The unclamp output is not connected correctly. 2. The unclamp confirmation signal or clamp confirmation signal is not input correctly.	1-1. The output target or destination of the unclamp output signal is wrong. - Check the output setting target using the data bank (external I/O signal). - Check that it is connected to the terminal block which is set by the parameter. 1-2. The logic of the unclamp output signal is inverse. - Machine parameter (system 2) Unclamp output signal 2-1. The input target and destination of the unclamp confirmation signal or clamp confirmation signal are wrong. - Check the input target using the data bank (external I/O signal). - Check that it is connected to the terminal block which is set by the parameter. 2-2. The logic of the unclamp confirmation signal or the clamp confirmation signal is inverse. - Machine parameter (system 2) Unclamp confirmation input signal, clamp confirmation input signal 2-3. Use dry contacts or two-wire proximity sensors for the unclamp confirmation signal and the clamp confirmation signal. *Check the following parameters to identify which of the unclamp confirmation signal and clamp confirmation signal is faulty. - Machine parameter (system 2) Clamp confirmation input signal
IO2052	4	2	*-axis not unclamped	The 8th axis cannot be unclamped even after 3 seconds has elapsed. 1. The unclamp valve does not operate. 2. The unclamp confirmation signal is not input. 3. When the machine parameter (system 2) <Optional 8th-axis> is set to <Yes(positioning only)>, the return distance to unclamp is too big.	1. The unclamping portion is faulty. 2. The unclamp confirmation signal is faulty. 3. Check the machine parameter (system 2) settings <8th-axis rapid feedrate(door open)> and <8th-axis position check pulse>.
IO2053	4	2	*-axis clamped	The unclamp confirmation signal was turned off while the 8th axis was turning.	The unclamp confirmation signal is faulty.
IO2054	4	2	*-axis unclamped	The clamp confirmation signal was turned off or the unclamp confirmation signal was turned on while the 8th axis stopped.	1. The unclamping portion is faulty. 2. The unclamp confirmation signal or clamp confirmation signal is faulty. *Check the following parameters to identify which of the unclamp confirmation signal and clamp confirmation signal is faulty. - Machine parameter (system 2) Clamp confirmation input signal

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO2055	4	2	Door operation timeout	DR_TOV_ERR signal (BY: Door time over) was detected. For the factory-installed ladder program, Automatic door took longer time for open/close operation than that was set for the user parameter (switch 1) "automatic door operation time".	Press the [RST] key to clear the alarm. Check the ladder. For the factory-installed ladder program: 1. Check the parameter setting. - User parameter (switch 1) automatic door operation time 2. Check the door limit switch. 3-1. Make sure that the door close valve operated. 3-2. Check the setting of the speed controller valve for the door close cylinder. 4-1. Make sure that the door open valve operated. 4-2. Check the setting of the speed controller valve for the door open cylinder. 5. Make sure that there are no chips or shavings between doors R and L. 6. Make sure that there are no chips or shavings between the door and machine cover. 7. The door sensor is faulty. Replace the door sensor. 8. The IL PCB is faulty. Replace the IL PCB. 9. The IO PCB is faulty. Replace the IO PCB. 10. The stroke learning operation may not have been executed for the automatic door. Perform the stroke learning operation again.
IO2056	4	2	The door is abnormal. (Electric cylinder)	DRALM2_ERR signal (BY: Door abnormal 2 (electric cylinder)) was detected. For the factory-installed ladder program, electric cylinder error was detected.	1. Check the ladder if the content has changed from the factory default settings. Consult with the person who changed the ladder program. 2. Check the light status (LED) of the electric cylinder controller and then contact us.
IO2057	4	2	The tool washing liquid surface is abnormal.	1. TW_LQLVERR signal (BY: Tool wash coolant level error) was detected. For the factory-installed ladder program, when tool washing was turned on, coolant level was under the dischargeable lower limit of the accumulator. 2. The wrong value has been set in the machine parameter.	Check the ladder. For the factory-installed ladder program, 1-1. The tool washing filter must be replaced or cleaned. 1-2. The tool wash coolant level sensor is faulty. Replace it. 2. Make sure that the machine parameter (system 1) <Cyclone filter> is set correctly.
IO2058	4	2	Tool washing liquid surface sensor is abnormal.	The tool wash coolant level sensor is faulty.	1 Replace the tool wash coolant level sensor. 2. Press the [CLT.P] key.
IO2059	4	2	*-axis clamp sensor error	The clamp confirmation input signal is not input correctly during QT (4th axis) unclamp operation.	Check the clamp confirmation input signal. If it occurs repeatedly, contact the nearest BROTHER approved service dealer.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
IO2060	4	2	*-axis unclamp sensor error	The unclamp confirmation input signal is not input correctly during QT (4th axis) clamp operation.	Check the unclamp confirmation input signal. If it occurs repeatedly, contact the nearest BROTHER approved service dealer.
IO2061	4	2	Line filter error	CTSLFER signal (BY: Line filter error) was detected. For the factory-installed ladder program, the line filter differential pressure SW was turned on with medium pressure (CT coolant) on.	1. Recover using the [RST] key. 2. Back wash by M496.
IO2062	4	2	The automatic door operated when it was locked.	The automatic door opened when it was attempted to be locked.	Press the [RST] key to clear the alarm. Do not open the automatic door just before operating the machine.
IO2063	4	2	The door is abnormal. (Door controller)	DR_CTL_ERR signal (BY: Door abnormal (door controller)) was detected. For the factory-installed ladder program, the limit switch on the door opening (closing) end is abnormal at the completion of door operation (electric cylinder type automatic door).	Press the [RST] key to reset the alarm. If the alarm occurs every time the door operates, check the ladder program. Contact us and report the conditions of the error when it occurs in a factory-installed ladder program. 1. Check the door limit switch. 2. Make sure that there are no chips or shavings caught between the door and the machine cover. 3. The door sensor is faulty. Replace the door sensor. 4. The IL PCB is faulty. Replace the IL PCB. 5. The IO PCB is faulty. Replace the IO PCB.
IO2064	4	2	The door is abnormal. (Home return)	DR_CTL_ERR signal (BY: Door abnormal (door controller)) was detected. For the factory-installed ladder program, the limit switch on the door opening (closing) end is abnormal at the completion of door operation (electric cylinder type automatic door).	Press the [RST] key to reset the alarm. If the alarm occurs every time the door operates, check the ladder program. Contact us and report the conditions of the error when it occurs in a factory-installed ladder program. 1. Check the door limit switch. 2. Make sure that there are no chips or shavings caught between the door and the machine cover. 3. The door sensor is faulty. Replace the door sensor. 4. The IL PCB is faulty. Replace the IL PCB. 5. The IO PCB is faulty. Replace the IO PCB.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO2065	4	2	Rapid traverse override connection error	There is RAPID TRAVERSE OVERRIDE switch error, or KEY PCB error.	Check the rapid traverse override connector. If the error still occurs even when connected properly, the RAPID TRAVERSE OVERRIDE switch, or the KEY PCB, is faulty. When not using the rapid traverse override, if the "Prevent rapid traverse override" in the user parameter (switch 1) is set to "1:Yes," then the rapid traverse override can be disabled and this alarm can also be cancelled with the [RST] key, etc.
IO2066	4	2	Cutting feedrate override connection error	There is FEEDRATE OVERRIDE switch error, or KEY PCB error.	Check the cutting feedrate override connector. If the error still occurs when connected properly, the cutting feedrate override switch, or the KEY PCB, is faulty. When not using the rapid traverse override, if the "Prevent cutting feedrate override" in the user parameter (switch 1) is set to "1:Yes," then the cutting feedrate override can be disabled and this alarm can also be cancelled with the [RST] key, etc.
IO2067	4	2	Spindle override connection error	There is SPINDLE OVERRIDE switch error, or KEY PCB error.	Check the spindle override connector. If the error still occurs even when connected properly, the spindle override switch, or the KEY PCB, is faulty. When not using the spindle override, if the "Prevent spindle override" in the user parameter (switch 1) is set to "1:Yes," then the spindle override can be disabled and this alarm can also be cancelled with the [RST] key, etc.
IO2068	4	2	Rapid traverse override not connected	The rapid traverse override connector is not connected.	Check the rapid traverse override connector. If the error still occurs even when connected properly, the RAPID TRAVERSE OVERRIDE switch, or the KEY PCB, is faulty. When not using the rapid traverse override, if the "Prevent rapid traverse override" in the user parameter (switch 1) is set to "1:Yes," then the rapid traverse override can be disabled and this alarm can also be cancelled with the [RST] key, etc.
IO2069	4	2	The door is abnormal. (Electric cylinder type 2)	The DRALM2_ERR signal (BY: Door error 2 (Electric cylinder)) was detected. An electric cylinder error was detected in the factory set ladder program.	1. Check the ladder if the content has changed from the factory default settings. Consult with the person who changed the ladder program. 2. Check the light status (LED) of the electric cylinder controller and then contact us.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO2070	4	2	Thermal error (cyclone coolant drawn up)	The CFOHERR signal (BY: Thermal error (cyclone coolant drawn up)) was detected. For the factory-installed ladder program, 1. There is an overload on the pump for the coolant. 2. The set value for the thermal relay is faulty. 3. There is an open-phase on the power source for the pump.	1-1. Remove the cause of the overload on the pump. - Cutting fluid with high viscosity - Clogged pipe 1-2. Press the reset button, for the thermal relay FC12 located inside the C protection unit, to reset the trip. 1-3. Operate the machine after turning the power supply OFF for 20 minutes. 2. The set value is faulty for the thermal relay FC12, located inside the C protection unit. 3-1. Check the wiring to the pump. 3-2. Press the reset button, for the thermal relay FC12 located inside the C protection unit, to reset the trip.
IO2071	4	2	ATC arm position error	- The ATC arm is not at the zero point. -All data was input, or the start data (adjusted absolute values) from the backup memory recovery operation was input.Encoder position data does not match the machine coordinates.	1. When the alarm <<The machine coordinates may be out of position.>> is triggered, perform the expired battery recovery operation. 2. When another cause (besides 1. above) applies, change to manual operation mode, and enable the recovery operation on the ATC arm position recovery screen. Then, while pressing the [RELSE] key, press the [RST] key to reset the alarm.Hereafter, follow the recovery procedure and perform the operations accordingly.
IO3500	3	3	The door is abnormal. (Electric cylinder)	DRALM1_ERR signal (BY: Door abnormal 1 (electric cylinder)) was detected. For the factory-installed ladder program, electric cylinder abnormality was detected.	1. Check the ladder if the content has changed from the factory default settings. Consult with the person who changed the ladder program. 2. Check the light status (LED) of the electric cylinder controller and then contact us.
IO3501	3	3	The door is abnormal. (Electric cylinder type 2)	DRALM1_ERR signal (BY: Door abnormal 1 (electric cylinder)) was detected. For the factory-installed ladder program, electric cylinder abnormality was detected.	1. Check the ladder if the content has changed from the factory default settings. Consult with the person who changed the ladder program. 2. Check the light status (LED) of the electric cylinder controller and then contact us.
IO4000	3	2	Center-through-coolant error 6	CT_ERR6 signal (BY: CT error 6) was detected. For the factory-installed ladder program, the clean tank water level sensor (empty) was turned ON.	Check the ladder. For the factory-installed ladder program, the clean tank is empty. 1. Check the clean tank level of the high pressure device. Replenish if insufficient. 2. Check the coolant tank level. Replenish if insufficient.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO4001	3	2	Center-through-coolant error 7	CT_ERR7 signal (BY: CT error 7) was detected. For the factory-installed ladder program, 1. The high pressure device control panel is not powered. 2. The wiring of the high pressure pump relay KP10 is abnormal. 3. The high pressure pump relay KP10 is faulty.	Check the ladder. For the factory-installed ladder program, 1. The high pressure device control panel may not be powered. Check that the Q8 breaker in the machine control panel is not tripped. If it is tripped, turn the breaker ON after removing the cause of the trip. 2. The wiring from the machine control panel to the high pressure device control panel may be broken. - Wiring connected to the TBHP PCB in the machine control panel - Wiring from the machine control panel to the high pressure device control panel - Wiring in the high pressure device control panel 2-1. Check for wiring error and correct it according to the installation manual. 2-2. Check the wiring for breakage. If broken, replace it. 3. The high pressure pump relay KP10 in the high pressure device control panel may be faulty. 3-1. Check the operation of KP10. Replace it when it is malfunctioning. 3-2. Check KP10 for weld. If welded, replace it.
IO4002	3	2	Abnormal temperature rise in electrical cabinet	The temperature sensor on the NC PCB detected abnormality.	1. Check that the ambient temperature is 40 degrees or less. 2. Check that the cooling fan does not stop. 3. Check the machine parameter. 4. The NC PCB is faulty. Replace the NC PCB.
IO4003	3	2	Line filter error	CTSLFER signal (BY: Line filter error) was detected. For the factory-installed ladder program, the line filter differential pressure SW was turned on with medium pressure (CT coolant) on.	1. Recover using the [RST] key. 2. Back wash by M496.
IO4004	3	2	Back wash filter error	CTSRVFER signal (BY: Back wash filter error) was detected. For the factory-installed ladder program, the back wash filter differential pressure SW was turned on with medium pressure (CT coolant) on.	Recover using the [RST] key. Back wash by M496. If it still occurs repeatedly, contact the nearest BROTHER approved service dealer.
IO4005	3	2	The coolant level has dropped.	COOLLLOWER signal (BY: Coolant low level) was detected. For the factory-installed ladder program, coolant amount is reduced.	Increase coolant amount.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO4006	3	2	Center-through-coolant error 3	CT_ERR3 signal (BY: CT error 3) was detected. For the factory-installed ladder program, the line filter is clogged.	Check the ladder. For the factory-installed ladder program, check for clogging of the line filter. 1. Clean or replace the filter element. 2. Check the piping of the line filter. If clogged, clean or replace it.
IO4007	3	2	Door abnormal (sensor error)	DR_SN_ERR signal (BY: Door abnormal (sensor error)) was detected. For the factory-installed ladder program, both of the limit switches on the door open end and close end were turned on (automatic door specifications).	Check the ladder. For the factory-installed ladder program: 1. Make sure that there are no chips or shavings caught between the door and machine cover. 2. Check the door limit switch. 3. The door sensor wiring is short-circuited. 4. The door sensor is faulty. Replace the door sensor. 5. The IL PCB is faulty. Replace the IL PCB. 6. The IO PCB is faulty. Replace the IO PCB. After removing the causes listed above, press the [RST] key to clear the alarm.
IO4008	3	2	Door operation timeout	DR_TOV_ERR signal (BY: Door time over) was detected. For the factory-installed ladder program, Automatic door took longer time for open/close operation than that was set for the user parameter (switch 1) "automatic door operation time".	Press the [RST] key to clear the alarm. Check the ladder. For the factory-installed ladder program: 1. Check the parameter setting. - User parameter (switch 1) automatic door operation time 2. Check the door limit switch. 3-1. Make sure that the door close valve operated. 3-2. Check the setting of the speed controller valve for the door close cylinder. 4-1. Make sure that the door open valve operated. 4-2. Check the setting of the speed controller valve for the door open cylinder. 5. Make sure that there are no chips or shavings between doors R and L. 6. Make sure that there are no chips or shavings between the door and machine cover. 7. The door sensor is faulty. Replace the door sensor. 8. The IL PCB is faulty. Replace the IL PCB. 9. The IO PCB is faulty. Replace the IO PCB. 10. The stroke learning operation may not have been executed for the automatic door. Perform the stroke learning operation again.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO4009	3	2	The door is abnormal. (Electric cylinder)	DRALM1_ERR signal (BY: Door abnormal 2 (electric cylinder)) was detected. For the factory-installed ladder program, electric cylinder abnormality was detected.	1. Check the ladder if the content has changed from the factory default settings. Consult with the person who changed the ladder program. 2. Check the light status (LED) of the electric cylinder controller and then contact us.
IO4010	3	2	The door is abnormal (Electric cylinder type 2)	DRALM1_ERR signal (BY: Door abnormal 2 (electric cylinder)) was detected. For the factory-installed ladder program, electric cylinder abnormality was detected.	1. Check the ladder if the content has changed from the factory default settings. Consult with the person who changed the ladder program. 2. Check the light status (LED) of the electric cylinder controller and then contact us.
IO4011	3	2	The door is abnormal. (Door controller)	DR_CTL_ERR signal (BY: Door abnormal (door controller)) was detected. For the factory-installed ladder program, the limit switch on the door opening (closing) end is abnormal at the completion of door operation (electric cylinder type automatic door).	Press the [RST] key to reset the alarm. If the alarm occurs every time the door operates, check the ladder program. For the factory-installed ladder program: 1. Make sure that there are no chips or shavings caught between the door and the machine cover. 2. Check the limit switch on the door. 3. The door sensor is faulty. Replace the door sensor. 4. The IL PCB is faulty. Replace the IL PCB. 5. The IO PCB is faulty. Replace the IO PCB.
IO4012	3	2	The door is abnormal. (Home return)	DR_CTL_ERR signal (BY: Door abnormal (door controller)) was detected. For the factory-installed ladder program, the limit switch on the door opening (closing) end is abnormal at the completion of door operation (electric cylinder type automatic door).	Press the [RST] key to reset the alarm. If the alarm occurs every time the door operates, check the ladder program. Contact us and report the conditions of the error when it occurs in a factory-installed ladder program. 1. Check the door limit switch. 2. Make sure that there are no chips or shavings caught between the door and the machine cover. 3. The IL PCB is faulty. Replace the IL PCB. 4. The IO PCB is faulty. Replace the IO PCB.
IO5000	2	3	The door is abnormal. (Electric cylinder)	DRALM1_ERR signal (BY: Door abnormal 1 (electric cylinder)) was detected. For the factory-installed ladder program, electric cylinder abnormality was detected.	1. Check the ladder if the content has changed from the factory default settings. Consult with the person who changed the ladder program. 2. Check the light status (LED) of the electric cylinder controller and then contact us.
IO5001	2	3	The door is abnormal (Electric cylinder type 2)	DRALM1_ERR signal (BY: Door abnormal 1 (electric cylinder)) was detected. For the factory-installed ladder program, electric cylinder abnormality was detected.	1. Check the ladder if the content has changed from the factory default settings. Consult with the person who changed the ladder program. 2. Check the light status (LED) of the electric cylinder controller and then contact us.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO5500	2	2	Pressure high (automatic oiling)	The pressure was not reduced even just before supplying oil.	1. Press the [RST] key. 2. Check the machine parameter. 3 The lubrication pump is faulty. 4 The pressure sensor is faulty.
IO5501	2	2	Pressure low (automatic oiling)	Pressure did not rise even though the lubrication pump was turned on.	1. Lubrication oil leaks in the track. Check the piping. 2 The lubrication pump is faulty. 3 The pressure sensor is faulty. 4. Check the machine parameter.
IO5502	2	2	Spindle motor fan locked	The spindle motor fan was locked and the thermal of the fan was turned on.	The spindle motor fan is not rotating. 1. Make sure that the motor breaker FC2 is ON. 2. Clean the oil adhering to the spindle motor fan. 3. Check the connection for the XSPKF connector on the IO PCB. 4. Check the connection for the XSFOH connector in spindle motor terminal box.
IO5503	2	2	Tool wash filter error	TW_FLTWERR signal (BY: Tool wash filter error) was detected. For the factory-installed ladder program, 1 The tool wash filter was clogged. 2. The clog detection sensor adjustment is faulty.	Check the ladder. For the factory-installed ladder program, 1. The tool washing filter must be replaced or cleaned. 2. Check the set value of the clog detection sensor.
IO5504	2	2	Tool wash filter clogging is predicted.	The number of checks for low fluid level IO_TCLS_OK (Tool cleaning fluid level sensor) at the end of the tool cleaning discharge has exceeded the user parameter (switch 1) setting for <<No. of tool wash filter clogging warning checks (Type 4)>>. For the factory-installed ladder program, 1. The tool wash filter was clogged. 2. The tool change was carried out consecutively while the ATC falls at a slow speed. 3. Tool change was carried out consecutively even though the timing for the AT-axis was slow or late.	Check the ladder. For the factory-installed ladder program, 1. The tool washing filter must be replaced or cleaned. 2. Review the user parameter (switch 1) <No. of tool wash filter clogging warning checks (Type 4)> or the ATC falling speed. 3. Review the user parameter (switch 1) <No. of tool wash filter clogging warning checks (Type 4)> or the <ATC arm turn speed ratio 1/2>.
IO5505	2	2	Duplicated errors of expansion input board	The same signal name is set to several terminal blocks via an external input signal.	Search for terminals that are redundant or duplicated using the "Overlapping terminal check" ([F5] key) on the external input signal screen, and delete accordingly.
IO5506	2	2	Center-through-coolant error 3	CT_ERR3 signal (BY: CT error 3) was detected. For the factory-installed ladder program, the line filter is clogged.	Check the ladder. For the factory-installed ladder program, check for clogging of the line filter. 1. Clean or replace the filter element. 2. Check the piping of the line filter. If clogged, clean or replace it.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO5507	2	2	Line filter error	CTSLFER signal (BY: Line filter error) was detected. For the factory-installed ladder program, the line filter differential pressure SW was turned on with medium pressure (CT coolant) on.	1. Recover using the [RST] key. 2. Back wash by M496.
IO5508	2	2	Automatic greasing signal abnormal	*Pump stop signal did not turned on even just before greasing.	1. Press the [RST] key. 2. Check the parameter setting. Machine parameter (system 1) Automatic greasing pause 1, Automatic greasing pause 2, Automatic greasing abnormal pressure monitoring time 3. The greasing pump is faulty. 4. The automatic greasing device is faulty.
IO5509	2	2	Automatic greasing discharge abnormal	A discharge error of the greasing device was detected.	1. Check the parameter setting. Machine parameter (system 1) Automatic greasing abnormal pressure monitoring time 2. Grease leaks in the track. Check the piping. 3. The greasing pump is faulty. 4. The automatic greasing device is faulty.
IO5510	2	2	Grease was not supplied automatically	Greasing was not executed even though the greasing pump was turned on.	1. Check the parameter setting. Machine parameter (system 1) Automatic greasing completion monitoring time 2. Grease leaks in the track. Check the piping. 3. The greasing pump is faulty. 4. The automatic greasing device is faulty.
IO5511	2	2	Chip conveyor relay error	1. The wiring around the chip conveyor protection relay is disconnected. 2. There is a chip conveyor protection relay failure or malfunction.	1. Check the wiring connection. 2. The chip conveyor protection relays KFP4 and KFP5 are adhering. Replace the corresponding relay.
IO5512	2	2	Door abnormal (sensor error)	DR_SN_ERR signal (BY: Door abnormal (sensor error)) was detected. For the factory-installed ladder program, both of the limit switches on the door open end and close end were turned on (automatic door specifications).	Check the ladder. For the factory-installed ladder program: 1. Make sure that there are no chips or shavings caught between the door and machine cover. 2. Check the door limit switch. 3. The door sensor wiring is short-circuited. 4. The door sensor is faulty. Replace the door sensor. 5. The IL PCB is faulty. Replace the IL PCB. 6. The IO PCB is faulty. Replace the IO PCB. After removing the causes listed above, press the [RST] key to clear the alarm.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO5513	2	2	Door operation timeout	DR_TOV_ERR signal (BY: Door time over) was detected. For the factory-installed ladder program, Automatic door took longer time for open/close operation than that was set for the user parameter (switch 1) "automatic door operation time".	Press the [RST] key to clear the alarm. Check the ladder. For the factory-installed ladder program: 1. Check the parameter setting. - User parameter (switch 1) automatic door operation time 2. Check the door limit switch. 3-1. Make sure that the door close valve operated. 3-2. Check the setting of the speed controller valve for the door close cylinder. 4-1. Make sure that the door open valve operated. 4-2. Check the setting of the speed controller valve for the door open cylinder. 5. Make sure that there are no chips or shavings between doors R and L. 6. Make sure that there are no chips or shavings between the door and machine cover. 7. The door sensor is faulty. Replace the door sensor. 8. The IL PCB is faulty. Replace the IL PCB. 9. The IO PCB is faulty. Replace the IO PCB. 10. The stroke learning operation may not have been executed for the automatic door. Perform the stroke learning operation again.
IO5514	2	2	The door is abnormal. (Electric cylinder)	DRALM1_ERR signal (BY: Door abnormal 2 (electric cylinder)) was detected. For the factory-installed ladder program, electric cylinder abnormality was detected.	1. Check the ladder if the content has changed from the factory default settings. Consult with the person who changed the ladder program. 2. Check the light status (LED) of the electric cylinder controller and then contact us.
IO5515	2	2	The door is abnormal. (Electric cylinder type 2)	DRALM1_ERR signal (BY: Door abnormal 2 (electric cylinder)) was detected. For the factory-installed ladder program, electric cylinder abnormality was detected.	1. Check the ladder if the content has changed from the factory default settings. Consult with the person who changed the ladder program. 2. Check the light status (LED) of the electric cylinder controller and then contact us.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO5516	2	2	The door is abnormal. (Door controller)	DR_CTL_ERR signal (BY: Door abnormal (door controller)) was detected. For the factory-installed ladder program, the limit switch on the door opening (closing) end is abnormal at the completion of door operation (electric cylinder type automatic door).	Press the [RST] key to reset the alarm. If the alarm occurs every time the door operates, check the ladder program. For the factory-installed ladder program: 1. Make sure that there are no chips or shavings caught between the door and the machine cover. 2. Check the limit switch on the door. 3. The door sensor is faulty. Replace the door sensor. 4. The IL PCB is faulty. Replace the IL PCB. 5. The IO PCB is faulty. Replace the IO PCB.
IO5517	2	2	The door is abnormal. (Home return)	DR_CTL_ERR signal (BY: Door abnormal (door controller)) was detected. For the factory-installed ladder program, the limit switch on the door opening (closing) end is abnormal at the completion of door operation (electric cylinder type automatic door).	Press the [RST] key to reset the alarm. If the alarm occurs every time the door operates, check the ladder program. Contact us and report the conditions of the error when it occurs in a factory-installed ladder program. 1. Check the door limit switch. 2. Make sure that there are no chips or shavings caught between the door and the machine cover. 3. The IL PCB is faulty. Replace the IL PCB. 4. The IO PCB is faulty. Replace the IO PCB.
IO5518	2	2	Chip auger relay error	1. The circuit around the chip auger protection relay is disconnected. 2. The chip auger protection relay is defective.	1. Check the wiring connection. 2. The chip auger protection relays KFC2 and KFC3 are adhering. Replace the corresponding relay.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO7000	1	2	P1-axis not clamped	The P1 axis cannot be clamped even after 3 seconds has elapsed. 1. The unclamp output is not connected correctly. 2. The unclamp confirmation signal or clamp confirmation signal is not input correctly.	1-1. The output target or destination of the unclamp output signal is wrong. - Check the output target using the data bank (external I/O signal). - Check that it is connected to the terminal block which is set by the parameter. 1-2. The logic of the unclamp output signal is inverse. - Machine parameter (PLC) P1 axis unclamp output signal 2-1. The input target and destination of the unclamp confirmation signal or clamp confirmation signal are wrong. - Check the input setting target using the data bank (external I/O signal). - Check that it is connected to the terminal block which is set by the parameter. 2-2. The logic of the unclamp confirmation signal or the clamp confirmation signal is inverse. - Machine parameter (PLC) P1 axis unclamp confirmation input signal, P1 axis clamp confirmation input signal 2-3. Use dry contacts or two-wire proximity sensors for the unclamp confirmation signal and the clamp confirmation signal. *Check the following parameter to identify which of the unclamp confirmation signal and clamp confirmation signal is faulty. - Machine parameter (PLC) P1 axis clamp confirmation input signal
IO7001	1	2	P1-axis not unclamped	The P1 axis cannot be unclamped even after 3 seconds has elapsed. 1. The unclamp valve does not operate. 2. The unclamp confirmation signal is not input.	1. The unclamping portion is faulty. 2. The unclamp confirmation signal is faulty.
IO7002	1	2	P1-axis clamped	The unclamp confirmation signal was turned off while the P1 axis was turning.	The unclamp confirmation signal is faulty.
IO7003	1	2	P1- axis unclamped	The clamp confirmation signal was turned off or the unclamp confirmation signal was turned on while P1 axis stopped.	1. The unclamping portion is faulty. 2. The unclamp confirmation signal or clamp confirmation signal is faulty. *Check the following parameters to identify which of the unclamp confirmation signal and clamp confirmation signal is faulty. - Machine parameter (PLC) P1 axis clamp confirmation input signal

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO7004	1	2	P2- axis not clamped	The P2 axis cannot be clamped even after 3 seconds has elapsed. 1. The unclamp output is not connected correctly. 2. The unclamp confirmation signal or clamp confirmation signal is not input correctly.	1-1. The output target or destination of the unclamp output signal is wrong. - Check the output target using the data bank (external I/O signal). - Check that it is connected to the terminal block which is set by the parameter. 1-2. The logic of the unclamp output signal is inverse. - Machine parameter (PLC) P2 axis unclamp output signal 2-1. The input target and destination of the unclamp confirmation signal or clamp confirmation signal are wrong. - Check the input setting target using the data bank (external I/O signal). - Check that it is connected to the terminal block which is set by the parameter. 2-2. The logic of the unclamp confirmation signal or the clamp confirmation signal is inverse. - Machine parameter (PLC) P2 axis unclamp confirmation input signal, P2 axis clamp confirmation input signal 2-3. Use dry contacts or two-wire proximity sensors for the unclamp confirmation signal and the clamp confirmation signal. *Check the following parameter to identify which of the unclamp confirmation signal and clamp confirmation signal is faulty. - Machine parameter (PLC) P2 axis clamp confirmation input signal
IO7005	1	2	P2- axis not unclamped	The P2 axis cannot be unclamped even after 3 seconds has elapsed. 1. The unclamp valve does not operate. 2. The unclamp confirmation signal is not input.	1. The unclamping portion is faulty. 2. The unclamp confirmation signal is faulty.
IO7006	1	2	P2- axis clamped	The unclamp confirmation signal was turned off while the P2 axis was turning.	The unclamp confirmation signal is faulty.
IO7007	1	2	P2- axis unclamped	The clamp confirmation signal was turned off or the unclamp confirmation signal was turned on while P2 axis stopped.	1. The unclamping portion is faulty. 2. The unclamp confirmation signal or clamp confirmation signal is faulty. *Check the following parameters to identify which of the unclamp confirmation signal and clamp confirmation signal is faulty. - Machine parameter (PLC) P2 axis clamp confirmation input signal

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO7008	1	2	P3- axis not clamped	The P3 axis cannot be clamped even after 3 seconds has elapsed. 1. The unclamp output is not connected correctly. 2. The unclamp confirmation signal or clamp confirmation signal is not input correctly.	1-1. The output target or destination of the unclamp output signal is wrong. - Check the output target using the data bank (external I/O signal). - Check that it is connected to the terminal block which is set by the parameter. 1-2. The logic of the unclamp output signal is inverse. - Machine parameter (PLC) P3 axis unclamp output signal 2-1. The input target and destination of the unclamp confirmation signal or clamp confirmation signal are wrong. - Check the input setting target using the data bank (external I/O signal). - Check that it is connected to the terminal block which is set by the parameter. 2-2. The logic of the unclamp confirmation signal or the clamp confirmation signal is inverse. - Machine parameter (PLC) P3 axis unclamp confirmation input signal, P3 axis clamp confirmation input signal 2-3. Use dry contacts or two-wire proximity sensors for the unclamp confirmation signal and the clamp confirmation signal. *Check the following parameter to identify which of the unclamp confirmation signal and clamp confirmation signal is faulty. - Machine parameter (PLC) P3 axis clamp confirmation input signal
IO7009	1	2	P3- axis not unclamped	The P3 axis cannot be unclamped even after 3 seconds has elapsed. 1. The unclamp valve does not operate. 2. The unclamp confirmation signal is not input.	1. The unclamping portion is faulty. 2. The unclamp confirmation signal is faulty.
IO7010	1	2	P3- axis clamped	The unclamp confirmation signal was turned off while the P3 axis was turning.	The unclamp confirmation signal is faulty.
IO7011	1	2	P3- axis unclamped	The clamp confirmation signal was turned off or the unclamp confirmation signal was turned on while P3 axis stopped.	1. The unclamping portion is faulty. 2. The unclamp confirmation signal or clamp confirmation signal is faulty. *Check the following parameters to identify which of the unclamp confirmation signal and clamp confirmation signal is faulty. - Machine parameter (PLC) P3 axis clamp confirmation input signal

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO7012	1	2	P4- axis not clamped	The P4 axis cannot be clamped even after 3 seconds has elapsed. 1. The unclamp output is not connected correctly. 2. The unclamp confirmation signal or clamp confirmation signal is not input correctly.	1-1. The output target or destination of the unclamp output signal is wrong. - Check the output target using the data bank (external I/O signal). - Check that it is connected to the terminal block which is set by the parameter. 1-2. The logic of the unclamp output signal is inverse. - Machine parameter (PLC) P4 axis unclamp output signal 2-1. The input target and destination of the unclamp confirmation signal or clamp confirmation signal are wrong. - Check the input setting target using the data bank (external I/O signal). - Check that it is connected to the terminal block which is set by the parameter. 2-2. The logic of the unclamp confirmation signal or the clamp confirmation signal is inverse. - Machine parameter (PLC) P4 axis unclamp confirmation input signal, P4 axis clamp confirmation input signal 2-3. Use dry contacts or two-wire proximity sensors for the unclamp confirmation signal and the clamp confirmation signal. *Check the following parameter to identify which of the unclamp confirmation signal and clamp confirmation signal is faulty. - Machine parameter (PLC) P4 axis clamp confirmation input signal
IO7013	1	2	P4- axis not unclamped	The P4 axis cannot be unclamped even after 3 seconds has elapsed. 1. The unclamp valve does not operate. 2. The unclamp confirmation signal is not input.	1. The unclamping portion is faulty. 2. The unclamp confirmation signal is faulty.
IO7014	1	2	P4- axis clamped	The unclamp confirmation signal was turned off while the P4 axis was turning.	The unclamp confirmation signal is faulty.
IO7015	1	2	P4- axis unclamped	The clamp confirmation signal was turned off or the unclamp confirmation signal was turned on while P4 axis stopped.	1. The unclamping portion is faulty. 2. The unclamp confirmation signal or clamp confirmation signal is faulty. *Check the following parameters to identify which of the unclamp confirmation signal and clamp confirmation signal is faulty. - Machine parameter (PLC) P4 axis clamp confirmation input signal
IO7500	1	1	Pot not at upper limit	An axis operation was executed without the pot at the upper end.	Press the [ATC] key to move the pot to the upper end.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO7501	1	1	Center-through-coolant error 8	CT_ERR8 signal (BY: CT error 8) was detected. For the factory-installed ladder program, The back wash filter is clogged.	Check the ladder. For the factory-installed ladder program, check for clogging of the back wash filter. 1. Clean or replace the filter element. 2. Check the piping of the back wash filter. If clogged, clean or replace it.
IO7502	1	1	Control box door open	The control panel door is open. Be sure to close the door to prevent the entry of oil mist.	Close the control panel door.
IO7503	1	1	Tool wash filter error	TW_FLTWERR signal (BY: Tool wash filter error) was detected. For the factory-installed ladder program, 1 The tool wash filter was clogged. 2. The clog detection sensor adjustment is faulty.	Check the ladder. For the factory-installed ladder program, 1. The tool washing filter must be replaced or cleaned. 2. Check the set value of the clog detection sensor.
IO7504	1	1	The tool wash filter has been confirmed.	The tool wash filter was checked for clogging.	
IO7505	1	1	The tool wash filter cannot be confirmed.	The tool wash filter could not be checked for clogging.	Check the set value of the machine parameter (system 1). - Tool wash filter check time - Z-axis lowering speed when checking tool wash filter
IO7506	1	1	The tool washing liquid surface is abnormal.	TW_LQLVERR signal (BY: Tool wash coolant level error) was detected. For the factory-installed ladder program, when tool washing was turned on, the coolant level was under the dischargeable lower limit of the accumulator.	Check the ladder. For the factory-installed ladder program, 1. The tool washing filter must be replaced or cleaned. 2 The tool wash coolant level sensor is faulty. Replace it.
IO7507	1	1	The tool washing liquid surface sensor is normal.	This alarm occurs when the coolant level sensor operation is found to be normal as a result of M295 (Tool wash level sensor failure diagnosis) execution.	
IO7508	1	1	Tool wash filter clogging is predicted.	The number of checks for low fluid level IO_TCLS_OK (Tool cleaning fluid level sensor) at the end of the tool cleaning discharge has exceeded the user parameter (switch 1) setting for <No. of tool wash filter clogging warning checks (Type 4)>. For the factory-installed ladder program, 1. The tool wash filter was clogged. 2. The tool change was carried out consecutively while the ATC falls at a slow speed. 3. Tool change was carried out consecutively even though the timing for the AT-axis was slow or late.	Check the ladder. For the factory-installed ladder program, 1. The tool washing filter must be replaced or cleaned. 2. Review the user parameter (switch 1) <No. of tool wash filter clogging warning checks (Type 4)> or the ATC falling speed. 3. Review the user parameter (switch 1) <No. of tool wash filter clogging warning checks (Type 4)> or the <ATC arm turn speed ratio 1/2>.

No.	Stop level	Reset level	Alarm message	Cause	Solution
IO7509	1	1	Time to change batteries	The battery voltage is reduced.	Replace all the dry batteries. Replace them after turning off the power.
IO7510	1	1	It is in the manual mode of chip conveyor.	SCRWMNLER signal (BY: Chip conveyor manual mode) was detected. For the factory-installed ladder program, the [Automatic/manual mode change] switch on the chip conveyor control panel is in the "manual" position.	Check the ladder. For the factory-installed ladder program, Place the [Automatic/manual mode change] switch in the "automatic" position.
IO7511	1	1	Rapid feed override switch is off the position.	The rapid feed override switch is off the scale.	1. Adjust the rapid feed override switch to the scale. 2. When the error still occurs even if the rapid feed override switch is adjusted to the scale, the switch is defective. Contact the nearest BROTHER approved service dealer.
IO7512	1	1	Feedrate override switch is off the position.	The feedrate override switch is off the scale.	1. Adjust the feedrate override switch to the scale. 2. When the error still occurs even if the feedrate override switch is adjusted to the scale, the switch is defective. Contact the nearest BROTHER approved service dealer.
IO7513	1	1	Spindle override switch is off the position.	The spindle override switch is off the scale.	1. Adjust the spindle override switch to the scale. 2. When the error still occurs even if the spindle override switch is adjusted to the scale, the switch is defective. Contact the nearest BROTHER approved service dealer.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SP0000	5	3	Indirect specif. err. in sending data for auto notification	The ladder made an invalid specification when indirectly specifying the sending data for the auto notification.	Check the PLC ladder program.
SP0500	5	2	Ladder stopped during operation (high-speed ladder 1).	Checked that the corresponding ladder program stopped during memory/MDI operations.	Press the [RST] key. Find the cause of ladder program stop. Start the ladder program.
SP0501	5	2	Ladder stopped during operation (standard ladder 1).	Checked that the corresponding ladder program stopped during memory/MDI operations.	Press the [RST] key. Find the cause of ladder program stop. Start the ladder program.
SP0502	5	2	Ladder stopped during operation (high-speed ladder 2).	Checked that the corresponding ladder program stopped during memory/MDI operations.	Press the [RST] key. Find the cause of ladder program stop. Start the ladder program.
SP0503	5	2	Ladder stopped during operation (standard ladder 2).	Checked that the corresponding ladder program stopped during memory/MDI operations.	Press the [RST] key. Find the cause of ladder program stop. Start the ladder program.
SP0504	5	2	No ladder program (high-speed ladder 1)	The corresponding ladder program is not in the system.	If the ladder program is backed up, enter the program from the external I/O screen. Enter the factory-set ladder program.
SP0505	5	2	No ladder program (standard ladder 1)	The corresponding ladder program is not in the system.	If the ladder program is backed up, enter the program from the external I/O screen. Enter the factory-set ladder program.
SP0506	5	2	No ladder program (high-speed ladder 2)	The corresponding ladder program is not in the system.	If the ladder program is backed up, enter the program from the external I/O screen. Enter the factory-set ladder program.
SP0507	5	2	No ladder program (standard ladder 2)	The corresponding ladder program is not in the system.	If the ladder program is backed up, enter the program from the external I/O screen. Enter the factory-set ladder program.
SP0508	5	2	Ladder prog. mismatch error (high-speed ladder 1)	1. After being edited, the ladder program is not written to RUN side. Or power failure occurred during writing. 2. After the ladder program was entered via communication, power was turned off during reset.	Write the ladder program from the PLC screen of the external I/O screen. Then start the program.
SP0509	5	2	Ladder program mismatch error (standard ladder 1)	1. After being edited, the ladder program is not written to RUN side. Or power failure occurred during writing. 2. After the ladder program was entered via communication, power was turned off during reset.	Write the ladder program from the PLC screen of the external I/O screen. Then start the program.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SP0510	5	2	Ladder prog. mismatch error (high-speed ladder 2)	1. After being edited, the ladder program is not written to RUN side. Or power failure occurred during writing. 2. After the ladder program was entered via communication, power was turned off during reset.	Write the ladder program from the PLC screen of the external I/O screen. Then start the program.
SP0511	5	2	Ladder program mismatch error (standard ladder 2)	1. After being edited, the ladder program is not written to RUN side. Or power failure occurred during writing. 2. After the ladder program was entered via communication, power was turned off during reset.	Write the ladder program from the PLC screen of the external I/O screen. Then start the program.
SP0512	5	2	Ladder program exec. error (high-speed ladder 1)	Writing /RUN of ladder program was specified, but failed.	Write the ladder program from the PLC screen of the external I/O screen. Then start the program.
SP0513	5	2	Ladder program execution error (standard ladder 1)	Writing /RUN of ladder program was specified, but failed.	Write the ladder program from the PLC screen of the external I/O screen. Then start the program.
SP0514	5	2	Ladder program exec. error (high-speed ladder 2)	Writing /RUN of ladder program was specified, but failed.	Write the ladder program from the PLC screen of the external I/O screen. Then start the program.
SP0515	5	2	Ladder program execution error (standard ladder 2)	Writing /RUN of ladder program was specified, but failed.	Write the ladder program from the PLC screen of the external I/O screen. Then start the program.
SP0516	5	2	Ladder program abnormal (high-speed ladder 1)	Abnormal ladder file was entered from external I/O. The entered ladder file was deleted.	Check the input PLC ladder program.
SP0517	5	2	Ladder program abnormal (standard ladder 1)	Abnormal ladder file was entered from external I/O. The entered ladder file was deleted.	Check the input PLC ladder program.
SP0518	5	2	Ladder program abnormal (high-speed ladder 2)	Abnormal ladder file was entered from external I/O. The entered ladder file was deleted.	Check the input PLC ladder program.
SP0519	5	2	Ladder program abnormal (standard ladder 2)	Abnormal ladder file was entered from external I/O. The entered ladder file was deleted.	Check the input PLC ladder program.
SP0520	5	2	PLC operation time limit over (high-speed ladder 1)	The scantime of the ladder program was longer than 1000 msec.	Check the PLC ladder program.
SP0521	5	2	PLC operation time limit over (standard ladder 1)	The scantime of the ladder program was longer than 1000 msec.	Check the PLC ladder program.
SP0522	5	2	PLC operation time limit over (high-speed ladder 2)	The scantime of the ladder program was longer than 1000 msec.	Check the PLC ladder program.
SP0523	5	2	PLC operation time limit over (standard ladder 2)	The scantime of the ladder program was longer than 1000 msec.	Check the PLC ladder program.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SP0524	5	2	Double coil error	There is an output for the output OM that is the same in the ladder program.	Check the PLC ladder program.
SP0525	5	2	Ladder write version mismatch (High-speed ladder 1)	The ladder program is written with a different system version.	Perform the ladder program write operation from the PLC screen on the external I/O screen. Then, start it up.
SP0526	5	2	Ladder write version mismatch (Standard speed ladder 1)	The ladder program is written with a different system version.	Perform the ladder program write operation from the PLC screen on the external I/O screen. Then, start it up.
SP0527	5	2	Ladder write version mismatch (High-speed ladder 2)	The ladder program is written with a different system version.	Perform the ladder program write operation from the PLC screen on the external I/O screen. Then, start it up.
SP0528	5	2	Ladder write version mismatch (Standard speed ladder 2)	The ladder program is written with a different system version.	Perform the ladder program write operation from the PLC screen on the external I/O screen. Then, start it up.
SP0529	5	2	PLC expanded function mismatch	A command is used that is not supported by the <Expanded function> in the PLC ladder program. 1. A command is used that is not supported within the <Expanded function range> on the PLC menu screen. 2. The value specified for the <Expanded function range> on the PLC menu screen does not match the value specified for the <Expanded function> in the ladder program. 3. The values specified for the <Expanded function> in all the ladder programs do not match.	1-1. Check the PLC ladder program and then press the [RST] key. 1-2. Execute version upgrade. 2-1. Check the PLC ladder program and then press the [RST] key. 2-2. Check the <Expanded function range> on the PLC menu screen, and then press the [RST] key. 3. Input all ladder programs that have the same values specified for the <Expanded function>, and then press the [RST] key.
SP0530	5	2	PLC expanded function error	A command is used that is not supported by the <Expanded function> in the PLC ladder program. 1. A command is used that is not supported within the <Expanded function range> on the PLC menu screen. 2. The value specified for the <Expanded function range> on the PLC menu screen does not match the value specified for the <Expanded function> in the ladder program.	1-1. Check the PLC ladder program and then press the [RST] key. 1-2. Execute version upgrade. 2-1. Check the PLC ladder program and then press the [RST] key. 2-2. Check the <Expanded function range> on the PLC menu screen, and then press the [RST] key.
SP0531	5	2	Indirect specif. err. in sending data for auto notification	The ladder made an invalid specification when indirectly specifying the sending data for the auto notification.	Check the PLC ladder program.
SP1500	4	3	Indirect specif. err. in sending data for auto notification	The ladder made an invalid specification when indirectly specifying the sending data for the auto notification.	Check the PLC ladder program.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SP2000	4	2	PLC watch dog timeout (High-speed ladder 1)	The scantime of the corresponding ladder program was longer than WDT monitor time.	Check the PLC ladder program.
SP2001	4	2	PLC watch dog timeout (Standard ladder 1)	The scantime of the corresponding ladder program was longer than WDT monitor time.	Check the PLC ladder program.
SP2002	4	2	PLC watch dog timeout (High-speed ladder 2)	The scantime of the corresponding ladder program was longer than WDT monitor time.	Check the PLC ladder program.
SP2003	4	2	PLC watch dog timeout (Standard ladder 2)	The scantime of the corresponding ladder program was longer than WDT monitor time.	Check the PLC ladder program.
SP2004	4	2	Abnormal memory (High-speed ladder 1)	The corresponding ladder program failed to allocate required memory.	Contact the nearest BROTHER approved service dealer.
SP2005	4	2	Abnormal memory (Standard ladder 1)	The corresponding ladder program failed to allocate required memory.	Contact the nearest BROTHER approved service dealer.
SP2006	4	2	Abnormal memory (High-speed ladder 2)	The corresponding ladder program failed to allocate required memory.	Contact the nearest BROTHER approved service dealer.
SP2007	4	2	Abnormal memory (High-speed ladder 2)	The corresponding ladder program failed to allocate required memory.	Contact the nearest BROTHER approved service dealer.
SP2008	4	2	Function execution error (High-speed ladder 1)	The corresponding ladder program failed to execute system functions.	Contact the nearest BROTHER approved service dealer.
SP2009	4	2	Function execution error (Standard ladder 1)	The corresponding ladder program failed to execute system functions.	Contact the nearest BROTHER approved service dealer.
SP2010	4	2	Function execution error (High-speed ladder 2)	The corresponding ladder program failed to execute system functions.	Contact the nearest BROTHER approved service dealer.
SP2011	4	2	Function execution error (Standard ladder 2)	The corresponding ladder program failed to execute system functions.	Contact the nearest BROTHER approved service dealer.
SP2012	4	2	BCD/BIN conversion error (High-speed ladder 1)	The corresponding ladder program caused BCD/BIN conversion data error.	Check the PLC ladder program.
SP2013	4	2	BCD/BIN conversion error (Standard ladder 1)	The corresponding ladder program caused BCD/BIN conversion data error.	Check the PLC ladder program.
SP2014	4	2	BCD/BIN conversion error (High-speed ladder 2)	The corresponding ladder program caused BCD/BIN conversion data error.	Check the PLC ladder program.
SP2015	4	2	BCD/BIN conversion error (Standard ladder 2)	The corresponding ladder program caused BCD/BIN conversion data error.	Check the PLC ladder program.
SP2016	4	2	Divided by zero (High-speed ladder 1)	The corresponding ladder program performed division by zero.	Check the PLC ladder program.
SP2017	4	2	Divided by zero (Standard ladder 1)	The corresponding ladder program performed division by zero.	Check the PLC ladder program.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SP2018	4	2	Divided by zero (High-speed ladder 2)	The corresponding ladder program performed division by zero.	Check the PLC ladder program.
SP2019	4	2	Divided by zero (Standard ladder 2)	The corresponding ladder program performed division by zero.	Check the PLC ladder program.
SP2020	4	2	Command code is abnormal. (High-speed ladder 1)	The corresponding ladder program is destroyed.	Contact the nearest BROTHER approved service dealer.
SP2021	4	2	Command code is abnormal. (Standard ladder 1)	The corresponding ladder program is destroyed.	Contact the nearest BROTHER approved service dealer.
SP2022	4	2	Command code is abnormal. (High-speed ladder 2)	The corresponding ladder program is destroyed.	Contact the nearest BROTHER approved service dealer.
SP2023	4	2	Command code is abnormal. (Standard ladder 2)	The corresponding ladder program is destroyed.	Contact the nearest BROTHER approved service dealer.
SP2024	4	2	No label specified by JMP, JE or CALL (High-speed ladder 1)	There is no specific label of JMP, JE, CALL in the corresponding ladder program.	Check the PLC ladder program.
SP2025	4	2	No label specified by JMP, JE or CALL (Standard ladder 1)	There is no specific label of JMP, JE, CALL in the corresponding ladder program.	Check the PLC ladder program.
SP2026	4	2	No label specified by JMP, JE or CALL (High-speed ladder 2)	There is no specific label of JMP, JE, CALL in the corresponding ladder program.	Check the PLC ladder program.
SP2027	4	2	No label specified by JMP, JE or CALL (Standard ladder 2)	There is no specific label of JMP, JE, CALL in the corresponding ladder program.	Check the PLC ladder program.
SP2028	4	2	CALL-RET, etc. not in a set (high-speed ladder 1)	CALL-RET, MC-MCR, FOR-NEXT are not made into a set in the corresponding ladder program.	Check the PLC ladder program.
SP2029	4	2	CALL-RET, etc. not in a set (standard ladder 1)	CALL-RET, MC-MCR, FOR-NEXT are not made into a set in the corresponding ladder program.	Check the PLC ladder program.
SP2030	4	2	CALL-RET, etc. not in a set (high-speed ladder 2)	CALL-RET, MC-MCR, FOR-NEXT are not made into a set in the corresponding ladder program.	Check the PLC ladder program.
SP2031	4	2	CALL-RET, etc. not in a set (standard ladder 2)	CALL-RET, MC-MCR, FOR-NEXT are not made into a set in the corresponding ladder program.	Check the PLC ladder program.
SP2032	4	2	Access to OM outside the area (High-sp. ladder 1)	The corresponding ladder program accessed out-of-range OM.	Check the PLC ladder program.
SP2033	4	2	Access to OM outside the area (Standard ladder 1)	The corresponding ladder program accessed out-of-range OM.	Check the PLC ladder program.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SP2034	4	2	Access to OM outside the area (High-sp. ladder 2)	The corresponding ladder program accessed out-of-range OM.	Check the PLC ladder program.
SP2035	4	2	Access to OM outside the area (Standard ladder 2)	The corresponding ladder program accessed out-of-range OM.	Check the PLC ladder program.
SP2036	4	2	MC, CALL, FOR nesting too deep (hi-speed ladder 1)	Nestings of MC, CALL, and FOR in the corresponding ladder program are too deep.	Check the PLC ladder program.
SP2037	4	2	MC, CALL, FOR nesting too deep (standard ladder 1)	Nestings of MC, CALL, and FOR in the corresponding ladder program are too deep.	Check the PLC ladder program.
SP2038	4	2	MC, CALL, FOR nesting too deep (hi-speed ladder 2)	Nestings of MC, CALL, and FOR in the corresponding ladder program are too deep.	Check the PLC ladder program.
SP2039	4	2	MC, CALL, FOR nesting too deep (standard ladder 2)	Nestings of MC, CALL, and FOR in the corresponding ladder program are too deep.	Check the PLC ladder program.
SP2040	4	2	DFC command execution failed (high-speed ladder 1)	DFC command execution was failed in the corresponding ladder program.	Check the PLC ladder program.
SP2041	4	2	DFC command execution failed (standard ladder 1)	DFC command execution was failed in the corresponding ladder program.	Check the PLC ladder program.
SP2042	4	2	DFC command execution failed (high-speed ladder 2)	DFC command execution was failed in the corresponding ladder program.	Check the PLC ladder program.
SP2043	4	2	DFC command execution failed (standard ladder 2)	DFC command execution was failed in the corresponding ladder program.	Check the PLC ladder program.
SP2044	4	2	The size is exceeded. (High-speed ladder 1)	Compiled code of the corresponding ladder program exceeded the designated size (compile code storage range).	Check the PLC ladder program.
SP2045	4	2	The size is exceeded. (Standard ladder 1)	Compiled code of the corresponding ladder program exceeded the designated size (compile code storage range).	Check the PLC ladder program.
SP2046	4	2	The size is exceeded. (High-speed ladder 2)	Compiled code of the corresponding ladder program exceeded the designated size (compile code storage range).	Check the PLC ladder program.
SP2047	4	2	The size is exceeded. (Standard ladder 2)	Compiled code of the corresponding ladder program exceeded the designated size (compile code storage range).	Check the PLC ladder program.
SP2048	4	2	No PLC (high-speed ladder 1)	The corresponding ladder program is not in the system.	If the ladder program is backed up, enter the ladder program from the external I/O screen. Enter the factory-set ladder program.
SP2049	4	2	No PLC (Standard ladder 1)	The corresponding ladder program is not in the system.	If the ladder program is backed up, enter the ladder program from the external I/O screen. Enter the factory-set ladder program.
SP2050	4	2	No PLC (High-speed ladder 2)	The corresponding ladder program is not in the system.	If the ladder program is backed up, enter the ladder program from the external I/O screen. Enter the factory-set ladder program.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SP2051	4	2	No PLC (Standard ladder 2)	The corresponding ladder program is not in the system.	If the ladder program is backed up, enter the ladder program from the external I/O screen. Enter the factory-set ladder program.
SP2052	4	2	Ladder program stopped (high-speed ladder 1)	Ladder program was stopped. 1. Ladder program stop operation was executed. 2. Execution was suspended due to ladder program writing operation. 3. Ladder program stop was detected by NC observation function. 4. Ladder program operation stopped due to another alarm being triggered.	1. Press the [RST] key after starting the ladder program. 2. Check that the ladder program is being executed. Press the [RST] key. 3. Press the [RST] key after starting the ladder program. 4. Remove the cause of the other alarm that was triggered at the same time.
SP2053	4	2	Ladder program stopped (standard ladder 1)	Ladder program was stopped. 1. Ladder program stop operation was executed. 2. Execution was suspended due to ladder program writing operation. 3. Ladder program stop was detected by NC observation function. 4. Ladder program operation stopped due to another alarm being triggered.	1. Press the [RST] key after starting the ladder program. 2. Check that the ladder program is being executed. Press the [RST] key. 3. Press the [RST] key after starting the ladder program. 4. Remove the cause of the other alarm that was triggered at the same time.
SP2054	4	2	Ladder program stopped (high-speed ladder 2)	Ladder program was stopped. 1. Ladder program stop operation was executed. 2. Execution was suspended due to ladder program writing operation. 3. Ladder program stop was detected by NC observation function. 4. Ladder program operation stopped due to another alarm being triggered.	1. Press the [RST] key after starting the ladder program. 2. Check that the ladder program is being executed. Press the [RST] key. 3. Press the [RST] key after starting the ladder program. 4. Remove the cause of the other alarm that was triggered at the same time.
SP2055	4	2	Ladder program stopped (standard ladder 2)	Ladder program was stopped. 1. Ladder program stop operation was executed. 2. Execution was suspended due to ladder program writing operation. 3. Ladder program stop was detected by NC observation function. 4. Ladder program operation stopped due to another alarm being triggered.	1. Press the [RST] key after starting the ladder program. 2. Check that the ladder program is being executed. Press the [RST] key. 3. Press the [RST] key after starting the ladder program. 4. Remove the cause of the other alarm that was triggered at the same time.
SP2056	4	2	Indirect specif. err. in sending data for auto notification	The ladder made an invalid specification when indirectly specifying the sending data for the auto notification.	Check the PLC ladder program.
SP3500	3	3	Indirect specif. err. in sending data for auto notification	The ladder made an invalid specification when indirectly specifying the sending data for the auto notification.	Check the PLC ladder program.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SP4000	3	2	A ladder program is outputting outside of the writing range	There is an M or D in the global OM that is outputting outside of the writing range.	Check the PLC ladder program or check the writing range for the <Power failure retention/write range setting>.
SP4001	3	2	Indirect specif. err. in sending data for auto notification	The ladder made an invalid specification when indirectly specifying the sending data for the auto notification.	Check the PLC ladder program.
SP5000	2	3	Indirect specif. err. in sending data for auto notification	The ladder made an invalid specification when indirectly specifying the sending data for the auto notification.	Check the PLC ladder program.
SP5500	2	2	Indirect specif. err. in sending data for auto notification	The ladder made an invalid specification when indirectly specifying the sending data for the auto notification.	Check the PLC ladder program.
SP6000	2	1	Indirect specif. err. in sending data for auto notification	The ladder made an invalid specification when indirectly specifying the sending data for the auto notification.	Check the PLC ladder program.
SP6500	1	3	Indirect specif. err. in sending data for auto notification	The ladder made an invalid specification when indirectly specifying the sending data for the auto notification.	Check the PLC ladder program.
SP7000	1	2	Indirect specif. err. in sending data for auto notification	The ladder made an invalid specification when indirectly specifying the sending data for the auto notification.	Check the PLC ladder program.
SP7500	1	1	Ladder not working exists (high-speed ladder 1)	Operation start was attempted, but the corresponding ladder program stops.	Check the PLC ladder program.
SP7501	1	1	Ladder not working exists (standard ladder 1)	Operation start was attempted, but the corresponding ladder program stops.	Check the PLC ladder program.
SP7502	1	1	Ladder not working exists (high-speed ladder 2)	Operation start was attempted, but the corresponding ladder program stops.	Check the PLC ladder program.
SP7503	1	1	Ladder not working exists (standard ladder 2)	Operation start was attempted, but the corresponding ladder program stops.	Check the PLC ladder program.
SP7504	1	1	Standard ladder 1 is not found	Power failure retention/write range setting was attempted when standard ladder 1 was not found.	Standard ladder 1 is necessary for power failure retention/write range setting. Enter standard ladder 1.
SP7505	1	1	Ladder write version mismatch (High-speed ladder 1)	The ladder program is written with a different system version.	Perform the ladder program write operation from the PLC screen on the external I/O screen. Then, start it up.
SP7506	1	1	Ladder write version mismatch (Standard speed ladder 1)	The ladder program is written with a different system version.	Perform the ladder program write operation from the PLC screen on the external I/O screen. Then, start it up.
SP7507	1	1	Ladder write version mismatch (High-speed ladder 2)	The ladder program is written with a different system version.	Perform the ladder program write operation from the PLC screen on the external I/O screen. Then, start it up.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SP7508	1	1	Ladder write version mismatch (Standard speed ladder 2)	The ladder program is written with a different system version.	Perform the ladder program write operation from the PLC screen on the external I/O screen. Then, start it up.
SP7509	1	1	A ladder program is outputting outside of the writing range	There is an M or D in the global OM that is outputting outside of the writing range.	Check the PLC ladder program or check the writing range for the <Power failure retention/write range setting>.
SP7510	1	1	Indirect specif. err. in sending data for auto notification	The ladder made an invalid specification when indirectly specifying the sending data for the auto notification.	Check the PLC ladder program.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SM0000	5	3	Machine setting error	The CF and machine identification signal does not match the NC's internal information.	1. If the NC PCB has been replaced and there is an internal data bank change, check for a data setting error. 2. If the IO PCB has been replaced and this alarm still triggers, make sure that all connectors on the IO PCB have been connected or that the latest IO PCB is being used. If operation does not restore, the PCB may be faulty. Therefore, replace it with the latest IO PCB. 3. If the CF has been replaced and this alarm still triggers, check the machine and the software version on the version screen. If the wrong CF is connected, the machine model name does not display correctly. If the wrong name appears, replace the CF with the correct one.
SM0001	5	3	Servo PWR off cannot be confirmed.	Servo PWR off cannot be confirmed after resetting abso. encoder counter. Servo off cannot be confirmed although the servo parameter setting has been changed.	The servo amplifier is defective. Contact the nearest BROTHER approved service dealer.
SM0002	5	3	Adjusted abso. data setting error	The value set for adjusted machine coordinates is not within the setting range.	Set the value correctly for [Adjusted MC coord.] on the <Reset abso. encoder counter> screen, and adjust the machine again.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SM0003	5	3	Machine param. setting error (SYS 2)	The following cause descriptions are divided according to their auxiliary number. 1 - The <Quick table> values do not match the machine configuration. 2 - The number of pulses for additional axis obtained from <*-axis gear ratio> and the number of pulses per motor rotation exceeds the acceptable value. 3 - The parameter <*-axis clamp mechanism> is not set to <0: Type 1> on an additional axis equipped with a brake mechanism. (Versions before WX0*.07.000) The parameter <*-axis clamp mechanism> is set to <1: Type 2> on an additional axis equipped with a brake mechanism. (Excluding A-axis on M140X2) (Versions after WX0*.07.000) 4 - When the <*-axis option> for the axis is set to <ON (For positioning)>, the <Expansion I/O board station number for *-th-axis positioning> is set to <0> or has not been set. 5 - The <*-axis option> setting for the axis does not match the machine configuration.	Set the parameters of Machine Parameters (System 2) to the correct values and turn power on again.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM0004	5	3	Machine param. setting error (SYS 3)	The following cause descriptions are divided according to their auxiliary number. 1 - The pulses of magazine obtained from [Magazine gear ratio], [Distance between magazine pots], etc. exceeds acceptable value. 2 - The value obtained by multiplying "[Magazine gear ratio (numerator)] divided by [Magazine gear ratio (denominator)]" by the number of pulses per motor rotation is not an integer. 3 - The distance per magazine rotation obtained from [Tool storage capacity] and [Distance between magazine pots] exceeds acceptable value. 4 - The number of pulses for the AT-axis, obtained from the <AT-axis gear ratio (numerator)> and <AT-axis gear ratio (denominator)>, exceeds the acceptable value. 5 - When the <AT-axis gear ratio (numerator)> / <AT-axis gear ratio (denominator)> is multiplied by the number of pulses per motor rotation, and the product or value is not an integer.	Set the parameters of Machine Parameters (System 3) to the correct values and turn power on again.
SM0005	5	3	Machine param. setting error (SYS 1)	The following cause descriptions are divided according to their auxiliary number. 1 - An invalid value is set to either or both of [*-axis stroke 1 (-)] and [Distance to Z-axis zero] for [Bearing position on *-axis fixed side] and [Bearing position on free side of * axis].	Set the parameters of Machine Parameters (System 1) to the correct values and turn power on again.
SM0006	5	3	Machine param. setting error (PLC)	The following cause descriptions are divided according to their auxiliary number. 1 - The pulse of PLC axis obtained from [P* axis lead/gear ratio], etc. exceeds acceptable value. 2 - A brake signal which is used in another axis is set to [P* axis brake].	Set the parameters of Machine Parameters (PLC) to the correct values and turn power on again.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM0007	5	3	The machine coordinates cannot be calculated.	1. The pulse value converted from the machine coordinates exceeds the acceptable value. 2. Machine coordinates calculated from motor feedback position exceeds the acceptable value. 3. Unable to calculate pitch error compensation from motor feedback position.	The PCB may be defective. Contact the nearest BROTHER approved service dealer.
SM0008	5	3	The X-axis brake-off cannot be confirmed.	X-axis brake OFF is not detected after X-axis servo is turned on.	The PCB may be defective. Contact the nearest BROTHER approved service dealer.
SM0009	5	3	The Y-axis brake-off cannot be confirmed.	Y-axis brake OFF is not detected after Y-axis servo is turned on.	The PCB may be defective. Contact the nearest BROTHER approved service dealer.
SM0010	5	3	The Z-axis brake-off cannot be confirmed.	Z-axis brake OFF is not detected after Z-axis servo is turned on.	The PCB may be defective. Contact the nearest BROTHER approved service dealer.
SM0011	5	3	The *-axis brake-off cannot be confirmed.	4-axis brake OFF is not detected after 4-axis servo is turned on.	The PCB may be defective. Contact the nearest BROTHER approved service dealer.
SM0012	5	3	The *-axis brake-off cannot be confirmed.	5-axis brake OFF is not detected after 5-axis servo is turned on.	The PCB may be defective. Contact the nearest BROTHER approved service dealer.
SM0013	5	3	The *-axis brake-off cannot be confirmed.	6-axis brake OFF is not detected after 6-axis servo is turned on.	The PCB may be defective. Contact the nearest BROTHER approved service dealer.
SM0014	5	3	The *-axis brake-off cannot be confirmed.	7-axis brake OFF is not detected after 7-axis servo is turned on.	The PCB may be defective. Contact the nearest BROTHER approved service dealer.
SM0015	5	3	The *-axis brake-off cannot be confirmed.	8-axis brake OFF is not detected after 8-axis servo is turned on.	The PCB may be defective. Contact the nearest BROTHER approved service dealer.
SM0016	5	3	The *-axis brake-off cannot be confirmed.	Magazine brake OFF is not detected after magazine servo is turned on.	The PCB may be defective. Contact the nearest BROTHER approved service dealer.
SM0017	5	3	The P1-axis brake-off cannot be confirmed.	P1-axis brake OFF is not detected after P1-axis servo is turned on.	The PCB may be defective. Contact the nearest BROTHER approved service dealer.
SM0018	5	3	The P2-axis brake-off cannot be confirmed.	P2-axis brake OFF is not detected after P2-axis servo is turned on.	The PCB may be defective. Contact the nearest BROTHER approved service dealer.
SM0019	5	3	The P3-axis brake-off cannot be confirmed.	P3-axis brake OFF is not detected after P3-axis servo is turned on.	The PCB may be defective. Contact the nearest BROTHER approved service dealer.
SM0020	5	3	The P4-axis brake-off cannot be confirmed.	P4-axis brake OFF is not detected after P4-axis servo is turned on.	The PCB may be defective. Contact the nearest BROTHER approved service dealer.
SM0021	5	3	Therm. displacement offset parameter setting error	Wrong value has been set to the thermal distortion compensation parameter.	Check the set value of thermal distortion compensation parameter.
SM0022	5	3	Simple panel communication data is abnormal. (*)	Reserved	Reserved
SM0023	5	3	Memory expansion option error	Option and OPSEL are inconsistent.	1. Check the connection with OPSEL. 2. Check the option.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM0024	5	3	Too many options	Option and OPSEL are inconsistent.	1. Check the connection with OPSEL. 2. Check the option.
SM0025	5	3	PLC-axis unavailable	The IL PCB does not support the PLC-axis.	
SM0026	5	3	Servo amplifier version error	An attempt was made to update the version using a firmware file that does not correspond to the servo amplifier being used.	Store only the latest servo amplifier firmware file in the memory card.
SM0027	5	3	The *-axis brake-off cannot be confirmed.	The AT-axis brake OFF status cannot be confirmed after the servo turns ON for the AT-axis.	The PCB may be defective. Contact the nearest BROTHER approved service dealer.
SM0500	5	2	Machine stopped with door open	1. The door was opened during operation while the [DOOR INTERLOCK] switch was enabled. 2. The door was opened during operation while the [Pendant valid] switch for the teaching pendant was set to [Valid] and the [ENABLE] switch was turned OFF. 3. Door opened while the door was locked.	Close the door and press the [RST] key. For machines equipped with an automatic door specification, press the [AUTOMATIC DOOR-CLOSE] switch while pressing the [RELSE] key. Make sure that the door lock operates correctly.
SM0501	5	2	Spindle rotation position error	«Arm type ATC» The ATC arm was not at zero point during spindle rotation. «Turret type ATC» Raising the Z-axis from the zero point was attempted with the spindle servo ON (this does not occur when the spindle is oriented to 0 degrees).	«Arm type ATC» Press the [RST] key to reset the alarm. «Turret type ATC» Press the [RST] key to reset the alarm.
SM0502	5	2	*-axis position shifted	1. The motor shaft has shifted due to machining load. 2. The axis has shifted during clamping more than the setting for [4th-axis position check pulse]. 3. Brake power is too low.	1. Check the air pressure. 2. Reduce the cutting load. 3. Check that the setting for [[4th-axis position check pulse] (machine parameter - system 2) is not too small. 4. Check that the brake is applied correctly.
SM0503	5	2	*-axis position shifted	1. The motor shaft has shifted due to machining load. 2. The axis has shifted during clamping more than the setting for [5th-axis position check pulse]. 3. Brake power is too low.	1. Check the air pressure. 2. Reduce the cutting load. 3. Check that the setting for [5th-axis position check pulse] (machine parameter - system 2) is not too small. 4. Check that the brake is applied correctly.
SM0504	5	2	*-axis position shifted	1. The motor shaft has shifted due to machining load. 2. The axis has shifted during clamping more than the setting for [6th-axis position check pulse]. 3. Brake power is too low.	1. Check the air pressure. 2. Reduce the cutting load. 3. Check that the setting for [6th-axis position check pulse] (machine parameter - system 2) is not too small. 4. Check that the brake is applied correctly.
SM0505	5	2	*-axis position shifted	1. The motor shaft has shifted due to machining load. 2. The axis has shifted during clamping more than the setting for [7th-axis position check pulse]. 3. Brake power is too low.	1. Check the air pressure. 2. Reduce the cutting load. 3. Check that the setting for [7th-axis position check pulse] (machine parameter - system 2) is not too small. 4. Check that the brake is applied correctly.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM0506	5	2	*-axis position shifted	1. The motor shaft has shifted due to machining load. 2. The axis has shifted during clamping more than the setting for [8th-axis position check pulse]. 3. Brake power is too low.	1. Check the air pressure. 2. Reduce the cutting load. 3. Check that the setting for [8th-axis position check pulse] (machine parameter - system 2) is not too small. 4. Check that the brake is applied correctly.
SM0507	5	2	Spindle in-position check timeout	The axis did not enter the in-position width after completing operation.	Check that the axis does not vibrate after it has stopped. [I/O] [2.SERVO] DEVIATION
SM0508	5	2	X-axis in-position check timeout	The axis did not enter the in-position width after completing operation.	1. Check the values of machine parameters (system 1). [X-axis in-position width] [X-axis in-position check timeout period] 2. Check that the axis does not vibrate after it has stopped. [I/O] [2.SERVO] DEVIATION
SM0509	5	2	Y-axis in-position check timeout	The axis did not enter the in-position width after completing operation.	1. Check the values of machine parameters (system 1). [Y-axis in-position width] [Y-axis in-position check timeout period] 2. Check that the axis does not vibrate after it has stopped. [I/O] [2.SERVO] DEVIATION
SM0510	5	2	Z-axis in-position check timeout	The axis did not enter the in-position width after completing operation.	1. Check the values of machine parameters (system 1). [Z-axis in-position width] [Z-axis in-position check timeout period] 2. Check that the axis does not vibrate after it has stopped. [I/O] [2.SERVO] DEVIATION
SM0511	5	2	*-axis in-position check timeout	The axis did not enter the in-position width after completing operation.	1. Check the values of machine parameters (system 1). [4th-axis in-position width] [4th-axis in-position check timeout period] 2. Check that the axis does not vibrate after it has stopped. [I/O] [2.SERVO] DEVIATION
SM0512	5	2	*-axis in-position check timeout	The axis did not enter the in-position width after completing operation.	1. Check the values of machine parameters (system 1). [5th-axis in-position width] [5th-axis in-position check timeout period] 2. Check that the axis does not vibrate after it has stopped. [I/O] [2.SERVO] DEVIATION
SM0513	5	2	*-axis in-position check timeout	The axis did not enter the in-position width after completing operation.	1. Check the values of machine parameters (system 1). [6th-axis in-position width] [6th-axis in-position check timeout period] 2. Check that the axis does not vibrate after it has stopped. [I/O] [2.SERVO] DEVIATION

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM0514	5	2	*-axis in-position check timeout	The axis did not enter the in-position width after completing operation.	1. Check the values of machine parameters (system 1). [7th-axis in-position width] [7th-axis in-position check timeout period] 2. Check that the axis does not vibrate after it has stopped. [I/O] [2.SERVO] DEVIATION
SM0515	5	2	*-axis in-position check timeout	The axis did not enter the in-position width after completing operation.	1. Check the values of machine parameters (system 1). [8th-axis in-position width] [8th-axis in-position check timeout period] 2. Check that the axis does not vibrate after it has stopped. [I/O] [2.SERVO] DEVIATION
SM0516	5	2	*-axis in-position check timeout	The axis did not enter the in-position width after completing operation.	1. Check the values of machine parameters (system 1). [Magazine in-position width] [Magazine in-position check timeout period] 2. Check that the axis does not vibrate after it has stopped. [I/O] [2.SERVO] DEVIATION
SM0517	5	2	*-axis servo power off	When door interlock is valid, the door was opened while the 4th-axis was in motion.	Press the [RST] key.
SM0518	5	2	*-axis servo power off	When door interlock is valid, the door was opened while the 5th-axis was in motion.	Press the [RST] key.
SM0519	5	2	*-axis servo power off	When door interlock is valid, the door was opened while the 6th-axis was in motion.	Press the [RST] key.
SM0520	5	2	*-axis servo power off	When door interlock is valid, the door was opened while the 7th-axis was in motion.	Press the [RST] key.
SM0521	5	2	*-axis servo power off	When door interlock is valid, the door was opened while the 8th-axis was in motion.	Press the [RST] key.
SM0522	5	2	*-axis index timeout	The QT axis does not contact the stopper when it moves at low speed within the time set for the parameter.	Check the setting for machine parameters (system 2). [Pause after 4th-axis stop] [Pause until 4th-axis stops] [4th-axis stop current ref. value]
SM0523	5	2	*-axis current error	The set current exceeded the limit set for the parameter while the QT axis is contacting the stopper.	1. Check the setting for machine parameters (system 2). [4th-axis stop current detection range] 2. Check the jump wiring for the X3 and X4 motor amplifier connectors, and make sure that everything is normal and that the wires are not reversed.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SM0524	5	2	Spindle command timeout	Spindle command could not be completed within the time set for [Spindle command timeout period] (machine parameter - system 1).	Check the setting for machine parameter (system 1). [Spindle command timeout period]. Pressing the [RST] key resets the error. The NC PCB is defective. If this error occurs repeatedly, contact the nearest BROTHER approved service dealer.
SM0525	5	2	Spindle position shifted	The spindle position has shifted more than the setting for [Spindle orientation tolerance] (machine parameter 1 / system 1) after spindle orientation was completed.	1. Check the setting for [Spindle orientation tolerance] (machine parameter 1 / system 1). 2. Contact the nearest BROTHER approved service dealer.
SM0526	5	2	The servo parameter is being changed.	A servo parameter is changed again during when it is being changed at resetting.	Press the [RST] key to clear the alarm.
SM0527	5	2	Machine param. setting error (pitch error offset)	The following cause descriptions are divided according to their auxiliary number. 1 - The data "Divisions for the *-th Axis Pitch Error Correction Interval" is incorrect.	Set the machine parameter (pitch error) correctly.
SM0528	5	2	Pitch error data setting error	Cumulative pitch error compensation exceeds the acceptable value.	Set the machine parameter (pitch error compensation) correctly and press the [RST] key.
SM0529	5	2	The machine coordin. have exceeded a limit value.	Machine coordinates exceeded the acceptable value.	1. Reset abso. encoder counter. 2. Move the axis to within the allowable range and press the [RST] key.
SM0530	5	2	Magazine position deviation amount is too large.	«Arm type ATC» Does not occur. «Turret type ATC» The magazine could not turn as instructed.	1. Check that the total tool weight is not too high. 2. Set [1. Valid] for [Turret type magazine low-speed turn mode], and conduct warm-up operation. 3. Contact the nearest BROTHER approved service dealer.
SM0531	5	2	The arm swivel motion time is short.	The time taken for arm turning is less than the setting for <ATC arm turn minimum time> (machine parameter 3).	1. The ATC home position sensor is faulty. 2. Check the setting for <ATC arm turn minimum time> (machine parameter 3). 3. Set the inverter frequency to the specified value.
SM0532	5	2	Tapping motion is unavailable.	A tap command is instructed during an error.	Press the [RST] key to clear the alarm.
SM0533	5	2	Motor insulation resist. monit. value is abnormal.	Insulation resistance monitor value is abnormal in motor insulation resistance measurement.	Motor insulation resistance has lowered. Check that which of insulation resistance between the motor or cable has lowered and plan the replacement by following the instruction manual. Press the [RST] key to clear the alarm. (After the alarm is cleared, normal operation is possible.)

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM0534	5	2	Motor insulation resist. monitor value is warned.	Insulation resistance monitor value is in alarm status in motor insulation resistance measurement.	Motor insulation resistance has lowered. Check that which of insulation resistance between the motor or cable has lowered and plan the replacement by following the instruction manual. Press the [RST] key to clear the alarm. (After the alarm is cleared, normal operation is possible.)
SM0535	5	2	Tool holder has collided with the spindle head.	The tool holder collided with the spindle head.	Recover in accordance with "Recovery of grip error" in the instruction manual.
SM0536	5	2	Speed error when door is open (X-axis)	The X-axis moved at a speed exceeding a certain limit speed when the door was open.	Press the [RST] key.
SM0537	5	2	Speed error when door is open (Y-axis)	The Y-axis moved at a speed exceeding a certain limit speed when the door was open.	Press the [RST] key.
SM0538	5	2	Speed error when door is open (Z-axis)	The Z-axis moved at a speed exceeding a certain limit speed when the door was open.	Press the [RST] key.
SM0539	5	2	Speed error when door is open (*-axis)	The 4th-axis moved at a speed exceeding a certain limit speed when the door was open.	Press the [RST] key.
SM0540	5	2	Speed error when door is open (*-axis)	The 5th-axis moved at a speed exceeding a certain limit speed when the door was open.	Press the [RST] key.
SM0541	5	2	Speed error when door is open (*-axis)	The 6th-axis moved at a speed exceeding a certain limit speed when the door was open.	Press the [RST] key.
SM0542	5	2	Speed error when door is open (*-axis)	The 7th-axis moved at a speed exceeding a certain limit speed when the door was open.	Press the [RST] key.
SM0543	5	2	Speed error when door is open (*-axis)	The 8th-axis moved at a speed exceeding a certain limit speed when the door was open.	Press the [RST] key.
SM0544	5	2	Speed error when door is open (*-axis)	The 9th-axis moved at a speed exceeding a certain limit speed when the door was open.	Press the [RST] key.
SM0545	5	2	Speed error when door is open (P1-axis)	The P1-axis moved at a speed exceeding a certain limit speed when the door was open.	Press the [RST] key.
SM0546	5	2	Speed error when door is open (P2-axis)	The P2-axis moved at a speed exceeding a certain limit speed when the door was open.	Press the [RST] key.
SM0547	5	2	Speed error when door is open (P3-axis)	The P3-axis moved at a speed exceeding a certain limit speed when the door was open.	Press the [RST] key.
SM0548	5	2	Speed error when door is open (P4-axis)	The P4-axis moved at a speed exceeding a certain limit speed when the door was open.	Press the [RST] key.
SM0549	5	2	Servo power on timeout	Servo was not turned on within the servo on processing time.	Check connection of the servo amplifier. Contact the maker to describe the failure.
SM0550	5	2	Curve speed error	The speed of each axis has exceeded the maximum speed in cutting feed mode.	1. Check the program start point, end point, and arc center. 2. Check that [Smooth path offset cancel angle] (user parameter/high accuracy) is not too large.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM0551	5	2	The adjusted absolute values could not be automatically set.	1. An alarm (Stop level 5) was triggered when the adjusted absolute values automatically began to set. 2. A command was sent to the amplifier, but it was not completed after the specified time had elapsed (positioning amplifier).	1. Reset the alarm before proceeding. 2. Check the communication cable connection. The servo amplifier may be faulty. Replace the servo amplifier.
SM0552	5	2	User param. setting error (switch 2)	The following cause description is listed according to its auxiliary number: 1- The set value for the "5th (6th, 7th or 8th) -axis stroke (-) (+)" is out of the acceptable range when the "Optional 5th (6th, 7th or 8th) -axis" is configured to "Yes (positioning only)."	Set the user parameters (Switch 2) to the correct values.
SM0553	5	2	The machine coordinates cannot be calculated.	1. The battery voltage is reduced. 2. The pulse value converted from the machine coordinates exceeds the acceptable value. 3. Machine coordinates calculated from motor feedback position exceeds the acceptable value. 4. Unable to calculate pitch error compensation from motor feedback position.	First, perform the expired battery recovery operation. The PCB may be defective. Contact the nearest BROTHER approved service dealer.
SM0554	5	2	Thread cutting command timeout	The lathe spindle did not reach the specified speed even after the machine parameter <Thread cutting command timeout period> had elapsed for the thread cutting command.	1. Check the machine parameter <Thread cutting command timeout period> setting. 2. The servo amplifier may be faulty. Replace the servo amplifier.
SM0555	5	2	Enable switch OFF	The [ENABLE] switch was turned OFF during operation.	Press the [RST] key.
SM0556	5	2	The tapping synchronous error is too big.	1. Tolerance during tapping operation exceeded [Synchronized tapping error limit] (machine parameter/system 1). 2. Invalid tapping command was issued.	1-1. Check the value for [Synchronized tapping error limit] (machine parameter/system 1). 1-2. Check whether the machining conditions are within the capacity. Press the [RST] key to recover. 2. Check the program.
SM0557	5	2	The machine coordinates have exceeded a limit value.	Machine coordinates exceeded the range of -800° to 800°.	Press the [RST] key to clear the alarm. Thereafter, press the [+4] or [-4] key while holding down the [RELSE] key, and move the axis to within the range of -800° to +800°.
SM0558	5	2	X-axis collision detection	1. Collision was detected on the X-axis. 2. This alarm was triggered at the same time due to a servo error, but it may have been a misdetection when the alarm was triggered.	1. After pressing the [RST] key, move the feed axis to a safe position while checking the machine status. 2. First, restore operation based on the servo error when a servo error is triggered with another alarm or error.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM0559	5	2	Y-axis collision detection	1. Collision was detected on the Y-axis. 2. This alarm was triggered at the same time due to a servo error, but it may have been a misdetection when the alarm was triggered.	1. After pressing the [RST] key, move the feed axis to a safe position while checking the machine status. 2. First, restore operation based on the servo error when a servo error is triggered with another alarm or error.
SM0560	5	2	Z-axis collision detection	1. Collision was detected on the Z-axis. 2. This alarm was triggered at the same time due to a servo error, but it may have been a misdetection when the alarm was triggered.	1. After pressing the [RST] key, move the feed axis to a safe position while checking the machine status. 2. First, restore operation based on the servo error when a servo error is triggered with another alarm or error.
SM0561	5	2	Tool collision during ATC	1. The tool holder collided with the end of the spindle. 2. This alarm was triggered at the same time due to a servo error.	1-1. If the tool has fallen off, follow the instructions below to restore operation. 1-1-1. Make sure that there is no error with the grip, the holder, the spindle's taper, the pull stud, the inside of the spindle and cover. 1-2. If the tool has not fallen off, follow the instructions below to restore operation. 1-2-1. Place a box or cushion, etc., to protect the tool and machine because a tool can fall off during the recovery operation. 1-2-2. Press the [+Z] key while holding down the [RELSE] key, and raise the Z-axis to a position where the tool can be removed. 1-2-3. Make sure that there is no error with the grip, the holder, the spindle's taper, the pull stud, the inside of the spindle and cover, etc. 1-2-4. Press the [-Z] key while holding down the [RELSE] key to lower the Z-axis. Contact the nearest BROTHER approved service dealer if another cause is identified that is not noted in the cause column. 2. First, restore operation based on the servo error when a servo error is triggered with another alarm or error. In addition, set the <Tool collision detection function during ATC> in the user parameter (switch 1) to <0: Disable>.
SM0568	5	2	[MASTER ON] turned OFF during machine operation	The hardware condition (IO_INVTER) signal for the [MASTER ON] was turned OFF during machine operation.	Press the [RST] key.
SM0571	5	2	Machine param. setting error (SYS 1)	The <Chip auger> was set to <1: Yes> on a machine model not equipped with a chip auger.	Set the machine parameters (System 1) to the correct values.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM0572	5	2	*-axis in-position check timeout	The axis did not enter the in-position width after completing operation.	1. Check the machine parameter (system 3) setting for <AT-axis in-position width> and <AT-axis in-position check timeout period>. 2. Make sure that the axis does not vibrate after it has stopped. <Input/output> <2. Servo controller> position deviation
SM0573	5	2	*-axis position shifted	The position shifted more than the <AT-axis position check pulse>.	Make sure that the machine parameter (system 3) <AT-axis position check pulse> is not too small.
SM0574	5	2	Speed error when door is open (*-axis)	The axis moved at a speed exceeding a certain speed limit while the door was open.	Press the [RST] key.
SM0575	5	2	Side door lock restriction - Machine stop	1. The [DOOR INTERLOCK] switch was changed from "O" (disable) to "I" (enable) when the loading system is attempting to open the side door. 2. While the [ENABLE] switch for the teaching pendant is OFF, the [ENABLE PENDANT] switch was changed from "O" (disable) to "I" (enable) when the loading system is attempting to open the side door.	1. Change the [DOOR INTERLOCK] switch to "O" (disable) and press the [RST] key. 2. Change the [ENABLE PENDANT] switch to "O" (disable) and press the [RST] key.
SM2000	4	2	Return distance too long	The return distance is too large when the servo motor is turned ON. (Note) After the absolute encoder is reset or after the automatic setting is executed for the adjusted absolute values, this alarm may trigger when the servo is first turned ON. This is not an error.	Press the [RST] key.
SM2001	4	2	Spindle command error	1. The next spindle command was sent while still processing the current spindle command. (Excluding orientation command while executing spindle stop command) 2. Turning the arm was attempted before spindle orientation had been completed.	Press the [RST] key to clear the alarm. The NC PCB is defective. If this error occurs repeatedly, contact the nearest BROTHER approved service dealer.
SM2002	4	2	*-axis position shifted	The shift in the position of the 4th-axis exceeded the tolerance during clamping.	1. Reduce the machining load. 2. Check that the setting for [4th-axis position check pulse] (machine parameter/system 2) is not too small.
SM2003	4	2	*-axis position shifted	1. The shift in the motor shaft exceeded the tolerance during 5th-axis clamping. 2. Brake power is too low.	1-1 Reduce the machining load. 1-2 Check that the setting for [5th-axis position check pulse] (machine parameter/system 2) is not too small. 2. Check that the brake is applied correctly.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2004	4	2	*-axis position shifted	1.The shift in the motor shaft exceeded the tolerance during 6th-axis clamping. 2. Brake power is too low.	1-1 Reduce the machining load. 1-2 Check that the setting for [6th-axis position check pulse] (machine parameter/system 2) is not too small. 2. Check that the brake is applied correctly.
SM2005	4	2	*-axis position shifted	1.The shift in the motor shaft exceeded the tolerance during 7th-axis clamping. 2. Brake power is too low.	1-1 Reduce the machining load. 1-2 Check that the setting for [7th-axis position check pulse] (machine parameter/system 2) is not too small. 2. Check that the brake is applied correctly.
SM2006	4	2	*-axis position shifted	1.The shift in the motor shaft exceeded the tolerance during 8th-axis clamping. 2. Brake power is too low.	1-1 Reduce the machining load. 1-2 Check that the setting for [8th-axis position check pulse] (machine parameter/system 2) is not too small. 2. Check that the brake is applied correctly.
SM2007	4	2	Z-axis position error	《Arm type ATC》 1. The Z-axis coordinate made an attempt to issue a Z-axis command when the Z-axis was above its zero point position. 2. A Z-axis was not at the ATC zero point when arm turning is operated. 《Turret type ATC》 The Z-axis command or M498/M499 (tool washing ON/OFF) was used when the Z-axis was above the zero point.	《Arm type ATC》 1. Press the [RST] key. 2. Press the [-Z] key with the [RELSE] key held down to lower the Z-axis below the Z-axis zero point. 《Turret type ATC》 1.Press the [RST] key. 2.Press the [-Z] key with the [RELSE] key held down to lower the Z-axis below the Z-axis zero point.
SM2008	4	2	Spindle rotation position error	《Arm type ATC》 A spindle command was used when the ATC arm was not at zero point. 《Turret type ATC》 Rotating spindle was attempted when the Z-axis was above the zero point.	《Arm type ATC》 1. Press the [RST] key to reset the alarm. 《Turret type ATC》 1. Press the [RST] key to reset the alarm. 2. Press the [-Z] key with the [RELSE] key held down to lower the Z-axis below the Z-axis home position.
SM2009	4	2	Magazine number error	- The magazine was not indexed. - All data was input, or the start data (adjusted absolute values) from the backup memory recovery operation was input.Encoder position data does not match the machine coordinates.	1.When the alarm <<The machine coordinates may be out of position.>> is triggered, perform the expired battery recovery operation. 2. When another cause (besides 1. above) applies, switch to manual operation mode, and press the [RST] key while pressing the [RELSE] key.While holding down the [RELSE] key, press the [MAG.F] key or the [MAG.R] key to move the magazine to the correct position.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2010	4	2	Magazine index error	A tool was not indexed correctly, but rotating the magazine was attempted or a Z-axis command was used.	The ATC cam sensor is OFF. 1.Change to manual mode, and press the [RST] key with the [RELSE] key held down. 2.With the [RELSE] key held down, press the [MAG.F] key or the [MAG.R] key to move the magazine to the correct position.
SM2011	4	2	Magazine position error	Rotating magazine was attempted when the Z axis was not at the ATC zero point.	The Z-axis coordinate is shifted from the ATC zero point coordinate. 1.Press the [RST] key to reset the alarm. 2.Press the [-Z] key with the [RELSE] key held down to lower the Z-axis below the Z-axis zero point.
SM2012	4	2	ATC command error	A number exceeding the number of tools set in the magazine was required by ATC command.	The NC PCB is defective. Contact the nearest BROTHER approved service dealer.
SM2013	4	2	Stroke exceeded (+X)	The axis has exceeded its stroke limit (+X). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 1.)
SM2014	4	2	Stroke exceeded (-X)	The axis has exceeded its stroke limit (+X). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 1.)
SM2015	4	2	Stroke exceeded (+Y)	The axis has exceeded its stroke limit (+Y). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 1.)
SM2016	4	2	Stroke exceeded (-Y)	The axis has exceeded its stroke limit (-Y). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 1.)
SM2017	4	2	Stroke exceeded (+Z)	The axis has exceeded its stroke limit (+Z). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 1.)
SM2018	4	2	Stroke exceeded (-Z)	The axis has exceeded its stroke limit (-Z). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 1.)
SM2019	4	2	Stroke exceeded (+*)	The axis has exceeded its stroke limit (+4). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2020	4	2	Stroke exceeded (-*)	The axis has exceeded its stroke limit (-4). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2021	4	2	Stroke exceeded (+*)	The axis has exceeded its stroke limit (+5). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2022	4	2	Stroke exceeded (-*)	The axis has exceeded its stroke limit (-5). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2023	4	2	Stroke exceeded (+*)	The axis has exceeded its stroke limit (+6). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2024	4	2	Stroke exceeded (-*)	The axis has exceeded its stroke limit (-6). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2025	4	2	Stroke exceeded (+*)	The axis has exceeded its stroke limit (+7). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2026	4	2	Stroke exceeded (-*)	The axis has exceeded its stroke limit (-7). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2027	4	2	Stroke exceeded (+*)	The axis has exceeded its stroke limit (+8). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2028	4	2	Stroke exceeded (-*)	The axis has exceeded its stroke limit (-8). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2029	4	2	Limit exceeded (+X)	1. The axis has exceeded its <stroke limit (+X)> in MDI or memory operation mode. 2. The axis exceeded its <Stroke limit (+X)> while the axis moved to a safe position during a pallet turn. 3. The axis exceeded its <Stroke limit (+X)> while the column moved during a tool change.	1. Press the [RST] key, and then correct the program. (Check the user parameter (switch 2) <Stroke limit 1 (+X)>) 2. Check the user parameter (switch 2) <Stroke limit 1 (+X)> 3. Check the user parameter (switch 1) <X-axis position when changing tool in manual mode>. (Check the user parameter (switch 2) <Stroke limit 1 (+X)>)
SM2030	4	2	Limit exceeded (-X)	1. The axis has exceeded its <stroke limit (-X)> in MDI or memory operation mode. 2. The axis exceeded its <Stroke limit (-X)> while the axis moved to a safe position during a pallet turn. 3. The axis exceeded its <Stroke limit (-X)> while the column moved during a tool change.	1. Press the [RST] key, and then correct the program. (Check the user parameter (switch 2) <Stroke limit 1 (-X)>) 2. Check the user parameter (switch 2) <Stroke limit 1 (-X)> 3. Check the user parameter (switch 1) <X-axis position when changing tool in manual mode>. (Check the user parameter (switch 2) <Stroke limit 1 (-X)>)
SM2031	4	2	Limit exceeded (+Y)	1. The axis has exceeded its <stroke limit> (+Y) in MDI or memory operation mode. 2. The axis exceeded its <Stroke limit (+Y)> while the axis moved to a safe position during a pallet turn. 3. The axis exceeded its <Stroke limit (+Y)> while the column moved during a tool change.	1. Press the [RST] key, and then correct the program. (Check the user parameter (switch 2) <Stroke limit 1 (+Y)>) 2. Check the user parameter (switch 2) <Stroke limit 1 (+Y)> 3. Check the user parameter (switch 1) <Y-axis position when changing tool in manual mode>. (Check the user parameter (switch 2) <Stroke limit 1 (+Y)>)
SM2032	4	2	Limit exceeded (-Y)	1. The axis has exceeded its <stroke limit (-Y)> in MDI or memory operation mode. 2. The axis exceeded its <Stroke limit (-Y)> while the axis moved to a safe position during a pallet turn. 3. The axis exceeded its <Stroke limit (-Y)> while the column moved during a tool change.	1. Press the [RST] key, and then correct the program. (Check the user parameter (switch 2) <Stroke limit 1 (-Y)>) 2. Check the user parameter (switch 2) <Stroke limit 1 (-Y)> 3. Check the user parameter (switch 1) <Y-axis position when changing tool in manual mode>. (Check the user parameter (switch 2) <Stroke limit 1 (-Y)>)
SM2033	4	2	Limit exceeded (+Z)	The axis has exceeded its stroke limit (+Z) in MDI or memory operation mode.	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2034	4	2	Limit exceeded (-Z)	The axis has exceeded its stroke limit (-Z) in MDI or memory operation mode.	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2035	4	2	Limit exceeded (+*)	The axis has exceeded its stroke limit (+4) in MDI or memory operation mode.	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2036	4	2	Limit exceeded (-*)	The axis has exceeded its stroke limit (-4) in MDI or memory operation mode.	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2037	4	2	Limit exceeded (+*)	The axis has exceeded its stroke limit (+5) in MDI or memory operation mode.	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2038	4	2	Limit exceeded (-*)	The axis has exceeded its stroke limit (-5) in MDI or memory operation mode.	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2039	4	2	Limit exceeded (+*)	The axis has exceeded its stroke limit (+6) in MDI or memory operation mode.	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2040	4	2	Limit exceeded (-*)	The axis has exceeded its stroke limit (-6) in MDI or memory operation mode.	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2041	4	2	Limit exceeded (+*)	The axis has exceeded its stroke limit (+7) in MDI or memory operation mode.	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2042	4	2	Limit exceeded (-*)	The axis has exceeded its stroke limit (-7) in MDI or memory operation mode.	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2043	4	2	Limit exceeded (+*)	The axis has exceeded its stroke limit (+8) in MDI or memory operation mode.	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2044	4	2	Limit exceeded (-*)	The axis has exceeded its stroke limit (-8) in MDI or memory operation mode.	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2045	4	2	Stroke exceeded 2	The axis has entered stroke 2 (restricted travel range). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 1 and machine parameter/system 2.)
SM2046	4	2	Limit exceeded 2	The axis has entered stroke limit 2 (not allowed travel range) in MDI or memory operation mode.	Press the [RST] key, and then correct the program. (Check the machine parameter/system 2.)
SM2047	4	2	Thread cutting speed error	1. Z-axis speed could not obtained from X- and Y-axes feedrates and radius in helical threading operation. 2. The feedrate of helical linear axis exceeded the limit when helical command was specified.	1. Check the end point, center point, and feedrate of threading operation after pressing the [RST] key. 2. Check the helical command after pressing the [RST] key.
SM2048	4	2	No thread cutting function	Helical thread cutting was attempted on a machine not equipped with this option.	Purchase the option.
SM2049	4	2	Calculation error	1.Coordinates cannot be calculated where specified. 2.Measurement result could not be calculated in automatic workpiece measurement mode.	Press the [RST] key, and the check the program.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2050	4	2	Tool breakage error	Tool breakage was detected.	the <Magazine tool> screen. Press the [RST] key. If a spare tool is registered, operation can be resumed. If a spare tool is not registered, replace the tool, set the tool length and life, and then resume operation. * When the [RST] key is pressed, tool life is automatically cleared to zero (for countdown method) or to "999999" (for count up method). * If the tool is not broken, the tool breakage detector needs adjustment.
SM2051	4	2	Large error in measured value (1)	The measurement result of automatic workpiece measurement has exceeded [Measurement tolerance 1] of [Automatic workpiece measurement] (user parameter).	Press the [RST] key, and then check the measurement position in the program.
SM2052	4	2	Large error in measured value (2)	The difference in the result between the current and previous automatic workpiece measurements has exceeded [Measurement tolerance 2] of [Automatic workpiece measurement] (user parameter).	Press the [RST] key, and then check the measurement position in the program.
SM2053	4	2	No measurement data	There is no measurement data although a command that uses measurement data (G10, G120, G168) was specified.	Press the [RST] key, and the check the program.
SM2054	4	2	No tool in magazine	The tool required for operation is not registered in the magazine.	Register the tool in the magazine.
SM2055	4	2	Capped	Changing the current tool with one in a capped pot was attempted, or a tool removal command that results in a capped pot moving to the spindle position was used.	Perform ATC with a pot which is not cap-designated.
SM2056	4	2	No A-axis option	Running an A-axis program was attempted although the machine is not equipped with the A-axis option.	
SM2057	4	2	No B-axis option	Running an B-axis program was attempted although the machine is not equipped with the B-axis option.	
SM2058	4	2	No C-axis option	Running an C-axis program was attempted although the machine is not equipped with the C-axis option.	
SM2059	4	2	No empty pot	There is no pot available when changing the tool from Standard to Large, or vice versa.	Prepare an empty pot.
SM2060	4	2	Invalid pot	An invalid pot was specified when performing ATC.	Perform ATC with a pot other than invalid pots.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2061	4	2	Pallet position error	Tool breakage detection (Type 3) was executed before indexing the target pallet inside had been completed.	Execute Tool breakage detection (Type 3) after indexing the target pallet inside has been completed.
SM2062	4	2	ATC tool setting error	ATC tool setting is not correct.	Set the ATC tool correctly.
SM2063	4	2	Outer pallet additional axis cannot be specified.	Activating the additional axis of the pallet opposite the operation pallet was attempted.	Press the [RST] key, and then check the program.
SM2064	4	2	No auto workpc. measurement func.	Automatic workpiece measurement was attempted although the machine is not equipped with this option.	Purchase the automatic workpiece measurement option or change the program.
SM2065	4	2	End mill tapping speed error	End milling speed is faster than tapping speed when an end milling/tapping command is used.	Press the [RST] key, and then check the program.
SM2066	4	2	No const. speed area for end milling	There is no constant speed range in the end milling section when an end milling/tapping command is used.	Press the [RST] key, and then check the program.
SM2067	4	2	No const. speed area for tapping	There is no constant speed range in the tapping section when an end milling/tapping command is used.	Press the [RST] key, and then check the program.
SM2068	4	2	End mill tapping	A normal spindle command was used during end milling and tapping.	Press the [RST] key to clear the alarm.
SM2069	4	2	XY-axes position error	ATC, synchronized tap return, and inching were performed when the X- or Y-axis was out of the stroke.	1.Press the [RST] key to reset the alarm. 2.Return the XY axes to within their stroke limits.
SM2070	4	2	X-axis position error	Operating the machine was attempted when the X axis was within the ATC operation range.	1.Press the [RST] key to reset the alarm. 2.Return the X axes to within their stroke limits.
SM2071	4	2	Tapping command not possible during spindle rotation	An end milling/tapping command was used during normal spindle rotation.	Press the [RST] key to clear the alarm.
SM2072	4	2	Excessive thermal distortion	Automatic thermal distortion compensation has exceeded the limit (30 mm).	See the parameter setting section in the Instruction Manual, and change the parameter setting so that automatic thermal distortion compensation is performed under the appropriate conditions.
SM2073	4	2	No auto thermal compen. func.(X)	Automatic X-axis thermal compensation was attempted although the machine is not equipped with this option.	Correct the program.
SM2074	4	2	No auto thermal compen. func.(Y)	Automatic Y-axis thermal compensation was attempted although the machine is not equipped with this option.	Correct the program.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2075	4	2	No auto thermal compen. func.(Z)	Automatic Z-axis thermal compensation was attempted although the machine is not equipped with this option.	Correct the program.
SM2076	4	2	Compensating diameter	The specified G or M code cannot be used during cutter compensation.	Correct the program.
SM2077	4	2	During rotational transformation	The specified G code cannot be used during rotational transformation.	Correct the program.
SM2078	4	2	Cutter compensation error	1. Arc command or circular cutting command was used after cutter compensation was started or canceled. 2. The current position was specified when cutter compensation was started in conversation mode. 3. ATC command and cutter compensation command were specified simultaneously with G18 and G19 modal. 4. The thread cutting was specified and the cutter compensation was cancelled.	Press the [RST] key, and then check the program.
SM2079	4	2	Cutter compensation too large	Program radius is smaller than the cutter compensation when cutting an inner arc.	Press the [RST] key, and then check the program.
SM2080	4	2	No R or Z command	There is no R or Z point in the canned cycle.	Press the [RST] key, and then set the R or Z point.
SM2081	4	2	No pitch data	NC: Pitches or threads are not specified although tapping is specified. Conversation: The pitch data is not set in the tool list when tapping.	Press the [RST] key, and then set the pitch data.
SM2082	4	2	Program restart error	1. Program restart was attempted when positioning was not completed. After automatic workpiece measurement or skip command is specified, program restart is impossible during positioning command (G90G0XYZ, etc.) 2. G52/G92 (coordinate system setting), scaling, and rotational transformation were performed when positioning was not completed during program restart. 3. Program restart was attempted in a block with smooth path offset function valid. 4. The specified sequence number is not found. 5. Program restart was attempted while the lathe spindle is rotating.	1. Restart the program after a positioning command which is specified before automatic workpiece measurement or skip command or after automatic workpiece measurement or skip command. 2. Restart the program before automatic workpiece measurement or skip command. 3. Restart the program in a block with smooth path offset function invalid. 4. Check the specified sequence number. 5. Restart the program after stopping the lathe spindle.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2083	4	2	Canned cycle address error	1.The address of the canned cycle was not specified. 2.The value specified for the canned cycle is not correct.	Press the [RST] key, and then check the program.
SM2084	4	2	Auto workpc. meas. address error	1.The address necessary for automatic workpiece measurement was not specified. 2.The specified addresses cannot be accessed.	Press the [RST] key, and then check the program.
SM2085	4	2	Macro command error	1. The macro command format is not correct. 2. A value has been substituted which exceeds the macro variable range. 3. An attempt was made to execute the macro command (WHILE) during extended memory operation or tape operation. 4. An attempt was made to execute external output during MDI operation or tape operation.	Press the [RST] key, and then check the program.
SM2086	4	2	Center through coolant on	M496 was used while CTS was on.	Press the [RST] key, and then correct the program.
SM2087	4	2	Tool length range error	Tool length is not set correctly for [Tool breakage detection option] (type 2). This error occurs when the length of the tool subject to breakage detection is as below. 1.(Tool length – Tool breakage detection offset) < (Breakage detection reference position) 2. (Tool length – Tool breakage detection offset) > (Breakage detection reference position + Breakage motor stroke)	Check the parameter settings. <User parameter> [Tool breakage detection option] [Tool breakage detection offset] <Tool list> [Tool length offset]
SM2088	4	2	Magazine index command error	Value of [Manual tool change position offset] (machine parameter/system 1) is faulty.	Set [Manual tool change position offset] (machine parameter/system 1) to a value which is the same or less number of tools to be stored in the magazine.
SM2089	4	2	Spindle tool cannot be specified	The tool specified by the M5xx command is attached to the spindle.	Return the spindle tool to the magazine, and then use the M5xx command.
SM2090	4	2	Unregistered miscellaneous function	1.Executing a non-registered miscellaneous function was attempted. 2.Executing a miscellaneous function not permitted for MDI operation was attempted.	Press the [RST] key, and then correct the miscellaneous function code.
SM2091	4	2	Invalid axis specified	The axis specified in the program is not correct.	Check the program.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2092	4	2	Door open	1. The door was open when ATC arm was attempted to be turned. 2. The door was open immediately after memory operation was started.	Close the door, and then press the [RST] key. 1. Check the door limit switch. 2. Make sure that there are no chips or shavings caught between the door and the machine cover. 3. The door sensor is faulty. Replace the door sensor. 4. The IL PCB is faulty. Replace the IL PCB. 5. The IO PCB is faulty. Replace the IO PCB.
SM2093	4	2	Cannot measure in incremental mode	Automatic workpiece measurement was attempted in incremental mode.	Change to absolute mode, and then perform automatic workpiece measurement.
SM2094	4	2	External program number error	Memory operation was executed when the external program selection signal was No.10000 or later.	Set the external program selection signal to No. 9999 or earlier.
SM2095	4	2	Arc radius error has exceeded the limit.	Radius error of the arc, helical thread cutting, cone or spiral has exceeded the user parameter setting.	1. Press the [RST] key, and then correct the program. 2. Press the [RST] key, and then correct the user parameter setting.
SM2096	4	2	Feedrate not specified	Cutting command was specified but the feedrate is not set.	Set feedrate before specifying cutting command.
SM2097	4	2	Wrong tool diameter	The diameter of the spare tool differs from that of the main tool.	Specify the main tool correctly on the <Magazine> screen, or check the tool data.
SM2098	4	2	Start position cannot be set	1. Start point setting was attempted when positioning was not completed. Start point cannot be set from when measurement operation is performed to when positioning command is specified. 2. Start point is set to a process with smooth path offset function invalid.	1. Set start point after positioning command which is specified before or after measurement operation. 2. Set start point to a process with smooth path offset function invalid.
SM2099	4	2	Memory operation mode	G17 to G19 was executed via MDI intervention during memory operation.	Press the [RST] key.
SM2100	4	2	Unable to prepare next tool	Conversation language machining data for arm type ATC models was input to operate a turret type ATC model.	Assign the tools again.
SM2101	4	2	Press [M.LCK] to disable machine lock.	Changing tool by specifying the pot number (M701 ~ M749) was attempted with machine lock ON in memory operation mode (conversation language program).	Disable machine lock. (Changing tool by specifying the pot number is not possible during machine lock.)

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2102	4	2	Pallet turn restricted range error	1. A pallet turn operation was attempted when the X-/Y-axis was in the pallet turn restricted range. 2. A travel operation was attempted that would move the X-/Y-axis into the pallet turn restricted range while the pallet was not indexed. 3. When the user parameter (quick table) <Pallet 1 loading> or <Pallet 2 loading> is set to <At 1st tool change>, the R-point position for the first tool change command (G100/M06) is issued at a position lower than the machine parameter (system 1) <R-point lower limit for pallet loading during tool change>.	1-1. Press the [RST] key to reset the alarm. 1-2. Check the program. *Check the value for [Pallet turn restricted distance (Y axis)] (machine parameter / system 1), and check the values for the X- and Y-axes to ensure that they are in a safe position when turning the pallet. 1-3. Move the X-/Y-axis in manual mode to a point outside of the pallet turn restricted range. 2-1. Press the [RST] key to reset the alarm. 2-2. Check the program. *Check the value for [Pallet turn restricted distance (Y axis)] (machine parameter / system 1), and check the values for the X- and Y-axes to ensure that they are in a safe position when turning the pallet. 2-3. Index the pallet in manual mode. 3-1. Press the [RST] key and correct the program. 3-2. Press the [RST] key, and change the user parameter (switch 1) setting for <ATC simultaneous operation start position> or <ATC reference tool length>. 3-3. Press the [RST] key, and change the setting to <0:At operation start> for the user parameter (quick table) <Pallet 1 loading> or <Pallet 2 loading>.
SM2103	4	2	ATC arm turn restricted range error	Arm turn was attempted when the X-axis was in ATC arm turn restricted range. (Except when the X-, Y-, and Z-axes are at the ATC zero point.)	1. Press the [RST] key to reset the alarm. 2. Press [-X] while holding down the [RELSE] key to move the X-axis to the position of [ATC arm turn restricted distance (X axis)] (machine parameter/system 1) or farther away from X-axis ATC position.
SM2104	4	2	Servo power off	1. The [EMERGENCY] switch was pressed during machine operation. 2. A door was opened during machine operation while the [DOOR INTERLOCK] switch was enabled ("I").	1. Release the [EMERGENCY STOP] switch, and then press the [RST] key. 2. Close the door, and press the [RST] key.
SM2105	4	2	Pot position error	Pot position is incorrect.	1. Switch to manual operation mode press the [RST] key with the [RELSE] key held down. 2. Then, press the [ATC] key with the [RELSE] key held down to move the magazine pot to the correct position.
SM2106	4	2	Editing	Program contents were changed when the program was loaded.	Check that the "During communication" mark is not displayed, and then press the [RST] key.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2107	4	2	Magazine position shifted	The position of the magazine has shifted during stop.	Check the machine parameter (SYSTEM 3). [Magazine position check pulse]
SM2108	4	2	ATC return position error	When the ATC arm stopped midway, the X-, Y- or Z-axis, pot position or pot shutter position is not in a position where the ATC arm can turn.	Recovery is not possible in ATC arm position recovery mode. Use ATC maintenance mode to restore. Only maintenance personnel who are trained can operate in ATC maintenance mode.
SM2109	4	2	Additional axis operation prohibited range error	Additional axis turn was attempted in manual operation mode when the Z-axis was in the additional axis operation restricted range.	Move the Z-axis out of the additional axis operation restricted range after pressing the [RST] key.
SM2110	4	2	Tool type error	When spare tool is used, the type is different from main tool.	Specify main tool correctly on the magazine screen or check tool data.
SM2111	4	2	Tool data spindle rotational frequency error	Spindle rotation was attempted at a speed of exceeding the maximum spindle speed registered in tool data.	Press the [RST] key, and the check the program.
SM2112	4	2	The tapping synchronous error is too big.	Tolerance during tapping operation exceeded [Synchronized tapping error limit] (machine parameter/system 1). 2. Invalid tapping command was issued.	1-1. Check the value for [Synchronized tapping error limit] (machine parameter/system 1). 1-2. Check whether the machining conditions are within the capacity. Press the [RST] key to recover. 2. Check the program.
SM2113	4	2	High-accuracy B execution error	Dry run was turned off just before high accuracy mode B function and high accuracy mode B function became valid with offset amount still remaining.	Start operation without dry run offset remaining after pressing the [RST] key.
SM2114	4	2	An external output command is being executed.	The mode was changed during data output via external output command.	Press the [RST] key. Change the mode after data output via external output command is completed.
SM2115	4	2	The high-accuracy B is being executed.	Ladder switched dry run while high accuracy B function was valid.	The operation described in the left cannot be executed during high accuracy B function. Execute after high accuracy B operation is completed.
SM2116	4	2	Therm. displacement offset parameter setting error	Incorrect value is set for parameters related to automatic thermal distortion compensation.	Check the settings for parameters related to automatic thermal distortion compensation.
SM2117	4	2	Time is out before a swivel of the ATC arm.	1. ATC arm couldn't move within the time set to ATC arm pre-turn monitor time. 2. The ATC zero point sensor is not turned off.	1-1. Check the value of ATC arm pre-turn monitor time. 1-2. The inverter is faulty. 1-3. The ATC motor is faulty. 1-4. The machine is faulty. 2. The ATC zero point sensor is faulty.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2118	4	2	Spline function execution error	1. Intervention of manual operation was performed just before the smooth path offset function started, and operation was started with intervention amount remaining on the X-, Y-, and Z-axes. 2. Dry run was turned off just before smooth path offset function and smooth path offset function became valid with offset amount still remaining.	1. Start operation without intervening manual operation after pressing the [RST] key. 2. Start operation without dry run offset remaining after pressing the [RST] key.
SM2119	4	2	Feedrate error	1. F command was not specified during cutting feed per rotation or was specified with the limit value exceeded. 2. The spindle does not rotate during cutting feed per rotation. 3. A command for thread cutting was issued when the lathe spindle was not rotating. 4. The speed in the cutting direction in the thread cutting cycle exceeded the limit value.	1. Start operation without intervening manual operation after pressing the [RST] key. 2. Start operation without dry run offset remaining after pressing the [RST] key. 3-4. Press the [RST] key and then check the program.
SM2120	4	2	Max. speed exceeded	The commanded speed per rotation exceeds maximum actual cutting travel speed.	Press the [RST] key, and then check the program or set [Max. actual cutting travel speed] (user parameter/switch 1).
SM2121	4	2	Measurement impossible because alarm occurring	An alarm (stop level 5) occurs when motor insulation resistance measurement is started.	Execute measurement after resetting the alarm.
SM2122	4	2	Axis travel command error	A command was specified to a traveling axis.	NC PCB is defective. Replace NC PCB. Contact the nearest BROTHER approved service dealer.
SM2123	4	2	Spindle speed error	S command was not specified when spindle rotation was executed or was specified with the limit value exceeded.	Press the [RST] key, and then check the program.
SM2124	4	2	Carry out spindle run-in in manual operation mode.	Immediately after machine installation or long-time recess, the spindle run-in operation must be executed.	Press the [RST] key. Change to the manual mode, and execute the spindle run-in operation.
SM2125	4	2	The spindle running-in has not been completed yet.	Spindle rotation command was specified without spindle running-in completed.	Press the [RST] key. Change to the manual mode, and execute the spindle run-in operation.
SM2126	4	2	No tool is mounted in the spindle.	1. There was no tool at tool clamping in tool change operation. 2. Tool fell at tool clamping in tool change operation.	Insert a tool in the pot and perform ATC after pressing the [RST] key.
SM2127	4	2	Check the runout of the spindle tool.	Tool was mounted with runout at tool clamping in tool change operation.	Recover in accordance with "Recovery of spindle tool runout" in the instruction manual.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2128	4	2	No interrupt type macro function	An attempt was made to use the interrupt type macro function even though the machine is not equipped with that function option.	Purchase the option.
SM2129	4	2	OPSEL mounted	OPSEL is mounted.	Please contact us if you encounter a situation.
SM2130	4	2	The high-accuracy mode overloads the X axis.	Current command exceeded the acceptable value during high-accuracy mode.	Decrease the cutting load and operation patterns.
SM2131	4	2	The high-accuracy mode overloads the Y axis.	Current command exceeded the acceptable value during high-accuracy mode.	Decrease the cutting load and operation patterns.
SM2132	4	2	The high-accuracy mode overloads the Z axis.	Current command exceeded the acceptable value during high-accuracy mode.	Decrease the cutting load and operation patterns.
SM2133	4	2	Stroke exceeded 3	The axis has entered stroke 3 (restricted travel range). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 1 and machine parameter/system 2.)
SM2134	4	2	Stroke exceeded 4	The axis has entered stroke 4 (restricted travel range). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 1 and machine parameter/system 2.)
SM2135	4	2	Stroke exceeded 5	The axis has entered stroke 5 (restricted travel range). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 1 and machine parameter/system 2.)
SM2136	4	2	Spindle rotating	After the spindle rotates in manual mode, a constant peripheral speed control command was issued upon restarting the program during MDI operation.	Press the reset key, and issue a peripheral speed control command after the spindle stops.
SM2137	4	2	*-axis overload in high accuracy mode	The current command value in high accuracy mode exceeds the tolerance.	Lower the operation pattern and cutting conditions to operate.
SM2138	4	2	*-axis overload in high accuracy mode	The current command value in high accuracy mode exceeds the tolerance.	Lower the operation pattern and cutting conditions to operate.
SM2139	4	2	*-axis overload in high accuracy mode	The current command value in high accuracy mode exceeds the tolerance.	Lower the operation pattern and cutting conditions to operate.
SM2140	4	2	*-axis overload in high accuracy mode	The current command value in high accuracy mode exceeds the tolerance.	Lower the operation pattern and cutting conditions to operate.
SM2141	4	2	*-axis overload in high accuracy mode	The current command value in high accuracy mode exceeds the tolerance.	Lower the operation pattern and cutting conditions to operate.
SM2142	4	2	Excessive thermal distortion on X-axis	The automatic thermal distortion compensation value has exceeded the compensation limit.	1. Refer to the parameter setting section in the Operation Manual, and change the parameter setting so that automatic thermal distortion compensation is performed under the appropriate conditions. 2. Review the operation conditions (is X-axis travel often too extreme). 3. Check the load on the load meter while the X-axis is stopped, and make sure that there is no interference.

Chapter 11 Alarms

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2143	4	2	Excessive thermal distortion on Y-axis	The automatic thermal distortion compensation value has exceeded the compensation limit.	1. Refer to the parameter setting section in the Operation Manual, and change the parameter setting so that automatic thermal distortion compensation is performed under the appropriate conditions. 2. Review the operation conditions (is Y-axis travel often too extreme). 3. Check the load on the load meter while the Y-axis is stopped, and make sure that there is no interference.
SM2144	4	2	Excessive thermal distortion on Z-axis	The automatic thermal distortion compensation value has exceeded the compensation limit.	1. Refer to the parameter setting section in the Operation Manual, and change the parameter setting so that automatic thermal distortion compensation is performed under the appropriate conditions. 2. Review the operation conditions (is Z-axis travel often too extreme). 3. Check the load on the load meter while the Z-axis is stopped, and make sure that there is no interference.
SM2145	4	2	Thread cutting execution error	The dry run offset caused the travel distance to change to 0 in the cutting direction for taper thread cutting or the finishing operation.	Press the [RST] key.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2146	4	2	Special setting error (Rapid feed adjustment)	1. When <Adjustment> was set for the <Rapid feed special setting selection>, the following set values for the <Adjustment> parameter on the special selection (rapid feed adjustment) screen are greater than the set values for the <Standard> setting. -<Z-axis rapid traverse rate (when ATC rises 2)> -<Z-axis rapid traverse rate (when ATC falls 2)> -<*-axis rapid feed constant FFGN1 > -<Z-axis rapid feed constant FFGN3> 2. When <Adjustment> was set for the <Rapid feed special setting selection>, the following set values for the <Adjustment> parameter on the special selection (rapid feed adjustment) screen are less than the set values for the <Standard> setting. -<*-axis rapid feed constant TVI1> -<*-axis rapid feed constant TVI2> -<*-axis lower limit time constant> 3. When <Adjustment> was set for the <Rapid feed special setting selection>, the <Adjustment> setting for the rapid feed acceleration is greater than the <Standard> setting. 4. The value that is set is different for <Z-ax.rpd. control type 1 > and <Z-ax.rpd. control type 2>.	1. Change the set value for the <Adjustment> parameter to the correct value, and press the [RST] key. 2. If the adjustment is not carried out, set the <Rapid feed special setting selection> to the <Standard> setting.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2147	4	2	Special setting error (Rapid feed adj. - Lin. interp. type)	1. When <Adjustment> was set for the <Rapid feed special setting selection>, the following set values for the <Adjustment> parameter on the special selection (rapid feed adjustment - linear interpolation type) screen are greater than the set values for the <Standard> setting. -<Z-axis rapid traverse rate (when ATC rises 2)> -<Z-axis rapid traverse rate (when ATC falls 2)> -<*-axis rapid feed constant FFGN5 > -<Z-axis rapid feed constant FFGN6> 2. When <Adjustment> was set for the <Rapid feed special setting selection>, the following set values for the <Adjustment> parameter on the special selection (rapid feed adjustment 2) screen are less than the set values for the <Standard> setting. -<*-axis rapid feed constant TVI6> -<*-axis rapid feed constant TVI7> -<*-axis lower limit time constant> 3. When <Adjustment> was set for the <Rapid feed special setting selection>, the <Adjustment> setting for the rapid feed acceleration is greater than the <Standard> setting. 4. The value that is set is different for <Z-axis rapid feed control type 3 > and <Z-axis rapid feed control type 4>.	1. Change the set value for the <Adjustment> parameter to the correct value, and press the [RST] key. 2. If the adjustment is not carried out, set the <Rapid feed special setting selection> to the <Standard> setting.
SM2148	4	2	Positioning command error for linear interpolation type	When the user parameter (switch 1) <Positioning method> is set to <Linear interpolation type - Positioning method 1>, a command was issued which exceeds the number of additional axes that are available for simultaneous linear interpolation commands on the machine.	Press the [RST] key, and the check the program.
SM2149	4	2	There is no 3D smoothing function.	3D smoothing start/end code was executed even though there is no 3D smoothing option.	Purchase the option.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2150	4	2	3D smoothing start position error	The coordinate used to execute the 3D smoothing start code is different from the <Start position X/Y/Z> for the 3D smoothing parameter.	Set the appropriate value in the <Automatic start position settings> or set directly in <Start position X/Y/Z>.
SM2151	4	2	3D data not smoothed	3D smoothing start code was executed while running a program with 3D data that is not smoothed.	Execute the 3D smoothing data operation.
SM2152	4	2	Program restart error from saved position	The return height operation could not be carried out to the saved position due to the operation status information. Example when return height operation is not possible - Call level of the sub program was changed - Call number for the call source in the sub program was changed - Saved block can no longer execute because macro variable value was changed	Press the [RST] key and perform one of the following. - Correct the program. - Specify the execution pointer, or restart the program that specifies the sequence number.
SM2153	4	2	Pitch data error	The pitch during a tapping operation was less than the <Minimum tapping pitch> in the machine parameter (system 1).	Press the [RST] key and check the program and pitch data.
SM2154	4	2	No spindle tool during ATC (when ATC rises)	1. There was no tool at tool unclamping in the tool change operation. 2. This alarm was triggered at the same time due to a servo error.	1-1. Make sure that a tool has not fallen off into the machine or stuck into a workpiece. 1-2. If a tool has fallen off, make sure that there is no error with the holder, the spindle's taper, the pull stud, the inside of the spindle and cover, etc. 2. First, restore operation based on the servo error when a servo error is triggered with another alarm or error. Contact the nearest BROTHER approved service dealer if another cause is identified that is not noted in the cause column. To continue to use the machine, set the <No tool detection function (when ATC rises)> in the user parameter (switch 1) to <0: Disable>.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2155	4	2	Spindle tool misaligned during ATC	1. Tool was mounted over the spindle drive key at tool clamping in tool change operation. 2. Changes in the spindle's clamp mechanism due to aging may cause misdetection. 3. This alarm was triggered at the same time due to a servo error, but it may have been a misdetection when the alarm was triggered.	1-1. Check whether the end of the tool holder is over the spindle drive key. 1-2. Place a box or cushion, etc., underneath to protect the tool and machine because a tool can fall off during the recovery operation. 1-3. Press the [RST] key. 1-4. Press the [ATC] key to move to the ATC zero point. 1-5. Make sure that there is no error with the holder, the spindle's taper, the pull stud, the inside of the spindle and cover, etc. 2. Please contact BROTHER. To continue to use the machine, set the <Tool over key detection function> in the user parameter (switch 1) to <0: Disable>. 3. First, restore operation based on the servo error when a servo error is triggered with another alarm or error. Contact the nearest BROTHER approved service dealer if another cause is identified that is not noted in the cause column. To continue to use the machine, set the <No tool detection function (when ATC rises)> in the user parameter (switch 1) to <0: Disable>.
SM2156	4	2	Spindle tool missing	1. The tool came off the spindle. 2. This alarm was triggered at the same time due to a servo error.	1-1-1. Make sure that a tool has not fallen off into the machine or stuck into a workpiece. 1-1-2. Make sure that there is no error with the holder, the spindle's taper, the pull stud, the inside of the spindle and cover, etc. 1-2. Under certain machining conditions, the tool might be misdetected as missing due to an external force being applied. In this situation, set the <Tool missing detection function> in the user parameter (switch 1) to <0: Disable> and use. 2. First, restore operation based on the servo error when a servo error is triggered with another alarm or error. Contact the nearest BROTHER approved service dealer if another cause is identified that is not noted in the cause column. To continue to use the machine, set the <Tool missing detection function> in the user parameter (switch 1) to <0: Disable>.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2172	4	2	No spindle tool during ATC (when ATC falls)	1. There was no tool at tool clamping in the tool change operation. 2. Tool fell at tool clamping in the tool change operation. 3. This alarm was triggered at the same time due to a servo error, but it may have been a misdetection when the alarm was triggered.	1. Make sure that the tool registered in the ATC tool (Details) screen is fitted onto the magazine. 2-1. Make sure that a tool has not fallen off into the machine or stuck into a workpiece. 2-2. If a tool has fallen off, make sure that there is no error with the holder, the spindle's taper, the pull stud, the inside of the spindle and cover, etc. 3. First, restore operation based on the servo error when a servo error is triggered with another alarm or error. Contact the nearest BROTHER approved service dealer if another cause is identified that is not noted in the cause column. To continue to use the machine, set the <No tool detection function (when ATC rises)> in the user parameter (switch 1) to <0: Disable>.
SM2194	4	2	User param. setting error (switch 1)	The following cause descriptions are divided according to their auxiliary numbers: 1. <Probe detection signal method> is set to <1: Method 2>. <1: Method 2> cannot be set on the machine being used. Versions before IO-4REV.P do not support this.	1. Set the <Probe detection signal method> to <0: Method 1>, and press the [RST] key.
SM2195	4	2	ATC monitoring parameter has not been set correctly	The automatic setting for the ATC monitoring parameter has not been executed. Or, the ATC monitoring setting is not correct.	Contact BROTHER.
SM2197	4	2	Side door lock restricted	An operation was performed to lock (such as from the [DOOR INTERLOCK] switch being enabled (“ ”)) the side door when the loading system is attempting to open the side door.	Press the [RST] key.
SM2198	4	2	Machining load maximum(*)	Machining load exceeded the <Max. machining load> in the monitoring section for the tool indicated by the auxiliary number.	The tool may be worn and/or chipped. Press the [RST] key and perform one of the following. - Replace with a new tool. - Make sure that the machining load monitor parameter is correctly set.
SM2199	4	2	Machining load monitor parameter setting error (common)	The following cause descriptions are divided according to their auxiliary numbers: 1. <Machining load monitor function> is set to <1: Enable>. The machining load monitor function is not available on the machine being used.	Change the parameter to the correct value, and press the [RST] key.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM2200	4	2	Stroke exceeded 6	The axis has entered stroke 6 (restricted travel range). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 1 and machine parameter/system 2.)
SM2201	4	2	Stroke exceeded 7	The axis has entered stroke 7 (restricted travel range). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 1 and machine parameter/system 2.)
SM2202	4	2	Stroke exceeded 8	The axis has entered stroke 8 (restricted travel range). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 1 and machine parameter/system 2.)
SM2203	4	2	Stroke exceeded 9	The axis has entered stroke 9 (restricted travel range). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 1 and machine parameter/system 2.)
SM2204	4	2	Stroke exceeded 10	The axis has entered stroke 10 (restricted travel range). (Excluding manual reset operation.)	Press the [RST] key, and then correct the program. (Check the machine parameter/system 1 and machine parameter/system 2.)
SM2205	4	2	Involute interpolation error limit exceeded	There is no end point on the involute curve that passes through the start point in the involute interpolation command. The margin of error, between the involute curve that passes through the start point and the end position on the involute curve that passes through the end point, is greater than the set value for the "Involute interpolation error limit" in the user parameter (switch 1).	1. Press the [RST] key and check the program. 2. Press the [RST] key, and then correct the user parameter value for "Involute interpolation error limit" in the user parameter (switch 1).
SM4000	3	2	Spindle speed error	An S-command was not used when rotating the spindle or the spindle speed limit was exceeded.	Press the [RST] key, and the check the program.
SM4001	3	2	Specified G code cannot be used.	1. G code command which cannot be used was specified. 2. G code command which is prohibited in MDI was specified.	Press the [RST] key, and the check the program.
SM4002	3	2	Invalid command	G/M code command which cannot be used simultaneously in the same block was specified.	Press the [RST] key, and the check the program.
SM4003	3	2	Specified M code cannot be used.	1.The specified M-code cannot be used. 2.The specified M-code is not allowed in MDI mode. 3.There are multiple G-codes in the same block, although they cannot be used simultaneously. 4.There are multiple M-codes in the same block, although they cannot be used simultaneously.	Press the [RST] key, and the check the program.
SM4004	3	2	Invalid character command	The specified characters cannot be used for NC programs.	Press the [RST] key, and the check the program.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM4005	3	2	Same code cannot be used.	Two or more M codes are used in the same block.	Check the program after pressing the [RST] key. To specify two or more M codes in the same block, set [Multiple M codes in one block] (user parameter/switch 1) to [1:Yes]. Then up to three codes can be specified.
SM4006	3	2	Invalid data	The specified format is not correct.	Press the [RST] key, and the check the program.
SM4007	3	2	Command data range error	1. The specified data exceeds the address range. 2. An incremental command on the additional axis for positioning was issued that exceeds 800 degrees.	Press the [RST] key, and the check the program.
SM4008	3	2	Characters per block exceeded	The number of characters in one block exceeds 128.	Press the [RST] key, and the check the program.
SM4009	3	2	No end of program code	M02 or M30 is not included in the program.	Press the [RST] key, and the check the program.
SM4010	3	2	Arc command error	1. There is no radius command when using the arc command. 2. Commands were used incorrectly. (e.g. I, J, or K command and R command are used simultaneously.) 3. Radius of both start point and end point is 0. 4. There is no I or D command when circular cutting command is used, or I command was set to 0. 5. R command was used when spiral or cone command was used. 6. Positive and negative of Q command value and the amount of radius change reverse when spiral or cone command is used. 7. Positive and negative of the amount of change per rotation of linear axis in IJK command and moving direction of linear axis reverse when spiral or cone command is used. 8. The tool length offset or the tool position compensation command is issued when using the circular interpolation modal.	Press the [RST] key, and the check the program.
SM4011	3	2	Subprogram call error	1. Nesting of subprogram call exceeded the specified value. 2. A program which was called from a called subprogram was called. 3. A sequence number which is the same as the called sequence number was called repeatedly. 4. The subprogram file failed to open.	1. Press the [RST] key, and then check the program. 2. Check the cause of the alarm that occurred at the same time.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SM4012	3	2	Subprogram number error	1. The subprogram number is not written in the program correctly. 2. The sequence number to be called is not written.	Press the [RST] key, and then check the subprogram number or the sequence number.
SM4013	3	2	Modal subprogram return error	1. G67 was used although the current mode is not macro modal call mode. 2. Executing G67 was attempted while calling modal sub program.	Press the [RST] key, and the check the program.
SM4014	3	2	Feedrate error	1. F command was not specified during cutting feed or was specified with the limit value exceeded. 2. The feedrate of helical linear axis exceeded the limit when helical interpolation command was specified. 3. Speed command was not specified when synchronized tap return was executed or was specified with the limit value exceeded. 4. A command was executed where the screw pitch was calculated as 0 (cutting feedrate / spindle speed), when G74 or G84 was executed for the feed rate per minute.	1. Check the program after pressing the [RST] key. 2. Check the helical interpolation command after pressing the [RST] key. 3. Check the speed command after pressing the [RST] key. 4. After pressing the [RST] key, check the feedrate command or the speed command.
SM4015	3	2	Coordinate calc. address error	The address necessary for coordinate calculation is not specified.	Press the [RST] key, and the check the program.
SM4016	3	2	Arc mode	The specified G or M code is not allowed in arc mode.	Press the [RST] key, and the check the program.
SM4017	3	2	No T-code command	A T-command is not used before G100 or M06 is executed.	Press the [RST] key, and the check the program.
SM4018	3	2	Subprogram return error	A return number was specified by a sub program return command when the number of repeats is set for the sub program call.	Press the [RST] key, and the check the program.
SM4019	3	2	No applicable sequence	1. The sequence number which was specified in subprogram call/return command is not found. 2. The sequence number which was specified in macro command (GOTO) is not found. 3. The sequence number which was specified in program restart (sequence search) is not found.	Press the [RST] key, and the check the program.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM4020	3	2	High accuracy A invalid command	The specified command cannot be used in high accuracy mode A. Ex: - An additional axis travel command was carried out. - An interrupt type macro was used. - Involute interpolation command (G02.2 and G03.2) was carried out.	Press the [RST] key, and the check the program.
SM4021	3	2	High accuracy A invalid modal	High accuracy mode A start command was used in a modal state where it cannot be started.	Press the [RST] key, and the check the program.
SM4022	3	2	Max. speed exceeded	1. The speed specified in the command for feedrate per rotation exceeds <Max. actual cutting travel speed>. (Note) When the additional axis travels using the conversation language, this may occur even if the speed in the operation display does not exceed the parameter value. This is because the operation display shows the peripheral speed but the actual speed is checked. 2. The feedrate of helical linear axis exceeded the <Max. actual cutting travel speed> when the helical interpolation command was issued.	Press the [RST] key, and then check the program or set [Max. actual cutting travel speed] (user parameter/switch 1).
SM4023	3	2	Mirror image mode	The specified G code is not allowed in mirror image mode.	Press the [RST] key, and the check the program.
SM4024	3	2	Scaling	The specified G code is not allowed in scaling mode.	Press the [RST] key, and the check the program.
SM4025	3	2	Scaling address error	1. Scaling was attempted for the invalid axis. 2. The item set for [Scaling] (user parameter/switch 1) does not match the address where magnification is specified in the program.	Press the [RST] key, and the check the program.
SM4026	3	2	Selected plane error	The specified G-code is not allowed for a plane other than the XY plane.	Press the [RST] key, and the check the program.
SM4027	3	2	Two or more axes are specified.	Two or more axes were specified for continuous skip.	Press the [RST] key, and the check the program.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM4028	3	2	Load weight setting error	1. When the machine model is equipped with a movable table, this error is triggered if the actual table load weight exceeds the <Table load weight> (user parameter) + tolerance for weight error. 2. This error is triggered if the command value for the quick table current exceeds the tolerance.	1. Make sure that the weight load on the table is the same as the user parameter <Table load weight>. 2. If the user parameter <Positioning method> is changed, the estimated load weight may be greater than the actual load weight. Press the [I/O] key and then select <Slave status display> under <1.I/O>. Set the user parameter <Table load weight> to the value displayed as the <Estimated results>.
SM4029	3	2	Coordinate transformation error	Coordinate transformation cannot be executed correctly (e.g. The amount of travel becomes 0 when scaling is applied.)	Press the [RST] key, and the check the program.
SM4030	3	2	Turn external program select off.	Tape operation was attempted when the external program selection is ON.	Turn the external program signal off.
SM4031	3	2	Comment error	Control out/in codes are not in the same block.	Press the [RST] key, and the check the program.
SM4032	3	2	No intersection	1. The intersection could not be calculated when cutting inside the workpiece during cutter compensation. 2. The intersection could not be calculated for the chamfering / corner R command.	Press the [RST] key, and the check the program.
SM4033	3	2	The tool data control method is different.	G/M code command which cannot be used in the selected tool data management method was specified.	Check the program and setting of [Tool data management method] (user parameter/switch 1) after pressing the [RST] key.
SM4034	3	2	Simultaneous command unable M code	M code command which cannot be used in the same block was specified.	Check the program and setting of [Multiple M codes in one block] (user parameter/switch 1) after pressing the [RST] key.
SM4035	3	2	CTS unable tool	CTS ON was attempted to a CTS unable tool.	Change to a CTS able tool and turn CTS on.
SM4036	3	2	Comm. issued to area other than (tool) data area.	The offset amount and wear offset are out of the tool data range.	Press the [RST] key, and the check the program.
SM4037	3	2	Interrupted program call error	1. Interruption signal was detected when a subprogram specified for interruption program was executed with G65/G66/M98. 2. Interruption signal was detected when a subprogram which is designated for interruption program and operates as the parent program was executed.	Press the [RST] key, and the check the program.
SM4038	3	2	High-accuracy B unable modal	High accuracy mode B start command was specified in a modal state where it cannot be started.	Press the [RST] key, and the check the program.
SM4039	3	2	High-accuracy B unable command	The specified command cannot be used in high accuracy mode B.	Press the [RST] key, and the check the program.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM4040	3	2	Too many external output command digit.	The range of data that can be output exceeded the limit when macro variables were output via BPRNT command.	Press the [RST] key, and the check the program.
SM4041	3	2	POPEN is unable.	BPRNT or DPRNT command was sent although POPEN command was not sent.	Press the [RST] key, and the check the program.
SM4042	3	2	No subprogram (*)	1. In extended memory operation, the designated subprogram does not exist in the same folder of the main program. 2. In tape operation, the designated subprogram does not exist in the root folder of the internal memory.	Press the [RST] key, and the check the program.
SM4043	3	2	Too many subprograms	The number of sub programs to be called exceeds the limit.	Reduce the number of sub programs.
SM4044	3	2	A command unable to select the lathe spindle	1. While lathe spindle was selected, axis travel command was specified to the additional axis corresponding to lathe spindle. 2. While lathe spindle was selected, clamp/ unclamp commands were specified to the additional axis corresponding to lathe spindle. 3. While lathe spindle was selected, canned cycle was specified. 4. While the lathe spindle was selected, an axis travel command is issued in incremental mode on the additional axis which corresponds to the lathe spindle on the same block as the spindle selection command (M141). 5. While the lathe spindle was selected, an interpolation command is issued on the additional axis which corresponds to the lathe spindle on the same block as the spindle selection command (M141). 6. G/M code command was issued which cannot be used when lathe spindle is selected.	Press the [RST] key, and the check the program.
SM4045	3	2	Selection of a spindle is abnormal.	M code command which cannot be used for the spindle being selected was attempted to be executed.	Press the [RST] key, and the check the program.
SM4046	3	2	Communicating	When communication was not finished after operation finished with M30 in tape operation (general communication device), tape operation restart was attempted.	Start tape operation after communication is finished.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SM4047	3	2	There is no P address.	Address P is not specified in a command which requires address P.	Specify address P.
SM4048	3	2	Different from pallet program	A program different from the one set for the user parameter (quick table) was used for tape operation.	Check the user parameter (quick table) for the program number of each pallet or operation program number.
SM4049	3	2	W/o smooth path offset function	Smooth path offset function command was specified even though the machine is not equipped with Smooth path offset function option.	Purchase the option.
SM4050	3	2	W/o turning function	Turning command was specified although the machine is not equipped with turning function.	Turning function cannot be used.
SM4051	3	2	Cannot use intrpl. command for positioning on addnl. Axis	Interpolation command was issued to the additional axis set up for positioning.	Correct the program.
SM4052	3	2	No spindle speed clamp command	The constant peripheral speed control command (G96) was executed before issuing a maximum spindle speed clamp command using the spindle speed clamp command G92.	Execute the constant peripheral speed control command G96 after specifying the maximum spindle speed clamp using the spindle speed clamp command G92.
SM4053	3	2	Press the [RST] key	Multiple alarms were triggered during the reset operation.	Press the [RST] key.
SM4054	3	2	Tool length offset cancel error	G49 with no Z-axis command or the H0 command block was executed when the parameter [Travel of X, Y or Z axis when tool length/tool position offset is changed] is set to [Type 1] and when the parameter [Error check when traveling during tool length/tool position offset cancel] is set to [Yes]. Ex: (Alarm is triggered in N2) N1 G90 G43 H1 (When H1 is not 0) N2 G49 (Note) When the firmware version is earlier than WX00.03.012, the alarm is not triggered in the above program.	1. Correct the program. 1-1. Add Z-axis command to G49 or H0 command block. 1-2. Use G28/G30 and G53 to add a Z-axis safe position operation to G49 or the block right before the H0 command. 2. Make sure that there is no travel interference during a tool length offset cancel based on G49 or H0 command execution. Set the user parameter [Error check when traveling during tool length/tool position offset cancel] to [1: No] (Configure the setting to [1: No] to make it compatible with CNC-B00 operation).

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM4055	3	2	Tool position offset cancel error	G49 with no X-, Y- and Z-axes command or the H0 command block was executed when the parameter [Travel of X, Y or Z axis when tool length/tool position offset is changed] is set to [Type 1] and when the parameter [Error check when traveling during tool length/tool position offset cancel] is set to [Yes]. Ex: (Alarm is triggered in N2) N1 G90 G143 H1 (When H1 is not 0) N2 G49 (Note) When the firmware version is earlier than WX00.03.012, the alarm is not triggered in the above program.	1. Correct the program. 1-1. Add X-, Y- and Z-axes command to G49 or H0 command block. 1-2. Use G28/G30 and G53 to add an X-, Y- and Z-axes safe position operation to G49 or the block right before the H0 command. 2. Make sure that there is no travel interference during a tool position offset cancel on the X-, Y- and Z-axes based on G49 or H0 command execution. Set the user parameter [Error check when traveling during tool length/tool position offset cancel] to [1: No].
SM4056	3	2	Rotary fixture offset command error	When the current rotary fixture offset is calculated: 1. There is no axis option that corresponds to the <Axis for calculation> in the standard rotary fixture offset. 2. The <Axis for calculation> in the standard rotary fixture offset is not set to either <Tilt axis*> or <Rotation axis*>. 3. The <Axis for calculation> in the standard rotary fixture offset is not set to the pair: <Tilt axis*> and <Rotation axis*>.	Set the correct values for <Axis for calculation> in the standard rotary fixture offset for 1. through 3.
SM4057	3	2	Rotary fixture offset in progress	1. A G code command was issued that is prohibited during rotary fixture offset. 2. An additional axis command was issued in the helical screw interpolation during rotary fixture offset.	Press the [RST] key and check the program.
SM4058	3	2	Failed to search macro control statement.	The program could not be read during operation.	1. Press the [RST] key. 2. For tape operation, check the memory card again.
SM4059	3	2	No G code macro program (*)	The program that is called in the G code macro does not exist in the <G code macro program folder>.	1. Press the [RST] key, and then check the program. 2. Press the [RST] key and check the <Program No.> in the <G code macro>.
SM4060	3	2	No M code macro program (*)	The program that is called in the M code macro does not exist in the <M code macro program folder>.	1. Press the [RST] key, and then check the program. 2. Press the [RST] key and check the <Program No.> in the <M code macro>.
SM4061	3	2	G code macro program No. not set	The program No. is not set for the G code macro that is used in the program.	Press the [RST] key and check the <Program No.> in the <G code macro>.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM4062	3	2	M code macro program No. not set	The program No. is not set for the M code macro that is used in the program.	Press the [RST] key and check the <Program No.> in the <M code macro>.
SM4063	3	2	Macro program call not possible	1. The macro program was called by the G code or M code in MDI. 2. After the G code macro call is executed, when the <Call type> for the G/M code is set to <1: Type 2>, the macro program is called by the G code by the time the G67 command is executed.	Press the [RST] key and check the program.
SM4064	3	2	No subprogram in the G code macro folder (*)	The specified subprogram does not exist in the G code macro folder during the extended memory operation or tape operation.	Press the [RST] key and check the program.
SM4065	3	2	No subprogram in the M code macro folder (*)	The specified subprogram does not exist in the M code macro folder during the extended memory operation or tape operation.	Press the [RST] key and check the program.
SM4066	3	2	No rotary fixture offset function	There is no option for the rotary fixture offset function but an order was issued for rotary fixture offset function.	Purchase the option to use this function.
SM4067	3	2	Arc angle error during inner side cutting	There is an arc angle between the start point and end point on the program path for the cutting command when cutting the inner side during cutter/nose R compensation. There is another arc angle between the start point and end point in the tool center path after being offset. The difference between those two arc angles is greater than 180°.	Press the [RST] key and check the program.
SM4068	3	2	No relief amount	There is no relief amount after cutting in a lathe system operation.	Press the [RST] key and check the program.
SM4069	3	2	Cutting amount error in thread cutting cycle	1. The amount of cutting in the thread cutting cycle is greater than the cutting distance or the relief amount. 2. The position of the cutting direction after finishing, in the thread cutting cycle, exceeded the start point of the cycle operation.	Set the user parameters <Cutting amount in thread cutting cycle> and <Cutting angle in thread cutting cycle> to a smaller value, or change the start point in the cycle operation.
SM4070	3	2	Cutting angle error in thread cutting cycle	The cutting angle is too small for the taper angle in the taper thread cutting cycle.	Set an appropriate value for the user parameter <Thread cutting cycle cutting angle>.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM4071	3	2	Taper amount too large	The position which was only shifted by the taper amount toward the cutting direction from the cutting end point in the lathe system cycle exceeded the start point of the cycle operation.	Press the [RST] key and check the program.
SM4072	3	2	Initial infeed amount error in thread cutting cycle	The initial infeed amount in a complex thread cutting cycle is greater than the thread height.	Press the [RST] key and check the program.
SM4073	3	2	Finishing allowance error in thread cutting cycle	The finishing allowance in a complex thread cutting cycle is greater than the thread height.	Press the [RST] key, and then check the program. Or, set an appropriate value for the user parameter <Complex thread cutting cycle finishing allowance>.
SM4074	3	2	Thread height error in thread cutting cycle	The thread height in a complex thread cutting cycle is greater than the approach distance.	Press the [RST] key and check the program.
SM4075	3	2	Thread cutting command error	1. 3 axis commands for X, Y and Z were issued in thread cutting (G33, G392, G396). 2. Only 1 axis command for X, Y and Z was issued in the thread cutting cycle (G392 and G376). 3. A thread cutting cycle command was issued in which the travel distance for the cutting direction is 0 and there is a taper amount.	Press the [RST] key and check the program.
SM4076	3	2	Thread cutting command not possible in feed rate per minute	A thread cutting command was issued during the feed rate per minute (G94) modal. A thread cutting command can only be issued during the feed rate per rotation (G95) modal.	Press the [RST] key and check the program.
SM4077	3	2	Thread cutting cycle address error	There is no P or Q address in the complex thread cutting cycle.	Press the [RST] key and check the program.
SM4078	3	2	Cutting amount error in thread cutting cycle	The travel distance for the cutting direction in the complex thread cutting cycle is shorter than the length calculated by the following equation. $P \times \tan(D/2)$ P: Thread height D: Thread angle	Press the [RST] key and check the program.
SM4079	3	2	Arc angle error when cutter compensation direction changed	When the offset direction has been changed during cutter/nose R compensation in an arc command, the arc angle for the start and end points in the tool center path after being offset is greater than 180° due to the arc angle for the start and end points set in the programmed path.	Press the [RST] key and check the program.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SM4080	3	2	3D smoothing data load error	A program with 3D smoothed values was loaded even though 1, 2, 3, 102, 103, 202 and 203 were registered to the <G code No.>, or even though 380 to 387 and 389 were registered to the <M code No.> in the <G/M code macro parameter>.	Delete <G code No.> 1, 2, 3, 102, 103, 202 and 203 and <M code No.> 380 to 387 and 389 are registered in the <G/M code macro parameter>.
SM4081	3	2	Unavailable modal for machining mode specification	The M298 command was issued in a modal status where the M298 (machining mode specification) command cannot be issued.	Press the [RST] key and check the program.
SM4082	3	2	3D smoothing command error	A 3D smoothing start/end code command was issued using macro variables, the calculation function and the optional block skip.	After pressing the [RST] key, correct the program and execute the 3D smoothing data operation.
SM4083	3	2	Machining mode specification error	The L7 (Adjustment A), L8 (Adjustment B) or L9 (Adjustment C) command was issued in M278 without setting the <Standard mode> in the user parameter (high accuracy (adjustment)).	Press the [RST] key and check the program. Or, check the setting for <Standard mode> in the user parameter (high accuracy (adjustment)).
SM4085	3	2	Parameter set. error in feature coordinate set. function	1. When the user parameter (Feature coordinate setting) was set to <Tilt axis*> or <Rotation axis*> for an axis where the machine parameter (system 2) <Optional *-axis> was set to <Not equipped>, a command was issued for the feature coordinate setting function. 2. A reserve value has been set to one of the user parameters (feature coordinate setting) <Forward direction for the coordinate system>.	Change the following parameters to the correct values, and press the [RST] key.
SM4086	3	2	No feature coordinate setting function	The feature coordinate setting function command was specified even though the machine is not equipped with the feature coordinate setting function option.	Purchase the option.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM4087	3	2	Feature coordinate command error	1. The command for the feature coordinate setting function cannot be executed in this modal status. 2. An attempt was made to cancel the feature coordinate manufacturing mode before turning OFF the function that was turned ON while in feature coordinate manufacturing mode. In addition, before the feature coordinate manufacturing mode is cancelled, an attempt was made to turn OFF the function that was turned ON when feature coordinate manufacturing mode was started. 3. A feature coordinate setting function command was specified when there was a shift in the local coordinate system. 4. All addresses I, J and K specified for the angle were omitted. 5. A feature coordinate setting function command was specified in the interrupt program. 6. An attempt was made to move an axis outside of the stroke range. 7. An R command was carried out for the tool change command (G100) before the feature coordinate index on an arm type ATC mechanism. 8. A command was carried out that cannot be executed before the axis travel command is issued for the G90 modal to the additional axis (selected as the lathe spindle).	Press the [RST] key, and the check the program.
SM4088	3	2	Feature coordinate manufacturing mode engaged	A command was specified that cannot be executed while in feature coordinate manufacturing mode. Ex: - A command was issued to rotate an additional axis. - An interrupt type macro command was specified. - A command was issued to select a lathe spindle. - A command was issued to change the workpiece coordinate system.	Press the [RST] key, and the check the program.
SM4089	3	2	Command not possible during inverse time feed	G/M code command was specified which cannot be used while in the inverse time feed (G93) modal.	Press the [RST] key, and the check the program.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM4090	3	2	Machining load maximum(*)	Machining load exceeded the <Max. machining load> in the monitoring section for the tool indicated by the auxiliary number.	The tool may be worn and/or chipped. Press the [RST] key and perform one of the following. - Replace with a new tool. - Make sure that the machining load monitor parameter is correctly set.
SM4091	3	2	Machining load minimum(*)	Machining load was under the <Min. machining load> in the monitoring section for the tool indicated by the auxiliary number.	The tool may be broken or machining was accidentally done twice. Press the [RST] key and perform one of the following. - Replace with a new tool. - Check the condition of the workpiece. - Make sure that the machining load monitor parameter is correctly set.
SM4092	3	2	Cutter compensation interference check error	It was detected that the infeed may be too great on the path offset by the cutter/nose R compensation.	Check the program and cutter compensation amount. If the alarm continues to trigger even though there is no excessive infeed, set the <Cutter compensation interference check> in the user parameter (switch 1) to <0:No>.
SM4093	3	2	Invalid command in machining mode	The specified command cannot be used in the selected machining mode. Ex: - An additional axis travel command was carried out. - An interrupt type macro was used. - Involute interpolation command (G02.2 and G03.2) was carried out.	Press the [RST] key and check the program. Check the setting for the <Machining mode> selection in the user parameter (high accuracy (adjustment)).
SM4094	3	2	Invalid command in accuracy specification mode	The command cannot be used in the specified accuracy mode. Ex: - An additional axis travel command was carried out. - An interrupt type macro was used. - Involute interpolation command (G02.2 and G03.2) was carried out.	Press the [RST] key and check the program. Check the setting for the <Accuracy spec. mode> selection in the user parameter (high accuracy (accuracy specification)).

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM4095	3	2	Involute interpolation command error	1. A command was issued for an axis that is not inside the selected plane. 2. The I, J and K commands were not carried out. 3. All the I, J and K commands had a value of 0 when they were carried out. 4. The R command had a value of 0 or less when it was carried out. 5. A command was issued where the start and end points are on the inside of the base circle. 6. A command was issued where the start and end points are equidistant from the center of the base circle. 7. A command with 100 or more rotations was issued.	Press the [RST] key and check the program.
SM4096	3	2	Invalid command for involute interpolation modal	G and M code commands were issued but are prohibited while in involute interpolation mode.	Press the [RST] key and check the program.
SM4097	3	2	There is no involute interpolation function	The involute interpolation command was specified even though the machine is not equipped with the involute interpolation function option.	Purchase the option.
SM4098	3	2	Involute interpolation cutter compensation error	The difference in the travel angle is 180 degrees or greater between the program path and the path after being offset for an involute interpolation command while carrying out cutter compensation / nose R compensation.	Press the [RST] key, and the check the program.
SM4099	3	2	An external output command is being executed.	The mode was changed during data output via external output command.	Press the [RST] key. Change the mode after data output via external output command is completed.
SM5500	2	2	Time to change batteries	The battery voltage for maintaining the absolute encoder data is low.	While the power switch is turned ON, replace all the batteries in the battery compartment (GB1) on the left side of the control box.
SM5501	2	2	Machining load maximum(*)	Machining load exceeded the <Max. machining load> in the monitoring section for the tool indicated by the auxiliary number.	The tool may be worn and/or chipped. Press the [RST] key and perform one of the following. - Replace with a new tool. - Make sure that the machining load monitor parameter is correctly set.
SM5502	2	2	Machining load minimum(*)	Machining load was under the <Min. machining load> in the monitoring section for the tool indicated by the auxiliary number.	The tool may be broken or machining was accidentally done twice. Press the [RST] key and perform one of the following. - Replace with a new tool. - Check the condition of the workpiece. - Make sure that the machining load monitor parameter is correctly set.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM7000	1	2	There is no P1-axis option.	P1-axis travel command was executed in ladder program when [Optional P1-axis] (machine parameter/PLC) was set to [Not equipped].	Turn the CTRL_P1 signal (BY:P1axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P1signal (BY:P1axis control) ON, and turn the RESET_P1 signal (BY:P1axis reset) ON.
SM7001	1	2	There is no P2-axis option.	P2-axis travel command was executed in ladder program when [Optional P2-axis] (machine parameter/PLC) was set to [Not equipped].	Turn the CTRL_P2 signal (BY:P2axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P2signal (BY:P2axis control) ON, and turn the RESET_P2 signal (BY:P2axis reset) ON.
SM7002	1	2	There is no P3-axis option.	P3-axis travel command was executed in ladder program when [Optional P3-axis] (machine parameter/PLC) was set to [Not equipped].	Turn the CTRL_P3 signal (BY:P3axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P3signal (BY:P3axis control) ON, and turn the RESET_P3 signal (BY:P3axis reset) ON.
SM7003	1	2	There is no P4-axis option.	P4-axis travel command was executed in ladder program when [Optional P4-axis] (machine parameter/PLC) was set to [Not equipped].	Turn the CTRL_P4 signal (BY:P4axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P4signal (BY:P4axis control) ON, and turn the RESET_P4 signal (BY:P4axis reset) ON.
SM7004	1	2	No comm. receivable as P1-ax contr sig is invalid.	P1-axis travel command was executed in ladder program when P1-axis control signal was off.	Turn the CTRL_P1 signal (BY:P1axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P1signal (BY:P1axis control) ON, and turn the RESET_P1 signal (BY:P1axis reset) ON.
SM7005	1	2	No comm. receivable as P2-ax contr sig is invalid.	P2-axis travel command was executed in ladder program when P2-axis control signal was off.	Turn the CTRL_P2 signal (BY:P2axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P2signal (BY:P2axis control) ON, and turn the RESET_P2 signal (BY:P2axis reset) ON.
SM7006	1	2	No comm. receivable as P3-ax contr sig is invalid.	P3-axis travel command was executed in ladder program when P3-axis control signal was off.	Turn the CTRL_P3 signal (BY:P3axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P3signal (BY:P3axis control) ON, and turn the RESET_P3 signal (BY:P3axis reset) ON.
SM7007	1	2	No comm. receivable as P4-ax contr sig is invalid.	P4-axis travel command was executed in ladder program when P4-axis control signal was off.	Turn the CTRL_P4 signal (BY:P4axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P4signal (BY:P4axis control) ON, and turn the RESET_P4 signal (BY:P4axis reset) ON.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM7008	1	2	Command is rejected (P1 in process)	P1-axis travel command was executed in ladder program when P1-axis was operating or causing alarm.	Turn the CTRL_P1 signal (BY:P1axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P1signal (BY:P1axis control) ON, and turn the RESET_P1 signal (BY:P1axis reset) ON.
SM7009	1	2	Command is rejected (P2 in process)	P2-axis travel command was executed in ladder program when P2-axis was operating or causing alarm.	Turn the CTRL_P2 signal (BY:P2axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P2signal (BY:P2axis control) ON, and turn the RESET_P2 signal (BY:P2axis reset) ON.
SM7010	1	2	Command is rejected (P3 in process)	P3-axis travel command was executed in ladder program when P3-axis was operating or causing alarm.	Turn the CTRL_P3 signal (BY:P3axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P3signal (BY:P3axis control) ON, and turn the RESET_P3 signal (BY:P3axis reset) ON.
SM7011	1	2	Command is rejected (P4 in process)	P4-axis travel command was executed in ladder program when P4-axis was operating or causing alarm.	Turn the CTRL_P4 signal (BY:P4axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P4signal (BY:P4axis control) ON, and turn the RESET_P4 signal (BY:P4axis reset) ON.
SM7012	1	2	Stroke exceeded (+P1)	The axis has exceeded [P1-axis stroke(+)] (user parameter/PLC) (Excluding manual reset operation.)	Turn the CTRL_P1 signal (BY:P1axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P1signal (BY:P1axis control) ON, and turn the RESET_P1 signal (BY:P1axis reset) ON.
SM7013	1	2	Stroke exceeded (-P1)	The axis has exceeded [P1-axis stroke(-)] (user parameter/PLC) (Excluding manual reset operation.)	Turn the CTRL_P1 signal (BY:P1axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P1signal (BY:P1axis control) ON, and turn the RESET_P1 signal (BY:P1axis reset) ON.
SM7014	1	2	Stroke exceeded (+P2)	The axis has exceeded [P2-axis stroke(+)] (user parameter/PLC) (Excluding manual reset operation.)	Turn the CTRL_P2 signal (BY:P2axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P2signal (BY:P2axis control) ON, and turn the RESET_P2 signal (BY:P2axis reset) ON.
SM7015	1	2	Stroke exceeded (-P2)	The axis has exceeded [P2-axis stroke(-)] (user parameter/PLC) (Excluding manual reset operation.)	Turn the CTRL_P2 signal (BY:P2axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P2signal (BY:P2axis control) ON, and turn the RESET_P2 signal (BY:P2axis reset) ON.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM7016	1	2	Stroke exceeded (+P3)	The axis has exceeded [P3-axis stroke(+)] (user parameter/PLC) (Excluding manual reset operation.)	Turn the CTRL_P3 signal (BY:P3axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P3signal (BY:P3axis control) ON, and turn the RESET_P3 signal (BY:P3axis reset) ON.
SM7017	1	2	Stroke exceeded (-P3)	The axis has exceeded [P3-axis stroke(-)] (user parameter/PLC) (Excluding manual reset operation.)	Turn the CTRL_P3 signal (BY:P3axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P3signal (BY:P3axis control) ON, and turn the RESET_P3 signal (BY:P3axis reset) ON.
SM7018	1	2	Stroke exceeded (+P4)	The axis has exceeded [P4-axis stroke(+)] (user parameter/PLC) (Excluding manual reset operation.)	Turn the CTRL_P4 signal (BY:P4axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P4signal (BY:P4axis control) ON, and turn the RESET_P4 signal (BY:P4axis reset) ON.
SM7019	1	2	Stroke exceeded (-P4)	The axis has exceeded [P4-axis stroke(-)] (user parameter/PLC) (Excluding manual reset operation.)	Turn the CTRL_P4 signal (BY:P4axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P4signal (BY:P4axis control) ON, and turn the RESET_P4 signal (BY:P4axis reset) ON.
SM7020	1	2	P1-axis servo power off	The servo was turned off during P1-axis travel.	Turn the CTRL_P1 signal (BY:P1axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P1signal (BY:P1axis control) ON, and turn the RESET_P1 signal (BY:P1axis reset) ON.
SM7021	1	2	P2-axis servo power off	The servo was turned off during P2-axis travel.	Turn the CTRL_P2 signal (BY:P2axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P2signal (BY:P2axis control) ON, and turn the RESET_P2 signal (BY:P2axis reset) ON.
SM7022	1	2	P3-axis servo power off	The servo was turned off during P3-axis travel.	Turn the CTRL_P3 signal (BY:P3axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P3signal (BY:P3axis control) ON, and turn the RESET_P3 signal (BY:P3axis reset) ON.
SM7023	1	2	P4-axis servo power off	The servo was turned off during P4-axis travel.	Turn the CTRL_P4 signal (BY:P4axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P4signal (BY:P4axis control) ON, and turn the RESET_P4 signal (BY:P4axis reset) ON.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM7024	1	2	P1-axis position shifted	1. The motor shaft was shifted by machining load. 2. The shift of the axis exceeded [P1-axis position confirmation pulse] during clamping. 3. Brake power is too low.	Turn the CTRL_P1 signal (BY:P1axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P1signal (BY:P1axis control) ON, and turn the RESET_P1 signal (BY:P1axis reset) ON. 1. Check the air pressure. 2. Reduce the machining load. 3. Check that the setting for [P1-axis position confirmation pulse] (machine parameter/PLC) is not too small. 4. Check that the brake is applied correctly.
SM7025	1	2	P2-axis position shifted	1. The motor shaft was shifted by machining load. 2. The shift of the axis exceeded [P2-axis position confirmation pulse] during clamping. 3. Brake power is too low.	Turn the CTRL_P2 signal (BY:P2axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P2signal (BY:P2axis control) ON, and turn the RESET_P2 signal (BY:P2axis reset) ON. 1. Check the air pressure. 2. Reduce the machining load. 3. Check that the setting for [P2-axis position confirmation pulse] (machine parameter/PLC) is not too small. 4. Check that the brake is applied correctly.
SM7026	1	2	P3-axis position shifted	1. The motor shaft was shifted by machining load. 2. The shift of the axis exceeded [P3-axis position confirmation pulse] during clamping. 3. Brake power is too low.	Turn the CTRL_P3 signal (BY:P3axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P3signal (BY:P3axis control) ON, and turn the RESET_P3 signal (BY:P3axis reset) ON. 1. Check the air pressure. 2. Reduce the machining load. 3. Check that the setting for [P3-axis position confirmation pulse] (machine parameter/PLC) is not too small. 4. Check that the brake is applied correctly.
SM7027	1	2	P4-axis position shifted	1. The motor shaft was shifted by machining load. 2. The shift of the axis exceeded [P4-axis position confirmation pulse] during clamping. 3. Brake power is too low.	Turn the CTRL_P4 signal (BY:P4axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P4signal (BY:P4axis control) ON, and turn the RESET_P4 signal (BY:P4axis reset) ON. 1. Check the air pressure. 2. Reduce the machining load. 3. Check that the setting for [P4-axis position confirmation pulse] (machine parameter/PLC) is not too small. 4. Check that the brake is applied correctly.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM7028	1	2	P1-axis in-position check timeout	The axis did not reach in-position after operation was finished.	Turn the CTRL_P1 signal (BY:P1axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P1signal (BY:P1axis control) ON, and turn the RESET_P1 signal (BY:P1axis reset) ON. 1. Check the values of [P1-axis in-position width] and [P1-axis in-position check timeout period] (machine parameter/PLC). 2. Check that the axis does not vibrate after it has stopped. [Input/output] [2. Servo controller] position deviation
SM7029	1	2	P2-axis in-position check timeout	The axis did not reach in-position after operation was finished.	Turn the CTRL_P2 signal (BY:P2axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P2signal (BY:P2axis control) ON, and turn the RESET_P2 signal (BY:P2axis reset) ON. 1. Check the values of [P2-axis in-position width] and [P2-axis in-position check timeout period] (machine parameter/PLC). 2. Check that the axis does not vibrate after it has stopped. [Input/output] [2. Servo controller] position deviation
SM7030	1	2	P3-axis in-position check timeout	The axis did not reach in-position after operation was finished.	Turn the CTRL_P3 signal (BY:P3axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P3signal (BY:P3axis control) ON, and turn the RESET_P3 signal (BY:P3axis reset) ON. 1. Check the values of [P3-axis in-position width] and [P3-axis in-position check timeout period] (machine parameter/PLC). 2. Check that the axis does not vibrate after it has stopped. [Input/output] [2. Servo controller] position deviation
SM7031	1	2	P4-axis in-position check timeout	The axis did not reach in-position after operation was finished.	Turn the CTRL_P4 signal (BY:P4axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P4signal (BY:P4axis control) ON, and turn the RESET_P4 signal (BY:P4axis reset) ON. 1. Check the values of [P4-axis in-position width] and [P4-axis in-position check timeout period] (machine parameter/PLC). 2. Check that the axis does not vibrate after it has stopped. [Input/output] [2. Servo controller] position deviation

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM7032	1	2	P1-axis speed command error	When P1-axis travel command was executed in ladder program, the specified speed was faster than [Rapid feed P1-axis] (machine parameter/PLC), or was zero.	Turn the CTRL_P1 signal (BY:P1axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P1signal (BY:P1axis control) ON, and turn the RESET_P1 signal (BY:P1axis reset) ON.
SM7033	1	2	P2-axis speed command error	When P2-axis travel command was executed in ladder program, the specified speed was faster than [Rapid feed P2-axis] (machine parameter/PLC), or was zero.	Turn the CTRL_P2 signal (BY:P2axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P2signal (BY:P2axis control) ON, and turn the RESET_P2 signal (BY:P2axis reset) ON.
SM7034	1	2	P3-axis speed command error	When P3-axis travel command was executed in ladder program, the specified speed was faster than [Rapid feed P3-axis] (machine parameter/PLC), or was zero.	Turn the CTRL_P3 signal (BY:P3axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P3signal (BY:P3axis control) ON, and turn the RESET_P3 signal (BY:P3axis reset) ON.
SM7035	1	2	P4-axis speed command error	When P4-axis travel command was executed in ladder program, the specified speed was faster than [Rapid feed P4-axis] (machine parameter/PLC), or was zero.	Turn the CTRL_P4 signal (BY:P4axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P4signal (BY:P4axis control) ON, and turn the RESET_P4 signal (BY:P4axis reset) ON.
SM7036	1	2	P1-axis position command error	When P1-axis travel command was executed in ladder program, the specified position was greater than upper limit.	Turn the CTRL_P1 signal (BY:P1axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P1signal (BY:P1axis control) ON, and turn the RESET_P1 signal (BY:P1axis reset) ON.
SM7037	1	2	P2-axis position command error	When P2-axis travel command was executed in ladder program, the specified position was greater than upper limit.	Turn the CTRL_P2 signal (BY:P2axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P2signal (BY:P2axis control) ON, and turn the RESET_P2 signal (BY:P2axis reset) ON.
SM7038	1	2	P3-axis position command error	When P3-axis travel command was executed in ladder program, the specified position was greater than upper limit.	Turn the CTRL_P3 signal (BY:P3axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P3signal (BY:P3axis control) ON, and turn the RESET_P3 signal (BY:P3axis reset) ON.
SM7039	1	2	P4-axis position command error	When P4-axis travel command was executed in ladder program, the specified position was greater than upper limit.	Turn the CTRL_P4 signal (BY:P4axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P4signal (BY:P4axis control) ON, and turn the RESET_P4 signal (BY:P4axis reset) ON.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM7040	1	2	The P1-axis servo cannot be ON.	P1-axis servo is not turned on even though four seconds have elapsed.	Turn the CTRL_P1 signal (BY:P1axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P1signal (BY:P1axis control) ON, and turn the RESET_P1 signal (BY:P1axis reset) ON.
SM7041	1	2	The P2-axis servo cannot be ON.	P2-axis servo is not turned on even though four seconds have elapsed.	Turn the CTRL_P2 signal (BY:P2axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P2signal (BY:P2axis control) ON, and turn the RESET_P2 signal (BY:P2axis reset) ON.
SM7042	1	2	The P3-axis servo cannot be ON.	P3-axis servo is not turned on even though four seconds have elapsed.	Turn the CTRL_P3 signal (BY:P3axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P3signal (BY:P3axis control) ON, and turn the RESET_P3 signal (BY:P3axis reset) ON.
SM7043	1	2	The P4-axis servo cannot be ON.	P4-axis servo is not turned on even though four seconds have elapsed.	Turn the CTRL_P4 signal (BY:P4axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P4signal (BY:P4axis control) ON, and turn the RESET_P4 signal (BY:P4axis reset) ON.
SM7044	1	2	The P1-axis servo OFF command is detected.	The servo off command was detected during P1-axis travel.	Turn the CTRL_P1 signal (BY:P1axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P1signal (BY:P1axis control) ON, and turn the RESET_P1 signal (BY:P1axis reset) ON.
SM7045	1	2	The P2-axis servo OFF command is detected.	The servo off command was detected during P1-axis travel.	Turn the CTRL_P2 signal (BY:P2axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P2signal (BY:P2axis control) ON, and turn the RESET_P2 signal (BY:P2axis reset) ON.
SM7046	1	2	The P3-axis servo OFF command is detected.	The servo off command was detected during P1-axis travel.	Turn the CTRL_P3 signal (BY:P3axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P3signal (BY:P3axis control) ON, and turn the RESET_P3 signal (BY:P3axis reset) ON.
SM7047	1	2	The P4-axis servo OFF command is detected.	The servo off command was detected during P1-axis travel.	Turn the CTRL_P4 signal (BY:P4axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P4signal (BY:P4axis control) ON, and turn the RESET_P4 signal (BY:P4axis reset) ON.
SM7048	1	2	External panel setting error (*)	IP address of external panel 1 overlapped with that of external panel 2.	Correct each parameter of communication parameter (external panel) and turn the power on again.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM7049	1	2	Simple panel calibration error (*)	Reserved	Reserved
SM7050	1	2	Magazine panel calibration error (*)	Reserved	Reserved
SM7051	1	2	Teaching pendant panel calibration error (*)	An error occurred during teaching pendant calibration.	1. Perform calibration again. 2. The teaching pendant is faulty. Contact the nearest BROTHER approved service dealer and report the conditions when the error occurred.
SM7052	1	2	Magazine panel communication data is abnormal. (*)	Reserved	Reserved
SM7053	1	2	Teaching pendant communic. data is abnormal. (*)	An error was detected in teaching pendant communication data.	Check the connection with teaching pendant.
SM7054	1	2	No comm receivable as P1-axis servo OFF is active.	When P1-axis attempted to start travel, P1-axis servo off command was being specified.	Turn the CTRL_P1 signal (BY:P1axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P1signal (BY:P1axis control) ON, and turn the RESET_P1 signal (BY:P1axis reset) ON.
SM7055	1	2	No comm receivable as P2-axis servo OFF is active.	When P2-axis attempted to start travel, P2-axis servo off command was being specified.	Turn the CTRL_P2 signal (BY:P2axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P2signal (BY:P2axis control) ON, and turn the RESET_P2 signal (BY:P2axis reset) ON.
SM7056	1	2	No comm receivable as P3-axis servo OFF is active.	When P3-axis attempted to start travel, P3-axis servo off command was being specified.	Turn the CTRL_P3 signal (BY:P3axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P3signal (BY:P3axis control) ON, and turn the RESET_P3 signal (BY:P3axis reset) ON.
SM7057	1	2	No comm receivable as P4-axis servo OFF is active.	When P4-axis attempted to start travel, P4-axis servo off command was being specified.	Turn the CTRL_P4 signal (BY:P4axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P4signal (BY:P4axis control) ON, and turn the RESET_P4 signal (BY:P4axis reset) ON.
SM7058	1	2	No command receivable as P1-axis stop is active.	When P1-axis attempted to start travel, P1-axis stop command was being specified.	Turn the CTRL_P1 signal (BY:P1axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P1signal (BY:P1axis control) ON, and turn the RESET_P1 signal (BY:P1axis reset) ON.
SM7059	1	2	No command receivable as P2-axis stop is active.	When P2-axis attempted to start travel, P2-axis stop command was being specified.	Turn the CTRL_P2 signal (BY:P2axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P2signal (BY:P2axis control) ON, and turn the RESET_P2 signal (BY:P2axis reset) ON.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM7060	1	2	No command receivable as P3-axis stop is active.	When P3-axis attempted to start travel, P3-axis stop command was being specified.	Turn the CTRL_P3 signal (BY:P3axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P3signal (BY:P3axis control) ON, and turn the RESET_P3 signal (BY:P3axis reset) ON.
SM7061	1	2	No command receivable as P4-axis stop is active.	When P4-axis attempted to start travel, P4-axis stop command was being specified.	Turn the CTRL_P4 signal (BY:P4axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P4signal (BY:P4axis control) ON, and turn the RESET_P4 signal (BY:P4axis reset) ON.
SM7062	1	2	OPSEL mounted	OPSEL is mounted.	Please contact us if you encounter a situation.
SM7063	1	2	No command is receivable as the door is open (P1-axis)	P1-axis travel command was executed in ladder program while the door corresponding to [P1 install. position] (machine parameter) was open.	Turn the CTRL_P1 signal (BY:P1axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P1signal (BY:P1axis control) ON, and turn the RESET_P1 signal (BY:P1axis reset) ON.
SM7064	1	2	No command is receivable as the door is open (P2-axis)	P2-axis travel command was executed in ladder program while the door corresponding to [P2 install. position] (machine parameter) was open.	Turn the CTRL_P2 signal (BY:P2axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P2signal (BY:P2axis control) ON, and turn the RESET_P2 signal (BY:P2axis reset) ON.
SM7065	1	2	No command is receivable as the door is open (P3-axis)	P3-axis travel command was executed in ladder program while the door corresponding to [P3 install. position] (machine parameter) was open.	Turn the CTRL_P3 signal (BY:P3axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P3signal (BY:P3axis control) ON, and turn the RESET_P3 signal (BY:P3axis reset) ON.
SM7066	1	2	No command is receivable as the door is open (P4-axis)	P4-axis travel command was executed in ladder program while the door corresponding to [P4 install. position] (machine parameter) was open.	Turn the CTRL_P4 signal (BY:P4axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P4signal (BY:P4axis control) ON, and turn the RESET_P4 signal (BY:P4axis reset) ON.
SM7067	1	2	Speed error when door is open (P1-axis)	The P1-axis moved at a speed exceeding a certain limit speed when the door was open.	Turn the CTRL_P1 signal (BY:P1axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P1signal (BY:P1axis control) ON, and turn the RESET_P1 signal (BY:P1axis reset) ON.
SM7068	1	2	Speed error when door is open (P2-axis)	The P2-axis moved at a speed exceeding a certain limit speed when the door was open.	Turn the CTRL_P2 signal (BY:P2axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P2signal (BY:P2axis control) ON, and turn the RESET_P2 signal (BY:P2axis reset) ON.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM7069	1	2	Speed error when door is open (P3-axis)	The P3-axis moved at a speed exceeding a certain limit speed when the door was open.	Turn the CTRL_P3 signal (BY:P3axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P3signal (BY:P3axis control) ON, and turn the RESET_P3 signal (BY:P3axis reset) ON.
SM7070	1	2	Speed error when door is open (P4-axis)	The P4-axis moved at a speed exceeding a certain limit speed when the door was open.	Turn the CTRL_P4 signal (BY:P4axis control) OFF, and press the [RST] key. In addition, turn the CTRL_P4signal (BY:P4axis control) ON, and turn the RESET_P4 signal (BY:P4axis reset) ON.
SM7071	1	2	A different key was pressed	The [START] switch or the [PALLET START] switch was pressed while another key was pressed at the same time. Or, the mode was changed while pressing another key at the same time.	Press the [RST] key. If you did not perform an operation while pressing another key at the same time, then replace the KEY PCB.
SM7072	1	2	Command not accepted because P1-axis servo is disabled	P1-axis travel command was executed in ladder program while P1-axis servo was disabled.	Press the [RST] key.
SM7073	1	2	Command not accepted because P2-axis servo is disabled	P2-axis travel command was executed in ladder program while P2-axis servo was disabled.	Press the [RST] key.
SM7074	1	2	Command not accepted because P3-axis servo is disabled	P3-axis travel command was executed in ladder program while P3-axis servo was disabled.	Press the [RST] key.
SM7075	1	2	Command not accepted because P4-axis servo is disabled	P4-axis travel command was executed in ladder program while P4-axis servo was disabled.	Press the [RST] key.
SM7076	1	2	ATC monitoring parameter error	The parameter was not calculated correctly due to one of the following causes. 1. Tool was not mounted correctly. 2. Z-axis torque raised abnormally close to the ATC zero point.	1-1. Make sure that the tool is fitted onto the magazine for <Magazine No. with tool>. 1-2. Make sure that no tool is fitted onto the magazine for <Magazine No. without tool>. 2. Refer to the sections <Replace Z-axis motor and coupling,> <Replace Z-axis ball screw> and <Replace Z-axis guide> in the Maintenance Manual to make sure that the parts are correctly assembled.
SM7082	1	2	Spindle taper not specified	The spindle taper specification is not set.	Set the spindle taper specification.
SM7083	1	2	Machining load maximum(*)	Machining load exceeded the <Max. machining load> in the monitoring section for the tool indicated by the auxiliary number.	The tool may be worn and/or chipped. Press the [RST] key and perform one of the following. - Replace with a new tool. - Make sure that the machining load monitor parameter is correctly set.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM7084	1	2	Machining load minimum(*)	Machining load was under the <Min. machining load> in the monitoring section for the tool indicated by the auxiliary number.	The tool may be broken or machining was accidentally done twice. Press the [RST] key and perform one of the following. - Replace with a new tool. - Check the condition of the workpiece. - Make sure that the machining load monitor parameter is correctly set.
SM7500	1	1	Memory operation mode	1. MDI operation was attempted during a pause in memory operation.	1. Execute MDI operation after memory operation is completed.
SM7501	1	1	MDI operation mode	1. Memory operation was attempted during a pause in MDI operation. 2. Changing the mode was attempted during MDI operation.	1. Execute memory operation after MDI operation is finished. 2. Perform mode switching operation after MDI operation is finished.
SM7502	1	1	Handle mode	1. An operation that should be performed in manual mode was attempted in handle mode. 2. User operation is in progress on the teaching pendant.	1. Set the axis selection switch of the handle to [OFF]. 2. Set the [ENABLE PENDANT] switch for the teaching pendant to disable ([O]).
SM7503	1	1	Spindle stopped with door open	The door was opened during spindle rotation.	Close the door, and press the [START] key. The spindle starts rotating from the position it stopped.
SM7504	1	1	Tap time constant too short	The tap time constant set by the M code is less than the one calculated by the NC, relative to the tap rotation.	Use the M code to set the appropriate time constant for tap rotation.
SM7505	1	1	Pallet position error	1. Memory operation or external additional axis command was executed without the pallet indexed. 2. Memory operation was executed with a pallet which was different from the [PALLET] key indexed.	1. Select manual operation mode and index a pallet using the [P.INDX] key. 2. Check the [PALLET] key.
SM7506	1	1	Press the [PALLET START] switch	Unloading the inner pallet was attempted before removing workpieces from the outer pallet.	Remove workpieces from the outer pallet, and then press the [PALLET START] key.
SM7507	1	1	Different from pallet program	1. An attempt was made to restart a program that is different from the user parameter (quick table) settings. 2. The program number for pallet 1 or pallet 2 was not set and an attempt was made to start after changing the [PALLET] key setting to "1↔2."	1. Check the pallet program folder or the program number for each pallet in the user parameter (quick table), or the operation program. 2. Set both the program number for pallet 1 and pallet 2 in the user parameter (quick table), or don't configure the parameter setting.
SM7508	1	1	Spindle lock signal on	Spindle operation was executed when external input signal SPLOCK was ON.	Operate after turning off the input signal.
SM7509	1	1	ATC lock signal on	ATC operation was executed when external input signal ATCLCK was ON.	Operate after turning off the input signal.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM7510	1	1	XY-axes lock signal on	X- or Y-axis operation was executed when external input signal XYLOCK was ON.	Operate after turning off the input signal.
SM7511	1	1	Z-axis lock signal on	Z-axis operation was executed when external input signal ZLOCK was ON.	Operate after turning off the input signal.
SM7512	1	1	*-ax lock signal on	4-axis operation was executed when external input signal 4LOCK was ON.	Operate after turning off the input signal.
SM7513	1	1	*-ax lock signal on	5-axis operation was executed when external input signal 5LOCK was ON.	Operate after turning off the input signal.
SM7514	1	1	*-ax lock signal on	6-axis operation was executed when external input signal 6LOCK was ON.	Operate after turning off the input signal.
SM7515	1	1	*-ax lock signal on	7-axis operation was executed when external input signal 7LOCK was ON.	Operate after turning off the input signal.
SM7516	1	1	*-ax lock signal on	8-axis operation was executed when external input signal 8LOCK was ON.	Operate after turning off the input signal.
SM7517	1	1	Mode change prohibit signal on	Changing the mode was attempted when external input signal MDLOCK was ON.	Operate after turning off the input signal.
SM7518	1	1	Door locked	Open and close the automatic door when the door interlock is valid.	
SM7519	1	1	Area sensor obstructed	DR_ARSN_ERR signal was detected in the ladder . If the ladder has not been changed, the area sensor was obstructed during auto door operation.	Check the ladder.
SM7520	1	1	Manual operation mode	Recovery operation was attempted to be enabled during manual operation.	Enable recovery operation after manual operation is finished.
SM7521	1	1	No 4-axis option	4-axis operation was attempted although the machine is not equipped with this option.	
SM7522	1	1	No 5-axis option	5-axis operation was attempted although the machine is not equipped with this option.	
SM7523	1	1	No 6-axis option	6-axis operation was attempted although the machine is not equipped with this option.	
SM7524	1	1	No 7-axis option	7-axis operation was attempted although the machine is not equipped with this option.	
SM7525	1	1	No 8-axis option	8-axis operation was attempted although the machine is not equipped with this option.	
SM7526	1	1	During ATC	Changing to another mode was attempted during ATC.	Mode cannot be switched during tool change. Switch the mode after tool change operation is finished.
SM7527	1	1	During chamfering/corner R oper.	MDI operation was attempted during chamfering/corner R operations in memory operation.	Perform MDI operation after chamfering/corner R operations are finished.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM7528	1	1	Restarting invalid	Tape operation was attempted when [1: Repeat] is set for [Program repeat] (user parameter/switch 1).	Set [0: Stop] for [Program repeat].
SM7529	1	1	No MDI execution program	The [START] key was pressed in MDI mode although the MDI program is empty, or all the MDI data required for MDI start is not yet input (conversation type).	Input the MDI program or input the MDI data (conversation type).
SM7530	1	1	Outer pallet additional axis operating	The [PALLET START] switch was pressed while the outer pallet additional axis was operating.	Pallet cannot be turned during outer pallet additional axis operation. Operate after outer pallet additional axis operation is finished.
SM7531	1	1	There is no quick table.	Pallet index or manual operation/external start operation of external index was performed. (FT model)	These operations cannot be performed in FT model. If it occurred in QT model, check the machine parameter (system 2).
SM7532	1	1	[PALLET START] button pressed	The outer pallet additional axis switch was pressed while external start was reserved.	Cancel external start reservation, or use the axis after operation is completed.
SM7533	1	1	Indexing prohibit signal on	The outer pallet additional axis switch was pressed while external input signal OUTLCK was on.	Operate after turning off external input signal OUTLCK.
SM7534	1	1	Outer pallet operating	The outer pallet additional axis switch was pressed during memory operation on the outer pallet (e.g. pallet turn M code is used after starting memory operation).	Use the key after operation is completed.
SM7535	1	1	Cannot start from midpoint of prog.	Memory operation was executed for external pallet after start position was set for program restart.	Program cannot be restarted for external pallet.
SM7536	1	1	Stroke exceeded (+*)	The outer pallet additional axis has exceeded its stroke limit (+5).	Check the stroke of user parameter (switch 2) or the amount of outer pallet index.
SM7537	1	1	Stroke exceeded (-*)	The outer pallet additional axis has exceeded its stroke limit (-5).	Check the stroke of user parameter (switch 2) or the amount of outer pallet index.
SM7538	1	1	Stroke exceeded (+*)	The outer pallet additional axis has exceeded its stroke limit (+6).	Check the stroke of user parameter (switch 2) or the amount of outer pallet index.
SM7539	1	1	Stroke exceeded (-*)	The outer pallet additional axis has exceeded its stroke limit (-6).	Check the stroke of user parameter (switch 2) or the amount of outer pallet index.
SM7540	1	1	Stroke exceeded (+*)	The outer pallet additional axis has exceeded its stroke limit (+7).	Check the stroke of user parameter (switch 2) or the amount of outer pallet index.
SM7541	1	1	Stroke exceeded (-*)	The outer pallet additional axis has exceeded its stroke limit (-7).	Check the stroke of user parameter (switch 2) or the amount of outer pallet index.
SM7542	1	1	Stroke exceeded (+*)	The outer pallet additional axis has exceeded its stroke limit (+8).	Check the stroke of user parameter (switch 2) or the amount of outer pallet index.
SM7543	1	1	Stroke exceeded (-*)	The outer pallet additional axis has exceeded its stroke limit (-8).	Check the stroke of user parameter (switch 2) or the amount of outer pallet index.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM7544	1	1	Turn external program select off.	Program restart was attempted although the program number of external program selection signal was different from that of the program to be restarted.	Turn off the external program selection signal.
SM7545	1	1	Mirror image mode	MDI operation was attempted during mirror image mode in memory operation.	Execute MDI with mirror image mode canceled.
SM7546	1	1	Scaling	MDI operation was attempted during scaling in memory operation.	Execute MDI with scaling canceled.
SM7547	1	1	Compensating diameter	Axis travel command execution was attempted in MDI operation during cutter compensation in memory operation.	Execute MDI operation after cutter compensation is completed.
SM7548	1	1	Close the automatic door.	Memory operation was attempted when the machine is equipped with an automatic door, the [MANU/AUTO] switch is set to [AUTO], and the automatic door is not fully closed or open.	Perform the following and then start memory operation. 1. Set the [MANU/AUTO] switch to [MANU]. 2. Press the [OPEN/CLOSE] switch to open or close the door. 3. Set the [MANU/AUTO] switch to [AUTO].
SM7549	1	1	High accuracy A invalid command	Memory operation using high accuracy mode A was changed to MDI operation, and a command that cannot be used for high accuracy mode A was executed in MDI operation.	Cancel high accuracy mode A, and then perform MDI operation.
SM7550	1	1	Machine is operating.	Opening or closing the automatic door was attempted during operation (when the door should be locked).	Stop the machine, and then operate the automatic door.
SM7551	1	1	Additional axis operation prohibited range error	Additional axis turn was attempted in manual operation mode when the Z-axis was in the additional axis operation restricted range.	Move the Z-axis out of the restricted travel range for the additional axis.
SM7552	1	1	Program number error	Memory operation was attempted when program number was not set.	Check the program number of each pallet of the user parameter (quick table) or the program number of slave station communication.
SM7553	1	1	No program (*)	The designated program is not found. The number in brackets () indicates the designated program number.	Check the program number.
SM7554	1	1	Wrong tool type	ATC was attempted with different types of tools in manual operation mode.	Perform ATC with the same type of tools.
SM7555	1	1	Invalid pot	An invalid pot was specified when performing ATC.	Only a cap can be set to an invalid pot. Perform ATC with a pot other than invalid pots.
SM7556	1	1	Capped	ATC was attempted with a pot which is cap-designated.	Perform ATC with a pot which is not cap-designated.
SM7557	1	1	The PLC axis control signal is ON.	PLC axis manual operation was attempted when PLC axis control signal was ON.	Operate PLC axis after turning off the PLC axis control signal.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM7558	1	1	There is no P1-axis option.	Running an P1-axis program was attempted although the machine is not equipped with the P1-axis option.	
SM7559	1	1	There is no P2-axis option.	Running an P2-axis program was attempted although the machine is not equipped with the P2-axis option.	
SM7560	1	1	There is no P3-axis option.	Running an P3-axis program was attempted although the machine is not equipped with the P3-axis option.	
SM7561	1	1	There is no P4-axis option.	Running an P4-axis program was attempted although the machine is not equipped with the P4-axis option.	
SM7562	1	1	Communicating	Editing the program or data was attempted during communication of the same data.	Start editing when communication is completed.
SM7563	1	1	No direction data	The program was run without setting the automatic centering measurement direction.	Set the measurement direction.
SM7564	1	1	Setting data error	The data is not set for the required items.	Set the data correctly.
SM7565	1	1	Rotational transformation mode	MDI operation was attempted during coordinate rotation in memory operation.	Execute MDI with rotational transformation canceled.
SM7566	1	1	Time to change batteries	The battery voltage for maintaining the absolute encoder data is low.	While the power switch is turned ON, replace all the batteries in the battery compartment (GB1) on the left side of the control box.
SM7567	1	1	An external output command is being executed.	The mode was changed during data output via external output command.	Change the mode after data output via external output command is completed.
SM7568	1	1	Interrupt type macro execution is being prepared.	Mode switching was attempted from when interruption signal was detected until when interruption program was started.	Mode cannot be switched during interrupt type macro execution preparation. Switch modes after interrupt type macro execution preparation is finished.
SM7569	1	1	Interrupt type macro post-process being executed.	Mode switching was attempted from when interruption program execution was completed until when the program flow returned to original program.	Mode cannot be switched during post-processing of interrupt type macro. Switch modes during post-processing of interrupt type macro.
SM7570	1	1	Operation prohibition signal on magazine panel ON.	Reserved	Reserved
SM7571	1	1	The simple panel is being edited.	Reserved	Reserved
SM7572	1	1	Restart from the middle of the program prohibited.	Program restart from the middle was attempted when restart operation in middle of program is prohibited.	Check [Prohibit restart operation in middle of program] (user parameter/switch 1).

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM7575	1	1	Spindle rotating	1. An attempt was made to enable the recovery operation during spindle rotation. 2. The constant peripheral speed control command was issued during memory operation after spindle rotation in manual mode.	1. Enable the recovery operation after the manual operation is finished. 2. Issue the command after stopping the spindle, or issue the constant peripheral speed control command and the spindle rotation command on the same block.
SM7576	1	1	Command interrupted in high-accuracy mode B.	Travel command was interrupted during operation in high accuracy mode B.	Check the value of [Speed clamp time B] (machine parameter/high accuracy).
SM7577	1	1	There is no simple panel calibration data.	Reserved	Reserved
SM7578	1	1	There is no magazine panel calibration data.	Reserved	Reserved
SM7579	1	1	There is no teaching pendant calibration data.	Calibration data of the teaching pendant is not found.	Perform calibration.
SM7580	1	1	The simple panel calibration data is abnormal.	Reserved	Reserved
SM7581	1	1	The magazine panel calibration data is abnormal.	Reserved	Reserved
SM7582	1	1	The teaching pendant calibration data is abnormal.	Calibration data of the teaching pendant is not correct.	Perform calibration.
SM7583	1	1	Simple panel data error (*)	Reserved	Reserved
SM7584	1	1	Magazine panel data error (*)	Reserved	Reserved
SM7585	1	1	Teaching pendant data error (*)	An error was detected in teaching pendant data.	Please contact us if you encounter a situation.
SM7586	1	1	Carry out spindle run-in in manual operation mode.	Immediately after machine installation or long-time recess, the spindle run-in operation must be executed.	Change to the manual mode, and execute the spindle run-in operation.
SM7587	1	1	Remove the tool from the spindle.	With a tool loaded in the spindle, an attempt to execute the spindle run-in operation was made.	1. Check whether a tool is loaded in the spindle. If a tool is loaded in the spindle, remove the tool. 2. Check the ATC Tool screen.
SM7588	1	1	Step travel is excessive	Step travel was attempted when step travel of the rotary axis exceeded 1°.	Reduce step travel of the rotary axis to less than 1°.
SM7589	1	1	Lathe spindle is rotating	1. M410/M411 command was specified during lathe spindle rotation. 2. Recovery/inspection operation was attempted during lathe spindle rotation.	1. Specify after the lathe spindle is stopped. 2. Perform recovery/inspection operation after the lathe spindle is stopped.
SM7590	1	1	Synchr. tapping return restoration position error	Synchronized tap return was commanded when Z-axis coordinates are above the Z-axis return coordinates.	Check the Z-axis return coordinate set values.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM7591	1	1	Pallet turn speed was restricted	1. The table load weight exceeds the set value. 2. The viscosity of the lubricant has increased due to cold temperatures.	1. Make sure that the table load weight is not too heavy. 2. Press the [P.INDX] key in manual operation mode to turn the pallet approximately 10 times. 3. Contact us about the status of the error.
SM7592	1	1	Servo disabled for axis required for operation	1. An attempt was made to perform a JOG, STEP or handle operation while the servo was disabled for the selected axis. 2. An attempt was made to perform a spindle rotation or stop operation while the servo was disabled for the S-axis. 3. An attempt was made to perform a tool change operation while the servo was disabled for either the S-, X-, Y-, Z-, M- or AT-axis. 4. An attempt was made to perform a magazine forward or reverse rotation while the servo was disabled for the Z- or M-axis. 5. An attempt was made to perform a pallet turn operation while the servo was disabled for the Z- or 4th-axis. 6. An attempt was made to perform a machine zero return operation while the servo was disabled for the X-, Y-, Z-, 4th-, 5th-, 6th-, 7th-, 8th-axis.	《Arm type ATC》 1. Enable the servo for the selected axis. 2. Enable the servo for the S-axis. 3. Enable the servos for the S-, X-, Y-, Z-, M- and AT-axes. 4. Enable the servo for the M-axis. 5. Enable the servo for the Z- and 4th-axis. 6. Enable the servo for the X-, Y-, Z-, 4th-, 5th-, 6th-, 7th-, 8th-axis. 《Turret type ATC》 1. Enable the servo for the selected axis. 2. Enable the servo for the S-axis. 3. Enable the servos for the S-, Z- and M-axes. 4. Enable the servo for the M-axis. 5. Enable the servo for the Z- and 4th-axis. 6. Enable the servo for the X-, Y-, Z-, 4th-, 5th-, 6th-, 7th-, 8th-axis.
SM7593	1	1	3D smoothing section currently running	An attempt was made to change to another mode besides editing mode during operation even though the 3D smoothing section is currently running.	
SM7594	1	1	Enter start position.	Start position was not entered for the program restart and an attempt was made to start up.	Enter the start position for program restart.
SM7595	1	1	3D data is being smoothed.	An operation was performed that cannot be carried out while the 3D data is being smoothed.	Operate after the 3D smoothing operation is complete.
SM7596	1	1	ATC monitor. parameter setting - Start position error	The ATC monitoring parameter cannot be automatically set when the Z-axis machine coordinate is above the zero point.	Press the [RELSE] and [-Z] keys to move the machine coordinate down from the Z-axis zero point. Then, execute the automatic setting for the ATC monitoring parameter again.
SM7597	1	1	ATC monitoring parameter setting - Speed error	1. Rapid traverse override was not 100% when the automatic setting started for the ATC monitoring parameter. 2. The speed was changed while the ATC monitoring parameter was being automatically set.	After setting the rapid traverse override to 100%, execute the automatic setting for the ATC monitoring parameter again. Do not change the speed while the ATC monitoring parameter is being automatically set.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM7598	1	1	ATC monitoring parameter setting - Magazine error	1. A magazine number with a tool and a magazine number without a tool were not set. 2. The same number has been set for a magazine number with a tool and a magazine number without a tool.	Correctly set the magazine number with a tool and the magazine number without a tool.
SM7605	1	1	Feature coordinate manufacturing mode engaged	A command was specified that cannot be executed while in feature coordinate manufacturing mode. Ex: - An attempt was made to change modes to MDI operation mode. - An attempt was made to rotate the additional axis in manual mode. - An attempt was made to select the lathe spindle.	
SM7606	1	1	Side door lock restricted	An operation was performed to lock (such as from the [DOOR INTERLOCK] switch being enabled (“ ”)) the side door when the loading system is attempting to open the side door.	After the loading system closes the side door, perform the operation.
SM7608	1	1	Restricted travel range error for AT-axis operation	1. A jog feed, step feed or handle operation was carried out inside the AT-axis restricted travel range. 2. When the AT-axis stopped within the AT-axis restricted travel range, a jog feed, handle operation or a step feed- which does not rotate an amount exceeding the AT-axis restricted travel range- was carried out.	1. Operate the AT-axis to move it outside of the AT-axis restricted travel range. To move it through the AT-axis restricted travel range, use the step feed to rotate it an amount that exceeds the AT-axis restricted travel range. 2. Use the step feed to rotate it an amount that exceeds the AT-axis restricted travel range.
SM7609	1	1	Machining load could not be measured	Machining load could not be measured due to one of the following reasons. - Spindle override and feedrate override were set to 100% while the machining load is being monitored. - The function is disabled on the common parameter <Machining load monitor function>. - The machining load monitor function has been disabled by the TLDSWOFF signal.	Perform one of the following. - Set the spindle override and the feedrate override to 100%. Or, set the common parameter <Machining load monitor during override> to <1: Monitor>. - Set the common parameter <Machining load monitor function> to <1: Enable>. - Cancel the machining load monitor function from being disabled by the TLDSWOFF signal.

Chapter 11 Alarms

No.	Stop level	Reset level	Alarm message	Cause	Solution
SM7610	1	1	*-axis clamp retry attempt was made (*)	An attempt to re-clamp was made because it was out of position more than machine parameter (system 2) <7th-axis position check pulse> after the clamp was finished. The auxiliary number shows the number of retry attempts.	If this occurs often, contact the nearest BROTHER approved service dealer. If this alarm is triggered by a clamp command (M431, M445), it is possible that there is an impact from the machining load during the unclamp that is right before. We recommend decreasing the machining load or machining while clamped.
SM9000	3	2	Macro error	This is an alarm caused by the macro (system variables) alarm display function (the function that displays the alarm number and alarm message set in the program).	Check the program.
SM9300	1	1		This is an alarm caused by the macro (system variables) alarm display function (the function that displays the alarm message set in the program).	Check the program.

11.7.8 SL

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL0000	5	3	Machine param. setting error (SYS 1)	The following cause description is listed according to its auxiliary number: 1 - The "Tool storage capacity" is not set. Or, an incorrect value is set.	Set the machine parameters (System 1) to the correct values and turn the power on again.
SL0001	5	3	Memory is not formatted	The memory has never been cleared or the data is damaged.	Output all data to an external device and format the memory.
SL0002	5	3	Machine param. setting error (PLC)	The following description is divided by auxiliary number. 1- [P* axis brake] and [P* axis clamp mechanism] are inconsistent.	Correct machine parameter (PLC) and turn the power on again.
SL0003	5	3	Local CPU temperature error	Local CPU temperature exceeds 85 degrees.	Turn power off for thirty minutes before resuming operation. When recurring, the local PCB may be defective. Replace the LOCAL PCB.
SL0004	5	3	Local CPU temperature error	Local CPU temperature exceeds 110 degrees.	Turn power off for thirty minutes before resuming operation. When recurring, the local PCB may be defective. Replace the LOCAL PCB.
SL0005	5	3	The 3D smoothing function cannot be enabled.	The local PCB does not support the 3D smoothing function.	1. Disable the 3D smoothing function option. 2. Replace the local PCB.
SL0006	5	3	Additional axis configuration error	1. <Yes> is set for the <Optional 4th(/5th/6th/7th/8th)-axis> on a machine model without a QT and is restricted from adding 1 or more additional axes. 2. <Yes> is set for the <Optional 5th(/6th/7th/8th)-axis> on a machine model with a QT and is restricted from adding 1 or more additional axes. 3. <Yes> is set for 2 or more axes as the <Optional 4th(/5th/6th/7th/8th)-axis> on a machine model without a QT and is restricted from adding 2 or more additional axes. 4. <Yes> is set for 2 or more axes as the <Optional 5th(/6th/7th/8th)-axis> on one pallet side for a machine model with a QT and is restricted from adding 2 or more additional axes.	1. Adding additional axes is restricted. Check the machine parameter (system 2) setting for the <Optional 4th(/5th/6th/7th/8th)-axis>. 2. Contact a BROTHER Industries dealer.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL0007	5	3	Main CPU operating system mismatch	The machine identification signal does not match the CPU's operating system information.	The following cause description is listed according to its auxiliary number: 1. When the NC PCB is replaced and an alarm is triggered, use the <VUP2> command to perform the version update operation again. 2. Replace again with the new revised NC PCB. 3. If an alarm is triggered after replacing the NC PCB or after the version update operation, perform the force version update operation.
SL0008	5	3	High speed processing mode 2 could not be enabled	This machine model does not support high speed processing mode 2.	1. Disable the high speed processing mode 2 option.
SL0500	5	2	No machine parameter data	There is no machine parameter data.	Create the machine parameter and turn power off and then on again.
SL0501	5	2	No PLC axis option		
SL0502	5	2	W/o turning function		
SL0503	5	2	Ladder operation not possible (high-speed ladder 2)	The program is inside of a ladder that cannot be operated.	Configure the corresponding ladder program to only issue the END command.
SL0504	5	2	Special setting error (Rapid feed adjustment)	When <Adjustment> was set for the <Rapid feed special setting selection>, there are some items in the <Adjustment> parameter that have not been set for the special selection (rapid feed adjustment) screen.	1. Change the set value for the <Adjustment> parameter to the correct value, and press the [RST] key. 2. If the adjustment is not carried out, set the <Rapid feed special setting selection> to the <Standard> setting.
SL0505	5	2	Special setting error (Rapid feed special setting selection)	1. <Level 1>, <Level 2> or <Level 3> has been set for the <Rapid feed special setting selection>. <Level 1>, <Level 2> and <Level 3> cannot be set for the <Rapid feed special setting selection> on your machine.	Change the <Rapid feed special setting selection> to the <Standard> setting, and press the [RST] key.
SL0506	5	2	Special setting error (Rapid feed adj. - Lin. interp. type)	When <Adjustment> was set for the <Rapid feed special setting selection>, there are some items in the <Adjustment> parameter that have not been set for the special selection (Rapid feed adjustment - Linear interpolation type) screen.	1. Change the set value for the <Adjustment> parameter to the correct value, and press the [RST] key. 2. If the adjustment is not carried out, set the <Rapid feed special setting selection> to the <Standard> setting.
SL1500	4	3	User param. setting error (Tool break. detect)	The following description is divided by auxiliary number. 1- [Tool breakage detection option] and machine configuration are inconsistent.	Correct each parameter of the user parameter (tool breakage detection) and turn the power on again.
SL1501	4	3	Machine param. setting error	Machine parameter setting is incorrect.	Contact the nearest BROTHER approved service dealer.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL2000	4	2	No user parameter data	There is no user parameter data.	Create a user parameter, and then turn the power off and then on again.
SL2001	4	2	No tool in tool list	The tool to be used for machining is not registered in the tool list.	Press the [RST] key, and then set the tool in the tool list.
SL2002	4	2	Tap size error	The denominator of the inch screw is zero (0).	Press the [RST] key, and then set the tap size correctly.
SL2003	4	2	No workpiece zero data	There is no workpiece coordinate zero point data.	Create workpiece coordinate zero point data or input it from an external device.
SL2004	4	2	No tool data	There is no tool data.	Create tool data or input it from an external device.
SL2005	4	2	No macro variable data	There is no macro variable data.	Create macro variable data or input it from an external device.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL2006	4	2	Machine param. setting error (SYS 2)	The following cause descriptions are divided according to their auxiliary numbers: 1 - The "Optional 4th-axis" is set to "Not equipped" on the QT machine. 2 - The "4th-axis address" is not set to the "QT" address for the QT machine. 3 - The "5th (6th, 7th or 8th)-axis position" is set to "No pallet" on a machine model equipped with a QT. Or, the "5th (6th, 7th or 8th)-axis position" is configured to another setting besides "No pallet" on a machine model that is not equipped with a QT. 4 - The "4th-axis clamp mechanism" value and the "Quick table" value do not match. 5 - The "4th (5th, 6th, 7th or 8th)-axis address" value and the additional axis for the A, B or C address on the individual table are redundant. An additional axis for the 3rd-axis or higher is set to the individual table on a machine model equipped with a QT. 6 - The "Optional 5th (6th, 7th or 8th) -axis" value is set to more than two "lathe spindle(s)" on the individual table. 7 - "QT" is set to the "4th-axis address" on another machine besides the QT machine. 8 - When the <Optional 5th-axis> is configured to another setting besides <Yes (positioning only)> or when the <Optional 7th-axis> is configured to another setting besides <Yes (lathe spindle)> on a M140X1. 9 - The "Optional 5th (6th, 7th or 8th) -axis" on the QT device is set to "Yes (lathe spindle)." 10 - When the <Optional 5th-axis> is set to <No> or to <Yes (lathe spindle)>, or when the <Optional 7th-axis> is configured to another setting besides <Yes (lathe spindle)> on an M series machine model (besides M140X1).	Set the machine parameters (System 2) to the correct values.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL2007	4	2	User param. setting error (switch 2)	The following cause descriptions are divided according to their auxiliary numbers. 1 - The "4th-axis stroke control" is set to "No" on the QT machine. 2 - The "4th-axis stroke(-) ," the "4th-axis stroke(+)," "Stroke limit for 4th-axis(-)" and the "Stroke limit for 4th-axis(+)" are invalid on the QT machine. 3 - When the "Optional 5th (6th, 7th or 8th) -axis" is set to "Yes(lathe spindle)," the "5th (6th, 7th or 8th) -axis stroke control" is set to "Yes." 4 - On a M140X1, when the "Optional 5th (6th, 7th or 8th) -axis" is set to "Yes(positioning only)," the "5th (6th, 7th or 8th) -axis stroke control" is set to "No." 5 - The "4th (5th, 6th, 7th or 8th) -axis stroke control" is set to "No" on the additional axis which is set by the "Additional axis stroke*." 6 - The set value for the <External A(B)-axis indexing(-)/(+)> is out of the acceptable range when the <Optional 5th(6th, 7th or 8th)-axis> is set to <Yes (positioning only)>.	Set the user parameters (Switch 2) to the correct values.
SL2008	4	2	Specified memory range exceeded	The specified memory range has been exceeded.	Press the [RST] key.
SL2009	4	2	Left- or right-hand thread?	The torsion direction is not set in the tool list when tapping.	Press the [RST] key, and then set the torsion direction.
SL2010	4	2	Machine param. setting error (SYS 1)	The following cause descriptions are divided according to their auxiliary numbers: 2 - The wrong value is set for the <Spindle air blow/Tool wash control method>. 3 - The <Automatic oiling function> and the <Automatic greasing function> are set together. 4 - One of the values for <Marked position for X- (Y- or Z-) axis adjustment> is not set. 5 - The wrong value has been set for <Operation in ATC range>. 6 - The wrong value has been set for <Cyclone filter>. 7 - The wrong value has been set for <Tool wash start position>.	Set the machine parameters (System 1) to the correct values.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL2011	4	2	User param. setting error (switch 1)	The following description is divided by auxiliary number. 1. <Conversation/NC language change> is set to <0: Conversation>. It cannot be set due to the machine specification. 2. <Z-axis travel method during tool change> is set to <1: Type 2>. <1: Type 2> cannot be set on the machine being used. 3. The <Positioning method> is set to <1: Linear interpolation type - Positioning method 1> even though there are items that have not been set in the linear interpolation type positioning parameter for the machine parameter (system 1) or (system 2). Or, the <Positioning method> is set to <2: Linear interpolation type - Positioning method 2> even though there are items included that have not been set in the linear interpolation type positioning parameter for the machine parameter (system 1). 4. The <Automatic door error level> is set to <2: Level 3> or <3: Level 2> on a machine model not equipped with a QT. <2: Level 3> or <3: Level 2> cannot be set on a machine model not equipped with a QT. 5. <1: 8 Mbyte> has been set for <Program load size>. The local PCB is incompatible. 6. <No tool detection function (when ATC rises)> or <No tool detection function (when ATC falls)> is set to <1: Method 1> or <2: Method 2>. <1: Method 1> and <2: Method 2> cannot be set on the machine being used. 7. <Tool over key detection function> is set to <1: Method 1> or <2: Method 2>. <1: Method 1> and <2: Method 2> cannot be set on the machine being used. 8. <1: Enable> is set for the <Tool collision detection function during ATC>. <1: Enable> cannot be set on the machine being used.	1. Set <Conversation/NC language change> to <1: NC> and then press the [RST] key. 2. Set the <Z-axis travel method during tool change> to <0: Type 1> and then press the [RST] key. 3. Set the <Positioning method> to the correct value and then press the [RST] key. If there are items included that have not been set in the linear interpolation type positioning parameter for the machine parameter (system 1), then set the parameter to <0: Non-linear interpolation type positioning>. If all items have been set, then set the parameter to <0: Non-linear interpolation type positioning> or to <2: Linear interpolation type - Positioning method 2>. Set the user parameters (switch 1) to the correct values, and then press the [RST] key. 4. Set <Automatic door error level> to <0: Level 5> or <1: Level 4>, and then press the [RST] key. 5. Set <0: 2 Mbyte> for the <Program load size>, and turn ON the power again. 6. Set the <No tool detection function> to <0: Disable> and press the [RST] key. 7. Set the <Tool over key detection function> to <0: Disable> and press the [RST] key. 8. Set the <Tool collision detection function during ATC> to <0: Disable> and press the [RST] key.

No.	Stop level	Reset level	Alarm message	Cause	Solution
				9. <Tool missing detection function> is set to <1: Method 1> or <2: Method 2>. <1: Method 1> and <2: Method 2> cannot be set on the machine being used. 10. Another setting besides <0:M252 (high)> has been set for <Tap accel. setting>. The settings <1: M253 (medium)> and <2: M254 (low)> cannot be set on the machine being used. 11. A value is set that falls outside of the setting range for <ATC arm turn speed ratio 1 (M422)> or <ATC arm turn speed ratio 2 (M423)>. 12. The value set for <Number of blocks to check cutter compensation interference> is larger than the <Maximum blocks to check cutter compensation interference> in the machine parameter (system 1). Or, there is no set value for <Number of blocks to check cutter compensation interference>. 990. <Condition to offset tool length/tool position> is set to <1: Type 2> respectively.	9. Set the <Tool missing detection function> to <0: Disable> and press the [RST] key. 10. Set <0: M252 (high)> to the <Tap accel. setting>, and then press the [RST] key. 11. Set a value greater than the machine parameter (system 3) <Lower limit for ATC arm turn speed ratio>, and then press the [RST] key. 12. Set a value less than the <Maximum blocks to check cutter compensation interference> in machine parameter (system 1), and then press the [RST] key. 990. Set <Condition to offset tool length/tool position> to <0: Method 1>, and press the [RST] key.
SL2012	4	2	Too many tools assigned	The number of tools used in the program is larger than the number of tool pots.	Press the [RST] key, and then check the program.
SL2013	4	2	T-code range error	The T-command is not within the specified range.	Check the T-command in the program.
SL2014	4	2	Program too large	The total size of the main and sub programs has exceeded the capacity.	Adjust the total size of the main and sub programs, or use the main program only.
SL2015	4	2	No tool in spindle	Tap job was attempted in MDI mode when the tool was not attached to the spindle.	Press the [RST] key. Attach the tool to the spindle and then set this tool for [ATC tool] on the screen.
SL2016	4	2	No pitch data	NC: Pitches or threads are not specified although tapping is specified. Conversation: The pitch data is not set in the tool list when tapping.	Press the [RST] key, and then set the pitch data
SL2017	4	2	Editing	Editing data already being edited was attempted.	Check that the “During communication” mark is not displayed, and then press the [RST] key.
SL2018	4	2	There is no machine data 11 (SYSC89).	Machine data 11 (SYSC89) is not found.	Enter machine data 11 (SYSC89).
SL2019	4	2	No external I/O signal data	External I/O signal data is not found.	Create I/O signal data or enter it from outside.
SL2020	4	2	No communication parameter data	Communication parameter data is not found.	Create communication parameter data or enter it from outside.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL2021	4	2	Machine param. setting error (PLC)	The following cause description is listed according to its auxiliary number: 1 - The "P1 (P2, P3 or P4) install. position" is set to "Individual" on the QT machine. Or, the "P1 (P2, P3 or P4) install. position" is set to "Table 1" or "Table 2" on another machine besides the QT machine.	Set the machine parameters (PLC) to the correct values.
SL2022	4	2	No operation level data	There is no operation level data, or it is corrupt.	Create operation level data, or input data from an external source.
SL2023	4	2	Param. not set could not be configured to factory-set values.	During the firmware version update, the factory-set values could not be configured for the parameters which have not be set.	1. Set the appropriate values for the user parameter, machine parameter and communication parameter that are not set. 2. Insert the latest factory-set parameter file into the USB memory, and update the firmware again.
SL2024	4	2	No data for G/M code macro parameter		
SL2025	4	2	Parameter setting error for rotary fixture offset	The following cause description is listed according to its auxiliary number. When equipped with the rotary fixture offset function (option): 1. The <*Axis option> is set to an axis with the setting <None> for <Tilt axis*> or <Rotation axis*>. 2. The axis used for the <Tilt axis*> or <Rotation axis*> is set to another <Tilt axis*> or <Rotation axis*>. 3. The <*axis install. position> is different than the additional axis specified for the pair: <Tilt axis*> and <Rotation axis*>. 4. The <Forward direction for the coordinate system on the tilt axis*> and <Forward direction for the coordinate system on the rotation axis*> are set on the same plane.	Change the following parameters to the correct values, and press the [RST] key. 1-3. <Tilt axis> and <Rotation axis> 4. <Forward direction for the coordinate system on the tilt axis*> and <Forward direction for the coordinate system on the rotation axis*>

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL2026	4	2	User parameter setting error (high accuracy)	1. A value other than <0> is set for both user parameters (high accuracy (adjustment and accuracy specification)) <Machining mode> and <Accuracy specification mode>. 2. A negative value is set for <Smoothness> when <Standard mode> is set to <2. Rough> in the user parameter (high accuracy (adjustment)). 3. <0: OFF> is set for <Standard mode> when the <Machining mode> in the user parameter (high accuracy (adjustment)) is set to one of the following <7. Adjustment A>, <8. Adjustment B> or <9. Adjustment C>.	1. Both user parameters (high accuracy (adjustment and accuracy specification)) <Machining mode> and <Accuracy specification mode> cannot be enabled together. Only use one of them. 2. Set the <Smoothness> in the user parameter (high accuracy (adjustment)) to a value greater than 0, and then press the [RST] key. 3. Check the setting for <Standard mode> in the user parameter (high accuracy (adjustment)).
SL2029	4	2	Parameter set. error in feature coordinate set. function	The following cause descriptions are divided according to their auxiliary numbers: When equipped with the option: feature coordinate setting function 1. The axis used for the <Tilt axis*> or <Rotation axis*> is set to another <Tilt axis*> or <Rotation axis*>. 2. The <*axis install. position> is different than the additional axis specified for the pair: <Tilt axis*> and <Rotation axis*>. 3. The <Forward direction for the coordinate system on the tilt axis*> and <Forward direction for the coordinate system on the rotation axis*> are set on the same plane. 4. When the machine parameter (system 2) <4th-axis address> was set to <3: QT>, and <1: 4th-axis> was set to <Tilt axis*> or <Rotation axis*>.	1.2. Set the user parameters (feature coordinate setting) <Tilt axis> and <Rotation axis> to the correct values, and the press the [RST] key. 3. Set the user parameters (feature coordinate setting) <Forward direction for the coordinate system on the tilt axis*> and <Forward direction for the coordinate system on the rotation axis*> to the correct values, and the press the [RST] key. 4. Set the machine parameter (system 2) <4th-axis address> and the user parameters (feature coordinate setting) <Tilt axis*> and <Rotation axis*> to the correct values, and the press the [RST] key.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL2030	4	2	User parameter setting error (quick table)	The following cause descriptions are divided according to their auxiliary number. 1. <1:At 1st tool change> is set to <Pallet 1 loading> or <Pallet 2 loading>. When R650X2 40 MG is equipped with an option to extend the turning diameter and <Move to safe position when turning pallet> (machine parameter (system 1)) is set to <1: Type 2>, the parameter cannot be set to <1:At 1st tool change>.	1. Set <0: At operation start> for <Pallet 1 loading> or <Pallet 2 loading>, and then press the [RST] key.
SL2031	4	2	Machine param. setting error (SYS 3)	The following cause descriptions are divided according to their auxiliary numbers: 1 - The wrong value has been set for <Return angle 1 for AT-axis> or <Return angle 2 for AT-axis>.	Set the machine parameters (System 3) to the correct values.
SL4000	3	2	Mark not found	Delimiters in the data have been lost.	1. Press the [RST] key. Delete the data, and then edit new data. 2. The NC PCB is defective. The CF is defective.
SL4001	3	2	Tool changed during operation	The tool data of the tool being used was changed.	Press the [RST] key, and then restart the program.
SL4002	3	2	Maintenance notice setting error	The setting for [Type] on the <Maintenance notice> screen is not within the range.	Check the setting on the screen, and set the type again.
SL4003	3	2	Required data not found	There is no specified data.	Check the program number or data bank number.
SL4004	3	2	Not formatted.	An input/output, deletion or memory change command was issued for an unformatted memory card.	Press the [RST] key and then format the memory card.
SL4005	3	2	Tool automatic assignment error	Tool assignment was not completed due to either of the following reasons when automatic assignment was performed. 1. Required tool not found 2. ? cannot be calculated	1. Start operation after checking the machining data, tool pattern, tap drill diameter, cutting conditions, and tool list, and correcting them if necessary. 2. Start operation after checking the machining data and correcting them if necessary.
SL4006	3	2	The tool is different from the one used before.	When [Check tools to be used when assigned automatically] (user parameter/switch 1) was set "Yes", a tool different from the previously used one was assigned and operation was attempted.	Assign a tool while checking the tool on Program edit - Operation preparation.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL4007	3	2	Error of the kind of screw thread	1. A program whose thread type is different from the machine was attempted to be operated. 2. A program for old machine model was attempted to be operated without tool reallocation.	1. Check the setting of [Change screw type] (machine parameter/system 1) and operate using a program whose thread type is appropriate to the machine. 2. Operate after tool reassignment.
SL4008	3	2	The machining data is corrupted.	Destroyed machining data was attempted to be operated or edited	Delete the machining data.
SL4009	3	2	Tool allocation failed when operation started	At automatic tool assignment, -9998 assignment (to give priority to current tools) could not be performed.	1. Start operation after checking the machining data, tool pattern, tap drill diameter, cutting conditions, and tool list, and correcting them if necessary. 2. Start operation after checking the machining data and correcting them if necessary.
SL4010	3	2	W/o smooth path offset function	The [Smooth path offset function] (user parameter) is enabled, but the "Smooth path offset" option has not been purchased.	Disable [Smooth path offset function] (user parameter/high accuracy) or purchase the option.
SL4011	3	2	Backup memory saving error	Backup memory could not be saved due to its defectiveness.	1. Restore the backup memory. 2. After clearing the backup memory, enter all data. 3. The NC PCB is defective.
SL4012	3	2	Ladder program could not be written.	1. An attempt was made to write a ladder program, which had been input with external input/output (Data bank), while the ladder program was writing with PLC editor. 2. An attempt was made to write a ladder program which had a command that could not be used in the current settings.	1. Press the [RST] key and write it after the writing with PLC editor is completed. 2. Check / correct the commands so that they can be used in the ladder program and in the <Expanded function range> on the PLC menu screen. Then, press the [RST] key and input the ladder program again.
SL4013	3	2	Press the [RST] key	Multiple alarms were triggered during the reset operation.	Press the [RST] key.
SL4014	3	2	No applicable sequence	1. The sequence number which was specified in subprogram call/return command is not found. 2. The sequence number which was specified in macro command (GOTO) is not found. 3. The sequence number which was specified in program restart (sequence search) is not found.	Press the [RST] key, and the check the program.
SL4015	3	2	Memory overflow (Memory card)	Insufficient memory.	Press the [RST] key. Replace the memory card.
SL4016	3	2	External output file cannot be created	The maximum number of external output files already exists in the memory card/FTP server.	Press the [RST] key. Check the external output files in the memory card/FTP server.
SL4017	3	2	3D smoothing data version error	This version does not support 3D smoothing data.	After pressing the [RST] key, check the set values on the <3D smoothing preparations> screen, and execute the 3D smoothing data operation.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SL4018	3	2	3D smoothing program error	3D smoothing data does not match the program.	After pressing the [RST] key, execute the 3D smoothing data operation.
SL4019	3	2	3D smoothing parameter error	3D smoothing data does not match the parameter.	After pressing the [RST] key, check the set values on the <3D smoothing preparations> screen, and execute the 3D smoothing data operation.
SL4020	3	2	3D smoothing path data error	There is a 3D smoothing path data error.	After pressing the [RST] key, execute the 3D smoothing data operation.
SL5500	2	2	Local CPU temperature error	Local CPU temperature exceeds 82 degrees for one minute.	Turn power off for thirty minutes before resuming operation.
SL5501	2	2	G code macro duplicate error	A G code number is duplicated and registered in the G/M code macro.	Check for and delete any G code numbers that are duplicated in the G code macro screen.
SL5502	2	2	M code macro duplicate error	An M code number is duplicated and registered in the G/M code macro.	Check for and delete any M code numbers that are duplicated in the M code macro screen.
SL5503	2	2	Local CPU temperature error	Local CPU temperature exceeds 98 degrees for ten seconds.	Turn power off for thirty minutes before resuming operation.
SL7000	1	2	Memory overflow	Insufficient directory or data area	Decrease the program size when editing the program, or delete other programs to increase memory space, and then retry operation. Press the [RST] key.
SL7001	1	2	Battery has expired. (Local)	Local of back up battery has expired.	1. Set the current date in the user parameter (switch 1). 2. Check the battery and replace it.
SL7002	1	2	Machine data (SYSC89) input is invalid.	Machine parameters could not be entered.	Contact the nearest BROTHER approved service dealer.
SL7003	1	2	There is no machine data (SYSC89).	Machine data 11 (SYSC89) is not found.	Enter machine data 11 (SYSC89).
SL7004	1	2	Machine unit system switching error	Machine unit system switching was failed.	Check that the data bank has abnormal error.
SL7005	1	2	In special operation level	Special operation level is valid.	Contact the nearest BROTHER approved service dealer.
SL7006	1	2	Program restart error from saved position	The return height operation could not be carried out to the saved position due to the operation status information. Example when return height operation is not possible - Call level of the sub program was changed - Call number for the call source in the sub program was changed - Saved block can no longer execute because macro variable value was changed	Press the [RST] key and perform one of the following. - Correct the program. - Specify the execution pointer, or restart the program that specifies the sequence number.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7007	1	2	ATC monitoring parameter has not been set	The automatic setting for the ATC monitoring parameter has not been executed. Or, the ATC monitoring setting is not correct.	1. Contact BROTHER. 2. Set the following user parameters (switch 1) <No tool detection function (when ATC rises)>, <No tool detection function (when ATC falls)>, <Tool over key detection function> and <Tool missing detection function> to <0: Disable>.
SL7500	1	1	Input data error	Data set was not within the specified range.	Set data within the range.
SL7501	1	1	No copy source data	There is no program or data bank to be copied.	Check the program number or data bank number.
SL7502	1	1	Data exists at copy destination.	The program or data bank to be copied already exists.	Copy to a different number, or delete the existing program or data bank.
SL7503	1	1	Data is write-protected.	1. Changing data was attempted while the [DATA PROTECTION] switch was set to [ON]. 2. Ladder program edit was attempted while "PLC ladder program protection" (user parameter/switch 1) is set to "1: On".	1. Set the [DATA PROTECTION] switch to [OFF], and then change the data. 2. Set "PLC ladder program protection" (user parameter/switch 1) to "0: Off" and edit the ladder program.
SL7504	1	1	Too many characters	The number of characters in one block exceeds 128.	Reduce the number of characters.
SL7505	1	1	Running program	Editing or deleting a file used for the program currently running was attempted.	Edit or delete the program after operation is completed.
SL7506	1	1	Check error	A check error occurred during input in check mode via communication.	
SL7507	1	1	Editing area exceeded	Contents being edited have exceeded the edit area (20M byte).	If this error occurs during editing, split the program. If this error occurs at the start of editing, the program is too large and cannot be edited.
SL7508	1	1	This tool is already registered.	The tool set for [ATC tool] is already registered for another pot.	Delete the registered tool, and then set the desired tool.
SL7509	1	1	Too many tools registered	Registering more than 30 tools in one tool group was attempted.	Up to 30 tools can be registered in one group.
SL7510	1	1	Tool is registered to next pot.	Setting a large tool was attempted although a tool is registered to the adjacent pot.	Delete the tool registered to the adjacent pot, or register the large tool to another pot.
SL7511	1	1	Deletion position error	Deleting data was attempted at a position where is deletion not possible.	Move the cursor to a position where deletion is possible, and then delete the data.
SL7512	1	1	Prog. edit prohibit signal on	Changing data was attempted although the program edit prohibit signal was on.	Turn the signal off, and then change the data.
SL7513	1	1	Edit prohibit signal on	Changing data was attempted although the edit prohibit signal was on.	Turn the signal off, and then change the data.
SL7514	1	1	Set position error	The data was set to a position where data setting is not possible.	The data cannot be set for that position.
SL7515	1	1	No subprogram	The sub program is not set although it is necessary for the program to be run.	Set the sub program, or delete the sub program call job.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7516	1	1	No tools assigned	Tool assignment has not been completed.	Assign the tools.
SL7517	1	1	Machining sequence setting error	Machining sequence is not 1 to 5, or it is not set.	Set the machining sequence correctly.
SL7518	1	1	No program	There is no program to be run.	Check the program number.
SL7519	1	1	Too many digits	Data set exceeds the limit for the number of digits.	Set the data within the limit for the number of digits.
SL7520	1	1	Has a minus sign	Negative data was set where not permitted.	Set positive data.
SL7521	1	1	Copy import position error	Copy import was attempted at an incorrect position.	Move the cursor to the correct position (job position), and perform copy import.
SL7522	1	1	No search data	There is no data to be searched for.	Check the data.
SL7523	1	1	Too many jobs	Creating a 100th job was attempted.	Create the program using up to 99 jobs.
SL7524	1	1	Specified subprogram exists.	The sub program you are trying to create already exists.	Create the sub program using another number, or delete the sub program from the current number and use this number to create the sub program.
SL7525	1	1	Change position error	Changing data was attempted at a position changing is not possible.	Move the cursor to the correct position, and then change the data.
SL7526	1	1	Cannot delete during insertion	Deleting a program was attempted during insertion.	Press the [INS] key again to stop insertion, or delete the program after insertion is completed.
SL7527	1	1	Insertion position error	Inserting a program was attempted at a position insertion is not possible.	Move the cursor to the correct position, and then insert the program.
SL7528	1	1	Cursor movement position error	The cursor was moved to a different position during program insertion.	Insert all data, and then move the cursor.
SL7529	1	1	Copy insertion position error	Copy insertion was attempted at a position insertion is not possible.	Move the cursor to the correct position, and then copy and insert the data.
SL7530	1	1	Too many operations	A 100th operation was created when editing the operation program.	Create the program using up to 99 operations.
SL7531	1	1	Too many lines	A 100th line was created when editing a contouring or chamfering job.	Create the program using up to 99 lines.
SL7532	1	1	Workpiece zero set position error	Setting the workpiece zero point was attempted at the incorrect position.	Move the cursor to the X position of the workpiece zero point, and then set the zero point.
SL7533	1	1	Cannot search during insertion	Searching was tried during insertion.	Press the [INS] key again to stop insertion, or start searching after insertion is completed.
SL7534	1	1	Too many schedules	More than 99 programs were set for a schedule program.	Create the schedule program using up to 99 programs.
SL7535	1	1	Copy not imported	Copy was not imported although copy insertion was attempted.	Import the data to be copied.
SL7536	1	1	Too many workpieces	More than 99 workpieces were inserted.	Create the program using up to 99 workpieces.
SL7537	1	1	No tool in tool list	A tool that is not registered in the tool list was set to the magazine.	Register the tool in the tool list.
SL7538	1	1	Too many points	The number of point random drilling holes has exceeded 12.	The number of drilling holes should be within 12.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7539	1	1	No ATC return data	Referring to the ATC return data was attempted although it does not exist.	
SL7540	1	1	Data error (program)	A program with a different unit system was set during editing or editing in operation.	Set the program with the same unit system.
SL7541	1	1	Wrong tool length	The tool length becomes 0 or less when automatically setting the tool length.	Check [Automatic tool length compensation] (user parameter /switch 1).
SL7542	1	1	No tool list data	There is no tool list data.	Set the tool list.
SL7543	1	1	No tool pattern data	There is no tool pattern data.	Set the tool pattern.
SL7544	1	1	No tapping drill diameter data	There is no tap drill diameter data.	Set the tap drill diameter.
SL7545	1	1	No cutting condition data	There is no cutting condition data.	Set the cutting condition.
SL7546	1	1	Workpiece material setting error	Workpiece material is not set.	Set the workpiece material.
SL7547	1	1	Required tool not found	There is no tool that corresponds to the desired machining when assigning tools.	Check the machining data, tool pattern, tap drill diameter, cutting conditions, and tool list, and correct them if necessary.
SL7548	1	1	Too many tools	The number of assigned tools has exceeded the magazine capacity.	Change the program so that the number of assigned tools is within the magazine capacity.
SL7549	1	1	No job	The program does not contain jobs.	Check the machining data.
SL7550	1	1	No data in tool list	There is no tool data specified in the tool list.	Set data for the tool number specified in the tool list.
SL7551	1	1	Tool type error	The specified tool is not appropriate for machining.	Set the appropriate tool.
SL7552	1	1	Tap size error	The denominator of the inch screw is zero (0).	Set the tap size correctly.
SL7553	1	1	There are jobs without tools.	There are jobs with tools not set for the tool pattern when assigning tools.	Set the tool to the tool pattern, and then assign the tools.
SL7554	1	1	Illegal tool pattern	Tools required for machining are not set in the tool pattern when assigning tools.	Check the tool pattern, and correct it.
SL7555	1	1	? cannot be calculated.	Question mark cannot be calculated.	Correct the machining data.
SL7556	1	1	Depth too shallow	The tip of the tool contacts the bottom of the hole when the tool stops after drilling the center hole or chamfering.	Check the machining data and tool list, and correct them if necessary.
SL7557	1	1	Counter boring depth too shallow	The tool contacts the counter boring face when chamfering.	Check the machining data and tool list, and correct them if necessary.
SL7558	1	1	Wrong workpiece height	The maximum tool infeed depth calculated from the workpiece height is lower than the table surface.	Check the machining data and tool list, and correct them if necessary.
SL7559	1	1	Minor diameter too small	The minor diameter of the tool is too small for the machining data.	Check the machining data and tool list, and correct them if necessary.
SL7560	1	1	No tool length setting	The tool length is not set or is zero.	Check the machining data, tool list, tool pattern, cutting conditions, etc., and correct them if necessary.
SL7561	1	1	No tool life setting	The assigned tool is expired.	
SL7562	1	1	Tool too long	The tool length is too long.	

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7563	1	1	Minor diameter too large	The minor diameter is too large.	
SL7564	1	1	Wrong minor diameter	The minor diameter is not correct.	
SL7565	1	1	Minor diameter length too long	The minor diameter length is too long.	
SL7566	1	1	Wrong minor diameter length	The minor diameter length is not correct.	
SL7567	1	1	Minor diameter length too short	The minor diameter length is too short.	
SL7568	1	1	Tool length too short	The tool cannot reach the infeed depth.	
SL7569	1	1	Tool diameter too small	The tool diameter is too small.	
SL7570	1	1	Wrong tool diameter	The tool diameter is not correct.	
SL7571	1	1	Wrong center angle	The center angle is not correct.	
SL7572	1	1	Wrong chamfer angle	The chamfer angle is not correct.	
SL7573	1	1	Length of teeth too short	The length of teeth is too short.	
SL7574	1	1	Wrong torsion direction	The tap torsion direction is not correct.	
SL7575	1	1	Wrong thread type	The tap type is not correct.	
SL7576	1	1	Wrong thread size	The tap size is not correct.	
SL7577	1	1	Wrong pitches or threads	The tap pitch or thread is not correct.	Check the machining data, tool list, tool pattern, cutting conditions, etc., and correct them if necessary.
SL7578	1	1	Drilling depth not enough	The drilling depth is not deep enough.	Check the machining data, tool list, tool pattern, cutting conditions, etc., and correct them if necessary.
SL7579	1	1	No tool number	Tool check was attempted with the cursor placed in a section where the tool number is not entered.	Enter the tool number or move the cursor to a section where the tool number is entered.
SL7580	1	1	Tool diameter too large	The tool diameter is too large.	Check the machining data, tool list, tool pattern, cutting conditions, etc., and correct them if necessary.
SL7581	1	1	Pocket milling not performed	Chamfering was attempted before pocket milling. (Corner rounding cutter is at the top of the program.)	
SL7582	1	1	Interfered (side)	The tool contacts the side of the workpiece when chamfering.	
SL7583	1	1	Interfered (bottom)	The tool contacts the bottom of the workpiece when chamfering.	
SL7584	1	1	Eye hole tool is attached.	This tool cannot be used for a blind hole.	Check the machining data, tool list, tool pattern, cutting conditions, etc., and correct them if necessary.
SL7585	1	1	Tool not matching cutting cond.	The tool set for tool assignment differs from the tool set for the cutting condition.	
SL7586	1	1	Corner R too large	Corner R is larger than half the shorter axis of X or Y in the pocket milling square pattern.	
SL7587	1	1	Approach radius too small	The approach radius (in program) is less than the tool radius in the chamfering job.	Check the program.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7588	1	1	Wrong machine unit system	A program with a different unit system was set during editing or editing in operation.	Set the program with the same unit system.
SL7589	1	1	No auto workpc. measurement func.	Automatic workpiece measurement setting was attempted although the machine is not equipped with this option	Purchase the option.
SL7590	1	1	No threading function	Helical thread cutting was attempted although the machine is not equipped with this option.	Purchase the option.
SL7591	1	1	No subprogram (*)	The specified sub program does not exist. The figure in brackets () is the specified program number.	Set the sub program, or delete the sub progr
SL7592	1	1	Subprogram number error	1. The sub program number is not written correctly in the program. 2. The sequence number to be called is not written.	Press the [RST] key, then check the sub program number or the sequence number in the program.
SL7593	1	1	Tool number not set	Setting the group number was attempted although the tool number is not set on the <ATC tool> screen.	Set the tool number and then set the group number.
SL7594	1	1	Too many blocks	The copied or cut section is too large.	Reduce the copy or cut section.
SL7595	1	1	Memory overflow	1. All data areas or directories are in use. 2. There is insufficient memory for the firmware version update.	Decrease the program size when editing the program, or delete other programs or data banks and waveform data to increase memory space, and then retry operation.
SL7596	1	1	No tool registered in group	The tool is not registered to the tool group specified in the program.	Check the tool group setting.
SL7597	1	1	No recovery instruction data	There is no recovery instruction data provided for the specified alarm.	
SL7598	1	1	Program is write-protected.	Programs from 8000 to 8999 are write-protected.	Release the write-protection.
SL7599	1	1	Cannot be copied.	Copying the data to the current data bank was attempted.	Copy the data to a number other than the number of the current data bank.
SL7600	1	1	No item selected.	The program or data bank number is not selected.	Set the number before input or output.
SL7601	1	1	Item exists at input destination.	Input was attempted although there is an item selected on the NC side.	Unselect the item before input.
SL7602	1	1	Item exists at output destination.	Output was attempted although there is an item selected on the output destination side.	Unselect the item before output.
SL7603	1	1	Cannot be deleted.	Cannot be deleted.	
SL7604	1	1	Not formatted.	An input/output, deletion or memory change command was issued for an unformatted memory card.	Format the memory card.
SL7605	1	1	Invalidate checksum.	[1: On] is set for [Checksum] (communication parameter) when a multiple-program file is input.	Set [0: Off] for [Checksum].

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7606	1	1	This program cannot be converted.	The selected program cannot be converted by G-code conversion.	
SL7608	1	1	Workpiece quantity setting error	The number of workpieces is not set when assigning tools.	Set the number of workpieces.
SL7609	1	1	No step data	Zero (0) is set for the step condition (one of the cutting conditions) or it is not set when converting a conversation program to an NC program.	Set the cutting condition correctly.
SL7610	1	1	? not calculated	Question mark calculation is not completed when converting a conversation program to an NC program.	Check that the line data is set correctly, and then assign the tools.
SL7611	1	1	No B-axis option	1. Running the B-axis program was attempted when converting a conversation program to an NC program, although the machine is not equipped with the B-axis option. 2. Automatic B-axis setting was attempted at the workpiece coordinate zero point although the machine is not equipped with the B-axis option.	
SL7612	1	1	Next tool in magazine not available	The next tool is not set to the magazine when converting a conversation program to an NC program.	Set the next tool to the magazine.
SL7613	1	1	Wrong data type	Running a different type of program was attempted when converting a conversation program to an NC program.	Check the program.
SL7614	1	1	No tool in spindle	The tool is not attached to the spindle when converting a conversation program to an NC program.	Set the correct tool to the spindle.
SL7615	1	1	Specified G code not found	The specified G-code is not found when converting a conversation program to an NC program.	Check the program.
SL7616	1	1	Mark not found	Delimiters in the data have been lost when converting a conversation program to an NC program.	Delete the data, and create the data again.
SL7617	1	1	Program too large	Machining data size is larger than the buffer capacity when converting a conversation program to an NC program.	Split the machining program. Split programs can be run sequentially using the schedule program function.
SL7618	1	1	No A-axis option	1. Running an A-axis program was attempted when converting a conversation program to an NC program, although the machine is not equipped with the A-axis option. 2. Automatic A-axis setting was attempted at the workpiece coordinate zero point although the machine is not equipped with the A-axis option.	

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7619	1	1	0 holes (*)	When converting a conversation program to an NC program, "0" was specified for the hole pattern. The figure in brackets () is the job number.	Set the number of holes.
SL7620	1	1	Tap size error (*)	When converting a conversation program to an NC program, the denominator of the inch screw is zero (0). The figure in brackets () is the job number.	Set the tap size correctly.
SL7621	1	1	0 divisions (*)	When converting a conversation program to an NC program, "0" was specified for pattern divisions. The figure in brackets () is the job number.	Set the number of divisions correctly.
SL7622	1	1	Cutter compensation error (*)	When converting a conversation program to an NC program, the arc command is used to start or cancel cutter compensation. The figure in brackets () is the job number.	Use the linear command start or cancel cutter compensation.
SL7623	1	1	No pitch data (*)	When converting a conversation program to an NC program, zero (0) is set for the tapping pitch data in the tool list or it is not set. The figure in brackets () is the job number.	Set the pitch data.
SL7624	1	1	Spindle speed not specified (*)	When converting a conversation program to an NC program, [Peripheral speed] (cutting condition) is not set or the value is too small. The figure in brackets () is the job number.	Set the cutting condition correctly.
SL7625	1	1	Feedrate not specified (*)	When converting a conversation program to an NC program, [Feed] (cutting condition) is not set or the value is too small. The figure in brackets () is the job number.	Set the cutting condition correctly.
SL7626	1	1	Position cannot be calculated (*)	When converting a conversation program to an NC program, the specified value exceeds +-9999.999. The figure in brackets () is the job number.	Check the program.
SL7627	1	1	Too many lines (*)	When converting a conversation program to an NC program, the number of lines exceeds 99. ("0" is set for [Cutting width] or the value is too small, or the tool diameter is too small). The figure in brackets () is the job number.	Set [Cutting width] and [Tool dia.] correctly.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7628	1	1	Too many infeeds (*)	When converting a conversation program to an NC program, the infeed setting is too small. (99 is the maximum). The figure in brackets () is the job number.	Set [Infeed] correctly.
SL7629	1	1	Corner cannot be calculated (*)	When converting a conversation program to an NC program, corner R or corner C for contouring or chamfering cannot be calculated. The figure in brackets () is the job number.	Correct the corner R or C section where the alarm occurred.
SL7630	1	1	Calculation error (*)	When converting a conversation program to an NC program, 1.The additional axis feed rate could not be calculated. 2.The machining path could not be calculated. 3.The tool data could not be calculated for tool assignment. The figure in brackets () is the job number.	1.2.Check the program, and set the correct data. 3. Check the tool list, and set the correct data.
SL7631	1	1	Pocket milling not possible (*)	When converting a conversation program to an NC program, the size of the pocket is not correct. 1.The tool diameter is larger than the pattern diameter for circle patterns. 2.The tool diameter is larger than the shorter stroke of X or Y for square patterns. 3.The tool radius is larger than the corner R for square patterns. (Excluding when corner R = 0). 4.The tool diameter is larger than the track width for track patterns. The figure in brackets () is the job number.	Use a tool whose diameter is smaller than the pot diameter.
SL7632	1	1	Chamfering not possible (*)	When converting a conversation program to an NC program, there is a chamfering function at the top of the pocket milling job. The figure in brackets () is the job number.	Assign the functions so that chamfering comes after milling.
SL7633	1	1	ATC not possible (*)	When converting a conversation program to an NC program, there is a tool change code in a miscellaneous function before or after cutting, or in a miscellaneous function in the motion call job. The figure in brackets () is the job number.	Delete the tool change code.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7634	1	1	Too many holes (*)	When converting a conversation program to an NC program, the number of holes in the pattern exceeds 999. The figure in brackets () is the job number.	Set the number of holes again.
SL7635	1	1	Spindle speed exceeded (*)	When converting a conversation program to an NC program, the specified spindle speed exceeded the setting for [Max. spindle speed] (machine parameter 1). The figure in brackets () is the job number.	Check the program.
SL7636	1	1	Feedrate exceeded (*)	When converting a conversation program to an NC program, the feed rate exceeded the setting for [Max. cutting feedrate] or [Rapid feedrate] (machine parameter 1). The figure in brackets () is the job number.	Set the correct data.
SL7637	1	1	Cutter compensation too large (*)	When converting a conversation program to an NC program, the program radius is smaller than the cutter compensation when cutting the inner arc. The figure in brackets () is the job number.	Set the program radius to be larger than the cutter compensation.
SL7638	1	1	Invalid approach specified (*)	When converting a conversation program to an NC program, [Center] is set for the cutter compensation method of line or arc approach. The figure in brackets () is the job number.	Correct the approach method.
SL7639	1	1	Invalid release specified (*)	When converting a conversation program to an NC program, [Center] is set for the cutter compensation method of line or arc release. The figure in brackets () is the job number.	Correct the release method.
SL7640	1	1	No cutting length (*)	When converting a conversation program to an NC program, there is no cutting length for the approach or release line. The figure in brackets () is the job number.	Correct the program.
SL7641	1	1	Not a valid B-axis program (*)	When converting a conversation program to an NC program, a B-axis command was used in a program that does not contain a B-axis. The figure in brackets () is the job number.	Change the program.
SL7642	1	1	No axis specified (*)	When converting a conversation program to an NC program, the axis data is not set. The figure in brackets () is the job number.	Set the axis data and then use the travel command.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7643	1	1	Invalid axis specified (*)	When converting a conversation program to an NC program, simultaneous 3-axes cutting was attempted using an axes other than X, Y, and Z, or simultaneous 4- or more axes cutting was attempted. The figure in brackets () is the job number.	Simultaneous 3-axes cutting is not possible when an additional axis is included.
SL7644	1	1	B-axis radius not specified (*)	When converting a conversation program to an NC program, the radius is not set for cutting using the B-axis. The figure in brackets () is the job number.	Specify the radius.
SL7645	1	1	No ATC return height (*)	When converting a conversation program to an NC program, the ATC return height is not registered in the program. The figure in brackets () is the job number.	Set the ATC return height.
SL7646	1	1	B-axis command range exceeded (*)	When converting a conversation program to an NC program, the specified travel range of the B-axis was not within 0 ~ 359.999. The figure in brackets () is the job number.	Check the program.
SL7647	1	1	Thread cutting speed error (*)	When converting a conversation program to an NC program, the Z-axis speed could not be calculated from the XY-axes feed rate and radius for helical thread cutting. The figure in brackets () is the job number.	Check the end point, center, and feed rate for thread cutting.
SL7648	1	1	No thread cutting function (*)	When converting a conversation program to an NC program, helical thread cutting was attempted although the machine is not equipped with this option. The figure in brackets () is the job number.	Purchase the option.
SL7649	1	1	Tool breakage detection pos. error (*)	When converting a conversation program to an NC program, tool breakage detection was attempted at a position below the return height. The figure in brackets () is the job number.	Check [Tool breakage detection return position] (user parameter 1) or lower the setting for [Return height (job change)] and [Return H (coord. change)].
SL7650	1	1	Rotational transformation not possible (*)	When converting a conversation program to an NC program, the XY positions after rotational transformation cannot be calculated because the X- or Y-axis machining coordinate positions are not determined. The figure in brackets () is the job number.	Correct the program, and retry operation. (Set both XY travel positions in the first motion call job in the program.)

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7651	1	1	A-axis radius not specified (*)	When converting a conversation program to an NC program, the radius is not set for cutting using the A-axis.	Specify the radius.
SL7652	1	1	A-axis command range exceeded (*)	When converting a conversation program to an NC program, the specified travel range of the A-axis was not within 0 ~ 359.999. The figure in brackets () is the job number.	Check the program.
SL7653	1	1	Not a valid A-axis program (*)	When converting a conversation program to an NC program, an A-axis command was used in a program that does not contain an A-axis. The figure in brackets () is the job number.	Change the program.
SL7654	1	1	Cannot measure in incremental mode (*)	When converting a conversation program to an NC program, automatic workpiece measurement was attempted in incremental mode. The figure in brackets () is the job number.	Change to absolute mode, and then perform automatic workpiece measurement.
SL7655	1	1	Cannot measure during RT (*)	When converting a conversation program to an NC program, automatic workpiece measurement was attempted in rotational transformation mode. The figure in brackets () is the job number.	Cancel rotational transformation, and then perform automatic workpiece measurement.
SL7656	1	1	Cannot measure in cutter comp. mode (*)	When converting a conversation program to an NC program, automatic workpiece measurement was attempted in cutter compensation mode. The figure in brackets () is the job number.	Cancel cutter compensation, and then perform automatic workpiece measurement.
SL7657	1	1	No auto workpc. measurement func.(*)	When converting a conversation program to an NC program, automatic workpiece measurement setting was attempted although the machine is not equipped with this option. The figure in brackets () is the job number.	Purchase the option.
SL7658	1	1	Tool type error (*)	When converting a conversation program to an NC program, the specified tool is not appropriate for machining. The figure in brackets () is the job number.	Set the tool appropriate for machining.
SL7659	1	1	No tool length setting (*)	When converting a conversation program to an NC program, the tool length is not set or is zero. The figure in brackets () is the tool number.	Check the machining data, tool list, tool pattern, cutting conditions, etc., and correct them if necessary.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7660	1	1	Outside diameter of drill not set (*)	When converting a conversation program to an NC program, the outer diameter of the drill to be used or the minor diameter of drills are not set in the tool list or are zero. The figure in brackets () is the tool number.	Set the drill outer diameter in the tool list.
SL7661	1	1	Left- or right-hand thread? (*)	When converting a conversation program to an NC program, the torsion direction for tapping is not set in the tool list. The figure in brackets () is the tool number.	Set the torsion direction.
SL7662	1	1	Tool life end (*)	When converting a conversation program to an NC program, the life of the current tool or the tool being replaced has expired. The figure in brackets () is the tool number.	Change the tool, and set the tool life.
SL7663	1	1	No tool in tool list (*)	When converting a conversation program to an NC program, the tool used for machining is not registered in the tool list. The figure in brackets () is the tool number.	Set the tool in the tool list.
SL7664	1	1	There are jobs without tools. (*)	When converting a conversation program to an NC program, and when assigning tools, there are jobs where there is no tool set for the tool pattern. The figure in brackets () is the job number.	Set the tool for the tool pattern, and then assign tools.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7665	1	1	Conversation/NC conversion error(*)	An error occurred when converting a conversation program to an NC program. The figure in brackets () indicates the cause. 6: An error occurred when calculating the additional axis speed. 7: Calculating the additional axis speed is not possible because the position of the additional axis is not determined. 9: An error occurred when opening, reading, or writing the file. 10: An error occurred while opening, reading or writing the data from or to the temporary file. ThThe number of files has exceeded the limit (1024 files) because temporary files were created. 14: Measurement number was used although measurement operation was not performed. 15:Memory is insufficient. 16: A motion where the position cannot be determined was used, meaning the data cannot be converted correctly. This error mainly occurs when tapping or drilling after zero point return because the return point is not fixed.	Check the program and correct it if necessary. 6. Check the program. 7. Perform positioning to determine the position. 9. Contact the nearest BROTHER approved service dealer. 10. Delete other programs to increase free space, and then retry operation. 14. Include a measurement motion in the program before using the measurement number. 15. Reduce the size of the program you are editing and then quit the program or delete other programs to increase free space and then retry operation. 16. Insert a positioning operation between the zero machine return operation and the tapping operation, and set the position.
SL7667	1	1	Same number already specified	Input/output was attempted by specifying the program from the list as well by the program number.	Deselect one before operation.
SL7668	1	1	[No] is set for [Parameter change]	Operation is not possible when [0: Write-protected] is set for [Parameter change].	Set [1: Writable] for [Parameter change] on I/O screen.
SL7669	1	1	Extended memory operation mode	Editing a file related to the program running in extended memory operation mode was attempted.	Complete operation before editing.
SL7670	1	1	Too many subprograms	The number of sub programs to be called exceeds the limit.	Reduce the number of sub programs.
SL7671	1	1	No C-axis option	Automatic C-axis setting was attempted at the workpiece coordinate zero point although the machine is not equipped with the B-axis option.	
SL7672	1	1	Start position cannot be set	The start point is not within the setting range.	Check the program.
SL7673	1	1	Memory operation data not created	Memory operation data is not created.	Assign tools.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7674	1	1	No schedule program	The specified schedule program does not exist.	Check the schedule program number.
SL7675	1	1	Spindle speed not specified	The spindle speed is not set although a rotation command is used in MDI mode.	Set the spindle speed, and then use the rotation command.
SL7676	1	1	No axis specified	The axis data is not set although a travel command is used in MDI mode.	Set the axis data, and then use the travel command.
SL7677	1	1	A-axis command range exceeded	The A-axis angle setting is not correct.	Check the program.
SL7678	1	1	B-axis command range exceeded	The B-axis angle setting is not correct.	Check the program.
SL7679	1	1	Invalid axis command	Simultaneous 3-axes cutting was attempted using axes other than X, Y, and Z, or simultaneous 4- or more axes cutting was attempted.	Simultaneous 3-axes cutting is not possible when an additional axis is included.
SL7680	1	1	Feedrate not specified	The feed rate is not specified although a cutting command is used in MDI mode.	Set the feed rate, and then use the cutting command.
SL7681	1	1	Radius not set for A-ax cutting	The radius is not specified when cutting using the A-axis.	Specify the radius.
SL7682	1	1	Radius not set for B-ax cutting	The radius is not specified when cutting using the B-axis.	Specify the radius.
SL7683	1	1	Z-axis not specified for tapping	Tapping command was used without setting the Z-axis data in MDI mode.	Set the Z-axis.
SL7684	1	1	XYAB-axes specified for tapping	Tapping was applied by setting the X, Y, A, and B axes in MDI mode.	The X, Y, A, and B axes cannot be used for tapping.
SL7685	1	1	Feed not specified	The feed type is not set in MDI mode.	Set the feed type.
SL7686	1	1	Ladder program write-protected	Ladder program was attempted to be input with external input/output (Data bank) while "PLC ladder program protection" (user parameter/switch 1) is set to "1: On".	Set "PLC ladder program protection" (user parameter/switch 1) to "0: Off".
SL7687	1	1	Communication format error (header/footer)	NC program with header/footer was attempted to input while "header/footer" is set to [0: Not equipped] for communication parameter, or NC program without header/footer was attempted to input while "header/footer" is set to [1: Equipped].	Check the communication parameter or NC program.
SL7688	1	1	Specified M code cannot be used. (*)	An M code that cannot be used (next tool preparation) was found when conversation program machining data was converted to G-code. The figure in brackets () indicates the process number.	Check the program.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7689	1	1	Cannot move to ATC pos. during diameter compensation(*)	The tool change position of the X or Y axis is set in the program currently running in compensation mode. (The tool change position is set on the operation preparation screen.) The figure in brackets () is the process number.	Cancel diameter compensation.
SL7690	1	1	Cannot move to ATC pos. during rotational transformation(*)	This error occurs when converting a conversation program to an NC program. The tool change position of the X or Y axis is set in the program currently running in rotational transformation mode. (The tool change position is set on the operation preparation screen.) The figure in brackets () is the process number.	Cancel rotational transformation.
SL7691	1	1	There are miscellaneous func. that cannot be converted(*)	There are miscellaneous functions (M1, M2, M6, M98, M99) that cannot be converted when converting a conversation program to an NC program. The figure in brackets () is the process number.	Delete the miscellaneous functions causing the error.
SL7692	1	1	There is no tool for initializing tool life	Although initializing the tool life has been executed on the ATC tool screen, there are no expired tools.	
SL7693	1	1	Editing tool list	Initializing the tool life was attempted on the ATC tool screen during editing the tool list.	Complete the editing before initializing the tool life.
SL7694	1	1	Cannot be initialized	Although initializing the tool life has been attempted on the ATC tool screen, it cannot be initialized because the initial life has not been set. (This error occurs only when [Type 1] is set for [Tool life counting method] in user parameter.)	Set the tool life on the tool list.
SL7695	1	1	Addition not possible	Addition was attempted for the value displayed as "*****" on the macro variables screen.	
SL7696	1	1	Alarm history detail data writing error (*)	Output of alarm log detail file was failed.	Contact the nearest BROTHER approved service dealer.
SL7697	1	1	Alarm history detail data reading error	Reading of alarm log detail file was failed.	Contact the nearest BROTHER approved service dealer.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7698	1	1	There is no alarm history detail data.	1. The alarm history details have been omitted and were not recorded due to the large quantity of alarms that occurred simultaneously. 2. No file exists with a detailed description for the alarm that has been selected.	1. Refer to the detailed description of another alarm that occurred almost at the same time. 2. Report the conditions when the error or alarm occurred.
SL7699	1	1	The alarm history detail data is corrupted.	Alarm log detail file to be read is destroyed.	Contact the nearest BROTHER approved service dealer.
SL7700	1	1	Alarm history detail access error (*)	An error occurred during operation related to alarm log detail file. The number in brackets () indicates the cause. Numbers and details are as follows. 0. A file was attempted to be output with the same name as that of an existing alarm log detail file.	Contact the nearest BROTHER approved service dealer.
SL7701	1	1	This is an unusable folder name.	Unusable or existing folder name was specified when a folder was created.	Specify a valid folder name. The following folder names cannot be used. - Other than usable characters (alphanumeric characters and period) - A period is used for the first or last character, or consecutive periods exist. - Only numeric characters are used. - 0 program name - Conversation program name - Extension is ".NC", ".TC", ".LDR", or ".CMT".
SL7702	1	1	T-code area error	T command is out of the designated range.	Check T commands in the program.
SL7703	1	1	No folder can be made.	1. A folder was attempted to be created in a folder. 2. A backup folder could not be created during upgrade operation.	
SL7704	1	1	The same program number cannot be copied.		
SL7705	1	1	The same program number cannot be checked.		
SL7706	1	1	Transfer to the same folder is unavailable.		
SL7707	1	1	Copy to and check of same folder not available.		
SL7708	1	1	A program is already existing.		
SL7709	1	1	This is a program number of different type.	The specified program number exists in another unit system.	Specify another program number, or delete a program with the same number which exists in another unit system.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7710	1	1	Edit cancelled due to error of a comment sentence.	The comment sentence of a program being edited could not be saved because it was not correct. 1. Mode switching from program edit mode was performed. 2. Mode switching was performed while a program which was not being operated was edited in edit during operation mode. 3. Mode switching was performed while a program which was being operated was edited in edit during operation mode with [Save when mode of current program is changed] (user parameter) set to [1:Overwrite save].	
SL7711	1	1	Automatic setting items lack.	[Cutter compensation], [Peripheral speed (m/min)], and [Feed per rotation (mm/rotation)] are not set correctly.	Set correct data.
SL7712	1	1	Automatic setting calculation is unable.	The sum of [Cutter compensation] and [Cutter wear offset] is 0 or less.	Set correct data.
SL7713	1	1	Communication parameter setting error	The following description is divided by auxiliary number. 1- [Sending data code] is [EIA] and [Character length] is [7 bits]. 2- The same value is set to [Port No.] and [PLC port number]. 3- The same value is set to [IP address] and [DNS server address 1]/[DNS server address 2].	Set correct data.
SL7714	1	1	Program folder error	A program was set which was not from the folder that is linked to [Pallet program folder] (user parameter/quick table) when the pallet selection was not turned "OFF" while in memory operation mode.	Check [Pallet program folder] (user parameter/quick table).
SL7715	1	1	This is a folder name that cannot be input.	A folder with an unusable name was attempted to be entered.	Change the folder name of the connected device to the one that can be entered, or select a folder that can be entered.
SL7716	1	1	No folder can enter the current folder position.	When the current folder position was not the root, a folder was attempted to be entered.	A folder cannot be entered in a folder. Set the current folder position on NC side as the root.
SL7717	1	1	There was a subfolder unable to input.	A folder including a subfolder was attempted to be entered.	The subfolder was not entered. Move the current folder on the connected device side and reenter the subfolder.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7718	1	1	Interrupted program number error	The program number called by the interrupt type macro is invalid.	Correct the program so that it can call the correct program number.
SL7719	1	1	Zero return not performed	Coordinate value setting was attempted using automatic setting when dog type machine zero return was not completed.	Perform machine zero return in manual operation mode.
SL7720	1	1	No tool in magazine	A tool required for operation is not set in the magazine.	Set the tool on the <ATC tool> screen correctly.
SL7721	1	1	No measurement data	Automatic setting (parameter) was attempted although there is no measurement data.	Perform measurement using the automatic centering function.
SL7722	1	1	No auto thermal compen. func.	Opening the compensation data screen was attempted although the machine is not equipped with the automatic thermal compensation function.	
SL7723	1	1	Tape operation mode	Master station communication was attempted during tape operation.	Perform communication after tape operation is completed.
SL7724	1	1	No Z-axis measurement function	This machine is not equipped with Z-axis measurement function.	This machine is not equipped with this function.
SL7725	1	1	Not all items are set	All items required to determine the display position for drawing are not entered.	Set all items.
SL7726	1	1	Check the tool in the spindle.	Already NC tool was set for spindle when setting machining start point for Z axis in conversation mode.	Set to spindle for conversation tool or eliminate the tool of the spindle.
SL7727	1	1	Scheduled program running	Schedule program change was attempted during schedule program operation.	Change the program after schedule program or stop it.
SL7728	1	1	Program number error	The specified program number cannot be set.	Set the correct program number.
SL7729	1	1	No program (*)	The specified program does not exist. The figure in brackets () is the specified program number.	Check the program number.
SL7730	1	1	Invalid pot	An invalid pot was specified when performing ATC.	Only "Cap" can be set for the invalid pot.
SL7731	1	1	Different from pallet program	A program different from the one set for the user parameter (quick table) was run in memory operation mode.	Check the user parameter (quick table) for the program number for each pallet.
SL7732	1	1	This is a folder name that cannot be checked.	Check was not performed because the folder name was unusable.	Change the folder name of the connected device to the one that can be checked, or select a folder that can be checked.
SL7733	1	1	There was a subfolder that cannot be checked.	A folder including a subfolder was attempted to be checked.	The subfolder was not checked. Move the current folder on the connected device side and recheck the subfolder.
SL7734	1	1	There was a file that cannot be removed.	There is a file other than program (machining data) in the target folder.	Delete on the Folder operation screen. In NC language mode, delete again after pressing [Display of all files].
SL7735	1	1	Custom screen data abnormal (ID) (Simple/mag.)	Reserved	Reserved

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7736	1	1	Custom screen data abnormal (Arr. pos) (Simp/mag)	Reserved	Reserved
SL7737	1	1	Custom screen data abnormal (Dupl arr) (Simp/mag)	Reserved	Reserved
SL7738	1	1	Custom screen data abnormal (Ref addr) (Simp/mag)	Reserved	Reserved
SL7739	1	1	Custom screen data abnormal (Wrt addr) (Simp/mag)	Reserved	Reserved
SL7740	1	1	Custom screen data abnormal (Set area) (Simp/mag)	Reserved	Reserved
SL7741	1	1	Custom screen data abnormal (No. items) (Simp/mag)	Reserved	Reserved
SL7742	1	1	Cust screen data abnormal (No. letters) (Simp/mag)	Reserved	Reserved
SL7743	1	1	Custom screen data abnormal (No. pages) (Simp/mag)	Reserved	Reserved
SL7744	1	1	Cust screen data abnormal (No. controls)(Simp/mag)	Reserved	Reserved
SL7745	1	1	Custom screen data abnormal (ID) (Teach)	A control identifier was found before the first page identifier was found. An unrecognizable identifier was found.	Correct and enter the data on the Custom screen of the teaching pendant.
SL7746	1	1	Custom screen data abnormal (Arrang. pos.) (Teach)	The set value of starting point line number/starting point column number is out of range. Control which was larger than 1x1 in size lay offscreen.	Correct and enter the data on the Custom screen of the teaching pendant.
SL7747	1	1	Custom screen data abnormal (Dupl arrang.) (Teach)	Groups or controls other than groups overlap one another.	Correct and enter the data on the Custom screen of the teaching pendant.
SL7748	1	1	Custom screen data abnormal (Ref. address) (Teach)	Reference address of button/lamp is out of range. Reference address of digital switch/data display is out of range. Reference address was set to an odd-numbered address when data width was 32 bits.	Correct and enter the data on the Custom screen of the teaching pendant.
SL7749	1	1	Custom screen data abnormal (Writ. address)(Teach)	Notification address/write address is out of range. Write address was set to an odd-numbered address when data width was 32 bits.	Correct and enter the data on the Custom screen of the teaching pendant.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7750	1	1	Custom screen data abnormal (Setting area) (Teach)	The set value of size/dot/data width is out of range. Dot was set to 24 dots when the size of the control was not 2x2, 2x4, 3x3, or 3x5. Data width was set to 32 bits when the size of the control was not 1x2 or 2x4.	Correct and enter the data on the Custom screen of the teaching pendant.
SL7751	1	1	Custom screen data abnormal (No. of items) (Teach)	Setting items are insufficient.	Correct and enter the data on the Custom screen of the teaching pendant.
SL7752	1	1	Custom screen data abnormal (No. letters) (Teach)	The number of characters of the title exceeds 30 one-byte characters. Total number of characters in character strings for the group arranged on one page exceeds 128 one-byte characters. Total number of characters in character strings for the control besides the group arranged on one page exceeds 256 one-byte characters.	Correct and enter the data on the Custom screen of the teaching pendant.
SL7753	1	1	Custom screen data abnormal (No. of pages) (Teach)	There are nine or more page identifiers.	Correct and enter the data on the Custom screen of the teaching pendant.
SL7754	1	1	Custom screen data abnormal (No. controls) (Teach)	There are 16 or more groups to be arranged on one page. There are 16 or more controls other than groups to be arranged on one page.	Correct and enter the data on the Custom screen of the teaching pendant.
SL7755	1	1	External alarm setting error	Cancel level 1 or 4 was set for the external alarms of Stop level 3 to 5. The specified cancel level is treated as level 2.	Set correct data.
SL7756	1	1	W/o external I/O signal key function		
SL7757	1	1	Special operation level is released		
SL7758	1	1	Version update file is not found.	The version update file is not found in the memory card.	Store any of the following version update files in the memory card. - Firmware file - Servo amplifier firmware file - Help data file - Firmware file for teaching pendant
SL7759	1	1	Firmware files are duplicated.	There are multiple servo amplifier firmware files for one servo amplifier in the memory card.	Store only the latest servo amplifier firmware file in the memory card.
SL7760	1	1	Memory overflow (Memory card)	Insufficient memory.	Replace the memory card.
SL7761	1	1	Set value outside of setting range	A set value was entered that falls outside of the setting range.	Check the set value or the setting range.
SL7762	1	1	No G code macro program (*)	The program that is called in the G code macro does not exist in the <G code macro program folder>.	1. Check the program. 2. Check the <Program No.> in the <G code macro>.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7763	1	1	No M code macro program (*)	The program that is called in the M code macro does not exist in the <M code macro program folder>.	1. Check the program. 2. Check the <Program No.> in the <M code macro>.
SL7764	1	1	G code macro program No. not set	The program No. is not set for the G code macro that is used in the program.	Check the <Program No.> in the <G code macro>.
SL7765	1	1	M code macro program No. not set	The program No. is not set for the M code macro that is used in the program.	Check the <Program No.> in the <M code macro>.
SL7766	1	1	Interrupt type macro command not possible	An M code command that cannot be issued was issued.	Check the program.
SL7767	1	1	Memory operation mode	An operation which cannot be done during the memory operation mode was performed.	Perform the operation after the memory operation is finished.
SL7768	1	1	MDI operation mode	An operation which cannot be done during the MDI operation mode was performed.	Perform the operation after the MDI operation is finished.
SL7769	1	1	No subprogram in the G code macro folder (*)	The specified subprogram is not in the G code macro folder. The program No. was not in the parenthesis ().	Set the G code macro program, or delete the G code macro program call block.
SL7770	1	1	No subprogram in the M code macro folder (*)	The specified subprogram is not in the M code macro folder. The program No. was not in the parenthesis ().	Set the M code macro program, or delete the M code macro program call block.
SL7771	1	1	Tool assignment error (*)	When converting a conversation program to an NC program, a job was added for which the tool cannot be used because the tool assignment was forced. The figure in brackets () is the job number.	Delete the corresponding data, and assign the tool again.
SL7772	1	1	User param. setting error (switch 3)	1. A modal with a large number of characters was set for one of the following user parameters (switch 3): <G code 4, 8, 12 and 16 for modal display> or <M code 4, 8, 12 and 16 for modal display>. 2. The next position for the modal display with a large number of characters has not been set to <0: Hide>.	1. Check the setting for one of the following user parameters (switch 3): <G code 4, 8, 12 and 16 for modal display> or <M code 4, 8, 12 and 16 for modal display>. 2. Set to <0: Hide> for the next position of the modal display with a large number of characters.
SL7773	1	1	Version updated Turn power OFF	After the version was updated, an attempt was made again to update the version.	Turn the power back ON and update the version again.
SL7774	1	1	3D data not smoothed	A program with 3D data that is not smoothed was selected in memory operation mode.	Execute the 3D smoothing data operation.
SL7775	1	1	3D smoothing data load error	An attempt was made to execute the 3D smoothing data operation even though 1, 2, 3, 102, 103, 202 and 203 were registered to the <G code No.>, or even though 380 to 387 and 389 were registered to the <M code No.> in the <G/M code macro parameter>.	Delete <G code No.> 1, 2, 3, 102, 103, 202 and 203 and <M code No.> 380 to 387 and 389 are registered in the <G/M code macro parameter>.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7776	1	1	3D data cannot be smoothed	An attempt was made to execute the 3D smoothing data operation even though the <M code No.> 380 through 387 and 389 were registered in the <G/M code macro parameter>.	Delete the <M code No.> 380 through 387 and 389 that are registered in the <G/M code macro parameter>.
SL7777	1	1	3D smoothing unit system error	1. When the 3D smoothing data operation was executed, one of the following does not match: the current machine unit system, program unit or minimum unit setting. 2. The current machine unit system is different from the unit when the 3D smoothing data operation was saved.	1. Execute the 3D smoothing data operation. 2-1. Change the user parameter (switch 1) <Machine unit system>. 2-2. Delete the 3D smoothing data and set the 3D smoothing parameter.
SL7778	1	1	Auto. start pos. settings for 3D smoothing could not set.	1. The tool path simulation function could not be used to reach the specified 3D smoothing start code. 2. The 3D smoothing start position could not be identified.	
SL7779	1	1	The 3D smoothing function cannot be enabled.	The local PCB does not support the 3D smoothing function.	After replacing the local PCB, enable the 3D smoothing function.
SL7780	1	1	There is no 3D smoothing function.	1. A program was selected on the <Program list (3D smoothing)> screen even though there is no 3D smoothing option. 2. A program was selected that included 3D smoothing start/end code, even though there is no 3D smoothing option.	Purchase the option.
SL7781	1	1	3D smoothing data version error	This version does not support the 3D smoothing data. Too many items were ignored, and items in the wrong range were displayed as "not set".	Check the set values on the <3D smoothing preparations> screen, and execute the 3D smoothing data operation.
SL7782	1	1	3D smoothing parameter setting error	One of the set values in the 3D smoothing parameter is outside of the range.	Check the set values on the <3D smoothing preparations> screen.
SL7783	1	1	Comm. parameter setting error (Auto notification function)	The following cause descriptions are divided according to their auxiliary numbers: 1- The combination of auto notification function parameters <Type of sending data>, <Initial address for sending data> and <Sending data length> is invalid. 2- When setting a value from 3 to 5 for the <Stop level for error when sending data is specified indirectly>, even if the <Cancel level for error when sending data is specified indirectly> is set to "1", it is processed as a cancel level 2.	Set the data with the correct parameters.

No.	Stop level	Reset level	Alarm message	Cause	Solution
SL7784	1	1	External output file cannot be created	The external output file for the maximum capacity already exists in the memory card.	Check the external output file in the memory card.
SL7785	1	1	Feature coordinate manufacturing mode engaged	An operation was performed that cannot be executed while in feature coordinate manufacturing mode. Ex: - An attempt was made to change the workpiece coordinate zero in the data bank.	

11.7.9 CM

No.	Stop level	Reset level	Alarm message	Cause	Solution
CM4000	3	2	DR signal off	DR signal turned off during communication.	Check the communication parameter or the state of the connected device.
CM4001	3	2	Connected to wrong device	1. Tape operation was attempted with "1: NC COMM device" selected for "Connected to" as the external I/O. 2. When the communication parameter "External output - Output destination" was set to "0: Master stat. output destination", an attempt was made for an external output command while the external I/O was "Connected" to "1:NC COMM device".	1. Change the external I/O connection setting so that it is "Connected" to "0: General COMM device" or "2: Memory card". 2. Change the external I/O to another connection setting so that it is not "Connected" to "1:NC COMM device". Or, change the communication parameter "External output - Output destination" to another setting that is not "0: Master stat. output destination".
CM4002	3	2	Press [RST] after COMM recovery	Communication was attempted while resetting the communication port.	Press the [RST] key after communication recovery time has elapsed.
CM4003	3	2	Communication timeout	The non-response time has exceeded the response monitoring time.	Check the communication parameter or the state of the connected device.
CM4004	3	2	Received invalid data	Invalid data was received when not converting invalid data.	Check the communication parameter or data.
CM4005	3	2	Communication format error	Communication header, data, or footer was not recognized.	Check the communication parameter or data.
CM4006	3	2	Communication TV check error	The number of characters in one block was not correct when the program was received.	Check the communication parameter or data.
CM4007	3	2	CS signal off	The CS signal is off when sending flow control DC1 and DC3 codes.	Check the communication parameter or the state of the connected device.
CM4008	3	2	Receive buffer overflow	Receiving buffer has overflowed.	Check the communication parameter or the state of the connected device.
CM4009	3	2	Communication framing error	Stop bit of the receiving character was not detected.	Check the communication parameter.
CM4010	3	2	Communication parity error	Parity error occurred in the receiving character.	Check the communication parameter.
CM4011	3	2	Communication overrun error	Characters are missing from received data.	Check the communication parameter.
CM4012	3	2	Communicating	1. Tape operation was attempted in master communication mode 2. Tape operation was attempted again although communication was not completed after tape operation (general communication device) ended due to M30. 3. External output was instructed in master communication mode.	1. Quit the external program I/O menu and then perform tape operation. 2. Wait until communication is completed, and then perform tape operation. 3. Quit the external program I/O menu and then begin external output.
CM4013	3	2	Press the [RST] key after communication is completed.	Communication parameter could not be reflected in slave communicating when the [RST] key was pressed.	Press the [RST] key after slave communicating is completed.

No.	Stop level	Reset level	Alarm message	Cause	Solution
CM4014	3	2	Communication format error (header/footer)	Entering an NC program with header/footer was attempted when [0: No] is set for [Header/footer] (user parameter / communication), or entering an NC program without header/footer was attempted when [1: Yes] is set.	Check the communication parameter or the NC program.
CM4015	3	2	The memory card cannot be identified.	The memory card reader is not connected.	1. Press the [RST] key and then check that the memory card is correctly installed in the drive. 2. Press the [RST] key and then check that the memory card is not damaged. 3. If several storage devices are connected to the USB ports, connect only one, and then press the [RST] key. 4. LOCAL PCB is defective. Replace LOCAL PCB.
CM4016	3	2	Memory card error	1. An error associated with the memory card occurred. 2. The memory card is used by another function.	1. Press the [RST] key and then recheck the memory card. 2. After pressing the [RST] key and the other functions are complete, execute the operation again.
CM4017	3	2	Multiple memory cards cannot be connected.	Multiple memory devices (e.g. memory card reader) are connected to the USB port.	Connect only one memory device to the USB port and then press the [RST] key.
CM4018	3	2	Unusable format (Memory card)	The format of the floppy disk or memory card is not supported.	Press the [RST] key and then connect a formatted memory device.
CM4019	3	2	Different from pallet program	A program different from the one set for the user parameter (quick table) was used for tape operation.	Check the user parameter (quick table) for the program number of each pallet or operation program number.
CM4020	3	2	POPEN is unable.	BPRNT or DPRNT command was sent although POPEN command was not sent.	Press the [RST] key, and then check the program.
CM4021	3	2	Write protect (Memory card)	Output, deletion, or format was used for a write-protected memory card.	Press the [RST] key. Release the write-protection on the memory card.
CM4022	3	2	Folder for connected device not found	When inputting and outputting data via an FTP connection, the folder for the connected device could not be found.	Check the folder for the connected device.
CM4023	3	2	Cannot connect to the server	1. An invalid FTP server name is set. 2. There is a problem with the DNS server or network environment.	1. Check the host name for the connected device. Set the correct FTP server name. 2. Check the status of the network cable or the connected device and execute again.
CM7000	1	2	Ladder file under edit/communication exists	1. External editing software operates ladder file. 2. Some ladder files have not been written yet after the files were entered via communication.	1. Perform communication again after operation with external editing software is completed. 2. Perform communication again after pressing the [RST] key and writing ladder files.
CM7500	1	1	Editing communication data	The data to be sent is being edited.	Send the data after editing is completed.

No.	Stop level	Reset level	Alarm message	Cause	Solution
CM7501	1	1	Communication parameter error	[1. EIA] is set for [Sending data code] when starting maintenance communication. [0: 7 bits] is set for [Character length] and [1. EIA] is set for [Sending data code] when communicating at port A or B.	Check the communication parameter.
CM7502	1	1	DR signal off	The DR signal turned off during communication.	Check the communication parameter or the state of the connected device.
CM7503	1	1	CS signal off	The CS signal is off when sending flow control DC1 and DC3 codes.	Check the communication parameter or the state of the connected device.
CM7504	1	1	Receive buffer overflow	Receiving buffer has overflowed.	Check the communication parameter or the state of the connected device.
CM7505	1	1	Communication parity error	Parity error occurred in the received characters.	Check the communication parameter.
CM7506	1	1	Communication overrun error	Characters are missing from received data.	Check the communication parameter.
CM7507	1	1	Communication framing error	The stop bit of the received characters could not be detected.	Check the communication parameter.
CM7508	1	1	Communication timeout	The non-response time has exceeded the response monitoring time.	Check the communication parameter or the state of the connected device.
CM7509	1	1	Communication format error	Communication header, data, or footer was not recognized.	Check the communication parameter or data.
CM7510	1	1	Receive header error	The format of the header received via communication is not correct.	Check the header format.
CM7511	1	1	Receive data name error	The data name of the header received via communication is not correct.	Check the header data name.
CM7512	1	1	Receive response abnormal end	An abnormal end response was received for the command sent via communication.	Check the connected device.
CM7513	1	1	Receive data ignored (symbol)	An invalid symbol was detected in the data received via communication, causing the corresponding record to be ignored.	Check the symbol in the data.
CM7514	1	1	Receive data ignored (ascend)	The symbols in the data received via communication are not in ascending order, causing the corresponding record to be ignored.	Check the sequence of symbols in the data.
CM7515	1	1	Receive data converted (qty.)	The number of items composing one record in the data received via communication is not correct. Excessive items are ignored, and insufficient items are not set.	Check the record format.

No.	Stop level	Reset level	Alarm message	Cause	Solution
CM7516	1	1	Receive data converted (range)	1. The range of items in the data received via communication is not correct, so they are converted. The converted sections are not set. 2. The item length (digits) in the data received via communication is not correct, so the data was converted. The converted places are data for the valid digits.	Check the value of each item.
CM7517	1	1	Communication data edit error	An edit error occurred while processing the communication data.	
CM7518	1	1	Too many receive directories	The number of received directories exceeds 1024.	The number of directories should be 1024 or less.
CM7519	1	1	Communication TV check error	The number of characters in one block is not correct when receiving a program.	Check the communication parameter or data.
CM7520	1	1	Received invalid data	Invalid data was received when not converting invalid data.	Check the communication parameter or data.
CM7521	1	1	Communication check sum error	The checksum of the received data is not correct.	Set the checksum correctly, or set [0: Off] for [Checksum] (communication user parameter).
CM7522	1	1	Receive command abnormal end (*)	A command received via TC slave communication ended abnormally. Refer the "Completion code list" of Chapter "External program I/O" in the Data manual for the value in brackets ().	Check the cause and send the command again.
CM7523	1	1	Communication interface error	An error occurred in the interface with the communication device.	The local PCB is defective. Contact the nearest BROTHER approved service dealer.
CM7524	1	1	Communication version check error	The version of the data to be sent differs from the system version.	Edit the corresponding data again.
CM7525	1	1	Program sent/received (slave)	Sending and receiving the program was performed via slave communication.	
CM7526	1	1	Multiple files cannot be specified.	1. A file other than a multiple-program file is selected. 2. Several multiple-program files are selected	1. Deselect the file. 2. Select only one multiple-program file.
CM7527	1	1	Slave communicating (Ethernet)	The file to be transferred is being used by the Ethernet.	Transfer the file once communication with the slave is finished.
CM7528	1	1	Comm. param. set error (IP address)	An invalid IP address is set.	Set the correct IP address.
CM7529	1	1	Comm. param. set error (mask bit)	An invalid mask bit is set.	Set the correct mask bit.
CM7530	1	1	Comm. param. set error (gateway)	An invalid default gateway is set.	Set the correct default gateway.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
CM7531	1	1	Ethernet is disconnected.	Communication with the connected device was interrupted during slave communication.	Check the connected device.
CM7532	1	1	Ethernet communication error (*)	An Ethernet error occurred during communication.	Check the network cable or the state of the connected device, and retry communication.
CM7533	1	1	Old program cannot be checked.	An old program was selected when inputting programs in check mode.	
CM7534	1	1	File access error (*)	An error occurred during access to the file.	Contact the nearest BROTHER approved service dealer.
CM7535	1	1	Memory card cannot be recognized.	Memory card is not connected.	Check the connection of memory card.
CM7536	1	1	Main station is being communicating. (Memory card)	1. The file to be transferred is being used by the memory card. 2. Maintenance and communication mode is enabled.	1. Transfer the file once communication with the master is finished. 2. Disable maintenance and communication mode before transferring the file.
CM7537	1	1	Following station is being communicating. (FTP)	The file to be transferred is being used by FTP.	Transfer the file once communication with the slave is finished.
CM7538	1	1	The main station is being communicating. (FTP)	The file to be transferred is being used by FTP.	Transfer the file once communication with the master is finished.
CM7539	1	1	Main station is being communicating. (Serial port)	The file to be transferred is being used by the serial port.	Transfer the file once communication with the master is finished.
CM7540	1	1	Folowing station is communicating. (Serial port)	The file to be transferred is being used by the serial port.	Transfer the file once communication with the slave is finished.
CM7541	1	1	Sending failed	1. File or folder with invalid name was attempted to be output via communication. 2. An unauthorized file or folder was attempted to be accessed.	1. Check the file name and folder name. 2. Check the access permission for the file or folder.
CM7542	1	1	Receiving failed	File or folder with invalid name was attempted to be entered via communication	Check the file name and folder name.
CM7543	1	1	Deleting failed	1. File or folder with invalid name was attempted to be deleted. 2. An unauthorized file or folder was attempted to be accessed.	1. Check the file name and folder name. 2. Check the access permission for the file or folder.

No.	Stop level	Reset level	Alarm message	Cause	Solution
CM7544	1	1	Renaming failed	1. File or folder with invalid name was attempted to be renamed. Or, an invalid name was specified for a file or folder name. 2. An unauthorized file or folder was attempted to be accessed.	1. Check the file name and folder name. 2. Check the access permission for the file or folder.
CM7545	1	1	Failed to stop ladder program	Failed to stop ladder program when entering it.	Enter the ladder program after ending the monitor mode of external editing software.
CM7546	1	1	Memory card error	1. An error associated with the memory card occurred. 2. The memory card is used by another function.	1. Recheck the memory card. 2. After the other functions are complete, execute the operation again.
CM7547	1	1	Write protect (Memory card)	Output, deletion, or format was used for a write-protected memory card.	Release write-protection.
CM7548	1	1	Multiple memory cards cannot be connected.	Multiple memory devices (e.g. memory card reader) are connected to the USB port.	Connect only one memory device to the USB port.
CM7549	1	1	Unusable format (Memory card)	The format of the floppy disk or memory card is not supported.	Use a memory device formatted.
CM7550	1	1	Name resolution of destination is failed.	1. An invalid FTP server name is set. 2. The DNS server and network environment have a problem.	1. Check the destination host name. Set a correct FTP sever name. 2. Check the condition of the network cable or the connected device and execute again.
CM7551	1	1	Communication format error (header/footer)	NC program with header/footer was attempted to input while "header/footer" is set to [0: Not equipped] for communication parameter, or NC program without header/footer was attempted to input while "header/footer" is set to [1: Equipped].	Check the communication parameter or NC program.
CM7552	1	1	3D smoothing data cannot be input.	An attempt was made to input 3D smoothing data which includes a program that does not exist in the NC directory.	First, input the program for the 3D smoothing data.
CM7553	1	1	3D smoothing data cannot be output.	An attempt was made to output a program that contains 3D smoothing data using an invalid file name.	Specify <O8999.NC> as the file name from <O0001.NC>. Or, delete the 3D smoothing data.
CM7554	1	1	3D smoothing data version error	This version does not support the 3D smoothing data that was received via communication. Too many items were ignored, and items in the wrong range were changed to "not set". The 3D smoothing path data and the shape characteristics were deleted.	Check the set values on the <3D smoothing preparations> screen, and execute the 3D smoothing data operation.

No.	Stop level	Reset level	Alarm message	Cause	Solution
CM7555	1	1	3D smoothing data ignored (symbol)	An invalid symbol was detected in the 3D smoothing data received via communication, causing the corresponding record to be ignored and deleting the 3D smoothing path data as well as the shape characteristics.	Check the set values on the <3D smoothing preparations> screen, and execute the 3D smoothing data operation.
CM7556	1	1	3D smoothing data ignored (symbol ascending order)	Symbols were not in ascending order in the 3D smoothing data received via communication, causing the corresponding record to be ignored and deleting the 3D smoothing path data as well as the shape characteristics.	Check the set values on the <3D smoothing preparations> screen, and execute the 3D smoothing data operation.
CM7557	1	1	3D smoothing data ignored (item range error)	The item range is invalid in the 3D smoothing data received via communication, causing the items to be changed to "not set" and deleting the 3D smoothing path data as well as the shape characteristics.	Check the set values on the <3D smoothing preparations> screen, and execute the 3D smoothing data operation.
CM7558	1	1	3D smoothing data ignored (item number mismatch)	The number of items is wrong for 1 record in the 3D smoothing data received via communication. Too many items were ignored, and items that were missing were changed to "not set". The 3D smoothing path data and the shape characteristics were deleted.	Check the set values on the <3D smoothing preparations> screen, and execute the 3D smoothing data operation.
CM7559	1	1	3D smoothing path data and shape characteristics deleted.	The 3D smoothing path data and the shape characteristics were deleted because there was an error in the 3D smoothing path data.	Execute the 3D smoothing data operation.
CM7560	1	1	Folder for connected device not found	When inputting and outputting data to maintain and transfer data, the folder for the connected device could not be found.	Use the folder operations to check the folder for the connected device.
CM7561	1	1	Name resolution for auto notification destination failed	1. An invalid host name is set. 2. There is a problem with the DNS server or network environment.	1. Check the notification host name. Set the correct host name. 2. Check the status of the network cable or the connected device and execute again.
CM7562	1	1	Ether. comm. err (*) connecting to auto notific. Destin	An Ethernet communication error occurred when connecting to the auto notification destination.	Check the status of the network cable or the auto notification device, and retry communication.
CM7563	1	1	Auto notification transmission error (*)	An error occurred when sending the auto notification.	Launch the application for receiving auto notifications on the destination device. If the application is already launched, check the settings, and then shut it down and try re-launching it.
CM7564	1	1	Invalid file name	There are characters included that cannot be specified for the file name.	Check the specified file name.

No.	Stop level	Reset level	Alarm message	Cause	Solution
CM7565	1	1	Logged in under common user	The user logged into the machine under a common user for FTP or communication protocol type 2 on a NC communication device.	

11.7.10 ES

No.	Stop level	Reset level	Alarm message	Cause	Solution
ES0000	5	3	Simple panel communication error (*)	Reserved	Reserved
ES0001	5	3	Magazine panel communication error (*)	Reserved	Reserved
ES0002	5	3	Teaching pendant communication error (*)	Communication with teaching pendant failed.	Check connection of the teaching pendant.
ES0003	5	3	Response data receiving timeout (Simple/panel)	Reserved	Reserved
ES0004	5	3	Response data receiving timeout (Magazine panel)	Reserved	Reserved
ES0006	5	3	Simple panel error message (*)	Reserved	Reserved
ES0007	5	3	Magazine panel error message (*)	Reserved	Reserved
ES0008	5	3	Teaching pendant error message (*)	Received an error response from teaching pendant.	Contact the nearest BROTHER approved service dealer.
ES0500	5	2	Teaching pendant removed		Press the [RST] key.
ES4000	3	2	There is no external panel 1.	Unable to communicate with external panel 1.	Check connection of the external panel 1.
ES4001	3	2	There is no external panel 2.	Unable to communicate with external panel 2.	Check connection of the external panel 2.
ES5500	2	2	Teaching pendant removed		Press the [RST] key.
ES7500	1	1	External panel 1 communication error (*)	Failed to communicate with external panel 1.	Check connection of the external panel 1.
ES7501	1	1	External panel 2 communication error (*)	Failed to communicate with external panel 2.	Check connection of the external panel 2.
ES7502	1	1	Response data receiving timeout (External panel 1)	Failed to receive response from external panel 1 within the specified time period.	Check connection of the external panel 1.
ES7503	1	1	Response data receiving timeout (External panel 2)	Failed to receive response from external panel 2 within the specified time period.	Check connection of the external panel 2.
ES7504	1	1	External panel 1 error message (*)	Received error response from external panel 1.	Please contact us if you encounter a situation.
ES7505	1	1	External panel 2 error message (*)	Received error response from external panel 2.	Please contact us if you encounter a situation.
ES7506	1	1	External panel 1 communication timeout (Character)	External panel 1 communication error.	Please contact us if you encounter a situation.
ES7507	1	1	External panel 2 communication timeout (Character)	External panel 2 communication error.	Please contact us if you encounter a situation.
ES7508	1	1	External panel 1 communication timeout (Boring)	External panel 1 communication error.	Please contact us if you encounter a situation.
ES7509	1	1	External panel 2 communication timeout (Boring)	External panel 2 communication error.	Please contact us if you encounter a situation.

No.	Stop level	Reset level	Alarm message	Cause	Solution
ES7510	1	1	External panel 1 communication timeout (Protocol)	External panel 1 communication error.	Please contact us if you encounter a situation.
ES7511	1	1	External panel 2 communication timeout (Protocol)	External panel 2 communication error.	Please contact us if you encounter a situation.

11.7.11 FC

No.	Stop level	Reset level	Alarm message	Cause	Solution
FC0000	5	3	User param. setting error (PROFIBUS DP)	The following description is divided by auxiliary number. 1- The I/O size set to the I/O setting exceeds 128 bytes. 2- The value set to the I/O setting is invalid.	Set the user parameters (PROFIBUS DP) to the correct values and turn the power on again.
FC0001	5	3	User param. setting error (CC-Link slave)	The following cause description is listed according to its auxiliary number: 1 - The "Slave station category" is not set to "Remote device station." 2 - The "Station number" is not set to a value within the setting range. 3 - The "Occupied number of stations" is not set to a value within the setting range. 4 - The total of the "Station number" and the "Occupied number of stations" is set to a value greater than 65.	Set each user parameter (CC-Link slave) to the correct value and turn the power ON again.
FC0002	5	3	User param. setting error (CC-Link master)	The following cause description is listed according to its auxiliary number: 1. The total for the <Self-station number> and the <Occupied number of stations (for intelligent device stations)> is set to a value greater than 64 stations.	Set each user parameter (CC-Link slave) to the correct value and turn the power ON again.
FC2000	4	2	User param. setting error (CC-Link master)	The following cause descriptions are divided according to their auxiliary numbers: 1. The <CC-Link version> is set to <1: Ver.2> when the <Self-station number> is set to <1-64>.	Set each user parameter (CC-Link master) to the correct value.

No.	Stop level	Reset level	Alarm message	Cause	Solution
FC4000	3	2	CC-Link configuration setting error	In the external I/O signal settings, there is an error with the set value for the CC-Link configuration parameter. The following cause descriptions are divided according to their auxiliary numbers: 1. In the CC-Link configuration, <Specify reserved station> is set to <0: Specify 0 pts.> and <Specify I/O assignment> is set to <1: Specify 0 pts.> when the user parameter (CC-Link master) <CC-Link version> is set to <1: Ver.2>. 2. In the CC-Link configuration, <By station category> is not set to <0: Remote I/O station> and <Specify I/O assignment> is set to one of the following: <2: Specify 8 pts.>, <3: Specify 16 pts.> or <4: Specify mix of 8 & 16 pts.> when the user parameter (CC-Link master) <CC-Link version> is set to <1: Ver.2>. 3. When the total occupied stations exceeds 32. 4. When the valid range is exceeded for the signals that can be assigned. 5. When a parameter is set or saved while the valid stations are not linked. 6. When a valid station remote output and remote input leading address are not in multiples of 16. 7. In the CC-Link configuration, when <By station category> is not set to <2: Intelligent device station>, or when a station, where <Specify reserved station> is set to <1: Yes>, is specified in the user parameter (CC-Link master) <Standby master station number> 8. When the remote I/O stations, the remote device stations and the intelligent device stations exceed the maximum number of connected stations	Check the set value for the CC-Link configuration parameter.

No.	Stop level	Reset level	Alarm message	Cause	Solution
				9. In the CC-Link configuration, when <By station category> is set to <2: Intelligent device station> for all the stations, and the total for <Transient send (or receive) buffer size> exceeds 2048 words	
FC5000	2	3	Field bus watch dog (FC)	No response from the FC PCB within the specified time period.	Please contact us if you encounter a situation.
FC5001	2	3	Field bus watch dog (NC)	No request to the FC PCB within the specified time period.	Please contact us if you encounter a situation.
FC5002	2	3	FC board communication error	FC PCB communication is abnormal.	Please contact us if you encounter a situation.
FC5003	2	3	The FC board communication is abnormal. (*)	FC PCB data is abnormal.	Please contact us if you encounter a situation.
FC5004	2	3	Field bus watch dog (CM)	No response from the CM PCB within the specified time period.	Please contact us if you encounter a situation.
FC5005	2	3	CM PCB communication error (*)	CM PCB communication is abnormal.	Please contact us if you encounter a situation.
FC5006	2	3	CM PCB error (*)	CM PCB data is abnormal.	Please contact us if you encounter a situation.
FC5500	2	2	Interface board error (CM) (*)	- There is an error with the parameter set value. - here is an interface board error.	- Check the parameter set value for the user parameter (CC-Link master) and the CC-Link configuration. - Please contact us if you encounter a situation.
FC7000	1	2	Transient parameter error	There is an error with the set value for the transient send parameter.	Check the set value for the PLC OM.
FC7500	1	1	Field bus communication stop	Field bus communications stopped due to an error, etc.	Remove the cause of the stopped communication.
FC7501	1	1	User param. setting error (CC-Link master)	The following cause description is listed according to its auxiliary number: 1. An attempt was made to carry out the line test when <Standby master station number> was not set to <0>.	Set each user parameter (CC-Link master) to the correct value.

11.7.12 OM

No.	Stop level	Reset level	Alarm message	Cause	Solution
0000	5	3	Resetting from PLC is disabled	Reset signals from PLC and external input signals are disabled due to the special operation. Special operation refers to pressing the following 4 buttons simultaneously: [EMERGENCY] switch, [FEED HOLD] switch, [START] switch, and [RELSE] key.	Turn the power off and then on again.
0001	5	3	Turn power off.	1. A power OFF level item has been changed in the servo parameter. 2. A process required for turning ON the power again has been carried out.	Turn the power off and then on again.
0002	5	3	Reset absolute encoder counter.	Abso. encoder counter was reset.	Turn the power off and then on again.
0003	5	3	PLC axis has been stopped.		
0004	5	3	Version is being updated.		
0005	5	3	Power failure hold area is changed	Power failure hold area is changed.	Do not run ladder programs before turning the power on again. Turn the power on again.
0500	5	2	Door open	1. The door was opened when the [DOOR INTERLOCK] switch was enabled. 2. The door was opened when the [Pendant valid] switch for the teaching pendant was set to [Valid] and the [ENABLE] switch was turned OFF.	1-1. Close the door. 1-2. Disable the door interlock. 1-3. Check the door limit switch. 1-4. Make sure that there are no chips or shavings caught between the door and the machine cover. 1-5. The door sensor is faulty. Replace the door sensor. 1-6. The IL PCB is faulty. Replace the IL PCB. 1-7. The IO PCB is faulty. Replace the IO PCB. (When also using the BV7) 1-8. DRRLY PCB is faulty. Replace the DRRLY PCB. 2-1. Close the door. 2-2. Press the [ENABLE] switch. 2-3. Switch the [Pendant valid] switch to [Invalid].
0501	5	2	Standby mode	The servo controller has switched off and entered standby mode because the machine has not been operated for the preset time.	Perform any of the following operations, and the servo controller turns on and operation is enabled. 1-1. Press any key on the operation panel 1-2. Press [START] or [PALLET START] 2. Change the setting for [Standby mode time] (user parameter 1/switch 1).
0502	5	2	Adjusted abso. data has been set.	Adjusted counter value has been set.	

Chapter 11 Alarms

No.	Stop level	Reset level	Alarm message	Cause	Solution
0503	5	2	Press the [RST] key	A reset level parameter has been changed.	Press the [RST] key.
0504	5	2	Reset absolute encoder counter.	The absolute encoder counter was reset.	Press the [RST] key to reset the alarm.
0505	5	2	[MASTER] is off.	1. EMG Stop switch was pressed. 2. External EMG Stop signal is open. 3. EMG SW ON command (a communication command) is received.	1. Press the [RST] key after removing the cause of the stop and releasing the EMG Stop SW. 2. Close the input terminal contacts and input the external reset signals. Check the following wiring. IL PCB, XTEMSIN 1-2, 3-4 and 5-6 3. Send the EMG SW OFF command (a communication command).
0506	5	2	Enter your password		Enter the data protection password.
0507	5	2	Changed to the ATC maintenance mode.	Changed to the ATC maintenance mode.	Press the [RST] key.
0508	5	2	The PLC-axis has been stopped.		
0509	5	2	During measurement of motor insulation resistance	This alarm occurs during measurement of motor insulation resistance. Turn off the servo relay.	
0510	5	2	All ladder programs are stopped	1. Ladder programs have been changed in the master station communication and the slave station communication. 2. Ladder program operation stopped due to another alarm being triggered.	1. Press the [RST] key. 2. Remove the cause of the other alarm that was triggered at the same time.
0511	5	2	Motor insulation resistance measurement start	This alarm occurs at the start of measurement of motor insulation resistance. Turn off the servo motors of all axes.	
0512	5	2	Position adjusted	The axis position has been adjusted.	Press the [RST] key.
0513	5	2	*-axis automatically adjusted	The QT-axis has been automatically adjusted.	Press the [RST] key.
0514	5	2	*-axis adjustment check again	The QT-axis has been adjusted again. The adjustment needs to be checked again.	Press the [RST] key, and then press the [RELSE] + [R] keys.
0515	5	2	[ENABLE] switch was turned OFF	The [ENABLE PENDANT] switch for the teaching pendant is set to [Valid], and the [ENABLE] switch was turned OFF while the axis was traveling and the door was opened.	1. Close the door. 2. Press the [ENABLE] switch. 3. Set the [ENABLE PENDANT] switch to [Invalid].
0516	5	2	Change to the mode for enabling and disabling the servo	Change to the mode for enabling and disabling the servo.	Press the [RST] key.

No.	Stop level	Reset level	Alarm message	Cause	Solution
0519	5	2	[MASTER ON] was turned OFF	The hardware condition (IO_INVTER) signal for the [MASTER ON] is OFF. When the [MASTER ON] function is used and an alarm (ALM signal) is triggered on this machine, this alarm may also trigger.	Remove the cause of the hardware condition (IO_INVTER) signal for [MASTER ON], and then press the [MASTER ON] switch.
1500	4	3	Turn power off.	1. An important parameter, related to the automatic thermal distortion compensation, has been changed. 2. A process required for turning ON the power again has been carried out.	Turn the power off and then on again.
1501	4	3	The PLC-axis has been stopped.		
2000	4	2	Counter 1 end	Workpiece counter 1 has reached the setting.	
2001	4	2	Counter 2 end	Workpiece counter 2 has reached the setting.	
2002	4	2	Counter 3 end	Workpiece counter 3 has reached the setting.	
2003	4	2	Counter 4 end	Workpiece counter 4 has reached the setting.	
2004	4	2	Tool life end (*)	The life of the current tool or the tool about to be used has expired. The figure in brackets () is the tool number.	Press the [RST] key. Change the tool and then set the tool life.
2005	4	2	Version update has been completed		
2006	4	2	Help data version update has been completed		
2007	4	2	The PLC-axis has been stopped		
2008	4	2	Press the [RST] key	A critical parameter was changed.	Press the [RST] key.
2009	4	2	Handle operation switch OFF	While the handle axis selection is not set to <OFF>, the [HANDLE OPERATION] switch is turned OFF when the door is opened.	1. Press the [HANDLE OPERATION] switch. 2. Close the door. 3. Turn the axis selection switch for the handle to <OFF>.
4000	3	2	Press the [RST] key	A critical parameter was changed.	Press the [RST] key.
4001	3	2	Check ATC tool.	1. After the ATC recovery operation is complete, the recovery operation was disabled. 2. Power was shut off while the ATC arm is turning.	Check the ATC tool, and then press the [RST] key.
4002	3	2	Assign tools. (*)	Operation was attempted when the tool list data is newer than the operation program in conversation mode.	Assign the tools again.

No.	Stop level	Reset level	Alarm message	Cause	Solution
4003	3	2	Switch to manual operation mode for restoration.	1. A stop level 5 alarm was triggered during the tapping operation. 2. A connector for the encoder is disconnected. 3. Absolute data for the encoder was cleared because the encoder battery has expired.	1. Switch to manual operation mode, and perform the synchronized tap return operation. 2. Make sure that the encoder connector is not disconnected. 3. Switch to manual operation mode, and perform the expired battery recovery operation.
4004	3	2	Assign tools.	Conversation language machining data created by new version software was entered for operation.	Assign the tools again.
4005	3	2	Check the tapping tool in the spindle.	1. When the power is turned OFF during tapping and then the power is turned back ON 2. After a stop level 5 alarm is triggered during tapping when the synchronized tap return does not execute and the power is turned OFF and ON 3. After a synchronized tap return recovery operation has been completed when the recovery operation is disabled 4. After a synchronized tap return recovery operation has been completed when the power is turned OFF and ON	The tapping tool on the spindle may be stuck in the workpiece. If it is stuck, perform a synchronized tap return in manual operation mode.
4006	3	2	The detachable panel has been forcibly removed.	Reserved	Reserved
4007	3	2	The PLC axis has been stopped.		
4008	3	2	The PLC axis has been stopped.		
4009	3	2	Backup memory has been saved	The backup memory has been saved successfully.	
4010	3	2	Check the cause of alarm and recovery procedure (SM4054)	The alarm <<Tool length offset cancel error>> has been triggered.	Check the recovery procedure for the alarm <<Tool length offset cancel error>>.
4011	3	2	Check the cause of alarm and recovery procedure (SM4055)	The alarm <<Tool position offset cancel error>> has been triggered.	Check the recovery procedure for the alarm <<Tool position offset cancel error>>.
4012	3	2	Make sure all axes are operating in manual operation mode	The mode for enabling and disabling the servo was disengaged.	Press the [RST] key and make sure that all axes operate.

No.	Stop level	Reset level	Alarm message	Cause	Solution
4013	3	2	The machine coordinates may be out of position.	All data was input, or the start data (adjusted absolute values) from the backup memory recovery operation was input. When old start data is input, the encoder position data for the magazine, ATC arm and additional axes (when set to no stroke) may not match the machine coordinates.	1. Press the [RST] key. 2. Check the magazine index position. 2-1. When a <<Magazine number error>> is triggered, make sure that the tool number that is indexed matches the machine state. If it does not match, perform the expired battery recovery operation. 2-2. When a <<Magazine number error>> is triggered, perform the expired battery recovery operation (do not perform the recovery operation using the [RELSE] key + [MAG.F] key or the [RELSE] key + [MAG.R] key). 2-3. When the magazine is not in the index position, the magazine zero point needs to be re-adjusted. Contact the nearest BROTHER approved service dealer. 3. When an arm type ATC mechanism is used, check the ATC arm position. 3-1. When an <<ATC arm position error>> is triggered, perform the expired battery recovery operation (do not perform the ATC arm position recovery operation). 3-2. When the ATC arm is not at the zero point, the ATC arm zero point must be re-adjusted. Contact the nearest BROTHER approved service dealer. 4. Make sure that the machine coordinates for the additional axis are correct. 4-1. When the machine coordinates for the additional axis are not correct, re-adjust the machine zero point.
5500	2	2	Oil level low (automatic oiling)	Insufficient lubricant.	1) Supply oil. 2) Press the [RST] key.
5501	2	2	Change grease (automatic greasing)	Grease has run out.	Change the grease cartridge, and press the [RST] key.
7000	1	2	Machine parameter was not deleted.	[0: Write-protected] was set for [Parameter change] when formatting or deleting all data banks.	Change to [1: Writable] on the <Input/Output> screen to delete machine parameters.
7001	1	2	*-axis servo power off	1. The door was opened while the door interlock was enabled. (The 4th-axis servo power was turned off.) 2. The [ENABLE PENDANT] switch for the teaching pendant is set to [Valid], and the [ENABLE] switch was turned OFF while the axis was traveling and the door was opened.	1. Close the door and the alarm is cleared. 2-1. Close the door. 2-2. Press the [ENABLE] switch. 2-3. Set the [ENABLE PENDANT] switch to [Invalid].

No.	Stop level	Reset level	Alarm message	Cause	Solution
7002	1	2	*-axis servo power off	1.The door was opened while the door interlock was enabled. (The 5th-axis servo power was turned off.) 2.The [ENABLE PENDANT] switch for the teaching pendant is set to [Valid], and the [ENABLE] switch was turned OFF while the axis was traveling and the door was opened.	1.Close the door and the alarm is cleared. 2-1.Close the door. 2-2.Press the [ENABLE] switch. 2-3.Set the [ENABLE PENDANT] switch to [Invalid].
7003	1	2	*-axis servo power off	1.The door was opened while the door interlock was enabled. (The 6th-axis servo power was turned off.) 2.The [ENABLE PENDANT] switch for the teaching pendant is set to [Valid], and the [ENABLE] switch was turned OFF while the axis was traveling and the door was opened.	1.Close the door and the alarm is cleared. 2-1.Close the door. 2-2.Press the [ENABLE] switch. 2-3.Set the [ENABLE PENDANT] switch to [Invalid].
7004	1	2	*-axis servo power off	1.The door was opened while the door interlock was enabled. (The 7th-axis servo power was turned off.) 2.The [ENABLE PENDANT] switch for the teaching pendant is set to [Valid], and the [ENABLE] switch was turned OFF while the axis was traveling and the door was opened.	1.Close the door and the alarm is cleared. 2-1.Close the door. 2-2.Press the [ENABLE] switch. 2-3.Set the [ENABLE PENDANT] switch to [Invalid].
7005	1	2	*-axis servo power off	1.The door was opened while the door interlock was enabled. (The 8th-axis servo power was turned off.) 2.The [ENABLE PENDANT] switch for the teaching pendant is set to [Valid], and the [ENABLE] switch was turned OFF while the axis was traveling and the door was opened.	1.Close the door and the alarm is cleared. 2-1.Close the door. 2-2.Press the [ENABLE] switch. 2-3.Set the [ENABLE PENDANT] switch to [Invalid].
7006	1	2	P1-axis servo power off	1.The door was opened while the door interlock was enabled. (The P1-axis servo power was turned off.) 2.The [ENABLE PENDANT] switch for the teaching pendant is set to [Valid], and the [ENABLE] switch was turned OFF while the axis was traveling and the door was opened.	1.Close the door and the alarm is cleared. 2-1.Close the door. 2-2.Press the [ENABLE] switch. 2-3.Set the [ENABLE PENDANT] switch to [Invalid].

No.	Stop level	Reset level	Alarm message	Cause	Solution
7007	1	2	P2-axis servo power off	1.The door was opened while the door interlock was enabled. (The P2-axis servo power was turned off.) 2.The [ENABLE PENDANT] switch for the teaching pendant is set to [Valid], and the [ENABLE] switch was turned OFF while the axis was traveling and the door was opened.	1.Close the door and the alarm is cleared. 2-1.Close the door. 2-2.Press the [ENABLE] switch. 2-3.Set the [ENABLE PENDANT] switch to [Invalid].
7008	1	2	P3-axis servo power off	1.The door was opened while the door interlock was enabled. (The P3-axis servo power was turned off.) 2.The [ENABLE PENDANT] switch for the teaching pendant is set to [Valid], and the [ENABLE] switch was turned OFF while the axis was traveling and the door was opened.	1.Close the door and the alarm is cleared. 2-1.Close the door. 2-2.Press the [ENABLE] switch. 2-3.Set the [ENABLE PENDANT] switch to [Invalid].
7009	1	2	P4-axis servo power off	1.The door was opened while the door interlock was enabled. (The P4-axis servo power was turned off.) 2.The [ENABLE PENDANT] switch for the teaching pendant is set to [Valid], and the [ENABLE] switch was turned OFF while the axis was traveling and the door was opened.	1.Close the door and the alarm is cleared. 2-1.Close the door. 2-2.Press the [ENABLE] switch. 2-3.Set the [ENABLE PENDANT] switch to [Invalid].
7010	1	2	Configured parameters that were not set to factory-set values.	During the firmware version update, the factory-set values were configured for the parameters which have not be set.	
7011	1	2	Memory extension function is enabled		Press the [RST] key.
7012	1	2	Smallest setting unit is enabled		Press the [RST] key.
7013	1	2	High accuracy B / spline is enabled		Press the [RST] key.
7014	1	2	Interrupt type macro is enabled		Press the [RST] key.
7015	1	2	Turning function is enabled		Press the [RST] key.
7016	1	2	Rotary fixture offset is enabled		Press the [RST] key.
7017	1	2	High speed processing mode (1 msec) is enabled		Press the [RST] key.
7018	1	2	Execute the automatic oiling/greasing cycle ON command	Automatic oiling or automatic greasing error was detected.	After replenishing the lubricant or grease, execute the automatic oiling / greasing cycle ON command in MDI operation. NC language: M478 Conversation language: Miscellaneous function 478

No.	Stop level	Reset level	Alarm message	Cause	Solution
7019	1	2	Scheduled notice to change relocat. detect. device batteries	It is time to change the batteries on the relocation detection device.	Follow the procedure below to replace the batteries for the relocation detection device. 1. Leave the power switch turned ON, and remove all the batteries from the relocation detection device. 2. Check if the alarm <<Change batteries on relocat. detect. device with power ON>> triggers. 3. Install new batteries for the relocation detection device. 4. Press the [RST] key and check if the alarm <<Change batteries on relocat. detect. device with power ON>> resets.
7020	1	2	3D smoothing is enabled.		Press the [RST] key.
7030	1	2	High speed processing mode 2 (0.5 msec) is enabled		Press the [RST] key.
7031	1	2	Feature coordinate setting function is enabled		Press the [RST] key.
7032	1	2	ATC monitoring parameter was set		
7033	1	2	Machining load could not be measured(*)	The following cause descriptions are divided according to their auxiliary numbers: 0. The parameter number used for the machining load monitor has not been specified. 1 to 99. Machining load could not be measured due to one of the following reasons on the tool indicated by the auxiliary number. - The measurement time is shorter than the <Time constant> for the machining load monitor parameter. - There is no section with a constant speed because the tapping or drilling has a short machining time, etc. - There is no cutting travel. - A command was issued to select the lathe spindle.	0. Specify the E address following the M code for turning ON the machining load monitor. Or, register the tool for monitoring the machining load on the ATC tool screen. 1 to 99. Perform one of the following. - Set the <Time constant> for the machining load monitor parameter to the correct value. - When the drilling or tapping has a short machining time, set the machining load monitor parameter <Load monitor method> to <1: Average> before using. - Revise the program.
7034	1	2	Involute interpolation function is enabled		Press the [RST] key.
7035	1	2	Relocation was detected	The relocation detection device detected that the machine has been relocated.	Check the request code and contact the sales dealer.
7500	1	1	Counter 1 approaching setting	Workpiece counter 1 reached the end signal setting.	
7501	1	1	Counter 2 approaching setting	Workpiece counter 2 reached the end signal setting.	

No.	Stop level	Reset level	Alarm message	Cause	Solution
7502	1	1	Counter 3 approaching setting	Workpiece counter 3 reached the end signal setting.	
7503	1	1	Counter 4 approaching setting	Workpiece counter 4 reached the end signal setting.	
7504	1	1	Door open	The front door on an FT model opened when the user parameter <Display alarm when door is open> was set to <1: Yes>. When the user parameter <Display alarm when door is open> was set to <0: No>, a run operation was performed under the following conditions with [DOOR INTERLOCK] switch disabled. - Manual operation with outer and inner doors open (including non-indexed QT-axis) (QT model). - Manual operation with the front door open (FT model). - Startup by memory operation with Door Interlock Switch turned ON and the front door open (FT model) of no-automatic door specification or FT model of automatic door specification where Manual-Automatic selector switch was selected to Manual.	If the user parameter “Display Alarm when Door is Open” is set to [1: Yes], close the indicated door. If the user parameter “Display Alarm when Door is Open” is set to [0: No], close the indicated door, and press any key. 1. Check the door limit switch. 2. Make sure that there are no chips or shavings caught between the door and the machine cover. 3. The door sensor is faulty. Replace the door sensor. 4. The IL PCB is faulty. Replace the IL PCB. 5. The IO PCB is faulty. Replace the IO PCB.
7505	1	1	Inner door open	When [Display alarm when door is open] (user parameter) is set to [1:Yes], the outer door is closed and the inner door is open (QT model).	Close the corresponding door.

No.	Stop level	Reset level	Alarm message	Cause	Solution
7506	1	1	Outer door open	When [Display alarm when door is open] (user parameter) is set to [1:Yes], the outer door opened (QT model). When [Display alarm when door is open] (user parameter) is set to [0:No], 1. Pallet turn/clamp (or unclamp) was attempted when the safety device was valid and the outer door was open, or machine operation was attempted when the pallet was not indexed yet. 2. Spindle rotation, ATC, magazine turn, tapping, etc. were attempted when the safety device was invalid, the outer door was open, and the pallet was not indexed yet. 3. Pallet turn was attempted in memory operation when the safety device was invalid, the outer door was open, and single operation was OFF. 4. A command to the external pallet was specified in memory operation when the safety device was invalid and the outer door was open. 5. External index operation was attempted when when the safety device was valid and the outer door was open (QT model). 6. Manual operation was performed when [Door interlock] switch was valid and the outer door was open (QT model). 7. Memory operation was started when [Door interlock] switch was valid and the outer door was open on a machine which is not automatic door specification, or is automatic door specification but [MANUAL-AUTO] selector switch is set to [MANUAL] (QT model).	Close the outer door when [Display alarm when door is open] (user parameter) is set to [1:Yes]. Press any key after closing the outer door when [Display alarm when door is open] (user parameter) is set to [0:No]. C. Specify a command after closing the outer door.
7507	1	1	Spare tool used	A spare tool was used because the tool life has expired during machining.	

No.	Stop level	Reset level	Alarm message	Cause	Solution
7508	1	1	Tool life ending (*)	The tool life has reached the setting that signals the end of tool life. The figure in brackets () is the tool number.	
7509	1	1	Select axis.	The additional axis change switch is not set after power is turned on.	Set the additional axis select switch.
7510	1	1	Sending completed (*)	The program shown in brackets () was sent to an external device via slave communication.	
7511	1	1	Sending completed (*)	The data shown in brackets () was sent to an external device via slave communication.	
7512	1	1	Receiving completed (*)	The program shown in brackets () was received from an external device via slave communication.	
7513	1	1	Receiving completed (*)	The data shown in brackets () was received from an external device via slave communication.	
7514	1	1	Spindle override on	Spindle override is not 100%.	Set the spindle override to 100%.
7515	1	1	Cutting feedrate override on	Cutting feed rate override is not 100%.	Set the cutting feed rate override to 100%.
7516	1	1	Rapid feed override on	1. Rapid traverse override is not 100%. 2. When the user parameter (switch 1) <Stop using feedrate override during rapid traverse> is set to <1: Enabled>, the feedrate override changed to <0%>.	1. Change the rapid traverse override to 100%. 2-1. Change the feedrate override to another setting beside 0%. 2-2. Set the user parameter (switch 1) <Stop using feedrate override during rapid traverse> to <0: Disabled>, and press the [RST] key.
7517	1	1	Side door open	The side door opened when the user parameter <Display alarm when door is open> was set to <1: Yes>. An attempt was made to operate the machine with the side door open when the user parameter <Display alarm when door is open> was set to <0: No>.	If the user parameter “Display Alarm when Door is Open” is set to [1: Yes], close the side door. If the user parameter “Display Alarm when Door is Open” is set to [0: No], close the side door and then run the machine.
7518	1	1	Communication command end (*)	The program shown in brackets () was deleted via NC slave communication.	
7519	1	1	Deletion completed (*)	The program shown in brackets () was deleted via NC slave communication.	
7520	1	1	Deletion completed (*)	The data shown in brackets () was deleted via NC slave communication.	
7521	1	1	Assign tools.	Tool assignment data was changed when tool assignment was completed.	Assign the tools again.
7522	1	1	Calculate ?.	The data associated with ? calculation was changed when ? calculation was completed.	

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No.	Stop level	Reset level	Alarm message	Cause	Solution
7523	1	1	Tool change position data was lost.	ATC return data was initialized because the machining sequence was changed.	
7524	1	1	Set machining sequence.	Editing the ATC return data was attempted although the machining sequence is not set.	
7525	1	1	Restarting program	1. Occurs when restarting a program. 2. Attempted to change operation mode during program restart. 3. Attempted to start the system during program restart.	1. The operator message is cleared when the program restarts. 2. Change the mode after program restart. 3. Start the system after program restart.
7526	1	1	Operation end	Tape operation (general communication device) is completed.	
7527	1	1	DB overwritten in memory oper. mode	When converting a conversation program to an NC program, [Convert] is set for [Tool data].	Check the parameter.
7528	1	1	Running automatic coolant recovery	This message is displayed at the start of operation during the time set for [Coolant delay time] when [Automatic coolant off time] is enabled.	The alarm is cleared after the time set for [Coolant delay time] has elapsed.
7529	1	1	Unable to convert breakage detection data	There was a tool breakage detection command when conversation program machining data was converted to G-code.	Add a tool breakage detection command to the NC program converted to G-code.
7530	1	1	Maintenance (*)	Current value for the item in brackets () exceeded the target value on the <Maintenance notice> screen.	Check the items in brackets () and reset the current value to zero.
7531	1	1	Prohibition of operating an add. outside axis ON.	Occurs when the following operation is executed when the user parameter/switch 1 "Prevent external operation for additional axes" is set to [1:Yes]: - Controlled the outside additional axis. - Controlled additional axes with the pallet not yet indexed.	Index a pallet using the [P.INDX] key.
7532	1	1	Restoration completed, so invalidate restoration.	Recovery operation is completed.	
7533	1	1	Restore the ATC arm position.	An attempt was made to enable recovery operation, which was not an ATC arm position recovery, even though the ATC arm was not at zero point.	Recover the ATC arm position first.
7534	1	1	Recover tool clamp failure.	An attempt was made to enable recovery operation, which was not a tool misclamp recovery, even though the tool holder was not properly clamped by the spindle.	Recover tool clamp failure first.

No.	Stop level	Reset level	Alarm message	Cause	Solution
7535	1	1	Invalidate the restoration operation.	1. The mode was switched to another mode besides manual operation mode, or a numerical or a character key was pressed, while the recovery/inspection function is enabled. 2. A position adjustment operation was attempted on the next axis while the recovery function is enabled.	Disable the recovery/inspection function first.
7536	1	1	The waveform data is being stored.	M238 was commanded during file saving after specifying End of Waveform Display Measurement {[F7] (End) key or M239}.	Command M238 after saving the waveform display measurement file.
7537	1	1	The waveform data has not been recorded.	1. Failed to record waveform data. 2. Too many data; unable to save waveform data.	Press any key. Check the parameters.
7538	1	1	Operation other than restoration op is prohibited.	An operation other than recovery was commanded when recovery must be performed.	Finish recovery before commanding any other operation.
7539	1	1	Switch to manual operation mode for restoration.	An alarm occurred. Recovery is necessary.	Press any key to release the alarm. Select the manual mode and follow the recovery procedure.
7540	1	1	Press [M.LCK] to disable machine lock.	An attempt was made to operate the machine while the machine lock was left ON.	Press [Machine Lock] key to unlock before operation.
7541	1	1	Operating	An operation which cannot be done during machine operation was performed.	Perform the operation after the machine operation is completed.
7542	1	1	Editing	An operation which cannot be done during edit was performed. The operation target is currently edited.	Perform the operation after edit is completed.
7543	1	1	Communicating	An operation which cannot be done during communication was performed. The operation target is currently communicating.	Perform the operation after the communication is completed.
7544	1	1	Required data not found	The specified data does not exist.	1. Check the program number or data bank number. 2. Check the file in the memory card.
7545	1	1	Machine param. change disabled	Inputting machine parameters via communication was attempted when [0: Write-protected] is set for [Parameter change].	To input machine parameters, select [1. I/O] on the <I/O> screen, and set [1: Writable] for [Parameter change].
7546	1	1	Initializing tool life	Editing the tool list or communication was performed while the initialize screen for ATC tool life was displayed.	Close the initialize screen for tool life before editing the tool list or communication.
7547	1	1	The waveform data is being measured.	M238 was specified again during waveform measurement.	Execute M238 after completing waveform measurement.
7548	1	1	Dry run offset amount (*)	With dry run ON, the set value of dry run offset is lower than the dry run offset lower limit.	Check the parameters.

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No.	Stop level	Reset level	Alarm message	Cause	Solution
7549	1	1	The operation history cannot be recorded.	Operation history records exceeded the upper limit. The oldest record will be deleted.	
7550	1	1	No operation is possible under the current level.	The attempted operation is prohibited in the current operation level.	Check the current operation level on the Operation Level, Input/Output screen.
7551	1	1	The detachable panel can be removed.	Reserved	Reserved
7552	1	1	Operation level setting (*)	Operation level was changed.	
7553	1	1	The password is wrong.		
7554	1	1	The password has not been set.		
7555	1	1	Tool assignment is being carried out.	Tool reassignment was executed at operation start because [Assign tools when starting memory operation] was set to [Yes].	
7556	1	1	Operation disable data has been skipped.		
7557	1	1	The current connection disables a selection.		
7558	1	1	Restart by a sequence search	Sequence number was entered by sequence search.	
7559	1	1	Already specially formatted	Special format was attempted to a special-formatted memory card.	
7560	1	1	This is not an external input signal key.	The memory card is not special-formatted.	Create an external input signal key after performing special format.
7561	1	1	External input signal key cannot be identified.	The special-formatted memory card is not connected.	Check that the special-formatted memory card is connected correctly.
7562	1	1	Cannot use manual op. for positioning on additional axis	1. An attempt was made to operate the additional axis used for positioning with a manual intervention operation during axis travel of said axis in memory operation mode. 2. Perform handle operations on the additional axis used for positioning.(M140X1)	1. Wait until the axis travel operation has finished before operating the axis. 2. Operate using the JOG key.
7563	1	1	The magazine panel is being operated.	Reserved	Reserved
7564	1	1	During the key check mode	Operation was started in key check mode.	
7565	1	1	During the ATC maintenance mode	An operation was performed that cannot be executed while in ATC maintenance mode.	Exit ATC maintenance mode.
7566	1	1	During automatic alignment mode	An operation which cannot be done in the automatic centering mode was performed.	Perform the operation after selecting a screen other than the automatic centering screen on the programming screen.

No.	Stop level	Reset level	Alarm message	Cause	Solution
7567	1	1	During the edit mode in operation	An operation which cannot be done in the edit mode in operation was performed.	Operate in a mode other than the edit mode in operation.
7568	1	1	During external input/output operation	An operation which cannot be done when the external input/output screen is displayed was performed.	Perform the operation after selecting a screen other than the external input/output screen on the programming screen.
7569	1	1	During operation of directory display	An operation which cannot be done during directory display operation was performed.	Perform the operation after selecting a screen other than the directory display screen on the programming screen.
7570	1	1	Set the memory card as the connecting destination.	Reserved	Reserved
7571	1	1	The version is being updated.	An operation which cannot be done during version update was performed.	Perform the operation after the completion of version update.
7572	1	1	The version update is being displayed.	An operation which cannot be done when the version update screen is displayed was performed.	Perform the operation after exiting the version update screen, or after the completion of version update.
7573	1	1	The PLC is being edited.	An operation which cannot be done during PLC edit was performed.	Perform the operation after the completion of PLC edit.
7574	1	1	The kind of screw differs.	The thread types are different between the machining data and the forcibly assigned tool.	Modify the machining data or tool list, or assign a correct tool.
7575	1	1	Memory operation program number has been changed.	When the user parameter/switch 1 "Set Same Number to Edit/memory Operation Programs" is set to [1:Yes], the system advises the operator that the memory operation mode was selected after changing the edit program number, and the operation program number was changed to the edit program number.	
7576	1	1	Reconnect the teaching pendant.	The teaching pendant is not connected.	Reconnect the teaching pendant.
7577	1	1	The external input terminal is duplicated.	Overlapping terminals exist in the external input signal.	Modify the overlapping terminals and press the [RST] key.
7578	1	1	The smooth path offset function is valid.	Attempted the following operations when the smooth path offset function is valid: 1. Mode switching 2. Dry run switching	These operations are not available when the smooth path offset function is valid. Perform them after disabling the smooth path offset function.
7579	1	1	The lathe spindle is being selected.	Attempted to jog or move by handle operation the axis that is functioning as the lathe spindle.	You may not jog or move by handle operation the axis that is functioning as the lathe spindle. Select other axis to continue operation.
7580	1	1	Memory operation mode	The [L.SP] key was used while the memory was operating (includes stop status).	This operation is not available in memory operation. To do this operation, first finish the memory operation.
7581	1	1	MDI operation mode	The [L.SP] key was used while the MDI was operating (includes stop status).	This operation is not available in MDI operation. To do this operation, first finish the MDI operation.

No.	Stop level	Reset level	Alarm message	Cause	Solution
7582	1	1	Comment sent. contains letter unable to be input.	There is a character that cannot be entered by key operation after control out (before control out or after control in).	Insert the characters that cannot be entered into a comment or delete them.
7583	1	1	Measure the motor insulation resistance.	This alarm occurs after the expiration of "Motor insulation resistance measurement interval" of the user parameter since the last measurement. The alarm is always displayed until motor insulation resistance is measured. If the status is reset on the motor insulation resistance measurement screen, the operator message disappears. However, it appears again when the power is turned on next time.	
7584	1	1	During measurement of motor insulation resistance	1. This alarm occurs when mode switching is attempted with the To measure motor insulation resistance screen selected. 2. This alarm occurs when manual operation is attempted during motor insulation resistance measurement.	
7585	1	1	Data not found	1. The search string or replace string was not found. 2. The search for block could not be executed because the program being edited and the program that is running are different or because the memory operation is not being used.	1. Change the character string. 2. Execute program during memory operation.
7586	1	1	Replacement performed too many times	The number of strings that can be replaced in one operation has exceeded the limit.	Reduce the number of strings to be replaced.
7587	1	1	There is no corresponding function.	This function cannot be used in the machine configuration.	
7588	1	1	Automatic alignment mode	1. Attempted to turn on machine lock during automatic centering. 2. Attempted to switch operation modes during automatic centering.	1. Turn on machine lock after centering. 2. Switch the mode after centering.
7589	1	1	During measurement	1. Attempted to switch operation modes during measuring. 2. Attempted to switch dry run during measuring.	These operations are not available when the system is measuring. Wait until measurement completes.
7590	1	1	Resetting	Operations were commanded during resetting/machine lock.	The machine may not be operated during resetting/machine lock. Wait until resetting is completed.

No.	Stop level	Reset level	Alarm message	Cause	Solution
7591	1	1	Servo power on	Performed manual operations that are related to an axis for which servo control is ON.	No manual operation is available during servo control ON. Wait until the end of the servo control.
7592	1	1	The feed hold switch is being pushed down.	Performed operations while the feed hold switch is being pressed.	No operation is available when the feed hold switch is pressed. Check that the switch is not pressed.
7593	1	1	The pallet selection key is OFF	Attempted to start automatic operation by external start with the pallet select key OFF.	Set the pallet select key to any position excepting OFF.
7594	1	1	Key oper. prohibit signal on	Keys are not accepted because the key operation prohibit signals are ON.	Turn the signals off to enable key entry.
7595	1	1	Performing tool breakage detection	The following operations are performed during tool breakage detection when the user parameter (Tool Breakage Detection) "Tool Breakage Detection Option" is set to [3: Type 3]: 1. Mode switching 2. Dry run switching	These operations are not available during tool breakage detection. Wait until detection is completed.
7596	1	1	During high-accuracy B	Dry run was switched when the high accuracy mode B function is valid.	Dry run switching is not permitted in the high accuracy mode B. Wait until the end of this mode.
7597	1	1	Constant peripheral speed is being controlled	An operation was performed that cannot be executed while the constant peripheral speed is being controlled.	Cancel constant peripheral speed control.
7598	1	1	Outside door 1 open	Outside PLC axis was attempted to be moved when outside door 1 was open.	Close the outside door 1 and move the outside PLC axis.
7599	1	1	Outside door 2 open	Outside PLC axis was attempted to be moved when outside door 2 was open.	Close the outside door 2 and move the outside PLC axis.
7600	1	1	Execute synchronized tap return	An attempt was made to enable recovery operation, which was not a synchronized tap return, even though the synchronized tap return was not restored.	1. First, perform a synchronized tap return. 2. If synchronized tap return cannot be performed first, press the [F3] (synchronized tap return reset) key on the synchronized tapping return recovery screen to reset synchronized tapping recovery. Then, perform the necessary recovery operations.
7601	1	1	Password cannot be unregistered.	A password was attempted to be unregistered at an operation level where password being unregistered is not allowed .	
7602	1	1	Recovery complete	Recovery operation is completed.	
7603	1	1	Version update not required		
7604	1	1	Selection of a spindle is abnormal.	An operation was performed that cannot be executed on the spindle that is currently selected.	Check the spindle that is currently selected.
7605	1	1	Not in program edit mode	The operation cannot be performed when the program edit mode is not enabled.	Enable the program edit mode before performing this operation.

No.	Stop level	Reset level	Alarm message	Cause	Solution
7606	1	1	Drawing	An operation was performed that cannot be executed during the drawing operation.	Perform the operation after the drawing is finished.
7607	1	1	Changing to ATC maintenance mode	An operation was performed that cannot be executed while changing to ATC maintenance mode.	Disengage the ATC maintenance mode.
7608	1	1	Position adjustment is not complete	An unnecessary attempt to adjust the position was made on the axis required for position adjustment.	
7609	1	1	*-axis adjustment check completed	The QT-axis adjustment has been checked.	
7610	1	1	Perform the expired battery recovery operation	An attempt was made to enable recovery operation, which was not an expired battery recovery operation, even though the encoder position data does not match the machine coordinates (because the battery is expired).	First, perform the expired battery recovery operation.
7611	1	1	Turn ON the power again before operating	An operation was performed that cannot be executed until the power is turned ON again.	
7612	1	1	Editing / Communicating	An attempt to carry out an operation was made even though the operation cannot be carried out while editing or communicating. The targeted operation is currently being communicated or edited.	Perform the operation once the editing or communication is finished.
7613	1	1	Start signal is ON	On the automatic door specification, the door automatically closed and an attempt was made to operate the program, but the program could not be operated because the start signal is being input.	Turn OFF the start signal. Operation will start.
7614	1	1	Pallet start signal is ON	On the automatic door specification, the door automatically closed and an attempt was made to operate the program, but the program could not be operated because the pallet start signal is being input.	Turn OFF the pallet start signal. Operation will start.
7615	1	1	Restore the spindle speed prior to manual operation after startup	The speed/peripheral speed prior to manual operation was restored for the spindle or lathe spindle.	
7616	1	1	Sub folder is being displayed	An operation was performed which cannot be carried out while the sub folder is being displayed. - Program list screen for graph settings of tool path simulation - Program list screen for memory operation	Close the sub folder display before operating.
7617	1	1	Chinese is unavailable	Chinese is unavailable in this specification.	Please select and use another language.

No.	Stop level	Reset level	Alarm message	Cause	Solution
7618	1	1	Sequence search in progress	1. An attempt was made to change the mode while a sequence search was in progress. 2. An attempt was made to perform a start operation while a sequence search was in progress.	1. Change the mode after the sequence search was completed. 2. Perform a start operation after the sequence search was completed.
7619	1	1	High-speed ladder 2 is write-protected	An attempt was made to transfer data and input into high-speed ladder 2, but the "Parameter change" is set to "Write-protected". An attempt was made to edit high-speed ladder 2, but the "Parameter change" is set to "Write-protected".	To input or edit high-speed ladder 2, set "Parameter change" to "Writable" at the I/O screen.
7620	1	1	Standard ladder 2 is write-protected	An attempt was made to transfer data and input into standard ladder 2, but the "Parameter change" is set to "Write-protected". An attempt was made to edit standard ladder 2, but the "Parameter change" is set to "Write-protected".	To input or edit standard ladder 2, set "Parameter change" to "Writable" at the I/O screen.
7621	1	1	Custom screen displaying	An operation was carried out which cannot be done while the custom screen is displayed on the teaching pendant.	Change the teaching pendant to another screen besides the custom screen and perform the operation.
7622	1	1	Signal data recorded		
7623	1	1	Basic axis servo data recorded		
7624	1	1	PLC axis servo data recorded		
7625	1	1	S-axis servo is disabled	Setting for S-axis servo is disabled.	Disengage the mode for enabling and disabling the servo.
7626	1	1	X-axis servo is disabled	Setting for X-axis servo is disabled.	Disengage the mode for enabling and disabling the servo.
7627	1	1	Y-axis servo is disabled	Setting for Y-axis servo is disabled.	Disengage the mode for enabling and disabling the servo.
7628	1	1	Z-axis servo is disabled	Setting for Z-axis servo is disabled.	Disengage the mode for enabling and disabling the servo.
7629	1	1	*-axis servo is disabled	Setting for 4th-axis servo is disabled.	Disengage the mode for enabling and disabling the servo.
7630	1	1	*-axis servo is disabled	Setting for 5th-axis servo is disabled.	Disengage the mode for enabling and disabling the servo.
7631	1	1	*-axis servo is disabled	Setting for 6th-axis servo is disabled.	Disengage the mode for enabling and disabling the servo.
7632	1	1	*-axis servo is disabled	Setting for 7th-axis servo is disabled.	Disengage the mode for enabling and disabling the servo.
7633	1	1	*-axis servo is disabled	Setting for 8th-axis servo is disabled.	Disengage the mode for enabling and disabling the servo.
7634	1	1	*-axis servo is disabled	Setting for M-axis servo is disabled.	Disengage the mode for enabling and disabling the servo.
7635	1	1	P1-axis servo is disabled	Setting for P1-axis servo is disabled.	Disengage the mode for enabling and disabling the servo.
7636	1	1	P2-axis servo is disabled	Setting for P2-axis servo is disabled.	Disengage the mode for enabling and disabling the servo.

No.	Stop level	Reset level	Alarm message	Cause	Solution
7637	1	1	P3-axis servo is disabled	Setting for P3-axis servo is disabled.	Disengage the mode for enabling and disabling the servo.
7638	1	1	P4-axis servo is disabled	Setting for P4-axis servo is disabled.	Disengage the mode for enabling and disabling the servo.
7639	1	1	Changing to the mode for enabling and disabling servo	An operation was performed that cannot be executed while changing to the mode for enabling and disabling the servo.	Exit the mode for enabling and disabling the servo.
7640	1	1	Currently in mode for enabling and disabling servo	An operation was performed that cannot be executed while currently in the mode for enabling and disabling the servo.	Exit the mode for enabling and disabling the servo.
7641	1	1	Relocation detection status has been reset		
7642	1	1	Relocation status detected - temporarily released	The relocation detection status did not reset.	Check the request code and contact the sales dealer.
7643	1	1	Reading enabled for external encoder on *axis	Reading is enabled for external encoder on 4th-axis.	Check the machine parameter.
7644	1	1	Reading enabled for external encoder on *axis	Reading is enabled for external encoder on 5th-axis.	Check the machine parameter.
7645	1	1	Reading enabled for external encoder on *axis	Reading is enabled for external encoder on 6th-axis.	Check the machine parameter.
7646	1	1	Reading enabled for external encoder on *axis	Reading is enabled for external encoder on 7th-axis.	Check the machine parameter.
7647	1	1	Reading enabled for external encoder on *axis	Reading is enabled for external encoder on 8th-axis.	Check the machine parameter.
7648	1	1	The operation status has been saved.		
7649	1	1	Outside of specification range.	Attempted to exceed the range for an allowable start position during extended memory operation.	
7650	1	1	Graph cannot be executed	A setting besides <0:2 Mbyte> has been set for the user parameter (switch 1) <Program load size>.	Set <0:2 Mbyte> for the user parameter (switch 1) <Program load size>.
7651	1	1	Adding additional axes is restricted on this machine(*)		
7652	1	1	Adding additional axes is not restricted on this machine		
7663	1	1	Screenshot was taken		

No.	Stop level	Reset level	Alarm message	Cause	Solution
7664	1	1	Automatic door - Machine zero return in progress	The DR_ORGMV_ERR signal (BY: Automatic door - Machine zero return in progress) was detected. After the power turned ON, an alarm is triggered at the first door operation in the factory set ladder program. The automatic door does not operate for a certain time or operates slowly while an alarm is triggered.	Press the [RST] key to reset the alarm. If the alarm occurs every time the door operates, check the ladder program. Contact us and report the conditions of the error when it occurs in a factory-installed ladder program. 1. Check the door limit switch. 2. Make sure that there are no chips or shavings caught between the door and the machine cover. 3. The door sensor is faulty. Replace the door sensor. 4. The IL PCB is faulty. Replace the IL PCB. 5. The IO PCB is faulty. Replace the IO PCB.
7665	1	1	Waveform data is saved	When <Save standby at measurement start> is set to <1: Yes>, an alarm is triggered while the waveform data is being saved.	
7666	1	1	*-axis servo is disabled	Setting for AT-axis servo is disabled.	Disengage the mode for enabling and disabling the servo.
7667	1	1	Loading system alarm triggered	An alarm is triggered on the loading system.	Check the loading system status.
7668	1	1	Axis operation lock will be temporarily disengaged.		
7669	1	1	Use [RELSE] + [G] keys to restore ATC arm position	1. ATC arm position is not at the zero point. 2. ATC arm moved to 180 degrees. 3. Pot position or spindle position was moved. 4. When the ATC arm was not between 175 and 185 degrees, one of the following key combinations for recovery was pressed: Spindle position recovery ([RELSE] + [H]), pot down ([RELSE] + [D]) or pot shutter open ([RELSE] + [P]).	Press the [RELSE] + [G] keys and follow instructions for the alarm or operator message.
7670	1	1	Use [RELSE] + [H] keys to restore spindle position	1. The Y-axis moved following the Y-axis machine parameter (system 1) <Safe position for Y-axis when restoring operation>, and a spindle orientation operation was carried out. 2. The keys ([RELSE] + [G]) for the ATC arm position recovery were pressed when the spindle was not close to the zero point. 3. The keys ([RELSE] + [G]) for the ATC arm position recovery were pressed when the Y-axis was not close to the ATC position.	Press the [RELSE] + [H] keys and follow instructions for the alarm or operator message.

Chapter 11 Alarms

No.	Stop level	Reset level	Alarm message	Cause	Solution
7671	1	1	Use [RELSE] + [D] keys to lower pot	1. The keys ([RELSE] + [G]) for the ATC arm position recovery were pressed when the pot position was not in the <Bottom> position.	Press the [RELSE] + [D] keys and follow instructions for the alarm or operator message.
7672	1	1	Use [RELSE] + [P] keys to open pot shutter	1. The keys ([RELSE] + [G]) for the ATC arm position recovery were pressed when the pot shutter was not in the <Open> position. 2. The keys ([RELSE] + [D]) for pot down were pressed when the pot shutter was not in the <Open> position.	Press the [RELSE] + [P] keys and follow instructions for the alarm or operator message.
7673	1	1	Select the tool status	1. A recovery operation was carried out when the status of the tool that is attached to the spindle, ATC arm or pot was not selected.	Check the status of the tools that are attached to the spindle, ATC arm and pot, and then press the corresponding function key.

11.8 List of servo error

11.8.1 Servo errors

- Servo errors: Alarms which are generated by the servos (amps).
- Servo warnings: Warnings which are generated by the servos (amps).
- Servo communication errors: Alarms which are generated when data such as servo parameters are received. (SDO communication errors)
- Servo status errors: Generated when data from the servos has not been received correctly.
The causes of errors and the methods of resolving them are the same for all axes.

(Supplement)

Explanation of table contents

- Cancel levels ○: Reset ×: Power supply OFF
- Messages displayed as supplementary messages (XXX): If the space is blank, the supplementary number is displayed.

11.8.1.1 Servo errors

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
15	NC power failure error	0Fh	NC power failure error - NC is detected as being stopped.	Check the power supply, and fix any problems such as power interruptions or voltage drops.	×
16	Invalid frame 0	10h	Invalid frame error on port 0 Rx - Invalid frame is received continuously from port 0. 1. Faulty communications cable. 2. Faulty servo amplifier or NC PCB.	1. Make sure that there are no poor or faulty contacts on the communications cable. 2. Replace the servo amplifier or NC PCB.	○
17	Invalid frame 1	11h	Invalid frame error on port 1 Rx - Invalid frame is received continuously from port 1. 1. Faulty communications cable. 2. Faulty servo amplifier or NC PCB.	1. Make sure that there are no poor or faulty contacts on the communications cable. 2. Replace the servo amplifier or NC PCB.	○
18	CRC error 0	12h	CRC error on port 0 Rx - Continuous error triggered on port 0 Rx. 1. Faulty communications cable. 2. Faulty servo amplifier or NC PCB.	1. Make sure that there are no poor or faulty contacts on the communications cable. 2. Replace the servo amplifier or NC PCB.	○
19	CRC error 1	13h	CRC error on port 1 Rx - Continuous error triggered on port 1 Rx. 1. Faulty communications cable. 2. Faulty servo amplifier or NC PCB.	1. Make sure that there are no poor or faulty contacts on the communications cable. 2. Replace the servo amplifier or NC PCB.	○
20	Transmission error 0	14h	Error on port 0 Tx - Continuous error triggered on port 0 Tx. 1. Faulty communications cable. 2. Faulty servo amplifier or NC PCB.	1. Make sure that there are no poor or faulty contacts on the communications cable. 2. Replace the servo amplifier or NC PCB.	○
21	Transmission error 1	15h	Error on port 1 Tx - Continuous error triggered on port 1 Tx. 1. Faulty communications cable. 2. Faulty servo amplifier or NC PCB.	1. Make sure that there are no poor or faulty contacts on the communications cable. 2. Replace the servo amplifier or NC PCB.	○

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
24	Lost link 0	18h	Lost link on port 0 - Port 0/1 cable is broken or disconnected while servo is ON. The upstream power supply has been interrupted or turned OFF. 1. Broken or disconnected communications cable. 2. Poor or faulty contact between connector and terminals.	1. Replace the communications cable. 2. Make sure that there are no poor or faulty contacts on the communications cable.	○
25	Lost link 1	19h	Lost link on port 1 - Port 0/1 cable is broken or disconnected while servo is ON. The upstream power supply has been interrupted or turned OFF. 1. Broken or disconnected communications cable. 2. Poor or faulty contact between connector and terminals.	1. Replace the communications cable. 2. Make sure that there are no poor or faulty contacts on the communications cable.	○
26	Communication time out	1Ah	Communication timeout - Output data not received during the specified cycle time. The command could not be received during the time period specified by the communication timeout setting. (SM2 event: PDO output data cannot be received) 1. Faulty communications cable. 2. Faulty servo amplifier or NC PCB.	1. Make sure that there are no poor or faulty contacts on the communications cable. 2. Replace the servo amplifier or NC PCB.	○
33	Main circuit error	21h	Error in main circuit power device (power element error) - Drive module overcurrent. - Drive power supply error detected. - Drive module overheating. 1. There is a short-circuit on the wiring between the servo amplifier and servo motor for the U, V and W phases on the servo amplifier. Or, there is a ground fault on the U, V and W phases. 2. There is a short-circuit or ground fault on the U, V and W phases for the servo motor side. 3. Faulty circuit inside the servo amplifier. 4. Faulty control box cooling fan.	1. Check the wiring, and fix any problems such as short-circuits. 2. Replace the servo motor. 3. Replace the servo amplifier. 4. Replace the control box cooling fan.	○
34	Current detection error 0	22h	Current detection error 0 - Current detection value error. 1. Faulty circuit inside the servo amplifier. 2. Faulty servo parameter setting.	1. Replace the servo amplifier. 2. Set the correct servo parameters.	○
35	Current detection error 1	23h	Current detection error 1 - Current detection circuit error.	1. Replace the servo amplifier.	○
36	Current detection error 2	24h	Current detection error 2 - Communication error on current detection circuit. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	○

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
37	HWGOFF error 1	25h	Safe torque (thrust) OFF error 1 - Timing error for safe torque (thrust) OFF input. - Safe torque (thrust) OFF 1 and safe torque (thrust) OFF 2 input logic do not match. 1. Faulty circuit inside the servo amplifier. 2. Faulty IL PCB. 3. Faulty connection cable between servo amplifier and IL PCB. (When also using the BV7) 4. DRRLY PCB is faulty.	1. Replace the servo amplifier. 2. Replace the IL PCB. 3. Replace the cable. (When also using the BV7) 4. Replace the DRRLY PCB.	×
38	HWGOFF error 2	26h	Safe torque (thrust) OFF error 2 - Failure or malfunction on safe torque (thrust) OFF circuit. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	×
49	Converter error	31h	Converter power device error - Converter module overcurrent. - Converter power supply error. - Converter module overheating. 1. Faulty circuit inside the servo amplifier. 2. There is a P/N short-circuit on the main circuit power supply DC block for the servo amplifier. Or, there is a P/N ground fault.	1. Replace the servo amplifier. 2. Check the wiring, and fix any problems such as short-circuits.	○
50	Phase detection error	32h	Phase detection error - Phase detection circuit error. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	○
51	Power supply current detection error	33h	Power supply current detection error - Power supply current detection circuit error. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	×
65	Overload 1	41h	Overload 1 - Effective torque (thrust) exceeded. 1. Faulty circuit inside the servo amplifier. 2. Faulty circuit inside motor encoder. 3. Effective load monitor has exceeded rated torque (thrust). 4. Faulty servo parameter setting. 5. Servo motor holding brake is not released (faulty connection cable between servo amplifier and IL PCB, or faulty connection cable between servo amplifier and IL PCB servo motor, or faulty IL PCB). 6. Broken or disconnected servo motor connection cable.	1. Replace the servo amplifier. 2. Replace the servo motor. 3. Monitor the load with the effective load monitor, check if the effective torque (thrust) exceeds the rated value, and revise the operation or load conditions. 4. Set the correct servo parameters. 5. Replace the cable or the IL PCB. 6. Replace the cable.	○

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
66	Overload 2	42h	Overload 2 - Stall overload. 1. Faulty circuit inside the servo amplifier. 2. Faulty circuit inside motor encoder. 3. Faulty servo parameter setting. 4. Servo motor holding brake is not released (faulty connection cable between servo amplifier and IL PCB, or faulty connection cable between servo amplifier and IL PCB servo motor, or faulty IL PCB). 5. Broken or disconnected servo motor connection cable.	1. Replace the servo amplifier. 2. Replace the servo motor. 3. Set the correct servo parameters. 4. Replace the cable or the IL PCB. 5. Replace the cable.	○
67	Regeneration resistance overload	43h	Regeneration resistance overload - Regeneration resistance load rate has been exceeded. 1. Regenerative power supply capacity has been exceeded according to the internal regeneration resistance specifications. Load inertia (weight on table) is too great. Or, the takt time is too short. 2. Faulty circuit inside servo amplifier.	1. Revise the operation conditions. If the load inertia (weight on table) exceeds the specified range, set it to within the range. 2. Replace the servo amplifier.	○
69	Continuous rotation speed excessive	45h	Continuous rotation speed excessive - Average rotation speed has been exceeded. 1. The average rotation speed exceeds the maximum rotation speed in a continuous range. The takt time is too short. 2. Faulty circuit inside servo amplifier.	1. Revise the operation conditions. 2. Replace the servo amplifier.	○
70	Converter overload	46h	Converter overload - Effective value for power supply current is excessive. 1. Faulty circuit inside the servo amplifier. 2. There is a P/N short-circuit on the main circuit power supply DC block for the servo amplifier. Or, there is a P/N ground fault.	1. Replace the servo amplifier. 2. Check the wiring, and fix any problems such as short-circuits.	○
71	Converter is overcurrent.	47h	Converter overcurrent - When the cumulative time takes more than 1 ms for the power supply current (during regeneration) to achieve more than 120% of the converter power module capacity. 1. Servo amplifier internal circuit is faulty.	1. Replace the servo amplifier.	○

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
81	Amplifier temperature error	51h	Amplifier temperature error - Overheating detected for servo amplifier ambient temperature. 1. Faulty circuit inside the servo amplifier. 2. Control box cooling fan is stopped.	1. Replace the servo amplifier. 2. Replace the control box cooling fan.	○
82	Rush prevention resistor overheat	52h	Rush prevention resistor overheat - Overheating detected on rush prevention resistor. 1. Faulty circuit inside the servo amplifier. 2. Power supply turned ON too frequently. 3. Control box cooling fan is stopped.	1. Replace the servo amplifier. 2. Reduce the frequency for turning the power supply ON and OFF. 3. Replace the control box cooling fan.	○
83	DB resistor overheat	53h	Dynamic brake resistor overheating - Overheating detected on the dynamic brake resistor. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	○
84	Internal regeneration resistance overheat	54h	Internal regeneration resistance overheat - Overheating detected on the internal regeneration resistance. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	○
85	External error	55h	External error - Error in external regeneration resistor, etc. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	○
86	Main circuit overheat	56h	Main circuit power device overheating - Overheating detected on the drive module (15, 30 and 50 A). 1. There is a short-circuit on the wiring between the servo amplifier and servo motor for the U, V and W phases on the servo amplifier. Or, there is a ground fault on the U, V and W phases. 2. There is a short-circuit or ground fault on the U, V and W phases for the servo motor side. 3. Faulty circuit inside the servo amplifier.	1. Replace the cable. 2. Replace the servo motor. 3. Replace the servo amplifier.	○

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
87	Discharge resistance overheat	57h	Overheating on rush prevention and discharge resistance - Overheating detected on rush current prevention resistor detected. - Overheating detected on discharge resistor. 1. Faulty circuit inside the servo amplifier. 2. Power supply turned ON too frequently. 3. Power supply regeneration circuit output is excessive. 4. There is a P/N short-circuit on the main circuit power supply DC block for the servo amplifier. Or, there is a P/N ground fault.	1. Replace the servo amplifier. 2. Reduce the frequency for turning the power supply ON and OFF. 3. Revise the operation conditions. 4. Replace the cable.	○
97	Overvoltage	61h	Overvoltage - DC overvoltage on main circuit. 1. Faulty circuit inside the servo amplifier. 2. Main circuit power supply voltage is outside the specified range.	1. Replace the servo amplifier. 2. Make sure that the power supply voltage is within the specified range.	○
98	Main circuit power shortage	62h	Main circuit power shortage - DC voltage sag on main circuit. 1. Main circuit power supply voltage is below the specified range. 2. Faulty circuit inside the servo amplifier.	1. Make sure that the power supply voltage is within the specified range. 2. Replace the servo amplifier.	○
99	Main power supply open-phase	63h	Open-phase on main power supply - One phase in a 3 phase main circuit power supply is open. 1. One phase is not being input in a 3 phase input R, S and T. 2. Faulty circuit inside the servo amplifier.	1. Check the wiring, and fix any incorrect wiring. 2. Replace the servo amplifier.	○
100	Power supply frequency error	64h	Power supply frequency error - The power supply frequency is lower than 45 Hz or higher than 65 Hz. 1. The power supply frequency is outside the specified range. 2. Faulty circuit inside the servo amplifier.	1. Make sure that the power supply is within the specified range. 2. Replace the servo amplifier.	○
101	Power unbalance ratio error	65h	Power unbalance ratio error - The unbalance ratio for the power supply phases is 8% or more (unbalance ratio for power supply voltages is approximately 10%). 1. The power supply frequency is outside the specified range. 2. Faulty circuit inside the servo amplifier.	1. Make sure that the power supply is within the specified range. 2. Replace the servo amplifier.	○

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
102	Power supply voltage detection error	66h	Power supply voltage detection error - Power supply voltage detection circuit error. 1. The power supply is outside the specified range. 2. Faulty circuit inside the servo amplifier.	1. Make sure that the power supply is within the specified range. 2. Replace the servo amplifier.	×
103	Power overvoltage	67h	Power overvoltage - The peak value for the main circuit AC power voltage exceeded 375 V. (Excluding when the power supply regeneration circuit is operating.)	1. Make sure that the power supply voltage is within the specified range. 2. Replace the servo amplifier.	○
113	Control power shortage	71h	Insufficient control power supply voltage - The control input power supply voltage dropped or a transient power fault occurred. 1. Faulty circuit inside the servo amplifier. 2. Power supply input voltage is below the specified range.	1. Replace the servo amplifier. 2. Make sure that the power supply voltage is within the specified range.	○
114	12V power shortage	72h	Insufficient control power supply voltage 1 - +12 V voltage dropped on the switching power supply used for control. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	○
115	5V power shortage	73h	Insufficient control power supply voltage 2 - +5 V dropped on the switching power supply used for control. 1. Faulty circuit inside the servo amplifier. 2. Faulty external circuit.	1. Replace the servo amplifier. 2. Disconnect the connector and turn ON the power again. If there is no alarm, check the external circuit.	×
129	Encoder disconnected 1	81h	Encoder connector 1 disconnected 1. Broken or disconnected phase signal wire for the incremental encoders A, B and Z (faulty servo motor or servo amplifier). 2. Broken or disconnected power supply line.	1. Replace the servo motor or the servo amplifier. 2. Replace the cable.	×
131	Encoder disconnected 2	83h	Encoder connector 2 disconnected 1. Broken or disconnected phase signal wire for the fully-closed encoders A, B and Z (faulty servo motor or servo amplifier). 2. Broken or disconnected power supply line.	1. Replace the servo motor or the servo amplifier. 2. Replace the cable.	○
132	Encoder communication error	84h	Serial encoder communication error - CRC, SYNC or FORM command error detected during communication with sensors. 1. Faulty circuit inside motor encoder. 2. Error in encoder wiring.	1. Replace the servo motor. 2. Replace the cable.	×

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
133	Encoder initial error 1	85h	Encoder initial processing error 1 - Error in reading CS data for incremental encoder. - Absolute encoder initial processing error. - Broken or disconnected cable. 1. Error in encoder wiring. 2. Faulty circuit inside the servo amplifier. 3. Faulty circuit inside motor encoder.	1. Replace the cable. 2. Replace the servo amplifier. 3. Replace the servo motor.	×
134	CS error	86h	CS error - CS data position skipped. 1. Error in encoder wiring. 2. Faulty circuit inside the servo amplifier. 3. Faulty circuit inside motor encoder.	1. Replace the cable. 2. Replace the servo amplifier. 3. Replace the servo motor.	×
135	CS disconnected	87h	CS disconnected - Broken or disconnected CS signal line. 1. Error in encoder wiring. 2. Faulty circuit inside the servo amplifier. 3. Faulty circuit inside motor encoder.	1. Replace the cable. 2. Replace the servo amplifier. 3. Replace the servo motor.	×
137	Encoder initial error 2	89h	Encoder initial processing error 2 and absolute encoder is BUSY 1. Faulty circuit inside servo amplifier. 2. Faulty circuit inside motor encoder.	1. Replace the servo amplifier. 2. Replace the servo motor.	×
160	Encoder error 0	A0h	Serial encoder internal error 0 - No alarm status exists for the status information read from the encoder. 1. Malfunction caused by noise. 2. Faulty circuit inside motor encoder.	When this alarm is triggered even after turning the power OFF and ON again, check the following. 1. Make sure that the servo amplifier ground is grounded correctly. 2. Make sure that the encoder cable is shielded. 3. Replace the servo motor.	×
161	Battery is low (Symbol display)	A1h	Serial encoder internal error 1 - Battery alarm	Replace the backup batteries.	○
162	Encoder error 2	A2h	This alarm does not trigger.	This alarm does not trigger.	○
163	Encoder error 3	A3h	Serial encoder internal error 3 - Speed error	Replace the servo motor.	○
164	Encoder error 4	A4h	Serial encoder internal error 4 - EEPROM error (not on spindle) - Calibration memory access error (spindle and lathe spindle)	Replace the servo motor.	○

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
165	Encoder error 5	A5h	Serial encoder internal error 5 1. ST error. The absolute sensor and the ASIC internal shift register data do not match (not spindle). 2. Excessive load and position information became invalid. 3. Communication error occurred during axis operation, and position information became invalid. 4. Pulse error (spindle without CTS) 5. Single-turn counter error detected (spindle or lathe spindle with CTS)	Revise the operation conditions when an error occurs during operation. Reset the absolute encoder counter and readjust the position if the error occurs even when the power is turned OFF and back ON. Replace the servo motor if the error still occurs even after performing the recovery steps above..	○
166	Encoder error 6	A6h	Serial encoder internal error 6 - PS error - CS error 1. Multi-turn counter error detected (not spindle). 2. Error in reading temporary position after turning the power ON (spindle without CTS).	1. If the error is still triggered after turning the power OFF and ON, reset the absolute encoder counter and readjust the position. 2. Replace the servo motor.	○
169	Encoder error 9	A9h	Serial encoder internal error 9 - Operation temperature error	Replace the servo motor.	○
170	Encoder error 10	AAh	Serial encoder internal error 10 - Analog signal error 1	Replace the servo motor.	○
172	Encoder error 12	ACH	This alarm does not trigger.	This alarm does not trigger.	○
173	Encoder error 13	ADh	Serial encoder internal error 13 - Setting memory access error	Replace the servo motor.	○
174	Encoder error 14	Aeh	This alarm does not trigger.	This alarm does not trigger.	○
175	Encoder error 15	AFh	Serial encoder internal error 15 - Analog signal error 2	Replace the servo motor.	○
176	Version consistency error	B0h	NC PCB and servo amplifier software version combination do not match	Update the software to the latest version for the firmware.	×
177	DB relay adhesion error	B1h	Adhesion occurred on DB relay contact.	Replace the servo amplifier.	○
193	Overspeed	C1h	Overspeed - 120% of maximum motor speed exceeded. 1. Faulty circuit inside the servo amplifier. 2. Faulty circuit inside motor encoder.	1. Replace the servo amplifier. 2. Replace the servo motor.	○
194	Speed control error	C2h	Speed control error - The current command and acceleration sign do not match. 1. The U, V and W phase wiring between the servo amplifier and servo motor is incorrect. 2. The A and B phase wiring for the pulse encoder is incorrect. 3. The servo motor is vibrating (oscillating).	1. Replace the cable.	○

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
195	Speed FB error	C3h	Speed feedback error 1. Servo motor power line broken or disconnected. 2. Faulty circuit inside servo amplifier. 3. When opening/closing the door, the [DOOR INTERLOCK] switch was enabled/disabled at the same time.	1-1. Replace the cable. 1-2. Replace the servo motor. 2. Replace the servo amplifier. 3. Press the [RST] key. When opening/closing a door, do not enable/disable the [DOOR INTERLOCK] switch at the same time.	○
197	Model following error	C5h	Damping control error for model tracking - The operating pattern is not suitable for damping control during model tracking. 1. Faulty servo parameter setting.	1. Set the correct servo parameters.	○
198	Spindle speed decreased	C6h	Spindle speed decreased - Speed has dropped below setting value (main spindle only). 1. Main spindle load is excessive.	1. Revise the operation conditions.	○
199	Overspeed	C7h	Overspeed - The actual speed is faster than the internal speed command for the servo amplifier.	1. Set the correct servo parameters. 2. Replace the servo motor.	○
209	Position deviation large	D1h	Position deviation large - Position deviation exceeds setting value. 1. Load inertia is too great. 2. Holding brake is not released. 3. One or all of the U, V and W phase wiring between the servo amplifier and servo motor is disconnected. 4. Faulty circuit inside the servo amplifier. 5. Faulty circuit inside motor encoder. 6. Voltage has dropped for main circuit power supply.	1. Revise the load conditions (weight on table). 2. Replace the cable or the servo motor. 3. Replace the cable. 4. Replace the servo amplifier. 5. Replace the servo motor. 6. Make sure that the power supply capacity for the main circuit and wiring are sufficient.	○
210	Position command error 1	D2h	Position command error 1 - Position command exceeds the setting range. 1. Faulty servo amplifier or NC PCB.	1. Replace the servo amplifier or the NC PCB. 2. Set the correct servo parameters.	○
211	Position command error 2	D3h	Position command error 2 - A position command which exceeds the processing range was input. 1. Faulty servo amplifier or NC PCB.	1. Replace the servo amplifier or the NC PCB.	○

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
221	Insulation resistance detection error	DDh	Insulation resistance detection error - Command error or status error during insulation resistance detection. - Short-circuit malfunction or failure on the insulation resistance detection circuit. 1. Main circuit power supply was turned OFF. (Emergency stop switch was pressed.) 2. Hardware gate OFF input was turned OFF. (Door was opened.) 3. Motor is not stopped.	1. Release the emergency stop switch, and then reset the insulation resistance detection mode. 2. Close the door, and then set the insulation resistance detection mode again. 3. Make sure that the motor has stopped, and then set the insulation resistance detection mode again. 4. Replace the servo amplifier.	○
222	Turn power off.	DEh	Turn the power supply OFF. - A servo parameter setting was changed and the power supply needs to be turned ON and OFF in order for the change to take effect.	1. Turn the power supply OFF and back ON again.	×
223	Test mode ended	DFh	Test mode ended - End of test mode is detected. 1. This is an indication that the test mode has ended, and operation is normal. (After the test mode is executed, the deviation on the controller is expected to remain, and therefore an alarm is triggered.)	1. Reset the alarm to restore normal operation.	○
225	ROM error	E1h	EEPROM error - EEPROM error inside amplifier. 1. Faulty circuit inside the servo amplifier. (There was no response from EEPROM when the servo parameters are maintained.) (Faulty amplifier control PCB)	1. Replace the servo amplifier.	×
226	Check sum error	E2h	EEPROM check sum error - Check sum error over whole EEPROM range. 1. The correct values were not written to the CPU from the EEPROM in the servo amplifier. 2. There was an error in writing the values to EEPROM the last time the power supply was turned OFF.	1. Replace the servo amplifier. 2. Replace the servo amplifier.	×
227	Memory error 1	E3h	Memory error 1 - Error accessing RAM inside CPU. 1. The RAM inside the CPU could not be accessed normally. (Control parts for servo amplifier are faulty)	1. Replace the servo amplifier.	×

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
228	Memory error 2	E4h	Memory error 2 - Flash memory check sum error. 1. Faulty circuit inside the servo amplifier. (Flash memory program check sums did not match when the control power supply was turned ON.) (The firmware inside the servo amplifier CPU is corrupt.)	1. Replace the servo amplifier.	×
229	Parameter error 1	E5h	System parameter error 1 - System parameters are outside setting range. 1. The system parameter values are outside the setting range. 2. Faulty circuit inside the servo amplifier.	1-1. Check the servo amplifier model number. 1-2. Check and adjust the system parameter setting values. Turn the power supply OFF and back ON again, and make sure that no alarm is triggered. 2. Replace the servo amplifier.	×
230	Parameter error 2	E6h	System parameter error 2 - System parameter combination error. - The system parameters and amplifier combination do not match. 1-1. The system parameter setting values and actual hardware combination is incorrect. 1-2. The system parameter setting combination is incorrect. 2. Faulty circuit inside the servo amplifier.	1-1. Check the servo amplifier model number. 1-2. Check and adjust the system parameter setting values. Turn the power supply OFF and back ON again, and make sure that no alarm is triggered. 2. Replace the servo amplifier.	×
231	Motor parameter error	E7h	Motor parameter error - Motor parameter check sum error. 1. The correct values were not written to the CPU from the EEPROM in the servo amplifier. 2. There was an error in writing values to EEPROM when the motor parameters were changed.	1. Set the motor parameters again, and then turn the power OFF and back ON again. If the alarm occurs again, replace the servo amplifier. 2. Set the motor parameters again, and then turn the power OFF and back ON again. If the alarm occurs again, replace the servo amplifier.	×
232	CPU peripheral circuit error	E8h	CPU peripheral circuit error - Error accessing the CPU and peripheral devices. 1-1. Access to the CPU and peripheral devices was not completed successfully during initialization. 1-2. Faulty servo amplifier control PCB.	1. Replace the servo amplifier.	×

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
233	System code error	E9h	System code error - The control PCB code and sensor setting do not match. 1. The servo amplifier control PCB sensor and sensor setting value do not match (Faulty servo amplifier control PCB).	1. Replace the servo amplifier.	×
234	Motor code error	EAh	Motor code setting error - The motor code is outside the setting range. Faulty servo parameter setting.	Set the correct servo parameters.	×
235	Sensor code error	EBh	Sensor code setting error - The sensor code is outside the setting range. Faulty servo parameter setting.	Set the correct servo parameters.	×
236	Converter communication error	ECh	Converter communication error - Error in communication circuit for converter CPU. - Converter CPU error. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	×
237	Converter early error	EDh	Converter initialization processing error - Error accessing the RAM inside the converter CPU. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	×
238	Autoconfiguration error 1	EEh	Motor parameter autoconfiguration error 1 - The motor parameter autoconfiguration cannot be executed. 1. Faulty servo parameter setting. 2. Faulty circuit inside motor encoder.	1. Set the correct servo parameters. 2. Replace the servo motor.	×
239	Autoconfiguration error 2	EFh	Motor parameter autoconfiguration error 2 - There is an error in motor parameter autoconfiguration result. 1. Faulty servo parameter setting. 2. Faulty circuit inside motor encoder.	1. Set the correct servo parameters. 2. Replace the servo motor.	×
241	Task processing error	F1h	Task processing error - CPU interrupt processing error. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	×
242	Initial timeout	F2h	Initial timeout - Initialization processing was not completed within the specified time. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	×

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
244	FPGA error 1	F4h	Encoder receiving FPGA error 1 - Encoder receiving FPGA configuration is not complete - Encoder receiving FPGA configuration data is corrupt 1. Faulty circuit inside servo amplifier.	1. Replace the servo amplifier.	×
245	FPGA error 2	F5h	Encoder receiving FPGA error 2 - Error when accessing the encoder receiving FPGA - Clock stopped inside the encoder receiving FPGA - Request signal to encoder receiving FPGA interrupted - Servo ASIC failed - Mismatch identified in amplifier capacity between amplifier and encoder receiving FPGA - Parameter writing error for encoder receiving FPGA 1. Faulty circuit inside servo amplifier.	1. Replace the servo amplifier.	×
246	Servo amplifier does not match	F6h	There is a problem with the type of servo amplifier connected.	Check the type of servo amplifier.	×
247	Motor does not match	F7h	There is a problem with the type of motor connected.	Check the type of motor.	×
255	Flash time-out	FFh	Self-flash timeout - Self-flash write processing was not completed within the specified time. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	×
500	Positioning amplifier	00h	This occurs when a servo error is detected and the cause is not immediately clear. Thereafter, once the cause is determined, it occurs together with the servo error for that cause.	Follow the recovery procedure for servo errors that occur with other problems.	○
522	Software limit (+)	16h	Software limit error (+) - Moved beyond the software limit coordinate in the positive direction.	Move back inside the software limit. Set the correct parameters.	○
523	Software limit (-)	17h	Software limit error (-) - Did not reach the software limit coordinate in the negative direction.	Move back inside the software limit. Set the correct parameters.	○
524	Point data invalid	18 h	Point data invalid error - Moved to a point that is not registered.	Press the [RST] key.	○
527	Machine zero return error	1Bh	Machine zero return operation error - The deceleration time was too short during a machine zero return operation.	Press the [RST] key.	○

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
528	Position mismatch error	1Ch	Position data mismatch error - The encoder position cannot be identified correctly.	The machine needs to be readjusted. Press the [RELSE] and the [RST] keys. After releasing the alarm, the machine zero point needs to be readjusted.	○
533	Main circuit error	21h	Error in main circuit power device (power element error) - Drive module overcurrent. - Drive power supply error detected. - Drive module overheating. 1. There is a short-circuit on the wiring between the servo amplifier and servo motor for the U, V and W phases on the servo amplifier. Or, there is a ground fault on the U, V and W phases. 2. There is a short-circuit or ground fault on the U, V and W phases for the servo motor side. 3. Faulty circuit inside the servo amplifier. 4. Faulty control box cooling fan.	1. Check the wiring, and fix any problems such as short-circuits. 2. Replace the servo motor. 3. Replace the servo amplifier. 4. Replace the control box cooling fan.	○
534	Current detection error 0	22 h	Current detection error 0 - Current detection value error. 1. Faulty circuit inside the servo amplifier. 2. Faulty servo parameter setting.	1. Replace the servo amplifier. 2. Set the correct servo parameters.	○
535	Current detection error 1	23 h	Current detection error 1 - Current detection circuit error.	1. Replace the servo amplifier.	○
536	Current detection error 2	24h	Current detection error 2 - Communication error on current detection circuit. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	○
549	Overtravel (+)	31h	Forward overtravel - Power (DC +24 V) for external input signal is faulty. - External wiring is wrong. - Overtravel signal logic is wrong. - Faulty circuit inside the servo amplifier. - Overtraveled.	Set power for external input signal to (DC +24 V+/-10%). Check the wiring, and fix any problems. Replace the servo amplifier. Move to a valid range.	○
550	Overtravel (-)	32h	Reverse overtravel - The same as forward overtravel.		○

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
565	Overload 1	41h	Overload 1 - Effective torque (thrust) exceeded. 1. Faulty circuit inside the servo amplifier. 2. Faulty circuit inside motor encoder. 3. Effective load monitor has exceeded rated torque (thrust). 4. Faulty servo parameter setting. 5. Servo motor holding brake is not released (faulty connection cable between servo amplifier and IL PCB, or faulty connection cable between servo amplifier and IL PCB servo motor, or faulty IL PCB). 6. Broken or disconnected servo motor connection cable.	1. Replace the servo amplifier. 2. Replace the servo motor. 3. Monitor the load with the effective load monitor, check if the effective torque (thrust) exceeds the rated value, and revise the operation or load conditions. 4. Set the correct servo parameters. 5. Replace the cable or the IL PCB. 6. Replace the cable.	○
566	Overload 2	42h	Overload 2 - Stall overload. 1. Faulty circuit inside the servo amplifier. 2. Faulty circuit inside motor encoder. 3. Faulty servo parameter setting. 4. Servo motor holding brake is not released (faulty connection cable between servo amplifier and IL PCB, or faulty connection cable between servo amplifier and IL PCB servo motor, or faulty IL PCB). 5. Broken or disconnected servo motor connection cable.	1. Replace the servo amplifier. 2. Replace the servo motor. 3. Set the correct servo parameters. 4. Replace the cable or the IL PCB. 5. Replace the cable.	○
567	Regeneration resistance overload	43h	Continuous rotation speed excessive - Average rotation speed has been exceeded. 1. The average rotation speed exceeds the maximum rotation speed in a continuous range. The takt time is too short. 2. Faulty circuit inside servo amplifier.	1. Revise the operation conditions. 2. Replace the servo amplifier.	○
581	Amplifier temperature error	51h	Amplifier temperature error - Overheating detected for servo amplifier ambient temperature. 1. Faulty circuit inside the servo amplifier. 2. Control box cooling fan is stopped.	1. Replace the servo amplifier. 2. Replace the control box cooling fan.	○

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
582	Rush prevention resistor overheat	52h	Rush prevention resistor overheat - Overheating detected on rush prevention resistor. 1. Faulty circuit inside the servo amplifier. 2. Power supply turned ON too frequently. 3. Control box cooling fan is stopped.	1. Replace the servo amplifier. 2. Reduce the frequency for turning the power supply ON and OFF. 3. Replace the control box cooling fan.	○
583	DB resistor overheat	53h	Dynamic brake resistor overheating - Overheating detected on the dynamic brake resistor. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	○
584	Internal regeneration resistance overheat	54h	Internal regeneration resistance overheat - Overheating detected on the internal regeneration resistance. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	○
585	External error	55h	External error - Error in external regeneration resistor, etc. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	○
597	Overvoltage	61h	Overvoltage - DC overvoltage on main circuit. 1. Faulty circuit inside the servo amplifier. 2. Main circuit power supply voltage is outside the specified range.	1. Replace the servo amplifier. 2. Make sure that the power supply voltage is within the specified range.	○
598	Main circuit power shortage	62h	Main circuit power shortage - DC voltage sag on main circuit. 1. Main circuit power supply voltage is below the specified range. 2. Faulty circuit inside the servo amplifier.	1. Make sure that the power supply voltage is within the specified range. 2. Replace the servo amplifier.	○
599	Main power supply open-phase	63h	Open-phase on main power supply - One phase in a 3 phase main circuit power supply is open. 1. One phase is not being input in a 3 phase input R, S and T. 2. Faulty circuit inside the servo amplifier.	1. Check the wiring, and fix any incorrect wiring. 2. Replace the servo amplifier.	○
613	Control power shortage	71h	Insufficient control power supply voltage - The control input power supply voltage dropped or a transient power fault occurred. 1. Faulty circuit inside the servo amplifier. 2. Power supply input voltage is below the specified range.	1. Replace the servo amplifier. 2. Make sure that the power supply voltage is within the specified range.	○

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
614	12V power shortage	72h	Insufficient control power supply voltage 1 - +-12 V voltage dropped on the switching power supply used for control. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	○
629	Pulse signal error 1	81h	Phase A / Phase B pulse signal error - Problem with encoder wiring. - Faulty circuit inside motor encoder. - Faulty circuit inside the servo amplifier.	1. Replace the cable. 2. Replace the servo amplifier. 3. Replace the servo motor.	×
630	Abs. signal disconnection	82h	Absolute encoder signal disconnection - Problem with encoder wiring. - Faulty circuit inside motor encoder. - Faulty circuit inside the servo amplifier.	1. Replace the cable. 2. Replace the servo amplifier. 3. Replace the servo motor.	○
631	External encoder signal error	83h	External encoder phase A / B signal error - Problem with encoder wiring. - Faulty circuit inside motor encoder. - Faulty circuit inside the servo amplifier.	1. Replace the cable. 2. Replace the servo amplifier. 3. Replace the servo motor.	○
632	Encoder communication error	84h	Communication error between encoder and amplifier - Problem with encoder wiring. - Faulty circuit inside motor encoder. - Faulty circuit inside the servo amplifier.	1. Replace the cable. 2. Replace the servo amplifier. 3. Replace the servo motor.	○
633	Encoder initial error	85h	Encoder initial processing error - Error in reading CS data for incremental encoder. - Absolute encoder initial processing error. - Broken or disconnected cable. 1. Error in encoder wiring. 2. Faulty circuit inside the servo amplifier. 3. Faulty circuit inside motor encoder.	1. Replace the cable. 2. Replace the servo amplifier. 3. Replace the servo motor.	×
635	CS disconnected	87h	CS disconnected - Broken or disconnected CS signal line. 1. Error in encoder wiring. 2. Faulty circuit inside the servo amplifier. 3. Faulty circuit inside motor encoder.	1. Replace the cable. 2. Replace the servo amplifier. 3. Replace the servo motor.	×
645	Encoder command error	91h	Encoder command error - Problem with encoder wiring. - Faulty circuit inside motor encoder.	1. Replace the servo motor. 2. Replace the cable.	○

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
646	Encoder FORM error	92h	Encoder FORM error - Problem with encoder wiring. - Faulty circuit inside motor encoder.	1. Replace the servo motor. 2. Replace the cable.	○
647	Encoder SYNC error	93h	Encoder SYNC error - Problem with encoder wiring. - Faulty circuit inside motor encoder.	1. Replace the servo motor. 2. Replace the cable.	○
648	Encoder CRC error	94h	Encoder CRC error - Problem with encoder wiring. - Faulty circuit inside motor encoder.	1. Replace the servo motor. 2. Replace the cable.	○
661	Encoder error 1	A1h	Encoder error 1 - Faulty circuit inside motor encoder.	Replace the servo motor.	○
662	Battery is low (Symbol display)	A2h	Absolute encoder battery error - Low battery voltage.	Replace the backup batteries.	○
663	Encoder overheating	A3h	Serial encoder internal error 9 - Encoder inside servo motor overheated.	Replace the servo motor.	○
665	Encoder error 3	A5h	Serial encoder internal error 3 - Overspeed error.	Replace the servo motor.	○
666	Encoder error 4	A6h	Serial encoder internal error 6 - Multi-turn counter error detected.	Replace the servo motor.	○
667	Encoder error 5	A7h	Serial encoder internal error 5 1. ST error. The absolute sensor and the ASIC internal shift register data do not match (not spindle). 2. Excessive load and position information became invalid. 3. Communication error occurred during axis operation, and position information became invalid. 4. Pulse error (spindle without CTS) 5. Single-turn counter error detected (spindle or lathe spindle with CTS)	Revise the operation conditions when an error occurs during operation. Reset the absolute encoder counter and readjust the position if the error occurs even when the power is turned OFF and back ON. Replace the servo motor if the error still occurs even after performing the recovery steps above.	○
668	Encoder error 6	A8h	Encoder error 6 - Faulty circuit inside motor encoder.	Replace the servo amplifier.	○
669	Encoder error 9	A9h	Serial encoder internal error 9 - Operation temperature error	Replace the servo motor.	○
678	Encoder error 2	B2h	Encoder error 2 - Faulty circuit inside motor encoder.	Replace the servo amplifier.	○
679	Rotation amount counter error	B3h	Rotation amount counter error on absolute encoder - Faulty circuit inside motor encoder.	Replace the servo amplifier.	○
680	1 rotation counter error	B4h	Single rotation counter error on absolute encoder - Faulty circuit inside motor encoder.	Replace the servo amplifier.	○

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
681	Initial speed exceeded	B5h	Speed tolerance exceeded when absolute encoder power is ON - Faulty circuit inside motor encoder.	Replace the servo amplifier.	○
682	Encoder memory error	B6h	Encoder internal memory error - Faulty circuit inside motor encoder.	Replace the servo amplifier.	○
683	Acceleration error	B7h	Acceleration error - Faulty circuit inside motor encoder. - Motor rotation acceleration exceeds speed tolerance.	1. Replace the servo motor. 2. Set the correct servo parameters.	○
693	Overspeed	C1h	Overspeed - 120% of maximum motor speed exceeded. 1. Faulty circuit inside the servo amplifier. 2. Faulty circuit inside motor encoder.	1. Replace the servo amplifier. 2. Replace the servo motor.	○
694	Speed control error	C2h	Speed control error - The current command and acceleration sign do not match. 1. The U, V and W phase wiring between the servo amplifier and servo motor is incorrect. 2. The A and B phase wiring for the pulse encoder is incorrect. 3. The servo motor is vibrating (oscillating). 4. The U, V and W phase wiring between the servo amplifier and servo motor is incorrect.	1. Replace the cable.	○
695	Speed FB error	C3h	Speed feedback error 1. Servo motor power line broken or disconnected. 2. Faulty circuit inside servo amplifier. 3. When opening/closing the door, the [DOOR INTERLOCK] switch was enabled/disabled at the same time.	1-1. Replace the cable. 1-2. Replace the servo motor. 2. Replace the servo amplifier. 3. Press the [RST] key. When opening/closing a door, do not enable/disable the [DOOR INTERLOCK] switch at the same time.	○
709	Position deviation large	D1h	Position deviation large - Position deviation exceeds setting value. 1. Load inertia is too great. 2. Holding brake is not released. 3. One or all of the U, V and W phase wiring between the servo amplifier and servo motor is disconnected. 4. Faulty circuit inside the servo amplifier. 5. Faulty circuit inside motor encoder. 6. Voltage has dropped for main circuit power supply.	1. Revise the load conditions (weight on table). 2. Replace the cable or the servo motor. 3. Replace the cable. 4. Replace the servo amplifier. 5. Replace the servo motor. 6. Make sure that the power supply capacity for the main circuit and wiring are sufficient.	○

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
710	Position command error 1	D2h	Position pulse frequency error 1 - The digital filter settings command for the command pulse input was input.	Replace the servo amplifier or the NC PCB.	○
711	Position command error 2	D3h	Position pulse frequency error 2 - The command pulse input frequency is too high. - The electronic gear's setting value is too large.	Replace the servo amplifier or the NC PCB. Set the correct servo parameters.	○
723	Test mode ended	DFh	Test mode ended - End of test mode is detected. 1. This is an indication that the test mode has ended, and operation is normal. (After the test mode is executed, the deviation on the controller is expected to remain, and therefore an alarm is triggered.)	1. Reset the alarm to restore normal operation.	○
725	ROM error	E1h	EEPROM error - EEPROM error inside amplifier. 1. Faulty circuit inside the servo amplifier. (There was no response from EEPROM when the servo parameters are maintained.) (Faulty amplifier control PCB)	1. Replace the servo amplifier.	×
726	Check sum error	E2h	EEPROM check sum error - Check sum error over whole EEPROM range. 1. The correct values were not written to the CPU from the EEPROM in the servo amplifier. 2. There was an error in writing the values to EEPROM the last time the power supply was turned OFF.	1. Replace the servo amplifier. 2. Replace the servo amplifier.	×
727	Internal RAM error	E3h	Internal RAM error - Faulty circuit inside the servo amplifier.	Replace the servo amplifier.	×
728	CPU-ASIC processing error	E4h	Processing error between CPU and ASIC - Faulty circuit inside the servo amplifier.	Replace the servo amplifier.	×
729	Parameter error 1	E5h	System parameter error 1 - System parameters are outside setting range. 1. The system parameter values are outside the setting range. 2. Faulty circuit inside the servo amplifier.	1-1. Check the servo amplifier model number. 1-2. Check and adjust the system parameter setting values. Turn the power supply OFF and back ON again, and make sure that no alarm is triggered. 2. Replace the servo amplifier.	×

Chapter 11 Alarms

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
730	Parameter error 2	E6h	System parameter error 2 - System parameter combination error. - The system parameters and amplifier combination do not match. 1-1. The system parameter setting values and actual hardware combination is incorrect. 1-2. The system parameter setting combination is incorrect. 2. Faulty circuit inside the servo amplifier.	1-1. Check the servo amplifier model number. 1-2. Check and adjust the system parameter setting values. Turn the power supply OFF and back ON again, and make sure that no alarm is triggered. 2. Replace the servo amplifier.	×
731	Turn power off.	E7h	The user parameter (switch 2) setting <*-axis stroke control> was changed.	Turn the power supply OFF and back ON again.	×
732	Servo amplifier does not match	E8h	There is a problem with the type of servo amplifier connected.	Check the type of servo amplifier.	×
739	Point data error	EFh	Point data check sum error - Faulty circuit inside the servo amplifier.	Replace the servo amplifier.	×
741	Task processing error	F1h	Task processing error - CPU interrupt processing error. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	×
742	Initial timeout	F2h	Initial timeout - Initialization processing was not completed within the specified time. 1. Faulty circuit inside the servo amplifier.	1. Replace the servo amplifier.	×

11.8.1.2 Servo warnings

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm bit	Cause	Remedy
1	Amplifier overheat	0	Amplifier overheat warning - Temperatures that are 90 degrees or higher, or -10 degrees or lower, is detected by the servo amplifier's internal temperature sensor.	Revise the operation conditions.
3	Overload	2	Overload warning - The estimated value for a rise in the winding temperature exceeded the <Overload warning level>.	Revise the operation conditions.
4	Regenerative overload	3	Regeneration resistance overload warning - The alarm detection level for the regeneration resistance overload is 90% or higher.	Revise the operation conditions.
8	Position deviation large	7	Position deviation is too large - Position deviation warning setting value is being exceeded.	Revise the operation conditions.
10	Motor overheated	9	Servo motor overheat warning - The servo motor temperature is 130 degrees or higher, or the encoder temperature is 95 degrees or higher.	Revise the operation conditions.
15	(Symbol display)	14	Serial encoder battery warning - Battery voltage is below 3.0 V.	Replace the backup batteries.
16	Control power voltage reduction	15	Control power voltage drop - Control power supply is less than AC 147 V.	Replace the servo amplifier.
17	Converter is overloaded.	C0	Converter overload warning - Converter overload error is at 90% or higher.	Revise the operation conditions.
18	Power unbalance ratio	C1	Power unbalance ratio warning - The unbalance ratio for the power supply phases is 4% or more (unbalance ratio for power supply voltages is approximately 5%).	Revise the operation conditions.
19	Converter is overcurrent.	C2	Converter overcurrent warning - The power supply current while the power running is 120% or higher than maximum converter current.	Revise the operation conditions.
501	Amplifier overheat	0	Amplifier overheat warning - Temperatures that are 90 degrees or higher, or -10 degrees or lower, is detected by the servo amplifier's internal temperature sensor.	Revise the operation conditions.
503	Overload	2	Overload warning - The estimated value for a rise in the winding temperature exceeded the <Overload warning level>.	Revise the operation conditions.
504	Regenerative overload	3	Regeneration resistance overload warning - The alarm detection level for the regeneration resistance overload is 90% or higher.	Revise the operation conditions.
507	(Symbol display)	6	Serial encoder battery warning Refer to auxiliary number 15.	
508	Position deviation large	7	Position deviation is too large - Position deviation warning setting value is being exceeded.	Revise the operation conditions.

11.8.1.3 Servo communication errors

Supp. No.	Messages displayed as supplementary messages (XXX)	Abort code	Cause	Remedy	Cancel level
1		0503 0000h	Toggle bit did not change.	The servo amplifier may be faulty. Replace the servo amplifier.	×
2		0504 0000h	SDO protocol timeout.	The servo amplifier may be faulty. Replace the servo amplifier.	×
3		0504 0001h	Client/Server command code is invalid or unknown.	The servo amplifier may be faulty. Replace the servo amplifier.	×
4		0505 0005h	Outside memory range.	The servo amplifier may be faulty. Replace the servo amplifier.	×
5		0601 0000h	Incompatible object accessed.	The servo amplifier may be faulty. Replace the servo amplifier.	×
6		0601 0001h	Read attempt made on object that is write-only.	The servo amplifier may be faulty. Replace the servo amplifier.	×
7		0601 0002h	Write attempt made on object that is read-only.	The servo amplifier may be faulty. Replace the servo amplifier.	×
8		0602 0000h	Object does not exist in the object dictionary.	The servo amplifier may be faulty. Replace the servo amplifier.	×
9		0604 0041h	PDO mapping is not possible on object.	The servo amplifier may be faulty. Replace the servo amplifier.	×
10		0604 0042h	Number or data length of mapping object exceeds PDO limit.	The servo amplifier may be faulty. Replace the servo amplifier.	×
11		0604 0043h	Incompatibility in general parameters.	The servo amplifier may be faulty. Replace the servo amplifier.	×
12		0604 0047h	Incompatibility in general device content.	The servo amplifier may be faulty. Replace the servo amplifier.	×
13		0606 0000h	Access error due to hardware error.	The servo amplifier may be faulty. Replace the servo amplifier.	×
14		0607 0010h	Data type mismatch or service parameter length does not match.	The servo amplifier may be faulty. Replace the servo amplifier.	×

Supp. No.	Messages displayed as supplementary messages (XXX)	Abort code	Cause	Remedy	Cancel level
15		0607 0012h	Data type mismatch or service parameter length is too long.	The servo amplifier may be faulty. Replace the servo amplifier.	×
16		0607 0013h	Data type mismatch or service parameter length is too short.	The servo amplifier may be faulty. Replace the servo amplifier.	×
17		0609 0011h	Sub-index does not exist.	The servo amplifier may be faulty. Replace the servo amplifier.	×
18		0609 0030h	Parameter value range exceeded. (for writing access)	Check the servo parameter values. Replace the servo amplifier.	×
19		0609 0031h	Write parameter is too large.	The servo amplifier may be faulty. Replace the servo amplifier.	×
20		0609 0032h	Write parameter is too small.	The servo amplifier may be faulty. Replace the servo amplifier.	×
21		0609 0036h	Maximum value is smaller than minimum value.	The servo amplifier may be faulty. Replace the servo amplifier.	×
22		0800 0000h	General error.	The servo amplifier may be faulty. Replace the servo amplifier.	×
23		0800 0020h	Data cannot be sent to or stored in the application.	The servo amplifier may be faulty. Replace the servo amplifier.	×
24		0800 0021h	Data cannot be sent to or stored in the application due to local control.	The servo amplifier may be faulty. Replace the servo amplifier.	×
25		0800 0022h	Data cannot be sent to or stored in the application in the current device state.	The servo amplifier may be faulty. Replace the servo amplifier.	×
26		0800 0023h	Object dictionary does not exist.	The servo amplifier may be faulty. Replace the servo amplifier.	×

11.8.1.4 Servo status errors

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
1			Servo amplifier cannot be found.	Check the rotary switch for the servo amplifier. Check the connection on the communications cable. The servo amplifier may be faulty. Replace the servo amplifier.	×
2			Servo amplifier cannot be started.	The servo amplifier may be faulty. Replace the servo amplifier.	×
3			Parameter writing/reading operation timeout.	Check the connection on the communications cable. The servo amplifier may be faulty. Replace the servo amplifier.	×
4			Parameter mismatch.	Check the connection on the communications cable. The servo amplifier may be faulty. Replace the servo amplifier.	×
5			Periodic communication timeout.	Check the connection on the communications cable. The servo amplifier may be faulty. Replace the servo amplifier.	×
6			Command sent to the amplifier but not completed after the specified time has elapsed.	Check the connection on the communications cable. The servo amplifier may be faulty. Replace the servo amplifier.	×
7			Servo readiness for X-, Y- and Z-axes is ON but the main circuit power supply for the spindle is OFF.	Check the power supply voltage for the main circuit. Power supply capacity for supplied power is insufficient. Check the connection on the communications cable. Servo relay KS1 may be faulty. Replace the servo amplifier.	○
8			Servo readiness cannot be checked from the amplifier after the specified time has elapsed.	Check the connection on the communications cable. The servo amplifier may be faulty. Replace the servo amplifier. Make sure that the x14 connector on the servo amplifier is connected.	○
9			Error is returned from the amplifier when FoE is written.	Make sure that the file that is being written is correct. Check the connection on the communications cable. The servo amplifier may be faulty. Replace the servo amplifier.	×

Supp. No.	Messages displayed as supplementary messages (XXX)	Amp alarm code	Cause	Remedy	Cancel level
10			FoE writing operation is not complete even after the specified time has elapsed.	Check the connection on the communications cable. The servo amplifier may be faulty. Replace the servo amplifier.	×
11			Command was sent to the amplifier, but it was not completed even after the specified time had elapsed. (Amplifier for positioning)	Check the connection on the communications cable. The servo amplifier may be faulty. Replace the servo amplifier.	×
12			NC target position and the amplifier's command position do not match. (amplifier for positioning)	Check the connection on the communications cable. The servo amplifier may be faulty. Replace the servo amplifier.	×

11.9 ATC Recovery Procedure

If the magazine and the Z-axis are not in their normal positions due to an error or power failure or because the **[EMERGENCY]** switch was pressed during ATC operation, use the following procedure to restore ATC operation.

11.9.1 Turret Type ATC

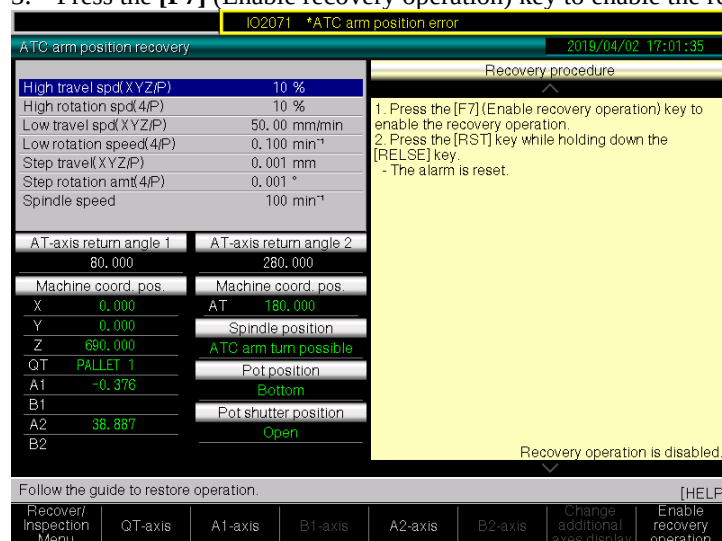
- Check the machine status.
If the <<Z-axis position error>> is triggered, proceed to step 2.
Select manual mode, and press the **[MAG.F]** key or the **[MAG.R]** key.
 - If the alarm <<Magazine position error>> is triggered:
→ Proceed to step 2.
 - If the alarm <<Magazine number error>> is triggered:
→ Proceed to step 3.
- The Z-axis coordinate has shifted from the ATC zero point coordinate.
 - Press the **[RST]** key while holding down the **[RELSE]** key. (Do not let go of the **[RELSE]** key.)
 - Press **[-Z]** key while holding down the **[RELSE]** key, and lower the Z-axis so that it is lower than the distance to Z-axis zero point.
→ Proceed to step 4.
- The magazine index position is not correct.
 - Press the **[RST]** key while holding down the **[RELSE]** key. (Do not let go of the **[RELSE]** key.)
 - While holding down the **[RELSE]** key, press the **[MAG.F]** key or the **[MAG.R]** key to move the magazine to the correct position.
→ Proceed to step 4.
- Make sure that the ATC has been restored.
Set the rapid traverse override to "2" and press the **[ATC]** key to execute the ATC.
If there is no abnormal noise and motion, gradually raise the rapid traverse override and execute the ATC.
Finally, switch to memory operation mode and perform continuous operation. If the ATC operates normally, the recovery procedure is complete.

If there is an abnormal noise and motion, contact Brother.

11.9.2 Arm Type ATC

11.9.2.1 Operation procedure

- If the alarm <<ATC arm position error>> is triggered, the operator message <<Switch to manual operation mode for restoration>> is displayed at the same time.
- Press the **[MANU]** key to change to manual operation mode, and then the ATC arm position recovery screen is displayed.
- Press the **[F7]** (Enable recovery operation) key to enable the recovery operation.



The manual conditions, AT-axis return angle, current machine coordinate position, pot position and pot shutter position are displayed on the left side of the screen. The instructions for the recovery operation are displayed on the right side.

Follow the instructions to restore operation.

Screen description

Item name	Description	
Manual conditions	Refer to the description on the manual conditions screen.	
AT-axis return angle 1	The recovery procedure is different when the AT-axis machine coordinate position falls between <AT-axis return angle 1> and <AT-axis return angle 2>. Refer to the description for the recovery procedure.	
AT-axis return angle 2		
Machine coord. pos.	The current machine coordinate position is displayed. The AT-axis is the ATC arm coordinate.	
Spindle position	ATC arm turn possible	The spindle is at a position from the zero point that is less than <Spindle orientation tolerance> (machine parameter (system 1)).
	ATC arm cannot turn	The spindle is not at a position from the zero point that is less than <Spindle orientation tolerance> (machine parameter (system 1)). (This also includes when the spindle position is undefined.)
Pot position	Bottom	The pot position is at the bottom.
	Top	The pot position is at the top.
	-	The pot position is not at the bottom or top.
Pot shutter position	Open	The pot shutter is open.
	Closed	The pot shutter is closed.
	-	The pot shutter is neither open nor closed.

Function screen description: **[F0]** (Recover/Inspection Menu) key

This key returns to the recover/inspection menu screen. When the **[F7]** key is used to enable recovery operation, the key input is disabled (grayed out).

[F1] to **[F5]** keys (Function keys for selecting the additional axis and PLC-axis)

These keys select the additional axis and PLC-axis to perform the JOG and stop feed and handle feed.

[F6] (Change additional axes display) key

The <Machine coordinate position> display for the additional axis (A1, A2, B1, B2 and C) and the PLC-axis (P1, P2, P3 and P4) follows and changes according to the user parameter (switch 1) <4th – 12th row display axis>. The function display also changes at the same time.

[F7] (Enable/Disable recovery operation) key

This key enables or disables the recovery operation. When the recovery operation is set to <Enable>, after changing to a mode that is not manual mode and after a number or letter key is input, the following operator message appears: <<Invalidate the restoration operation.>> and the key input is ignored. The recovery operation is ignored when it is set to <Disable>. The key is disabled (grayed out) once recovery operation is underway.

Next function **[F1]** (Previous step) key

The screen goes back to the previous recovery procedure. Keys are disabled (grayed out) on the other screens that are not steps 3 or 4 in the recovery procedure.

Next function **[F2]** (No tool) key to **[F4]** (Tool error) key

The recovery procedure is different depending on the selected function key. Follow the recovery procedure that is displayed when selecting a key. Keys are disabled (grayed out) on the other screens apart from step 3 in the recovery procedure.

11.9.2.2 Recovery procedure

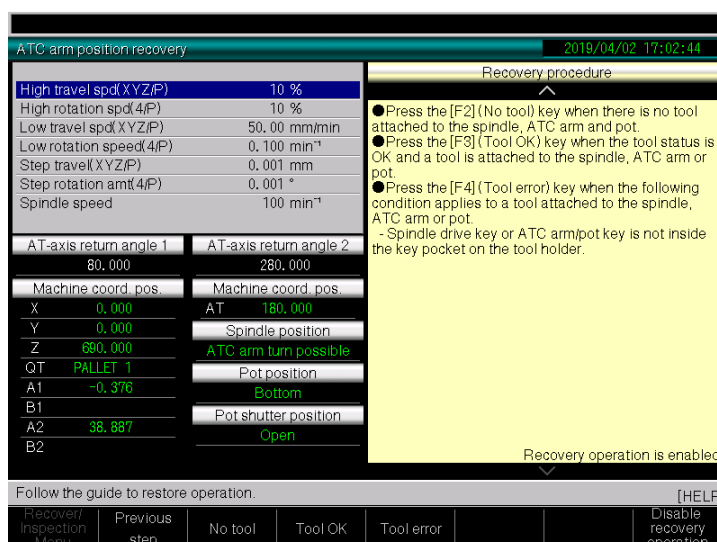
IO2071 *ATC arm position error		2019/04/02 17:01:54				
ATC arm position recovery						
High travel spd(XYZ/P)	10 %	Recovery procedure 1. Press the [F7] (Enable recovery operation) key to enable the recovery operation. 2. Press the [RST] key while holding down the [RELSE] key. - The alarm is reset. Recovery operation is enabled.				
High rotation spd(4/P)	10 %					
Low travel spd(XYZ/P)	50.00 mm/min					
Low rotation speed(4/P)	0.100 min ⁻¹					
Step travel(XYZ/P)	0.001 mm					
Step rotation amt(4/P)	0.001 °					
Spindle speed	100 min ⁻¹					
AT-axis return angle 1	80.000					
AT-axis return angle 2	280.000					
Machine coord. pos.	Machine coord. pos.					
X	0.000	AT	180.000			
Y	0.000	Spindle position				
Z	690.000	ATC arm turn possible				
QT	PALLET 1	Pot position				
A1	-0.376	Bottom				
B1		Pot shutter position				
A2	38.887	Open				
B2						
Follow the guide to restore operation. [HELP]						
Recover/ Inspection Menu	Previous step	No tool	Tool OK	Tool error		Disable recovery operation

When [>>] (Next function) key is pressed

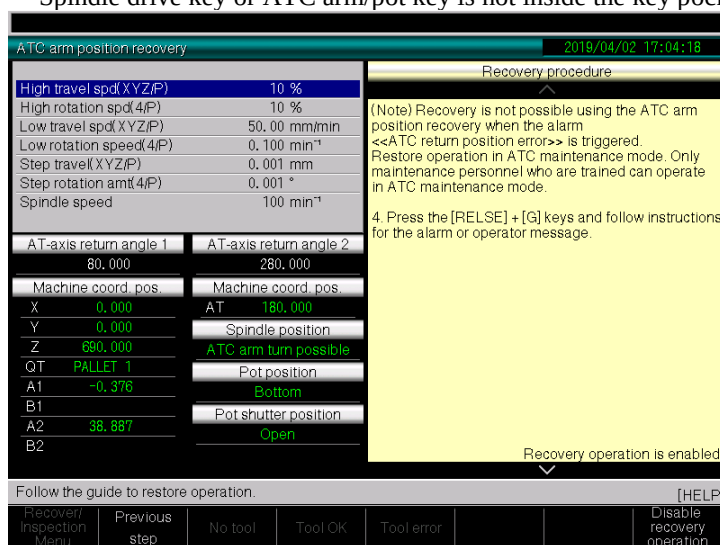
IO2071 *ATC arm position error		2019/04/02 17:01:54				
ATC arm position recovery						
High travel spd(XYZ/P)	10 %	Recovery procedure 1. Press the [F7] (Enable recovery operation) key to enable the recovery operation. 2. Press the [RST] key while holding down the [RELSE] key. - The alarm is reset. Recovery operation is enabled.				
High rotation spd(4/P)	10 %					
Low travel spd(XYZ/P)	50.00 mm/min					
Low rotation speed(4/P)	0.100 min ⁻¹					
Step travel(XYZ/P)	0.001 mm					
Step rotation amt(4/P)	0.001 °					
Spindle speed	100 min ⁻¹					
AT-axis return angle 1	80.000					
AT-axis return angle 2	280.000					
Machine coord. pos.	Machine coord. pos.					
X	0.000	AT	180.000			
Y	0.000	Spindle position				
Z	690.000	ATC arm turn possible				
QT	PALLET 1	Pot position				
A1	-0.376	Bottom				
B1		Pot shutter position				
A2	38.887	Open				
B2						
Follow the guide to restore operation. [HELP]						
Recover/ Inspection Menu	Previous step	No tool	Tool OK	Tool error		Disable recovery operation

1. Press the [F7] (Enable recovery operation) key to enable the recovery operation.
2. Press the [RST] key while holding down the [RELSE] key.
- The alarm is reset.

ATC arm position recovery		2019/04/02 17:02:13				
ATC arm position recovery						
High travel spd(XYZ/P)	10 %	Recovery procedure (Note) Recovery is not possible using the ATC arm position recovery when the alarm <<ATC return position error>> is triggered. Restore operation in ATC maintenance mode. Only maintenance personnel who are trained can operate in ATC maintenance mode. 3. Check the status of the tools that are attached to the spindle, ATC arm and pot. Press the [>>] (Next function) key when the functions below are not displayed. Recovery operation is enabled.				
High rotation spd(4/P)	10 %					
Low travel spd(XYZ/P)	50.00 mm/min					
Low rotation speed(4/P)	0.100 min ⁻¹					
Step travel(XYZ/P)	0.001 mm					
Step rotation amt(4/P)	0.001 °					
Spindle speed	100 min ⁻¹					
AT-axis return angle 1	80.000					
AT-axis return angle 2	280.000					
Machine coord. pos.	Machine coord. pos.					
X	0.000	AT	180.000			
Y	0.000	Spindle position				
Z	690.000	ATC arm turn possible				
QT	PALLET 1	Pot position				
A1	-0.376	Bottom				
B1		Pot shutter position				
A2	38.887	Open				
B2						
Follow the guide to restore operation. [HELP]						
Recover/ Inspection Menu	Previous step	No tool	Tool OK	Tool error		Disable recovery operation



3. Check the status of the tools that are attached to the spindle, ATC arm and pot. Press the [>>] (Next function) key when the functions below are not displayed.
 - Press the [F2] (No tool) key when there is no tool attached to the spindle, ATC arm and pot.
 - Press the [F3] (Tool OK) key when the tool status is OK and a tool is attached to the spindle, ATC arm or pot.
 - Press the [F4] (Tool error) key when the following condition applies to a tool attached to the spindle, ATC arm or pot.
 - Spindle drive key or ATC arm/pot key is not inside the key pocket on the tool holder.



4. Press the [RELSE] + [G] keys and follow instructions for the alarm or operator message.

In the ATC arm position recovery, perform one of the following recovery operations more than once depending on the AT-axis position, Y-axis position, spindle position, pot position or pot shutter position.

[RELSE] + [G] keys: ATC arm position recovery

When the ATC arm is at a position that is less than the <AT-axis return angle 1>, it turns in reverse and returns to 0 degrees.

When the ATC arm is at a position that is greater than the <AT-axis return angle 2>, it turns in the forward direction and returns to 0 degrees.

When the ATC arm is positioned between the <AT-axis return angle 1> and the <AT-axis return angle 2>, the [RELSE] + [G] keys must be pressed twice.

- When the [RELSE] + [G] keys are pressed the first time, the ATC arm takes the shortest path and moves to 180 degrees. If the ATC arm is already positioned at 180 degrees, this motion is skipped.

- When the **[RELSE]** + **[G]** keys are pressed the second time, the ATC arm turns in the forward direction and returns to 0 degrees.
However, when the **[F3]** (Tool OK) key is pressed in step 3, it does not turn unless the spindle position is <ATC arm turn possible>, the pot position is at the <Bottom> and the pot shutter position is <Open>.

When these conditions are not met, first, follow the operator message instructions and change the spindle position, the pot position and the pot shutter position. Then, press the **[RELSE]** + **[G]** keys again.

When the ATC arm has returned to 0 degrees, proceed to step 5.

[RELSE] + **[H]** keys: Spindle position recovery

When the **[RELSE]** + **[H]** keys are pressed twice, the Y-axis moves to a safe position, the spindle orientation is performed and the Y-axis returns to the ATC position.

- When the **[RELSE]** + **[H]** keys are pressed the first time, the Y-axis moves from the ATC position to the <Safe position for Y-axis when restoring operation> noted in the machine parameter (System 1). Then, the spindle is oriented to the 0 degree position.
- When the **[RELSE]** + **[H]** keys are pressed the second time, the Y-axis returns to the ATC position and the spindle servo turns OFF.

[RELSE] + **[P]** keys: Pot shutter open

The pot shutter opens.

[RELSE] + **[D]** keys: Pot down

The pot lowers down.

7532 Restoration completed, so invalidate restoration		2019/04/02 17:04:46	
ATC arm position recovery		Recovery procedure	
High travel spd(XYZ/P)	10 %	5. Press the [F7] (Disable recovery operation) key to disable the recovery operation. - The operator message <<Check ATC tool>> appears. 6. After checking the ATC tool, press the [RST] key and make sure that the alarm <<ATC arm position error>> does not trigger.	
High rotation spd(4/P)	10 %		
Low travel spd(XYZ/P)	50.00 mm/min		
Low rotation speed(4/P)	0.100 min ⁻¹		
Step travel(XYZ/P)	0.001 mm		
Step rotation amt(4/P)	0.001 °		
Spindle speed	100 min ⁻¹		
AT-axis return angle 1	80.000	AT-axis return angle 2	280.000
Machine coord. pos.		Machine coord. pos.	
X	0.000	AT	0.000
Y	0.000	Spindle position	
Z	690.000	ATC arm turn possible	
QT	PALLET 1	Pot position	Bottom
A1	-0.376	Pot shutter position	Open
B1			
A2	38.887		
B2			
Recovery operation is enabled.			
Follow the guide to restore operation. [HELP]			
Recovery/ Inspection Menu	Previous step	No tool	Tool OK
		Tool error	
			Disable recovery operation

5. Press the **[F7]** (Disable recovery operation) key to disable the recovery operation.
- The operator message <<Check ATC tool.>> appears.
6. After checking the ATC tool, press the **[RST]** key and make sure that the alarm <<ATC arm position error>> does not trigger.

11.9.2.3 Special notes

1. The ATC arm position recovery screen changes to the default screen for manual operation mode, from when the alarm <<ATC arm position error>> is triggered until the recovery operation is complete. Therefore, if the **[RST]** key is pressed in manual operation mode, or if the mode is changed to another mode and then changed back to manual operation mode again, the ATC arm position recovery screen is displayed.
2. When the alarm <<ATC arm position error>> is triggered during manual operation mode, the screen currently being displayed changes to the ATC arm position recovery screen.
3. After the alarm <<ATC arm position error>> is triggered and after enabling the recovery operation in manual operation mode, press the **[RELSE]** and **[RST]** keys to reset the alarm.
4. After pressing the **[RELSE]** and **[RST]** keys in manual operation mode to reset the alarm <<ATC arm position error>>, if an attempt is made to perform manual operation when the recovery operation is not complete, the operator message <<Operation other than restoration op is prohibited>> appears.
5. If the power is turned OFF due to a power failure while the ATC arm is turning during ATC operation, the ATC arm position recovery screen is displayed automatically when the power is turned ON.
6. When the recovery operation for the ATC arm position is complete, the operator message <<Restoration completed, so invalidate restoration.>> is triggered.
Unless the recovery operation is disabled, the user cannot change to another mode or input number or letter keys.
7. When recovery operation is disabled using the **[F7]** key, the operator message <<Check ATC tool.>> appears.
8. If the power is turned OFF during the following period, the operator message <<Check ATC tool.>> will appear when the power is turned ON the next time.
 - From when the alarm <<ATC arm position error>> is triggered until operation (6) is completed (when the alarm <<ATC arm position error>> is triggered in manual operation mode).
 - From when the ATC arm position recovery screen is displayed (by pressing the **[MANU]** key to change to manual operation mode) until operation (6) is completed (when the alarm <<ATC arm position error>> is triggered in another mode besides manual operation mode).
9. If **[F7]** (Enable recovery operation) is pressed during manual mode, the alarm <<Manual operation mode>> is triggered and the operation becomes invalid. Exit manual mode and then perform the operation.
10. When performing a manual operation (that is not a handle operation) while in handle mode, the alarm <<Handle mode>> is triggered and the operation is invalid. Set the axis selection switch for the handle to <<OFF>> and then perform manual operation.
11. If **[F7]** (Enable recovery operation) is pressed during spindle rotation, the alarm <<Spindle rotating>> appears and the operation becomes invalid. Stop spindle rotation and then perform the operation.
12. If **[F7]** (Enable recovery operation) is pressed during lathe spindle rotation, the alarm <<Lathe spindle is rotating>> appears and the operation becomes invalid. Stop spindle rotation and then perform the operation.
13. If the **[F7]** (Enable recovery operation) key is pressed and the encoder position information does not match the machine coordinates because the battery has expired, the following operator message appears: <<Perform the expired battery recovery operation>> and key operation is ignored.

11.10 Collision detection function

This function is available on the following machine models.

Machine model	S300X2/S500X2/S700X2/F600X1/R450X2/R650X2 The function can also be used on the following models if the software version is W*0*.08.010 or later. S300X1/S500X1/S700X1/ S300X1N/S500X1N/S700X1N
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(Procedure for checking if the function is enabled or disabled)

1. Press the **[ALARM]** key to display the <Current alarm / Operator message> screen.
2. Press the **[F1]** (Alarm log list) key to display the <Alarm log (List)> screen.
 - When the function is enabled
When the **[F5]** (Collision log list) key is pressed, the <Collision log list> screen is displayed.
 - When the function is disabled
When the **[F5]** (Collision log list) key is pressed, the operator message <There is no corresponding function.> displays.

11.10.1 Function

This function detects a collision between the spindle (or tool) and workpiece or jig, etc.

If a collision is detected, machine operation stops immediately.

The collision log saves the most recent 10 alarms that are triggered by collision detection.

11.10.2 Collision detection function

11.10.2.1 Collision detection alarm

When a collision is detected during axis travel, one of the following collision alarms is triggered.

The collision detection alarm is displayed on the collision log screen. Refer to "Chapter 11 - 11.10.3 Collision Log" for further details. The collision detection alarms, similar to the other alarms, are displayed in the alarm and operator message area, the current alarm screen and the alarm log screen.

Alarm message	Cause	Recovery procedure
X-axis collision detection	1. Collision was detected on the X-axis. 2. This alarm was triggered at the same time due to a servo error.	1. After pressing the [RST] key, move the feed axis to a safe position while checking the machine status. 2. First, restore operation based on the servo error when a servo error also is triggered.
Y-axis collision detection	1. Collision was detected on the Y-axis. 2. This alarm was triggered at the same time due to a servo error.	1. After pressing the [RST] key, move the feed axis to a safe position while checking the machine status. 2. First, restore operation based on the servo error when a servo error also is triggered.
Z-axis collision detection	1. Collision was detected on the Z-axis. 2. This alarm was triggered at the same time due to a servo error.	1. After pressing the [RST] key, move the feed axis to a safe position while checking the machine status. 2. First, restore operation based on the servo error when a servo error also is triggered.

- (Note 1) Even if no collision occurs between the spindle (or tool) and the workpiece or jig, etc., when a disturbance force greater than the standard values is detected due to a machine failure/malfunction or due to travel outside of the stroke range, then a collision detection alarm is triggered.
- (Note 2) When multiple axes collide at the same time, the collision alarm is triggered for the axis where the disturbance force (greater than the standard values) is initially detected.
- (Note 3) When a servo error is triggered and the cause is a communications error, the collision detection alarm may be triggered, but the alarm does not display on the collision log screen.

11.10.2.2 Special notes

1. Collision detection does not apply to axes that are disabled using the mode for enabling and disabling the servo.
2. When equipped with a turret type ATC, collision detection does not apply to the Z-axis if the Z-axis machine coordinate is at a position that is higher than the "Distance to Z-axis zero".
3. When equipped with an arm type ATC, collision detection does not apply to X-, Y- and Z-axes if the X- and Y-axes machine coordinates are at the ATC position and the Z-axis machine coordinate is at the zero point.

11.10.3 Collision log

11.10.3.1 List

A list of collision detection alarms is displayed.

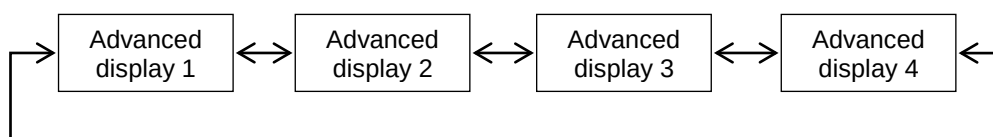
Press the **[F5]** (Collision log list) key on the alarm log (list) screen to display the collision log (list) screen.

Collision log (list)				
2016/10/07 16:47:05				
Alarm	Message	Lingo	Time	Date
SM0558	*X-axis collision detection	NC	19:34:32	2016/10/06
SM0558	*X-axis collision detection	NC	19:16:57	2016/10/06
SM0558	*X-axis collision detection	NC	19:15:39	2016/10/06
SM0560	*Z-axis collision detection	NC	18:55:43	2016/10/06
SM0560	*Z-axis collision detection	NC	18:30:45	2016/10/06
SM0560	*Z-axis collision detection	NC	18:29:35	2016/10/06
SM0560	*Z-axis collision detection	NC	18:25:43	2016/10/06
SM0560	*Z-axis collision detection	NC	18:25:22	2016/10/06
SM0560	*Z-axis collision detection	NC	18:25:17	2016/10/06
SM0560	*Z-axis collision detection	NC	18:24:56	2016/10/06
To display details: Move the cursor to the item and press [ENT].				
Current alarm	Alarm log list			Collision log list

11.10.3.2 Displaying advanced information

Use the **[CURSOR]** key to select the collision detection alarm that you wish to reference on the collision log (list) screen. After pressing the **[ENT]** key, the advanced information for the collision detection alarm is displayed.

Press the **[PAGE DOWN V]** and **[PAGE UP ^]** keys to scroll through the screens in the following order.



Advanced display 2

The machine coordinates, door and door interlock status and supplementary tool information are displayed.

(For NC language)

Collision log (advanced display 2)				2016/10/07 16:48:24	
Alarm		Message		Lingo	Time Date
SM0558		*X-axis collision detection		NC	19:34:32 2016/10/07
Mode		Screen	Edit program	Operation program	Job number
MDI		PRGRM	KA/0026	KA/0026	000001
Machine coordinate pos		Distance left	Door		Door interlock
X	-162.798	X	-37.201	Door	CloseValid
Y	-127.850	Y	0.000	Pendant enabled	
Z	310.122	Z	0.000	Side door	Close
C		C		PLC axis during travelEnable	
A		A			
B		B		Tool offsetCutter compensation	
C		C			
A		A			
Magazine	25.714			X	0.000
P1		P1		Y	0.000
P2		P2		Z	0.000
P3		P3			
P4		P4			
☐ Collision log (advanced display 1) ☒ Collision log (advanced display 3)					
Collision log list	Current alarm				

(For conversation language)

Collision log (advanced display 2)				2016/10/07 18:06:10		
Alarm		Message		Lingo	Time	Date
SM0558		*X-axis collision detection		Conv.	17:50:24	2016/10/07
Mode		Screen	Edit program	Operation program		Job number
MANU		MANU COND	0155	0155		01
Machine coordinate pos		Distance left		Door		Door interlock
X	-355.000	X	355.000	Door	Close	Valid
Y	0.000	Y	0.000	Pendant enabled		
Z	480.000	Z	0.000	Side door	Close	Enable
C		C		PLC axis during travel		
A		A				
B		B		Current motion		
C		C				
A		A				
Magazine	282.857					
P1		P1				
P2		P2				
P3		P3				
P4		P4				
				Tool length offset		
☐ Collision log (advanced display 1) ☒ Collision log (advanced display 3)						
Collision log list		Current alarm				Next

Item name	Description
Machine coordinate pos.	Displays the same description as in advanced display 1.
Distance left	Displays the same description as in advanced display 1.
Door	Displays the door's open/close status when a collision is detected. The door name varies depending on the machine model.
Door interlock	Displays the status of the [DOOR INTERLOCK] switch when a collision is detected.
Pendant enabled	Displays the status of the [ENABLE PENDANT] switch when a collision is detected.
Enable	Displays the status of the [ENABLE] switch when a collision is detected.
PLC axis during travel	Displays the PLC axis travel status when a collision is detected. Only the name of the axis that is travelling is displayed, and if the axis is stopped, the field is blank.
Current motion (*Conversation language)	Displays the motion when a collision is detected.
Tool offset (*NC language)	Displays the offset amount when the tool length offset or tool position compensation is enabled.
Tool length offset (*Conversation language)	Displays the offset amount when the tool length offset is enabled.
Cutter compensation	Displays the compensation amount when the cutter compensation is enabled.

Advanced display 3

The deviation and current limit/command are displayed.

Current limit/Current command display is shown only when the machine parameter (system 2)

<Quick table> is set to <2: Type 2>.

(Same in NC/conversation language)

Collision log (advanced display 3)			
Alarm	Message	Lingo	Time Date
SM0558	*X-axis collision detection	NC	19:34:32 2016/10/06
Deviation			
Spindle	0	P1	
X	-232	P2	
Y	-1	P3	
Z	-1	P4	
C			
A			
B			
C			
A			
Magazine	0		
☐ Collision log (advanced display 2) ☒ Collision log (advanced display 4)			
Collision log list	Current alarm		

Item name	Description
Deviation	Displays the deviation when a collision is detected.
Current limit / Current command	Displays the current limit / current command on the QT-axis when a collision is detected.

Advanced display 4

30 operations are displayed that came immediately before the collision was detected. The log status is displayed for 5 logs.

(Same in NC/conversation language)

Collision log (advanced display 4)				2016/10/11 15:46:18	
Alarm	Message		Lingo	Time	Date
SM0558	*X-axis collision detection		Conv.	15:28:58	2016/10/11
Operation log					
Time	Operation description	Time	Operation description	Time	Operation description
15:28:16	Down CURSOR	15:28:24	DATA BANK	15:28:43	EOB/ENT
15:28:16	0	15:28:25	ALARM	15:28:45	Down CURSOR
15:28:17	EOB/ENT	15:28:26	F1	15:28:45	Down CURSOR
15:28:17	F0	15:28:27	F5	15:28:46	EOB/ENT
15:28:18	F0	15:28:29	EOB/ENT	15:28:47	Down CURSOR
15:28:18	RST	15:28:30	PAGE Down	15:28:48	EOB/ENT
15:28:21	POS	15:28:36	DATA BANK	15:28:49	RST
15:28:22	Z RTN	15:28:38	MD1	15:28:50	MANU
15:28:22	RST	15:28:39	MEM	15:28:50	RST
15:28:24	ALARM	15:28:41	F1	15:28:59	Z RTN
Status log					
Status	Time		Error No./Program No.		
Standby mode	15:28:50 - 15:28:49				
Error occurred	15:28:49 - 15:28:49		SM2054 *		
Standby mode	15:28:49 - 15:28:47				
Error occurred	15:28:47 - 15:28:22		SM2054 *		
Standby mode	15:28:22 - 15:28:58				
☐ Collision log (advanced display 3)					
☒ Collision log (advanced display 1)					
Collision log list	Current alarm				

Item name	Description
Operation log	Time Operation description Displays 3 columns of the operation log for the operations right before a collision was detected. Displays the operations so that the newest ones are on the top-left and the oldest are on the bottom-right.
Status log	Status Time Error No./Program No. Displays the status log of the machine right before a collision was detected. The status field displays one of the following: error occurred, standby mode, running, power is off or stopped. In the error No./program No. field, the error number is displayed when an error is triggered, and the program number is displayed during operation.

11.11 ATC Monitoring Function

11.11.1 Function overview

The ATC monitoring function detects when a tool was not properly changed. It helps prevent machine and tool damage or a machining defect, caused by a mounting error on the tool holder or a missing tool holder during machining.

This function does not use a sensor. It monitors the load pattern on the Z-axis to detect tool mounting errors. When an abnormal load pattern is detected, an alarm is triggered. Machine operation is stopped and it outputs the information to the global OM in the built-in PLC. The function detects “No tool”, “Tool over key” or a “Tool collision during ATC” depending on the corresponding pattern.

This function is available on the following machine models.

Model	S300X2/S500X2/S700X2/F600X1/R450X2/R650X2 (excluding 40 MG) M200X3/M300X3
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The factory default settings are shown below for each detection function. When using this function, set the user parameters to another setting that is not <0: Disable>.

User parameter (Switch 1)	
<No tool detection function (when ATC rises)>	<0: Disable>
<No tool detection function (when ATC falls)>	<0: Disable>
<Tool over key detection function>	<0: Disable>
<Tool collision detection function during ATC>	<0: Disable>

11.11.1.1 No tool detection function (when ATC rises)

This function detects when there is no tool on the spindle during an unclamp operation on the tool holder during ATC. If a tool holder comes off of the spindle during machining, this function helps prevent machine damage caused by a collision between the spindle and the remaining tool. It only monitors when in memory operation mode.

To use this function, set the <No tool detection function (when ATC rises)> in the user parameter (switch 1) to <1: Method 1>. Or, to stop operation under specific timing conditions, set it to <2: Method 2> and use the built-in PLC function to create separate processing conditions when an alarm occurs.

During an unclamp operation on a tool holder, if a tool is not mounted, the machine operation follows the settings below.

<No tool detection function (when ATC rises)>	Alarm	Operation when alarm is triggered	Built-in PLC Global OM
<1: Method 1>	<<No spindle tool during ATC (when ATC rises)>> triggers	Machine operation stops at Z-axis zero point	Turns ON <BX1CC No tool detection (when ATC rises)>
<2: Method 2>	Does not trigger	(Operation continues)	Turns ON <BX1CC No tool detection (when ATC rises)>

11.11.1.2 No tool detection function (when ATC falls)

This function detects when there is no tool on the spindle during a clamp operation on the tool holder during ATC. If the user forgets to mount a tool onto the magazine, or if there is a tool clamp error, this function prevents machining when there is no tool. It only monitors when in memory operation mode.

To use this function, set the <No tool detection function (when ATC falls)> in the user parameter (switch 1) to <1: Method 1>. Or, to stop operation under specific timing conditions, set it to <2: Method 2> and use the built-in PLC function to create separate processing conditions when an alarm occurs.

During a clamp operation on a tool holder, if a tool is not mounted, the machine operation follows the settings below.

<No tool detection function (when ATC falls)>	Alarm	Operation when alarm is triggered	Built-in PLC Global OM
<1: Method 1>	<<No spindle tool during ATC (when ATC falls)>> triggers	Machine operation stops at Z-axis zero point	Turns ON <BX1CD No tool detection (when ATC falls)>
<2: Method 2>	Does not trigger	(Operation continues)	Turns ON <BX1CD No tool detection (when ATC falls)>

No tool detection function (when ATC rises) and no tool detection function (when ATC falls) both monitor when a tool number is set on the ATC tool screen during clamp/unclamp operations on the tool holder.

In order to perform a tool change in NC language on a magazine where there is no tool holder, change the tool number so it is not set on the ATC tool screen. Then, specify the magazine number (101-150) in the T code and perform the tool change. When using conversation language, change the tool number so it is not set on the ATC tool screen. Then, specify the magazine number (701-749) and perform the tool change using the signal output.

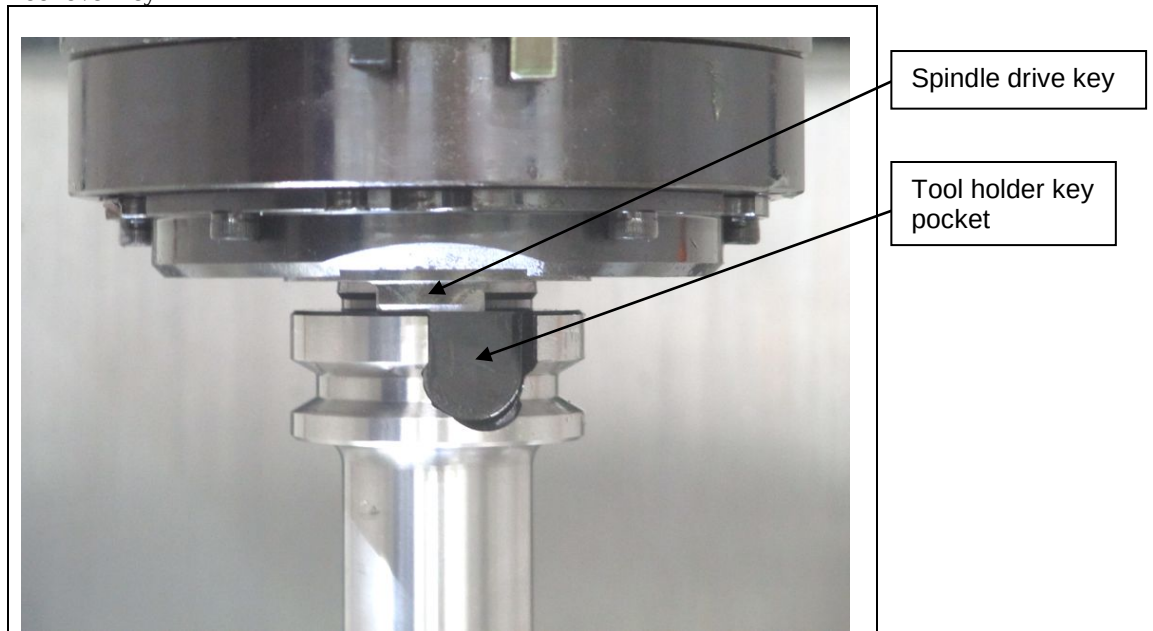
(Note 1) When using a canned cycle for the tool change, if moving to a position that is lower than the Z-axis zero point while performing a tool change, it may stop at a position that is below the Z-axis zero point. However, it will not move beyond the command position.

(Note 2) Refer to the PLC System Manual for further details on the global OM.

11.11.1.3 Tool over key detection function

If the key pocket on the tool holder is not aligned with the key on the magazine grip, the tool holder goes over the spindle drive key when the tool is clamped and machining will not work properly. This function detects when the tool goes over the spindle drive key (tool over key) during a clamp operation on the tool holder during ATC. It monitors when in memory, MDI and manual operation modes.

Tool over key



To use this function, set the <Tool over key detection function> in the user parameter (switch 1) to <1: Method 1>. Or, to stop operation under specific timing conditions, set it to <2: Method 2> and use the built-in PLC function to create separate processing conditions when an alarm occurs. When a tool holder goes over the spindle drive key, the machine operation follows the settings below.

<Tool over key detection function>	Alarm	Operation when alarm is triggered	Built-in PLC Global OM
<1: Method 1>	<<Spindle tool misaligned during ATC>> triggers	Machine operation stops at Z-axis zero point	Turns ON <BX1CE Tool over key detection>
<2: Method 2>	Does not trigger	(Operation continues)	Turns ON <BX1CE Tool over key detection>

(Note 1) When using a canned cycle for the tool change, if moving to a position that is lower than the Z-axis zero point while performing a tool change, it may stop at a position that is below the Z-axis zero point. However, it will not move beyond the command position.

(Note 2) Refer to the PLC System Manual for further details on the global OM.

11.11.1.4 Tool collision detection function during ATC

This function detects a collision between the tool holder and the end of the spindle during a clamp operation on the tool holder during ATC. If a tool holder is mounted onto the magazine at an angle, this function helps prevent machine damage caused by a collision between the tool holder and the end of the spindle. It monitors when in memory, MDI and manual operation modes.

To use this function, set the <Tool collision detection function during ATC> in the user parameter (switch 1) to <1: Enable>. When a collision is detected, the alarm <<Tool collision during ATC>> is triggered and operation immediately stops.

11.11.2 Enable conditions

Each detection function monitors operation according to the table below.

Detection function	Manual operation	MDI operation	Memory operation
No tool detection	(Does not monitor)	(Does not monitor)	Tool change command
Tool over key detection	ATC	Tool change command	Tool change command
Tool collision detection during ATC	ATC	Tool change command	Tool change command

(Note 1) The function is disabled when the machine lock is ON.

(Note 2) The function is disabled when a door is open.

11.11.3 Notes

(Note 1) When at least either the no tool detection or the tool over key detection is enabled, the tool change time may take longer.

(Note 2) Each detection function may not operate properly depending on the conditions. The following are examples of causes.

Cause	Countermeasure
Tool holder being used does not meet the limit requirements	Use a compliant tool holder per “3.7.2 Tool holder limits”.
ATC monitoring function has not been adjusted after replacing parts related to the Z-axis feed mechanism or ATC mechanism.	An adjustment must be performed after replacing parts. Contact Brother or the nearest authorized dealer.
ATC monitoring function has not been adjusted after adjusting the Z-axis zero point.	An adjustment must be performed after adjusting the Z-axis zero point. Contact Brother or the nearest authorized dealer.
Part related to the Z-axis feed mechanism or ATC mechanism has deteriorated or failed.	The part(s) must be replaced. Contact Brother or the nearest authorized dealer.
The rapid feed special setting is not set to <Standard> when the tool over key detection does not work properly.	Detection may not be possible with this function.
When tool collision during ATC does not detect properly, a parameter such as the Z-axis rapid feed time constant has been changed.	Return the changed parameter back to its factory default setting.

In addition, each detection function may not operate properly depending on the machine status, usage conditions and environment. The detection functions do not necessarily guarantee operation under all conditions.

To continue using the machine, set the following user parameters (switch 1): <No tool detection function (when ATC rises)>, <No tool detection function (when ATC falls)>, <Tool over key detection function> and <Tool collision detection function during ATC> to <0: Disable>.

(Note 3) Each detection function may not operate properly when there is a servo error. First, restore operation based on the servo error when a servo error is triggered with another alarm or error.

11.12 Tool Missing Detection Function

11.12.1 Function overview

The tool missing detection function monitors when a tool is missing or about to come off of the spindle. This function helps prevent damage caused by a collision between the spindle and the remaining tool. It monitors when the Z-axis rises in memory, MDI and manual operation modes.

If a tool gets stuck in the workpiece during tapping or other machining, the tool may come off from the spindle when the Z-axis rises thereafter. This function monitors when a tool comes off by the abnormal load that occurs on the Z-axis. When an abnormal load is detected, an alarm is triggered. Machine operation is stopped, it outputs the information to the global OM in the built-in PLC and it reports on the status.

This function is available on the following machine models.

Model	S300X2/S500X2/S700X2/F600X1/R450X2/R650X2 (excluding 40 MG) M200X3/M300X3
-------	--

The factory default value is <0: Disable>. To use this function, set the <Tool missing detection function> in the user parameter (switch 1) to <1: Method 1>. Or, to stop operation under specific timing conditions, set it to <2: Method 2> and use the built-in PLC function to create separate processing conditions when an alarm occurs.

When a tool holder is missing or about to come off of the spindle, the machine operation follows the settings below.

<Tool missing detection function>	Alarm	Operation when alarm is triggered	Built-in PLC Global OM
<1: Method 1>	<<Spindle tool missing>> triggers	Machine is stopped	Turns ON <BX1CF Tool missing detection>
<2: Method 2>	Does not trigger	(Operation continues)	Turns ON <BX1CF Tool missing detection>

(Note) Refer to the PLC System Manual for further details on the global OM.

11.12.2 Enable conditions

The tool missing detection is enabled when the Z-axis rises using manual operation mode or when the Z-axis rises using rapid feed/cutting feed in memory and MDI operation mode.

(Note) The function is disabled when the machine lock is ON.

11.12.3 Notes

(Note 1) The tool holder missing cannot be detected while machining without the Z-axis rising, such as in milling.

(Note 2) This detection function may not operate properly depending on the conditions. The following are examples of causes.

Cause	Countermeasure
Load is too big when machining with the Z-axis rising, such as in back boring.	Revise the machining method and/or machining conditions.
Part(s) related to the Z-axis feed mechanism or ATC mechanism has deteriorated or failed.	The part(s) must be replaced. Contact Brother or the nearest authorized dealer.

In addition, this function may not operate properly depending on the machine status, usage conditions and environment. This function does not necessarily guarantee operation under all conditions.

To continue to use the machine, set the <Tool missing detection function> in the user parameter (switch 1) to <0: Disable>.

(Note 3) This detection function may not operate properly when there is a servo error. First, restore operation based on the servo error when a servo error is triggered with another alarm or error.

11.13.3 Setting Machining Load Monitor Segments

11.13.3.1 M function (NC language mode only)

Use the M code below to enable or disable the machining load monitor function when in NC language mode.

M code	Description	Maximum machining load monitor	Minimum machining load monitor	Timing of axis travel	Modal / One shot
M340	Machining load monitor OFF	No	No	Before	Modal
M341	Machining load monitor ON	Yes	Yes	Before	
M342	Machining load monitor ON (Max. only)	Yes	No	Before	
M343	Machining load monitor ON (Min. only)	No	Yes	Before	

The M340 modal (function OFF) is enabled following the timing below.

- During tool change (G100/M06)
- Operation reset
- At power ON
- During a reset
- At operation end
- At memory operation start

When enabling a function with an M code, use the same number for the machining load monitor parameter as the tool number for the tool on the spindle to start the load monitor. In addition, the E address after the M code can be used to specify which machining load monitor parameter is used.

Command format

```
M340
M341 E__
M342 E__
M343 E__
```

E: Machining load monitor parameter number (1 to 99)

Ex:

In the example below, the maximum and minimum machining loads are monitored at the same time on line N3. At this time, the machining load is monitored using the No.4 machining load monitor parameter.

```
N1 G100 T4 Z20. M03 S10000 ;
N2 M341 ;           <-----Machining load monitor ON
N3 G01 X-10. ;      Load monitor executing (using parameter No.4)
N4 M340 ;           <-----Machining load monitor OFF
N5 Y-10. ;
```

In addition, the maximum and minimum machining loads are monitored at the same time as shown on line N3 in the example below. However, in this case, “1” is specified as the E address after M341 on line N2. Machining load monitor parameter No.1 is used for monitoring.

```
N1 G100 T4 Z20. M03 S10000 ;
N2 M341 E1 ;        <-----Machining load monitor ON
N3 G01 X-10. ;      Load monitor executing (using parameter No.1)
N4 M340 ;           <-----Machining load monitor OFF
N5 Y-10. ;
```

11.13.3.2 Auxiliary function (Conversation language mode only)

Use the auxiliary functions below to enable or disable the machining load monitor function when in conversation language mode.

Auxiliary function	Description	Maximum machining load monitor	Minimum machining load monitor	Timing of axis travel
340	Machining load monitor OFF	No	No	Before
341	Machining load monitor ON	Yes	Yes	Before
342	Machining load monitor ON (Max. only)	Yes	No	Before
343	Machining load monitor ON (Min. only)	No	Yes	Before

The functions are disabled at the following times.

- During tool change
- Operation reset
- At power ON
- During a reset
- At operation end
- At memory operation start

When using an auxiliary function, a machining load monitor parameter number cannot be specified if it is different from the number of the tool being used. When enabling a function with an auxiliary function, use the same number for the machining load monitor parameter as the tool number for the tool on the spindle to start the load monitor. Make sure that the tool number on the tool list screen and the parameter number for the machining load monitor match each other in order to use these functions.

Ex: When the drill is set to tool No.1 and the tap is set to tool No.5 on the tool list screen, the drill setting must be entered to data number 1 for the machining load monitor parameter and the tap setting must be entered to data number 5.

11.13.4 Machining Load Monitor Settings

This function is used to set the parameters at the setting screen in the data bank menu. Refer to “11.13.5 Examples when using machining load monitor function” for further details on each parameter setting.

11.13.4.1 Machining load monitor parameter (list) screen

Set the machining load monitor for one tool (or cutting conditions) in each column on the parameter screen. A maximum of 99 data items can be registered. Each item that can be set and its description is shown below.

Machining load monitor param.(list)					
2019/04/01 19:24:54					
Parameter number	01	02	03	04	05
Load monitor method	1	0	1	0	0
Time constant	10	100	10	1	100
Stop level for max.	3	0	0	1	0
Stop level for min.	2	0	0	0	0
Max. machining load	0.001	0.000	0.000	0.000	0.000
Min. machining load	999.999	0.000	0.000	0.000	0.000
Updated load [Nm]					
Peak value	0.000	0.000	0.000	0.000	0.000
Average	0.000	0.000	0.000	0.000	0.000
Load monitor method					
0:Peak value 1:Average [HELP]					
Completion mode	Parameter (List)	Common parameter	Copy	Paste	Initialization
					Search

(1) Load monitor method

Select the machining load monitor method.

0: Peak value

The threshold value is assessed based on the moving average value using the <Time constant> for the measured machining load (refer to figure below).

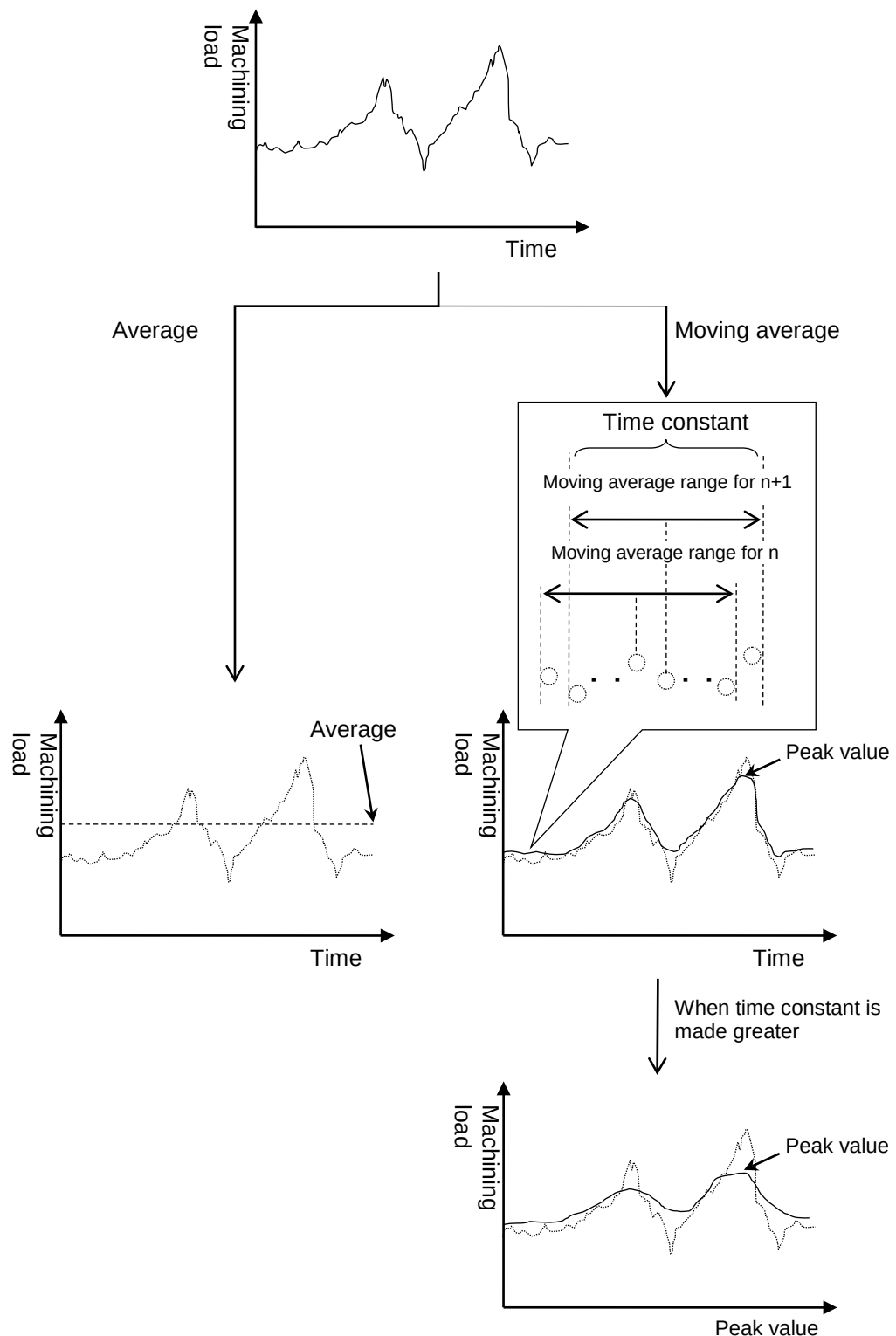
The maximum (peak value) in the moving average value is used for the threshold value.

The machining load is not measured while the spindle is accelerating/decelerating.

1: Average value

The threshold value is assessed based on the machining load average taken from the start of the machining load measurement until the end of the measurement (refer to figure below).

The machining load is measured while the spindle is accelerating/decelerating.



(2) Time constant

Set the time constant used for the moving average calculation (peak value calculation). (Refer to figure above)

The unit is in **[msec]** (1 to 999). Normally, the factory-default setting (100**[msec]**) can be used as is. Any instant fluctuation in the machining load can be detected easily by making the unit value smaller, but variation per measurement becomes more exaggerated. When making the unit value larger, variation per measurement becomes less and the machining load can be measured more consistently. However, any instant fluctuation in the machining load is harder to detect. When changing the unit value, perform test machining before actual machining and make sure that it detects correctly beforehand.

- (3) Stop level for max.
Set the stop level for the alarm <<Machining load maximum>> when the measured machining load in the monitoring segment exceeds the <Max. machining load>.
- 0: Level 1
Operation does not stop.
- 1: Level 2
Operation stops after 1 cycle.
- 2: Level 3
Operation stops at the block.
- 3: Level 4
Operation pauses. (instantly)
- (4) Stop level for min.
Set the stop level for the alarm <<Machining load minimum>> when the measured machining load in the monitoring segment falls below the <Min. machining load>.
- 0: Level 1
Operation does not stop.
- 1: Level 2
Operation stops after 1 cycle.
- 2: Level 3
Operation stops at the block.
- (5) Max. machining load
Set the threshold value to trigger the alarm <<Machining load maximum>>.
When <Max. machining load> is initialized (0 or blank field), the alarm <<Machining load maximum>> does not trigger. The unit is in **[Nm]** (0.000 to 999.999).
- (6) Min. machining load
Set the threshold value to trigger the alarm <<Machining load minimum>>.
When <Min. machining load> is initialized (0 or blank field), the alarm <<Machining load minimum>> does not trigger. The unit is in **[Nm]** (0.000 to 999.999).

Checking measurement results

The results from the actual machining load measurement are displayed at the bottom of the screen. The items and their descriptions are shown below. However, the parameter number fields are displayed in orange when the machining load is in the middle of being measured.

- (1) Updated load(Peak value)
Maximum machining load when the travel is averaged from the start of the machining load measurement until it is complete.
- (2) Updated load(Average)
Average machining load when measuring from the start of the machining load monitor until it is complete.

The machining load is carried out using both of the above methods regardless of the <Load monitor method> setting. Both items are updated on the screen (The parameter <Load monitor method> determines which of these measurement values is used for assessing the threshold value). The unit is in **[Nm]**.

Timing when threshold value is assessed and when measurement value is updated

For the machining load that is measured, the following describes the timing when the <Max. machining load> and <Min. machining load> are assessed and when the <Updated load (Average and peak value)> is updated.

- When changing from modals M341, M342 and M343 (monitor ON) to modal M340 (monitor OFF) (Ex: M341 → M340)
- When a tool change is carried out while modal M341, M342 or M343 (monitor ON) is engaged (Ex: M341 → G100 T3)
- When an M341, M342 or M343 (monitor ON) command is issued again while modal M341, M342 or M343 (monitor ON) is engaged (Ex: M341→ M342, M341 E1 → M341 E2)

When the <Load monitor method> is set to <0: Peak value>, the <Max. machining load> is constantly assessed during the M341 and M342 modals. If the alarm <<Machining load maximum>> is triggered, the <Updated load (Average and peak value)> is updated.

11.13.4.6 Paste parameter (Machining load monitor parameter (list) screen only)

- (1) Move the cursor to the column for the parameter numbers where you wish to paste the imported parameters.
- (2) Press the **[F4]** (Paste) key.
Parameter group 1, where the cursor is, changes to the parameters that are imported.

11.13.4.7 Initialize parameter (Machining load monitor parameter (list) screen only)

- (1) Move the cursor to the column for the parameter numbers that you wish to initialize.
- (2) Press the **[F5]** (Initialization) key.
Parameter group 1, where the cursor is, initializes.

11.13.4.8 Exit editing

After the setting is completed, press the **[F0]** (Completion mode) key to end and exit editing.

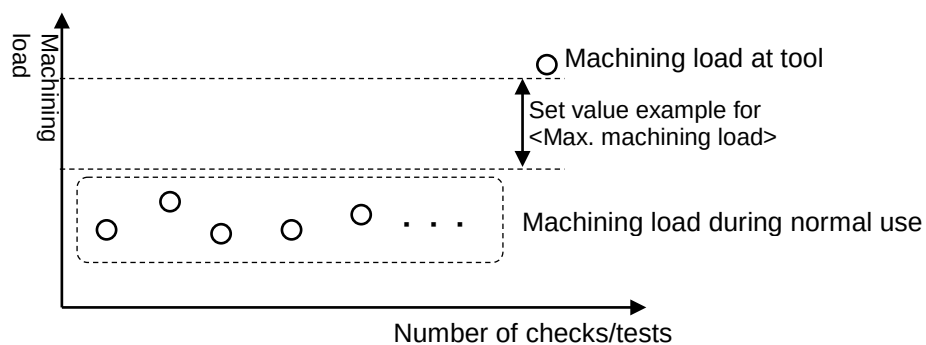
11.13.5 Examples When Using Machining Load Monitor Function

The parameter settings are different depending on how the user wishes to use the function. Refer to following examples.

11.13.5.1 Timing reference for tool change

If the tool status deteriorates due to wear or other factors, the machining load tends to increase. The customer can use this features to check and detect the right timing to change the tool according to the customer's preferences.

- (1) Follow the description in “11.13.2 Flow when using machining load monitor function” and perform the test machining multiple times. Then, record the most recent machining load as the normal or standard machining load.
- (2) Reproduce the tool status (when the tool status deteriorates) when the tool was changed (by the customer) and record that machining load.
- (3) Set the <Max. machining load> between the machining load during normal use and the machining load at the time of a tool change based on the values taken in (1) and (2) above.

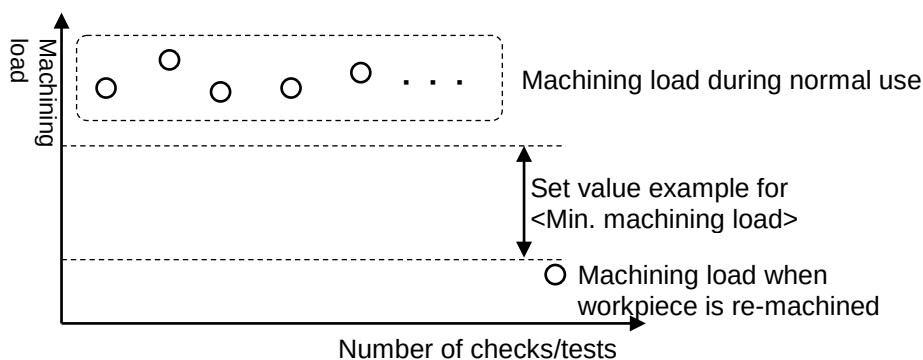


If there is no clear difference between the machining load under normal use and the machining load at the time of a tool change, this function will not be able to assess the timing appropriately. When the set value is changed based on “11.13.5.3 Parameter adjustment examples”, a difference in the machining load under normal use and at the time of a tool change may become more distinct and clear.

11.13.5.2 Improve machining to avoid re-machining workpieces

If the same workpiece is machined (re-machined) without changing tools, the spindle machining load decreases more than normal use. However, the machining load will not necessarily become 0 because of the contact between the tool and the workpiece, etc. If a workpiece needs to be re-machined, then the workpiece was not machined properly. Therefore, set the <Min. machining load> appropriately to ensure a decrease in the machining load is detected.

- (1) Follow the description in “11.13.2 Flow when using machining load monitor function” and perform the test machining multiple times. Then, record the most recent machining load as the normal or standard machining load.
- (2) Re-machine a workpiece that was already machined properly without changing the workpiece, and then record that machining load.
- (3) Set the <Min. machining load> between the machining load during normal use and the machining load when the workpiece was re-machined, based on the values taken in (1) and (2) above (refer to figure below).



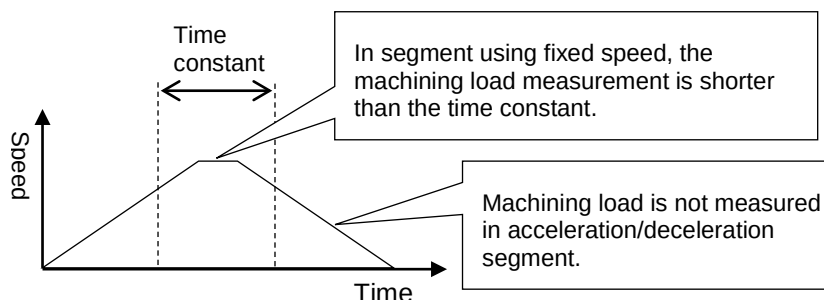
If there is no clear difference between the machining load under normal use and the machining load when a workpiece is re-machined, then this function cannot be used. When the set value is changed based on “11.13.5.3 Parameter adjustment examples”, a difference in the machining load under normal use and when a workpiece is re-machined may become more distinct and clear.

11.13.5.3 Parameter adjustment examples

When using this function, the parameter may need to be adjusted depending on the situation. Refer to following examples.

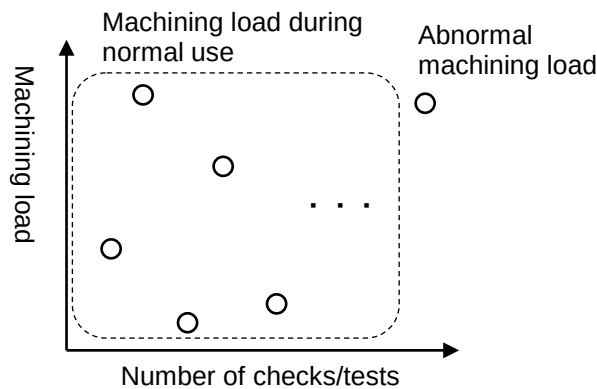
Monitoring the machining load when the segment using a fixed speed is short

When the <Load monitor method> is set to <0: Peak value>, the machining load is not measured because the load is easily affected by the torque during acceleration/deceleration when the spindle accelerates and decelerates. In addition, the operator message <<Machining load could not be measured>> appears when the machining load that is measured does not follow the time constant. As a result, if the operator message <<Machining load could not be measured>> appears, the cause may be that the machining load measurement is shorter than the time constant in the segment using a fixed speed (as shown in the figure below). Therefore, this may be fixed by setting <Load monitor method> to <1: Average>.

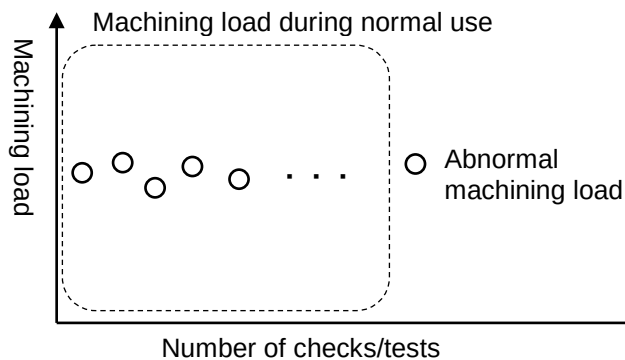


When a distinction cannot be made between normal and abnormal machining loads

If there is a big variation in the machining load as shown in the figure below and a distinction cannot be made between normal and abnormal machining loads, this distinction may be fixed or made clearer by making the <Time constant> bigger or by changing the <Load monitor method> to <1: Average>. If no distinction can be made even after changing this setting noted above, this function will not be able to assess appropriately.



Also, if there is little variation in the machining load as shown in the figure below and a distinction cannot be made between normal and abnormal machining loads, this distinction may be fixed or made clearer by making the <Time constant> smaller. If no distinction can be made even after changing this setting noted above, this function will not be able to assess appropriately.

When the machining load is not consistent

If the results based on the threshold value are not consistent, this could be attributed to a motor temperature change caused by the ambient temperature or by the heat produced by the motor. Refer to the following to determine whether this function can be used or not.

As a timing reference for a tool change, particularly when machining with a small machining load, be sure to check beforehand for any machining load changes caused by motor temperature fluctuation. For example, check the machining load in the morning when starting work (when the ambient temperature and motor temperature are low) and again during the day (when the ambient temperature and motor temperature are high). Then, consider revising the <Max. machining load>. If a clear distinction can be made between abnormal and normal machining loads regardless of the temperature, then after the threshold value is set to the appropriate value and this function can be used.

To improve machining to avoid re-machining workpieces, particularly when machining with a small machining load, be sure to check beforehand for any machining load changes caused by motor temperature fluctuation. For example, compare the machining load in the morning when starting work (when the ambient temperature and motor temperature are low) with the machining load during the day (when the ambient temperature and motor temperature are high). Then, consider revising the <Min. machining load>. If a clear distinction can be made between the machining load when a workpiece is re-machined and the machining load under normal use regardless of the temperature, then after the threshold value is set to the appropriate value and this function can be used.

Stop level setting examples when the maximum and minimum are reached

Refer to the following when setting the stop levels.

(Note) In the event that damage results after the alarms <<Machining load maximum>> and <<Machining load minimum>> are triggered, the stop triggered by this function is designed to prevent a problem from becoming worse. This function is not designed to prevent damage 100% of the time.

Stop level	Usage example
<0: Level 1> (No stop)	To check the effectiveness of the setting after setting <Max. machining load> and <Min. machining load> (when the user does not wish to stop operation for misdetections in order to adjust the settings during actual machining, etc.)
<1: Level 2> (Cycle stop)	To put a measure in place such as a tool change at the end of the program during operation. (Note 1)
<2: Level 3> (Block stop)	To stop quickly in order to minimize the impact or affect on the tool and workpiece. (Note 2)
<3: Level 4> (Immediate stop)	To stop operation as quickly as possible. (Note 3) (Note 4)

(Note 1) Machining is carried out until it is finished even if an alarm is triggered. Therefore, the setting may not prevent the spindle from becoming damaged or broken.

(Note 2) Operation stops during canned cycle machining when the tool and workpiece move to a position away from each other. During other machining that is not a canned cycle, there is a risk that the tool or workpiece may become damaged because the machine stops while the tool and workpiece are in contact.

(Note 3) During canned cycle tapping, operation stops after the tapping is complete.

(Note 4) There is a risk that the tool or workpiece may become damaged because the machine stops while the tool and workpiece are in contact.

11.13.6 PLC Signals

This function can be disabled by a PLC signal.

In addition, the following monitoring data for the machining load can be output via PLC signals.

<<Machining load maximum>> detection, <<Machining load minimum>> detection, machining load monitoring status, current machining load 1, current machining load 2 and the current parameter number being used

The current machining load 1 is the machining load calculated using a moving average with the <Time constant> for the machining load monitor parameter being used.

The current machining load 2 is the machining load before applying a moving average with the <Time constant> for the machining load monitor parameter. The value is always set regardless if the machining load monitor function is ON or OFF. The unit is in [Nm] (Display unit: 0.001 Nm). Refer to “Chapter 6 – 6.2 Global OM description” in the PLC System Manual for further details.

11.13.7 External Output Signals

The following monitoring data for the machining load can be output via external signals.

<<Machining load maximum>> detection, <<Machining load minimum>> detection and the machining load monitoring status

Refer to “Chapter 2 Data bank - 2.6 External I/O signals” in the Operation Manual (Data) for further details.

11.13.8 Macro Variables

The following items can be referenced using the macro system variables.

<Current machining load 1> and <Parameter number being used> and the corresponding <Load monitor method>, <Time constant>, <Stop level for max.>, <Stop level for min.>, <Max. machining load> and <Min. machining load>

Parameter No.1 to 99 <Updated load(Peak value)> and <Updated load(Average)>

The <Current machining load 1> is the machining load calculated using a moving average with the <Time constant> for the machining load monitor parameter being used.

Refer to “Chapter 6 Macros – 6.2.6 System variables” in the Programming Manual (NC language) for further details.

11.13.9 Waveform Display

The machining load waveform after applying the moving average can be displayed by selecting <3: Machining load 1> from the servo measurement items in the waveform display function. The displayed machining load is the machining load calculated using a moving average with the <Time constant> for the machining load monitor parameter being used.

However, the waveform does not display in the following situations (value is 0).

- When an axis that is not the <0: Spindle> is selected as the servo axis
- When the machining load is not measured, such as when M340 modal is engaged (function OFF) or when the spindle is accelerating or decelerating

The unit is in **[Nm]** (Display unit: 0.001 Nm).

The machining load waveform before applying the moving average can be displayed by selecting <4: Machining load 2> from the servo measurement items in the waveform display function. The displayed machining load is the machining load before applying the moving average with the <Time constant> for the machining load monitor parameter. The value is always displayed regardless if the machining load monitor function is ON or OFF.

However, the waveform does not display in the following situations (value is 0).

- When an axis that is not the <0: Spindle> is selected as the servo axis

The unit is in **[Nm]** (Display unit: 0.001 Nm).

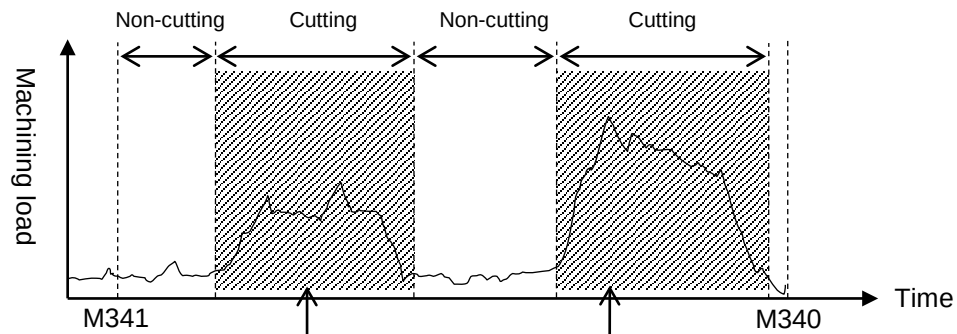
Refer to “Chapter 6 I/O – 6.1 Waveform display” in the Installation Manual for further details.

11.13.10 Machining Load Measurement Conditions

The machining load is measured when all of the following conditions are met.

- M341 to M343 modal is engaged (function is ON)
- Memory operation mode is engaged
- Cutting in progress
- Spindle rotating
- Dry run is not in progress
- Machine lock is not engaged
- Spindle is not accelerating/decelerating (Peak value only. The average is measured even if it is accelerating/decelerating.)

As a result, the machining load is measured and the machining load that is used for assessing the threshold value is only the diagonal stripe sections (as shown in the figure below). In addition, when the maximum machining load is monitored with <Load monitor method> set to <Peak value>, the threshold value is constantly assessed in the diagonal stripe sections noted below. In all other situations, the threshold value is assessed when the M340 is carried out (Refer to “Timing when threshold value is assessed and when measurement value is updated” under the section “11.13.4.1 Machining load monitor parameter (list) screen”).



Threshold value is constantly assessed during cutting when monitoring the maximum machining load with <Load monitor method> set to <0: Peak value>.

Threshold value is assessed at the same time as M340 using the machining load measured during cutting in the following situations:
 When monitoring the minimum machining load with <Load monitor method> set to <0: Peak value>
 When monitoring the maximum and minimum machining loads with <Load monitor method> set to <1: Average>.

11.13.11 Special Notes

- The machining load value that is measured is a guide for the user. This function is not designed for measuring the torque and force. The machining load that is displayed may be different from the actual machining load.
- When the machining conditions are changed, the machining load that is measured also changes. Make sure the <Max. machining load> and <Min. machining load> are set to the appropriate values.
- The machining load that is measured may change depending on factors such as fluctuation in the temperature, power voltage or motor temperature.
- The machining load that is measured may change if a component is replaced like the motor. Make sure the <Max. machining load> and <Min. machining load> are set to the appropriate values.
- To issue an M341, M342 or M343 command without the E address, register the tool for monitoring the machining load on the ATC tool screen. If there is no tool mounted on the spindle that is registered, the alarm <<Machining load could not be measured>> is triggered and the machining load monitor will not start.
- This function cannot be used in MDI operation mode. If an M340, M341, M342 or M343 command is issued, the alarm <<Specified M code cannot be used>> is triggered.
- When the lathe spindle is selected while the machining load monitor modal is engaged, the operator message <<Machining load could not be measured>> appears and the machining load monitor stops.
- The machining load monitor is not possible while the lathe spindle selection modal is engaged. If an M341, M342 or M343 command is issued, the alarm <<A command unable to select the lathe spindle>> is triggered.
- If an M341, M342 or M343 command is issued on a machine where the machining load monitor function is unavailable, it follows user parameter (switch 1) <Unregistered M-code>.
- When restarting a program in a segment where the machining load is monitored and when performing test machining, the machining conditions may change and the <Max. machining load> and <Min. machining load> may not be detected properly.
- If the program is stopped by pressing the **[FEED HOLD]** switch or by turning ON the **[SINGL]** key in memory operation, the machining load monitor is paused. Press the **[START]** switch to restart the machining load monitor. In this situation, when performing test machining, the machining conditions may change and the <Max. machining load> and <Min. machining load> may not be detected properly.
- If memory operation is paused by pressing the **[MANU]** key or by pressing the **[MDI]** key during memory operation, the machining load monitor is paused. When the **[MEM]** key is pressed to return back to memory operation mode and then the **[START]** switch is pressed to restore operation, the machining load monitor will restart. In this situation, when performing test machining, the machining conditions may change and the <Max. machining load> and <Min. machining load> may not be detected properly.
- When an M341, M342 or M343 modal is changed to the M340 modal using the reset operation, the <Max. machining load> and <Min. machining load> for the machining load being measured will not be assessed and the <Updated load (Average and peak value)> will not be updated.

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CHAPTER 12 (1)

COOLANT UNIT

- 1 Handling Precautions**
- 2 Functions**
- 3 Operation Procedure**

1 Handling Precautions

WARNING

If the maintenance cover is removed, coolant coming from inside the machine when it is operating may get into your eyes, or the tools or workpieces may shoot out causing injury.

The operator should visually check to make sure that the maintenance cover is attached before turning ON the power.

The operator should warn the supervisor if the maintenance cover is not attached.

The supervisor must attach the maintenance cover.

WARNING

If this is not observed, someone may trip over the machine, or nearby equipment, or slip on coolant and fall over while setup or maintenance work is being carried out.

Protective equipment such as safety footwear and helmets must be worn while working.

The area around the machine must be organized, tidy and clean before performing work.

Any excess length of cable should be removed so that there is no slack. If cables are running along the floor, they must be covered.

WARNING

The coolant may shoot out and injure your eyes while work is being carried out.

Always be sure to wear protective goggles.

If coolant gets into eyes, rinse with clean water and then seek medical advice.

This function's operation is performed in standard ladder 2 for the built-in PLC function.

The user can assume that there is a factory-set ladder program embedded already and operating for the user operations and machine operations noted below.

Do not change standard ladder 2 which is used for this coolant unit.

In addition, do not change the corresponding contact.

If the user changes the ladder program, the product will no longer be covered under the warranty.

In addition, do not perform the debug function (stop, ON/OFF for related contacts, etc.) for standard ladder 2 while this function is operating.

2 Functions

1. Coolant
The function of the coolant is to cool the tool and workpiece and to remove chips or shavings.
2. Chip shower
This function feeds chips, which build up inside the machining chamber during machining and cleaning, to the chip pan that is installed at the rear of the machine.

3 Operation Procedure

3.1 Coolant Pump

Turning ON the coolant pump

1. After pressing the [CLT.P] key on the operation panel, the key lamp lights up and the coolant motor starts.

Turning OFF the coolant pump

2. After pressing the [CLT.P] key on the operation panel, the key lamp turns off and the coolant motor stops.

* If the [EMERGENCY] switch is pressed, the hardware control turns it “OFF”.

3.1.1 Coolant Pump Wait Function

In the following situations, the coolant pump automatically turns OFF and the lamp for [CLT.P] key flashes.

- When the <Automatic coolant off time> has elapsed and the machine is not operating
- When the key is ON and a stop level 5 alarm is triggered that turns ON the ALM, or when the alarm <<Standby mode>> is triggered.
- If the [CLT.P] key is pressed when the key is OFF and an alarm is triggered that turns ON the ALM.

* <<*Door open>> is a stop level 5 alarm. The ALM does not turn ON because it is necessary to perform the CTS back wash and drain cycle.

Follow the procedure below to turn the coolant pump ON when the above conditions apply.

- Reset the stop level 5 alarm. Then press the [CLT.P] key twice to turn the pump ON again.
- Start (External startup) memory or MDI operation. (Note that operation starts right after the <Coolant delay time> has elapsed.)

3.2 Coolant Valve

Discharging the coolant

1. Turn ON the power for the machine.
2. Turn the coolant pump ON.
3. Press the **[MDI]** key to switch to the MDI operation mode screen.
4. Make sure that the front door is closed.*

For NC language

5. Press the **[CAN]** key to clear the editing area.
6. Use the number keys to enter "M8".

For conversation language

5. Use the **[CURSOR]** key to move the cursor to <Coolant>.
6. Use the number keys to enter "1" (1: ON) and press the **[ENT]** key.

7. Press the **[START]** switch to discharge the coolant.

- * If the front door is open, the coolant will not discharge.
(The valve is turned OFF using ladder control and hardware control when the **[DOOR INTERLOCK]** switch is enabled. The valve is turned OFF using ladder control when the **[DOOR INTERLOCK]** switch is disabled. If it discharges before opening the door, close the door and restart in MDI mode (because memory operation is not available) to open the valve using ladder control and to discharge again. (Note that this does not apply when the machine's factory default settings for the ladder have been overwritten.))

Stopping coolant discharge

1. Press the **[MDI]** key to switch to the MDI operation mode screen.

For NC language

2. Press the **[CAN]** key to clear the editing area.
3. Use the number keys to enter "M9".

For conversation language

2. Use the **[CURSOR]** key to move the cursor to <Coolant>.
3. Use the number keys to enter "0" (0: OFF) and press the **[ENT]** key.

4. Press the **[START]** switch to stop coolant discharge.

- * In the following situations, the valve turns "OFF" and the coolant stops.
 - When **[RST]** key is pressed.
 - When the **[M.LCK]** is ON.
 - When memory operation is finished.
 - When an M00/M01 command (NC language) is given, and when the program (conversation) is stopped.
 - When a stop level 5 alarm is triggered.
 - When <Coolant during tool change> is set to <0: Stop> and the tool is changed.
 - When the <Tool breakage detection coolant OFF> is set to <1: Yes> in conversation, and the tool breakage detection (Type 3) operation is activated.
- * In the following situations, the valve turns "ON" and the coolant discharges.
 - When the <Coolant during tool change> is set to <0: Stop> and the tool change operation is completed, after the coolant begins to discharge at the beginning of the tool change operation.
 - When restarting after the coolant begins to discharge before the program stop in conversation.
 - When the <Tool breakage detection coolant OFF> is set to <1: Yes> in conversation and the tool breakage detection operation is completed, after coolant begins to discharge when the tool breakage detection (Type 3) operation is activated.

3.3 Chip Shower

Discharging the chip shower

1. Turn ON the power for the machine.
2. Turn ON the **[CHP.F]** key on the operation panel. The key lamp lights up.
3. Press the **[MDI]** key to switch to the MDI operation mode screen.
4. Make sure that the front door is closed.

For NC language

5. Press the **[CAN]** key to clear the editing area.
6. Use the number keys to enter "M400".

For conversation language

5. Use the **[CURSOR]** keys to move the cursor to <Ext. signal>.
6. Use the number keys to enter "400" and press the **[ENT]** key.
7. After pressing the **[START]** switch, the chip shower motor starts and the chip shower begins to discharge.
8. After the **[CHP.F]** key is turned OFF, the key lamp turns off and the chip shower stops. If the **[CHP.F]** key is turned ON again, the chip shower begins to discharge again.
9. Press the **[RST]** key to stop the chip shower pump and the chip shower.
- * If the front door is open, the chip shower will not discharge.
(The pump power is turned OFF using ladder control and hardware control when the **[DOOR INTERLOCK]** switch is enabled. The pump power is turned OFF using ladder control when the **[DOOR INTERLOCK]** switch is disabled. If it discharges before opening the door, restart in MDI mode (because memory operation is not available) to discharge again using ladder control. (Note that this does not apply when the machine's factory default settings for the ladder have been overwritten.))

Stopping chip shower discharge

1. Press the **[MDI]** key to switch to the MDI operation mode screen.

For NC language

2. Press the **[CAN]** key to clear the editing area.
3. Use the number keys to enter "M401".

For conversation language

2. Use the **[CURSOR]** keys to move the cursor to <Ext. signal>.
3. Use the number keys to enter "401" and press the **[ENT]** key.
4. Press the **[START]** switch to stop the chip shower.
- * After the <Chip shower retention time> in the user parameter has elapsed, the chip shower turns OFF.

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CHAPTER 12 (2)

CTS DEVICE (COOLANT-THROUGH-SPINDLE)

- 1 Handling Precautions**
- 2 Functions**
- 3 Operation Procedure**

1 Handling Precautions

WARNING

If the maintenance cover is removed, coolant coming from inside the machine when it is operating may get into your eyes, or the tools or workpieces may shoot out causing injury.

The operator should visually check to make sure that the maintenance cover is attached before turning ON the power.

The operator should warn the supervisor if the maintenance cover is not attached.

The supervisor must attach the maintenance cover.

Take the necessary precautions even when the maintenance cover is attached, because dangerous situations, such as an explosion caused by combustible materials, cannot always be prevented.

WARNING

If this is not observed, someone may trip over the machine, or nearby equipment, or slip on coolant and fall over while setup or maintenance work is being carried out.

Protective equipment such as safety footwear and helmets must be worn while working.

The area around the machine must be organized, tidy and clean before performing work.

Any excess length of cable should be removed so that there is no slack. If cables are running along the floor, they must be covered.

WARNING

The coolant may shoot out and injure your eyes while work is being carried out.

Always be sure to wear protective goggles.

If coolant gets into eyes, rinse with clean water and then seek medical advice.

This function's operation is performed in standard ladder 2 for the built-in PLC function.

The user can assume that there is a factory-set ladder program embedded already and operating for the user operations and machine operations noted below.

Do not change standard ladder 2 which is used for this coolant unit.

In addition, do not change the corresponding contact.

If the user changes the ladder program, the product will no longer be covered under the warranty.

In addition, do not perform the debug function (stop, ON/OFF for related contacts, etc.) for standard ladder 2 while this function is operating.

2 Functions

The CTS device is to be used with a special tool and functions to discharge the coolant from the tip of the tool.

3 Operation Procedure

Discharging the coolant

1. Turn ON the power for the machine.
2. Turn the coolant pump ON.
3. Press the **[MDI]** key to switch to the MDI operation mode screen.
4. Make sure that the front door (other doors besides front door on models equipped with QT; front door and side door on models equipped with the side door option but without the QT) is closed.

For NC language

5. Press the **[CAN]** key to clear the editing area.
6. Use the number keys to enter "M494".

For conversation language

5. Use the **[CURSOR]** keys to move the cursor to <Ext. signal>.
6. Use the number keys to enter "494" and press the **[ENT]** key.
7. Press the **[START]** switch to discharge coolant from the tip of the tool.
- * If the front door (other doors besides front door on models equipped with QT; front door and side door on models equipped with the side door option but without the QT) is open, the coolant will not discharge.
The discharge is stopped using ladder control and hardware control when the **[DOOR INTERLOCK]** switch is enabled.
The discharge is stopped using ladder control when the **[DOOR INTERLOCK]** switch is disabled.
If it starts to discharge before opening a door, restart in MDI mode (because memory operation is not available) to open the valve using ladder control and to discharge again.
(Note that this does not apply when the machine's factory default settings for the ladder have been overwritten.)

Stopping coolant discharge

1. Press the **[MDI]** key to switch to the MDI operation mode screen.

For NC language

2. Press the **[CAN]** key to clear the editing area.
3. Use the number keys to enter "M495".

For conversation language

2. Use the **[CURSOR]** keys to move the cursor to <Ext. signal>.
3. Use the number keys to enter "495" and press the **[ENT]** key.
4. Press the **[START]** switch to stop coolant discharging from the tip of the tool.
- * For the following situation, the valve turns "OFF" and the CTS stops.
 - When **[RST]** key is pressed.
 - When the **[M.LCK]** is ON.
 - When memory operation is finished.
 - When an M00/M01 command (NC language) is given, and when the program (conversation) is stopped.
 - When a stop level 5 alarm is triggered.
 - When a tool change starts.
 - When **[ATC]** is operated in manual mode.
- * For the following situation, the valve turns "ON" and the CTS discharges.
 - When the CTS begins to discharge before the program stop in conversation and operation is restarted.
 - When the user parameter (switch 1) <Coolant return method for program stop> is set to <0: Method 1> in NC language and when the coolant discharges before the program stop, and operation is restarted.

- (Note) The coolant discharge automatically stops when a tool is changed in manual operation, MDI operation and memory operation. When the Z-axis raises to the ATC zero point and the coolant discharge stops at the same time, the tool change operation is carried out. However, the coolant discharge does not restore after the tool change operation.

3.1 Back Wash Cycle

The back wash cycle reverses the circulation of normal coolant flow and washes the filter. This function cannot be used when the machine parameter (system 1) <Cyclone filter> is set to <1: Equipped>.

1. Turn ON the power for the machine.
2. After pressing the **[CLT.P]** key on the operation panel, the key lamp lights up and the coolant motor starts.
3. Press the **[MDI]** key to switch to the MDI operation mode screen.

For NC language

4. Press the **[CAN]** key to clear the editing area.
5. Use the number keys to enter "M496".

For conversation language

4. Use the **[CURSOR]** keys to move the cursor to <Ext. signal>.
5. Use the number keys to enter "496" and press the **[ENT]** key.
6. Press the **[START]** switch to start the back wash cycle.

(Note 1) Even if the tool change operation is carried out during a back wash cycle, the cycle does not stop.

(Note 2) The alarm <<*Center through coolant on>> is triggered when a back wash cycle command is issued during coolant discharge from the CTS.

3.2 Drain Cycle

The drain cycle returns the coolant that is in the piping back to the coolant tank. This function cannot be used when the machine parameter (system 1) <Cyclone filter> is set to <1: Enable>.

1. Turn ON the power for the machine.
2. After pressing the **[CLT.P]** key on the operation panel, the key lamp lights up and the coolant motor starts.
3. Press the **[MDI]** key to switch to the MDI operation mode screen.

For NC language

4. Press the **[CAN]** key to clear the editing area.
5. Use the number keys to enter "M479".

For conversation language

4. Use the **[CURSOR]** keys to move the cursor to <Ext. signal>.
5. Use the number keys to enter "479" and press the **[ENT]** key.
6. Press the **[START]** switch to start the drain cycle.

(Note) When the user parameter (switch 1) <CTS depressurize check delay time> is not set to "0", the depressurize check will trigger the alarm again even if the alarm is reset with the **[RST]** key. As a result, set the <CTS depressurize check delay time> to 0, or to a larger value, so that the alarm <<CTS pressure has not lowered>> does not trigger, and then execute the M code command.

CHAPTER 12 (3)

AUTOMATIC OILING DEVICE

- 1 Functions
- 2 Alarms

1 Functions

This chapter describes the automatic lubrication system which regularly carries out lubrication on the ball screws and guides for the X-, Y- and Z-axes.

This system turns the pump ON and OFF repeatedly at the time intervals which have been previously set, so that lubricant can be supplied automatically.

2 Alarms

- Common

Alarm description	Stop / Cancel	Cause	Recovery procedure
Machine param. setting error (SYS 1)	(4,2)	The "Automatic oiling function" and the "Automatic greasing function" are set together.	Check the parameter value. Machine parameter (system 1) <Automatic oiling function> and <Automatic greasing function>
Execute the automatic oiling/greasing cycle ON command	(1,2)	Automatic oiling or automatic greasing error was detected.	After replenishing the lubricant or grease, execute the automatic oiling / greasing cycle ON command in MDI operation. NC language: M478 Conversation language: Miscellaneous function 478

- Automatic oiling function

Alarm description	Stop / Cancel	Cause	Recovery procedure
Pressure high (automatic oiling)	(2,2)	The pressure did not decrease before supplying oil.	1. Press the [RST] key. 2. Check the parameter value. Machine parameter (system 1) <Automatic oiling pause 1>, <Automatic oiling pause 2> and <Automatic greasing completion monitoring time> 3. The lubrication pump is defective. 4. The pressure sensor is defective.
Pressure low (automatic oiling)	(2,2)	The pressure did not increase even though the lubrication pump was turned on.	1. Check the parameter value. Machine parameter (system 1) <Automatic oiling monitoring time> 2. The lubricant is leaking in the piping. Check the piping. 3. The lubrication pump is defective. 4. The pressure sensor is defective.
Oil level low (automatic oiling)	(2,2)	Insufficient lubricant. (Alarm will not reset until the empty signal for automatic oiling turns ON.)	1. Replenish the lubricant. 2. Press the [RST] key to restore operation.

When one of the three alarms noted above is triggered:

- When alarm triggers: If the OILPMP signal (automatic oiling/greasing pump) was turned ON, the OIL PMP signal immediately turns OFF.
- After alarm is triggered: OILPMP signal does not turn ON.
- After the alarm is reset: Press the **[RST]** key to reset the alarm, and thereafter the timer begins counting following the interval set for the automatic oiling pause. If method 2 is used, the timer that counts while the STL signal is ON does not reset.

If one of the above alarms is triggered: <<Pressure high (automatic oiling)>> or <<Pressure low (automatic oiling)>>, the operation message <<Execute the automatic oiling/greasing cycle ON command>> appears.

- When the operator message appears: Memory operation is not available.
- After the operator message is reset: After executing the automatic oiling and greasing cycle ON in MDI operation mode, if the counts for automatic oiling pause 1 and automatic oiling pause 2 are cleared and the oiling operation is carried out normally, it resets. Thereafter, the OILPMP signal turns OFF and the count for automatic oiling pause 2 is carried out.

Operation after resetting alarm

After resetting alarm:

Automatic oiling: The timer begins counting for the interval set for the automatic oiling pause. Once that interval period elapses, the OILPMP signal turns ON.

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CHAPTER 12 (4)

AUTOMATIC GREASING DEVICE

- 1 Functions
- 2 Alarms

* This option is not available on the S500Z1, S700Z1, or R450Z1.

1 Functions

This automatic greasing device regularly lubricates the ball screws and linear guides for the X-, Y- and Z-axes.

(Note 1) Always use the following grease on this system.

Using another grease will cause a problem or malfunction on this system.

Description: Lube LHL-X100

(Note 2) The operating temperature under warranty for this system is: 0° to 40°C.

(Note 3) If the ambient temperature drops below 0°C, warm up the machine for approximately 10 to 15 minutes before starting your work. This is to reduce the viscosity for the automatic greasing.

2 Alarms

• Common

Alarm description	Stop / Cancel	Cause	Recovery procedure
Machine param. setting error (SYS 1)	(4,2)	The "Automatic oiling function" and the "Automatic greasing function" are set together.	Check the parameter value. Machine parameter (system 1) <Automatic oiling function> and <Automatic greasing function>
Execute the automatic oiling/greasing cycle ON command	(1,2)	Automatic oiling or automatic greasing error was detected.	After replenishing the lubricant or grease, execute the automatic oiling / greasing cycle ON command in MDI operation. NC language: M478 Conversation language: Miscellaneous function 478

• Automatic greasing function

Alarm description	Stop / Cancel	Cause	Recovery procedure
Automatic greasing signal abnormal	(2,2)	*Pump stop signal did not turn ON before greasing.	1. Press the [RST] key. 2. Check the parameter value. Machine parameter (system 1) <Automatic greasing pause 1>, <Automatic greasing pause 2> and <Automatic greasing abnormal pressure monitoring time> 3. The greasing pump is defective 4. The automatic greasing device is defective.
Automatic greasing discharge abnormal	(2,2)	A discharge error on the greasing device was detected.	1. Check the parameter value. Machine parameter (system 1) <Automatic greasing abnormal pressure monitoring time> 2. The grease is leaking in the piping. Check the piping. 3. The greasing pump is defective 4. The automatic greasing device is defective.
Grease was not supplied automatically	(2,2)	Greasing was not executed even though the greasing pump was turned ON.	1. Check the parameter value. Machine parameter (system 1) <Automatic greasing completion monitoring time> 2. The grease is leaking in the piping. Check the piping. 3. The greasing pump is defective 4. The automatic greasing device is defective.

Alarm description	Stop / Cancel	Cause	Recovery procedure
Change grease (automatic greasing)	(2,2)	Grease has run out. (Alarm will not reset until the empty signal for automatic greasing turns OFF.)	Change the grease cartridge, and press the [RST] key to restore operation.

When one of the three alarms noted above is triggered:

- When an alarm triggers: If the OILPMP signal was turned ON, the OIL PMP signal immediately turns OFF.
- After the alarm is triggered: OILPMP signal does not turn ON.
- After the alarm is reset: Press the **[RST]** key to reset the alarm, and thereafter the OILPMP signal for automatic greasing turns ON.

If one of the alarms above <<Automatic greasing signal abnormal>>, <<Automatic greasing discharge abnormal>> or <<Grease was not supplied automatically>> is triggered, the operator message <<Execute the automatic oiling/greasing cycle ON command>> appears.

- When the operator message appears: Memory operation is not available.
- After the operator message is reset: After executing the automatic oiling and greasing cycle ON in MDI operation mode, if the counts for automatic greasing pause 1 and automatic greasing pause 2 are reached and the greasing operation is carried out normally, it resets. Thereafter, the OILPMP signal turns OFF and the count for automatic greasing pause 2 is carried out.

Operation after resetting alarm

After resetting alarm:

Automatic greasing: OILPMP signal turns ON.

<<Change grease (automatic greasing)>> triggers when there is no grease and is a standard alarm when using the automatic greasing function. If greasing is carried out when the grease level is low, <<Change grease (automatic greasing)>> triggers before the greasing operation is complete. Since the pause interval for automatic greasing is set over a long period (5 hours), the greasing pump turns ON after resetting the alarm in order to avoid using the machine when the greasing has not been carried out for a long period of time (more than 10 hours) after resetting the alarm.

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CHAPTER 12 (5)

MANUAL PULSE GENERATOR (PSRD006 HANDLE)

- 1 Handling Precautions**
- 2 Functions**
- 3 External View**
- 4 Operation Procedure**

1 Handling Precautions

WARNING

If the door interlock switch is disabled, the machine will operate even when a door is open, and you may get caught or drawn into the machine.

The installer or setup personnel must never leave the machine unattended while the door interlock switch is disabled.

The supervisor must not let the operator use the machine when the door interlock switch is disabled.

The operator must check that the door interlock switch is enabled before operating the machine.

The operator must inform the supervisor if the door interlock key is attached.

The door interlock key should be kept by the supervisor.

If there is an inner door, lock the inner door with the key before using the machine.

If there is an inner door, do not open the inner door while the axis in the machining room is operating.

WARNING

There are movable parts inside the machine such as the spindle head and table, and your body may get caught on them if you go inside the machine.

Do not go inside the machine while the machine is operating.

The operator should visually check to make sure that there is nobody inside the machine before starting the machine.

Do not go inside the machine. If you need to go inside the machine, the supervisor must be notified.

If you go inside the machine, turn OFF the main power breaker, and then attach a padlock to the main power breaker so that the power cannot be turned ON.

A sign or notice should be placed near the operation panel of the machine to warn others that work is in progress.

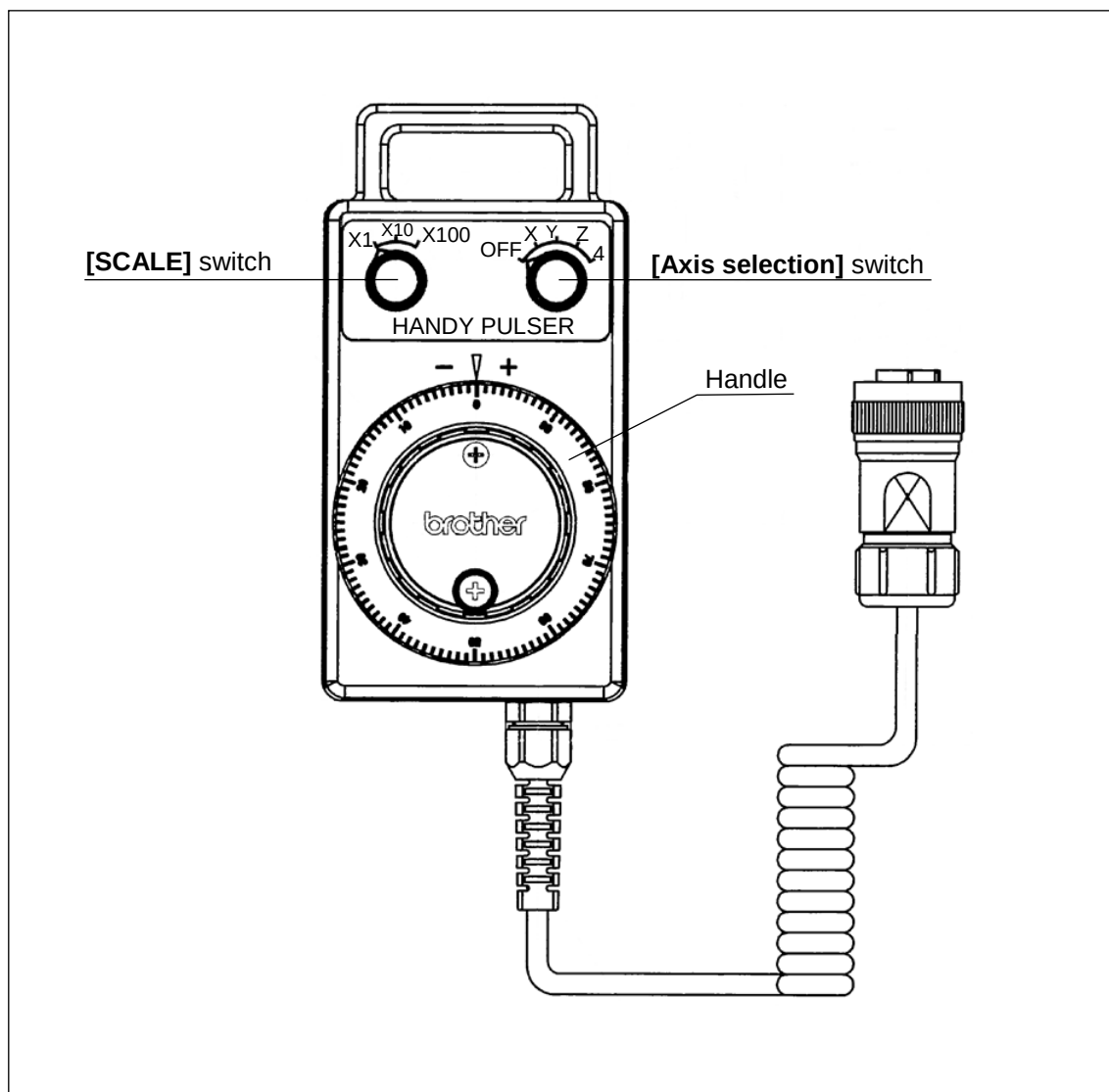
When the axis selection switch on the manual pulse generator is at any position other than OFF, manual operation on the main unit operation panel will be disabled. At this time, the message <<Handle mode>> will appear on the screen when an operation panel key such as an axis travel key is pressed.

When the door is open, the travel speed for the X-, Y- and Z-axes is restricted by the machine parameter setting <Rapid feedrate (door open)>, and the rotation speed for the A-, B- and C-axes is also restricted by the machine parameter setting <4th- to 8th-axis rapid feedrate (door open)>. As a result, if the operation is set too fast, the axis travel may not be carried out at the speed that was set. In addition, even if handle rotation is stopped in this situation, the motion will not stop right away and will continue to travel for the pulses that are still remaining internally, and then it will stop.

2 Functions

The manual pulse generator carries out axis travel manually based on the scaling for the X-, Y- and Z-axes.

3 External View



4 Operation Procedure

When using the manual pulse generator, follow the procedure below to move the axes.

1. Press the **[MANU]** key to switch to manual operation mode.
2. Use the **[Axis selection]** switch to select the axis (X-, Y-, Z- and 4-*) to be moved.
3. Use the **[SCALE]** switch to select the speed of axis travel.
4. Turn the handle to move the axis.

* **[Axis selection]** switch “4”

- When operating an additional axis and PLC-axis, set the **[Axis selection]** switch to “4” (additional axis). Thereafter, display the manual conditions screen and then press one of the keys between **[F0]** and **[F4]**, or press the **[F5]** (Change additional axes display) key and then press one of the keys between **[F0]** and **[F3]** in order to select the axis to be operated.
- When selecting an additional axis and PLC-axis, only 1 axis from the additional and PLC-axes can be selected. (An axis from both the additional and PLC-axes cannot be selected at the same time).
- When configuring multiple settings for the same axis in the user parameter <4th–12th row display axis>, even if one of the function keys that is displayed for that axis is pressed, the axis that operates when **[+4]** or **[-4]** is pressed is selected. Furthermore, all of the function keys for that axis are selected and highlighted.
- When the machine is equipped with a lathe spindle, if the **[Axis selection]** switch is set to “4” and the additional axis is selected as the lathe spindle, the handle operation is ignored and the operator message <<The lathe spindle is being selected>> appears.

Scaling and axis travel (X-, Y- and Z-axes using metric system)

Scale switch	Axis travel	
×1	0.001 mm/step	Travels 0.1 mm per 1 handle rotation
×10	0. 01 mm/step	Travels 1.0 mm per 1 handle rotation
×100	0.1 mm/step	Travels 10.0 mm per 1 handle rotation

Scale and axis travel (X-, Y- and Z-axes using inches)

Scale switch	Axis travel	
×1	0.0001 inch/step	Travels 0.01 inch per 1 handle rotation
×10	0.001 inch/step	Travels 0.1 inch per 1 handle rotation
×100	0.01 inch/step	Travels 1.0 inch per 1 handle rotation

Scale and axis travel (A-, B-, C- and QT-axes, same when using metric system/inches)

Scale switch	Axis travel	
×1	0.001°/step	Travels 0.1° per 1 handle rotation
×10	0.01°/step	Travels 1.0° per 1 handle rotation
×100	0.1°/step	Travels 10.0° per 1 handle rotation

(Note 1) There is no QT-axis function for S300X1, S500X1, S700X1, S1000X1, M140X1, M140X2, S300X2, S500X2, S700X2, F600X1, M200X3 or M300X3.

(Note 2) When the **[Axis selection]** switch on the handle is at any position other than OFF, manual operation on the main unit operation panel will be disabled.

At this time, the message <<Handle mode>> is displayed on the screen.

(Note 3) Refer to “V. PLC-Axis” in the PLC System Manual for further details about the PLC-axis scale and axis travel.

CHAPTER 12 (6)

AUTOMATIC DOOR (TYPE 1)

- 1 Handling Precautions**
- 2 Functions**
- 3 Operation Procedure**
- 4 Area Sensor**
- 5 Door Stop Conditions**
- 6 Alarms**

*** This option is not available on the S500Z1, S700Z1, or R450Z1.**

1 Handling Precautions

WARNING

If the door interlock switch is disabled, the machine will operate even when a door is open, and you may get caught or drawn into the machine.

The installer or setup personnel must never leave the machine unattended while the door interlock switch is disabled.

The supervisor must not let the operator use the machine when the door interlock switch is disabled.

The operator must check that the door interlock switch is enabled before operating the machine.

The operator must inform the supervisor if the door interlock key is attached.

The door interlock key should be kept by the supervisor.

If there is an inner door, lock the inner door with the key before using the machine.

If there is an inner door, do not open the inner door while the axis in the machining room is operating.

WARNING

When opening and closing a door, do not insert your fingers into the space between the door panels or between the door and the machine cover. Otherwise, your fingers may get caught causing injury.

Always be sure to hold the handle when opening and closing a door.

If you insert your fingers in between a door and the machine cover, they may get caught causing injury.

This function's operation is performed in standard ladder program 2 for the built-in PLC function.

The user can assume that there is a factory-set ladder program embedded already and operating for the user operations and machine operations noted below.

Do not change standard ladder program 2 which is used for this device.

In addition, do not change the corresponding contact.

If the user changes the ladder program, operation will no longer be covered under the warranty.

In addition, do not perform the debug function (stop, ON/OFF for related contacts, etc.) for standard ladder program 2 while this function is operating.

On a machine with an automatic door specification, the front door can be linked with a program or robot operation and can be opened and closed automatically.

If you get caught in between the door and the machine cover, you may become injured.

2 Functions

1. Open and close for automatic door
The front door is automatically opened and closed.
2. Area sensor
When the area sensor is obstructed or blocked during automatic door operation, the door operation immediately stops.

3 Operation Procedure

3.1 Operation Switch

1. **[AUTOMATIC DOOR MANU/AUTO] switch**
 - When using the “MANU” (Manual) setting
The **[AUTOMATIC DOOR OPEN/CLOSE]** switch is enabled. The automatic door does not open and close automatically in memory operation.
 - When using the “AUTO” (automatic) setting
The **[AUTOMATIC DOOR OPEN/CLOSE]** switch is disabled. The automatic opens and closes automatically in memory operation.
2. **[AUTOMATIC DOOR OPEN/CLOSE] switch**
The front door opens and closes. The **[AUTOMATIC DOOR MANU/AUTO]** switch is enabled when it is set to “MANU” (Manual).
 - When using the “OPEN” setting
Press the **[AUTOMATIC DOOR OPEN/CLOSE]** switch to open the front door. (Press and release the switch to open the door.)
 - When using the “CLOSE” setting
Press the **[AUTOMATIC DOOR OPEN/CLOSE]** switch to close the front door. (Press and release the switch to open the door.)

However, if the door lock is engaged even when the **[AUTOMATIC DOOR MANU/AUTO]** switch is enabled, the **[AUTOMATIC DOOR OPEN/CLOSE]** switch is disabled.
After turning ON the power, move the door slowly only once.

3.2 Automatic Door Control During Operation

If the **[AUTOMATIC DOOR MANU/AUTO]** switch is set to “AUTO” (Automatic), when the **[START]** switch is pressed in memory operation mode, the front door automatically closes and machining starts.

When not using a QT machine model

1. Switch to memory operation mode and press the **[START]** switch.
2. The front door is automatically closed.
3. The NC checks that the automatic door is closed, and then machining starts.
4. After machining is complete, the front door automatically opens.

(Note) When the **[AUTOMATIC DOOR MANU/AUTO]** switch is set to “MANU” (Manual), even if the **[START]** switch is pressed, the front door does not automatically close.

In addition, the automatic door opens after the M00/M01 command.

1. The M00/M01 command is issued during memory operation.
2. The automatic door will open.
3. Press the **[START]** switch.
4. The automatic door will close.
5. Machining resumes.

When using a QT machine model

[PALLET] key	
When "OFF"	1. Press the [START] switch. 2. The automatic door will close. 3. The NC checks that the automatic door is closed. 4. Machining starts. 5. Machining ends. 6. The automatic door will open.
When "1" or "2" is selected	1. The M00/M01 command is issued during memory operation. 2. The automatic door will open. 3. Press the [START] switch. 4. The automatic door will close. 5. Machining resumes.
When "1" or "2" is selected	1. Press the [PALLET START] switch. 2. The automatic door will close. 3. The NC checks that the automatic door is closed. 4. The table rotates. 5. Machining starts. 6. Machining ends. 7. The table rotates. 8. The pallet moves to the side where the workpiece is loaded. 9. The automatic door will open.
When "1" ↔ "2" is selected	1. Press the [PALLET START] switch. 2. The automatic door will close. 3. The NC checks that the automatic door is closed. 4. The table rotates and the other pallet moves to the side where the workpiece is loaded. 5. The automatic door opens and machining begins. 6. Press the [PALLET START] switch. 7. The automatic door will close. 8. The NC checks that the automatic door is closed. 9. The last machining operation ends. 10. The table rotates. 11. The automatic door opens and the next machining operation begins.

(Note 1) When "1", "2" or "1↔2" is selected on the [PALLET] key, the automatic door will not operate even if the [START] switch is used to start machining.

(Note 2) To open the door while the pallet start lamp is lit up (scheduled startup), set the [AUTOMATIC DOOR MANU/AUTO] switch to "MANU" (Manual) and press the "OPEN" switch. (When the door opens, the pallet start lamp turns off at the same time.) Thereafter, to close the door again, set the [AUTOMATIC DOOR MANU/AUTO] switch to "AUTO" (Automatic) and press the [PALLET START] switch. (The door closes, and the pallet start lamp lights up.)

3.3 Operation of Automatic Door Safety Device

Specification for automatic door operation

[AUTOMATIC DOOR MANU/AUTO] Switch	[DOOR INTERLOCK] Switch	Mode			
		Manual Operation	MDI Operation	Memory Operation	Program Editing
Manual	Disabled	Available	Available	Available	Available
	Enabled	Available*	Available*	Available*	Available*
Automatic	Disabled	Does not operate	Does not operate	Operates	Does not operate
	Enabled	Does not operate	Does not operate	Operates	Does not operate

* The [AUTOMATIC DOOR OPEN/CLOSE] switch is disabled while the door lock is engaged.

4 Area Sensor

When the area sensor is obstructed or blocked during automatic door operation, the alarm <<Area sensor obstructed>> is triggered and the door operation stops.

4.1 Automatic Door Control During Alarm <<Area sensor obstructed>>

1. Area sensor obstructed or blocked during door close operation
The automatic door is stopped.
2. Area sensor obstructed or blocked during door open operation
The user parameter setting <Area sensor pattern> can be used to either stop or not stop the automatic door operation. To stop automatic door operation, set the parameter to <1: Pattern 2 (Valid during door open/close operation)>.

- (Note)** In memory operation mode, when the area sensor is obstructed or blocked and operation stops while the automatic door is opening or closing, the alarm <<Close the automatic door>> is triggered upon restarting operation. Use the following procedure to restore operation.
1. Set the [AUTOMATIC DOOR MANU/AUTO] switch to “MANU” (Manual).
 2. Press the [AUTOMATIC DOOR OPEN/CLOSE] switch to open the door.
Or, to close it.
 3. Set the [AUTOMATIC DOOR MANU/AUTO] switch to “AUTO” (Automatic).

5 Door Stop Conditions

The door open and close operations stop when the following conditions apply.

- When an error is triggered on the machine.
- When the alarm <<The door is abnormal.>> or <<Door operation timeout>> is triggered.
- When the area sensor is obstructed or blocked.
- When the **[FEED HOLD]** switch is pressed.
- When the **[RST]** key is pressed.

5.1 Door Open/Close Operation During Alarm

Even if an alarm is triggered that prevents operation, the following procedure can be used to open or close the front door.

1. Set the **[AUTOMATIC DOOR MANU/AUTO]** switch to “MANU” (Manual).
2. Set the **[EMERGENCY]** switch to OFF.
3. Hold down the **[RELSE]** key and press the **[AUTOMATIC DOOR OPEN/CLOSE]** switch to open and close the front door. Release the **[RELSE]** key and turn ON the **[EMERGENCY]** switch to stop the door open/close operation.

6 Alarms

1. <<Door abnormal (sensor error)>>
The alarm is triggered when the limit switch on the automatic door is turned ON at the door close and door open positions. The user parameter <<Automatic door error level>> setting can be used to trigger this alarm.
2. The area sensor alarm <<Area sensor obstructed>> is triggered in the following situations.
(1) When the area sensor is obstructed or blocked during a door close operation.
(2) While the area sensor is obstructed or blocked, an attempt is made to open or close the door.
3. Alarm <<Door locked>> is triggered disabling the **[AUTOMATIC DOOR OPEN/CLOSE]** switch because the door lock is engaged
If the door is locked and the **[AUTOMATIC DOOR MANU/AUTO]** switch is set to “MANU” (Manual), this alarm is triggered when the **[AUTOMATIC DOOR OPEN/CLOSE]** switch is pressed.
4. <<Door operation timeout>>
This alarm is triggered when the automatic door open and close operations take more time than the time set in the user parameter <Automatic door operation time>. The user parameter <Automatic door error level> setting can be used to trigger this alarm.
5. <<The door is abnormal. (Electric cylinder)>>
This alarm is triggered when an automatic door (Electric cylinder type 1) error is detected. The user parameter <Automatic door error level> setting can be used to trigger this alarm.
6. <<The door is abnormal. (Electric cylinder type 2)>>
This alarm is triggered when an automatic door (Electric cylinder type 2) error is detected. The user parameter <Automatic door error level> setting can be used to trigger this alarm.
7. <<Automatic door - Machine zero return in progress>>
After the power turned ON, an alarm is triggered at the first door operation. The automatic door does not operate for a certain time or operates slowly while an alarm is triggered.

CHAPTER 12 (7)

INDICATORS

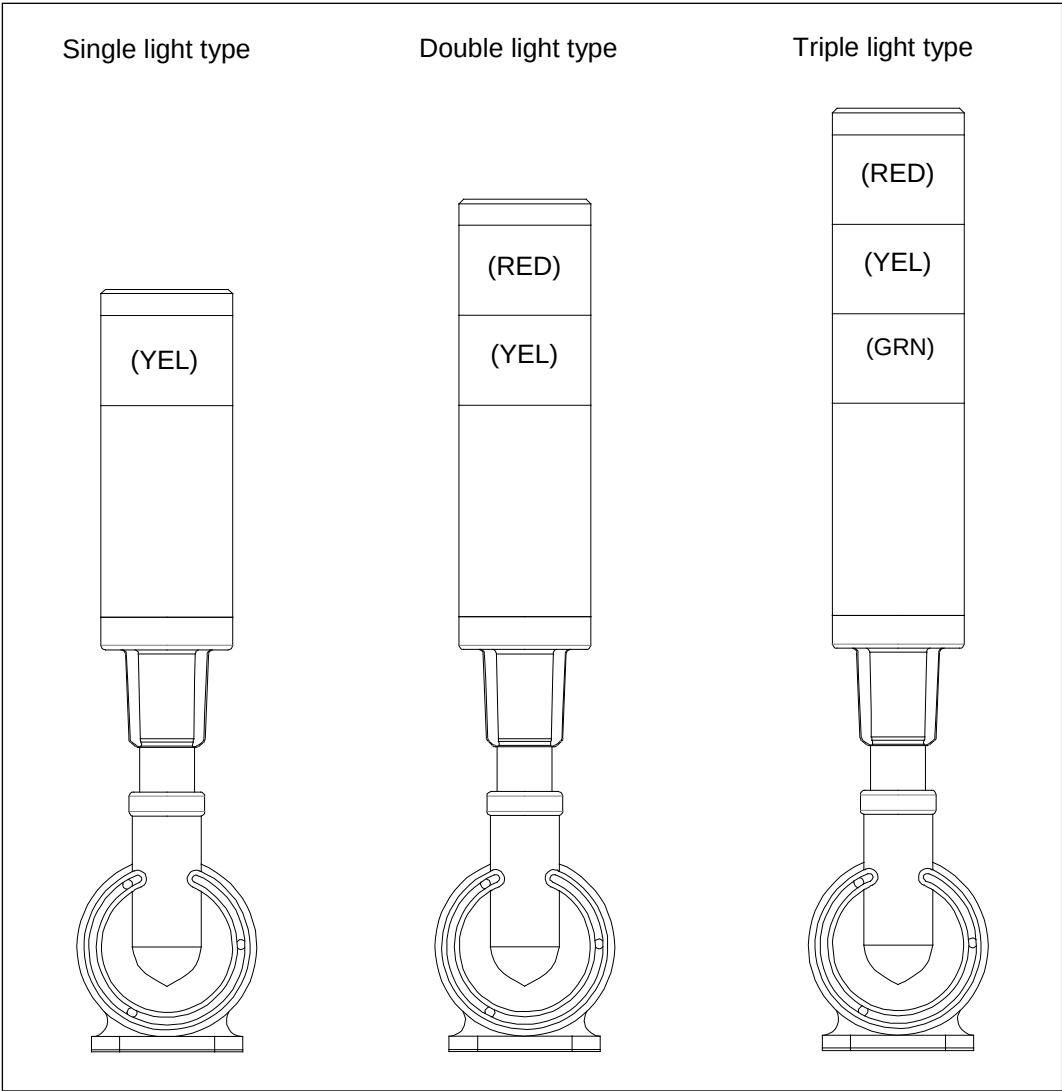
- 1 Functions
- 2 External View

1 Functions

The indicators can be used to inform an operator about the status of the machine (operation in progress, machining complete and error) at a remote location. In addition, the indicators light up in the following situations.

1. Red light (RED)
 - When the emergency stop switch is pressed.
 - When a servo OFF level alarm is triggered.
 - When the workpiece counter is finished (Note when the user parameter (switch 1) <Lamp status when count ends> is set to <2: Red>).
 2. Yellow light (YEL)
 - When memory operation is finished (M30 output).
 - When the workpiece counter is finished (Note when the user parameter (switch 1) <Lamp status when count ends> is set to <1: Yellow>).
 3. Green light (GRN)
 - When the memory operation and MDI operation are in progress (External output signal: Output of automatic operation starting up (STL)).
- * The red, yellow and green lamps light up when configured to the factory default settings. The red, yellow and green indicators are the same as the external I/O output signals <RED>, <YEL> and <GRN>.
- If the ladder is changed, the above does not necessarily apply.

2 External View



* The option for single and double light types is not available on the S500Z1, S700Z1 or R450Z1.

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CHAPTER 12 (8)

TOOL BREAKAGE DETECTOR

- 1** **Functions**
- 2** **Operation Procedure**
- 3** **Tool Breakage Alarm**

*** This option is not available on the S500Z1, S700Z1, or R450Z1.**

1 Functions

The tool breakage detector is a device which detects tool breakages when the tool touches a sensor. The tool detection sensors are attached to the top of the table within the range of the XY strokes. When a breakage is detected, the device moves the table or the spindle after cutting.

2 Operation Procedure

2.1 NC Language

During memory operation, tool breakage detection is performed for the sub program number when the “M203P_” command is issued.

Command format M203 P_; P : Sub program number

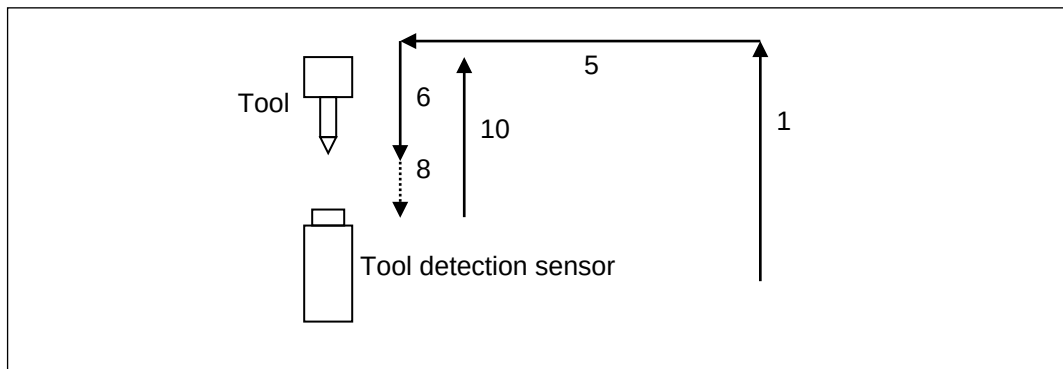
The user parameter (tool breakage detection) <Tool breakage detection OP> determines whether to execute tool breakage detection. If set to <3: Type 3>, the program specified by “P” is executed. This command is ignored when set to <0: Not equipped>.

- (Note 1) An alarm (<<Compensating diameter>>, <<During rotational transformation>>, <<Scaling>> or <<Mirror image mode>>) is triggered if the M203 command is issued during cutter compensation, rotational transformation, scaling or mirror imaging, respectively. No error is triggered while in the high accuracy mode A.
- (Note 2) An error is not triggered when M203 is included when the program restarts.
- (Note 3) <<Specified M code cannot be used>> is triggered when the command M203 is issued in MDI operation mode.
- (Note 4) The alarm <<Feature coordinate manufacturing mode engaged>> is triggered when an M203 command is issued while in feature coordinate machining mode.
- (Note 5) Any M203 characteristics not noted above are the same as M98 (sub program call).

An example of the sub program (breakage detection operation) called by M203 is shown below.

2.1.1 Sub Program Example

No.	Program	Description
1	G0Z_	Z-axis return
2	M19	Spindle orientation to 0 degrees
3	M9	Coolant OFF
4	M495	Center-through-coolant OFF
5	G0X_Y_	Positions the X- and Y-axes at the measurement position
6	G0Z_	Positions the Z-axis at the start position for tool breakage detection
7	M321	Detection signal OFF check
8	G1Z_F_	Pushes in the Z-axis direction (detection signal turns ON)
9	M120	Tool breakage error check (ON)
10	G0Z_	Z-axis return
11	M8	Coolant ON



- (Note 1) When the [RST] key is pressed after the <<*Tool breakage error>> is triggered, the life of the damaged tool is reset to “0”. However, if the <<*Tool breakage error>> is triggered in MDI operation mode, the life of the damaged tool does not reset to “0” even if the [RST] key is pressed.
- (Note 2) The machine does not operate while the machine lock is ON.
- (Note 3) The “M120” operation is not carried out while dry run is ON.

2.2 Conversation Language

The command for the tool breakage detection operation is issued using the signal output “203” in memory operation. The settings are configured using the signal output job for the machining data or using a miscellaneous function before or after the assigned cutting on the tool.

Signal output 203 203: Tool breakage detection ON

Tool breakage detection is carried out when changing tools after signal output 203 is executed. As a result, signal output 203 must be carried out to perform the detection while the tool is attached to the spindle (before changing to another tool).

Signal output 203 is disabled during a tool change and also when the following operations are performed.

- When the [RST] key is pressed.
- When the program end is executed (including program end in the middle of a scheduled program).
- When operation has been reset. *

* Operation is reset when one of the following operations is performed.

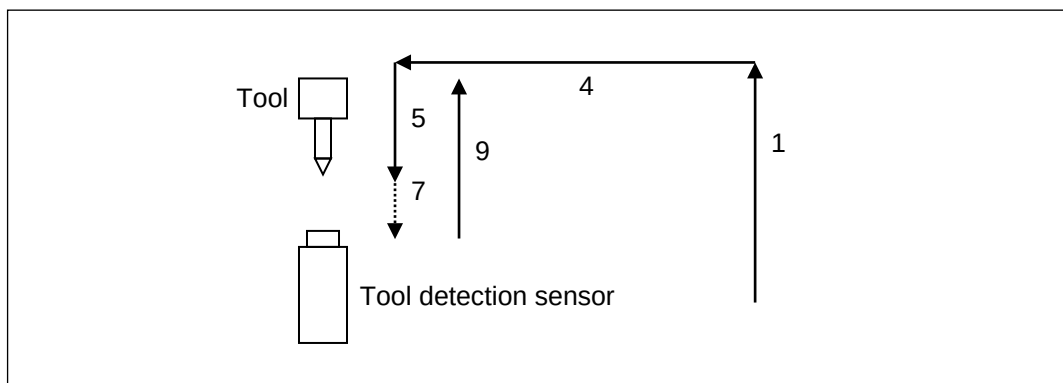
- When the [M.LCK] key is pressed.
- When ATC is used in manual mode.
- When a machine zero return (including external machine zero return) is performed.
- When a pallet index operation is performed.
- When a magazine turn (forward/reverse) is performed.

- (Note 1) When detection (not signal output 203 but detection before the tool change) is executed during cutter compensation, the alarm <<Compensating diameter>> is triggered.
- (Note 2) The signal output 203 command cannot be issued in MDI operation mode. (<<Input data error>> is triggered.)
- (Note 3) The dry run operation and mode cannot be changed while the detection is being executed. (The alarm <<Performing tool breakage detection>> is triggered.)
- (Note 4) The user parameter <<Tool breakage detection execution interval (No. of operations)>> is enabled for signal output 203. (The detection operation is carried out according to the value that is set.)

The following shows where the breakage detection operation is automatically inserted before the tool change.

2.2.1 Breakage Detection Operation

No.	Operation display	Description
1	Spindle orientation (0°) Machine zero return (Z)	Spindle orientation (0°) Z-axis machine zero return
2	Coolant OFF	When the user parameter <Tool breakage detection coolant OFF> is set to <1: Yes>, the OFF command is issued regardless of the coolant status.
3	CT coolant OFF	When the center-through-coolant function is enabled, the OFF command is issued regardless of the center-through-coolant status.
4	Positioning X_ Y_	Positioning is carried out according to the user parameters <Tool breakage detection position X-axis and Y-axis>.
5	Positioning Z_	Positioning is carried out according to the user parameters <Tool breakage detection position Z-axis> + <Tool breakage detection start distance>.
6	Detection signal OFF check	If the signal output is OFF, the alarm <<*Detection signal on>> is triggered.
7	Measurement feed Z_ Feed **** mm/min	The machine moves for the measurement according to the user parameters <Tool breakage detection position Z-axis> - <Tool breakage detection stop distance>. The feed speed is set using the user parameter <Tool breakage detection speed>.
8	Tool breakage check (ON)	If the signal output does not turn ON, the alarm <<*Tool breakage error>> is triggered.
9	Machine zero return (Z)	Z-axis machine zero return
10	Coolant ON	Coolant turns ON to restore operation if it was ON when “Coolant OFF” was executed in No.2 above. This operation is not carried out if it was in OFF status.
11	Tool change Tool No. * X_ Y_ Z_ Spindle speed ****	The tool is changed to the next tool.



- (Note 1) When the [RST] key is pressed after the <<*Tool breakage error>> is triggered, the life of the damaged tool is reset to “0”.
- (Note 2) The machine does not operate while the machine lock is ON.
- (Note 3) The “Tool breakage check (ON)” operation is not carried out while dry run is ON.
- (Note 4) When the signal output 203 command is issued, <Tool breakage detection ON> is displayed.
- (Note 5) After the signal output 203 is executed when a tool change command is issued for the same tool, the tool change is not carried out but the tool breakage detection is carried out.

- (Note 6) When No. 2 “Coolant OFF” has been executed, the No. 10 (Coolant ON), as explained, is executed to return the coolant to the original state. However, when the user parameter <Coolant during tool change> is set to <0: Stop>, it does not turn ON at this stage but it does turn ON after the tool change operation. (When the coolant turns ON after a tool change, the operation is not displayed.)
- (Note 7) The center-through-coolant does not automatically turn ON. To turn the center-through-coolant ON again, enter the signal output 494 for the machining data.
- (Note 8) When the tool change position is specified during operation preparations, the machine moves to the specified position after the breakage detection operation and then the tool is changed. As a result, if either the X-axis or the Y-axis is not set when specifying the position of the tool change, and then the breakage detection operation is carried out, the tool change position may vary.
- (Note 9) When an alarm is triggered during breakage detection (between No.1 and No.10 noted above), <<Performing tool breakage detection>> appears in the operation display for the alarm log.
- (Note 10) If the user parameter (quick table) <Pallet loading> is set to <1: At 1st tool change> on a QT machine model, the pallet is loaded at the same time as the tool change. In this situation, if the “Signal output 203” command is issued before the 1st tool change command, the <<Pallet position error>> is triggered during the 1st operation in the tool breakage detection.

2.2.2 G Code Conversion for Breakage Detection

When converting the signal output 203 to G code in conversation language, the location where signal output 203 is noted, is converted into a comment with brackets “(” and “)” and appears like: “(M203)”. The tool breakage detection operation is not converted into G code. In this situation, the alarm <<Unable to convert breakage detection data>> is triggered.

Conversion example

Program	G code conversion
Job 1 Axis travel Z0.00	(#JOB01 AXIS)
Job 2 Signal output 203	G90 G00 Z0. (M203) ← Comment
Job 3 Signal output 501	(#JOB03 SIGNL)
	G100 T01

- (Note) The signal output 203 is converted into a comment regardless whether the machine is equipped or not with the tool breakage detection option and regardless of the type.

3 Tool Breakage Alarm

3.1 Tool Breakage Error

When a tool breaks and the break is detected by the tool breakage detection, the <<Tool breakage error>> is triggered.

3.2 Reset Tool Breakage Error

If a break is detected by the tool breakage detection and the <<Tool breakage error>> is triggered, use the following procedure to reset the alarm and restore operation.

1. Press the **[RST]** key to reset the <<Tool breakage error>>.
2. Replace the broken tool with a new tool, reset the tool data and press the **[START]** switch.

CHAPTER 12 (9)

AUTOMATIC CENTERING UNIT

- 1 Functions**
- 2 Operation Procedure**
- 3 Data Settings for Automatic Centering**
- 4 Installing and Removing Probe**

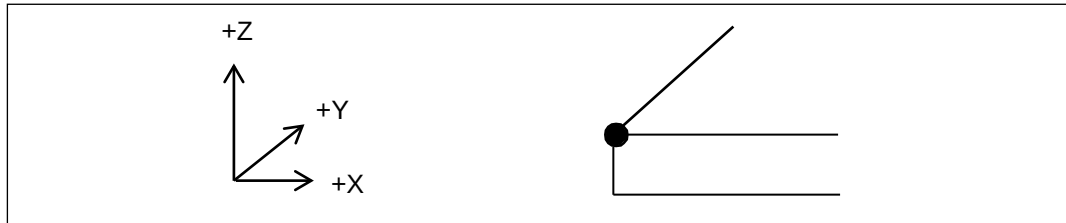
1 Functions

The automatic workpiece measurement function carries out automatic centering.

The following 5 shapes can be automatically centered.

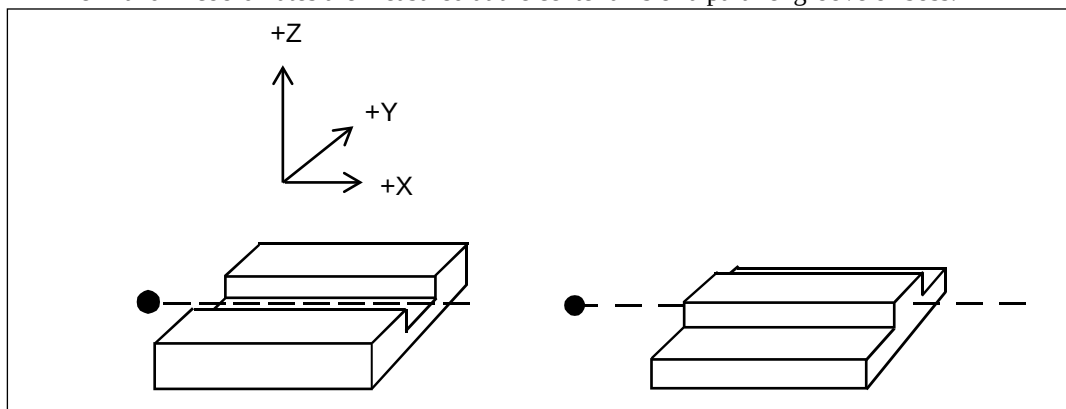
1. Corner

The X and Y coordinates on a corner are measured.



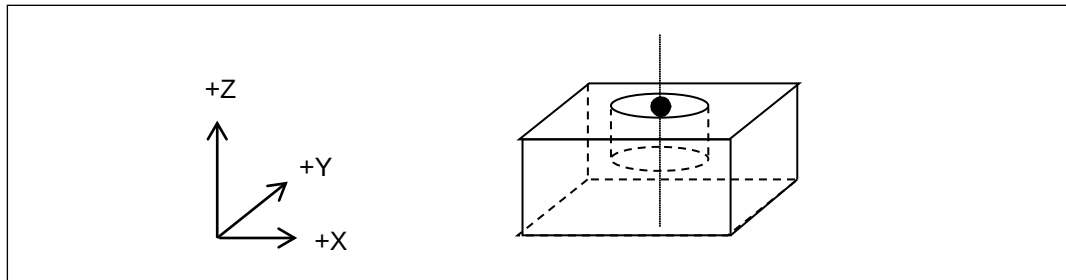
2. Parallel

The X and Y coordinates are measured at the center axis of a parallel groove or boss.



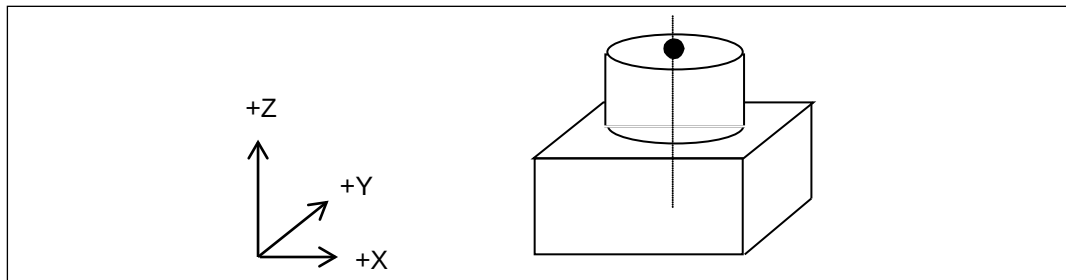
3. Circle center (hole)

The X and Y coordinates at the center of a hole are measured.

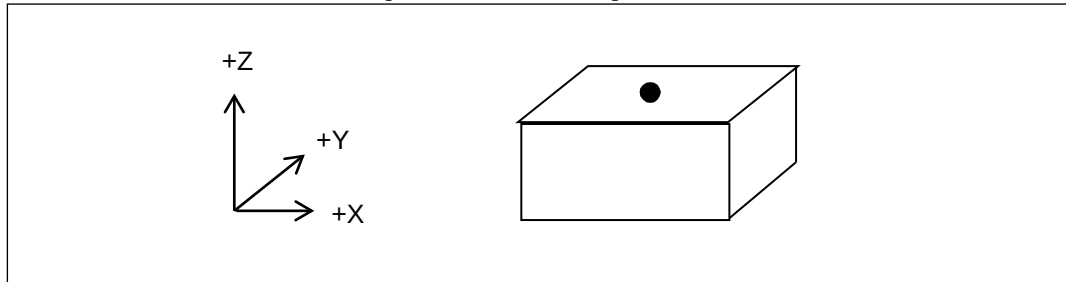


4. Circle center (boss)

The X and Y coordinates at the center of a circular boss are measured.



5. Z-axis height
The Z-axis coordinate on the top surface of the workpiece is measured.



6. Probe compensation (corner)
The compensation for the probe is measured when it is used to issue an automatic workpiece measurement command (G121 and G129).
7. Probe compensation (3 point circle)
The compensation for the probe is measured when it is used to issue an automatic workpiece measurement command (G124 and G125).

(Note 1) The measurement value is acquired by this function, and the true value varies depending on the delay that is unique to the probe. Therefore, contact the probe manufacturer and adjust it accordingly.

(Note 2) Make sure that no chips or shavings are stuck to the end of the measurement probe or on the measurement surface. In addition, make sure that there is no disturbance (caused by vibrations from outside of the machine) that adversely affects the machine. We cannot guarantee the measurement accuracy when there are chips or other factors that can cause inaccuracy.

2 Operation Procedure

2.1 Automatic Centering Screen

1. Press the **[MDI]** key to switch to the MDI operation screen.
2. Press the **[F0]** <Automatic centering> key to switch to the automatic centering screen.
3. Select the shape to be measured or probe compensation from the function keys on the automatic centering screen.

[F1]: Corner

[F2]: Parallel

[F3]: Circle (hole)

[F4]: Circle (boss)

[F5]: Z-axis height

[>>] (Next function) **[F1]:** Probe comp. (corner)

[>>] (Next function) **[F2]:** Probe comp. (3pt.circle)

Automatic centering screen (Ex: <Automatic centering (Corner)>)

Automatic centering(Corner)		2012/05/31 11:24:34	
Absolute coord. pos.	X 0.000 Y 0.000 Z 480.000	Guidance(Corner)	
Measurement result(Corner)	X Y	X Y S1 S2 S1 S2	
Measurement instructions(Corner)	X -32.094 Y -30.094 Direction +X X -26.746 Y -28.746 Direction -Y Height(H) 634.773 Return(R) 638.761	Z H R	
X mm			
[HELP]			
MDI operation	Corner	Parallel	Circle(hole)
		Circle(boss)	Z-axis height
		Automatic parameter setting	Capture position data

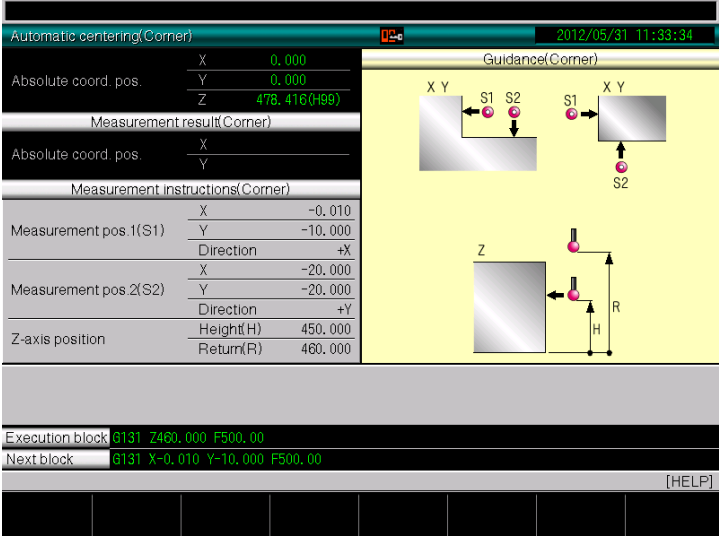
Automatic centering screen (Ex: <Automatic centering (Corner)> when the next function **[>>]** key is pressed)

Automatic centering(Corner)		2012/05/31 11:25:42	
Absolute coord. pos.	X 0.000 Y 0.000 Z 480.000	Guidance(Corner)	
Measurement result(Corner)	X Y	X Y S1 S2 S1 S2	
Measurement instructions(Corner)	X -32.094 Y -30.094 Direction +X X -26.746 Y -28.746 Direction -Y Height(H) 634.773 Return(R) 638.761	Z H R	
X mm			
[HELP]			
MDI operation	Probe comp. (corner)	Probe comp. (3pt.circle)	
		Automatic parameter setting	Capture position data

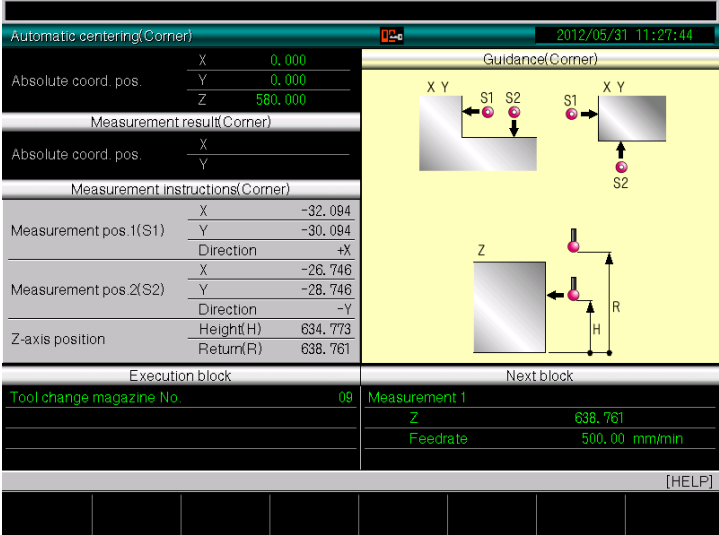
4. When the screen for setting the specified shape is displayed, set the measurement positions, the Z-axis position and other items*, and then press the **[START]** switch to execute automatic centering.

The following screen is displayed during operation. However, the screen display is different in the NC language mode and in conversation language mode.

(1) NC language



(2) Conversation language



Press the **[F0]** (MDI operation) key on the automatic centering screen to return to the MDI operation screen.

Setting items for the shapes and details of the centering operation are described in the following section.

[F6]: Automatic parameter setting

The **[F6]** (Automatic parameter setting) key is enabled for the probe compensation (corner and 3 point circle). After measurement, press the **[F6]** (Automatic parameter setting) key to set the probe compensation in the user parameter. If the key is pressed before measurement, the alarm <<No measurement data>> is triggered.

- * Press the **[F7]** (Capture position data) key, to automatically set the current absolute coordinates for the setting item where the cursor is. Press the **[F7]** (Capture position data) key to set the "X" and "Y" measurement position at the same time. "Height" and "Return height" for the Z-axis position are individually set.

- (Note 1) The mode cannot be changed during the automatic centering operation. (Including temporary stops or pauses and block stops)
After the [RST] key is pressed and the automatic centering operation has stopped, the mode can be changed.
- (Note 2) The automatic centering mode is enabled regardless of the modal status, except the status for G54 to G59 (workpiece coordinate selection).
- (Note 3) When in automatic centering mode and using the NC language, the tool length offset is automatically applied for the tool length on tool number 99. When using conversation language, the tool changes automatically to the centering tool registered to the ATC tools. (When there are multiple tools registered, the centering tool closest to the current position is selected.)
- (Note 4) If automatic centering is attempted during M142 modal on a machine equipped with a lathe spindle, the alarm <<Selection of a spindle is abnormal>> is triggered and the operation cannot be executed. Start operation after switching to M141 modal.

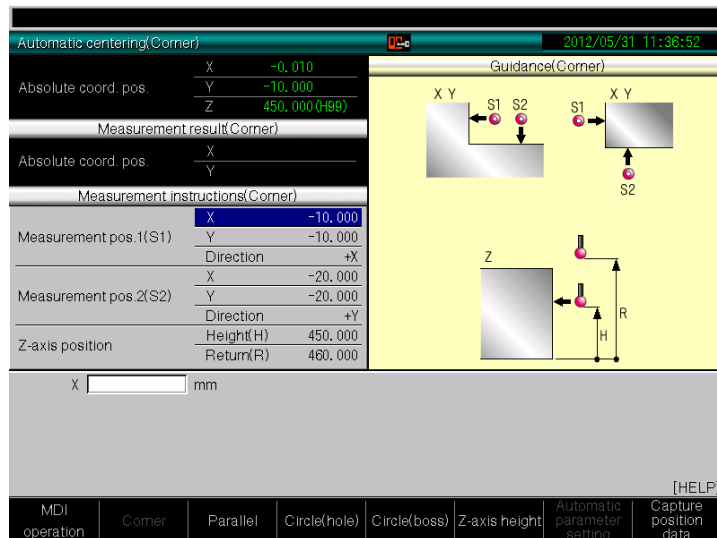
2.2 MDI Manual Mode

Press the [MANU] key with the [MDI] key held down on the automatic centering screen in order to enter MDI manual mode. In MDI manual mode, the [+X], [-X], [+Y], [-Y], [+Z], [-Z], [+4] and [-4] keys, or the handle operation is enabled. The current position can be captured at a given position using the [F7] (Capture position data) key.

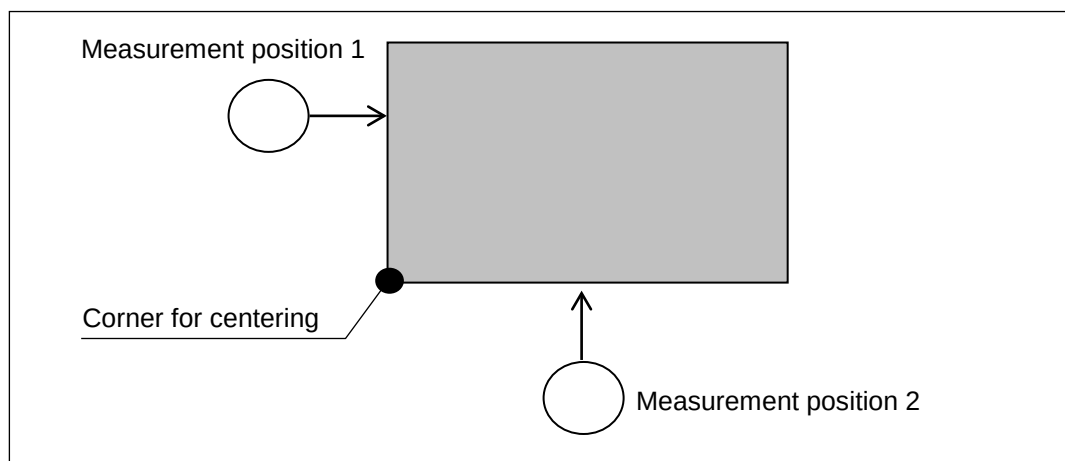
1. Press the [MANU] key with the [MDI] key held down on the automatic centering screen (LEDs on both keys light up).
2. Move the spindle to the measurement position or the Z-axis position using the [+X], [-X], [+Y], [-Y], [+Z] and [-Z] keys or the handle.
3. Move the cursor to the setting item and press the [F7] (Capture position data) key to automatically set the current absolute coordinates for the spindle. Press the [F7] (Capture position data) key to set the "X" and "Y" measurement positions at the same time. However, the "Height" and "Return height" for the Z-axis position are individually set.

To exit MDI manual mode, press the [MDI] key in MDI manual mode (the [+X], [-X], [+Y], [-Y], [+Z] and [-Z] keys or handle operation is disabled). The LED on the [MANU] key turns off.

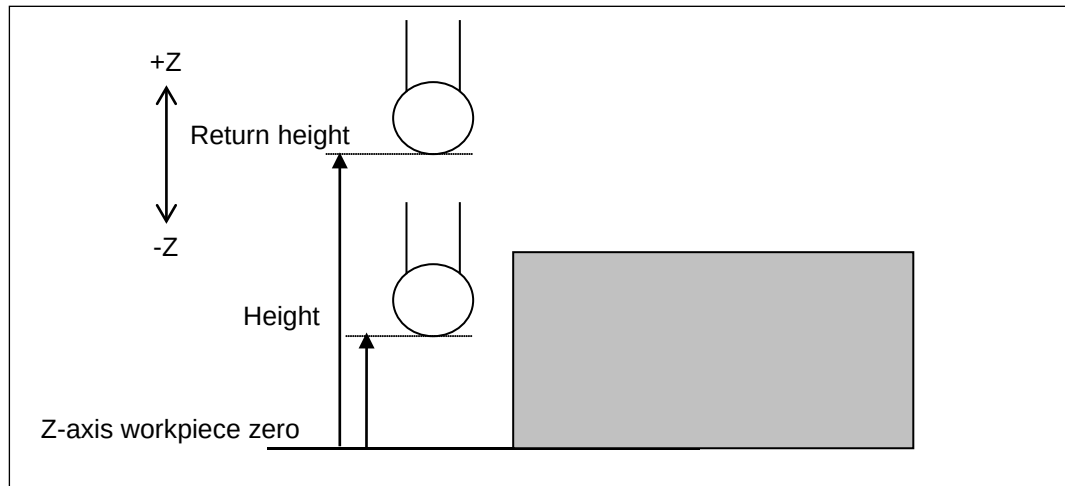
2.3 Corner: Measure X and Y Coordinates for Corner



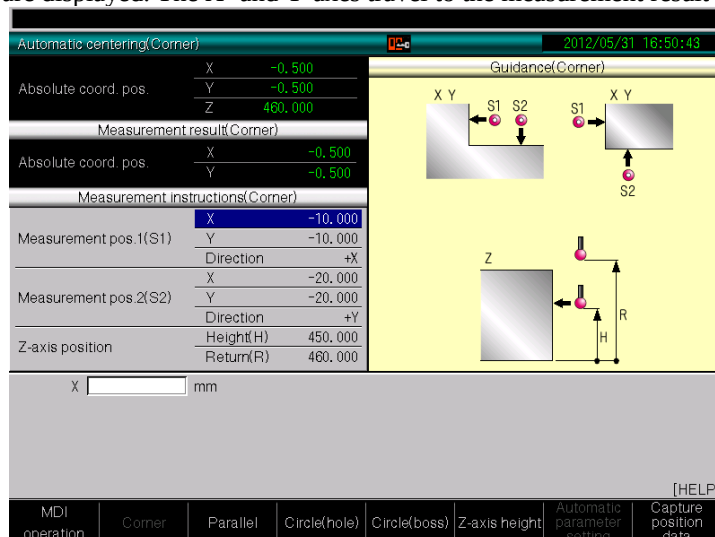
1. Make sure that parameter <Probe compensation 1> and <Probe compensation 2> are set.*
2. Press the **[F1]** (Corner) key on the automatic centering screen.
3. Set the values required for the measurement.
 - Measurement position 1 (X and Y)
Specify the start point for the centering operation using the X and Y absolute coordinates.
 - Measurement position 1 (Direction)
Specify the measurement direction (1: +X, 2: -X, 3: +Y and 4: -Y) from the position that is set in measurement position 1 (X and Y).
 - Measurement position 2 (X and Y)
Specify the start point for the centering operation using the X and Y absolute coordinates.
 - Measurement position 2 (Direction)
Specify the measurement direction (1: +X, 2: -X, 3: +Y and 4: -Y) from the position that is set in the measurement position (X and Y).



- Z-axis position (Height)
Specify the measurement height using the Z-axis absolute coordinates.
- Z-axis position (Return height)
Use the Z-axis absolute coordinates to specify the return height when moving from the measurement position 1 to measurement position 2 and the return height after the measurement is finished.



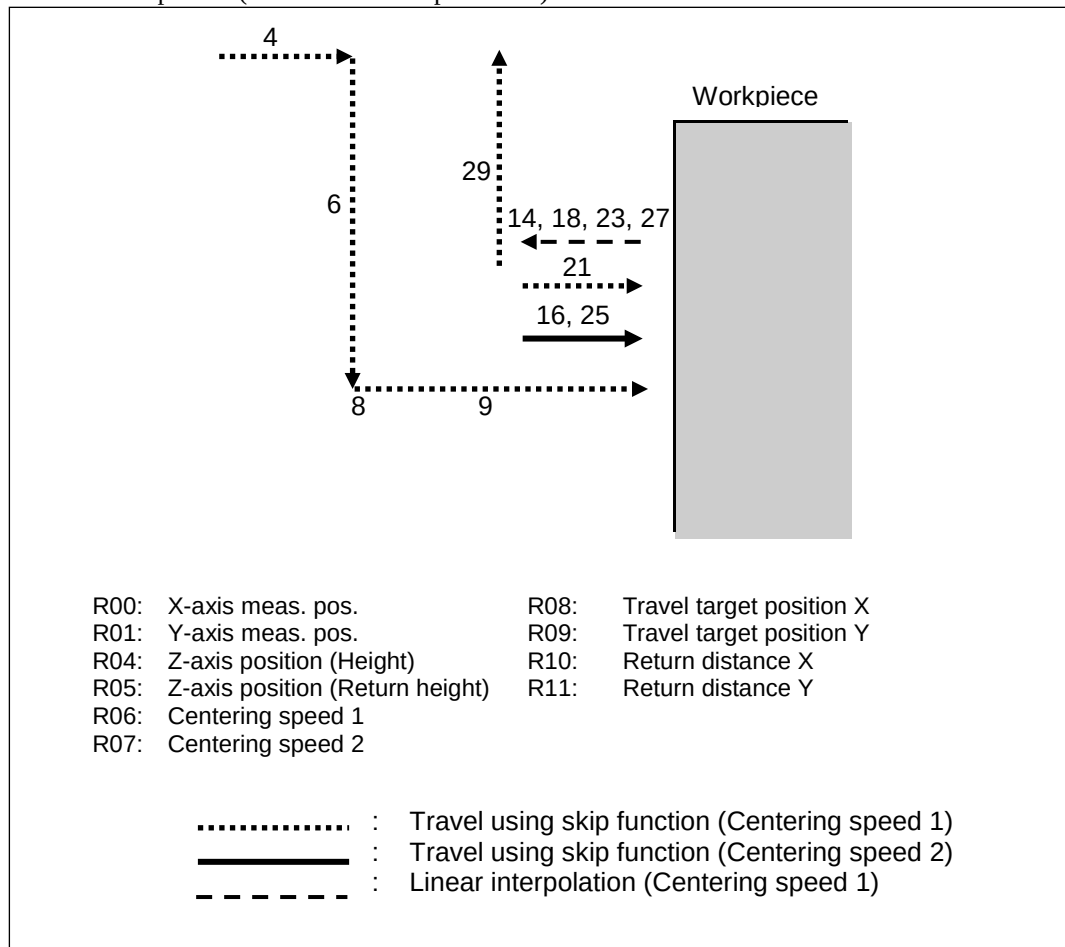
4. Press the **[START]** switch to execute the automatic centering operation. The automatic centering operation consists of the following four operations. The details of the operations are described in the following section.
 - (1) Execution of measurement start motion
 - (2) Execution of measurement pattern for measurement position 1
 - (3) Execution of measurement pattern for measurement position 2
 - (4) Travel to measurement result position
5. After the measurement is finished, the absolute coordinates for the measurement result position are displayed. The X- and Y-axes travel to the measurement result position.



- * Refer to "Chapter 12 (5) 2.9 Measure Probe Compensation (Corner)" for the parameter settings.

2.3.1 Operation Details

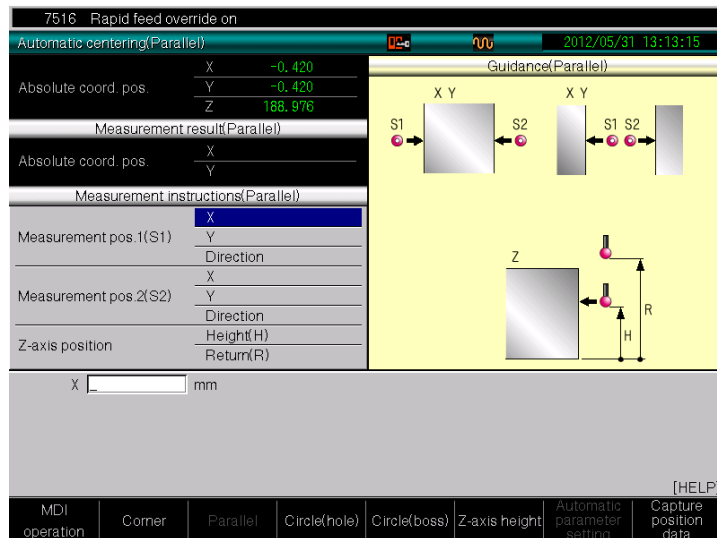
Measurement pattern (for measurement position 1)



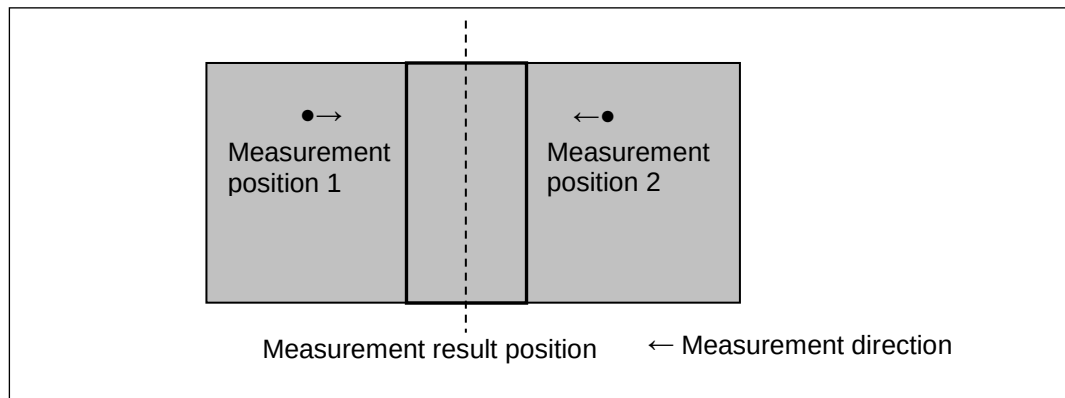
Step	NC language	Conversation language
(1) Execution of measurement start motion		
1		Machine zero return (Z)
2		Tool change pot No.xx
3	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(2) Execution of measurement pattern for measurement position 1		
4	G131 X R00 Y R01 F R06	Measurement 1 X R00 Y R01 Feed R06
5	M321	Detection signal OFF check
6	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
7	M321	Detection signal OFF check
8	M19	Spindle orientation (0°)
9	G131 X R08 Y R01 F R06	Measurement 1 X R08 Y R01 Feed R06
10	M320	Detection signal ON check
11	M310	Internal calculation 2
12	M302	Internal calculation 1
13	M302	Internal calculation 1
14	G01 X R10 Y R01 F R06	Cutting feed X R10 Y R01 Feed R06
15	M321	Detection signal OFF check
16	G131 X R08 Y R01 F R07	Measurement 1 X R08 Y R01 Feed R07
17	M310	Internal calculation 2
18	G01 X R10 Y R01 F R06	Cutting feed X R10 Y R01 Feed R06
19	M321	Detection signal OFF check
20	M111	Spindle orientation (180°)
21	G131 X R08 Y R01 F R06	Measurement 1 X R08 Y R01 Feed R06
22	M320	Detection signal ON check
23	G01 X R10 Y R01 F R06	Cutting feed X R10 Y R01 Feed R06
24	M321	Detection signal OFF check
25	G131 X R08 Y R01 F R07	Measurement 1 X R08 Y R01 Feed R07
26	M310	Internal calculation 2
27	G01 X R10 Y R01 F R06	Cutting feed X R10 Y R01 Feed R06

Step	NC language	Conversation language
28	M05	Spindle stop
29	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(3) Execution of measurement pattern for measurement position 2		
30	G131 X R00 Y R01 F R06	Measurement 1 X R00 Y R01 Feed R06
31	M321	Detection signal OFF check
32	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
33	M321	Detection signal OFF check
34	M19	Spindle orientation (0°)
35	G131 X R00 Y R09 F R06	Measurement 1 X R00 Y R09 Feed R06
36	M320	Detection signal ON check
37	M310	Internal calculation 2
38	M302	Internal calculation 1
39	M302	Internal calculation 1
40	G01 X R00 Y R11 F R06	Cutting feed X R00 Y R11 Feed R06
41	M321	Detection signal OFF check
42	G131 X R00 Y R09 F R07	Measurement 1 X R00 Y R09 Feed R07
43	M310	Internal calculation 2
44	G01 X R00 Y R11 F R06	Cutting feed X R00 Y R11 Feed R06
45	M321	Detection signal OFF check
46	M111	Spindle orientation (180°)
47	G131 X R00 Y R09 F R06	Measurement 1 X R00 Y R09 Feed R06
48	M320	Detection signal ON check
49	G01 X R00 Y R11 F R06	Cutting feed X R00 Y R11 Feed R06
50	M321	Detection signal OFF check
51	G131 X R00 Y R09 F R07	Measurement 1 X R00 Y R09 Feed R07
52	M310	Internal calculation 2
53	G01 X R00 Y R11 F R06	Cutting feed X R00 Y R11 Feed R06
54	M05	Spindle stop
55	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(4) Travel to measurement result position		
56	G131 X (Measurement result position) Y (Measurement result position) F R06	Measurement 1 X (Measurement result position) Y (Measurement result position) Feed R06
57	M321	Detection signal OFF check

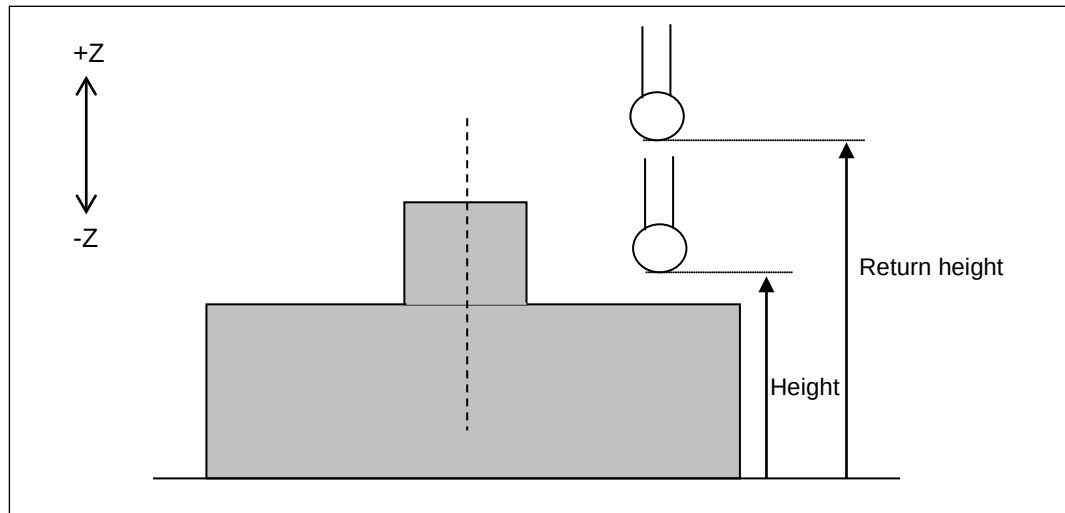
2.4 Parallel: Measure X and Y Coordinates at Center Axis of Parallel Groove or Boss



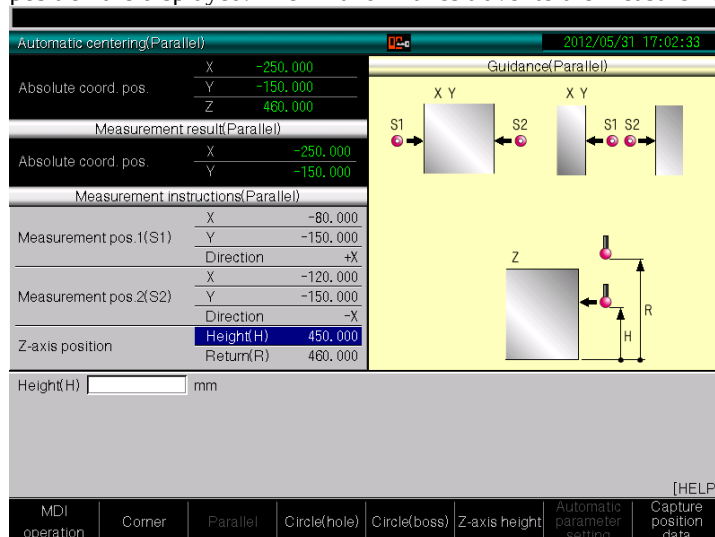
1. Press the **[F2]** (Parallel) key on the automatic centering screen.
2. Set the values required for the measurement.
 - Measurement position 1 (X and Y)
Specify the start point for the centering operation using the X and Y absolute coordinates.
 - Measurement position 1 (Direction)
Specify the measurement direction (1: +X, 2: -X, 3: +Y and 4: -Y) from the position that is set in measurement position 1 (X and Y).
 - Measurement position 2 (X and Y)
Specify the start point for the centering operation using the X and Y absolute coordinates.
 - Measurement position 2 (Direction)
Specify the measurement direction (1: +X, 2: -X, 3: +Y and 4: -Y) from the position that is set in measurement position 2 (X and Y).



- Z-axis position (Height)
Specify the measurement height using the Z-axis absolute coordinates.
- Z-axis position (Return height)
Use the Z-axis absolute coordinates to specify the return height when moving from the measurement position 1 to measurement position 2 and the return height after the measurement is finished.

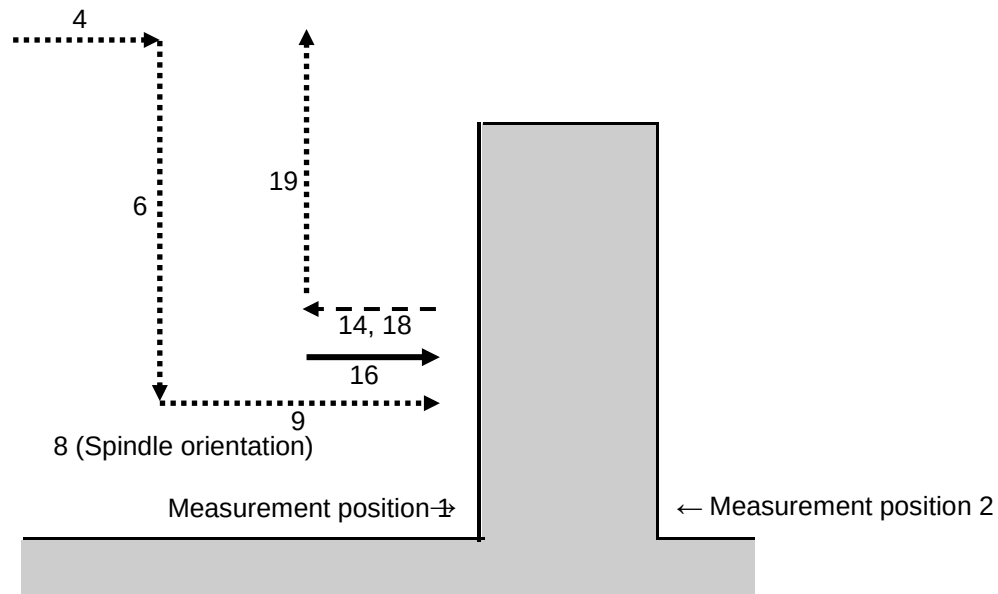


3. Press the **[START]** switch to execute the automatic centering operation. The automatic centering operation consists of the following four operations. The details of the operations are described in the following section.
 - (1) Execution of measurement start motion
 - (2) Execution of measurement pattern for measurement position 1
 - (3) Execution of measurement pattern for measurement position 2
 - (4) Travel to measurement result position
4. After the measurement is finished, the absolute coordinates for the measurement result position are displayed. The X- and Y-axes travel to the measurement result position.



2.4.1 Operation Details

Measurement pattern



R00: X-axis meas. pos.	R08: Target position X in positive direction
R01: Y-axis meas. pos.	R09: Target position Y in positive direction
R04: Z-axis position (Height)	R10: Return distance X in positive direction
R05: Z-axis position (Return height)	R11: Return distance Y in positive direction
R06: Centering speed 1	
R07: Centering speed 2	

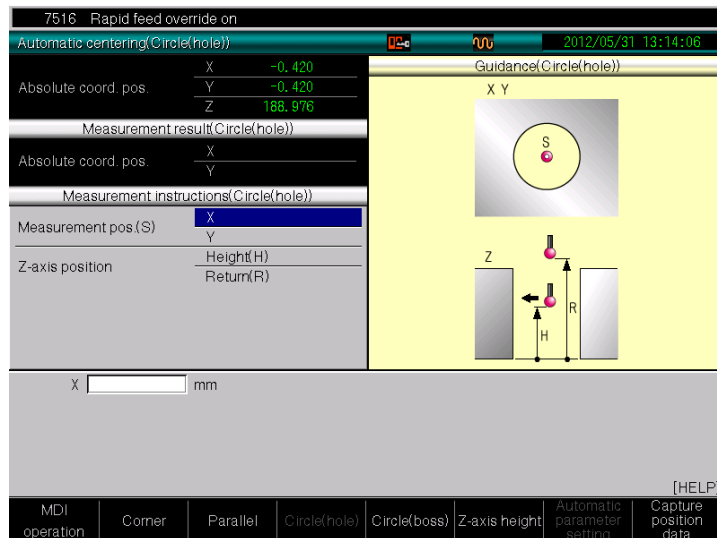
..... : Travel using skip function (Centering speed 1)
 ————— : Travel using skip function (Centering speed 2)
 - - - - - : Linear interpolation (Centering speed 1)

Step	NC language	Conversation language
(1) Execution of measurement start motion		
1		Machine zero return (Z)
2		Tool change pot No.xx
3	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(2) Execution of measurement pattern for measurement position 1		
4	G131 X R00 Y R01 F R06	Measurement 1 X R00 Y R01 Feed R06
5	M321	Detection signal OFF check
6	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
7	M321	Detection signal OFF check
8	M19	Spindle orientation (0°)
9	G131 X R08 Y R01 F R06	Measurement 1 X R08 Y R01 Feed R06
10	M320	Detection signal ON check
11	M310	Internal calculation 2
12	M302	Internal calculation 1
13	M302	Internal calculation 1
14	G01 X R10 Y R01 F R06	Cutting feed X R10 Y R01 Feed R06
15	M321	Detection signal OFF check
16	G131 X R08 Y R01 F R07	Measurement 1 X R08 Y R01 Feed R07
17	M310	Internal calculation 2
18	G01 X R10 Y R01 F R06	Cutting feed X R10 Y R01 Feed R06
19	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(3) Execution of measurement pattern for measurement position 2		
20	M321	Detection signal OFF check
21	G131 X R00 Y R01 F R06	Measurement 1 X R00 Y R01 Feed R06
22	M321	Detection signal OFF check
23	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06

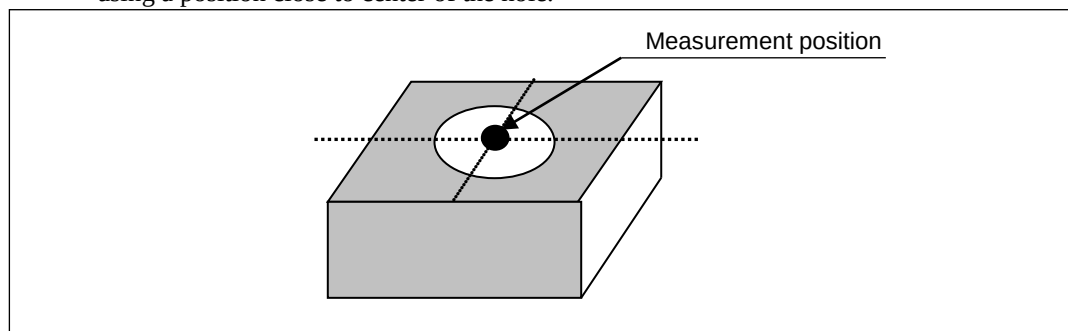
Step	NC language	Conversation language
24	M321	Detection signal OFF check
25	M111	Spindle orientation (180°)
26	G131 X R08 Y R01 F R06	Measurement 1 X R08 Y R01 Feed R06
27	M320	Detection signal ON check
28	M310	Internal calculation 2
29	M302	Internal calculation 1
30	M302	Internal calculation 1
31	G01 X R10 Y R01 F R06	Cutting feed X R10 Y R01 Feed R06
32	M321	Detection signal OFF check
33	G131 X R08 Y R01 F R07	Measurement 1 X R08 Y R01 Feed R07
34	M310	Internal calculation 2
35	G01 X R10 Y R01 F R06	Cutting feed X R10 Y R01 Feed R06
36	M05	Spindle stop
37	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(4) Travel to measurement result position		
38	G131 X (Measurement result position) Y (Measurement result position) F R06	Measurement 1 X (Measurement result position) Y (Measurement result position) Feed R06
39	M321	Detection signal OFF check

When the target position for the skip travel is in the “+X” direction or the “+Y” direction, M19 (Spindle orientation 0°) is used. When the target position for the skip travel is in the “-X” direction or the “-Y” direction, M111 (Spindle orientation 180°) is used.

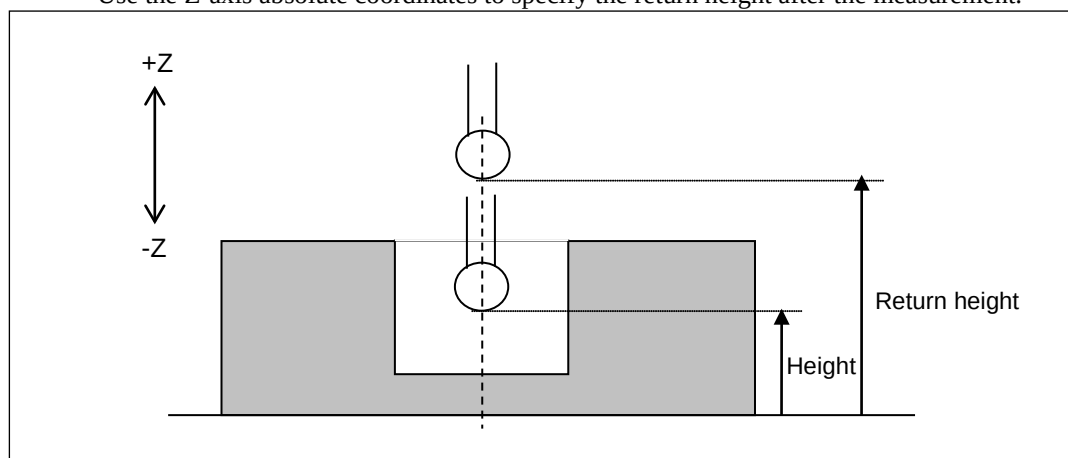
2.5 Circle (Hole): Measure X and Y Coordinates for Hole Center



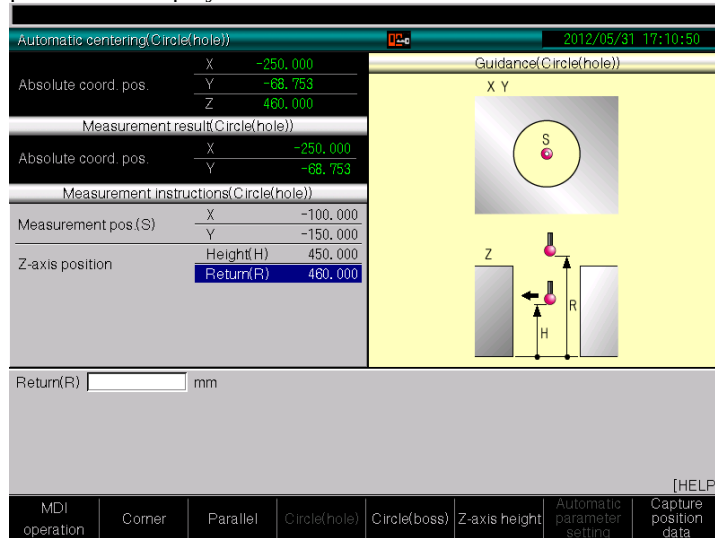
1. Press the **[F3]** (Circle (Hole)) key on the automatic centering screen.
2. Set the values required for the measurement.
 - Measurement position (X and Y)
Use the X and Y absolute coordinates to specify the start point for the centering operation using a position close to center of the hole.



- Z-axis position (Height)
Specify the measurement height using the Z-axis absolute coordinates.
- Z-axis position (Return height)
Use the Z-axis absolute coordinates to specify the return height after the measurement.

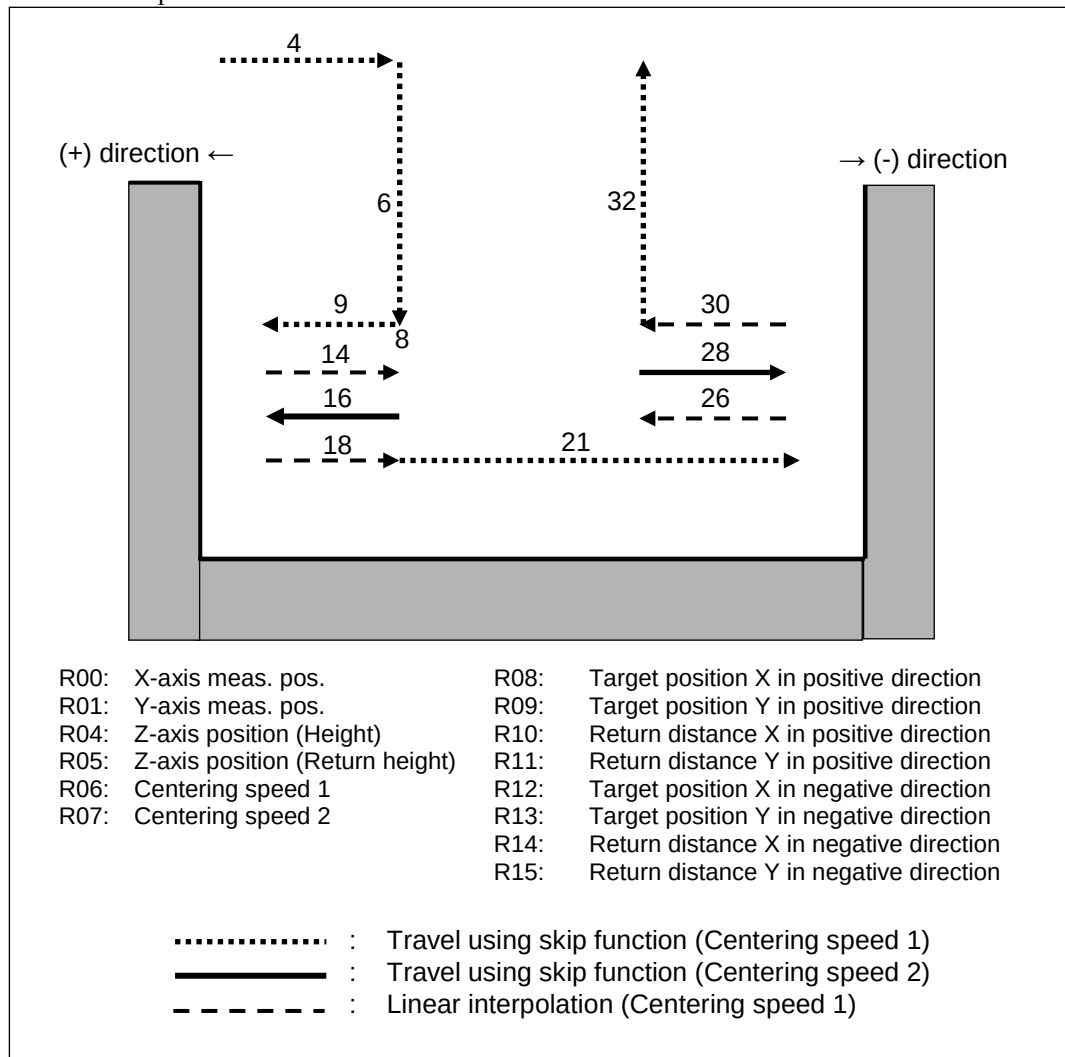


3. Press the **[START]** switch to execute the automatic centering operation. The automatic centering operation consists of the following five operations. The details of the operations are described in the following section.
 - (1) Execution of measurement start motion
 - (2) Execution of measurement pattern in X-axis direction for measurement position
 - (3) Execution of measurement pattern in Y-axis direction for position measured in (2)
 - (4) Execution of measurement pattern in X-axis direction for position measured in (3)
 - (5) Travel to measurement result position
4. After the measurement is finished, the absolute coordinates for the measurement result position are displayed. The X- and Y-axes travel to the measurement result position.



2.5.1 Operation Details

Measurement pattern

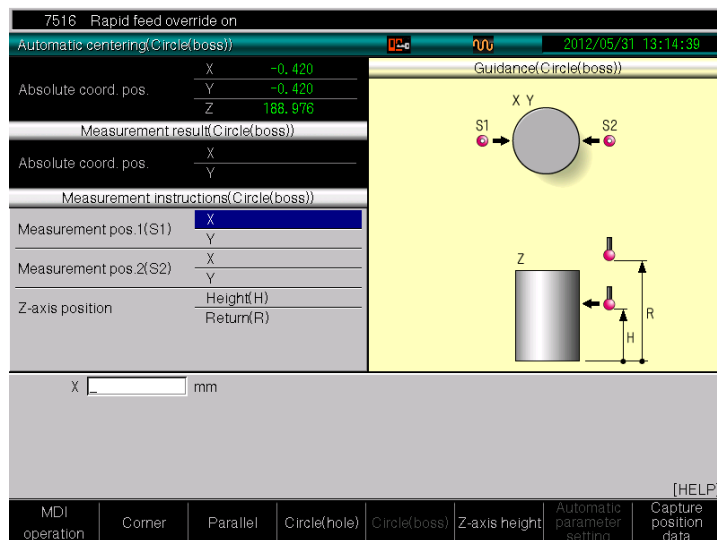


Step	NC language	Conversation language
(1) Execution of measurement start motion		
1		Machine zero return (Z)
2		Tool change pot No.xx
3	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(2) Execution of measurement pattern in X-axis direction for measurement position		
4	G131 X R00 Y R01 F R06	Measurement 1 X R00 Y R01 Feed R06
5	M321	Detection signal OFF check
6	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
7	M321	Detection signal OFF check
8	M19	Spindle orientation (0°)
9	G131 X R08 Y R01 F R06	Measurement 1 X R08 Y R01 Feed R06
10	M320	Detection signal ON check
11	M310	Internal calculation 2
12	M302	Internal calculation 1
13	M302	Internal calculation 1
14	G01 X R10 Y R01 F R06	Cutting feed X R10 Y R01 Feed R06
15	M321	Detection signal OFF check
16	G131 X R08 Y R01 F R07	Measurement 1 X R08 Y R01 Feed R07
17	M310	Internal calculation 2
18	G01 X R10 Y R01 F R06	Cutting feed X R10 Y R01 Feed R06
19	M321	Detection signal OFF check
20	M111	Spindle orientation (180°)
21	G131 X R12 Y R01 F R06	Measurement 1 X R12 Y R01 Feed R06
22	M320	Detection signal ON check
23	M310	Internal calculation 2

Step	NC language	Conversation language
24	M302	Internal calculation 1
25	M302	Internal calculation 1
26	G01 X R14 Y R01 F R06	Cutting feed X R14 Y R01 Feed R06
27	M321	Detection signal OFF check
28	G131 X R12 Y R01 F R07	Measurement 1 X R12 Y R01 Feed R07
29	M310	Internal calculation 2
30	G01 X R14 Y R01 F R06	Cutting feed X R14 Y R01 Feed R06
31	M05	Spindle stop
32	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(3) Execution of measurement pattern in Y-axis direction for position measured in (2) (measurement position 1)		
33	G131 X (Measurement result 1) Y R01 F R06	Measurement 1 X (Measurement result 1) Y R01 Feed R06
34	M321	Detection signal OFF check
35	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
36	M321	Detection signal OFF check
37	M19	Spindle orientation (0°)
38	G131 X (Measurement result 1) Y R09 F R06	Measurement 1 X (Measurement result 1) Y R09 Feed R06
39	M320	Detection signal ON check
40	M310	Internal calculation 2
41	M302	Internal calculation 1
42	M302	Internal calculation 1
43	G01 X (Measurement result 1) Y R11 F R06	Cutting feed X (Measurement result 1) Y R11 Feed R06
44	M321	Detection signal OFF check
45	G131 X (Measurement result 1) Y R09 F R07	Measurement 1 X (Measurement result 1) Y R09 Feed R07
46	M310	Internal calculation 2
47	G01 X (Measurement result 1) Y R11 F R06	Cutting feed X (Measurement result 1) Y R11 Feed R06
48	M321	Detection signal OFF check
49	M111	Spindle orientation (180°)
50	G131 X (Measurement result 1) Y R13 F R06	Measurement 1 X (Measurement result 1) Y R13 Feed R06
51	M320	Detection signal ON check
52	M310	Internal calculation 2
53	M302	Internal calculation 1
54	M302	Internal calculation 1
55	G01 X (Measurement result 1) Y R15 F R06	Cutting feed X (Measurement result 1) Y R15 Feed R06
56	M321	Detection signal OFF check
57	G131 X (Measurement result 1) Y R13 F R07	Measurement 1 X (Measurement result 1) Y R13 Feed R07
58	M310	Internal calculation 2
59	G01 X (Measurement result 1) Y R15 F R06	Cutting feed X (Measurement result 1) Y R15 Feed R06
60	M05	Spindle stop
61	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(4) Execution of measurement pattern in X-axis direction for position measured in (3) (measurement position 2)		
62	G131 X (Measurement result 1) Y (Measurement result 2) F R06	Measurement 1 X (Measurement result 1) Y (Measurement result 2) Feed R06
63	M321	Detection signal OFF check
64	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
65	M321	Detection signal OFF check
66	M19	Spindle orientation (0°)
67	G131 X R08 Y (Measurement result 2) F R06	Measurement 1 X R08 Y (Measurement result 2) Feed R06
68	M320	Detection signal ON check
69	M310	Internal calculation 2
70	M302	Internal calculation 1
71	M302	Internal calculation 1
72	G01 X R10 Y (Measurement result 2) F R06	Cutting feed X R10 Y (Measurement result 2) Feed R06
73	M321	Detection signal OFF check

Step	NC language	Conversation language
74	G131 X R08 Y (Measurement result 2) F R07	Measurement 1 X R08 Y (Measurement result 2) Feed R07
75	M310	Internal calculation 2
76	G01 X R10 Y (Measurement result 2) F R06	Cutting feed X R10 Y (Measurement result 2) Feed R06
77	M321	Detection signal OFF check
78	M111	Spindle orientation (180°)
79	G131 X R12 Y (Measurement result 2) F R06	Measurement 1 X R12 Y (Measurement result 2) Feed R06
80	M320	Detection signal ON check
81	M310	Internal calculation 2
82	M302	Internal calculation 1
83	M302	Internal calculation 1
84	G01 X R14 Y (Measurement result 2) F R06	Cutting feed X R14 Y (Measurement result 2) Feed R06
85	M321	Detection signal OFF check
86	G131 X R12 Y (Measurement result 2) F R07	Measurement 1 X R12 Y (Measurement result 2) Feed R07
87	M310	Internal calculation 2
88	G01 X R14 Y (Measurement result 2) F R06	Cutting feed X R14 Y (Measurement result 2) Feed R06
89	M05	Spindle stop
90	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(5) Travel to measurement result position		
91	G131 X (Measurement result 1) Y (Measurement result 2) F R06	Measurement 1 X (Measurement result 1) Y (Measurement result 2) Feed R06
92	M321	Detection signal OFF check

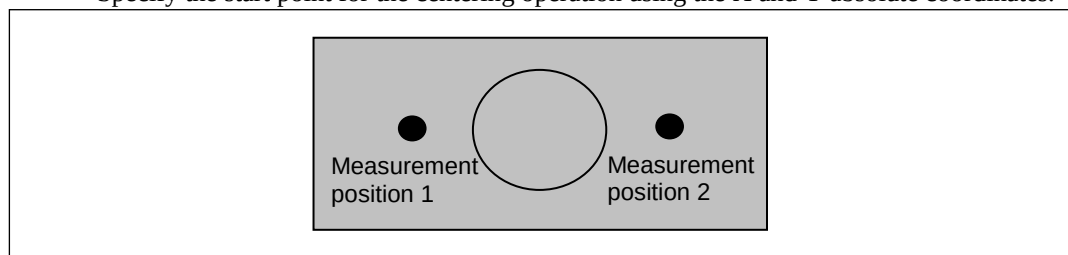
2.6 Circle (Boss): Measure X and Y Coordinates for Circular Boss Center



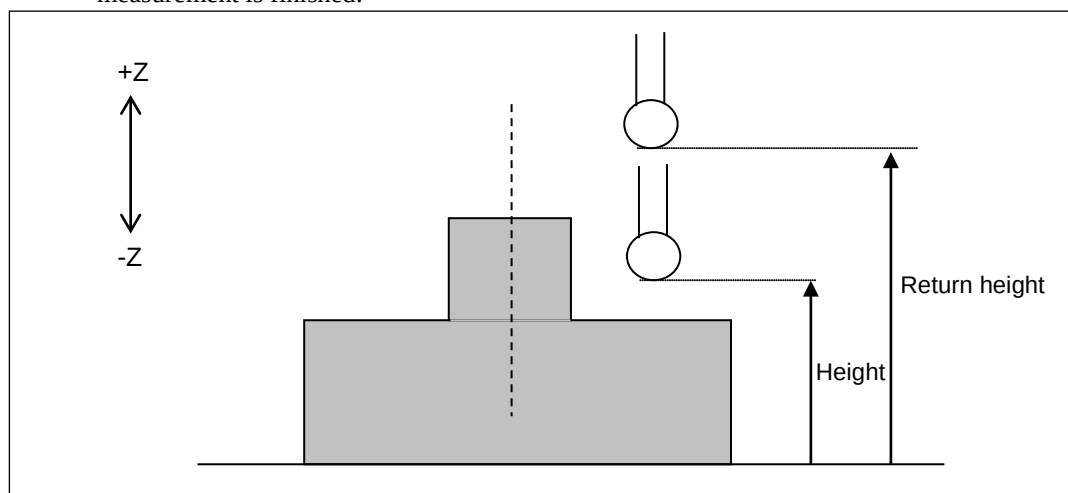
1. Press the **[F4]** (Circle (Boss)) key on the automatic centering screen.
2. Set the values required for the measurement.

- Measurement position 1 (X and Y)
- Measurement position 2 (X and Y)

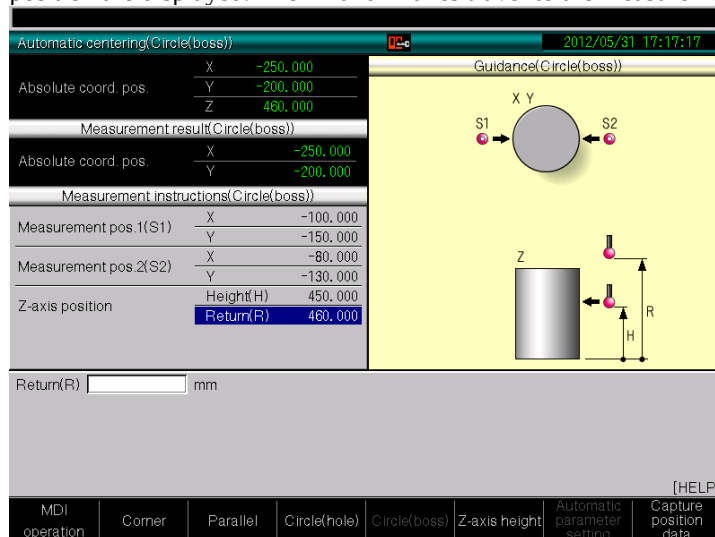
Specify the start point for the centering operation using the X and Y absolute coordinates.



- Z-axis position (Height)
Specify the measurement height using the Z-axis absolute coordinates.
- Z-axis position (Return height)
Use the Z-axis absolute coordinates to specify the return height when moving from the measurement position 1 to measurement position 2 and the return height after the measurement is finished.

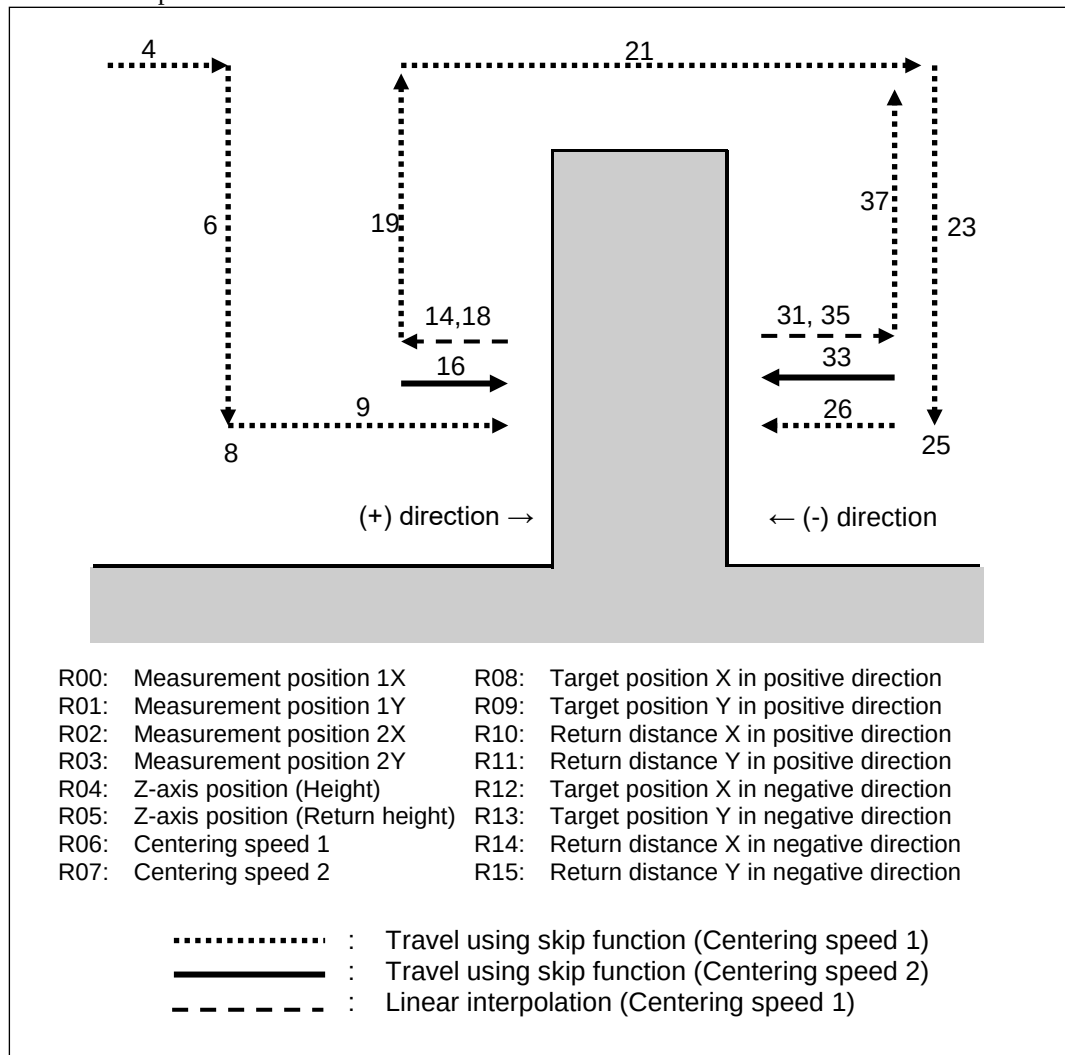


3. Press the **[START]** switch to execute the automatic centering operation. The automatic centering operation consists of the following five operations. The details of the operations are described in the following section.
 - (1) Execution of measurement start motion
 - Converts measurement position 1 and 2 into data for measuring the circle center
 - Measurement position 1 = (X1 and Y1) and measurement position 2 = (X2 and Y2)
 - Conversion position 1 = (min (X1 and X2) and (Y1+Y2)/2)
 - Conversion position 2 = (max (X1 and X2) and (Y1+Y2)/2)
 - Conversion position 3 = (3 X-axis meas. pos. and (Y1+Y2)/2 - (Conversion position 2X - Conversion position 1X)/2)
 - Conversion position 4 = (3 X-axis meas. pos. and (Y1+Y2)/2 + (Conversion position 2X - Conversion position 1X)/2)
 - Conversion position 5 = (Conversion position 1X and 4 Y-axis meas. pos.)
 - Conversion position 6 = (Conversion position 2X and 4 Y-axis meas. pos.)
 - (2) Execution of measurement pattern in X-axis direction for conversion positions 1 and 2
 - (3) Execution of measurement pattern in Y-axis direction for conversion positions 3 and 4
 - (4) Execution of measurement pattern in X-axis direction for conversion positions 5 and 6
 - (5) Travel to measurement result position
4. After the measurement is finished, the absolute coordinates for the measurement result position are displayed. The X- and Y-axes travel to the measurement result position.



2.6.1 Operation Details

Measurement pattern

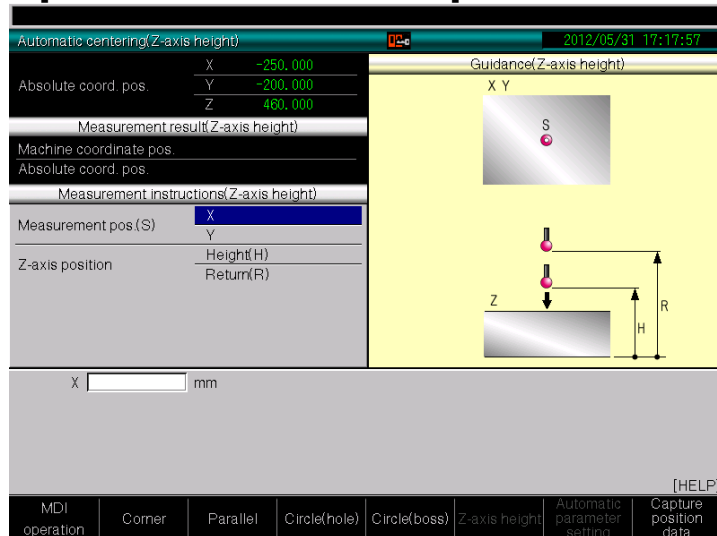


Step	NC language	Conversation language
(1) Execution of measurement start motion		
1		Machine zero return (Z)
2		Tool change pot No.xx
3	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(2) Execution of measurement pattern in X-axis direction for conversion positions 1 and 2		
4	G131 X R00 Y (R01+R03)/2 F R06	Measurement 1 X R00 Y (R01+R03)/2 Feed R06
5	M321	Detection signal OFF check
6	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
7	M321	Detection signal OFF check
8	M19	Spindle orientation (0°)
9	G131 X R08 Y (R01+R03)/2 F R06	Measurement 1 X R08 Y (R01+R03)/2 Feed R06
10	M320	Detection signal ON check
11	M310	Internal calculation 2
12	M302	Internal calculation 1
13	M302	Internal calculation 1
14	G01 X R10 Y (R01+R03)/2 F R06	Cutting feed X R10 Y (R01+R03)/2 Feed R06
15	M321	Detection signal OFF check
16	G131 X R08 Y (R01+R03)/2 F R07	Measurement 1 X R08 Y (R01+R03)/2 Feed R07
17	M310	Internal calculation 2
18	G01 X R10 Y (R01+R03)/2 F R06	Cutting feed X R10 Y (R01+R03)/2 Feed R06
19	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
20	M321	Detection signal OFF check

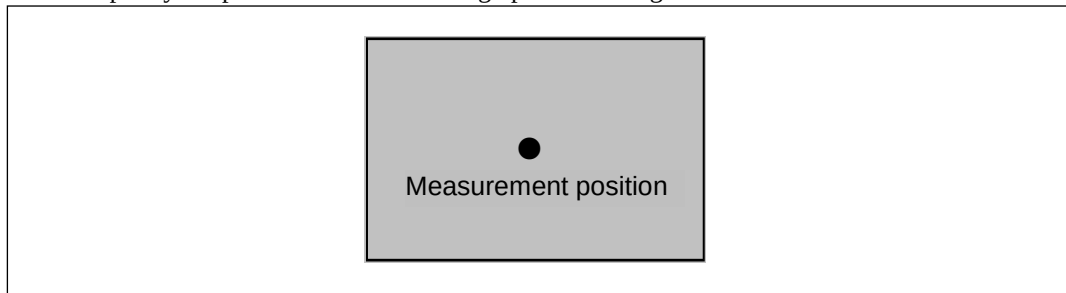
Step	NC language	Conversation language
21	G131 X R02 Y (R01+R03)/2 F R06	Measurement 1 X R02 Y (R01+R03)/2 Feed R06
22	M321	Detection signal OFF check
23	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
24	M321	Detection signal OFF check
25	M111	Spindle orientation (180°)
26	G131 X R12 Y (R01+R03)/2 F R06	Measurement 1 X R12 Y (R01+R03)/2 Feed R06
27	M320	Detection signal ON check
28	M310	Internal calculation 2
29	M302	Internal calculation 1
30	M302	Internal calculation 1
31	G01 X R14 Y (R01+R03)/2 F R06	Cutting feed X R14 Y (R01+R03)/2 Feed R06
32	M321	Detection signal OFF check
33	G131 X R12 Y (R01+R03)/2 F R07	Measurement 1 X R12 Y (R01+R03)/2 Feed R07
34	M310	Internal calculation 2
35	G01 X R14 Y (R01+R03)/2 F R06	Cutting feed X R14 Y (R01+R03)/2 Feed R06
36	M05	Spindle stop
37	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(3) Execution of measurement pattern in Y-axis direction for results (measurement position 1) measured in (2) and conversion positions 3 and 4		
38	G131 X (Measurement result 1) Y R01 F R06	Measurement 1 X (Measurement result 1) Y R01 Feed R06
39	M321	Detection signal OFF check
40	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
41	M321	Detection signal OFF check
42	M19	Spindle orientation (0°)
43	G131 X (Measurement result 1) Y R09 F R06	Measurement 1 X (Measurement result 1) Y R09 Feed R06
44	M320	Detection signal ON check
45	M310	Internal calculation 2
46	M302	Internal calculation 1
47	M302	Internal calculation 1
48	G01 X (Measurement result 1) Y R11 F R06	Cutting feed X (Measurement result 1) Y R11 Feed R06
49	M321	Detection signal OFF check
50	G131 X (Measurement result 1) Y R09 F R07	Measurement 1 X (Measurement result 1) Y R09 Feed R07
51	M310	Internal calculation 2
52	G01 X (Measurement result 1) Y R11 F R06	Cutting feed X (Measurement result 1) Y R11 Feed R06
53	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
54	M321	Detection signal OFF check
55	G131 X (Measurement result 1) Y R03 F R06	Measurement 1 X (Measurement result 1) Y R03 Feed R06
56	M321	Detection signal OFF check
57	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
58	M321	Detection signal OFF check
59	M111	Spindle orientation (180°)
60	G131 X (Measurement result 1) Y R13 F R06	Measurement 1 X (Measurement result 1) Y R13 Feed R06
61	M320	Detection signal ON check
62	M310	Internal calculation 2
63	M302	Internal calculation 1
64	M302	Internal calculation 1
65	G01 X (Measurement result 1) Y R15 F R06	Cutting feed X (Measurement result 1) Y R15 Feed R06
66	M321	Detection signal OFF check
67	G131 X (Measurement result 1) Y R13 F R07	Measurement 1 X (Measurement result 1) Y R13 Feed R07
68	M310	Internal calculation 2
69	G01 X (Measurement result 1) Y R15 F R06	Cutting feed X (Measurement result 1) Y R15 Feed R06
70	M05	Spindle stop
71	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06

Step	NC language	Conversation language
(4) Execution of measurement pattern in X-axis direction for results (measurement position 2) measured in (3) and conversion positions 5 and 6		
72	G131 X R00 Y (Measurement result 2) F R06	Measurement 1 X R00 Y (Measurement result 2) Feed R06
73	M321	Detection signal OFF check
74	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
75	M321	Detection signal OFF check
76	M19	Spindle orientation (0°)
77	G131 X R08 Y (Measurement result 2) F R06	Measurement 1 X R08 Y (Measurement result 2) Feed R06
78	M320	Detection signal ON check
79	M310	Internal calculation 2
80	M302	Internal calculation 1
81	M302	Internal calculation 1
82	G01 X R10 Y (Measurement result 2) F R06	Cutting feed X R10 Y (Measurement result 2) Feed R06
83	M321	Detection signal OFF check
84	G131 X R08 Y (Measurement result 2) F R07	Measurement 1 X R08 Y (Measurement result 2) Feed R07
85	M310	Internal calculation 2
86	G01 X R10 Y (Measurement result 2) F R06	Cutting feed X R10 Y (Measurement result 2) Feed R06
87	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
88	M321	Detection signal OFF check
89	G131 X R02 Y (Measurement result 2) F R06	Measurement 1 X R02 Y (Measurement result 2) Feed R06
90	M321	Detection signal OFF check
91	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
92	M321	Detection signal OFF check
93	M111	Spindle orientation (180°)
94	G131 X R12 Y (Measurement result 2) F R06	Measurement 1 X R12 Y (Measurement result 2) Feed R06
95	M320	Detection signal ON check
96	M310	Internal calculation 2
97	M302	Internal calculation 1
98	M302	Internal calculation 1
99	G01 X R14 Y (Measurement result 2) F R06	Cutting feed X R14 Y (Measurement result 2) Feed R06
100	M321	Detection signal OFF check
101	G131 X R12 Y (Measurement result 2) F R07	Measurement 1 X R12 Y (Measurement result 2) Feed R07
102	M310	Internal calculation 2
103	G01 X R14 Y (Measurement result 2) F R06	Cutting feed X R14 Y (Measurement result 2) Feed R06
104	M05	Spindle stop
105	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(5) Travel to measurement result position		
106	G131 X (Measurement result 3) Y (Measurement result 2) F R06	Measurement 1 X (Measurement result 3) Y (Measurement result 2) Feed R06
107	M321	Detection signal OFF check

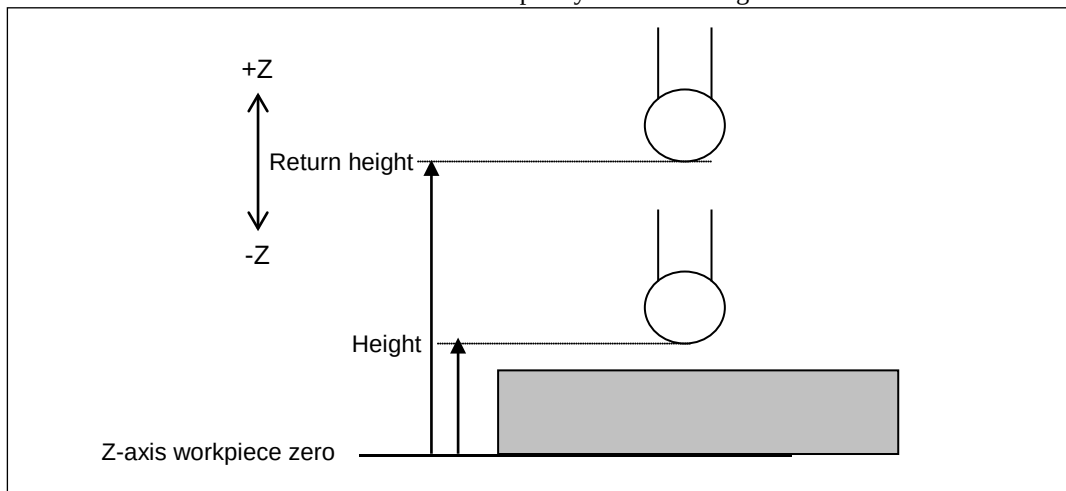
2.7 Z-axis Height: Measure Z-axis Coordinate on Top Surface of Workpiece



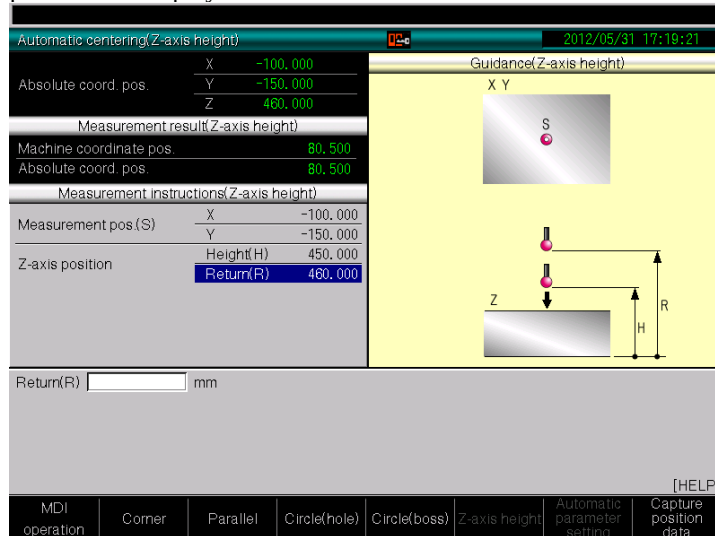
1. Press the **[F5]** (Z-axis height) key on the automatic centering screen.
2. Set the values required for the measurement.
 - Measurement position (X and Y)
Specify the position for the centering operation using the X and Y absolute coordinates.



- Z-axis position (Height)
Specify the height for starting the centering operation using the Z-axis absolute coordinate.
- Z-axis position (Return height)
Use the Z-axis absolute coordinates to specify the return height after the measurement.

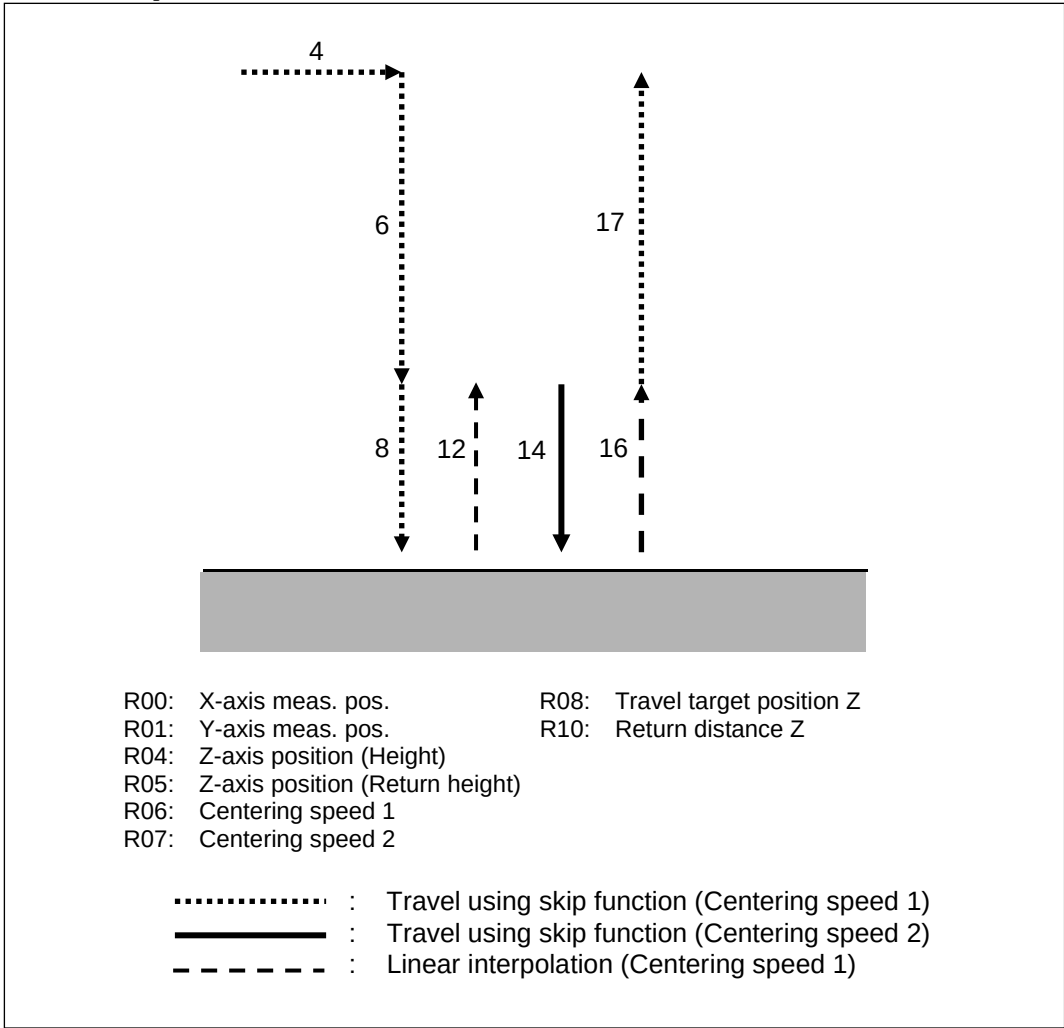


3. Press the **[START]** switch to execute the automatic centering operation. The automatic centering operation consists of the following two operations. The details of the operations are described in the following section.
 - (1) Execution of measurement start motion
 - (2) Execution of measurement pattern for the measurement position
4. After the measurement is finished, the absolute coordinates for the measurement result position are displayed.



2.7.1 Operation Details

Measurement pattern



Step	NC language	Conversation language
(1) Execution of measurement start motion		
1		Machine zero return (Z)
2		Tool change pot No.xx
3	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(2) Execution of measurement pattern for the measurement position		
4	G131 X R00 Y R01 F R06	Measurement 1 X R00 Y R01 Feed R06
5	M321	Detection signal OFF check
6	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
7	M321	Detection signal OFF check
8	G131 Z R08 F R06	Measurement 1 Z R08 Feed R06
9	M320	Detection signal ON check
10	M311	Internal calculation 3
11	M302	Internal calculation 1
12	G01 Z R10 F R06	Cutting feed Z R10 Feed R06
13	M321	Detection signal OFF check
14	G131 Z R08 F R07	Measurement 1 Z R08 Feed R07
15	M311	Internal calculation 3
16	G01 Z R10 F R06	Cutting feed Z R10 Feed R06
17	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06

2.8 Probe Compensation

The compensation for the probe is a setting requirement in the automatic measurement menu. If the probe compensation is not set, it causes a margin of error in the measurement results.

The probe compensation does not need to be set for the following items because it is not required.

- Automatic centering <Parallel>, <Circle (Hole)>, <Circle (Boss)> and <Z-axis height>
- Automatic measurements G122, G123, G126, G127 and G128

Items that require probe compensation 1 and 2

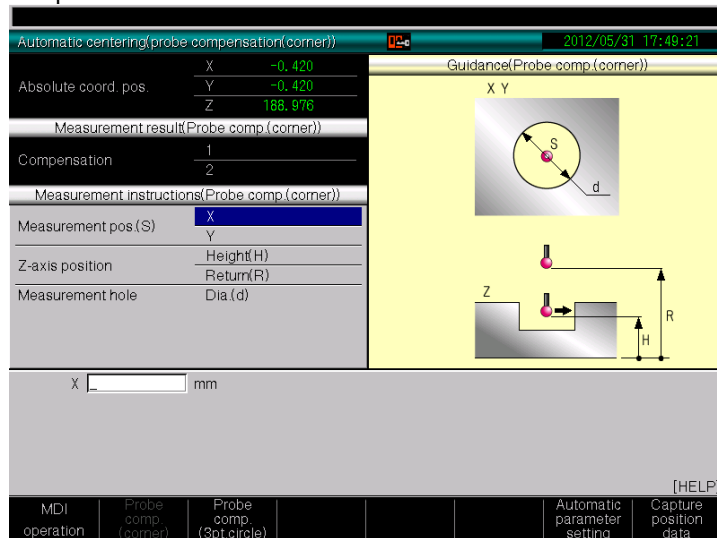
- Automatic measurements G121 and G129

Items that require probe compensation 3 and 4

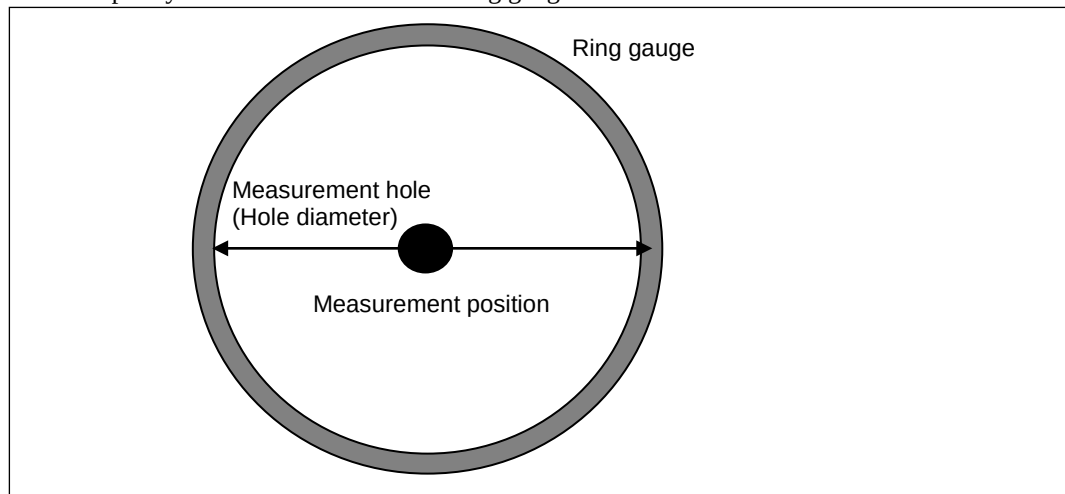
- Automatic measurements G124 and G125

2.9 Measure Probe Compensation (Corner)

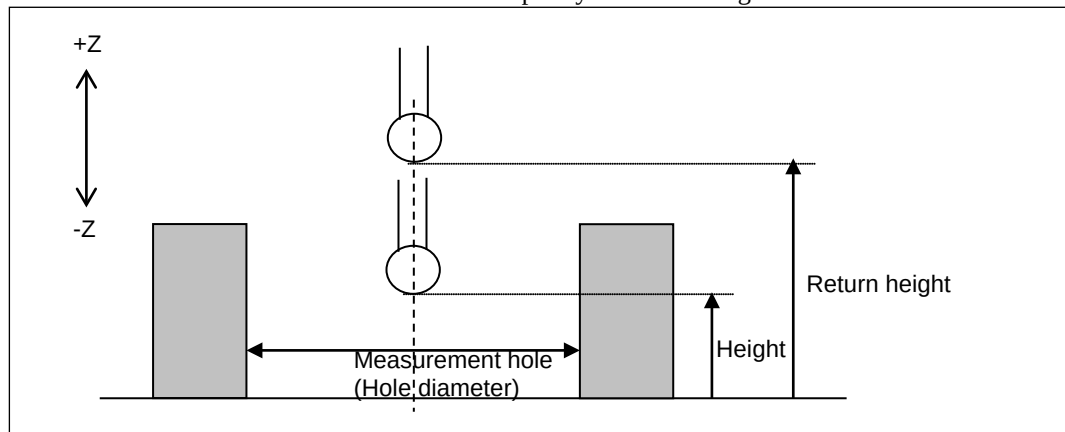
Measure the values that are set to the parameters <Probe compensation 1> and <Probe compensation 2>.



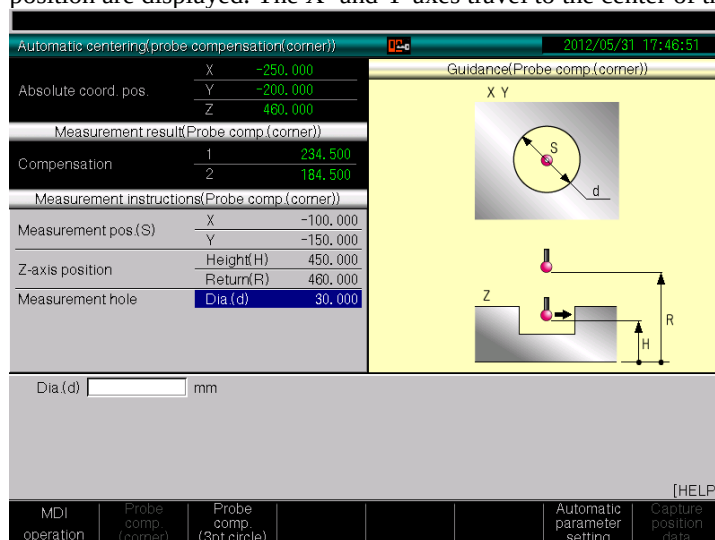
1. Have a ring gauge ready and available.
2. Press the [>>] (Next function) key on the automatic centering screen, and then select the [F1] (Probe comp. (corner)) key.
3. Set the values required for the measurement.
 - Measurement position (X and Y)
Use the X and Y absolute coordinates to specify the start point for the centering operation using a position close to center of the ring gauge.
 - Measurement hole (Hole diameter)
Specify the inner diameter on the ring gauge.



- Z-axis position (Height)
Specify the measurement height using the Z-axis absolute coordinates.
- Z-axis position (Return height)
Use the Z-axis absolute coordinates to specify the return height after the measurement.

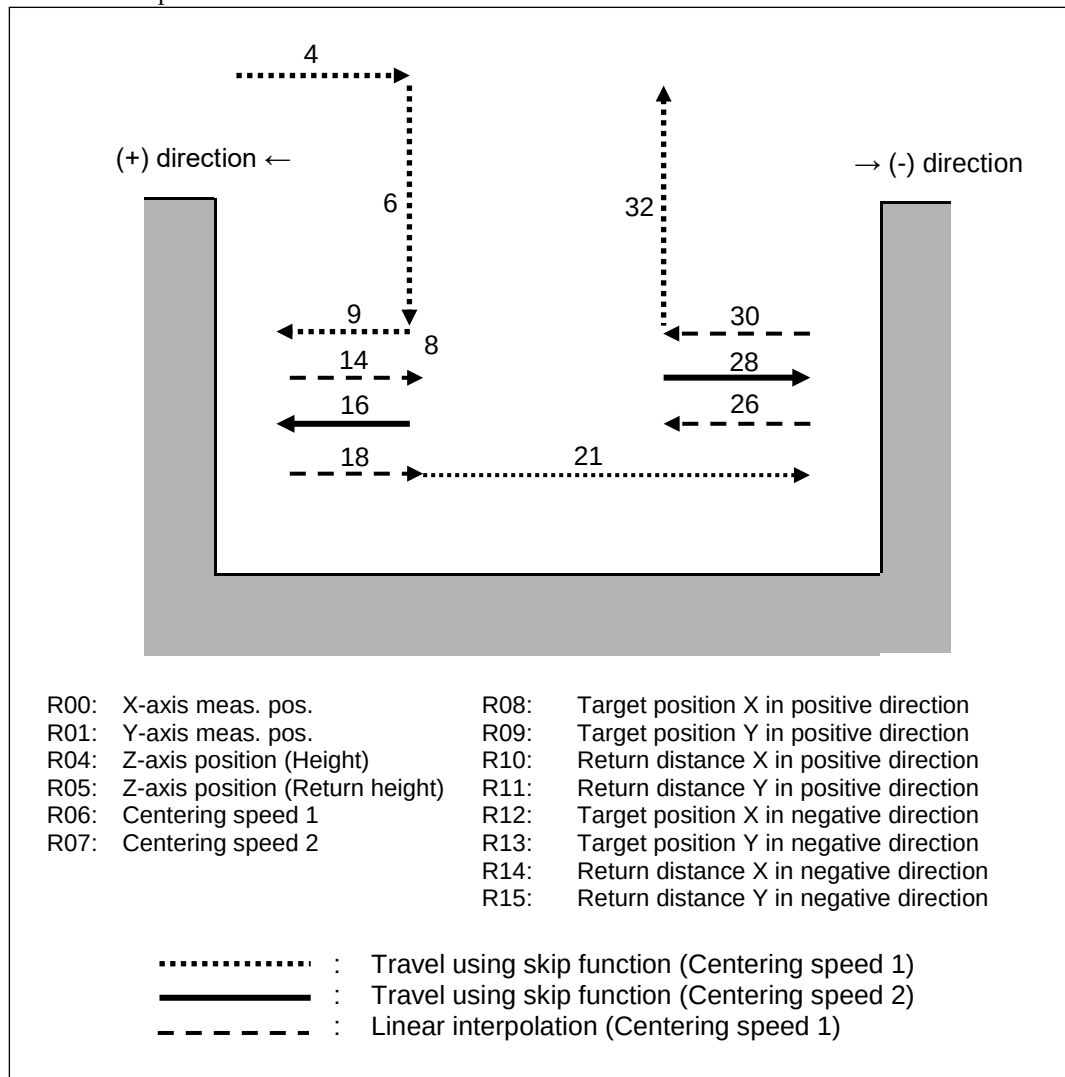


4. Press the **[START]** switch to execute the automatic centering operation. The automatic centering operation consists of the following five operations. The details of the operations are described in the following section.
 - (1) Execution of measurement start motion
 - (2) Execution of measurement pattern in X-axis direction for measurement position
 - (3) Execution of measurement pattern in Y-axis direction for position measured in (2)
 - (4) Execution of measurement pattern in X-axis direction for position measured in (3)
 - (5) Travel to measurement result position
5. After the measurement is finished, the absolute coordinates for the measurement result position are displayed. The X- and Y-axes travel to the center of the ring gauge.



2.9.1 Operation Details

Measurement pattern



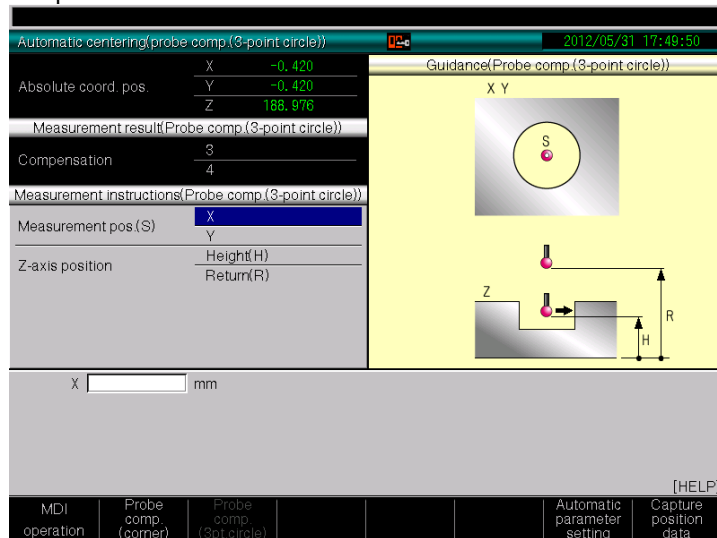
Step	NC language	Conversation language
(1) Execution of measurement start motion		
1		Machine zero return (Z)
2		Tool change pot No.xx
3	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(2) Execution of measurement pattern in X-axis direction for measurement position		
4	G131 X R00 Y R01 F R06	Measurement 1 X R00 Y R01 Feed R06
5	M321	Detection signal OFF check
6	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
7	M321	Detection signal OFF check
8	M19	Spindle orientation (0°)
9	G131 X R08 Y R01 F R06	Measurement 1 X R08 Y R01 Feed R06
10	M320	Detection signal ON check
11	M310	Internal calculation 2
12	M302	Internal calculation 1
13	M302	Internal calculation 1
14	G01 X R10 Y R01 F R06	Cutting feed X R10 Y R01 Feed R06
15	M321	Detection signal OFF check
16	G131 X R08 Y R01 F R07	Measurement 1 X R08 Y R01 Feed R07
17	M310	Internal calculation 2
18	G01 X R10 Y R01 F R06	Cutting feed X R10 Y R01 Feed R06
19	M321	Detection signal OFF check
20	M111	Spindle orientation (180°)
21	G131 X R12 Y R01 F R06	Measurement 1 X R12 Y R01 Feed R06
22	M320	Detection signal ON check

Step	NC language	Conversation language
23	M310	Internal calculation 2
24	M302	Internal calculation 1
25	M302	Internal calculation 1
26	G01 X R14 Y R01 F R06	Cutting feed X R14 Y R01 Feed R06
27	M321	Detection signal OFF check
28	G131 X R12 Y R01 F R07	Measurement 1 X R12 Y R01 Feed R07
29	M310	Internal calculation 2
30	G01 X R14 Y R01 F R06	Cutting feed X R14 Y R01 Feed R06
31	M05	Spindle stop
32	G131 Z R05 F R06	Measurement 1 Z R Feed R06
(3) Execution of measurement pattern in Y-axis direction for position measured in (2) (measurement position 1)		
33	G131 X (Measurement result 1) Y R01 F R06	Measurement 1 X (Measurement result 1) Y R01 Feed R06
34	M321	Detection signal OFF check
35	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
36	M321	Detection signal OFF check
37	M19	Spindle orientation (0°)
38	G131 X (Measurement result 1) Y R09 F R06	Measurement 1 X (Measurement result 1) Y R09 Feed R06
39	M320	Detection signal ON check
40	M310	Internal calculation 2
41	M302	Internal calculation 1
42	M302	Internal calculation 1
43	G01 X (Measurement result 1) Y R11 F R06	Cutting feed X (Measurement result 1) Y R11 Feed R06
44	M321	Detection signal OFF check
45	G131 X (Measurement result 1) Y R09 F R07	Measurement 1 X (Measurement result 1) Y R09 Feed R07
46	M310	Internal calculation 2
47	G01 X (Measurement result 1) Y R11 F R06	Cutting feed X (Measurement result 1) Y R11 Feed R06
48	M321	Detection signal OFF check
49	M111	Spindle orientation (180°)
50	G131 X (Measurement result 1) Y R13 F R06	Measurement 1 X (Measurement result 1) Y R13 Feed R06
51	M320	Detection signal ON check
52	M310	Internal calculation 2
53	M302	Internal calculation 1
54	M302	Internal calculation 1
55	G01 X (Measurement result 1) Y R15 F R06	Cutting feed X (Measurement result 1) Y R15 Feed R06
56	M321	Detection signal OFF check
57	G131 X (Measurement result 1) Y R13 F R07	Measurement 1 X (Measurement result 1) Y R13 Feed R07
58	M310	Internal calculation 2
59	G01 X (Measurement result 1) Y R15 F R06	Cutting feed X (Measurement result 1) Y R15 Feed R06
60	M05	Spindle stop
61	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(4) Execution of measurement pattern in X-axis direction for position measured in (3) (measurement position 2)		
62	G131 X (Measurement result 1) Y (Measurement result 2) F R06	Measurement 1 X (Measurement result 1) Y (Measurement result 2) Feed R06
63	M321	Detection signal OFF check
64	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
65	M321	Detection signal OFF check
66	M19	Spindle orientation (0°)
67	G131 X R08 Y (Measurement result 2) F R06	Measurement 1 X R08 Y (Measurement result 2) Feed R06
68	M320	Detection signal ON check
69	M310	Internal calculation 2
70	M302	Internal calculation 1
71	M302	Internal calculation 1
72	G01 X R10 Y (Measurement result 2) F R06	Cutting feed X R10 Y (Measurement result 2) Feed R06

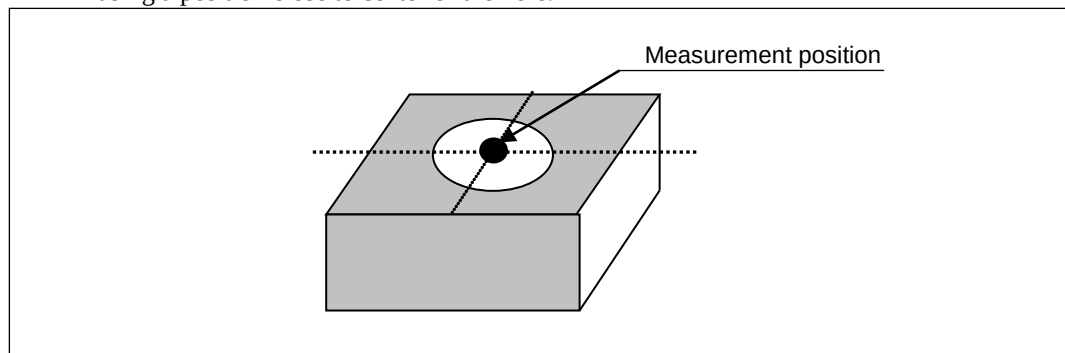
Step	NC language	Conversation language
73	M321	Detection signal OFF check
74	G131 X R08 Y (Measurement result 2) F R07	Measurement 1 X R08 Y (Measurement result 2) Feed R07
75	M310	Internal calculation 2
76	G01 X R10 Y (Measurement result 2) F R06	Cutting feed X R10 Y (Measurement result 2) Feed R06
77	M321	Detection signal OFF check
78	M111	Spindle orientation (180°)
79	G131 X R12 Y (Measurement result 2) F R06	Measurement 1 X R12 Y (Measurement result 2) Feed R06
80	M320	Detection signal ON check
81	M310	Internal calculation 2
82	M302	Internal calculation 1
83	M302	Internal calculation 1
84	G01 X R14 Y (Measurement result 2) F R06	Cutting feed X R14 Y (Measurement result 2) Feed R06
85	M321	Detection signal OFF check
86	G131 X R12 Y (Measurement result 2) F R07	Measurement 1 X R12 Y (Measurement result 2) Feed R07
87	M310	Internal calculation 2
88	G01 X R14 Y (Measurement result 2) F R06	Cutting feed X R14 Y (Measurement result 2) Feed R06
89	M05	Spindle stop
90	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(5) Travel to measurement result position		
91	G131 X (Measurement result 3) Y (Measurement result 2) F R06	Measurement 1 X (Measurement result 3) Y (Measurement result 2) Feed R06
92	M321	Detection signal OFF check

2.10 Measure Probe Compensation (3 Point Circle)

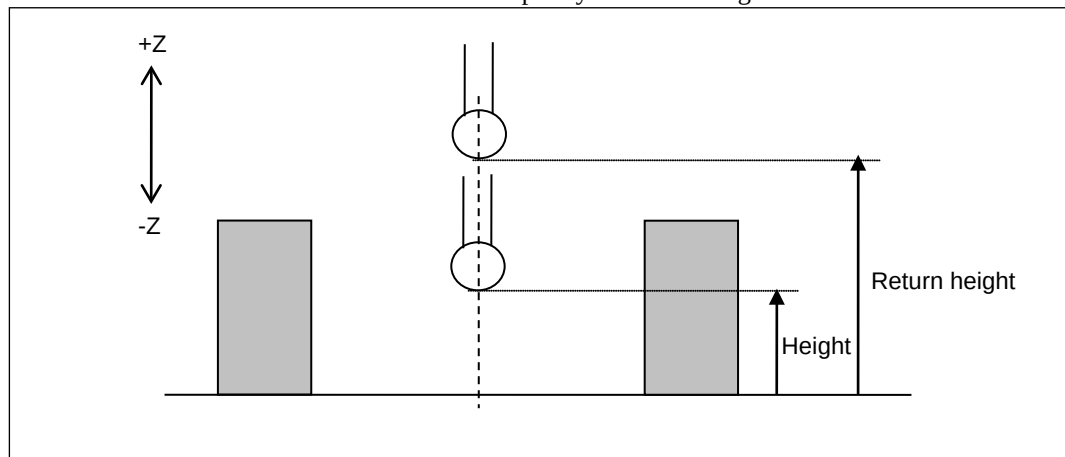
Measure the values that are set to the parameters <Probe compensation 3> and <Probe compensation 4>.



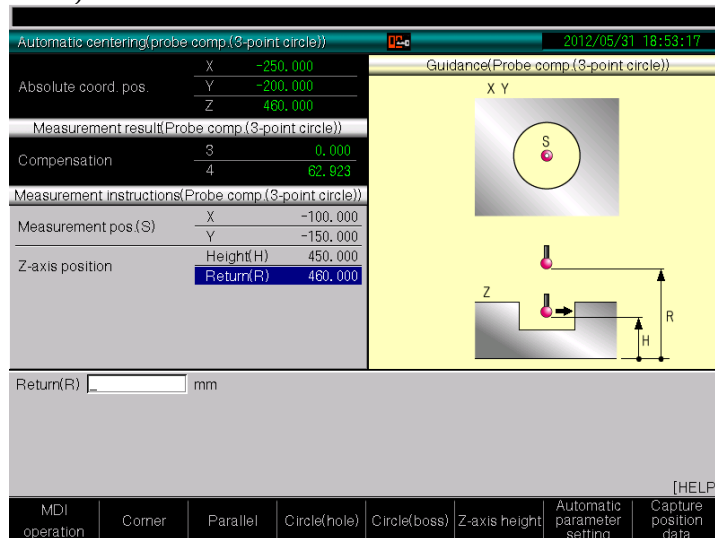
1. Press the [>>] (Next function) key on the automatic centering screen, and then select the [F2] (Probe comp. (3pt. circle)) key.
2. Set the values required for the measurement.
 - Measurement position (X and Y)
Use the X and Y absolute coordinates to specify the start point for the centering operation using a position close to center of the hole.



- Z-axis position (Height)
Specify the measurement height using the Z-axis absolute coordinates.
- Z-axis position (Return height)
Use the Z-axis absolute coordinates to specify the return height after the measurement.

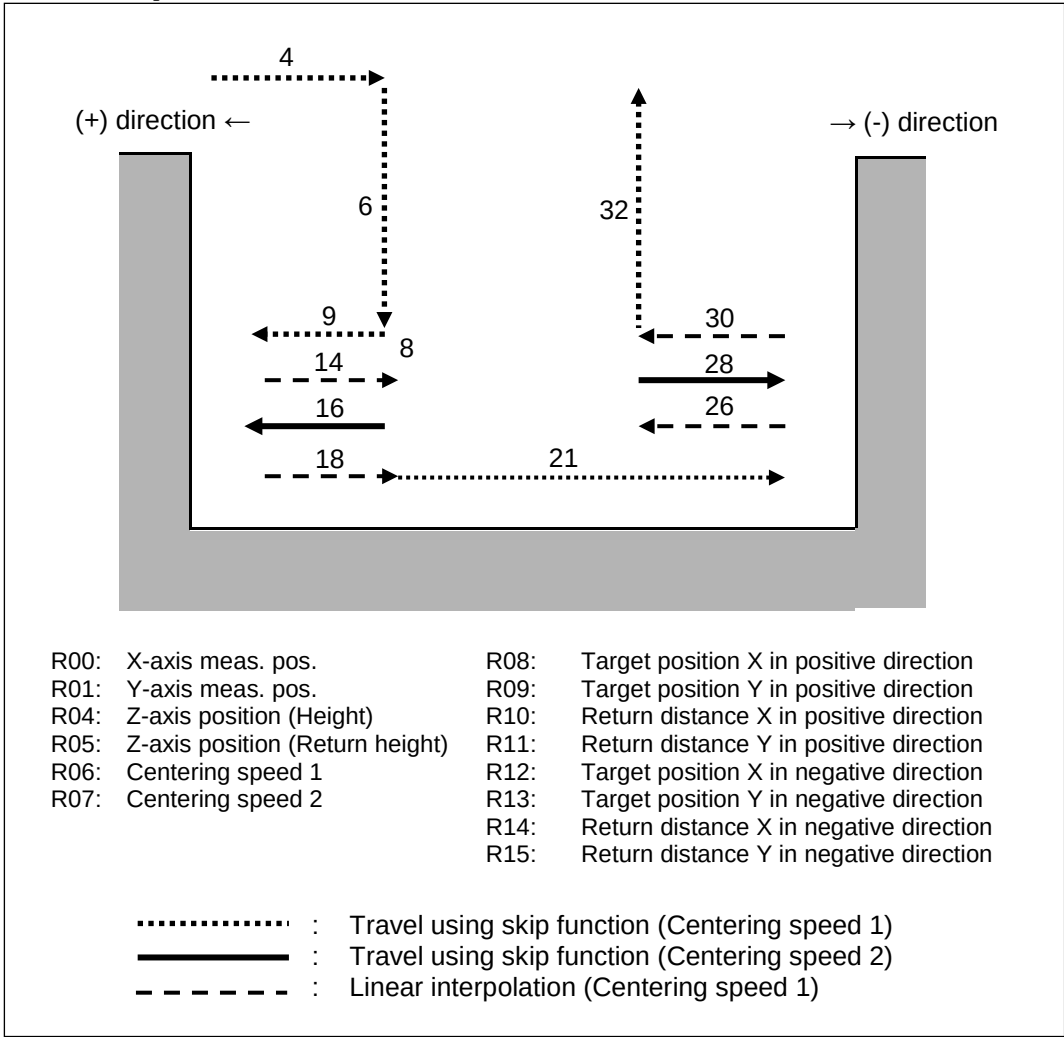


3. Press the **[START]** switch to execute the automatic centering operation. The automatic centering operation consists of the following seven operations. The details of the operations are described in the following section.
 - (1) Execution of measurement start motion
 - (2) Execution of measurement pattern 1 in X-axis direction for measurement position
 - (3) Execution of measurement pattern 1 in Y-axis direction for position measured in (2)
 - (4) Execution of measurement pattern 1 in X-axis direction for position measured in (3)
 - (5) Execution of measurement pattern 2 in 120° direction for position measured in (4)
 - (6) Execution of measurement pattern 2 in 240° direction for position measured in (4)
 - (7) Travel to measurement result position
4. After the measurement is finished, the absolute coordinates for the measurement result position are displayed. The X- and Y-axes travel to the measurement result position (hole center).

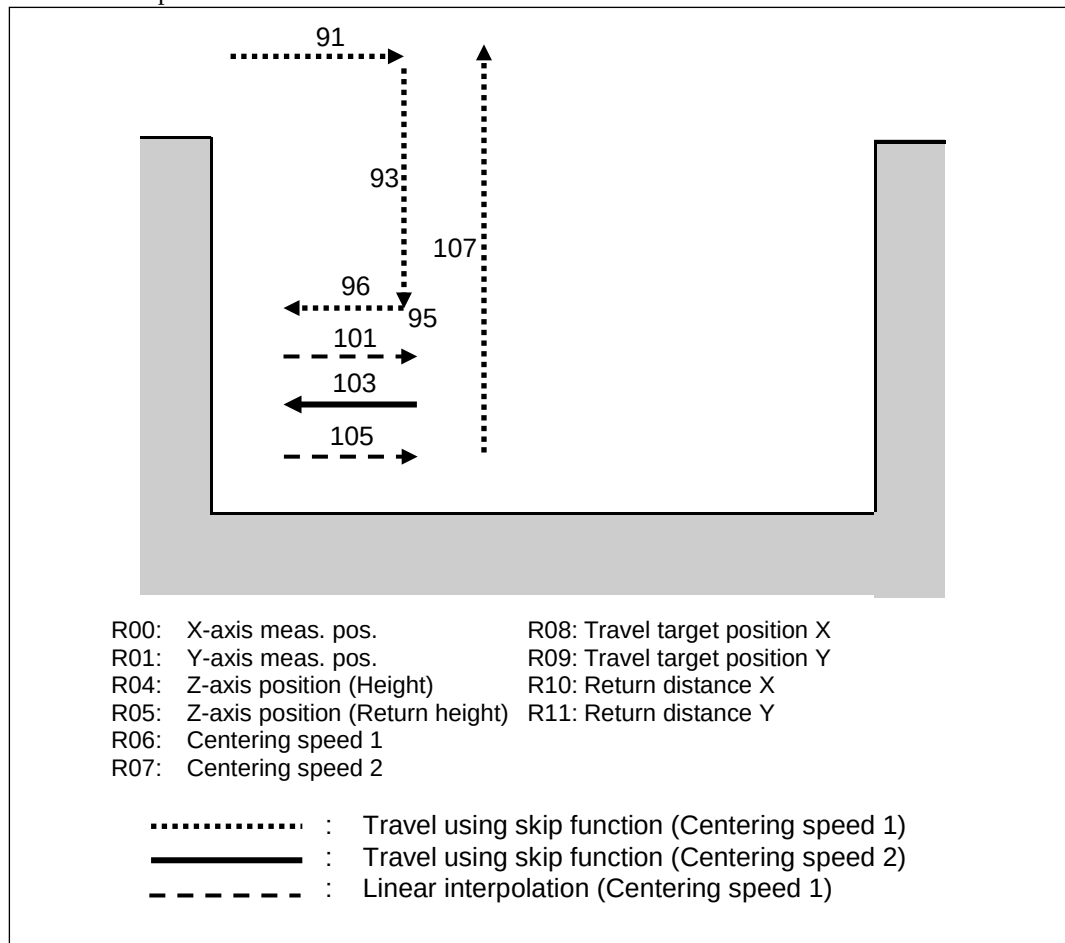


2.10.1 Operation Details

Measurement pattern 1



Measurement pattern 2



Step	NC language	Conversation language
(1) Execution of measurement start motion		
1		Machine zero return (Z)
2		Tool change pot No.xx
3	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(2) Execution of measurement pattern 1 in X-axis direction for measurement position		
4	G131 X R00 Y R01 F R06	Measurement 1 X R00 Y R01 Feed R06
5	M321	Detection signal OFF check
6	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
7	M321	Detection signal OFF check
8	M19	Spindle orientation (0°)
9	G131 X R08 Y R01 F R06	Measurement 1 X R08 Y R01 Feed R06
10	M320	Detection signal ON check
11	M310	Internal calculation 2
12	M302	Internal calculation 1
13	M302	Internal calculation 1
14	G01 X R10 Y R01 F R06	Cutting feed X R10 Y R01 Feed R06
15	M321	Detection signal OFF check
16	G131 X R08 Y R01 F R07	Measurement 1 X R08 Y R01 Feed R07
17	M310	Internal calculation 2
18	G01 X R10 Y R01 F R06	Cutting feed X R10 Y R01 Feed R06
19	M321	Detection signal OFF check
20	M111	Spindle orientation (180°)
21	G131 X R12 Y R01 F R06	Measurement 1 X R12 Y R01 Feed R06
22	M320	Detection signal ON check
23	M310	Internal calculation 2
24	M302	Internal calculation 1
25	M302	Internal calculation 1
26	G01 X R14 Y R01 F R06	Cutting feed X R14 Y R01 Feed R06
27	M321	Detection signal OFF check
28	G131 X R12 Y R01 F R07	Measurement 1 X R12 Y R01 Feed R07

Step	NC language	Conversation language
29	M310	Internal calculation 2
30	G01 X R14 Y R01 F R06	Cutting feed X R14 Y R01 Feed R06
31	M05	Spindle stop
32	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(3) Execution of measurement pattern 1 in Y-axis direction for position measured in (2) (measurement position 1)		
33	G131 X (Measurement result 1) Y R01 F R06	Measurement 1 X (Measurement result 1) Y R01 Feed R06
34	M321	Detection signal OFF check
35	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
36	M321	Detection signal OFF check
37	M19	Spindle orientation (0°)
38	G131 X (Measurement result 1) Y R09 F R06	Measurement 1 X (Measurement result 1) Y R09 Feed R06
39	M320	Detection signal ON check
40	M310	Internal calculation 2
41	M302	Internal calculation 1
42	M302	Internal calculation 1
43	G01 X (Measurement result 1) Y R11 F R06	Cutting feed X (Measurement result 1) Y R11 Feed R06
44	M321	Detection signal OFF check
45	G131 X (Measurement result 1) Y R09 F R07	Measurement 1 X (Measurement result 1) Y R09 Feed R07
46	M310	Internal calculation 2
47	G01 X (Measurement result 1) Y R11 F R06	Cutting feed X (Measurement result 1) Y R11 Feed R06
48	M321	Detection signal OFF check
49	M111	Spindle orientation (180°)
50	G131 X (Measurement result 1) Y R13 F R06	Measurement 1 X (Measurement result 1) Y R13 Feed R06
51	M320	Detection signal ON check
52	M310	Internal calculation 2
53	M302	Internal calculation 1
54	M302	Internal calculation 1
55	G01 X (Measurement result 1) Y R15 F R06	Cutting feed X (Measurement result 1) Y R15 Feed R06
56	M321	Detection signal OFF check
57	G131 X (Measurement result 1) Y R13 F R07	Measurement 1 X (Measurement result 1) Y R13 Feed R07
58	M310	Internal calculation 2
59	G01 X (Measurement result 1) Y R15 F R06	Cutting feed X (Measurement result 1) Y R15 Feed R06
60	M05	Spindle stop
61	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(4) Execution of measurement pattern 1 in X-axis direction for position measured in (3) (measurement position 2)		
62	G131 X (Measurement result 1) Y (Measurement result 2) F R06	Measurement 1 X (Measurement result 1) Y (Measurement result 2) Feed R06
63	M321	Detection signal OFF check
64	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
65	M321	Detection signal OFF check
66	M19	Spindle orientation (0°)
67	G131 X R08 Y (Measurement result 2) F R06	Measurement 1 X R08 Y (Measurement result 2) Feed R06
68	M320	Detection signal ON check
69	M310	Internal calculation 2
70	M302	Internal calculation 1
71	M302	Internal calculation 1
72	G01 X R10 Y (Measurement result 2) F R06	Cutting feed X R10 Y (Measurement result 2) Feed R06
73	M321	Detection signal OFF check
74	G131 X R08 Y (Measurement result 2) F R07	Measurement 1 X R08 Y (Measurement result 2) Feed R07
75	M310	Internal calculation 2
76	G01 X R10 Y (Measurement result 2) F R06	Cutting feed X R10 Y (Measurement result 2) Feed R06

Step	NC language	Conversation language
77	M321	Detection signal OFF check
78	M111	Spindle orientation (180°)
79	G131 X R12 Y (Measurement result 2) F R06	Measurement 1 X R12 Y (Measurement result 2) Feed R06
80	M320	Detection signal ON check
81	M310	Internal calculation 2
82	M302	Internal calculation 1
83	M302	Internal calculation 1
84	G01 X R14 Y (Measurement result 2) F R06	Cutting feed X R14 Y (Measurement result 2) Feed R06
85	M321	Detection signal OFF check
86	G131 X R12 Y (Measurement result 2) F R07	Measurement 1 X R12 Y (Measurement result 2) Feed R07
87	M310	Internal calculation 2
88	G01 X R14 Y (Measurement result 2) F R06	Cutting feed X R14 Y (Measurement result 2) Feed R06
89	M05	Spindle stop
90	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(5) Execution of measurement pattern 2 in 120° direction for position measured in (4) (measurement position 3)		
91	G131 X (Measurement result 3) Y (Measurement result 2) F R06	Measurement 1 X (Measurement result 3) Y (Measurement result 2) Feed R06
92	M321	Detection signal OFF check
93	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
94	M321	Detection signal OFF check
95	M19	Spindle orientation (0°)
96	G131 X R08 Y R09 F R06	Measurement 1 X R08 Y R09 Feed R06
97	M320	Detection signal ON check
98	M310	Internal calculation 2
99	M302	Internal calculation 1
100	M302	Internal calculation 1
101	G01 X R10 Y R11 F R06	Cutting feed X R10 Y R11 Feed R06
102	M321	Detection signal OFF check
103	G131 X R08 Y R09 F R06	Measurement 1 X R08 Y R09 Feed R07
104	M310	Internal calculation 2
105	G01 X R10 Y R11 F R06	Cutting feed X R10 Y R11 Feed R06
106	M05	Spindle stop
107	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(6) Execution of measurement pattern 2 in 240° direction for position measured in (4) (measurement position 3)		
108	G131 X (Measurement result 3) Y (Measurement result 2) F R06	Measurement 1 X (Measurement result 3) Y (Measurement result 2) Feed R06
109	M321	Detection signal OFF check
110	G131 Z R04 F R06	Measurement 1 Z R04 Feed R06
111	M321	Detection signal OFF check
112	M19	Spindle orientation (0°)
113	G131 X R08 Y R09 F R06	Measurement 1 X R08 Y R09 Feed R06
114	M320	Detection signal ON check
115	M310	Internal calculation 2
116	M302	Internal calculation 1
117	M302	Internal calculation 1
118	G01 X R10 Y R11 F R06	Cutting feed X R10 Y R11 Feed R06
119	M321	Detection signal OFF check
120	G131 X R08 Y R09 F R06	Measurement 1 X R08 Y R09 Feed R07
121	M310	Internal calculation 2
122	G01 X R10 Y R11 F R06	Cutting feed X R10 Y R11 Feed R06
123	M05	Spindle stop
124	G131 Z R05 F R06	Measurement 1 Z R05 Feed R06
(7) Travel to measurement result position		
125	G131 X (Measurement result 4) Y (Measurement result 4) F R06	Measurement 1 X (Measurement result 4) Y (Measurement result 4) Feed R06
126	M321	Detection signal OFF check

3 Data Settings for Automatic Centering

3.1 Tool Data

During the automatic centering operation, the tool length offset is always applied for the tool length on tool number 99. The tool length and tool diameter (probe diameter) for the probe to be used is set to tool data No. 99.

3.2 Probe Detection Signal Method

Set the following item in the “User parameter 1 (switch 1)”.

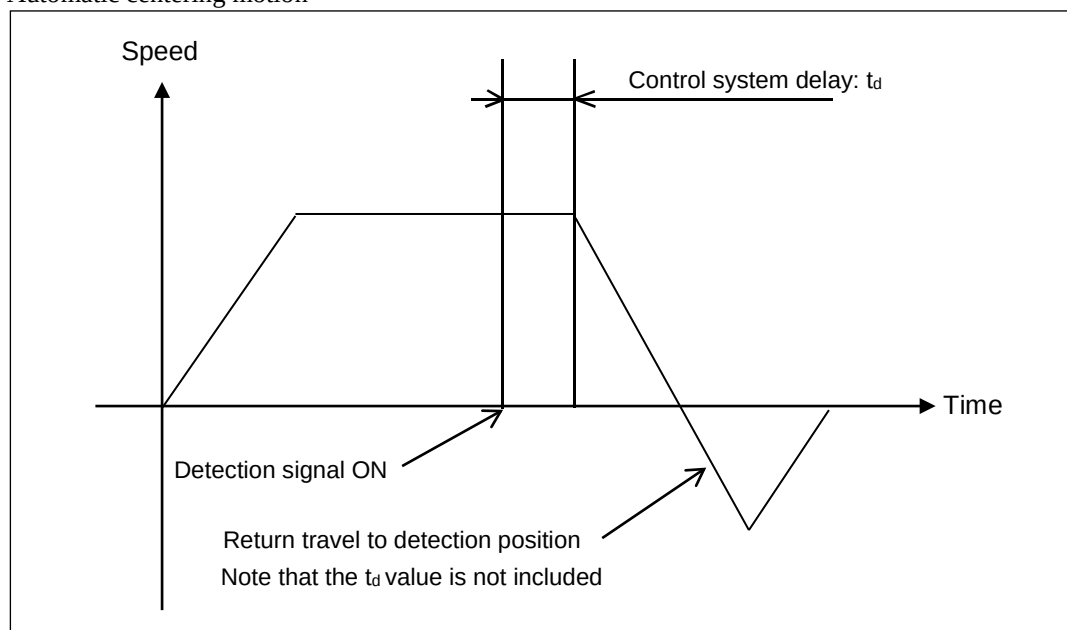
Item name	Setting range	Description
Probe detection signal method	0: Method 1 1: Method 2	Set the accuracy of the detection signal that is obtained from the touch probe. Even if the <1: Method 2> increases the travel speed of the probe more than <0: Method 1>, there is less variation in the measurement values. As a result, the speed can be set higher than <Automatic centering speed 2> and <Measurement speed 2>. However, <1: Method 2> cannot be set due to the machine structure, and an alarm is triggered.

3.3 Parameters

Parameters	Set value ([]: Default value)	Description
Automatic centering speed 1	1 to 5000 mm/min [500]	Sets the positioning speed for automatic centering and sets the axis travel speed in a rough measurement.
Automatic centering speed 2	1 to 5000 mm/min [10]	Sets the axis travel speed during the actual measurement for automatic centering.
Automatic centering return	0 to 99.999 mm [0.500]	Sets the return value before starting the actual measurement after the rough measurement has finished during automatic centering.
Automatic centering compensation	0 to 9.999 mm [0.000]	Sets the detection position compensation when performing automatic centering.

3.4 Specify Parameters

Automatic centering motion



1. <Automatic centering speed 1>

Relief amount of probe	= L (mm)
Skip feed time const. 1	= t (msec)
<Automatic centering speed 1>	= F ₁ (mm/min)
Delay in control system	= t _d (msec) = 6 (msec)
Delay unique to probe	= t _p (msec)

$$L \geq \underbrace{((F_1 \times t_d) \div (60 \times 1000))}_{\text{Overtravel amount due to delay in control system}} + \underbrace{((F_1 \times t/2) \div (60 \times 1000))}_{\text{Overtravel amount}} + \underbrace{((F_1 \times t_p) \div (60 \times 1000))}_{\text{Overtravel amount due to delay unique to probe}}$$

$$\therefore F_1 \leq ((120000 \times L) \div (8 + t + 2 \times t_p)) \div \underline{1.2}$$

Safety factor

2. <Automatic centering speed 2>

Tolerance in control system	= E (mm)
<Automatic centering speed 2>	= F ₂ (mm/min)
Margin of error in control system	= t _d (msec)
(When probe detection signal method = 0: Method 1)	
td(msec)=0.300 (msec)	
(When probe detection signal method = 1: Method 2)	
td(msec)=0.085 (msec)	
Margin of error unique to probe	= e _p (msec)

$$E \geq (F_2 \times t_d) \div (60 \times 100) + e_p$$

$$\therefore F_2 \leq 60000 \times (E - e_p) \div t_d$$

3. <Automatic centering return>

<Automatic centering return>	= L _b (mm)
Skip feed time const. 1	= t (msec)
<Automatic centering speed 2>	= F ₂ (mm/min)

 - (1) If the Z-axis is raised to a position that is too close, the probe may interfere with the workpiece because the shape of the workpiece varies. Therefore, a 1.0 mm setting is appropriate.
 - (2) When measuring, make the probe touch the workpiece at a regular speed (not accelerated speed).

Set to the larger setting between (1) and (2).

$$\therefore L_b \geq \text{MAX} \{1.0, F_2 \times t / 6000\}$$

4. <Automatic centering compensation>

Time delay from when the automatic centering tool touches the workpiece until the detection signal turns ON	= t _c (msec)
<Automatic centering speed 2>	= F ₂ (mm/min)
<Automatic centering compensation>	= L _c (mm)

$$\therefore L_c = t_c \times F_2 / 6000$$

5. <Skip feed time const.>

The constant is decided based on automatic centering speed 1 and the relief amount of the probe. Set to 100 msec. when there is no particular reason.

(Note) The set values, such as the automatic centering speed and automatic centering return distance after measurement, vary depending on the probe being used. Therefore, contact the probe manufacturer when deciding on the set values

4 Installing and Removing Probe

Refer to the section “3.7 Important Notes When Attaching Workpieces and Tools” in the Operation Manual for details on installing to and removing from the probe magazine.

CHAPTER 12 (10)

THERMAL DISTORTION COMPENSATION SYSTEM

- 1 Handling Precautions**
- 2 Principles**
- 3 Usage**
- 4 Parameters**
- 5 How to Confirm the Compensation Results**
- 6 Adjust the Compensation Value**

1 Handling Precautions

Precautions when enabling the function

This system executes the thermal distortion compensation based on previous operations.

(How many operations are recorded varies depending on the machine model. (*Reference))

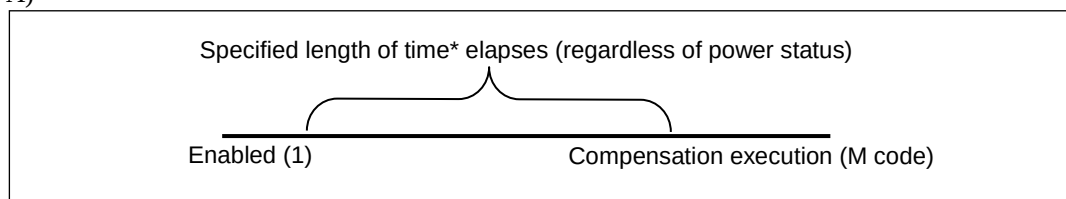
After the function is enabled, observe one of the following conditions when executing the compensation operation.

(A) Execute compensation once the specified length of time* has elapsed after the function was enabled.

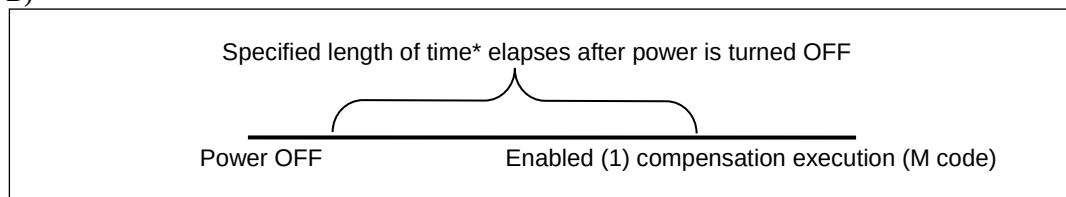
(B) Execute compensation once the specified length of time* has elapsed after the function was enabled while the power was turned OFF.

When the function is enabled, the past data is ignored. If the function is enabled under conditions not noted above, the machine status and the data will not match causing incorrect compensation because past data does not exist even when the machine actually undergoes thermal expansion.

A)



B)



* Specified length of time

S300X1/S500X1/S700X1/S1000X1/S300X2/S500X2/S700X2/F600X1	5 hours
M140X1/M140X2/R450X1/R650X1/ R450X2/R650X2/ M200X3/M300X3	9 hours

Usage after applying grease

After applying grease to the Z-axis, perform a test run covering the entire stroke for 10 to 20 minutes and then execute the thermal distortion compensation.

Features of the thermal distortion compensation system and usage precautions

1. Compensation is not performed from actual measurement values using sensors.
All compensation values are calculated by the NC unit, and therefore accuracy cannot be guaranteed at a level of several micrometers.
2. External factors such as differences in the coolant supply method and flow rate or fluctuations in the room temperature are not taken into account in the compensation. Therefore, those factors may affect the accuracy depending on machining conditions.
3. Parameters may vary depending on the operating conditions or environment of each machine. Therefore, parameters must be changed depending on those conditions.

2 Principles

Basic system configuration

This system is designed to compensate for thermal distortion in the machine motion for the axis directions noted below. The system is configured so that the amount of compensation is calculated only using the software. Sensors and other tools are not used to measure the thermal distortion.

Axes equipped with the compensation function

X-, Y- and Z-axes (all 3 axes come standard equipped)

The Z-axis is used as an example in the explanation below, but it also applies to the X- and Y-axes.

The signal output code for executing the compensation is different for the X- and Y-axes.

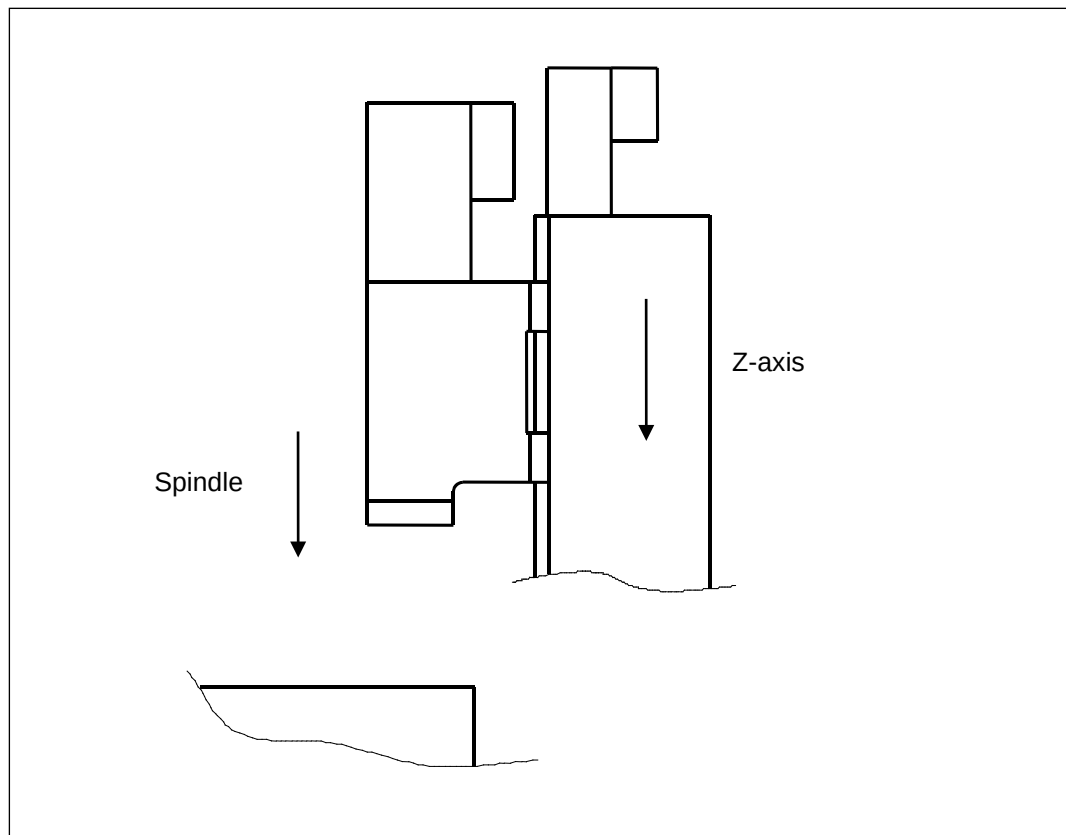
System features

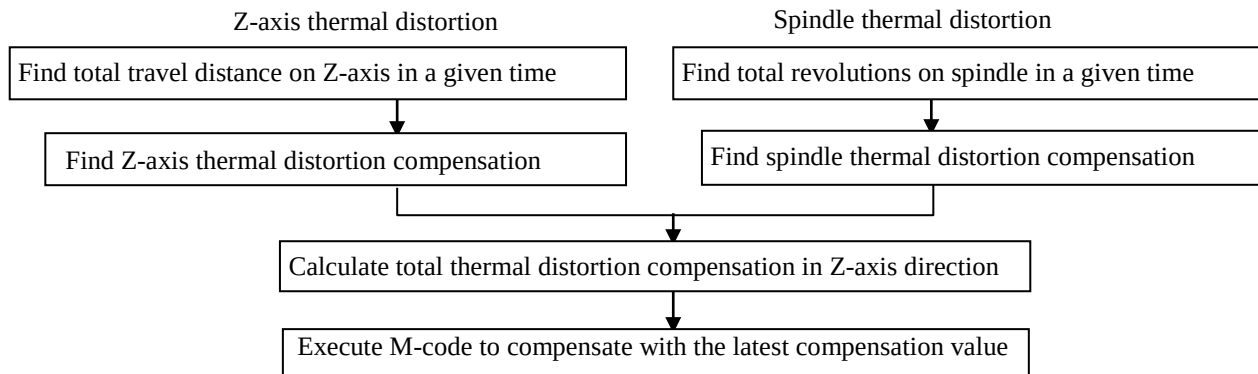
1. There is no cycle time loss when the thermal distortion compensation system executes.
2. Warming up the machine is not required when starting operation or when resuming operation after a pause.
3. Creating a program is easy.
4. No sensors are used on the table, thereby eliminating the need to consider interference when designing jigs.
5. No sensors are used, eliminating problems caused by sensor failure.

Operation principles

The compensation value is calculated every minute. The total amount of compensation in the Z-axis direction is calculated based on the formula (1) below, taking into account the Z-axis travel distance and the number of spindle rotations during said length of time.

Total amount of thermal distortion compensation in Z-axis direction = Z-axis thermal distortion + Spindle thermal distortion...Formula (1)





3 Usage

1. Enable automatic thermal distortion compensation.
(Refer to “5.2 Change operation status” for further details.)
↓
2. The NC unit automatically calculates the compensation value.
(Calculation only; the compensation is actually not carried out)
↓
- 3-1. Execute the compensation with M-code.
Execute M352*¹⁾ in memory operation.
(352 in the signal output job when using conversation language)
The most recent compensation value is applied at this point to perform compensation.
This value is used to perform compensation until the next M352 is executed.
More than two M352 commands can be entered in one program. In this situation, the compensation value is updated each time an M352 command is issued.

*1)
X-axis: M350
Y-axis: M351
Z-axis: M352
X-, Y- and Z-axes: M353
- 3-2. Automatically execute compensation
The compensation can be automatically executed at the start of memory operation without using an M-code. Set the user parameter <Auto thermal distortion compen. at memory operation start> (Automatic thermal distortion compensation) to <1: Yes>. When set to <1: Yes>, the compensation is automatically executed at the start of memory operation. Compensation is executed on axes that are <Operating>, and axes that are <Stopped> are not compensated. For example, if the X-, Y- and Z-axes are <Operating>, compensation is executed on the X-, Y- and Z-axes.

- (Note 1) Automatic compensation is executed only at the start in memory operation.**
Compensation cannot be executed after a pause, temporary stop or block stop.
Compensation cannot be executed in MDI operation mode.
- (Note 2) Compensation can also be used with M-code.**
Enter M-code into the program in order to apply the latest compensation value during memory operation.
- (Note 3) When the user parameter <Positioning method> is set to <0: Non-linear interpolation type positioning> and when <1: Linear interpolation type - Positioning method 1> or <2: Linear interpolation type - Positioning method 2> is set, the compensation is not the same even if the same program is executed.**

Refer to section “5. How to Confirm the Compensation Results” for details on understanding and confirming the compensation value.

Program example (NC language)

M352 (Note)-----

(.....COORDINATE DATA)

G90 G10 L2 P1 X0. Y0. Z0.

G90 G10 L2 P2 X-50.7 Y-48.214 Z150.

(.. TOOL LENG OFFSET DATA ..)

G90 G10 L10 P1 R200.

G90 G10 L10 P2 R200.

(..... NC DATA)

G40 G49 G64 G80 G90 G99

G54

(M8)

G91 G28 Z0

G90 G55

T1

N1 G100 G00 X-110. Y-51.6 M3 S9847 G43 H1 Z19.7

T2

M352

G01 X0. Y-51.6 F9650

X0. Y-45.2

X-97. Y-4.9

G02 X-94. Y-1.9 I3. J0.

If the M-code for executing the automatic thermal distortion compensation is set at the beginning of a program, machining continues with this compensation value until the next M-code is set.

Program example (Conversation mode)

Example when setting 352 in the signal output job code on the <Machining data> screen

Machining data		2013/09/12 16:34:58	
Program (VAM/3000) Job 01			
Workpiece zero	X	0.000	Y 0.000 Z 0.000
	A	? B	?
Return height (job change)	Height	50.000	
Workpieces		1	
Workpiece material		S45C	
Job01	Signal output	Code	352
Job02	Hole	Hole dia.	3.000
		Chamfer	0.000
		Point	
	Z-axis position	Bottom of hole	Blind Depth 20.000
		Height	30.000 Return height 35.000
	X-Y-axis position	X	-200.000 Y -200.000
		Peripheral speed	Peripheral speed 100% Feed 100%
Main program end			
Sub program end			
Code			
Signal output: 1 - 99,900 - 999,800 - 899 (PLC)			
Tool change code: 501 - 599 (Tool No.), 701 - 722 (Magazine or pot No.)			
Excludes 3, 4, 30			
Search: Job/line [HELP]			
Completion mode	Previous Job	Next Job	Search Add Set workpiece data Copy Job Paste Job

Example when setting 352 in the <Misc.func.before cutting> on the <Tool assignment> screen.

Tool assignment		2012/05/31 11:04:41	
Program (3000) **** 7777			
Job02	1. Center hole	Tool No.	01 Center drill
Hole		Tool length offset	Cutter compensation **
		Misc.func.before cutting	352 Misc.func.after cutting
		Dwell time	
	2. Hole	Tool No.	04 Drill 3.000
		Tool length offset	Cutter compensation **
		Misc.func.before cutting	Misc.func.after cutting
		Dwell time	
Program end			
Miscellaneous function code before cutting			
Signal output: 1 - 99,900 - 999,800 - 899 (PLC)			
Excluding 3,4,30, Tool change code (501 - 599, 701 - 721)			
Tool assignment is completed. ? calculation was performed. [HELP]			
Completion mode	Previous tool	Next tool	Change function Calculation result Search Tool check Assign tools

4 Parameters

User parameters (Automatic thermal distortion compensation)

Item name	Setting range	Description
Z-axis compensation	-999 to 999(%)	Adjusts the compensation value amount for the Z-axis.
Spindle compensation (Z-axis)	-999 to 999(%)	Adjusts the compensation value amount for the Z-axis spindle.
X-axis compensation	-999 to 999(%)	Adjusts the compensation value amount for the X-axis.
Spindle compensation (X-axis)	-999 to 999(%)	Adjusts the compensation value amount for the X-axis spindle.
Y-axis compensation	-999 to 999(%)	Adjusts the compensation value amount for the Y-axis.
Spindle compensation (Y-axis)	-999 to 999(%)	Adjusts the compensation value amount for the Y-axis spindle.
Auto thermal distortion compen. at memory operation start	0 to 1	This parameter is valid only when the machine is equipped with the automatic thermal distortion compensation software. If “1: Yes” is set on a machine that is not equipped, the parameter does not do anything. When set to “1: Yes”, the automatic thermal distortion compensation is executed automatically on the X-, Y- and Z-axes at the start of memory operation.
Lathe spindle compensation (Z-axis)	-999 to 999(%)	Adjusts the compensation value amount for the Z-axis spindle. This parameter is only valid for M series model machines.
Lathe spindle compensation (X-axis)	-999 to 999(%)	Adjusts the compensation value amount for the X-axis spindle. This parameter is only valid for M series model machines.
Lathe spindle compensation (Y-axis)	-999 to 999(%)	Adjusts the compensation value amount for the Y-axis spindle. This parameter is only valid for M series model machines.

5 How to Confirm the Compensation Results

The compensation results can be checked for each axis on the <Auto. thermal distortion comp.> screen under the production monitor menu.

Perform one of the following key operations on the production monitor menu screen to display the <Auto. thermal distortion comp.> screen:

- Press **[5]** and then **[ENT]**.
- Move the cursor to menu number 5 and press the **[ENT]** key.
- Press the **[F5]** (Auto. thermal distortion comp.) key.

Use the **[F1]** (X-axis comp.) key, the **[F2]** (Y-axis comp.) key or the **[F3]** (Z-axis comp.) key to select and confirm the desired axis.

Auto. thermal distortion comp (Z-axis)

2014/02/12 16:20:00

Auto. thermal distortion comp.

Operation status

Operating

Position setting

Position 1

0.000

Position 2

0.000

Current time compen

Compensation 1

0.000

Compensation 2

0.000

Remaining data

56

Time

No. of cal.

Z average speed

Displac. 1

Displac. 2

Sp. rotation sum

Sp. displac.

16:19:41

55

0.000

0.000

0.000

0

0.000

16:19:11

54

0.000

0.000

0.000

0

0.000

16:18:41

53

0.000

0.000

0.000

0

0.000

16:18:11

52

0.000

0.000

0.000

0

0.000

16:17:42

51

0.000

0.000

0.000

0

0.000

16:17:12

50

0.000

0.000

0.000

0

0.000

16:16:42

49

0.000

0.000

0.000

0

0.000

16:16:12

48

0.000

0.000

0.000

0

0.000

16:15:42

47

0.000

0.000

0.000

0

0.000

16:15:11

46

0.000

0.000

0.000

0

0.000

16:14:41

45

0.000

0.000

0.000

0

0.000

16:14:12

44

0.000

0.000

0.000

0

0.000

16:13:42

43

0.000

0.000

0.000

0

0.000

16:13:12

42

0.000

0.000

0.000

0

0.000

Production monitor menu

X-axis comp.

Y-axis comp.

Z-axis comp.

Change operation status.

Delete data

Position setting

[HELP]

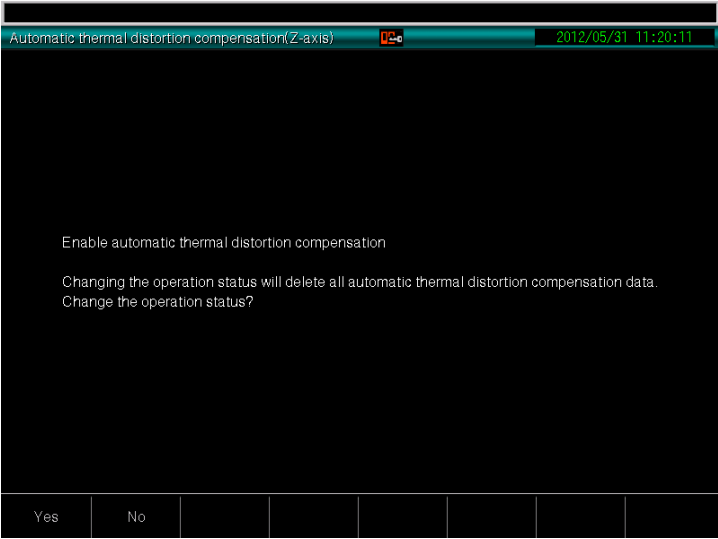
5.1 Description of Thermal Distortion Compensation Data

- (1) <Operation status>
Displays whether the automatic thermal distortion compensation function is ON or OFF.
- (2) <Position setting>
Displays the position set by **[F7]** (Position setting).
- (3) <Current time compen>
Displays the compensation for the position set by the position setting.
The compensation is equal to “Compensation for displacement of each axis” + “Compensation for spindle displacement”.
- (4) <Remaining data>
Displays the data that is used for calculation.
- (5) <Time>
Displays the time that is calculated from the compensation. The places that are highlighted in blue indicate where the time displacement is used to execute the compensation.
- (6) <No. of cal.>
The number of calculations is counted in the following order.
It counts from 0 to 65535 and then resets to 0.
- (7) <Average speed>
Displays the average travel speed (mm/calculation interval) from the number of calculations that is displayed.
- (8) <Displac. 1/2>
Displays the thermal distortion displacement (mm) for the number of calculations that is displayed at the position that is set by the position setting.
- (9) <Sp. rotation sum>
Displays the spindle rotation sum (rotations) for the number of calculations that is displayed.
On M series machine models, it displays the rotation sum for the spindle and lathe spindle.
In this situation, the “Spindle rotation sum / Lathe spindle rotation sum” is displayed on the screen.
- (10) <Sp. displac.>
Displays the thermal distortion compensation (mm) on the spindle for the number of calculations that is displayed.
On M series models, it displays the thermal distortion for the spindle and lathe spindle.
In this situation, the “Spindle thermal distortion / Lathe spindle thermal distortion” is displayed on the screen.

5.2 Change Operation Status

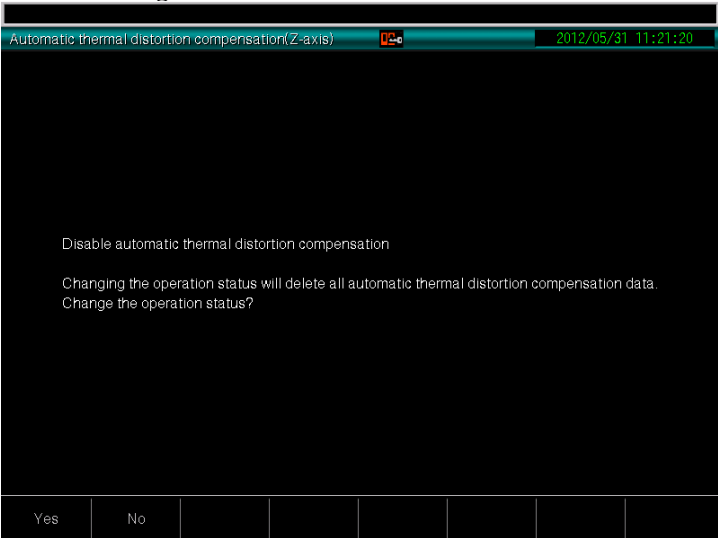
The automatic thermal distortion compensation function can be enabled and disabled for each axis.
Press the **[F4]** key.

When enabling



Press the **[F0]** key to enable and press the **[F1]** key to cancel enabling.
When the operation status is changed, the automatic thermal distortion compensation data is deleted.
When changing the operation status, turn the power OFF and then ON.

When disabling



Press the **[F0]** key to disable and press the **[F1]** key to cancel disabling.
When the operation status is changed, the automatic thermal distortion compensation data is deleted.
When changing the operation status, turn the power OFF and then ON.

5.3 Delete Compensation Data

The automatic thermal distortion compensation data can be deleted for each axis.

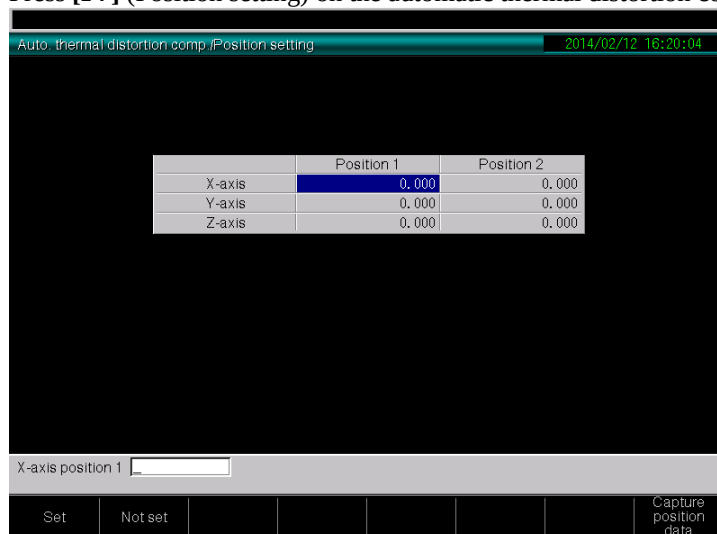
Press the **[F5]** key.



Press the **[F0]** key to delete the data, and press the **[F1]** key to cancel the deletion. When pressing the **[F0]** key, turn the power OFF and then ON.

5.4 Position Setting

Press **[F7]** (Position setting) on the automatic thermal distortion compensation screen.



- Use the cursor keys to move the cursor to the item to be set, enter a value and press the **[ENT]** key.
- Press **[F0]** (Set) to record the setting position, and then it will return to the previous screen.
- After the position is set, the display such as for displacement is deleted and the message <<Turn power off>> is displayed. (When the power is turned OFF and ON, the displacement for the newly set position is calculated.)
- Press **[F1]** (Not set) to delete the setting position, and then it will return to the previous screen.
- Press the **[F7]** (Capture position data) to capture or grab the machine coordinate for the cursor position.

6 Adjust the Compensation Value

To enhance the accuracy more than the basic parameter settings:

If there is a constant shift in the accuracy, the shift can be corrected or minimized by adjusting the compensation value in the user parameter: automatic thermal distortion compensation.

Use the following to calculate the value that is set in the parameter.

S500X1, S700X1, S1000X1, S500X2, S700X2

	When there is a shift in the (+) direction in the coordinates	When there is a shift in the (-) direction in the coordinates
X-axis	Calculate the value using the method in example 1	Calculate the value using the method in example 2
Y-axis	Calculate the value using the method in example 1	Calculate the value using the method in example 2
Z-axis	Calculate the value using the method in example 1	Calculate the value using the method in example 2

S300X1, M140X1, M140X2, S300X2, F600X1, M200X3, M300X3

	When there is a shift in the (+) direction in the coordinates	When there is a shift in the (-) direction in the coordinates
X-axis	Calculate the value using the method in example 2	Calculate the value using the method in example 1
Y-axis	Calculate the value using the method in example 1	Calculate the value using the method in example 2
Z-axis	Calculate the value using the method in example 1	Calculate the value using the method in example 2

R450X1, R650X1, R450X2, R650X2

	When there is a shift in the (+) direction in the coordinates	When there is a shift in the (-) direction in the coordinates
X-axis	Calculate the value using the method in example 2	Calculate the value using the method in example 1
Y-axis	Calculate the value using the method in example 2	Calculate the value using the method in example 1
Z-axis	Calculate the value using the method in example 1	Calculate the value using the method in example 2

These examples are for the Z-axis. To calculate the value for X-axis or Y-axis, replace the Z-axis with the respective axis and calculate accordingly.

Ex.1: When the 15 μm (a) compensation is never enough for target value

Check the displacement at that time.

Displacement of Z-axis is 0.025 (b) mm.

Displacement of spindle is 0.050 (c) mm.

Calculate the shift value for the compensation.

When there is a shift: $a / (b + c) = 15 / (25 + 50) = 0.2$ (20%)

Change the compensation rate.

<Z-axis compensation>: Enter 100 + 20 = 120%

<Spindle compensation (Z-axis)>: Enter 100 + 20 = 120%

Ex.2: When the 15 μm (a) compensation is always too much for target value

Check the displacement at that time.

Displacement of Z-axis is 0.025 (b) mm.

Displacement of spindle is 0.050 (c) mm.

Calculate the shift value for the compensation.

When there is a shift: $a / (b + c) = 15 / (25 + 50) = 0.2$ (20%)

Change the compensation rate.

<Z-axis compensation>: Enter 100 - 20 = 80%

<Spindle compensation (Z-axis)>: Enter 100 - 20 = 80%

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CHAPTER 12 (11)

MEMORY CAPACITY EXPANSION

1 Functions

1 Functions

Expansion the memory capacity is available to save more program and databank in the machine.

Standard type of memory capacity : 100M byte.

Memory capacity after memory expansion : 500M byte.

Note that there are same restrictions as standard memory.

1. Maximum number of files. : 4000
This number is sum of program and data bank.
2. Editable program size : 20M byte
3. Program size changing to expansion memory operation: 2M byte
For details of expansion memory operation, refer to “Chapter 7 Memory operation (NC language)”.

CHAPTER 12 (12)

CHIP CONVEYOR (HINGE + SCRAPER TYPE)

The description in this chapter is omitted because this product is not equipped with this function.

CHAPTER 12 (13)

COOLANT TANK WITH CHUTE

The description in this chapter is omitted because this product is not equipped with this function.

CHAPTER 12 (14)

ROTARY FIXTURE OFFSET FUNCTION

- 1 Overview**
- 2 User Parameter**
- 3 Manual Operation When Function is Enabled**
- 4 Rotary Fixture Offset Setting and Reference**
- 5 Special Notes**
- 6 Rotary Fixture Offset Motion Example**

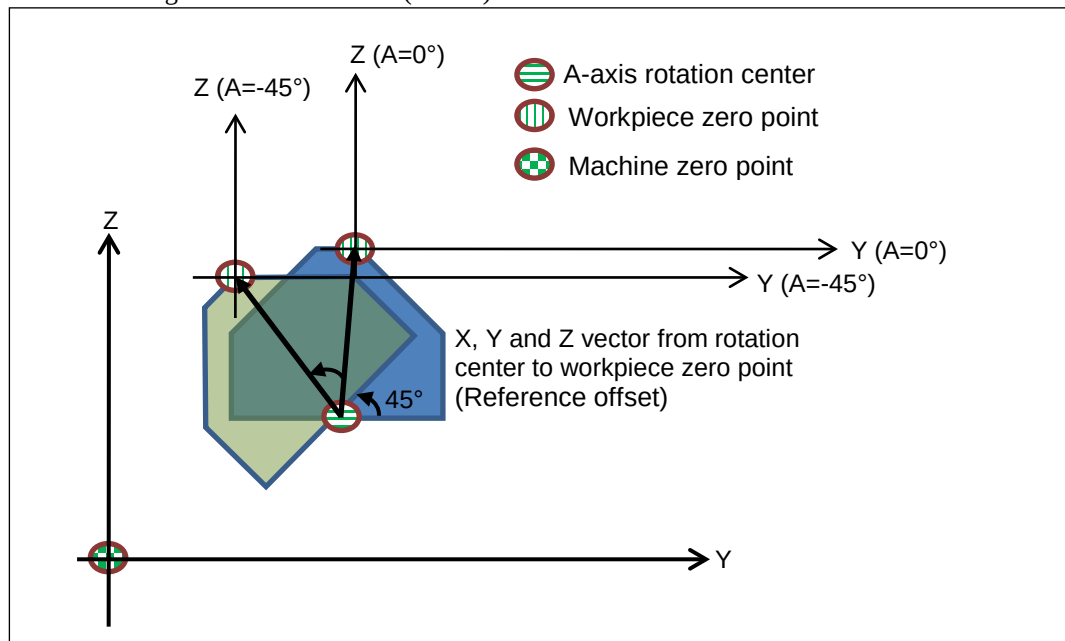
1 Overview

This function is available on machines with an index table.

* The option for rotary fixture offset function is required.

By setting the rotation center for the index table in the workpiece coordinate system (G54 to G59, G54.1, G92, G52 and external workpiece coordinate system shift) as well as setting the X, Y and Z vector from the rotation center to the workpiece zero point, when the table rotates, this function automatically calculates if the workpiece zero point has moved to another point in the machine coordinates. And, it sets a new workpiece coordinate system based on that position.

Ex: When using it on the index table (A-axis) for an X-axis turn



First, decide on the angle (reference angle - 0° in example above) that becomes the reference for index table A-axis. Next, set the X, Y and Z vectors (reference offset) from the rotation center to the workpiece zero point when the A-axis is at the reference angle. If calculating the offset for the A-axis, when the rotary fixture offset command is issued, the workpiece coordinates that are linked with the A-axis rotation are set.

For example, if a positioning command is issued to rotate the A-axis to -45°, the workpiece zero point is calculated from -45°. Therefore, that point becomes the new zero point and the workpiece coordinate system is set accordingly.

The reference angle, the reference offset and the axis for calculation are set for the reference rotary fixture offset. Up to 8 combinations can be registered for the reference rotary fixture offset.

When the “G54.2 Pn” command is issued during memory operation and MDI operation, the rotary fixture offset function is enabled. The reference rotary fixture offset uses a combination of set values for Pn (n:1 to 8) when the command is issued.

In addition, the current rotary fixture offset refers to the the vector from the rotation center to the workpiece zero point when the rotary fixture offset function is enabled.

2 User Parameter

Rotary fixture offset

Name	Setting range	Description
X-, Y- and Z-axes travel during current rotary fixture offset change	0: Type 1 1: Type 2	Sets whether to move the X-, Y- and Z-axes when issuing a G54.2 command or when issuing a rotation command on the additional axis while the rotary fixture offset function is enabled. When set to <0: Type 1>, the X-, Y- and Z-axes do not travel. The current machine coordinates do not change and the absolute coordinates change. When set to <1: Type 2>, the X-, Y- and Z-axes travel. The current absolute coordinates do not change and the machine coordinates change. Travel depends on the modal group which includes G00. However, when using circular interpolation modal, such as G02, and involute interpolation, it operates as G01.
Rotary fixture offset reset method	0: Type 1 1: Type 2	Sets whether to cancel the rotary fixture offset or not, when the [RST] key is pressed or when operation ends. When set to <0: Type 1>, the rotary fixture offset is canceled or reset (G54.2 modal becomes G54.2 P0.). In addition to the reset, axis travel is not carried out. When set to <1: Type 2>, the rotary fixture offset does not cancel or reset.
Tilt axis 1	0: Not specified 1: 4th-axis 2: 5th-axis 3: 6th-axis 4: 7th-axis 5: 8th-axis	Sets the tilt axis for the first group. When set to <0: Not specified> for <Rotation axis 1>, the rotary fixture offset is only calculated from <Tilt axis 1>. Set it to <0: Not specified> when not using the tilt axis.
Rotation axis 1	0: Not specified 1: 4th-axis 2: 5th-axis 3: 6th-axis 4: 7th-axis 5: 8th-axis	Sets the rotation axis for the first group. When an axis is set to <Rotation axis 1>, the rotary fixture offset will be calculated as a component that is installed on the axis that is set to <Tilt axis 1>. When set to <0: Not specified> for <Tilt axis 1>, the rotary fixture offset is only calculated from <Rotation axis 1>. Set it to <0: Not specified> when not using the rotation axis.
Tilt axis 2	0: Not specified 1: 4th-axis 2: 5th-axis 3: 6th-axis 4: 7th-axis 5: 8th-axis	Sets the tilt axis for the second group. When set to <0: Not specified> for <Rotation axis 2>, the rotary fixture offset is only calculated from <Tilt axis 2>. Set it to <0: Not specified> when not using the tilt axis.
Rotation axis 2	0: Not specified 1: 4th-axis 2: 5th-axis 3: 6th-axis 4: 7th-axis 5: 8th-axis	Sets the rotation axis for the second group. When an axis is set to <Rotation axis 2>, the rotary fixture offset will be calculated as a component that is installed on the axis that is set to <Tilt axis 2>. When set to <0: Not specified> for <Tilt axis 2>, the rotary fixture offset is only calculated from <Rotation axis 2>. Set it to <0: Not specified> when not using the rotation axis.
Tilt axis 3	0: Not specified 1: 4th-axis 2: 5th-axis 3: 6th-axis 4: 7th-axis 5: 8th-axis	Sets the tilt axis for the third group. When set to <0: Not specified> for <Rotation axis 3>, the rotary fixture offset is only calculated from <Tilt axis 3>. Set it to <0: Not specified> when not using the tilt axis.

Name	Setting range	Description
Rotation axis 3	0: Not specified 1: 4th-axis 2: 5th-axis 3: 6th-axis 4: 7th-axis 5: 8th-axis	Sets the rotation axis for the third group. When an axis is set to <Rotation axis 3>, the rotary fixture offset will be calculated as a component that is installed on the axis that is set to <Tilt axis 3>. When set to <0: Not specified> for <Tilt axis 3>, the rotary fixture offset is only calculated from <Rotation axis 3>. Set it to <0: Not specified> when not using the rotation axis.
Tilt axis 4	0: Not specified 1: 4th-axis 2: 5th-axis 3: 6th-axis 4: 7th-axis 5: 8th-axis	Sets the tilt axis for the fourth group. When set to <0: Not specified> for <Rotation axis 4>, the rotary fixture offset is only calculated from <Tilt axis 4>. Set it to <0: Not specified> when not using the tilt axis.
Rotation axis 4	0: Not specified 1: 4th-axis 2: 5th-axis 3: 6th-axis 4: 7th-axis 5: 8th-axis	Sets the rotation axis for the fourth group. When an axis is set to <Rotation axis 4>, the rotary fixture offset will be calculated as a component that is installed on the axis that is set to <Tilt axis 4>. When set to <0: Not specified> for <Tilt axis 4>, the rotary fixture offset is only calculated from <Rotation axis 4>. Set it to <0: Not specified> when not using the rotation axis.
Forward direction for the coordinate system on tilt axis 1 Forward direction for the coordinate system on rotation axis 1 Forward direction for the coordinate system on tilt axis 2 Forward direction for the coordinate system on rotation axis 2 Forward direction for the coordinate system on tilt axis 3 Forward direction for the coordinate system on rotation axis 3 Forward direction for the coordinate system on tilt axis 4 Forward direction for the coordinate system on rotation axis 4	0: Y→Z 1: Z→Y 2: Z→X 3: X→Z 4: X→Y 5: Y→X	Sets the forward direction for the coordinate system on each axis. 0: Y → Z - It rotates from the Y-axis forward direction to the Z-axis forward direction. 1: Z → Y - It rotates from the Z-axis forward direction to the Y-axis forward direction. 2: Z → X - It rotates from the Z-axis forward direction to the X-axis forward direction. 3: X → Z - It rotates from the X-axis forward direction to the Z-axis forward direction. 4: X → Y - It rotates from the X-axis forward direction to the Y-axis forward direction. 5: Y → X - It rotates from the Y-axis forward direction to the X-axis forward direction.

(Parameter setting example)

1. Tilt axis 1 to 4

When setting the rotation axis for 2 axes on Pallet 1 and 2 on the QT model

<Rotation axis 1>: 2: 5th-axis

<Rotation axis 2>: 3: 6th-axis

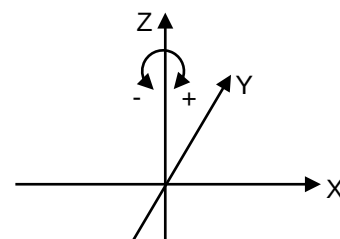
<Rotation axis 3>: 4: 7th-axis

<Rotation axis 4>: 5: 8th-axis

2. Forward direction for the coordinate system on tilt axis 1

Looking from the positive direction on the Z-axis,
clockwise is the positive direction for tilt axis 1

<Forward direction for the coordinate system on tilt axis 1>: 5 (Y → X)



3 Manual Operation When Function is Enabled

When the rotary fixture offset function is enabled, the workpiece coordinate system does not change even if the index table is rotated using manual operation. When a G54.2 command is executed or when an axis travel command on the rotation axis is executed during memory operation or MDI operation, the travel position is calculated for the workpiece zero point. And, the workpiece coordinate system is changed.

4 Rotary Fixture Offset Setting and Reference

4.1 Rotary Fixture Offset Setting and Reference on the Screen

The reference rotary fixture offset can be set at the reference rotary fixture offset screen. In addition, the current rotary fixture offset can be referenced at the position screen.

Refer to section “2.2 Workpiece coordinate zero (NC)” in the Operation Manual (Data) and section “4.1 Position” in the Operation Manual for further details.

4.2 Programmable Data Input

The reference rotary fixture offset settings can be changed while the program is running.

Refer to section “3.6 Programmable data input” in the Programming Manual (NC language) for further details.

4.3 System Variables

The reference rotary fixture offset settings can be referenced and overwritten using the system variables.

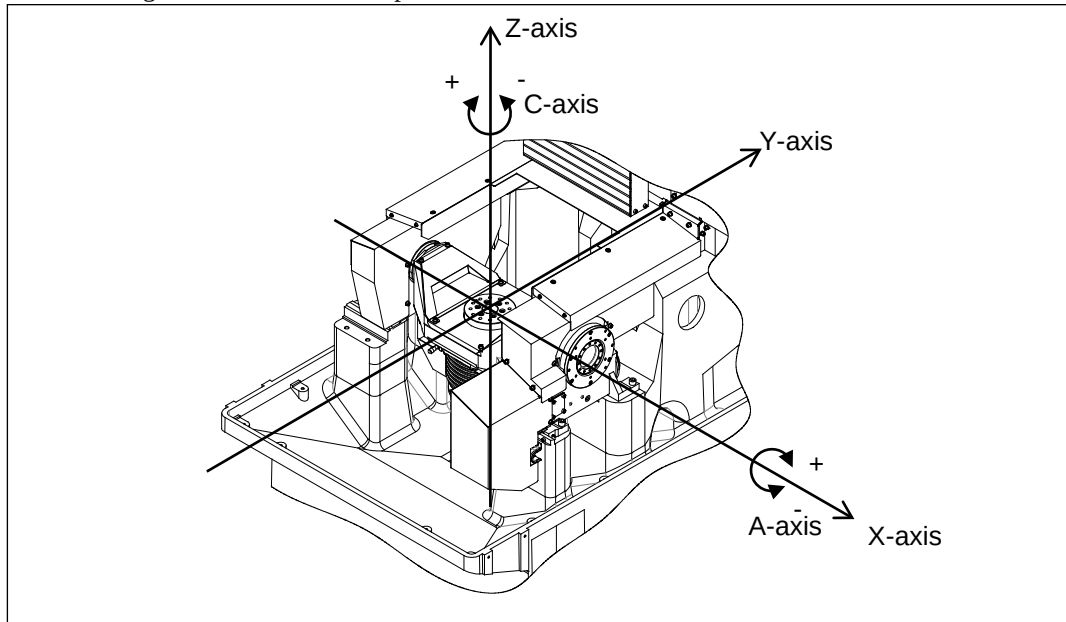
Refer to section “6.2 Variable Function” in the Programming Manual (NC language) for further details.

5 Special Notes

1. When setting up multiple groups (multiple group setting on a single side pallet for the QT model) for the tilt axis or rotation axis on the table to use the rotary fixture offset function, you must set up and use the reference rotary fixture offset for each group.
2. On the QT model, if the <Rotary fixture offset reset method> is set to <1 : Method 2>, the machine operates using the pallet coordinate system from the previous operation when switching between pallet 1 and pallet 2 operation.
When using a QT model, put a G54.2 command at the beginning of the program or set the <Rotary fixture offset reset method> to <0 : Method 1>.
3. The rotary fixture offset function is a coordinate conversion function. It does not offer compensation for indexing accuracy that is specific to each machine. Check the indexing accuracy, and correct it when necessary.
4. When <X-, Y- and Z-axes travel during current rotary fixture offset change> is set to <1: Type 2> and the X-, Y- and Z-axes travel in another modal besides G0, the alarm <<Feedrate not specified>> is triggered if there is no F command during G93 (inverse time feed).

6 Rotary Fixture Offset Motion Example

The following shows a motion example for a M140X2.



Machine parameter

<5th-axis address>: 0 (A)

<7th-axis address>: 2 (C)

User parameter

<Tilt axis 1>: 2 (5th-axis)

<Rotation axis 1>: 4 (7th-axis)

<Forward direction for the coordinate system on tilt axis 1>: 1 (Z → Y)

<Forward direction for the coordinate system on rotation axis 1>: 5 (Y → X)

Workpiece coordinate system selection (G54) = A-axis and C-axis rotation center

X = -100.000, Y = -300.000, Z = 30.000

A = 0.000, C = 0.000

Reference rotary fixture offset (G54.2 P1)

Reference angle (A, B, C) = (0.000, 0.000, 0.000)

Reference offset (X, Y, Z) = (5.000, 10.000, 20.000)

Axis for calculation = A, C

N1 G90 G54 G00 X0 Y0 Z200. A0. C0.

N2 G54.2 P1

N3 G00 A45.

N4 G01 X2. F1000

N5 G00 A60. C270.

N6 G54.2 P0

User parameter <X-, Y- and Z-axes travel during current rotary fixture offset change>: 0 (Type 1)

	Machine coordinates					Workpiece coordinates					Current rotary fixture offset		
	X	Y	Z	A	C	X	Y	Z	A	C	X	Y	Z
N1	-100.000	-300.000	230.000	0.000	0.000	0.000	0.000	200.000	0.000	0.000	0.000	0.000	0.000
N2	-100.000	-300.000	230.000	0.000	0.000	-5.000	-10.000	180.000	0.000	0.000	5.000	10.000	20.000
N3	-100.000	-300.000	230.000	45.000	0.000	-5.000	-21.213	192.929	45.000	0.000	5.000	21.213	7.071
N4	-93.000	-300.000	230.000	45.000	0.000	2.000	-21.213	192.929	45.000	0.000	5.000	21.213	7.071
N5	-93.000	-300.000	230.000	60.000	270.000	17.000	-19.821	194.330	60.000	270.000	-10.000	19.821	5.670
N6	-93.000	-300.000	230.000	60.000	270.000	7.000	0.000	200.000	60.000	270.000	0.000	0.000	0.000

User parameter <X-, Y- and Z-axes travel during current rotary fixture offset change>: 1 (Type 2)

	Machine coordinate					Workpiece coordinates					Current rotary fixture offset		
	X	Y	Z	A	C	X	Y	Z	A	C	X	Y	Z
N1	-100.000	-300.000	230.000	0.000	0.000	0.000	0.000	200.000	0.000	0.000	0.000	0.000	0.000
N2	-95.000	-290.000	250.000	0.000	0.000	0.000	0.000	200.000	0.000	0.000	5.000	10.000	20.000
N3	-95.000	-278.787	237.071	45.000	0.000	0.000	0.000	200.000	45.000	0.000	5.000	21.213	7.071
N4	-93.000	-278.787	237.071	45.000	0.000	2.000	0.000	200.000	45.000	0.000	5.000	21.213	7.071
N5	-108.000	-280.179	235.670	60.000	270.000	2.000	0.000	200.000	60.000	270.000	-10.000	19.821	5.670
N6	-108.000	-280.179	235.670	60.000	270.000	-8.000	19.821	205.670	60.000	270.000	0.000	0.000	0.000

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CHAPTER 12 (15)

SWITCH BOX FOR MAGAZINE TURNING

The description in this chapter is omitted because this product is not equipped with this function.

CHAPTER 12 (16)

HIGH SPEED PROCESSING MODE

- 1 Overview
- 2 Restrictions

1 Overview

This function is an option only for NC language mode. It is not available for customers that use conversation language.

When there is a fast feedrate for a command in the NC program, it may decelerate on cutting blocks that cover very small distances. This deceleration is influenced by the processing performance of the NC unit. This type of deceleration is often seen when there are successive commands on a block for linear interpolation for machining a shape, such as when there are a lot of cutting blocks with little travel on a section where there is a significant curve.

The high speed processing mode is used in situations as described above. This specification provides 2 times faster processing speed than the standard specification and maintains the original machining speed as much as possible regardless if there are cutting blocks that cover very small distances.

When the following commands are issued, the speed on a standard specification will decelerate. However, in high speed processing mode, it will maintain the command speed without decelerating.

Linear interpolation blocks with a speed of 4000 mm/min and a travel distance of 0.100 mm

Linear interpolation blocks with a speed of 8000 mm/min and a travel distance of 0.200 mm

Nevertheless, if a rapid feed command is issued for an extremely short distance, it may decelerate. However, this special specification will decelerate less than the standard specification.

2 Restrictions

When the high speed processing specification is selected, it has the following restrictions.

- Conversation language cannot be selected.
- High-speed ladder 2 in the built-in PLC cannot be used. In addition, high-speed ladder 2 requires changing to a ladder program for the END command only.
- The scan time for standard ladder 1 and 2 in the built-in PLC may exceed the maximum (double).
- The set values and valid range for high accuracy mode B in the machine parameter will partially change.

CHAPTER 12 (17)

A-AXIS CLAMP

The description in this chapter is omitted because this product is not equipped with this function.

CHAPTER 12 (18)

FIELD BUS

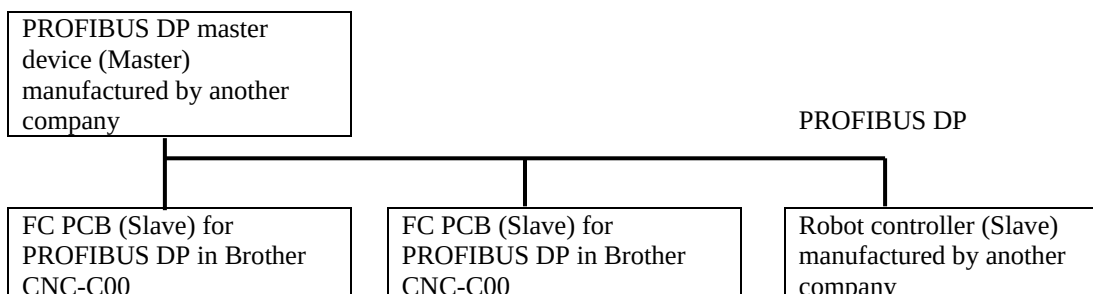
- 1 PROFIBUS DP (Slave)**
- 2 DeviceNet (Slave)**
- 3 CC-Link (Remote Device Station)**
- 4 CC-Link (Master Station)**

1 PROFIBUS DP (Slave)

1.1 PROFIBUS DP Function Overview

1.1.1 Function Overview

PROFIBUS DP is a fieldbus network that can connect various devices such as a CNC, PLC and robot controller. The PROFIBUS DP slave function is supported by connecting the FC PCB (= Fieldbus Conversion PCB) for the PROFIBUS DP to the CNC-C00.



The CNC-C00 supports the following PROFIBUS DP slave function.

Compatible version	PROFIBUS DP-V0
Number of I/O points	Input: 1024 bit Output: 1024 bit
Transfer rate	Automatic detection

1.1.2 Wiring Method

For further details, refer to “Chapter 11 FC PCB” in the Installation Manual.

1.2 Setting & Monitor Screen

1.2.1 Basic Settings for PROFIBUS DP

The basic settings can be configured for the PROFIBUS DP slave.

Procedure

1. Select <User parameter> on the data bank menu screen.
2. The following screen appears when <PROFIBUS DP> is selected from the function keys. Move the cursor to the item to be set, and enter the parameter.



Parameter list

Parameter name	Setting range	Description
<Node address>	0 - 126	Sets the node address.
<I/O setting 1 to 20>	Up to 4 characters	Sets the input size from the PROFIBUS DP to the NC and the output size from the NC to the PROFIBUS DP.

3. After the setting is completed, press the **[F0]** (Completion) key to complete and exit the editing.

(Note) After changing the parameter, the alarm <<Turn power off>> is triggered.

Set values for I/O settings 1 to 20

The data size, which is assigned to the 128 byte input and 128 byte output range, is defined. When multiple sizes are defined, the definitions are left-justified and arranged in order starting with the leading address.

When spacing the assigned addresses, adjust by adding dummy data to the spacing size definition.

Data size	Input setting NC→PROFIBUS	Output setting PROFIBUS→NC	Data size	Input setting NC→PROFIBUS	Output setting PROFIBUS→NC
1 byte	90	A0	1 word	D0	E0
2 bytes	91	A1	2 words	D1	E1
3 bytes	92	A2	3 words	D2	E2
4 bytes	93	A3	4 words	D3	E3
8 bytes	97	A7	8 words	D7	E7
12 bytes	9B	AB	12 words	DB	EB
16 bytes	9F	AF	16 words	DF	EF
20 bytes (Note 2)	4093	8093	20 words (Note 2)	40D3	80D3
32 bytes (Note 2)	409F	809F	32 words (Note 2)	40DF	80DF
64 bytes (Note 2)	40BF	80BF	64 words (Note 2)	40FF	80FF

Ex: When setting an 8 byte input to the input address 0x10, a 1 byte input to 0x18, a 2 word input to 0x1C, a 64 byte output to 0x40:

I/O setting 1:9F (dummy input for 16 bytes (0x10))
 I/O setting 2:97 (8 byte input)
 I/O setting 3:90 (1 byte input)
 I/O setting 4:92 (dummy data definition for 3 bytes)
 I/O setting 5:D1 (2 word input)
 I/O setting 6:80BF (dummy output for 64 bytes (0x40))
 I/O setting 7:80BF (64 byte output)
 I/O settings 8 to 20: Not set

(Note 1) The <<User parameter setting error (PROFIBUS DP)>> is triggered when turning ON the power in the following situations.

- When data other than the data specified in the table above is entered for input settings 1 through 20
- When the overall size of either the input and output definitions, which are set to the I/O settings 1 through 20, exceeds 128 bytes

(Note 2) There are cases when support does not work due to the master device. In these cases, the connection may have failed because of the data size settings.

Configure multiple settings using 16 byte/16 word (or less) data.

Ex: When using 64 word input, configure the settings as shown below on the NC unit.

I/O setting 1: DF (16 word input)
 I/O setting 2: DF (16 word input)
 I/O setting 3: DF (16 word input)
 I/O setting 4: DF (16 word input)

Configure the same size setting for the master device as well.

1.2.2 NC Input Signals

NC input signals* are assigned to the FC PCB that is used for the PROFIBUS DP (slave).

* NC input signals are the output signals when looking from the PROFIBUS DP master side.

Procedure

1. Select <External I/O signal> on the data bank menu screen.
2. Press the [>>] key until the function key selections change to the following display.

Completion mode	PROFIBUS DP -> NC	NC -> PROFIBUS DP	PROFIBUS DP status			Search	Next menu
-----------------	-------------------	-------------------	--------------------	--	--	--------	-----------

3. Press the [F1] (PROFIBUS DP → NC) key to display the following screen.

PROFIBUS DP slave PROFIBUS DP → NC 02-d 2013/04/18 19:07:41							
PROFIBUS DP slave FC board for PROFIBUS DP							
Signal	Signal	Signal	Signal	Signal	Signal	Signal	Signal
0x000 PLCX000-00F	0x008 PLCX000-00F	0x010 PLCX010-01F	0x018 PLCX010-01F	0x001 PLCX000-00F	0x009 PLCX000-00F	0x011 PLCX010-01F	0x019 PLCX010-01F
0x002 PLCX000-00F	0x00A PLCX000-00F	0x012 PLCX010-01F	0x01A PLCX010-01F	0x003 PLCX000-00F	0x00B PLCX000-00F	0x013 PLCX010-01F	0x01B PLCX010-01F
0x004 PLCX000-00F	0x00C PLCX000-00F	0x014 PLCX010-01F	0x01C PLCX010-01F	0x005 PLCX000-00F	0x00D PLCX000-00F	0x015 PLCX010-01F	0x01D PLCX010-01F
0x006 PLCX000-00F	0x00E PLCX000-00F	0x016 PLCX010-01F	0x01E PLCX010-01F	0x007 PLCX000-00F	0x00F PLCX000-00F	0x017 PLCX010-01F	0x01F PLCX010-01F
0x000 [] Check OK [] [] Can be deleted							
900. PLCX000-00F	910. PLCX0A0-0AF	920. PLCX140-14F	930. PLCX1E0-1EF	940. PLCX280-28F	950. PLCX320-32F	960. PLCX360-36F	970. PLCX400-40F
901. PLCX010-01F	911. PLCX0B0-0BF	921. PLCX150-15F	931. PLCX1F0-1FF	941. PLCX290-29F	951. PLCX330-33F	961. PLCX370-37F	971. PLCX410-41F
902. PLCX020-02F	912. PLCX0C0-0CF	922. PLCX160-16F	932. PLCX200-20F	942. PLCX2A0-2AF	952. PLCX340-34F	962. PLCX380-38F	972. PLCX420-42F
903. PLCX030-03F	913. PLCX0D0-0DF	923. PLCX170-17F	933. PLCX210-21F	943. PLCX2B0-2BF	953. PLCX350-35F	963. PLCX390-39F	973. PLCX430-43F
904. PLCX040-04F	914. PLCX0E0-0EF	924. PLCX180-18F	934. PLCX220-22F	944. PLCX2C0-2CF	954. PLCX360-36F	964. PLCX400-40F	974. PLCX440-44F
905. PLCX050-05F	915. PLCX0F0-0FF	925. PLCX190-19F	935. PLCX230-23F	945. PLCX2D0-2DF	955. PLCX370-37F	965. PLCX410-41F	975. PLCX450-45F
906. PLCX060-06F	916. PLCX100-10F	926. PLCX1A0-1AF	936. PLCX240-24F	946. PLCX2E0-2EF	956. PLCX380-38F	966. PLCX420-42F	976. PLCX460-46F
907. PLCX070-07F	917. PLCX110-11F	927. PLCX1B0-1BF	937. PLCX250-25F	947. PLCX2F0-2FF	957. PLCX390-39F	967. PLCX430-43F	977. PLCX470-47F
908. PLCX080-08F	918. PLCX120-12F	928. PLCX1C0-1CF	938. PLCX260-26F	948. PLCX300-30F	958. PLCX3A0-3AF	968. PLCX440-44F	978. PLCX480-48F
909. PLCX090-09F	919. PLCX130-13F	929. PLCX1D0-1DF	939. PLCX270-27F	949. PLCX310-31F	959. PLCX3B0-3BF	969. PLCX450-45F	979. PLCX490-49F
Completion mode	PROFIBUS DP -> NC	NC -> PROFIBUS DP	PROFIBUS DP slave status	Overlapping terminal check	Search	Next menu	

Move the cursor to the address to be set, and set the signal number to be assigned.

The output signals that can be assigned are indicated below.

- Signals noted in the external output signal list found in section “2.6.2.2 Data input” from the Operation Manual (Data).
- Setting range: 2000 to 4047 (PLCD0000 to PLCD2047)

(Note) If the same signal is repeated and assigned while in <PROFIBUS DP status>, the PROFIBUS DP status information becomes valid. Do not repeat and assign the signal.

Press the [▼] key to show the addresses that can be set and that are offset by 32 bits. The addresses can be assigned up to 0x3FFF. In addition, after keying in the address to be changed, press the [F6] (Search) key to move the cursor to the specified address.

Ex: Use the search function to move the cursor to 0x3F0

Input the [3], [F] and [0] keys, and press the [F6] key

(Note) The greyed out locations indicate that the FC PCB (PROFIBUS DP) is not connected or that the address falls outside of the usage range (addresses that are larger than the overall size of the input definition set in <I/O settings 1-20> for the PROFIBUS DP slave parameter). Even if the address is greyed out, the signal can be assigned but will not input into the NC.

* After the setting is completed, the alarm <<Press the [RST] key>> is triggered.

1.2.3 NC Output Signals

NC output signals* are assigned to the FC PCB that is used for the PROFIBUS DP (slave).

* NC output signals are the input signals when looking from the PROFIBUS DP master side.

Procedure

1. Select <External I/O signal> on the data bank menu screen.
2. Press the [>>] key until the function key selections change to the following display.

Completion mode	PROFIBUS DP -> NC	NC -> PROFIBUS DP	PROFIBUS DP status			Search	Next menu
-----------------	-------------------	-------------------	--------------------	--	--	--------	-----------

3. Press the [F2] (NC → PROFIBUS DP) key to display the following screen.

PROFIBUS DP slave NC -> PROFIBUS DP				2013/04/18 19:08:44	
PROFIBUS DP slave FC board for PROFIBUS DP					
Signal		Signal		Signal	
0x000	PLCY000-00F	0x008	PLCY000-00F	0x010	PLCY010-01F
0x001	PLCY000-00F	0x009	PLCY000-00F	0x011	PLCY010-01F
0x002	PLCY000-00F	0x00A	PLCY000-00F	0x012	PLCY010-01F
0x003	PLCY000-00F	0x00B	PLCY000-00F	0x013	PLCY010-01F
0x004	PLCY000-00F	0x00C	PLCY000-00F	0x014	PLCY010-01F
0x005	PLCY000-00F	0x00D	PLCY000-00F	0x015	PLCY010-01F
0x006	PLCY000-00F	0x00E	PLCY000-00F	0x016	PLCY010-01F
0x007	PLCY000-00F	0x00F	PLCY000-00F	0x017	PLCY010-01F
Can be deleted					
900. PLCY000-00F	910. PLCY0A0-0AF	920. PLCY140-14F	930. PLCY1E0-1EF	940. PLCY280-28F	950. PLCY320-32F
901. PLCY010-01F	911. PLCY0B0-0BF	921. PLCY150-15F	931. PLCY1F0-1FF	941. PLCY290-29F	951. PLCY330-33F
902. PLCY020-02F	912. PLCY0C0-0CF	922. PLCY160-16F	932. PLCY200-20F	942. PLCY2A0-2AF	952. PLCY340-34F
903. PLCY030-03F	913. PLCY0D0-0DF	923. PLCY170-17F	933. PLCY210-21F	943. PLCY2B0-2BF	953. PLCY350-35F
904. PLCY040-04F	914. PLCY0E0-0EF	924. PLCY180-18F	934. PLCY220-22F	944. PLCY2C0-2CF	954. PLCY360-36F
905. PLCY050-05F	915. PLCY0F0-0FF	925. PLCY190-19F	935. PLCY230-23F	945. PLCY2D0-2DF	955. PLCY370-37F
906. PLCY060-06F	916. PLCY100-10F	926. PLCY1A0-1AF	936. PLCY240-24F	946. PLCY2E0-2EF	956. PLCY380-38F
907. PLCY070-07F	917. PLCY110-11F	927. PLCY1B0-1BF	937. PLCY250-25F	947. PLCY2F0-2FF	957. PLCY390-39F
908. PLCY080-08F	918. PLCY120-12F	928. PLCY1C0-1CF	938. PLCY260-26F	948. PLCY300-30F	958. PLCY3A0-3AF
909. PLCY090-09F	919. PLCY130-13F	929. PLCY1D0-1DF	939. PLCY270-27F	949. PLCY310-31F	959. PLCY3B0-3BF
Completion mode	PROFIBUS DP -> NC	NC -> PROFIBUS DP	PROFIBUS DP slave status	Overlapping terminal check	Search
				Next menu	

Move the cursor to the address to be set, and set the signal number to be assigned.

The output signals that can be assigned are indicated below.

- Signals noted in the external output signal list found in section “2.6.2.2 Data input” from the Operation Manual (Data).
- Setting range: 2000 to 4047 (PLCD0000 to PLCD2047)

Press the [▼] key to show the addresses that can be set and that are offset by 32 bits. The addresses can be assigned up to 0x3FF. In addition, after keying in the address to be changed, press the [F6] (Search) key to move the cursor to the specified address.

Ex: Use the search function to move the cursor to 0x3F0
Input the [3], [F] and [0] keys, and press the [F6] key

(Note) The greyed out locations indicate that the FC PCB (PROFIBUS DP) is not connected or that the address falls outside of the usage range (addresses that are larger than the overall size of the output definition set in <I/O settings 1-20> for the PROFIBUS DP slave parameter. Even if the address is greyed out, the signal can be assigned but it will not output from the NC.

* After the setting is completed, the alarm <<Press the [RST] key>> is triggered.

1.2.4 Status

The FC PCB status information for the PROFIBUS DP can be assigned as a PLC input signal. The status information that can be referenced is shown below.

Status	Description		Content
STATUS	0 bit	Initialization completion flag	0: Not initialized 1: Initialized
	1 bit	Communicating flag	0: No communication 1: Communicating
	2 bit	Error flag	0: Normal 1: Error
	3 bit...15bit	Not used	0
ST_INIT	0 bit	FC PCB initialization flag	0: Initializing 1: Initialized
	1 bit	Communication standby	0: Normal 1: Standby
	2 bit...15 bit	Not used	0
ST_ERR	0 bit	(Duplicate node)	0 (Fixed)
	1 bit	Communication time out	0: Normal 1: Communication time out
	2 bit...14 bit	Not used	0
	15 bit	Other network errors	0: Normal 1: Network error triggered

Procedure

1. Select <External I/O signal> on the data bank menu screen.
2. Press the [>>] key until the function key selections change to the following display.

Completion Mode	PROFIBUS DP -> NC	NC -> PROFIBUS DP	PROFIBUS DP status			Search	Next menu
-----------------	-------------------	-------------------	--------------------	--	--	--------	-----------

3. Press the [F3] (PROFIBUS DP status) key to display the following screen.

Status	Signal
STATUS	
ST_INIT	
ST_ERR	

STATUS Check OK

999, RESERV
2000, PLCD0000
2001, PLCD0001
2002, PLCD0002
...
4045, PLCD2045
4046, PLCD2046
4047, PLCD2047

Completion mode	PROFIBUS DP -> NC	NC -> PROFIBUS DP	PROFIBUS DP slave status	Overlapping terminal check	Search	Next menu
-----------------	-------------------	-------------------	--------------------------	----------------------------	--------	-----------

Move the cursor to the address to be set, and enter the signal number to be assigned. However, only the signals noted below can be assigned to an address. The operator message <<Input data error>> is triggered if another signal, other than those signals noted, is assigned.

- 999 (RESERV)
- 2000 to 4047 (PLCD0000 to PLCD2047)

(Note 1) If the FC PCB (PROFIBUS DP) is not connected, the address is greyed out. Even if the address is greyed out, the signal can be assigned but will not input into the NC.

(Note 2) If the [F6] (Search) key and the [F7] (Next menu) key is greyed out on the PROFIBUS DP slave status screen, the keys cannot be input.

* After the setting is completed, the alarm <<Press the [RST] key>> is triggered.

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2. Press the **[F5]** (Field bus) key to display the following screen.

(Total of 3 pages)

Status	Description	
<Bus type>	Indicates the type of the fieldbus that is connected to the FC PCB.	
	PROFIBUS DP	Indicates that the PROFIBUS-DP is connected.
	Not connected	Indicates that the bus is not connected.
	Error	Displayed when there is an error.
<Status>	Indicates the current communication status.	
	Not connected	Indicates that the FC PCB is not connected.
	Initializing	Indicates that the FC PCB is initializing.
	Waiting for configuration	Indicates that communication is being established with the master device.
	Communicating	Indicates that the FC PCB is operating normally.
	Communication stop	Indicates that communication has stopped.
	Error	Indicates that there is an error.
<Address>	Indicates the node address that is set to the FC PCB.	
	0 to 126	When the profibus is connected
	0 to 63	When DeviceNet is connected
	1 to 64	When CC-Link is connected
	255	Indicates when not connected and when the address value is not valid.

Status	Description	
Transfer rate	Indicates the transfer rate that is set to the FC PCB.	
	9.6 kbps	Indicates the PROFIBUS DP transfer rate.
	31.25 kbps	
	45.45 kbps	
	19.2 kbps	
	93.75 kbps	
	187.5 kbps	
	500 kbps	
	1.5 Mbps	
	3 Mbps	
	6 Mbps	
	12 Mbps	
	125 kbps	Indicates the DeviceNet transfer rate.
	250 kbps	
	500 kbps	
	156 kbps	Indicates the CC-Link transfer rate.
	625 kbps	
	2.5 Mbps	
	5 Mbps	
	10 Mbps	
	(Blank field)	A blank field is displayed for any rate not noted above.
<System status>	Indicates the system status for the FC PCB.	
	0	Indicates that it is operating normally.
	Other than 0	Indicates that there is a system error on netX.
<Communication error>	Indicates that there is an FC PCB communication error.	
	0	Indicates that it is operating normally.
	Other than 0	Indicates that there is a communication error on netX.
<Commun. error counter>	0 to 999999	Indicates the number of communication errors that occurred.

3. Press the **[PAGE DOWN]** key to display the I/O signal status for the field bus.

(From page 2 on)

Input/output (field bus)				2013/04/18 19:10:53			
Input				Output			
FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210
1 0000 0000 0000 0000	17 0000 0000 0000 0000	2 0000 0000 0000 0000	18 0000 0000 0000 0000	1 0000 0000 0000 0000	17 0000 0000 0000 0000	2 0000 0000 0000 0000	18 0000 0000 0000 0000
3 0000 0000 0000 0000	19 0000 0000 0000 0000	4 0000 0000 0000 0000	20 0000 0000 0000 0000	3 0000 0000 0000 0000	19 0000 0000 0000 0000	4 0000 0000 0000 0000	20 0000 0000 0000 0000
5 0000 0000 0000 0000	21 0000 0000 0000 0000	6 0000 0000 0000 0000	22 0000 0000 0000 0000	5 0000 0000 0000 0000	21 0000 0000 0000 0000	6 0000 0000 0000 0000	22 0000 0000 0000 0000
7 0000 0000 0000 0000	23 0000 0000 0000 0000	8 0000 0000 0000 0000	24 0000 0000 0000 0000	7 0000 0000 0000 0000	23 0000 0000 0000 0000	8 0000 0000 0000 0000	24 0000 0000 0000 0000
9 0000 0000 0000 0000	25 0000 0000 0000 0000	10 0000 0000 0000 0000	26 0000 0000 0000 0000	9 0000 0000 0000 0000	25 0000 0000 0000 0000	10 0000 0000 0000 0000	26 0000 0000 0000 0000
11 0000 0000 0000 0000	27 0000 0000 0000 0000	12 0000 0000 0000 0000	28 0000 0000 0000 0000	11 0000 0000 0000 0000	27 0000 0000 0000 0000	12 0000 0000 0000 0000	28 0000 0000 0000 0000
13 0000 0000 0000 0000	29 0000 0000 0000 0000	14 0000 0000 0000 0000	30 0000 0000 0000 0000	13 0000 0000 0000 0000	29 0000 0000 0000 0000	14 0000 0000 0000 0000	30 0000 0000 0000 0000
15 0000 0000 0000 0000	31 0000 0000 0000 0000	16 0000 0000 0000 0000	32 0000 0000 0000 0000	15 0000 0000 0000 0000	31 0000 0000 0000 0000	16 0000 0000 0000 0000	32 0000 0000 0000 0000
FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210
Output 0000000000000000				Status			
FEDCBA9876543210				Field bus 033-064			
I/O menu	Main	Local	Slave status display	Extension I/O	Field bus	Version	

After the input bytes and output bytes for the bus that is set, "0" is used for the remaining bits.

1.3 GSD FILE

1.3.1 What is the GSD File?

The GSD file is a text format file (filename extension (.gsd)) with a description about the PROFIBUS DP slave device. The GSD file can be used to set the master parameter for the PROFIBUS DP master station.

1.3.2 GSD File for PROFIBUS DP on FC PCB

Refer to BRO0E32.gsd for the GSD file.

1.3.3 How to Obtain the GSD Files

To obtain the GSD files, download them from the Profibus International (PROFIBUS/PROFINET) website or use the PLC Programming Software installation CD.

Profibus International homepage (PROFIBUS/PROFINET)

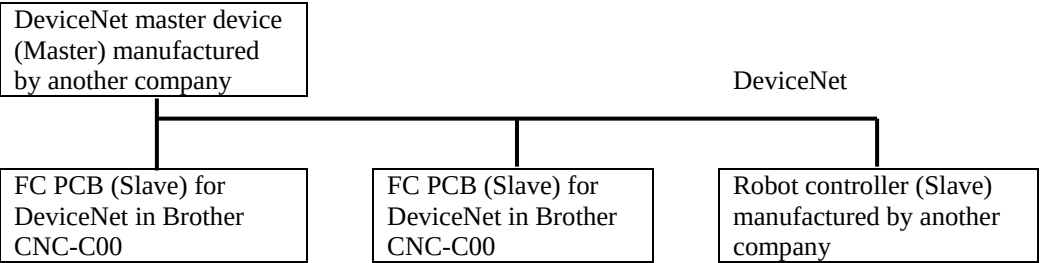
<http://www.profibus.com/>

2 DeviceNet (Slave)

2.1 DeviceNet Function Overview

2.1.1 Function Overview

DeviceNet is a field bus network that can connect various devices such as a CNC, PLC and robot controller. The DeviceNet slave function is supported by connecting the FC PCB (= Fieldbus Conversion PCB) for the DeviceNet to the CNC-C00.



The CNC-C00 supports the following DeviceNet slave function.

Transfer rate	125k/250k/500k bps (Parameter selection)
---------------	--

2.1.2 Wiring Method

For further details, refer to “Chapter 11 FC PCB” in the Installation Manual.

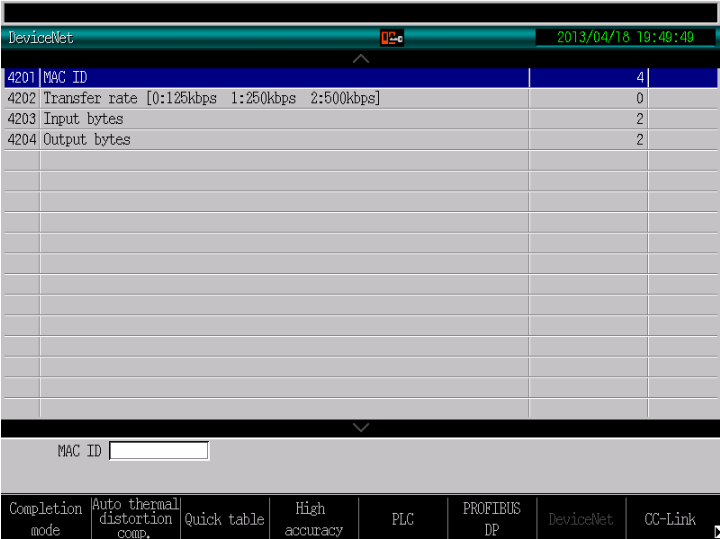
2.2 Setting & Monitor Screen

2.2.1 Basic Settings for DeviceNet

The basic settings can be configured for DeviceNet.

Procedure

1. Select <User parameter> on the data bank menu screen.
2. The following screen appears when <DeviceNet> is selected from the function keys. Move the cursor to the item to be set, and enter the parameter.



Parameter list

Parameter name	Setting range	Description
<MAC ID>	0 to 63	Sets the MAC ID for DeviceNet.
<Transfer rate>	0: 125kbps 1: 250kbps 2: 500kbps	Sets the transfer speed.
<Input bytes>	0 to 128	Sets the bytes that are input from DeviceNet into the NC. There is no data that is input from DeviceNet when set to 0.
<Output bytes>	0 to 128	Sets the bytes that are output from the NC into DeviceNet. There is no data that is output to DeviceNet when set to 0.

- After the setting is completed, press the **[F0]** (Completion mode) key to complete and exit the editing.

(Note) After changing the parameter, the alarm <<Turn power off>> is triggered.

2.2.2 NC Input Signals

NC input signals* are assigned to the FC PCB for the DeviceNet (slave).

* NC input signals are the output signals when looking from the DeviceNet master side.

Procedure

- Select <External I/O signal> on the data bank menu screen.
- Press the **[>>]** key until the function key selections change to the following display.

Completion mode	DeviceNet -> NC	NC -> DeviceNet	DeviceNet status			Search	Next menu
-----------------	-----------------	-----------------	------------------	--	--	--------	-----------

- Press the **[F1]** (DeviceNet → NC) key to display the following screen.

DeviceNet slave DeviceNet -> NC 2013/04/17 08:35:34							
DeviceNet slave FC board for DeviceNet							
Signal	Signal	Signal	Signal	Signal	Signal	Signal	Signal
0x000 PLCX000-00F	0x008 PLCX000-00F	0x010 PLCX010-01F	0x018 PLCX010-01F	0x001 PLCX000-00F	0x009 PLCX000-00F	0x011 PLCX010-01F	0x019 PLCX010-01F
0x002 PLCX000-00F	0x00A PLCX000-00F	0x012 PLCX010-01F	0x01A PLCX010-01F	0x003 PLCX000-00F	0x00B PLCX000-00F	0x013 PLCX010-01F	0x01B PLCX010-01F
0x004 PLCX000-00F	0x00C PLCX000-00F	0x014 PLCX010-01F	0x01C PLCX010-01F	0x005 PLCX000-00F	0x00D PLCX000-00F	0x015 PLCX010-01F	0x01D PLCX010-01F
0x006 PLCX000-00F	0x00E PLCX000-00F	0x016 PLCX010-01F	0x01E PLCX010-01F	0x007 PLCX000-00F	0x00F PLCX000-00F	0x017 PLCX010-01F	0x01F PLCX010-01F
0x000 [] Check OK [] []							
Can be deleted							
900. PLCX000-00F	910. PLCX0A0-0AF	920. PLCX140-14F	930. PLCX1E0-1EF	940. PLCX280-28F	950. PLCX320-32F	960. PLCX360-36F	970. PLCX400-40F
901. PLCX010-01F	911. PLCX0B0-0BF	921. PLCX150-15F	931. PLCX1F0-1FF	941. PLCX290-29F	951. PLCX330-33F	961. PLCX370-37F	971. PLCX410-41F
902. PLCX020-02F	912. PLCX0C0-0CF	922. PLCX160-16F	932. PLCX200-20F	942. PLCX2A0-2AF	952. PLCX340-34F	962. PLCX380-38F	972. PLCX420-42F
903. PLCX030-03F	913. PLCX0D0-0DF	923. PLCX170-17F	933. PLCX210-21F	943. PLCX2B0-2BF	953. PLCX350-35F	963. PLCX390-39F	973. PLCX430-43F
904. PLCX040-04F	914. PLCX0E0-0EF	924. PLCX180-18F	934. PLCX220-22F	944. PLCX2C0-2CF	954. PLCX360-36F	964. PLCX400-40F	974. PLCX440-44F
905. PLCX050-05F	915. PLCX0F0-0FF	925. PLCX190-19F	935. PLCX230-23F	945. PLCX2D0-2DF	955. PLCX370-37F	965. PLCX410-41F	975. PLCX450-45F
906. PLCX060-06F	916. PLCX100-10F	926. PLCX1A0-1AF	936. PLCX240-24F	946. PLCX2E0-2EF	956. PLCX380-38F	966. PLCX420-42F	976. PLCX460-46F
907. PLCX070-07F	917. PLCX110-11F	927. PLCX1B0-1BF	937. PLCX250-25F	947. PLCX2F0-2FF	957. PLCX390-39F	967. PLCX430-43F	977. PLCX470-47F
908. PLCX080-08F	918. PLCX120-12F	928. PLCX1C0-1CF	938. PLCX260-26F	948. PLCX300-30F	958. PLCX400-40F	968. PLCX440-44F	978. PLCX480-48F
909. PLCX090-09F	919. PLCX130-13F	929. PLCX1D0-1DF	939. PLCX270-27F	949. PLCX310-31F	959. PLCX410-41F	969. PLCX450-45F	979. PLCX490-49F
Completion mode	DeviceNet -> NC	NC -> DeviceNet	DeviceNet slave status	Overlapping terminal check	Search	Next menu	

Move the cursor to the address to be set, and set the signal number to be assigned.

The output signals that can be assigned are indicated below.

- Signals noted in the external output signal list found in section “2.6.2.2 Data input” from the Operation Manual (Data).
- Setting range: 2000 to 4047 (PLCD0000 to PLCD2047)

(Note) If the same signal is repeated and assigned while in <Device Net status>, the Device Net status information becomes valid. Do not repeat and assign the signal.

Press the [▼] key to show the addresses that can be set and that are offset by 32 bits. The addresses can be assigned up to 0x3FF. In addition, after keying in the address to be changed, press the [F6] (Search) key to move the cursor to the specified address.

Ex: Use the search function to move the cursor to 0x3F0
Input the [3], [F] and [0] keys, and press the [F6] key

(Note) If the address is greyed out, it indicates that the FC PCB (DeviceNet) is not connected or that the address falls outside of the usage range (addresses that are larger than the <Input bytes> for the DeviceNet slave parameter). Even if the address is greyed out, the signal can be assigned but it will not input into the NC.

* After the setting is completed, the alarm <<Press the [RST] key>> is triggered.

2.2.3 NC Output Signals

NC output signals* are assigned to the FC PCB for the DeviceNet (slave).

* NC output signals are the input signals when looking from the DeviceNet master side.

Procedure

1. Select <External I/O signal> on the data bank menu screen.
2. Press the [>>] key until the function key selections change to the following display.

Completion mode	DeviceNet -> NC	NC -> DeviceNet	DeviceNet status			Search	Next menu
-----------------	-----------------	-----------------	------------------	--	--	--------	-----------

3. Press the [F2] (NC → DeviceNet) key to display the following screen.

DeviceNet slave NC → DeviceNet				2013/04/17 09:36:28			
DeviceNet slave FC board for DeviceNet							
Signal		Signal		Signal		Signal	
0x000	PLCY000-00F	0x008	PLCY000-00F	0x010	PLCY010-01F	0x018	PLCY010-01F
0x001	PLCY000-00F	0x009	PLCY000-00F	0x011	PLCY010-01F	0x019	PLCY010-01F
0x002	PLCY000-00F	0x00A	PLCY000-00F	0x012	PLCY010-01F	0x01A	PLCY010-01F
0x003	PLCY000-00F	0x00B	PLCY000-00F	0x013	PLCY010-01F	0x01B	PLCY010-01F
0x004	PLCY000-00F	0x00C	PLCY000-00F	0x014	PLCY010-01F	0x01C	PLCY010-01F
0x005	PLCY000-00F	0x00D	PLCY000-00F	0x015	PLCY010-01F	0x01D	PLCY010-01F
0x006	PLCY000-00F	0x00E	PLCY000-00F	0x016	PLCY010-01F	0x01E	PLCY010-01F
0x007	PLCY000-00F	0x00F	PLCY000-00F	0x017	PLCY010-01F	0x01F	PLCY010-01F

0x000	Check OK						
Can be deleted							
900. PLCY000-00F	910. PLCY0A0-0AF	920. PLCY140-14F	930. PLCY1E0-1EF	940. PLCY280-28F	950. PLCY320-32F	960. PLCY360-36F	
901. PLCY010-01F	911. PLCY0B0-0BF	921. PLCY150-15F	931. PLCY1F0-1FF	941. PLCY290-29F	951. PLCY330-33F	961. PLCY370-37F	
902. PLCY020-02F	912. PLCY0C0-0CF	922. PLCY160-16F	932. PLCY200-20F	942. PLCY2A0-2AF	952. PLCY340-34F	962. PLCY380-38F	
903. PLCY030-03F	913. PLCY0D0-0DF	923. PLCY170-17F	933. PLCY210-21F	943. PLCY2B0-2BF	953. PLCY350-35F	963. PLCY390-39F	
904. PLCY040-04F	914. PLCY0E0-0EF	924. PLCY180-18F	934. PLCY220-22F	944. PLCY2C0-2CF	954. PLCY360-36F	964. PLCY400-40F	
905. PLCY050-05F	915. PLCY0F0-0FF	925. PLCY190-19F	935. PLCY230-23F	945. PLCY2D0-2DF	955. PLCY370-37F	965. PLCY410-41F	
906. PLCY060-06F	916. PLCY100-10F	926. PLCY1A0-1AF	936. PLCY240-24F	946. PLCY2E0-2EF	956. PLCY380-38F	966. PLCY420-42F	
907. PLCY070-07F	917. PLCY110-11F	927. PLCY1B0-1BF	937. PLCY250-25F	947. PLCY2F0-2FF	957. PLCY390-39F	967. PLCY430-39F	
908. PLCY080-08F	918. PLCY120-12F	928. PLCY1C0-1CF	938. PLCY260-26F	948. PLCY300-30F	958. PLCY3A0-3AF	968. PLCY440-44F	
909. PLCY090-09F	919. PLCY130-13F	929. PLCY1D0-1DF	939. PLCY270-27F	949. PLCY310-31F	959. PLCY3B0-3BF	969. PLCY450-45F	

Completion mode	DeviceNet → NC	NC → DeviceNet	DeviceNet slave	Overlapping terminal	Search	Next menu
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Move the cursor to the address to be set, and set the signal number to be assigned.

The output signals that can be assigned are indicated below.

- Signals noted in the external output signal list found in section “2.6.2.2 Data input” from the Operation Manual (Data).
- Setting range: 2000 to 4047 (PLCD0000 to PLCD2047)

Press the [▼] key to show the addresses that can be set and that are offset by 32 bits. The addresses can be assigned up to 0x3FF. In addition, after keying in the address to be changed, press the [F6] (Search) key to move the cursor to the specified address.

Ex: Use the search function to move the cursor to 0x3F0
Input the [3], [F] and [0] keys, and press the [F6] key

12 - 18 - 13

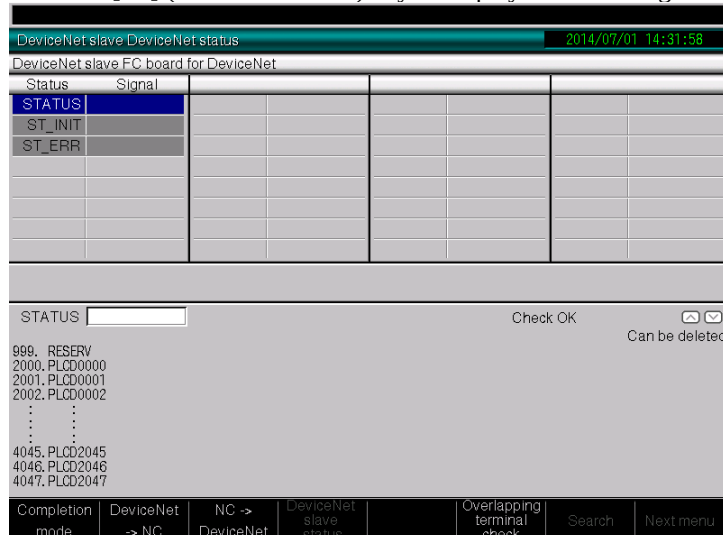
eCOM3OP12-18

The FC PCB status information for the DeviceNet can be assigned as a PLC input signal. The status information that can be referenced is shown below.

Status	Description		Content
STATUS	0 bit	Initialization completion flag	0: Not initialized 1: Initialized
	1 bit	Communicating flag	0: No communication 1: Communicating
	2 bit	Error flag	0: Normal 1: Error
	3 bit ...15 bit	Not used	0
ST_INIT	0 bit	FC PCB initialization flag	0: Initializing 1: Initialized
	1 bit	Communication standby	0: Normal 1: Standby
	2 bit ...15 bit	Not used	0
ST_ERR	0 bit	Duplicate node	0: Normal 1: Duplicate node
	1 bit	Communication time out	0: Normal 1: Communication time out
	2 bit ...14 bit	Not used	0
	15 bit	Other network errors	0: Normal 1: Network error triggered

1. Select <External I/O signal> on the data bank menu screen.
2. Press the [>>] key until the function key selections change to the following display.

Completion mode	DeviceNet -> NC	NC -> DeviceNet	DeviceNet status			Search	Next menu
-----------------	-----------------	-----------------	------------------	--	--	--------	-----------



- 999 (RESERV)
- 2000 to 4047 (PLCD0000 to PLCD2047)

(Note 2) If the [F6] (Search) key and the [F7] (Next menu) key is greyed out on the DeviceNet slave status screen, the keys cannot be input.

* After the setting is completed, the alarm << Press the **[RST]** key >> is triggered.

2.2.5 DeviceNet Monitor

The FC PCB, such as the communication status or the alarms, can be monitored.

Procedure

1. Select < I/O> on the I/O menu screen.

I/O menu	Main	Local	Slave status display	Extension I/O	Field bus		
----------	------	-------	----------------------	---------------	-----------	--	--

2. Press the **[F5]** (Field bus) key to display the following screen.

(Total of 3 pages)

Input/output (field bus)										2013/04/17 09:38:41	
Status											
FEDC BA98 7654 3210		FEDC BA98 7654 3210		FEDC BA98 7654 3210		FEDC BA98 7654 3210					
1	0000	0000	0000	0000	17	0000	0000	0110	0100	Bus type DeviceNet	
2	0000	0000	0000	0000	18	0000	0000	0000	0000	Status Communicating	
3	0000	0000	0000	0011	19	0000	0000	0000	0000	Address 2	
4	0000	0000	0000	0001	20	0000	0000	0000	0000	Transfer rate 125kbps	
5	0000	0000	0000	0000	21	1101	0000	1111	1011		
6	0000	0000	0000	0000	22	0000	0000	0000	0000	System status 0	
7	0000	0000	0000	0001	23	0000	0000	0000	0001	Communication error 0	
8	0000	0000	0111	0001	24	0000	0000	0000	0000	Commun. error counter 1	
9	0000	0000	0111	0000	25	0000	0000	0000	0000		
10	0000	0000	0000	0000	26	0000	0000	0000	0000		
11	0000	0000	0000	0111	27	0000	0000	0000	0000		
12	0000	0000	0000	0000	28	0000	0000	0000	0000		
13	0000	0000	0000	0100	29	0000	0000	0000	0000		
14	0000	0000	0000	0000	30	0000	0000	0000	0000		
15	0000	0000	0000	0000	31	0000	0000	0000	0000		
16	0000	0000	0000	0001	32	0000	0000	0000	0000		
FEDC BA98 7654 3210		FEDC BA98 7654 3210		FEDC BA98 7654 3210		FEDC BA98 7654 3210		FEDC BA98 7654 3210			
										☐ Field bus 033-064 ☒ Field bus 001-032	
I/O menu	Main	Local	Slave status display	Extension I/O	Field bus					Version	

Status	Description	
<Bus type>	Indicates the type of the field bus that is connected to the FC PCB.	
	DeviceNet	Indicates that DeviceNet is connected.
	Not connected	Indicates that the bus is not connected.
	Error	Displayed when there is an error.
<Status>	Indicates the current communication status.	
	Not connected	Indicates that the FC PCB is not connected.
	Initializing	Indicates that the FC PCB is initializing.
	Waiting for configuration	Indicates that communication is being established with the master device.
	Communicating	Indicates that the FC PCB is operating normally.
	Communication stop	Indicates that communication has stopped.
	Error	Indicates that there is an error.

Status	Description	
<Address>	Indicates that MAC ID for DeviceNet is set to the FC PCB.	
	0 to 63	When DeviceNet is connected
	255	Indicates when not connected and when the address value is not valid.
Transfer rate	Indicates the transfer rate that is set to the FC PCB.	
	125 kbps	Indicates the DeviceNet transfer rate.
	250 kbps	
	500 kbps	
	(Blank field)	A blank field is displayed for any rate not noted above.
<System status>	Indicates the system status for the FC PCB.	
	0	Indicates that it is operating normally.
	Other than 0	Indicates that there is a system error on netX.
<Communication error>	Indicates that there is an FC PCB communication error.	
	0	Indicates that it is operating normally.
	Other than 0	Indicates that there is a communication error on netX.
<Commun. error counter>	0 to 999999	Indicates the number of communication errors that occurred.

3. Press the **[PAGE DOWN]** key to display the I/O signal status for the field bus.
(From page 2 on)

Input/output (field bus)																2013/04/17 09:39:22	
Input								Output									
FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210		
1 0000 0000 0000 0000	17 0000 0000 0000 0000	2 0000 0000 0000 0000	18 0000 0000 0000 0000	3 0000 0000 0000 0000	19 0000 0000 0000 0000	4 0000 0000 0000 0000	20 0000 0000 0000 0000	5 0000 0000 0000 0000	21 0000 0000 0000 0000	6 0000 0000 0000 0000	22 0000 0000 0000 0000	7 0000 0000 0000 0000	23 0000 0000 0000 0000	8 0000 0000 0000 0000	24 0000 0000 0000 0000		
9 0000 0000 0000 0000	25 0000 0000 0000 0000	10 0000 0000 0000 0000	26 0000 0000 0000 0000	11 0000 0000 0000 0000	27 0000 0000 0000 0000	12 0000 0000 0000 0000	28 0000 0000 0000 0000	13 0000 0000 0000 0000	29 0000 0000 0000 0000	14 0000 0000 0000 0000	30 0000 0000 0000 0000	15 0000 0000 0000 0000	31 0000 0000 0000 0000	16 0000 0000 0000 0000	32 0000 0000 0000 0000		
FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210		
Output 0000000000000000																Status	
FEDCBA9876543210																Field bus 033-064	
I/O menu	Main	Local	Slave status display	Extension I/O	Field bus											Version	

After the input bytes and output bytes for the bus that is set, “0” is used for the remaining bits.

2.3 EDS File

2.3.1 What is the EDS File?

The Device Profile is a text format file (filename extension (.eds)) with a description about the slave device. The ESD file can be used to set the master parameter for the DeviceNet master station.

2.3.2 EDS File for DeviceNet on FC PCB

Refer to CNC-C00.eds for the EDS file.

2.3.3 How to Obtain the EDS Files

To obtain the EDS files, download them from the ODVA website or use the PLC Programming Software installation CD.

ODVA is an organization which supports network technology such as DeviceNet.

ODVA homepage

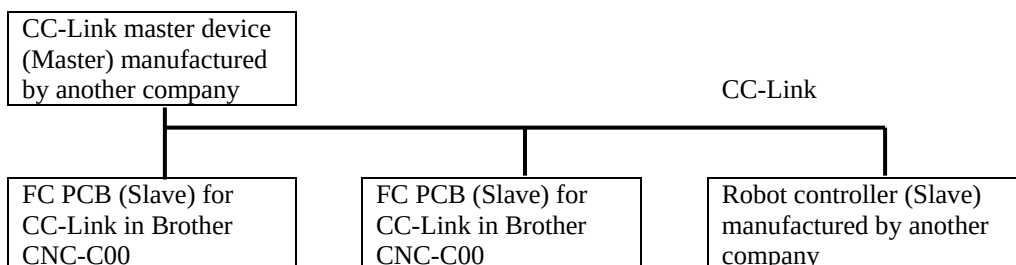
<http://www.odva.org/Home/tabid/53/Inq/en-US/Default.aspx>

3 CC-Link (Remote Device Station)

3.1 CC-Link Function Overview

3.1.1 Function Overview

CC-Link is a field bus network that can connect various devices such as a CNC, PLC and robot controller. The CC-Link slave function is supported by connecting the FC PCB (= Fieldbus Conversion PCB) for the DeviceNet to the CNC-C00.



The CNC-C00 supports the following CC-Link function.

CC-Link version	Ver.1.11
Compatible slave station category	Remote device station
Occupied number of offices	Stations 1 to 4 (Parameter selection)
Transfer rate	156k/625k/2.5M/5M/10M bps (Parameter selection)

3.1.2 Wiring Method

For further details, refer to “Chapter 11 FC PCB” in the Installation Manual.

3.2 Setting & Monitor Screen

3.2.1 Basic Settings for CC-Link

The basic settings can be configured for CC-Link parameters.

Procedure

1. Select <User parameter> on the data bank menu screen.
2. The following screen appears when <CC-Link> is selected from the function keys. Move the cursor to the item to be set, and enter the parameter.

The screenshot shows the 'CC-Link' parameter setting screen. At the top, it displays 'CC-Link' and a timestamp '2013/04/18 19:50:47'. Below this is a table of parameters:

4401	Station number	1
4402	Slave station category [0:Remote device station]	0
4403	Occupied number of stations	1
4404	Transfer rate [0:156kbps 1:625kbps 2:2.5Mbps 3:5Mbps 4:10Mbps]	1
4405	Specify data clear in case of communication error [0:Clear 1:Hold]	0

Below the table, there is a 'Station number' input field with a dropdown arrow. At the bottom, there is a navigation bar with buttons: Completion mode, Auto thermal distortion comp., Quick table, High accuracy, PLC, PROFIBUS DP, DeviceNet, and CC-Link.

Parameter list

Parameter name	Setting range	Description
<Station number>	1 to 64	Sets the station numbers.
<Slave station category>	<0: Remote device station>	The slave station category is fixed to <0:Remote device station>.
<Occupied number of stations>	1 to 4	Sets the number of occupied stations.
<Transfer rate>	0:156 kbps 1:625 kbps 2:2.5 Mbps 3:5 Mbps 4:10 Mbps	Sets the transfer speed.
<Specify data clear in case of communication error>	<0: Clear> <1: Hold>	Selects whether to clear or hold data received during a communication error.

- After the setting is completed, press the **[F0]** (Completion mode) key to complete and exit the editing.

(Note) After changing the parameter, the alarm <<Turn power off>> is triggered. <<User parameter setting error (CC-Link)>> is triggered when the slave station category is configured to another setting other than <0: Remote device station>.

3.2.2 NC Input Signals (Remote Output and Remote Register Output)

NC input signals* are assigned to the remote output (hereafter referred to as “RY”) on the FC PCB for the CC-Link. In addition, the NC input signals can be assigned to the remote register output (hereafter referred to as “RWw”).

* NC input signals are the output signals when looking from the CC-Link master side.

Procedure

- Select <External I/O signal> on the data bank menu screen.
- Press the **[>>]** key until the function key selections change to the following display.

Completion mode	RY/RWw (CC-Link → NC)	RX/RWw (NC → CC-Link)	CC-Link status				Next menu
-----------------	-----------------------	-----------------------	----------------	--	--	--	-----------

- Press the **[F1]** (RY/RWw (CC-Link → NC)) key to display the following screen.

CC-Link slave RY/RWw (CC-Link → NC) 2013/04/18 17:09:32					
FC board for CC-Link					
RY	Signal	RY	Signal	RY	Signal
RY000	PLCX000-00F	RY008	PLCX000-00F	RY010	PLCX010-01F
RY001	PLCX000-00F	RY009	PLCX000-00F	RY011	PLCX010-01F
RY002	PLCX000-00F	RY00A	PLCX000-00F	RY012	PLCX010-01F
RY003	PLCX000-00F	RY00B	PLCX000-00F	RY013	PLCX010-01F
RY004	PLCX000-00F	RY00C	PLCX000-00F	RY014	PLCX010-01F
RY005	PLCX000-00F	RY00D	PLCX000-00F	RY015	PLCX010-01F
RY006	PLCX000-00F	RY00E	PLCX000-00F	RY016	PLCX010-01F
RY007	PLCX000-00F	RY00F	PLCX000-00F	RY017	PLCX010-01F
Can be deleted					
RY000		Check OK			
900. PLCX000-00F	910. PLCX0A0-0AF	920. PLCX140-14F	930. PLCX1E0-1EF	940. PLCX280-28F	950. PLCX320-32F
901. PLCX010-01F	911. PLCX0B0-0BF	921. PLCX150-15F	931. PLCX1F0-1FF	941. PLCX290-29F	951. PLCX330-33F
902. PLCX020-02F	912. PLCX0C0-0CF	922. PLCX160-16F	932. PLCX200-20F	942. PLCX2A0-2AF	952. PLCX340-34F
903. PLCX030-03F	913. PLCX0D0-0DF	923. PLCX170-17F	933. PLCX210-21F	943. PLCX2B0-2BF	953. PLCX350-35F
904. PLCX040-04F	914. PLCX0E0-0EF	924. PLCX180-18F	934. PLCX220-22F	944. PLCX2C0-2CF	954. PLCX360-36F
905. PLCX050-05F	915. PLCX0F0-0FF	925. PLCX190-19F	935. PLCX230-23F	945. PLCX2D0-2DF	955. PLCX370-37F
906. PLCX060-06F	916. PLCX100-10F	926. PLCX1A0-1AF	936. PLCX240-24F	946. PLCX2E0-2EF	956. PLCX380-38F
907. PLCX070-07F	917. PLCX110-11F	927. PLCX1B0-1BF	937. PLCX250-25F	947. PLCX2F0-2FF	957. PLCX390-39F
908. PLCX080-08F	918. PLCX120-12F	928. PLCX1C0-1CF	938. PLCX260-26F	948. PLCX300-30F	958. PLCX3A0-3AF
909. PLCX090-09F	919. PLCX130-13F	929. PLCX1D0-1DF	939. PLCX270-27F	949. PLCX310-31F	959. PLCX3B0-3BF
Completion mode	RY/RWw (CC-Link → NC)	RX/RWw (NC → CC-Link)	CC-Link status	Overlapping terminal check	Search Next menu

Move the cursor to the RY to be set, and enter the signal number to be assigned. Note, for further details on the input signals that can be assigned to the RY, refer to section “2.6.2.2 Data input” in the Operation Manual (Data).

In addition, the [▼] key can be pressed to show the RY that is offset by 32 bits. The RY can be assigned up to RY06F. The RY that can actually be used in the <Occupied number of stations> setting is shown below.

Occupied number of stations	RY usage range (User range)
Station 1	RY000 to RY00F (System range for RY010 to RY01F)
Station 2	RY000 to RY02F (System range for RY030 to RY03F)
Station 3	RY000 to RY04F (System range for RY050 to RY05F)
Station 4	RY000 to RY06F (System range for RY070 to RY07F)

Press the [▼] key again to display the RWw signal assignment screen.

Move the cursor to the RWw to be set, and enter the signal number to be assigned. However, only the signals noted below can be assigned to the RWw. The operator message <<Input data error>> is triggered if another signal, other than those signals noted, is assigned.

- 999 (RESERV)
- 2000 to 4047 (PLCD0000 to PLCD2047)

(Note) If the same signal is repeated and assigned while in <CC-Link status>, the CC-Link status information becomes valid. Do not repeat and assign the signal.

Also, the RWw, which can actually be used in the <Occupied number of stations> setting for the CC-Link slave parameter, is shown below.

Occupied number of stations	RWw usage range
Station 1	RWw000 to RWw003
Station 2	RWw000 to RWw007
Station 3	RWw000 to RWw00B
Station 4	RWw000 to RWw00F

(Note) If the RYxxx/RWwxxx is greyed out, it indicates that the FC PCB (CC-Link) is not connected or that it falls outside of the usage range (Refer to the CC-Link slave parameter <Occupied number of stations> and section “3.2.4 Remote Input and Remote Output System Range”). Even if the RYxxx/RWwxxx is greyed out, the signal can be assigned but it will not input into the NC.

- * After the setting is completed, the alarm <<Press the [RST] key>> is triggered.

3.2.3 NC Output Signals (Remote Input and Remote Register Input)

NC input signals* are assigned to the remote input (hereafter referred to as “RX”) on the FC PCB for the CC-Link. In addition, the NC output signals can be assigned to the remote register input (hereafter referred to as “RWr”).

* NC output signals are the input signals when looking from the CC-Link master side.

Procedure

1. Select <External I/O signal> on the data bank menu screen.
2. Press the [>>] key until the function key selections change to the following display.

Completion mode	RY/RWw (CC-Link -> NC)	RX/RWw (NC -> CC-Link)	CC-Link status				Next menu
-----------------	------------------------	------------------------	----------------	--	--	--	-----------

3. Press the [F2] (RX/RWw (NC → CC-Link)) key to display the following screen.

CC-Link slave RX/RWw (NC → CC-Link)						2013/04/18 17:11:50	
FC board for CC-Link							
RX	Signal	RX	Signal	RX	Signal	RX	Signal
RX000	PLCY000-00F	RX008	PLCY000-00F	RX010	PLCY010-01F	RX018	PLCY010-01F
RX001	PLCY000-00F	RX009	PLCY000-00F	RX011	PLCY010-01F	RX019	PLCY010-01F
RX002	PLCY000-00F	RX00A	PLCY000-00F	RX012	PLCY010-01F	RX01A	PLCY010-01F
RX003	PLCY000-00F	RX00B	PLCY000-00F	RX013	PLCY010-01F	RX01B	PLCY010-01F
RX004	PLCY000-00F	RX00C	PLCY000-00F	RX014	PLCY010-01F	RX01C	PLCY010-01F
RX005	PLCY000-00F	RX00D	PLCY000-00F	RX015	PLCY010-01F	RX01D	PLCY010-01F
RX006	PLCY000-00F	RX00E	PLCY000-00F	RX016	PLCY010-01F	RX01E	PLCY010-01F
RX007	PLCY000-00F	RX00F	PLCY000-00F	RX017	PLCY010-01F	RX01F	PLCY010-01F
Can be deleted							
RX000				Check OK		OK Cancel	
				Can be deleted			
900. PLCY000-00F	910. PLCY0A0-0AF	920. PLCY140-14F	930. PLCY1E0-1EF	940. PLCY280-28F	950. PLCY320-32F	960. PLCY360-36F	970. PLCY400-40F
901. PLCY010-01F	911. PLCY0B0-0BF	921. PLCY150-15F	931. PLCY1F0-1FF	941. PLCY290-29F	951. PLCY330-33F	961. PLCY370-37F	971. PLCY410-41F
902. PLCY020-02F	912. PLCY0C0-0CF	922. PLCY160-16F	932. PLCY200-20F	942. PLCY2A0-2AF	952. PLCY340-34F	962. PLCY380-38F	972. PLCY420-42F
903. PLCY030-03F	913. PLCY0D0-0DF	923. PLCY170-17F	933. PLCY210-21F	943. PLCY2B0-2BF	953. PLCY350-35F	963. PLCY390-39F	973. PLCY430-43F
904. PLCY040-04F	914. PLCY0E0-0EF	924. PLCY180-18F	934. PLCY220-22F	944. PLCY2C0-2CF	954. PLCY360-36F	964. PLCY400-40F	974. PLCY440-44F
905. PLCY050-05F	915. PLCY0F0-0FF	925. PLCY190-19F	935. PLCY230-23F	945. PLCY2D0-2DF	955. PLCY370-37F	965. PLCY410-41F	975. PLCY450-45F
906. PLCY060-06F	916. PLCY100-10F	926. PLCY1A0-1AF	936. PLCY240-24F	946. PLCY2E0-2EF	956. PLCY380-38F	966. PLCY420-42F	976. PLCY460-46F
907. PLCY070-07F	917. PLCY110-11F	927. PLCY1B0-1BF	937. PLCY250-25F	947. PLCY2F0-2FF	957. PLCY390-39F	967. PLCY430-43F	977. PLCY470-47F
908. PLCY080-08F	918. PLCY120-12F	928. PLCY1C0-1CF	938. PLCY260-26F	948. PLCY300-30F	958. PLCY3A0-3AF	968. PLCY440-44F	978. PLCY480-48F
909. PLCY090-09F	919. PLCY130-13F	929. PLCY1D0-1DF	939. PLCY270-27F	949. PLCY310-31F	959. PLCY3B0-3BF	969. PLCY450-45F	979. PLCY490-49F
Completion mode	RY/RWw (CC-Link → NC)	RX/RWw (NC → CC-Link)	CC-Link status	Overlapping terminal check	Search	Next menu	

Move the cursor to the RX to be set, and enter the signal number to be assigned. Note, for further details on the external output signals that can be assigned to the RX, refer to section “2.6.2.2 Data input” in the Operation Manual (Data).

In addition, the [▼] key can be pressed to show the RX that is offset by 32 bits. The RX can be assigned up to RX06F. However, the RX, which can actually be used in the <Occupied number of stations> setting for the CC-Link slave parameter, is shown below.

Occupied number of stations	RX usage range (User range)
Station 1	RX000 to RX00F (System range for RX010 to RX01F)
Station 2	RX000 to RX02F (System range for RX030 to RX03F)
Station 3	RX000 to RX04F (System range for RX050 to RX05F)
Station 4	RX000 to RX06F (System range for RX070 to RX07F)

Press the [▼] key again to display the signal assignment screen for the remote register input (hereafter referred to as “RWr”).

Move the cursor to the RWr to be set, and enter signal number to be assigned. However, only the signals noted below can assigned to the RWr. The operator message <Input data error> is triggered if another signal, other than those signals noted, is assigned.

- 999 (RESERV)
- 2000 to 4047 (PLCD0000 to PLCD2047)

Also, the RWr, which can actually be used in the <Occupied number of stations> setting for the CC-Link slave parameter, is shown below.

Occupied number of stations	RWr usage range
Station 1	RWr000 to RWr003
Station 2	RWr000 to RWr007
Station 3	RWr000 to RWr00B
Station 4	RWr000 to RWr00F

(Note) If the RXxxx/RWrxxx is greyed out, it indicates that the FC PCB (CC-Link) is not connected or that it falls outside of the usage range (Refer to the CC-Link slave parameter <Occupied number of stations> and section “3.2.4 Remote Input and Remote Output System Range”). Even if the RXxxx/RWrxxx is greyed out, the signal can be assigned but it will not output from the NC.

- * After the setting is completed, the alarm <<Press the [RST] key>> is triggered.

3.2.4 Remote Input and Remote Output System Range

The 16 bit range noted below for the remote input (RX) and the remote output (RY) is used for the CC-Link system range. Therefore, it cannot be used for NC input and output.

Occupied number of stations	System range
Station 1	RX010 to RX01F RY010 to RY01F
Station 2	RX030 to RX03F RY030 to RY03F
Station 3	RX050 to RX05F RY050 to RY05F
Station 4	RX070 to RX07F RY070 to RY07F

FC PCB system range definition

The significance of each bit in the system range is shown below. (* changes depending on the occupied number of stations.)

For example, if there is 1 occupied station, the remote is ready for RX01B. If there are 4 occupied stations, the remote is ready for RX07B.

RX system range	Description	RY system range	Description
RX**0	Not used	RY**0	Not used
RX**1	"	RY**1	"
RX**2	"	RY**2	"
RX**3	"	RY**3	"
RX**4	"	RY**4	"
RX**5	"	RY**5	"
RX**6	"	RY**6	"
RX**7	"	RY**7	"
RX**8	"	RY**8	"
RX**9	"	RY**9	"
RX**A	"	RY**A	"
RX**B	Remote ready	RY**B	"
RX**C	Not used	RY**C	"
RX**D	"	RY**D	"
RX**E	"	RY**E	"
RX**F	"	RY**F	"

(Note) On the CC-Link master side, ensure that “Remote ready” is ON and then configure the settings so that the RX/RX communication in the user range is enabled.

3.2.5 Status

The FC PCB status information for the CC-Link can be assigned as a PLC input signal. The status information that can be referenced is shown below.

Status	Description		Content
STATUS	0 bit	Initialization completion flag	0: Not initialized 1: Initialized
	1 bit	Communicating flag	0: No communication 1: Communicating
	2 bit	Error flag	0: Normal 1: Error
	3 bit ...15 bit	Not used	0
ST_INIT	0 bit	FC PCB initialization flag	0: Initializing 1: Initialized
	1 bit	Communication standby	0: Normal 1: Standby
	2 bit ...15 bit	Not used	0
ST_ERR	0 bit	(Duplicate node)	0 (Fixed)
	1 bit	Communication time out	0: Normal 1: Communication time out
	2 bit ...14 bit	Not used	0
	15 bit	Other network errors	0: Normal 1: Network error triggered

Procedure

1. Select <External I/O signal> on the data bank menu screen.
2. Press the [>>] key until the function key selections change to the following display.

Completion mode	RY/RWw (CC-Link -> NC)	RX/RWr (NC -> CC-Link)	CC-Link status				Next menu
-----------------	------------------------	------------------------	----------------	--	--	--	-----------

3. Press the [F3] (CC-Link status) key to display the following screen.

Move the cursor to the status address to be set, and enter the signal number to be assigned. However, only the signals noted below can be assigned to each status address. The operator message <<Input data error>> is triggered if another signal, other than those signals noted, is assigned.

- 999 (RESERV)
- 2000 to 4047 (PLCD0000 to PLCD2047)

(Note 1) If the FC PCB (CC-Link) is not connected, the address is greyed out. Even if the address is greyed out, the signal can be assigned but it will not input into the NC.

(Note 2) If the [F6] (Search) key and the [F7] (Next menu) key is greyed out on the CC-Link slave parameter screen, the keys cannot be pressed.

* After the setting is completed, the alarm <<Press the [RST] key>> is triggered.

3.2.6 CC-Link Monitor

The FC PCB, such as the communication status or the alarms, can be monitored.

Procedure

1. Select < I/O > on the I/O menu screen.

I/O menu	Main	Local	Slave status display	Extension I/O	Field bus		
----------	------	-------	----------------------	---------------	-----------	--	--

2. Press the [F5] (Field bus) key to display the following screen.

(Total of 3 pages)

Input/output (field bus)																2013/04/18 17:15:37			
Status																			
FEDC BA98 7654 3210				FEDC BA98 7654 3210				Bus typeCC-Link StatusCommunicating Address5 Transfer rate156kbps System status0 Communication error0 Commun. error counter2											
1	0000	0000	0000	0000	17	0000	0000											0110	0100
2	0000	0000	0000	0000	18	0000	0000											0000	0000
3	0000	0000	0000	0001	19	0000	0000											0000	0000
4	0000	0000	0000	0001	20	0000	0000											0000	0001
5	0000	0000	0000	0000	21	0000	1110											1000	1101
6	0000	0000	0000	0000	22	0000	0000											0000	0000
7	0000	0000	0000	0001	23	0000	0000											0000	0010
8	0000	0000	1011	1001	24	0000	0000											0000	0000
9	0000	0000	1011	1000	25	0000	0000											0000	0000
10	0000	0000	0000	0000	26	0000	0000											0000	0000
11	0000	0000	0000	0111	27	0000	0000											0000	0000
12	0000	0000	0000	0000	28	0000	0000											0000	0000
13	0000	0000	0000	0100	29	0000	0000											0000	0000
14	0000	0000	0000	0000	30	0000	0000											0000	0000
15	0000	0000	0000	0000	31	0000	0000											0000	0000
16	0000	0000	0000	0001	32	0000	0000											0000	0000
FEDC BA98 7654 3210				FEDC BA98 7654 3210															
☐ Field bus033-064 ☒ Field bus001-032																			
I/O menu	Main	Local	Slave status display	Extension I/O	Field bus	Version													

Status	Description	
<Bus type>	Indicates the type of the field bus that is connected to the FC PCB.	
	CC-Link	Indicates that CC-Link is connected.
	Not connected	Indicates that the bus is not connected.
	Error	Displayed when there is an error.
<Status>	Indicates the current communication status.	
	Not connected	Indicates that the FC PCB is not connected.
	Initializing	Indicates that the FC PCB is initializing.
	Waiting for configuration	Indicates that communication is being established with the master device.
	Communicating	Indicates that the FC PCB is operating normally.
	Communication stop	Indicates that communication has stopped.
	Error	Indicates that there is an error.
<Address>	Indicates the station number that is set to the FC PCB.	
	1 to 64	When CC-Link is connected
	255	Indicates when not connected and when the address value is not valid.

Status	Description	
<Transfer rate>	Indicates the transfer rate that is set to the FC PCB.	
	156kbps	Indicates the CC-Link transfer rate.
	625kbps	
	2.5Mbps	
	5Mbps	
	10Mbps	
	(Blank field)	A blank field is displayed for any rate not noted above.
<System status>	Indicates the system status for the FC PCB.	
	0	Indicates that it is operating normally.
	Other than 0	Indicates that there is a system error on netX.
<Communication error>	Indicates that there is an FC PCB communication error.	
	0	Indicates that it is operating normally.
	Other than 0	Indicates that there is a communication error on netX.
<Commun. error counter>	0 to 999999	Indicates the number of communication errors that occurred.

3. Press the **[PAGE DOWN]** key to display the I/O signal status for the field bus.
(From page 2 on)

Input/output (field bus) 2013/04/18 17:16:51															
Input								Output							
FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210
1 0000 0000 0000 0000	17							1 0000 0000 0000 0000	17						
2 0000 0000 0000 0000	18							2 0000 0000 0000 0000	18						
3 0000 0000 0000 0000	19							3 0000 0000 0000 0000	19						
4 0000 0000 0000 0000	20							4 0000 0000 0000 0000	20						
5 0000 0000 0000 0000	21							5 0000 0000 0000 0000	21						
6 0000 0000 0000 0000	22							6 0000 0000 0000 0000	22						
7 0000 0000 0000 0000	23							7 0000 0000 0000 0000	23						
8 0000 0000 0000 0000	24							8 0000 1000 0000 0000	24						
9	25							9	25						
10	26							10	26						
11	27							11	27						
12	28							12	28						
13	29							13	29						
14	30							14	30						
15	31							15	31						
16	32							16	32						
FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210	FEDC BA98 7654 3210
Output 0000000000000000								Status							
FEDCBA9876543210								Field bus 033-064							
I/O menu	Main	Local	Slave status display	Extension I/O	Field bus										Version

After the input bytes and output bytes for the bus that is set, "0" is used for the remaining bits.

3.3 CSP File

3.3.1 What is the CSP File?

The CSP file is a text format file (filename extension (.csp)) with a description about the CC-Link slave device. The CSP file can be used to set the master parameter for the CC-Link master station.

3.3.2 CSP File for CC-Link on FC PCB

The CSP file that is used varies depending on the <Occupied number of stations> setting in the CC-Link slave parameter.

CC-Link slave parameter Occupied number of stations	CSP file name
1	CNC-C00RemoteDev_1_V1_10.CSP
2	CNC-C00RemoteDev_2_V1_10.CSP
3	CNC-C00RemoteDev_3_V1_10.CSP
4	CNC-C00RemoteDev_4_V1_10.CSP

3.3.3 CSP+ Files

3.3.3.1 What are CSP+ Files?

CSP files are highly descriptive profile specifications.

3.3.3.2 CSP+ File for CC-Link on FC PCB

The CSP+ file that is used varies depending on the <Occupied number of stations> setting in the CC-Link slave parameter.

Japanese

CC-Link slave parameter Occupied number of stations	CSP+ file name
1	CNC-C00RemoteDev_1_V1_10_ja.zip
2	CNC-C00RemoteDev_2_V1_10_ja.zip
3	CNC-C00RemoteDev_3_V1_10_ja.zip
4	CNC-C00RemoteDev_4_V1_10_ja.zip

English

CC-Link slave parameter Occupied number of stations	CSP+ file name
1	CNC-C00RemoteDev_1_V1_10_en.zip
2	CNC-C00RemoteDev_2_V1_10_en.zip
3	CNC-C00RemoteDev_3_V1_10_en.zip
4	CNC-C00RemoteDev_4_V1_10_en.zip

3.3.4 How to Obtain the CSP Files

To obtain the CSP or CSP+ files, download them from the CC-Link Partner Association website or use the PLC Programming Software installation CD.

CC-Link Partner Association homepage

<http://www.cc-link.org/>

4 CC-Link (Master Station)

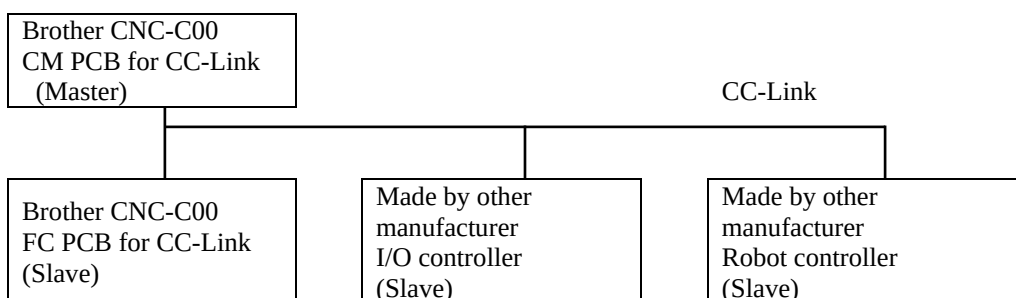
4.1 CC-Link Function Overview

4.1.1 Function Overview

This function is available on the following machine models.

Machine model	S300X2/S500X2/S700X2/F600X1/R450X2/R650X2/M200X3/M300X3
---------------	---

CC-Link is a field bus network that can connect various devices such as a CNC, PLC and robot controller. The CC-Link master function is supported by connecting the CC-Link Fieldbus Conversion PCB (hereafter referred to as CM PCB (CM: CC-Link Master)) to the CNC-C00.



The following table describes the master station and intelligent device stations.

Master station	This station holds the control information (parameters) and manages the overall network. One master station is needed for one network and its station number is set to 0.
Intelligent device station	This station sends (transmits in cycles) the master station in regular intervals I/O signals (unit: bit) and I/O data (unit: word). (Note)

(Note) The Brother specification cannot support transient communication on the CM PCB for intelligent device stations. In addition, support is only provided on Version 1.0.

The CC-Link master PCB supports the following CC-Link function.

CC-Link compatible versions	Ver.1.0/2.0 (Parameter selection)
Compatible slave station types	Remote I/O stations, remote device stations, intelligent device stations, local stations and standby master station
Maximum connected stations	32 stations
Transfer rate	156k/625k/2.5M/5M/10M bps (Parameter selection)
Extended cyclical communication	Single, double, quadruple and octuple (Parameter selection)

4.1.2 Wiring

Refer to "Chapter 11 CM PCB" in the Installation Manual for further details.

4.2 Setting Screen

4.2.1 Basic Settings for CC-Link Master Station

The basic parameter settings can be configured for the CC-Link master station.

<<Procedure>>

1. Select <User parameter> on the data bank menu screen.
2. The following screen appears when <CC-Link master> is selected from the function keys. Move the cursor to the item to be set, and enter the parameter.

Parameter	Value
10201 Transfer rate [0:156kbps 1:625kbps 2:2.5Mbps 3:5Mbps 4:10Mbps]	4
10202 Self-station number	0
10203 Interface board setting [0:Online (normal) 1:Offline]	0
Occupied number of stations (for intelligent device stations)	
10204 [0:1 station 1:2 stations 2:3 stations 3:4 stations]	0
10205 Standby master station number	21
10206 CC-Link version [0:Ver.1 1:Ver.2]	1
10207 Retry attempts	3
10208 Number of automatic return nodes	1
10209 Specify action when there is a driver error [0:Stop 1:Continue]	0
10210 Specify station data clear in case of any error [0:Clear 1:Hold]	0

Transfer rate

[HELP]

Completion mode Switch 1 Switch 2 Switch 3 Ref. position/Conversation zero point User registered misc. func. Tool breakage detection Automatic workpiece meas.

Parameter list

Parameter name	Setting range	Description
Transfer rate	0:156kbps 1:625kbps 2:2.5Mbps 3:5Mbps 4:10Mbps	Set transfer rate. The transfer rate setting for the line test is also included.
Self-station number	0: Master station 1 - 64: Intelligent device station	Set the station number for the NC unit.
Interface board setting	0: Online 1: Offline	Set interface board operation mode. Even if set to <0: Online>, it does not change to online status when no slave stations are set to the CC-Link configuration while the <Self-station number> is set to <0: Master station>.
Occupied number of stations (for intelligent device stations)	0: 1 station 1: 2 stations 2: 3 stations 3: 4 stations	Set the number of occupied stations when the NC unit is an intelligent device station. Valid only when the <Self-station number> is not set to <0:Master station>.
Standby master station number	0 to 32	Set the standby master station number connected to the NC unit. Set to 0 when standby master station is not connected. Valid when the <Self-station number> is set to <0: Master station>.
CC-Link version	0: Ver.1 1: Ver.2	Set the CC-Link version for the CM PCB. (Note 1) If the <Self-station number> is set to <1-64> and the <CC-Link version> is set to <1: Ver.2>, the error <<User param. setting error (CC-Link master)>> is triggered.

Parameter name	Setting range	Description
Retry attempts	1 to 7	Set the number of retry attempts when there is no response from the slave station. Valid when the <Self-station number> is set to <0:Master station>.
Number of automatic return nodes	1 to 10	Set the number of automatic return nodes within one link scan for a station with an error. Valid when the <Self-station number> is set to <0:Master station>.
Specify action when there is a driver error	0:Stop 1:Continue	Select whether the data link operation can be continued or not after a CC-Link driver error is detected. Valid when the <Self-station number> is set to <0:Master station>.
Specify station data clear in case of any error	0:Clear 1:Hold	Select whether to clear or hold input data from a station with an error. Valid when the <Self-station number> is set to <0: Master station>.

Parameter list

- After the setting is completed, press the **[F0]** key (Completion mode) to end and exit editing.

(Note 1) When the <CC-Link version> is set to <0: Ver.1>, only Ver.1.00 and Ver.1.10 are supported in the CC-Link standards.

When the <CC-Link version> is set to <1: Ver.2> and a remote I/O station is connected, the I/O data area (unit: word) is ignored.

(Note 2) When a parameter from 1 to 10 is changed, the operator message <<Press the [RST] key.>> appears.

4.2.2 Basic Settings for CC-Link Slave Station

The basic parameter settings can be configured for the CC-Link slave station when self-station is the master station.

<<Procedure>>

- Select <External I/O signal> on the data bank menu screen.
- Press the **[>>]** key until the function key selections change to the following display.

Completion mode	CC-Link config	CC-Link → NC(Master)	NC(Master) → CC-Link	CC-Link status	Copy	Paste	Delete CC-Link data
-----------------	----------------	----------------------	----------------------	----------------	------	-------	---------------------

- When the **[F1]** key (CC-Link configuration) is pressed, the following screen appears.

CC-Link configuration 2019/07/25 11:01:46

Slave station	01	02	03	04
Device name	DEVICE1	DEVICE2		DEVICE3
By station category	0	1		2
CC-Link version	0	0		1
Extended cyclical setting	0	0		0
Occupied number of stations		1		3
Transient send buffer size	0	0		128
Transient receive buffer size	0	0		128
Specify reserved station	0	0		0
Specify invalid station due to error	0	0		0
Specify I/O assignment	0	0		0

Device name Can be deleted

Completion mode External input signal External output signal External alarm Copy Paste Delete CC-Link data

In addition, when the [▽] key is pressed, the slave station number is offset by four a time. A maximum of 32 slave station numbers are displayed.

When the cursor is at the device name position, the [F5], [F6] and [F7] keys are valid. After the [F7] key is pressed, all the data for that station is deleted. In addition, the device name is the only item that can be deleted with the [DEL] key. Note that the following confirmation screen appears when the [F7] key is pressed.

Slave station01 Delete?							
Yes	No						

Pressing the [F0] (Yes) key deletes all the station number data.

When the [F1] (No) key is pressed, the delete operation is cancelled and the screen goes back to the previous screen.

When <1: 2 stations> or higher is set to the <Occupied number of stations>, only the set number of consecutive stations (including the setting station) are occupied. The occupied stations are reserved (grayed out). In this situation, parameter input is possible but the slave station is invalid. For example, when <By station category> for station No. 02 is set to <1: Remote device station> and the <Occupied number of stations> is set to <1: 2 stations>, then station No. 03 is occupied by station No. 02 and the next station that can be set is "04". When a parameter is set or saved while the valid stations are not linked, an error is triggered.

When the settings are saved, the setting content is checked. If there is an error in the setting, an alarm is triggered.

Parameter list

Parameter name	Setting range	Description
Device name	1 to 12 characters	Set the device name for the slave station.
By station category	0: Remote I/O station 1: Remote device station 2: Intelligent device station	Set the station category for the slave station. (Local station and standby master station are a type of intelligent device station.) If there is nothing entered for the station category, the items for this station number are processed as not set.
CC-Link version	0: Ver.1 1: Ver.2	Set the master station version. When <By station category> is set to <0: Remote I/O station>, the version <0: Ver.1> is used regardless of this parameter. The CC-Link version setting is also included for the line test (specific station). This is only valid when the user parameter (CC-Link master) <CC-Link version> is set to <1: Ver.2>.
Extended cyclical setting	0: Normal pts. 1: Double pts. 2: Quadruple pts. 3: Octuple pts.	Set the extended cyclical setting for the slave station. When the <Extended cyclical setting> is set to another setting besides <0: Normal pts.> while the <CC-Link version> is set to <0: Ver.1>, then this is ignored. This is only valid when the user parameter (CC-Link master) <CC-Link version> is set to <1: Ver.2>.
Occupied number of stations	0: 1 station 1: 2 stations 2: 3 stations 3: 4 stations	Set the occupied number of slave stations. An error is triggered when the total occupied stations exceeds 32. Note when <By station category> is set to <0: Remote I/O station>, <0: 1 station> is used for the occupied number of stations regardless of this parameter.

Parameter name	Setting range	Description
Transient send buffer size	1 - 512 (Word) 0:No setting	Set the maximum size assigned to each local station and intelligent device station for the buffer when sending in transient communication.
Transient receive buffer size	1 - 512 (Word) 0:No setting	Set the maximum size assigned to each local station and intelligent device station for the buffer when receiving in transient communication.
Specify reserved station	0:No 1:Yes	Specify the reserved station for the slave stations.
Specify invalid station due to error	0:No 1:Yes	Specify which station is invalid due to an error from error detection.
Specify I/O assignment	0: Depends on occupied number of stations 1: Specify 0 pts. 2: Specify 8 pts. 3: Specify 16 pts. 4: Specify mix of 8 &16 pts.	Set the unit for the signal points. This parameter is valid when the user parameter (CC-Link master) <CC-Link version> is set to <1: Ver.2>. When the user parameter (CC-Link master) <CC-Link version> is set to <0: Ver.1>, <0: Depends on occupied number of stations> is used regardless of this parameter setting. In addition, when the user parameter (CC-Link master) <CC-Link version> is set to <1: Ver.2>, an error is triggered if the following conditions apply. 1. When set to <1: Specify 0 pts.> while the <Specify reserved station> is set to <0: No> 2. <By station category> is not set to <0: Remote I/O station> and one of the following settings is used: <2: Specify 8 pts.>, <3: Specify 16 pts.> or <4: Specify mix of 8 &16 pts.>

4.2.2.1 Copy data

One data set for a slave station number can be copied to another station number.

<<Procedure>>

1. Move the cursor to the device name of the station number that has the data you wish to import.
2. Press the **[F5]** key (Copy).
Import the data set for the station number where the cursor is.
The imported data is saved until data from another station is imported or until the power is turned OFF.
3. Move the cursor to the station number where you wish to paste it.
4. Press the **[F6]** key (Paste).
The data set from the copy source is overwritten at the copy destination.

4.2.3 NC Input Signals (Remote Output and Remote Register Output)

NC input signals (input into a master station or intelligent device station) are assigned to the remote output (hereafter referred to as “RX”) for the CC-Link. In addition, the NC input signals can be assigned to the remote register output (hereafter referred to as “RWr”).

(Note) NC input signals are the output signals when looking from the CC-Link slave side.

The following is the procedure when the NC is the master station (<Self-station number> is set to <0>).

<<Procedure>>

1. Select <External I/O signal> on the data bank menu screen.
2. Press the [>>] key until the function key selections change to the following display.

The following displays when the <Self-station number> is set to <0>.

Completion mode	CC-Link config.	CC-Link → NC(Master)	NC(Master) → CC-Link	CC-Link status	Copy	Paste	Delete CC-Link data
-----------------	-----------------	----------------------	----------------------	----------------	------	-------	---------------------

3. When the [F2] key (CC-Link → NC (master)) is pressed, the following screen appears.

CC-Link master CC-Link → NC (master)							
2019/06/15 18:22:38							
RX/BWr.01 Slave station 01 ST1							
RX	Signal	RX	Signal	RX	Signal	RX	Signal
RX000	EXER22	RX008		RX010	PLCX010-01F	RX018	PLCX010-01F
RX001	EXER23	RX009		RX011	PLCX010-01F	RX019	PLCX010-01F
RX002	EXER24	RX00A		RX012	PLCX010-01F	RX01A	PLCX010-01F
RX003		RX00B		RX013	PLCX010-01F	RX01B	PLCX010-01F
RX004		RX00C		RX014	PLCX010-01F	RX01C	PLCX010-01F
RX005		RX00D		RX015	PLCX010-01F	RX01D	PLCX010-01F
RX006		RX00E		RX016	PLCX010-01F	RX01E	PLCX010-01F
RX007		RX00F		RX017	PLCX010-01F	RX01F	PLCX010-01F
RX000							
Check OK							
1. PR01	11. EXRST	21. EXER15	31. EXER25	41. EXER35	51. EXER45	Can be deleted	
2. PR02	12. EXORG	22. EXER16	32. EXER26	42. EXER36	52. EXER46		
3. PR04	13. EXZORG	23. EXER17	33. EXER27	43. EXER37	53. EXER47		
4. PR08	14. CTURN	24. EXER18	34. EXER28	44. EXER38	54. EXER48		
5. PR016	15. MFIN	25. EXER19	35. EXER29	45. EXER39	55. EXER49		
6. PR032	16. EXER10	26. EXER20	36. EXER30	46. EXER40	56. EXER50		
7. PR064	17. EXER11	27. EXER21	37. EXER31	47. EXER41	57. EXER51		
8. EXSTRT	18. EXER12	28. EXER22	38. EXER32	48. EXER42	58. EXER52		
9. OUTST	19. EXER13	29. EXER23	39. EXER33	49. EXER43	59. EXER53		
10. EXSTOP	20. EXER14	30. EXER24	40. EXER34	50. EXER44	60. EXER54		
Completion mode	CC-Link config.	CC-Link → NC(Master)	NC(Master) → CC-Link	CC-Link status	Overlapping terminal check	Search	Next menu

Move the cursor to the RX to be set, and enter the signal number to be assigned.

The subtitle slave station number and the device name are updated depending on the cursor position.

For further details on the external output signals that can be assigned to the RX, refer to section “2.6.2.2 Data input” in the Operation Manual (Data).

In addition, the [V] key can be pressed to show the RX that is offset by 32 bits.

The example below is a screen that is connected to a remote I/O station with 8 points and a remote I/O station (2nd) with 16 points. RX008 to RX00F shown on the screen are empty fields and grayed out. Even if they are set, they are ignored.

In this example, <Specify I/O assignment> for the first remote I/O station needs to be set to <4: Specify mix of 8 & 16 pts.>. The same also applies when a remote device station, intelligent device station or local station is set as the 2nd station. If <Specify I/O assignment> for the first remote I/O station is not set to <4: Specify mix of 8 & 16 pts.>, the alarm <<CC-Link configuration setting error>> is triggered.

CC-Link master CC-Link → NC (master)							
2019/06/15 18:25:23							
RX/BWr.01 Slave station 01 D1							
RX	Signal	RX	Signal	RX	Signal	RX	Signal
RX000		RX008		RX010		RX018	
RX001		RX009		RX011		RX019	
RX002		RX00A		RX012		RX01A	
RX003		RX00B		RX013		RX01B	
RX004		RX00C		RX014		RX01C	
RX005		RX00D		RX015		RX01D	
RX006		RX00E		RX016		RX01E	
RX007		RX00F		RX017		RX01F	
RX000							
Check OK							
1. PR01	11. EXRST	21. EXER15	31. EXER25	41. EXER35	51. EXER45	Can be deleted	
2. PR02	12. EXORG	22. EXER16	32. EXER26	42. EXER36	52. EXER46		
3. PR04	13. EXZORG	23. EXER17	33. EXER27	43. EXER37	53. EXER47		
4. PR08	14. CTURN	24. EXER18	34. EXER28	44. EXER38	54. EXER48		
5. PR016	15. MFIN	25. EXER19	35. EXER29	45. EXER39	55. EXER49		
6. PR032	16. EXER10	26. EXER20	36. EXER30	46. EXER40	56. EXER50		
7. PR064	17. EXER11	27. EXER21	37. EXER31	47. EXER41	57. EXER51		
8. EXSTRT	18. EXER12	28. EXER22	38. EXER32	48. EXER42	58. EXER52		
9. OUTST	19. EXER13	29. EXER23	39. EXER33	49. EXER43	59. EXER53		
10. EXSTOP	20. EXER14	30. EXER24	40. EXER34	50. EXER44	60. EXER54		
Completion mode	CC-Link config.	CC-Link → NC(Master)	NC(Master) → CC-Link	CC-Link status	Overlapping terminal check	Search	Next menu

The signals can be assigned up to RX3FF.

The system area is also included in the assignment area, but it is possible to assign signals following the directions in the manual for the slave device.

Note when a local station or intelligent device station is connected as a slave station, the last two bits in the assignment area are grayed out because the assignment is prohibited. They are ignored even when set. The same applies for RY as well.

In addition, after the page number (1 to 32: RX, 33 to 36: RWr) is entered, when the **[F6]** key (Search) is pressed, the screen changes to the specified page. The page number is displayed on the left side of the subtitle.

Press the **[▽]** key again to display the signal assignment screen for the remote register input (hereafter referred to as “RWr”).

The subtitle slave station number and the device name are updated depending on the cursor position.

CC-Link master: CC-Link -> NC (master) 2019/06/15 16:22:47

RX/RWr: 33 Slave station 01 ST1

RWr	Signal	RWr	Signal	RWr	Signal	RWr	Signal
RWr000	PLCD0768	RWr008	PLCD0776	RWr010	PLCD0784	RWr018	PLCD0792
RWr001	PLCD0769	RWr009	PLCD0777	RWr011	PLCD0785	RWr019	PLCD0793
RWr002	PLCD0770	RWr00A	PLCD0778	RWr012	PLCD0786	RWr01A	PLCD0794
RWr003	PLCD0771	RWr00B	PLCD0779	RWr013	PLCD0787	RWr01B	PLCD0795
RWr004	PLCD0772	RWr00C	PLCD0780	RWr014	PLCD0788	RWr01C	PLCD0796
RWr005	PLCD0773	RWr00D	PLCD0781	RWr015	PLCD0789	RWr01D	PLCD0797
RWr006	PLCD0774	RWr00E	PLCD0782	RWr016	PLCD0790	RWr01E	PLCD0798
RWr007	PLCD0775	RWr00F	PLCD0783	RWr017	PLCD0791	RWr01F	PLCD0799

RWr000 Check OK

Can be deleted

999, RESERV
2000, PLCD0000
2001, PLCD0001
2002, PLCD0002
:
:
:
:
4045, PLCD2045
4046, PLCD2046
4047, PLCD2047

Completion mode	CC-Link config.	CC-Link -> NC (Master)	NC (Master) -> CC-Link	CC-Link status	Overlapping terminal check	Search	Next menu
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Move the cursor to the RWr to be set, and enter the signal number to be assigned. However, only the signals noted below can be assigned to the RWr. The operator message <<Input data error>> appears if another signal, other than those signals noted, is assigned.

- 999 (RESERV)
- 2000 to 4047 (PLCD0000 to PLCD2047)

When the **[▽]** key is pressed, the RWr is displayed and offset by 32 words. Signals can be assigned up to RWr07F.

(Note 1) If the RXxxx/RWrxxx is grayed out, it indicates that the CM PCB (CC-Link) is not connected or that it falls outside of the usable range (Refer to the CC-Link slave parameters <By station category>, <Occupied number of stations> and section below “3.2.5 List of CC-Link points”).

Even if the RXxxx/RWrxxx is grayed out, the signal does not input from the slave station. Refer to “2.6.2.2 Data input” in the Operation Manual (Data) for further details about editing.

(Note 2) When a setting is changed, the operator message <<Press the [RST] key.>> appears.

The following is the procedure when the NC is an intelligent device station (<Self-station number> is not set to <0>).

<<Procedure>>

1. Select <External I/O signal> on the data bank menu screen.
2. Press the [>>] key until the function key selections change to the following display.
The follow display appears when <Self-station number> is not set to <0>. Editing is possible with the [F1] key (CC-Link configuration), but it does not output from the NC. The function keys apart from the [F1] key are the same when it is a master station.

Completion mode	CC-Link config.	CC-Link -> NC (ID station)	NC (ID station) -> CC-Link	CC-Link status	Overlapping terminal check	Search	Next menu
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3. When the [F2] key (CC-Link → NC (ID station)) is pressed, the following screen appears.
The signal address for the subtitle is the leading self-station address and is updated according to the <Self-station number>.

The same also applies to the RW_r, RY and RW_w signals.

CC-Link master CC-Link → NC (ID station)							
RX/RWr.01 Intelligent device station RX000							
RX	Signal	RX	Signal	RX	Signal	RX	Signal
+000	EXER22	+008		+010	PLCX010-01F	+018	PLCX010-01F
+001	EXER23	+009		+011	PLCX010-01F	+019	PLCX010-01F
+002	EXER24	+00A		+012	PLCX010-01F	+01A	PLCX010-01F
+003		+00B		+013	PLCX010-01F	+01B	PLCX010-01F
+004		+00C		+014	PLCX010-01F	+01C	PLCX010-01F
+005		+00D		+015	PLCX010-01F	+01D	PLCX010-01F
+006		+00E		+016	PLCX010-01F	+01E	PLCX010-01F
+007		+00F		+017	PLCX010-01F	+01F	PLCX010-01F
Addresses are relative to the leading self-station							
RX000				Check OK		Can be deleted	
1. PR01	11. EXRST	21. EXER15	31. EXER25	41. EXER35	51. EXER45		
2. PR02	12. EXORG	22. EXER16	32. EXER26	42. EXER36	52. EXER46		
3. PR04	13. EXZORG	23. EXER17	33. EXER27	43. EXER37	53. EXER47		
4. PR08	14. CTURN	24. EXER18	34. EXER28	44. EXER38	54. EXER48		
5. PR16	15. MFIN	25. EXER19	35. EXER29	45. EXER39	55. EXER49		
6. PR32	16. EXER10	26. EXER20	36. EXER30	46. EXER40	56. EXER50		
7. PR064	17. EXER11	27. EXER21	37. EXER31	47. EXER41	57. EXER51		
8. EXSTRT	18. EXER12	28. EXER22	38. EXER32	48. EXER42	58. EXER52		
9. OUTST	19. EXER13	29. EXER23	39. EXER33	49. EXER43	59. EXER53		
10. EXSTOP	20. EXER14	30. EXER24	40. EXER34	50. EXER44	60. EXER54		
Completion mode	CC-Link config.	CC-Link -> NC (ID station)	NC (ID station) -> CC-Link	CC-Link status	Overlapping terminal check	Search	Next menu

Press the [▽] key again to display the signal assignment screen for the following remote register input (hereafter referred to as “RW_r”).

Input (parameter referred to as "RWr")

CC-Link master CC-Link → NC (ID station)				2019/06/15 16:24:35			
RX/RWr.33 Intelligent device station RWr000							
RWr	Signal	RWr	Signal	RWr	Signal	RWr	Signal
+000	PLCD0768	+008	PLCD0776	+010	PLCD0784	+018	PLCD0792
+001	PLCD0769	+009	PLCD0777	+011	PLCD0785	+019	PLCD0793
+002	PLCD0770	+00A	PLCD0778	+012	PLCD0786	+01A	PLCD0794
+003	PLCD0771	+00B	PLCD0779	+013	PLCD0787	+01B	PLCD0795
+004	PLCD0772	+00C	PLCD0780	+014	PLCD0788	+01C	PLCD0796
+005	PLCD0773	+00D	PLCD0781	+015	PLCD0789	+01D	PLCD0797
+006	PLCD0774	+00E	PLCD0782	+016	PLCD0790	+01E	PLCD0798
+007	PLCD0775	+00F	PLCD0783	+017	PLCD0791	+01F	PLCD0799

Addresses are relative to the leading self-station

RWr000		Check OK	☒ ☐
		Can be deleted	
999. RESERV			
2000. PLCD0000			
2001. PLCD0001			
2002. PLCD0002			
:			
:			
:			
4045. PLCD2045			
4046. PLCD2046			
4047. PLCD2047			

Completion mode	CC-Link config	CC-Link → NC (ID station) / CC-Link	NC (ID station) → CC-Link	CC-Link status	Overlapping terminal check	Search	Next menu
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4.2.4 NC Output Signals (Remote Input and Remote Register Input)

NC output signals (output from a master station or intelligent device station) are assigned to the remote input (hereafter referred to as “RY”) for the CC-Link. In addition, the NC output signals can be assigned to the remote register input (hereafter referred to as “RWw”). The following does not include intelligent device stations.

(Note) NC output signals are the input signals when looking from the CC-Link slave side.

<<Procedure>>

1. Select <External I/O signal> on the data bank menu screen.
2. Press the [>>] key until the function key selections change to the following display.

Completion mode	CC-Link config.	CC-Link → NC(Master)	NC(Master) → CC-Link	CC-Link status	Copy	Paste	Delete CC-Link data
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3. When the [F3] key (NC (master) → CC-Link) is pressed, the following screen appears.

CC-Link master NC (master) → CC-Link				2019/06/15 18:22:53			
RY/RWw.01 Slave station01 ST1							
RY	Signal	RY	Signal	RY	Signal	RY	Signal
RY000	PLCY000-00F	RY008	PLCY000-00F	RY010	PLCY010-01F	RY018	PLCY010-01F
RY001	PLCY000-00F	RY009	PLCY000-00F	RY011	PLCY010-01F	RY019	PLCY010-01F
RY002	PLCY000-00F	RY00A	PLCY000-00F	RY012	PLCY010-01F	RY01A	PLCY010-01F
RY003	PLCY000-00F	RY00B	PLCY000-00F	RY013	PLCY010-01F	RY01B	PLCY010-01F
RY004	PLCY000-00F	RY00C	PLCY000-00F	RY014	PLCY010-01F	RY01C	PLCY010-01F
RY005	PLCY000-00F	RY00D	PLCY000-00F	RY015	PLCY010-01F	RY01D	PLCY010-01F
RY006	PLCY000-00F	RY00E	PLCY000-00F	RY016	PLCY010-01F	RY01E	PLCY010-01F
RY007	PLCY000-00F	RY00F	PLCY000-00F	RY017	PLCY010-01F	RY01F	PLCY010-01F

RY000		Check OK			
		Can be deleted			
1. M00	11. M24	21. M482	31. MEMMOD	41. ASTLV4	51. ZP4
2. M30/1	12. M28	22. M484	32. EXPRJUN	42. ASTLV5	52. ZP5
3. M30/2	13. MF	23. M486	33. N00K	43. TOOL	53. ZP6
4. M08	14. M900	24. M450	34. ALM	44. TOLPRE	54. ZP7
5. M11	15. M400	25. M451	35. ALM2	45. ORGFIN	55. ZP8
6. M12	16. M402	26. M455	36. ALMLV1	46. SPTURN	56. PFIN1
7. M14	17. M404	27. M456	37. ALMLV2	47. ZPX	57. PFIN2
8. M18	18. M406	28. AUTO	38. ASTLV1	48. ZPY	58. RSTOUT
9. M21	19. M408	29. STL	39. ASTLV2	49. ZPZ	59. TLBRKE
10. M22	20. M480	30. MEMOK	40. ASTLV3	50. ZPSP	60. RUNSTP

Completion mode	CC-Link config	CC-Link → NC(Master)	NC(Master) → CC-Link	CC-Link status	Overlapping terminal	Search	Next menu
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Move the cursor to the RY to be set, and enter the signal number to be assigned.

The subtitle slave station number and the device name are updated depending on the cursor position.

Note, for further details on the output signals that can be assigned to the RY, refer to section “2.6.2.2 Data input” in the Operation Manual (Data).

In addition, the [▽] key can be pressed to show the RY that is offset by 32 bits.

In addition, after the page number (1 to 32: RY, 33 to 36: RWw) is entered, when the [F6] key (Search) is pressed, the screen changes to the specified page. The page number is displayed on the left side of the subtitle.

Press the [▽] key again to display the RWw signal assignment screen.

The subtitle slave station number and the device name are updated depending on the cursor position.

CC-Link master NC (master) -> CC-Link 2019/06/15 16:22:59

RY/RWw33 Slave station01 ST1

RWw	Signal	RWw	Signal	RWw	Signal	RWw	Signal
RWw000	PLCD0512	RWw008	PLCD0520	RWw010	PLCD0528	RWw018	PLCD0536
RWw001	PLCD0513	RWw009	PLCD0521	RWw011	PLCD0529	RWw019	PLCD0537
RWw002	PLCD0514	RWw00A	PLCD0522	RWw012	PLCD0530	RWw01A	PLCD0538
RWw003	PLCD0515	RWw00B	PLCD0523	RWw013	PLCD0531	RWw01B	PLCD0539
RWw004	PLCD0516	RWw00C	PLCD0524	RWw014	PLCD0532	RWw01C	PLCD0540
RWw005	PLCD0517	RWw00D	PLCD0525	RWw015	PLCD0533	RWw01D	PLCD0541
RWw006	PLCD0518	RWw00E	PLCD0526	RWw016	PLCD0534	RWw01E	PLCD0542
RWw007	PLCD0519	RWw00F	PLCD0527	RWw017	PLCD0535	RWw01F	PLCD0543

RWw000 Check OK Can be deleted

999. RESERV
2000. PLCD0000
2001. PLCD0001
2002. PLCD0002
:
:
:
4045. PLCD2045
4046. PLCD2046
4047. PLCD2047

Completion mode	CC-Link config.	CC-Link -> NC(Master)	NC(Master) -> CC-Link	CC-Link status	Overlapping terminal check	Search	Next menu
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Move the cursor to the RWw to be set, and enter the signal number to be assigned. However, only the signals noted below can be assigned to the RWw. The operator message <<Input data error>> appears if another signal, other than those signals noted, is assigned.

- 999 (RESERV)
- 2000 to 4047 (PLCD0000 to PLCD2047)

When the [▽] key is pressed, the RWw is displayed and offset by 32 words. Signals can be assigned up to RWw07F.

(Note 1) If the RYxxx/RWwxxx is grayed out, it indicates that the CM PCB (CC-Link) is not connected or that it falls outside of the usable range (CC-Link slave parameter <By station category>, <Occupied number of stations> and section below “List of CC-Link points”). Even if the RXxxx/RWrxxx is grayed out, the signal can be assigned but the signal will not output to the slave station.

(Note 2) When a setting is changed, the operator message <<Press the [RST] key.>> appears.

4.2.5 List of CC-Link Points

The following includes a list of link points included in the system area, which is based on the CC-Link standards.

(Note) These points exclude header information in the extended cyclical setting for the bit area.

The extended cyclical setting becomes valid when the basic setting for the CC-Link master <CC-Link version> is set to <1: Ver.2>. The CC-Link points when set to <0: Ver.1> are the same as when the extended cyclical setting below is set to 1.

Extended cyclical setting		1	2	4	8
1 station occupied	Bit	RX 32 RY 32	RX 32 RY 32	RX 64 RY 64	RX 128 RY 128
	Word	RWr 4 RWw 4	RWr 8 RWw 8	RWr 16 RWw 16	RWr 32 RWw 32
2 stations occupied	Bit	RX 64 RY 64	RX 96 RY 96	RX 192 RY 192	RX 384 RY 384
	Word	RWr 8 RWw 8	RWr 16 RWw 16	RWr 32 RWw 32	RWr 64 RWw 64
3 stations occupied	Bit	RX 96 RY 96	RX 160 RY 160	RX 320 RY 320	RX 640 RY 640
	Word	RWr 12 RWw 12	RWr 24 RWw 24	RWr 48 RWw 48	RWr 96 RWw 96
4 stations occupied	Bit	RX 128 RY 128	RX 224 RY 224	RX 448 RY 448	RX 896 RY 896
	Word	RWr 16 RWw 16	RWr 32 RWw 32	RWr 64 RWw 64	RWr 128 RWw 128

(Note) The system area is different depending on the device specification of each manufacturer. For further details, refer to the device manual that is supplied by the manufacturer. If the NC is set as intelligent device station (when <Self-station number> is not set to 0), the last 16 points of the bit range cannot be used as they are part of the system range. This grays out and is ignored.

4.2.6 CC-Link Status

The CM PCB status information for the CC-Link can be assigned as a PLC input signal. The status information that can be referenced is shown below.

Status	Description		Content
STATUS	0 bit	Initialization completion flag	0: Not initialized 1: Initialized
	1 bit	Communicating flag	0: No communication 1: Communicating
	2 bit	Error flag	0: Normal 1: Error
	3 bit...15 bit	Not used	0
ST_INIT	0 bit	CM PCB initialization flag	0: Initializing 1: Initialized
	1 bit	Communication standby	0: Normal 1: Standby
	2 bit...15 bit	Not used	0
ST_ERR	0 bit	(Duplicate node)	0 (Fixed)
	1 bit	Communication timeout	0: Normal 1: Communication timeout
	2 bit...14 bit	Not used	0
	15 bit	Other network errors	0: Normal 1: Network error triggered
ST_BRD	0 bit...3 bit	Not used	0
	4 bit	LINE(ERROR)	0: Normal 1: Error
	5 bit	TIME(ERROR)	
	6 bit	PRM(ERROR)	
	7 bit	M/S(ERROR)	
	8 bit	SW(ERROR)	
	9 bit...10 bit	Not used	0
	11 bit	LOCAL	0: Normal 1: Error
	12 bit	S.MST	
	13 bit	MST	
	14 bit	ERR	
	15 bit	RUN	
ST_SET	0 bit	Station type	0: Master station / intelligent device station 1: Standby master station
	1 bit	Master station redundancy function	0: Not equipped 1: Equipped
	2 bit	Not used	0
	3 bit	Input status from data link error	0: Clear 1: Maintain
	4 bit...5 bit	Occupied number of stations	00: 1 station occupied 10: 2 stations occupied 11: 3 stations occupied 01: 4 stations occupied
	6 bit...9 bit	Not used	0
	10 bit...11 bit	Extended cyclical setting	00: Normal pts. 01: Double pts. 10: Triple pts. 11: Quadruple pts.
	12 bit...15 bit	Not used	0

Status		Description	Content
ST_S010	0 bit...15 bit	Station data link (for station numbers 1 to 16) (Note)	0: Normal 1: Data link error triggered
ST_S011	16 bit...31 bit	Station data link (for station numbers 17 to 32) (Note)	
ST_S020	0 bit...15 bit	Station watchdog timer error (for station numbers 1 to 16) (Note)	0: No watchdog timer error 1: Watchdog timer error
ST_S021	16 bit...31 bit	Station watchdog timer error (for station numbers 17 to 32) (Note)	
ST_S030	0 bit...15 bit	Station fuse meltdown (for station numbers 1 to 16) (Note)	0: Normal 1: Fuse meltdown
ST_S031	16 bit...31 bit	Station fuse meltdown (for station numbers 17 to 32) (Note)	
ST_S040	0 bit...15 bit	Station switch status (for station numbers 1 to 16) (Note)	0: No change 1: Changed
ST_S041	16 bit...31 bit	Station switch status (for station numbers 17 to 32) (Note)	
ST_S050	0 bit...15 bit	Transient communication status (for station numbers 1 to 16) (Note)	0: No transient communication error 1: Transient communication error
ST_S051	16 bit...31 bit	Transient communication status (for station numbers 17 to 32) (Note)	
ST_S060	0 bit...15 bit	Station number duplication (for station numbers 1 to 16) (Note)	0: Normal 1: Station number duplicated
ST_S061	16 bit...31 bit	Station number duplication (for station numbers 17 to 32) (Note)	
ST_S070	0 bit...15 bit	Mounted hardware matches parameter (for station numbers 1 to 16) (Note)	0: Normal 1: Mismatch error
ST_S071	16 bit...31 bit	Mounted hardware matches parameter (for station numbers 17 to 32) (Note)	
ST_S080	0 bit...15 bit	Line test results (for station numbers 1 to 16) (Note)	0: Normal 1: Error
ST_S081	16 bit...31 bit	Line test results (for station numbers 17 to 32) (Note)	
ST_S090	0 bit...15 bit	By station category (2) (for station numbers 1 to 16) (Note)	0: Ver.1 compatible slave station 1: Ver.2 compatible slave station
ST_S091	16 bit...31 bit	By station category (2) (for station numbers 17 to 32) (Note)	

Status		Description	Content
ST_S100	0 bit...15 bit	Mounted hardware matches parameter (2) (for station numbers 1 to 16) (Note)	0: Normal 1: Mismatch error
ST_S101	16 bit...31 bit	Mounted hardware matches parameter (2) (for station numbers 17 to 32) (Note)	
ST_RE00	0 bit...15 bit	Reserved data 00	
ST_RE01	0 bit...15 bit	Reserved data 01	
ST_RE02	0 bit...15 bit	Reserved data 02	
ST_RE03	0 bit...15 bit	Reserved data 03	
ST_RE04	0 bit...15 bit	Reserved data 04	
ST_RE05	0 bit...15 bit	Reserved data 05	
ST_RE06	0 bit...15 bit	Reserved data 06	
ST_RE07	0 bit...15 bit	Reserved data 07	
ST_RE08	0 bit...15 bit	Reserved data 08	

(Note) The relationship between each bit and the station number is the same as line numbers 9 to 28 for the status I/O signals under “4.3 Monitor screen”.

<<Procedure>>

1. Select <External I/O signal> on the data bank menu screen.
2. Press the [>>] key until the function key selections change to the following display.

Completion mode	CC-Link config	CC-Link -> NC(Master)	NC(Master) -> CC-Link	CC-Link status	Copy	Paste	Delete CC-Link data
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3. Press the [F4] key (Status) to display the following screen.

CC-Link master CC-Link status							
2019/06/15 16:23:05							
Status	Signal	Status	Signal	Status	Signal	Status	Signal
STATUS	PLCD0720	ST_S021	PLCD0728	ST_S061	PLCD0736	ST_S101	PLCD0744
ST_INIT	PLCD0721	ST_S030	PLCD0729	ST_S070	PLCD0737	ST_RE00	PLCD0745
ST_ERR	PLCD0722	ST_S031	PLCD0730	ST_S071	PLCD0738	ST_RE01	PLCD0746
ST_BRD	PLCD0723	ST_S040	PLCD0731	ST_S080	PLCD0739	ST_RE02	PLCD0747
ST_SET	PLCD0724	ST_S041	PLCD0732	ST_S081	PLCD0740	ST_RE03	PLCD0748
ST_S010	PLCD0725	ST_S050	PLCD0733	ST_S090	PLCD0741	ST_RE04	PLCD0749
ST_S011	PLCD0726	ST_S051	PLCD0734	ST_S091	PLCD0742	ST_RE05	PLCD0750
ST_S020	PLCD0727	ST_S060	PLCD0735	ST_S100	PLCD0743	ST_RE06	PLCD0751
STATUS Check OK OK Cancel Can be deleted							
999. RESERV 2000. PLCD0000 2001. PLCD0001 2002. PLCD0002 : : : 4045. PLCD2045 4046. PLCD2046 4047. PLCD2047							
Completion mode	CC-Link config	CC-Link -> NC(Master)	NC(Master) -> CC-Link	CC-Link status	Overlapping terminal check	Search	Next menu

Move the cursor to the status address to be set, and enter the signal number to be assigned. However, only the signals noted below can be assigned to each status address. The operator message <<Input data error>> appears if another signal, other than those signals noted, is assigned.

- 999 (RESERV)
- 2000 to 4047 (PLCD0000 to PLCD2047)

(Note 1) If the CM PCB (CC-Link) is not connected, the address is greyed out. Even if the address is greyed out, the signal can be assigned but it will not input into the NC.

(Note 2) In addition, the [F6] key (Search) and the [F7] key (Next menu) are greyed out on the CC-Link status screen, and the keys cannot be pressed.

(Note 3) When the setting is complete, the operator message <<Press the [RST] key.>> appears.

4.2.7 Intelligent Device Station

The CM PCB supports the following function for the intelligent device station.

- Cyclical communication (bit/word data)

<<Procedure>>

1. Select <User parameter> on the data bank menu screen.
2. Set the <Self-station number> to a setting that is not 0.
- Set <Occupied number of stations (for intelligent device stations)> as necessary.

(Note) When the setting is complete, the operator message <<Press the [RST] key.>> appears.

4.2.8 Transient Communication

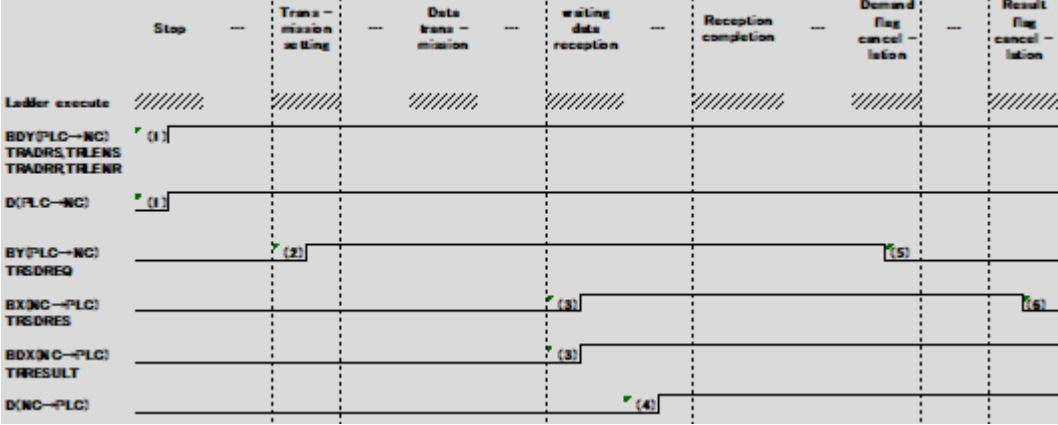
This function is used for extracting slave station data from the master station and using parameters (apart from the CC-Link ones) to set the slave station from the master station.

<<Procedure>>

1. Select <External I/O signal> on the data bank menu screen.
2. Press the [>>] key and then press the [F1] key (CC-Link configuration).
3. Set the <Transient send buffer size> and <Transient receive buffer size>.
4. Set the transient communication according to the data (D, BY and BDY) and the internal I/O relay saved in the PLC. Refer to “Chapter 6 – 6.2.3 to 6.2.6” in the PLC System Manual for further details.

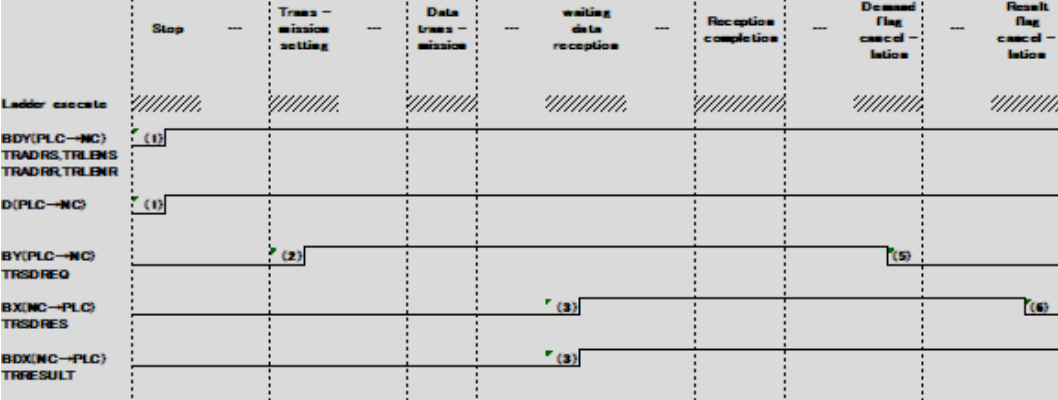
The timing chart for 4. in the <<Procedure>> is shown below.

Data reading for slave station



Turn (2) ON from the next scan time that was turned ON for (1).
Turn (5) OFF after receiving data for (3). Note that it is possible to read the receiving data for (4) from the next scan time after (3).

Data writing for slave station



(Note) Cyclical communication stops during transient communication.

4.3 Monitor Screen

The CM PCB information, such as the communication status or the alarms, can be monitored.

<<Procedure>>

1. Select <I/O> on the I/O menu screen.

I/O menu	Main	Local	Slave status display	Extension I/O	Field bus	Loading system	Version
----------	------	-------	----------------------	---------------	-----------	----------------	---------

2. Press the [F5] key (Field bus) to display the following screen.
(Total of 7 pages)

Input/output(field bus)										2019/06/15 16:23:30									
Status																			
FEDC BA98 7654 3210					FEDC BA98 7654 3210														
1	0100	0000	0000	0000	17	0000	0000	0000	0000	Bus type									
2	0000	0000	0000	0000	18	0000	0000	0000	0000	CC-Link master station									
3	0000	0000	0000	0000	19	0000	0000	0000	0000	Transfer rate									
4	0000	0000	0000	0000	20	0000	0000	0000	0000	Address									
5	0000	0000	0000	0000	21	0000	0000	0000	0000	Data link status									
6	0000	0000	0000	0000	22	0000	0000	0000	0000	Initial status									
7	0000	0000	0000	0000	23	0000	0000	0000	0000	Operation status									
8	0000	0000	0000	0000	24	0000	0000	0000	0000	Normal									
9	1111	1111	1111	1111	25	0000	0000	0000	0000	Mode setting status									
10	1111	1111	1111	1111	26	0000	0000	0000	0000	Online									
11	0000	0000	0000	0000	27	0000	0000	0000	0000	Total number of stations									
12	0000	0000	0000	0000	28	0000	0000	0000	0000	0									
13	0000	0000	0000	0000	29	0000	0000	0000	0000	Max. comm. station No.									
14	0000	0000	0000	0000	30	0000	0000	0011	0000	0									
15	0000	0000	0000	0000	31	0000	0000	0000	0000	No. of connecting nodes									
16	0000	0000	0000	0000	32	0000	0000	0000	0000	Error code									
FEDC BA98 7654 3210					FEDC BA98 7654 3210					Line test results									
										Line test error code									

Status	Description
<Bus type>	Indicates the type of the field bus that is connected to the CM PCB. <CC-Link master station> or <CC-Link ID station> is displayed.
<Transfer rate>	Indicates the transfer rate that is set for the CM PCB.
<Address>	Indicates the station number that is set for the CM PCB.
<Data link status>	Indicates the data link status for the self-station. (Initial status, parameter receive standby, data link in progress, data link stopped, no data link (no polling), no data link (line error), no data link (other), line test in progress, parameter being set, auto return processing and resetting)
<Operation status>	Indicates the operation status for the self-station. (Normal, transmit line error detect, parameter error detect, CRC error detection, timeout error detection, abort error detection, setting error detection, and other error detection)
<Mode setting status>	Indicates the mode setting for the self-station. (Online, offline)
<Total number of stations>	Indicates the last station number that is set in the parameter. 0 to 32 (Note)
<Max. comm. station No.>	Indicates the maximum number of stations that is set in the data link. 0 to 32 (Note)
<No. of connecting nodes>	Indicates the number of connecting nodes that is set in the data link. 0 to 32 (Note)
<Error code>	Indicates the error information on the I/F PCB that is integrated.

Status	Description
<Line test results>	Indicates the results from the line test. This field is blank when no test has been carried out.
<Line test error code>	Indicates the error code that prevents it from passing the line test.

(Note) Zero (0) is displayed when the bus type is set to <CC-Link ID station>.

The following shows the details about the status I/O signals on the left side of the screen. The error (SW area for dual port RAM on CM PCB) is displayed for each slave station.

Line No. for status I/O signal	Description	Content																											
1	Interface board status	The interface board status is stored. bit 4 LINE(ERROR) bit 5 TIME(ERROR) bit 6 PRM(ERROR) bit 7 M/S(ERROR) bit 8 SW(ERROR) bit B LOCAL bit C S.MST bit D MST bit E ERROR bit F RUN Not used for items not listed above.																											
2	Operation setting status	The operation setting status is stored. 0: OFF 1: ON bit 0 Station type 0: Master station / intelligent device station, 1: Standby master station bit 1 Master station redundancy function 0: Not equipped, 1: Equipped bit 3 Input status from station with data link error 0: Clear, 1: Maintain bit 4,5 Occupied number of stations 00: 1 station, 10: 2 stations, 11: 3 stations, 01: 4 stations bit A,B Extended cyclical setting 00: Normal pts. 01: Double pts. 10: Triple pts. 11: Quadruple pts. Not used for items not listed above.																											
3,4	Specified status for reserved station	The specified status for the reserved station is stored. 0: Non-reserved station 1: Reserved station bit <table border="1"><tr><td>F</td><td>E</td><td>D</td><td>C</td><td></td><td>3</td><td>2</td><td>1</td><td>0</td></tr><tr><td>16</td><td>15</td><td>14</td><td>13</td><td>to</td><td>4</td><td>3</td><td>2</td><td>1</td></tr><tr><td>32</td><td>31</td><td>30</td><td>29</td><td>to</td><td>20</td><td>19</td><td>18</td><td>17</td></tr></table> 1 to 32 in the table refers to the slave station number.	F	E	D	C		3	2	1	0	16	15	14	13	to	4	3	2	1	32	31	30	29	to	20	19	18	17
F	E	D	C		3	2	1	0																					
16	15	14	13	to	4	3	2	1																					
32	31	30	29	to	20	19	18	17																					
5,6	Specified status for invalid station due to error	The setting status for an invalid station (due to error) is stored. 0: Station besides an error invalid station 1: Error invalid station bit <table border="1"><tr><td>F</td><td>E</td><td>D</td><td>C</td><td></td><td>3</td><td>2</td><td>1</td><td>0</td></tr><tr><td>16</td><td>15</td><td>14</td><td>13</td><td>to</td><td>4</td><td>3</td><td>2</td><td>1</td></tr><tr><td>32</td><td>31</td><td>30</td><td>29</td><td>to</td><td>20</td><td>19</td><td>18</td><td>17</td></tr></table> 1 to 32 in the table refers to the slave station number.	F	E	D	C		3	2	1	0	16	15	14	13	to	4	3	2	1	32	31	30	29	to	20	19	18	17
F	E	D	C		3	2	1	0																					
16	15	14	13	to	4	3	2	1																					
32	31	30	29	to	20	19	18	17																					
7,8	Temporary error invalid status	The invalid status (due to a temporary error) is stored. 0: Normal 1: Temporary error invalid status bit <table border="1"><tr><td>F</td><td>E</td><td>D</td><td>C</td><td></td><td>3</td><td>2</td><td>1</td><td>0</td></tr><tr><td>16</td><td>15</td><td>14</td><td>13</td><td>to</td><td>4</td><td>3</td><td>2</td><td>1</td></tr><tr><td>32</td><td>31</td><td>30</td><td>29</td><td>to</td><td>20</td><td>19</td><td>18</td><td>17</td></tr></table> 1 to 32 in the table refers to the slave station number.	F	E	D	C		3	2	1	0	16	15	14	13	to	4	3	2	1	32	31	30	29	to	20	19	18	17
F	E	D	C		3	2	1	0																					
16	15	14	13	to	4	3	2	1																					
32	31	30	29	to	20	19	18	17																					

Line No. for status I/O signal	Description	Content																		
9,10	Station data link status	0: Normal 1: Data link error triggered bit F E D C 3 2 1 0 <table><tr><td>16</td><td>15</td><td>14</td><td>13</td><td>to</td><td>4</td><td>3</td><td>2</td><td>1</td></tr><tr><td>32</td><td>31</td><td>30</td><td>29</td><td>to</td><td>20</td><td>19</td><td>18</td><td>17</td></tr></table> 1 to 32 in the table refers to the slave station number.	16	15	14	13	to	4	3	2	1	32	31	30	29	to	20	19	18	17
16	15	14	13	to	4	3	2	1												
32	31	30	29	to	20	19	18	17												
11,12	Station watchdog timer error	0: No watchdog timer error 1: Watchdog timer error bit F E D C 3 2 1 0 <table><tr><td>16</td><td>15</td><td>14</td><td>13</td><td>to</td><td>4</td><td>3</td><td>2</td><td>1</td></tr><tr><td>32</td><td>31</td><td>30</td><td>29</td><td>to</td><td>20</td><td>19</td><td>18</td><td>17</td></tr></table> 1 to 32 in the table refers to the slave station number.	16	15	14	13	to	4	3	2	1	32	31	30	29	to	20	19	18	17
16	15	14	13	to	4	3	2	1												
32	31	30	29	to	20	19	18	17												
13,14	Station fuse meltdown	0: Normal 1: Fuse meltdown bit F E D C 3 2 1 0 <table><tr><td>16</td><td>15</td><td>14</td><td>13</td><td>to</td><td>4</td><td>3</td><td>2</td><td>1</td></tr><tr><td>32</td><td>31</td><td>30</td><td>29</td><td>to</td><td>20</td><td>19</td><td>18</td><td>17</td></tr></table> 1 to 32 in the table refers to the slave station number.	16	15	14	13	to	4	3	2	1	32	31	30	29	to	20	19	18	17
16	15	14	13	to	4	3	2	1												
32	31	30	29	to	20	19	18	17												
15,16	Station switch status	0: No change 1: Changed bit F E D C 3 2 1 0 <table><tr><td>16</td><td>15</td><td>14</td><td>13</td><td>to</td><td>4</td><td>3</td><td>2</td><td>1</td></tr><tr><td>32</td><td>31</td><td>30</td><td>29</td><td>to</td><td>20</td><td>19</td><td>18</td><td>17</td></tr></table> 1 to 32 in the table refers to the slave station number.	16	15	14	13	to	4	3	2	1	32	31	30	29	to	20	19	18	17
16	15	14	13	to	4	3	2	1												
32	31	30	29	to	20	19	18	17												
17,18	Transient communication status	0: No transient communication error 1: Transient communication error bit F E D C 3 2 1 0 <table><tr><td>16</td><td>15</td><td>14</td><td>13</td><td>to</td><td>4</td><td>3</td><td>2</td><td>1</td></tr><tr><td>32</td><td>31</td><td>30</td><td>29</td><td>to</td><td>20</td><td>19</td><td>18</td><td>17</td></tr></table> 1 to 32 in the table refers to the slave station number.	16	15	14	13	to	4	3	2	1	32	31	30	29	to	20	19	18	17
16	15	14	13	to	4	3	2	1												
32	31	30	29	to	20	19	18	17												
19,20	Station number duplication	0: Normal 1: Station number duplicated bit F E D C 3 2 1 0 <table><tr><td>16</td><td>15</td><td>14</td><td>13</td><td>to</td><td>4</td><td>3</td><td>2</td><td>1</td></tr><tr><td>32</td><td>31</td><td>30</td><td>29</td><td>to</td><td>20</td><td>19</td><td>18</td><td>17</td></tr></table> 1 to 32 in the table refers to the slave station number.	16	15	14	13	to	4	3	2	1	32	31	30	29	to	20	19	18	17
16	15	14	13	to	4	3	2	1												
32	31	30	29	to	20	19	18	17												
21,22	Mounted hardware matches parameter	0: Normal 1: Mismatch error bit F E D C 3 2 1 0 <table><tr><td>16</td><td>15</td><td>14</td><td>13</td><td>to</td><td>4</td><td>3</td><td>2</td><td>1</td></tr><tr><td>32</td><td>31</td><td>30</td><td>29</td><td>to</td><td>20</td><td>19</td><td>18</td><td>17</td></tr></table> 1 to 32 in the table refers to the slave station number.	16	15	14	13	to	4	3	2	1	32	31	30	29	to	20	19	18	17
16	15	14	13	to	4	3	2	1												
32	31	30	29	to	20	19	18	17												
23,24	Line test results	The line test results are displayed. 0: Normal 1: Error bit F E D C 3 2 1 0 <table><tr><td>16</td><td>15</td><td>14</td><td>13</td><td>to</td><td>4</td><td>3</td><td>2</td><td>1</td></tr><tr><td>32</td><td>31</td><td>30</td><td>29</td><td>to</td><td>20</td><td>19</td><td>18</td><td>17</td></tr></table> 1 to 32 in the table refers to the slave station number.	16	15	14	13	to	4	3	2	1	32	31	30	29	to	20	19	18	17
16	15	14	13	to	4	3	2	1												
32	31	30	29	to	20	19	18	17												
25,26	By station category (2)	The slave station compatible with Ver.2 is displayed. 0: Ver.1 compatible slave station 1: Ver.2 compatible slave station bit F E D C 3 2 1 0 <table><tr><td>16</td><td>15</td><td>14</td><td>13</td><td>to</td><td>4</td><td>3</td><td>2</td><td>1</td></tr><tr><td>32</td><td>31</td><td>30</td><td>29</td><td>to</td><td>20</td><td>19</td><td>18</td><td>17</td></tr></table> 1 to 32 in the table refers to the slave station number.	16	15	14	13	to	4	3	2	1	32	31	30	29	to	20	19	18	17
16	15	14	13	to	4	3	2	1												
32	31	30	29	to	20	19	18	17												

- Press the **[▽]** key to display the remote I/O signal status for the field bus.

Input/output (field bus)													2019/06/15 16:23:37										
Input												Output											
	FEDC	BA98	7654	3210		FEDC	BA98	7654	3210		FEDC	BA98	7654	3210		FEDC	BA98	7654	3210				
1	0000	0000	0000	0000	17	0000	0000	0000	0000		1	0000	0000	0000	0000	17	0000	0000	0000	0000			
2	0000	0000	0000	0000	18	0000	0000	0000	0000		2	0000	0000	0000	0000	18	0000	0000	0000	0000			
3	0000	0000	0000	0000	19	0000	0000	0000	0000		3	0000	0000	0000	0000	19	0000	0000	0000	0000			
4	0000	0000	0000	0000	20	0000	0000	0000	0000		4	0000	0000	0000	0000	20	0000	0000	0000	0000			
5	0000	0000	0000	0000	21	0000	0000	0000	0000		5	0000	0000	0000	0000	21	0000	0000	0000	0000			
6	0000	0000	0000	0000	22	0000	0000	0000	0000		6	0000	0000	0000	0000	22	0000	0000	0000	0000			
7	0000	0000	0000	0000	23	0000	0000	0000	0000		7	0000	0000	0000	0000	23	0000	0000	0000	0000			
8	0000	0000	0000	0000	24	0000	0000	0000	0000		8	0000	0000	0000	0000	24	0000	0000	0000	0000			
9	0000	0000	0000	0000	25	0000	0000	0000	0000		9	0000	0000	0000	0000	25	0000	0000	0000	0000			
10	0000	0000	0000	0000	26	0000	0000	0000	0000		10	0000	0000	0000	0000	26	0000	0000	0000	0000			
11	0000	0000	0000	0000	27	0000	0000	0000	0000		11	0000	0000	0000	0000	27	0000	0000	0000	0000			
12	0000	0000	0000	0000	28	0000	0000	0000	0000		12	0000	0000	0000	0000	28	0000	0000	0000	0000			
13	0000	0000	0000	0000	29	0000	0000	0000	0000		13	0000	0000	0000	0000	29	0000	0000	0000	0000			
14	0000	0000	0000	0000	30	0000	0000	0000	0000		14	0000	0000	0000	0000	30	0000	0000	0000	0000			
15	0000	0000	0000	0000	31	0000	0000	0000	0000		15	0000	0000	0000	0000	31	0000	0000	0000	0000			
16	0000	0000	0000	0000	32	0000	0000	0000	0000		16	0000	0000	0000	0000	32	0000	0000	0000	0000			
FEDC BA98 7654 3210					FEDC BA98 7654 3210					FEDC BA98 7654 3210					FEDC BA98 7654 3210								

Output

0000000000000000

FEDCBA9876543210

⏏ Status

⏏ Field bus

033-064

I/O menu	Main	Local	Slave status display	Extension I/O	Field bus	Loading system	Version

4.3.1 Line Test

This test is only valid when the basic setting <Self-station number> for the CC-Link master is set to <0> and when the <Interface board setting> is set to <0: Online>.

- Line test 1 Checks the communication status with all the connected slave stations.
- Line test 2 Checks the communication status with the specified connected slave station(s).

1. Select <I/O> on the I/O menu screen.

2020/04/16

2. Press the **[F5]** key (Field bus) to display the following screen.

Status			
FEBC BA98 7654 3210		FEBC BA98 7654 3210	
1	0100 0000 0000 0000	17	0000 0000 0000 0000
2	0000 0000 0000 0000	18	0000 0000 0000 0000
3	0000 0000 0000 0000	19	0000 0000 0000 0000
4	0000 0000 0000 0000	20	0000 0000 0000 0000
5	0000 0000 0000 0000	21	0000 0000 0000 0000
6	0000 0000 0000 0000	22	0000 0000 0000 0000
7	0000 0000 0000 0000	23	0000 0000 0000 0000
8	0000 0000 0000 0000	24	0000 0000 0000 0000
9	1111 1111 1111 1111	25	0000 0000 0000 0000
10	1111 1111 1111 1111	26	0000 0000 0000 0000
11	0000 0000 0000 0000	27	0000 0000 0000 0000
12	0000 0000 0000 0000	28	0000 0000 0000 0000
13	0000 0000 0000 0000	29	0000 0000 0000 0000
14	0000 0000 0000 0000	30	0000 0000 0011 0000
15	0000 0000 0000 0000	31	0000 0000 0000 0000
16	0000 0000 0000 0000	32	0000 0000 0000 0000
FEBC BA98 7654 3210		FEBC BA98 7654 3210	

Bus type: CC-Link master station
 Transfer rate:
 Address: 255
 Data link status: Initial status
 Operation status: Normal
 Mode setting status: Online
 Total number of stations: 0
 Max. comm. station No.: 0
 No. of connecting nodes: 0
 Error code: 0
 Line test results:
 Line test error code:

0:Line test(all stations) 1 - 32:Specific station

Field bus 161-192
 Field bus 001-032

I/O menu Main Local Slave status display Extension I/O Field bus Loading system Version

After entering 0 into the input field on the bottom left, press the **[ENT]** key to start the line test (for all stations).

In addition, when a number between 1 and 32 is entered and the **[ENT]** key is pressed, the line test (specific station) starts only for the station number that was entered.

After the line test is complete, the line test result (OK/NG) is displayed on the right side of the field bus screen.

If the result is NG (failed), then an error code for the line test is displayed.

(Note 1) A line test cannot be carried out when <0> is not set for the user parameter (CC-Link master) <Standby master station number>. When an attempt is made to perform a line test, the alarm <<User param. setting error (CC-Link master)>> is triggered.

(Note 2) When a line test is carried out at a slow speed, for example, when <0:156kbps> is set for the user parameter (CC-Link master) <Transfer rate>, there is a temporary data link and an operation status error triggers after the test, but this is no problem with the operation.

4.4 CSP+ Files

4.4.1 What are CSP+ Files?

CSP+ files contain data required for device startup, operation and maintenance. They are XML formatted files with the extension “.cspp” and are stored as zip files.

Parameters and other data can be easily set using GXworks2.

4.4.2 CSP+ Files for CC-Link on CM PCB

Japanese : 0x0300_SPEEDIO-ID_12_ja.zip

English : 0x0300_SPEEDIO-ID_12_en.zip

4.4.3 How to Obtain CSP+ Files

To obtain CSP+ files, download them from the CC-Link Partner Association website or use the PLC Programming Software installation CD.

CC-Link Partner Association website

<https://www.cc-link.org/>

CHAPTER 12 (19)

FEATURE COORDINATE SETTING FUNCTION

1 Feature Coordinate Setting Function

1 Feature Coordinate Setting Function

1.1 Overview

1.1.1 Introduction

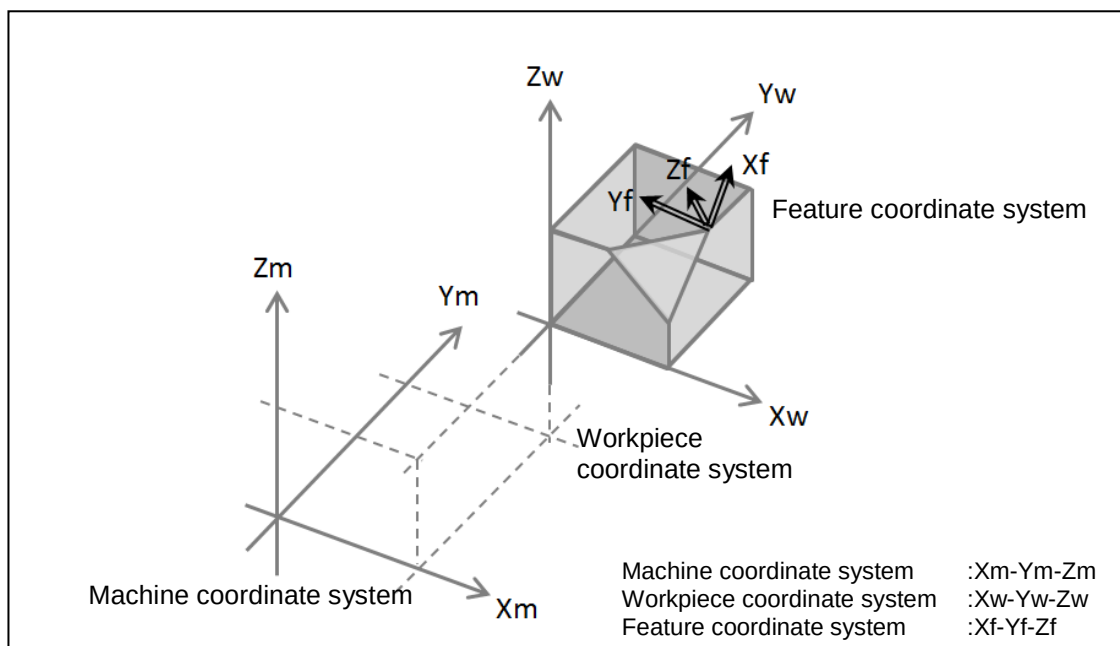
When machining on a feature coordinate, set a new coordinate system on this plane to issue position commands in order to make the program simpler.

This new coordinate system is called the feature coordinate system and is set in the workpiece coordinate system.

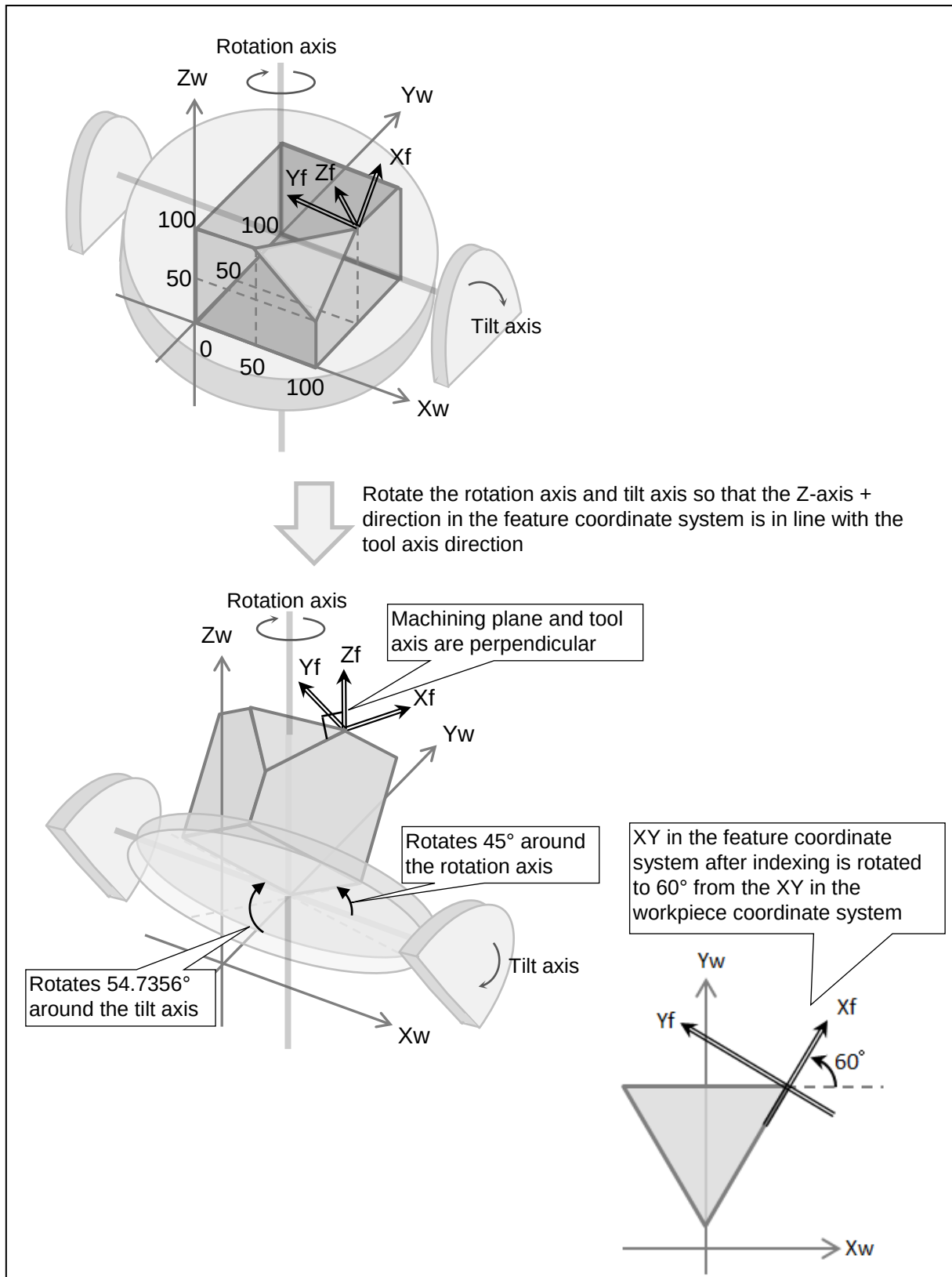
When the feature coordinate system is set, this is called feature coordinate machining mode. The commands are controlled as commands in the feature coordinate coordinate system until the feature coordinate machining mode is cancelled.

Feature coordinate index refers to when the additional axis is moved so that the Z-axis + direction in the feature coordinate system is in line with the tool axis direction.

Relationship between machine coordinate system, workpiece coordinate system and feature coordinate system



Feature coordinate index



1.1.2 Usage Conditions

The feature coordinate setting function is an option that is required to enable and use this function. This function can only be used in NC language mode.

1.2 Parameter Setting

1.2.1 Machine Parameter

System 2

	Name	Setting range	Description
Tilt axis 1	Rotation center X coordinate for tilt axis 1	-9999.999 to 9999.999 mm	Set the rotation center coordinate for the tilt axis and for the rotation axis using the machine coordinates.
	Rotation center Y coordinate for tilt axis 1		
	Rotation center Z coordinate for tilt axis 1		
Rotation axis 1	Rotation center X coordinate for rotation axis 1		
	Rotation center Y coordinate for rotation axis 1		
	Rotation center Z coordinate for rotation axis 1		
Tilt axis 2	Rotation center X coordinate for tilt axis 2		Use the machine coordinates to set the rotation center coordinate on the tilt axis and on the rotation axis for the additional axis on pallet 2 of a machine model equipped with a QT.
	Rotation center Y coordinate for tilt axis 2		
	Rotation center Z coordinate for tilt axis 2		
Rotation axis 2	Rotation center X coordinate for rotation axis 2		
	Rotation center Y coordinate for rotation axis 2		
	Rotation center Z coordinate for rotation axis 2		

- Refer to the user parameter (feature coordinate setting) when specifying the tilt axis and rotation axis.

1.2.2 User Parameter

Feature coordinate setting

Name	Setting range	Description
Tilt axis 1	0: Not specified 1: 4th-axis 2: 5th-axis 3: 6th-axis 4: 7th-axis 5: 8th-axis	Set which axis is a tilt axis or rotation axis.
Rotation axis 1		
Tilt axis 2	0: Not specified 1: 4th-axis 2: 5th-axis 3: 6th-axis 4: 7th-axis 5: 8th-axis	Set the additional axis as a tilt axis or rotation axis for pallet 2 on a machine model equipped with a QT.
Rotation axis 2		
Forward direction for the coordinate system on tilt axis 1	1: Z→Y 3: X→Z 5: Y→X	Set the forward direction for the coordinate system on each axis. 1: $Z \rightarrow Y$ - It rotates from the Z-axis forward direction to the Y-axis forward direction. 3: $X \rightarrow Z$ - It rotates from the X-axis forward direction to the Z-axis forward direction. 5: $Y \rightarrow X$ - It rotates from the Y-axis forward direction to the X-axis forward direction.
Forward direction for the coordinate system on rotation axis 1		
Forward direction for the coordinate system on tilt axis 2		
Forward direction for the coordinate system on rotation axis 2		

Name	Setting range	Description
Rotation center X coordinate offset for tilt axis 1	-9999.999 to 9999.999 mm	Set the offset from the rotation center coordinate that is set in the machine parameter (system 2) when necessary.
Rotation center Y coordinate offset for tilt axis 1		
Rotation center Z coordinate offset for tilt axis 1		
Rotation center X coordinate offset for rotation axis 1		
Rotation center Y coordinate offset for rotation axis 1		
Rotation center Z coordinate offset for rotation axis 1		
Rotation center X coordinate offset for tilt axis 2		
Rotation center Y coordinate offset for tilt axis 2		
Rotation center Z coordinate offset for tilt axis 2		
Rotation center X coordinate offset for rotation axis 2		
Rotation center Y coordinate offset for rotation axis 2		
Rotation center Z coordinate offset for rotation axis 2		
Absolute coordinate display format	0: :Workpiece coordinate. system 1: Feature coordinate system	Set the absolute coordinate display on the position screen to display the absolute coordinate values based on the workpiece coordinate system, or to display the absolute coordinate values based on the feature coordinate system.

(Note) The alarm <<Parameter set. error in feature coordinate set. function>> is triggered in the following situations.

- The axis used for the <Tilt axis*> or <Rotation axis*> is set to another <Tilt axis*> or <Rotation axis*>.
- The <*axis install. position> is different than the additional axis specified for the pair: <Tilt axis*> and <Rotation axis*>.
- The <Forward direction for the coordinate system on the tilt axis*> and <Forward direction for the coordinate system on the rotation axis*> are set on the same plane.

1.3 Screen Display

Current position				2018/05/11 14:12:58			
Program(5555)							
Relative coord. pos		Absolute coord. pos		Override		Load monitor	
X	-50.000	X	70.111(F)	Rapid feed	100 %	SP	
Y	-100.000	Y	40.875(F)	Cut. feed	100 %	X	
Z	250.000	Z	-28.887(F)	Spindle	100 %	Y	
B		B				Z	
A	0.000	A	0.000			B	
B	0.000	B	0.000			A	
C	0.000	C	0.000			B	
C		C				C	
Machine coord. pos		Remaining distance					
X	-50.000	X	0.000	Door		Spindle speed	
Y	-100.000	Y	0.000	Door	Close	0 min ⁻¹	
Z	250.000	Z	0.000			Feedrate	
B		B					
A	0.000	A	0.000	Door interlock		Magazine	
B	0.000	B	0.000	Valid		Tool No. 02	
C	0.000	C	0.000			Magazine No. 02	
C		C					
[HELP] ☒ Workpiece coordinate zero							
[HELP] ☒ Modal 1							
Current position	Modal 1	Modal 2	Soft limit	Enlargement	Workpiece coordinate zero	Select add. axes/change display	Counter reset

Absolute coordinate position:

In the user parameter (feature coordinate setting), the display can be selected to either show the values based on the workpiece coordinate system or the values based on the feature coordinate system.

(F) is shown on the right of the X-, Y- and Z-axes when displaying in the feature coordinate system.

Workpiece coordinate zero				2018/04/20 11:33:13			
Program(MY(0001))							
G54	G92	G52	External				
X 0.000	X 0.000	X 0.000	X				
Y 0.000	Y 0.000	Y 0.000	Y				
Z 0.000	Z 0.000	Z 0.000	Z				
A 0.000	A 0.000	A 0.000	A				
B 0.000	B 0.000	B 0.000	B				
C 0.000	C 0.000	C 0.000	C				
G54.2 P0	G54.2(Current)	Workpiece coordinate zero					
X	X	X	X				
Y	Y	Y	Y				
Z	Z	Z	Z				
A	A	A	A				
B	B	B	B				
C	C	C	C				
				[HELP] ☒ Enlargement			
				[HELP] ☒ Current position			
Current position	Modal 1	Modal 2	Soft limit	Enlargement	Workpiece coordinate zero	Select add axes/change display	Counter reset

Feature coordinate machining mode

Feature coordinate machining mode cancel

Workpiece coordinate zero				2018/04/20 10:04:21				
Program(0033)								
G58	G92	G52	External					
X 0.000	X 0.000	X 0.000	X					
Y 0.000	Y 0.000	Y 0.000	Y					
Z 0.000	Z 0.000	Z 0.000	Z					
A 0.000	A 0.000	A 0.000	A					
B 0.000	B 0.000	B 0.000	B					
C 0.000	C 0.000	C 0.000	C					
G58.2	Workpiece coordinate zero							
X	X	X	X					
Y	Y	Y	Y					
Z	Z	Z	Z					
A	A	A	A					
B	B	B	B					
C	C	C	C					
G53.1	Workpiece coordinate zero							
R	R	R	R					
				[HELP] ☒ Enlargement				
				[HELP] ☒ Current position				
Current position	Modal 1	Modal 2	Soft limit	Enlargement	Workpiece coordinate zero	Select add axes/change display	Counter reset	

G68.2:

The XYZ workpiece coordinate values (shift amount from workpiece coordinate zero) at zero in the feature coordinate system is displayed.

G53.1:

The amount that the additional axis is rotated for the feature coordinate index is displayed.

R shows the rotation amount (angle) from XY in the workpiece coordinate system for the XY in the feature coordinate system.

Workpiece coordinate zero:

The machine coordinate is displayed for the zero point position in the workpiece coordinate system that is currently being used for control.

The zero point in the workpiece coordinate system - the feature coordinate system setting is based on - and the machine coordinates that include each shift amount are displayed.

1.4 Operation Example

The following shows a motion for a M140X2.

Machine parameter

<5th-axis address>: 0 (A)

<7th-axis address>: 2 (C)

User parameter < Feature coordinate setting >

<Tilt axis 1>: 2 (5th-axis)

<Rotation axis 1>: 4 (7th-axis)

<Forward direction for the coordinate system on tilt axis 1>: 1 ($Z \rightarrow Y$)

<Forward direction for the coordinate system on rotation axis 1>: 5 ($Y \rightarrow X$)

<Rotation center X coordinate offset for tilt axis 1>: 0.000 (Note) Same for Y and Z coordinates

<Rotation center X coordinate offset for rotation axis 1>: 0.000 (Note) Same for Y and Z coordinates

Workpiece coordinate zero (G54)

X = -200.000, Y = -300.000, Z = 250.000, A = 0.000, C = 0.000

Program

G90 G54;

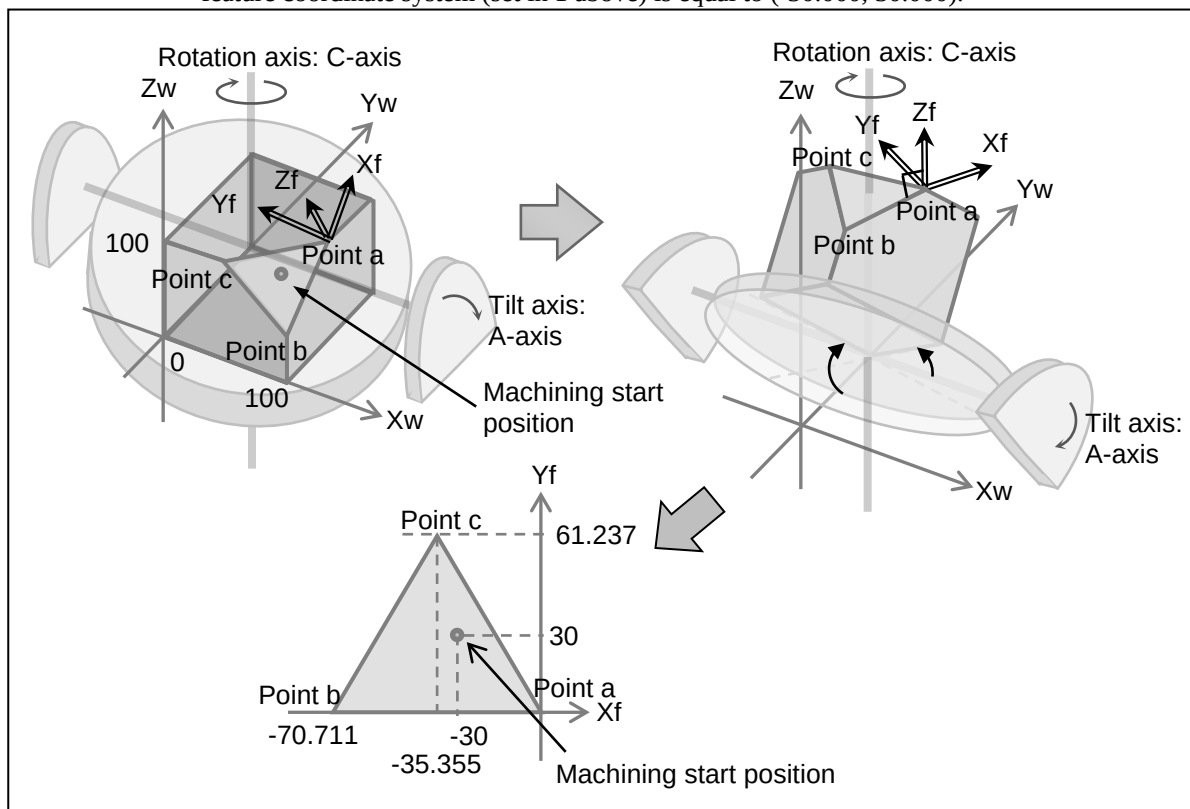
G68.2 X100. Y50. Z100. I45. J54.736 K60.; ← Feature coordinate setting command (1.)

G53.1; ← Feature coordinate index (2.)

G00 X-30. Y30.; ← Positioning up to machining start position (3.)

Program description

1. When the feature coordinate setting command (G68.2) is executed, the feature coordinate system (X_f , Y_f , Z_f) is set for point a which becomes the zero point.
2. When the feature coordinate index command (G53.1) is executed, the A-axis and C-axis rotate so that the X_f -axis + direction in the feature coordinate system (set in 1 above) is in line with the tool axis direction.
3. When the positioning command is executed, positioning is carried out so that (X_f , Y_f) in the feature coordinate system (set in 1 above) is equal to (-30.000, 30.000).



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CHAPTER 12 (20)

CYCLONE FILTER DEVICE

The description in this chapter is omitted because this product is not equipped with this function.

CHAPTER 12 (21)

CHIP AUGER (COIL CONVEYOR)

The description in this chapter is omitted because this product is not equipped with this function.

CHAPTER 12 (22)

TOOL CLEANING SYSTEM

- 1 Handling Precautions**
- 2 Function**
- 3 Operation**

1 Handling Precautions

WARNING

If the maintenance cover is removed, coolant coming from inside the machine when it is operating may get into your eyes, or the tools or workpieces may shoot out causing injury.

The operator should visually check to make sure that the maintenance cover is attached before turning ON the power.

The operator should warn the supervisor if the maintenance cover is not attached.

The supervisor must attach the maintenance cover.

WARNING

If this is not observed, someone may trip over the machine, or nearby equipment, or slip on coolant and fall over while setup or maintenance work is being carried out.

Protective equipment such as safety footwear and helmets must be worn while working.

The area around the machine must be organized, tidy and clean before performing work.

Any excess length of cable should be removed so that there is no slack. If cables are running along the floor, they must be covered.

WARNING

The coolant may shoot out and injure your eyes while work is being carried out.

Always be sure to wear protective goggles.

If coolant gets into eyes, rinse with clean water and then seek medical advice.

This function's operation is performed in standard ladder 2 for the built-in PLC function.

The user can assume that there is a factory-set ladder program embedded already and operating for the user operations and machine operations noted below.

Do not change standard ladder 2 which is used for this coolant unit.

In addition, do not change the corresponding contact.

If the user changes the ladder program, the product will no longer be covered under the warranty.

In addition, do not perform the debug function (stop, ON/OFF for related contacts, etc.) for standard ladder 2 while this function is operating.

2 Function

In this system, the coolant is cleaned as it passes through a line filter and is discharged from the spindle tip's perimeter. It keeps the tip of the spindle and the clamping surfaces of the tool holder clean.

3 Operation

1. Turn ON the power for the machine.
2. Turn ON the **[CLT.P]** key.
3. Press the **[MDI]** key to switch to the MDI operation mode screen.
4. Make sure that the front door (other doors besides front door on models equipped with QT; front door and side door on models equipped with the side door option but without the QT) is closed.

For NC language

5. Press the **[CAN]** key to clear the editing area.
6. Use the number keys to enter "G100T101".

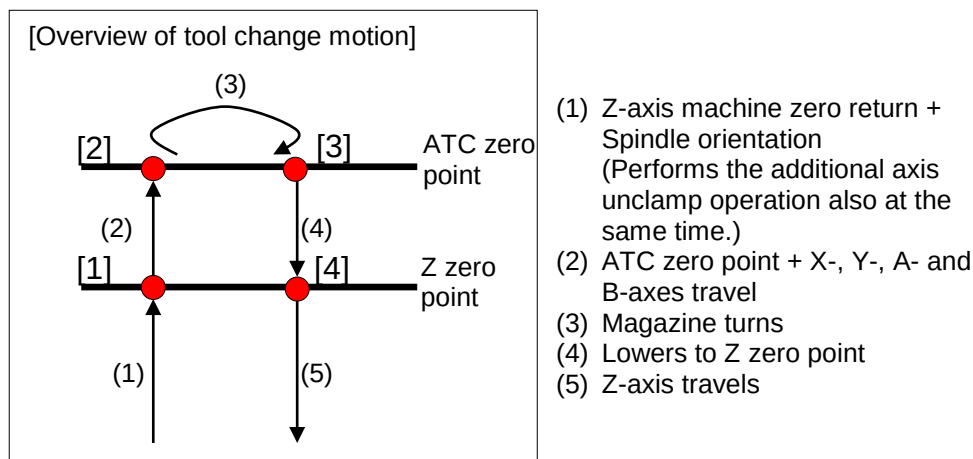
For conversation language

5. Use the **[CURSOR]** keys to move the cursor to <Tool>.
6. Use the number keys to enter "1" and press the **[ENT]** key.
7. Press the **[START]** switch to change the tool and discharge coolant while the ATC arm is turning.

3.1 Motion

3.1.1 Turret Type ATC

When the machine parameter (system 1) <Tool wash start position> is set to <0>:



Tool wash ON timing

- When the number of magazine turn pitches for one tool change operation (during memory operation and MDI operation) is less than the setting for the machine parameter (system 1) <Tool wash start position change offset>, then the tool wash coolant discharges following the timing indicated by [1].
- When the number of magazine turn pitches for one tool change operation (during memory operation and MDI operation) is greater than the setting for the machine parameter (system 1) <Tool wash start position change offset>, then the tool wash coolant discharges following the timing indicated by [2].
- When performing a tool change operation in manual mode, the tool wash coolant discharges following the timing indicated by [3].
- If the rapid traverse override is used for the timing in (2) or (3), the tool wash coolant discharge stops at the point when the override is used. Then, the tool wash coolant discharge starts again following the timing indicated by [3].

Tool wash OFF timing

- When the Z-axis goes beyond the Z zero point during Z-axis travel for (5), the tool wash coolant stops following the timing indicated by [4].
- If the rapid traverse override is used for the timing in (2) or (3), the tool wash coolant stops at the point when the override is used.

3.1.2 Arm Type ATC

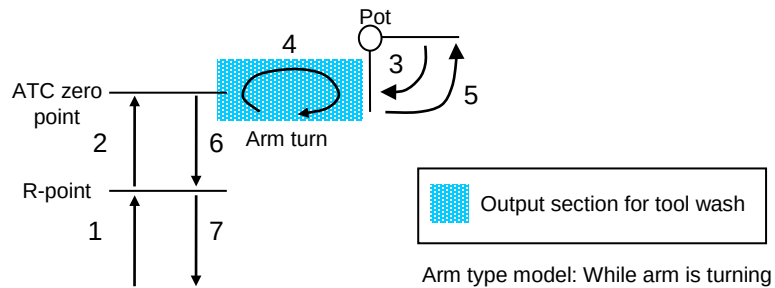
The tool wash coolant discharges when the tool is changed in MDI operation mode or in memory operation mode.

If in manual operation mode, the tool wash coolant discharges when the arm turns.

When the machine parameter (system 1) <Tool wash start position> is set to <6: Arm turn>:

The tool is washed following the timing below.

(The numbers in the figure below indicate the order of the operations. Excluding numbers 3 and 5, which refer to the motion of the X- and Y-axes.)



The tool wash is carried out after the machine parameter (system 1) <Pot lowering check time 1>.

CHAPTER 12 (23)

INVOLUTE INTERPOLATION FUNCTION

1 Involute Interpolation Function

1 Involute Interpolation Function

1.1 Overview

1.1.1 Introduction

This function enables the user to machine along an involute curve.

By using the involute interpolation, the user does not need to approximate minute or fine straight lines and arcs for an involute curve but can machine the curve with good accuracy.

In addition, the involute interpolation override function and high accuracy mode can be used to machine with even better accuracy.

1.1.2 Usage Conditions

The involute interpolation function option is required to enable and use this function.

This function can only be used in NC language mode.

1.2 Command Format

Command format

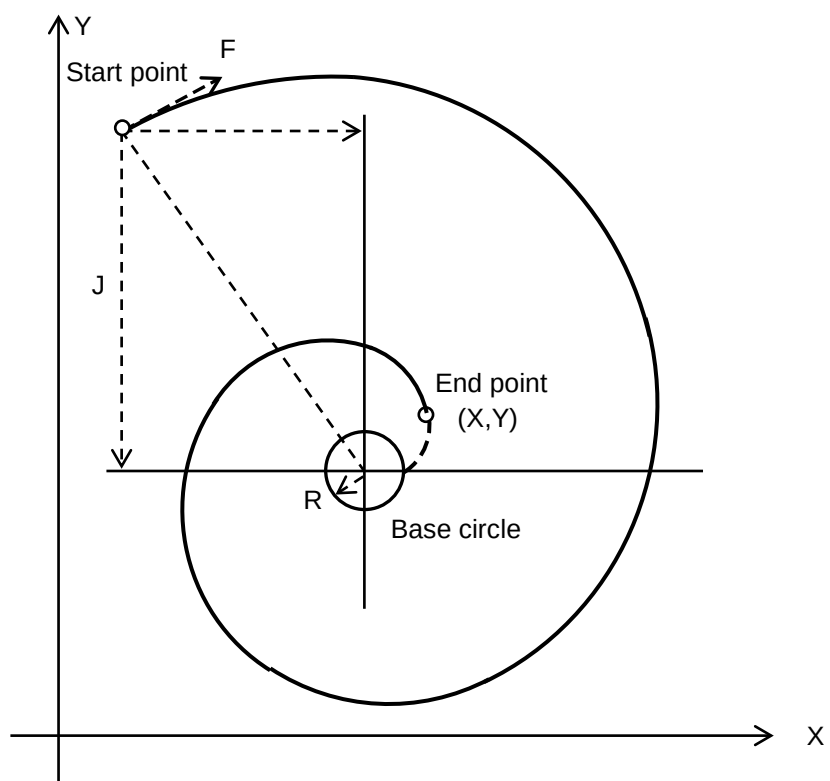
```
For X-Y plane:
{G17} G02.2 X_Y_I_J_R_F_;
{G17} G03.2 X_Y_I_J_R_F_;
For Z-X plane:
{G18} G02.2 Z_X_K_I_R_F_;
{G18} G03.2 Z_X_K_I_R_F_;
For Y-Z plane:
{G19} G02.2 Y_Z_J_K_R_F_;
{G19} G03.2 Y_Z_J_K_R_F_;
```

A description of the commands is shown below.

Rotation direction		G02.2	Clockwise (CW)
		G03.2	Counterclockwise (CCW)
End point	G90 mode	X, Y, Z	End position in workpiece coordinate system
	G91 mode	X	Distance in X-axis direction between the start and end points
		Y	Distance in Y-axis direction between the start and end points
		Z	Distance in Z-axis direction between the start and end points
Distance from start point to center point		I	Distance in X-axis direction between the start and center points
		J	Distance in Y-axis direction between the start and center points
		K	Distance in Z-axis direction between the start and center points
Base circle radius		R	Radius of base circle
Feedrate		F	Speed along tangent line of involute curve

On an XY plane, clockwise and counterclockwise refer to the rotation directions when looking at the negative direction from the positive direction on the Z-axis.

Ex: Involute interpolation command (G02.2) for G17 (X-Y) plane



1.2.1 Start Point and End Point

The start point on an involute curve is the current position.

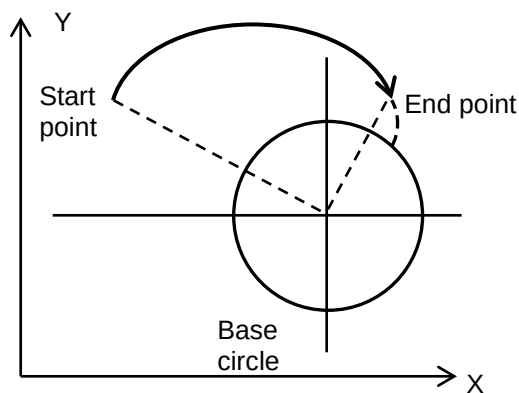
The end point on an involute curve follows the command that is based on the address X, Y and Z. When an end point command is not carried out, or when an axis that is not on the selected plane is used in the command, then the alarm <<Involute interpolation command error>> is triggered.

When the start point or end point on an involute curve is on the inside of the base circle, the alarm <<Involute interpolation command error>> is triggered because an involute curve is not required. The same alarm is triggered when the start and end points are inside the base circle after being corrected or compensated due to cutter compensation or nose R compensation.

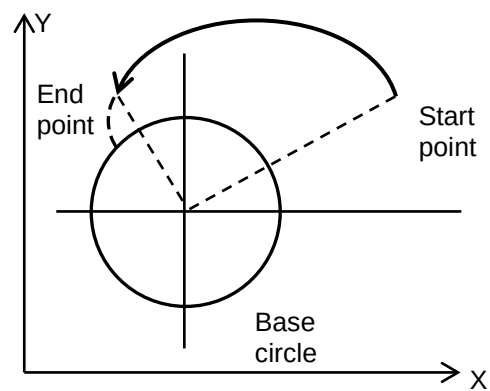
Depending on the relative distance between the center of the base circle, the start point and the end point, the involute curve will either curve toward the base circle or away from it.

The involute curve will curve toward the base circle when the end point is closer to the center of the base circle than the start point.

When using G02.2

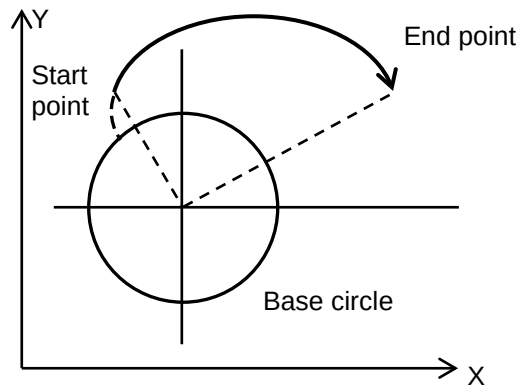


When using G03.2

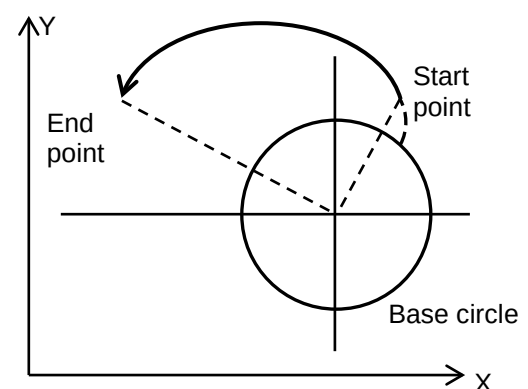


The involute curve will curve away from the base circle when the end point is further away from the center of the base circle than the start point.

When using G02.2



When using G03.2



When the start point and end point are equidistant from the center of the base circle, the alarm <<Involute interpolation command error>> is triggered because it cannot decide on a direction for the involute curve.

1.2.2 Base Circle Center Coordinate

I, J and K are used to issue a command for the center coordinate of the base circle. The command value is always an increment from the start point regardless of G90/G91. The alarm <<Involute interpolation command error>> is triggered in the following situations.

- When there is no I, J or K command
- When the command values for I, J and K are all zero (0)
- When a command was issued for an axis that is not inside the selected plane

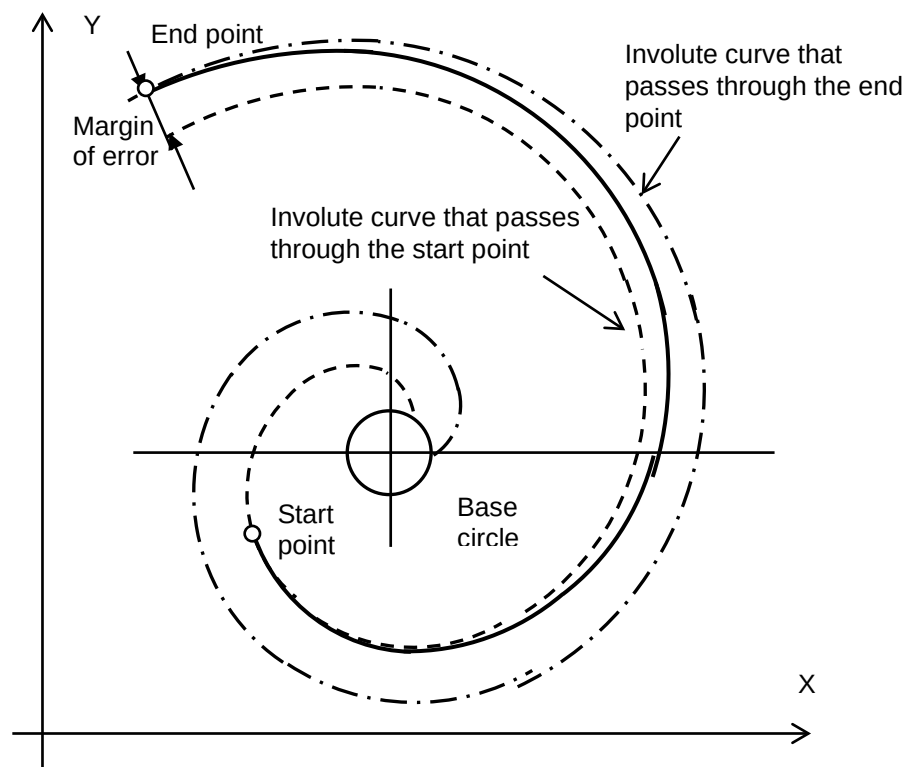
1.2.3 Base Circle Radius

The radius of the base circle uses a positive value in the R address to issue a command. The alarm <<Involute interpolation command error>> is triggered when the command value is less than zero (0).

1.3 Margin of Error for End Point on Involute Curve

When there is no end point for a command on the involute curve that passes through the start point, the following operations apply.

- When the margin of error, between the involute curve that passes through the start point and the end position of the involute curve that passes through the end point, is less than the user parameter (switch 1) <Allowable margin of error for involute interpolation>, then the curve heads toward the end point for the command.
- When the margin of error, between the involute curve that passes through the start point and the end position of the involute curve that passes through the end point, exceeds the user parameter (switch 1) <Involute interpolation error limit>, then the alarm <<Involute interpolation error limit exceeded>> is triggered.



1.4 Involute Interpolation Command During Cutter Compensation

An involute interpolation command can be issued during offset mode for cutter compensation (G41/G42) and nose R compensation (G141/G142). The involute interpolation during cutter compensation uses an imaginary circle for the start point and end point and carries out cutter compensation for that imaginary circle. Then, this function interpolates on the involute compensation curve relative to the results of the cutter compensation.

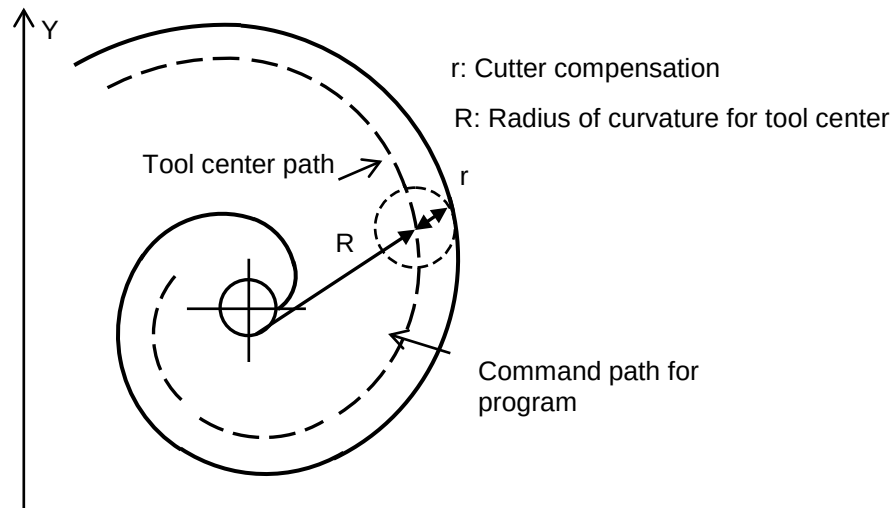
(Note 1) When an involute interpolation command is issued in startup and cancel mode, the alarm <<Cutter compensation error>> is triggered.

(Note 2) When the next block for the involute interpolation is a cutter compensation cancel command, the position that makes up a vertical vector to the end point of the involute interpolation becomes the end point.

1.4.1 Involute Interpolation Override

The radius of curvature is not fixed for an involute curve. Therefore, even if the speed for the tool center is constant or fixed, the speed on the cutting plane will not be constant and changes depending on the radius of curvature. In particular, the radius of curvature is small close to the base circle, and therefore that change becomes bigger.

The involute interpolation override function adjusts the actual feedrate to a speed that matches the radius of curvature, so that the speed on the cutting plane becomes the command value.



Cutter compensation	Override value	Actual feedrate
Inside workpiece	$R/(R + r)$	Command speed × Override value
Outside of workpiece	$R/(R - r)$	

(Note 1) When cutting the inside of the workpiece in cutter compensation mode, the override value may become extremely small as it gets closer to the base circle. The lower limit can be set in the user parameter (switch 1) <Involute interpolation override limit>. If the calculated override value is less than the <Involute interpolation override limit>, the override value is calculated using the <Involute interpolation override limit> value.

$$\text{Actual feedrate} = \text{Command speed} \times \frac{\text{<Involute interpolation override limit>}}{100}$$

(Note 2) When cutting the outside of the workpiece in cutter compensation mode, the actual feedrate is greater than the command speed. When the feedrate for each axis exceeds the following parameters, the speed is clamped at the following values.

- User parameter (switch 1) <Max. actual cutting travel speed>
- Machine parameter <System 1>: <X- to Z-axis max. cutting feedrate>
- Machine parameter <High accuracy>: <Max. feedrate A/B>

(Note 3) When the user parameter (switch 1) <Involute interpolation override limit> is zero (0), the involute interpolation override function is disabled. The override value is fixed at 100% and it always operates at the command speed.

1.5 Involute Interpolation Command in High Accuracy Mode

An involute interpolation command is possible in high accuracy mode (high accuracy A / high accuracy B / quick setting for high accuracy). The user can machine with even better accuracy by using the <Automatic arc deceleration function> in high accuracy mode.

(Note 1) An involute interpolation command is not possible with the fully automatic deceleration function in high accuracy mode A, nor is it possible in accuracy specification mode in the quick setting for high accuracy. When there is a command, the alarm <<High accuracy A invalid command>> and <<Invalid command in accuracy specification mode>> are triggered.

(Note 2) Smooth path offset function is cancelled when there is an involute interpolation command.

1.6 Restrictions

- A command is only possible when the start and end points for the involute interpolation command are less than 100 rotations from the starting point of the involute curve. When the command is more than 100 rotations, the alarm <<Involute interpolation command error>> is triggered.
- When the corner CR is specified in the preceding block, the alarm <<Specified G code cannot be used.>> is triggered.
- The corner CR cannot be specified for the involute interpolation. When a command is issued, the alarm <<Invalid data>> is triggered.
- An involute interpolation command is not possible when the mirror image is enabled (G51.1). When a command is issued, the alarm <<Mirror image mode>> is triggered.
- An involute interpolation command is not possible when the scaling is enabled (G51). When a command is issued, the alarm <<Scaling>> is triggered.
- After the feature coordinate setting, an involute interpolation command cannot be issued before the feature coordinate index. When a command is issued, the alarm <<Feature coordinate manufacturing mode engaged>> is triggered.
- An involute interpolation command is not possible while in the inverse time feed (G93) modal. If a command is issued, the alarm <<Command not possible during inverse time feed>> is triggered.
- When the following command is issued during the involute interpolation modal, the alarm <<Invalid command for involute interpolation modal>> is triggered.
 - Tool length offset (G43, G44) and tool position compensation (G143, G144)
 - Coordinate calculation function (G36 to G39)
 - Scaling (G51) and mirror image (G51.1)
 - Automatic workpiece measurement (G121 to G129)
 - Inverse time feed (G93)

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